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COURSE IN MATHEMATICS 6

for the IIT-JEE & Other Engineering Entrance Examinations

Calculus II



K. R. Choubey
Ravikant Choubey
Chandrakant Choubey

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PEARSON

COURSE IN MATHEMATICS

(FOR IIT JEE AND OTHER ENGINEERING ENTRANCE EXAMINATIONS)

CALCULUS TWO

K.R. CHOUBEY

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PEARSON

Chandigarh • Delhi • Chennai

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CONTENTS

Preface *v*

PART A INDEFINITE INTEGRAL

Lecture 1	**Simple Integrals**	**A.3—A.12**
Lecture 2	**Standard Integral Using Substitution**	**A.13—A.28**
Lecture 3	**Integration of Algebraic Expressions 1**	**A.29—A.46**
Lecture 4	**Integration of Algebraic Expressions 2**	**A.47—A.63**
Lecture 5	**Integration by Parts**	**A.65—A.77**
Lecture 6	**Reduction Formulae**	**A.79—A.91**
Lecture 7	**Integration of Different Forms**	**A.93—A.118**
Lecture 8	**Test Your Skills**	**A.119—A.138**

PART B DEFINITE INTEGRATION AND AREAS UNDER CURVE

Lecture 1	**Basics of Definite Integral**	**B.3—B.20**
Lecture 2	**Properties of Definite Integral**	**B.21—B.40**
Lecture 3	**Some more Properties of Definite Integral**	**B.41—B.64**
Lecture 4	**Estimation of Integral and other Definite Integrals**	**B.65—B.83**
Lecture 5	**Definite Integral as a Limit of Sum**	**B.85—B.94**
Lecture 6	**Area Under Curve 1**	**B.95—B.109**
Lecture 7	**Area Under Curve 2**	**B.111—B.137**
Lecture 8	**Test Your Skills**	**B.139—B.169**

PART C DIFFERENTIAL EQUATIONS

Lecture 1	**Order and Degree of the Differential Equation**	**C.3—C.22**
Lecture 2	**Solution of a Differential Equation**	**C.23—C.45**
Lecture 3	**Various Types of Differential Equations**	**C.47—C.63**
Lecture 4	**Solutions of Exact Differential Equations**	**C.65—C.88**
Lecture 5	**Test Your Skills**	**C.89—C.102**

PREFACE

When a new book is written on a well known subject like *Calculus* for class XI/XII Academics/AIEEE/IIT/ State engineering entrance exams and NDA, several questions arise like—why, what, how and for whom? What is new in it? How is it different from the other books? For whom is it meant? The answers to these questions are often not mutually exclusive. Neither are they entirely satisfactory except perhaps to the authors. We are certainly not under the illusion that there are no good books. There are many good books available in the market.

However, none of them caters specifically to the needs of students. Students find it difficult to solve most of the problems of any of the books in the absence of proper planning. This inspired us to write this book *Calculus II*, to address the requirements of students of class XI/XII CBSE and State Board Academics. In this book, we have tried to give a connected and simple account of the subject. It gives a detailed, lecture wise description of basic concepts with many numerical problems and innovative tricks and tips. Theory and problems have been designed in such a way that the students can themselves pursue the subject. We have also tried to keep this book self contained. In each lecture all relevant concepts, prerequisites and definitions have been discussed in a lucid manner and also explained with suitable illustrated examples including tests.

Due care has been taken regarding the Board (CBSE/State) examination need of students and nearly 100 per cent articles and problems set in various examinations including the IIT-JEE have been included.

The presentation of the subject matter is lecturewise, intelligent and systematic, the style is lucid and rational, and the approach is comprehensible with emphasis on improving speed and accuracy. The basic motive is to attract students towards the study of mathematics by making it simple, easy and interesting and on a day-to-day basis. The instructions and method for grasping the lectures are clearly outlined topic wise. The presentation of each lecture is planned for better experiential learning of mathematics which is as follows:

1. Basic Concepts: Lecture Wise
2. Solved Subjective Problems (XII Board (C.B.S.E./State): For Better Understanding and Concept Building of the Topic.
3. Unsolved Subjective Problems (XII Board (C.B.S.E./State): To Grasp the Lecture Solve These Problems.
4. Solved Objective Problems: Helping Hand.
5. Objective Problem: Important Questions with Solutions.
6. Unsolved Objective Problems (Identical Problems for Practice)
   For Improving Speed with Accuracy.

7. Worksheet: To Check Preparation Level
8. Assertion-Reason Problems: Topic Wise Important Questions and Solutions with Reasoning
9. Mental Preparation Test: 01
10. Mental Preparation Test: 02
11. Topic Wise Warm Up Test: 01: Objective Test
12. Topic Wise Warm Up Test: 02: Objective Test
13. Objective Question Bank Topic Wise: Solve These to Master.

This book will serve the need of the students of class XI/XII board, NDA, AIEEE and SLEEE (state level engineering entrance exam) and IIT-JEE. We suggest each student to attempt as many exercises as possible without looking up the solutions. However, one should not feel discouraged if one needs frequent help of the solutions as there are many questions that are either tough or lengthy. Students should not get frustrated if they fail to understand some of the solutions in the first attempt. Instead they should go back to the beginning of the solution and try to figure out what is being done At the end of every topic, some harder problems with 100 per cent solutions and Question Bank are also given for better understanding of the subject.

There is no end and limit to the improvement of the book. So, suggestions for improving the book are always welcome.

We thank our publisher, Pearson Education for their support and guidance in completing the project in record time.

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Part A

Indefinite Integral

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LECTURE



1 Simple Integrals

BASIC CONCEPTS

DEFINITION

1. Integral of a Function

$$\int f(x)dx = F(x) \Leftrightarrow \frac{d}{dx}[F(x)] = f(x)$$

The process of finding the integral of a function is called **integration** and the given function is called **integrand**. It is obvious to note that the operation of integration is inverse operation of differentiation.

Hence integral of a function is also named as **anti-derivative** of that function.

2. Basic Theorems on Integration

If $f(x)$, $g(x)$ are two functions of a variable x and k is a constant, then:

(i) $\int kf(x)dx = k\int f(x)dx$

(ii) $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$

(iii) $\int \left(\frac{d}{dx} f(x) \right) dx = f(x) = \frac{d}{dx}(\int f(x)dx)$

3. Standard Integrals (Based on Differentiation)

1. $\int 0\, dx = \text{constant}$
2. $\int x^n dx = \frac{x^{n+1}}{n+1} + c (n \neq -1), \int \frac{dx}{x^{n+1}} = \frac{x^{-n}}{-n} + c$
3. $\int \frac{dx}{x^2} = -\frac{1}{x}$
4. $\int \frac{dx}{(ax+b)^2} = \frac{-1}{a(ax+b)}$
5. $\int \frac{dx}{\sqrt{x}} = 2\sqrt{x}$
6. $\int \frac{dx}{\sqrt{ax+b}} = \frac{2\sqrt{ax+b}}{a}$
7. $\int \frac{1}{x} dx = \log|x| + c, \int \frac{dx}{x^n} = -\frac{1}{(n-1)x^{n-1}}, n \neq 1$
8. $\int e^x dx = e^x + c$
9. $\int a^x dx = a^x / \log a + c$
10. $\int \sin x dx = -\cos x + c$
11. $\int \cos x dx = \sin x + c$
12. $\int \tan x dx = \log|\sec x| + c \quad \text{or} -\log|\cos x|$
13. $\int \cot x dx = \log|\sin x| + c$
14. $\int \sec x dx = \log(\sec x + \tan x) = \log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$
15. $\int \operatorname{cosec} x dx = \log(\operatorname{cosec} x - \cot x) = \log \tan \frac{x}{2}$
16. $\int \sec^2 x dx = \tan x + c$
17. $\int \operatorname{cosec}^2 x dx = -\cot x + c$
18. $\int \sec x \tan x dx = \sec x + c$
19. $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$

20. $\int \sec x \operatorname{cosec} x dx = \log \tan x + c$

21. $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$ or $-\cos^{-1} x + c$ for $|x| < 1$

22. $\int \frac{dx}{\sqrt{x^2-a^2}} = \log[x + \sqrt{x^2-a^2}]$ or $\cos h^{-1}(x/a)$

23. $\int \frac{dx}{\sqrt{x^2+a^2}} = \log[x + \sqrt{x^2+a^2}]$ or $\sin h^{-1}\left(\frac{x}{a}\right)$

4. **Non-Integrable or Inexpressible Integrals** If an integral $\int f(x)dx$ is expressed in terms of elementary functions then integral is known as computable (or "expressible"). But if an integral $\int f(x)dx$ is not expressed in terms of elementary functions then integral is known as "Inexpressible" or that is "can not be found".

Therefore the integrals

$\int e^{x^2} dx, \int e^{-x^2} dx, \int \sqrt{\sin x} dx, \int \sqrt{\cos x} dx,$

$\int x \tan x \, dx, \int \frac{\sin x}{x} dx, \int \frac{\cos x}{x} dx,$

$\int \frac{dx}{\ln x}, \int \left(\sqrt{1+x^3}\right) dx, \int \sin x^2 dx,$

$\int \sqrt{(1-k^2 \sin^2 x)} dx,$

$\int \cos x^2 dx, \int \frac{x^2}{1+x^5} dx,$

$\int \frac{dx}{\sqrt{(1-x^2)(1-k^2x)}} (k^2 < 1), \int x^x dx$

are said to be "inexpressible" or non-integrable.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate: $\int \frac{2\cos x}{3\sin^2 x} dx$

[AI-2008]

Solution

$$\frac{2}{3}\int \operatorname{cosec} x . \cot x dx = -\frac{2}{3}\operatorname{cosec} x + c$$

2. Evaluate: $\int \frac{e^{5\log x} - e^{4\log x}}{e^{3\log x} - e^{2\log x}} dx$

Solution

$$\int \frac{x^5 - x^4}{x^3 - x^2} dx = \int x^2 dx = \frac{x^3}{3} + c$$

3. If $f(x)$ is the integral of $\frac{2\sin x - \sin 2x}{x^3}$, $x \neq 0$, find $\lim_{x \to 0} f'(x)$.

Solution

Given $f(x) = \int \frac{2\sin x - \sin 2x}{x^3} dx, x \neq 0$

$\therefore$ By def. $f'(x) = \frac{2\sin x - \sin 2x}{x^3}, x \neq 0.$

$\therefore$ $\lim_{x \to 0} f'(x) = \lim_{x \to 0} \frac{2\sin x - \sin 2x}{x^3}$

$= \lim_{x \to 0} \frac{2\sin x - 2\sin x \cos x}{x^3}$

$= 2 . \lim_{x \to 0} \frac{\sin x}{x} . \lim_{x \to 0} \frac{1-\cos x}{x^2}$

$= 2.1 . \lim_{x \to 0} \frac{1-\cos x}{x^2} . \frac{1+\cos x}{1+\cos x}$

$= 2 \lim_{x \to 0} \frac{\sin^2 x}{x^2(1+\cos x)}$

$= 2\left(\lim_{x \to 0} \frac{\sin x}{x}\right)^2 . \frac{1}{1+\lim_{x \to 0} \cos x}$

$= 2.1^2 . \frac{1}{1+1} = \frac{2}{2} = 1$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Integrate the following with respect to x.
$\frac{x^3+3x^2+4}{\sqrt{x}}$

2. Integrate the following with respect to x
(i) $\frac{6}{\sqrt{1-x^2}}+3\sec^2 x$
(ii) $\frac{\sin x}{\sin(a-\alpha)}$

3. Integrate the following trigonometrical functions with respect to x.
(i) $\frac{\sin^3 x+\cos^3 x}{\sin^2 x\times\cos^2 x}$
(ii) $\sqrt{1+\sin 2x}$

4. Evaluate $\int\frac{\cos(x-\alpha)}{\cos x}dx$

5. Evaluate (i) $\int e^{\log_e x}dx$ (ii) $\int e^{-\log_e x}dx$

[MP-98]

6. Evaluate $\int(\tan x+\cot x)^2 dx$.

[MP-2000]

7. Evaluate $\int\left(\frac{\sec x}{\sec x-\tan x}\right)dx$

[MP-88, 93, 96, 97, 99]

8. Evaluate $\int\sec^2(7-4x)dx$

[CBSE-2010]

ANSWERS

1. $\frac{2}{7}x^{7/2}+\frac{6}{5}x^{5/2}+8\sqrt{x}$
2. (i) $6\sin^{-1}x+3\tan x$ (ii) $\frac{-\cos x}{\sin(a-\alpha)}$
3. (i) $\sec x-\operatorname{cosec} x$ (ii) $\sin x+\cos x$
4. $x\cos\alpha+\sin\alpha\log(\sec x)$
5. (i) $x^2/2$ (ii) $\log_e x$
6. $\tan x-\cot x+c$
7. $\tan x+\sec x$
8. $\frac{\tan(7x-4)}{4}$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. If $f'(x)=\sqrt{x}$ and $f(1)=2$ then $f(x)$ is equal to
(a) $\frac{3}{2}x^{3/2}$ (b) $\frac{3}{2}x^{3/2}+4/3$
(c) $\frac{2}{3}x^{3/2}$ (d) $\frac{2}{3}x^{3/2}+4/3$

[VIT-2005]

Solution

(d) $f'(x)=\sqrt{x}\Rightarrow f(x)=\frac{2}{3}x^{3/2}+c$

But $f(1)=2\Rightarrow 2=\frac{2}{3}+c\Rightarrow c=\frac{4}{3}$

$\therefore\quad f(x)=\frac{2}{3}x^{3/2}+\frac{4}{3}$

2. $x = f'(t)\cos t + f(t)\sin t$

$y = -f''(t)\sin t + f'(t)\cos t,$

then $\int\left[\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2\right]^{1/2} dt$ equals

(a) $f'(t)+f'(t)+c$
(b) $f''(t)+f'''(t)+c$
(c) $f(t)+f''(t)+c$
(d) $f'(t)-f''(t)+c$ ***[SCRA-99]***

Solution

(c) $\frac{dx}{dt} = f'''(t)\cos t - f''(t)\sin t + f''(t)\sin t + f'(t)\cos t$

$= [f'''(t)+f'(t)]\cos t$

$\frac{dy}{dt} = -f'''(t)\sin t - f''(t)\cos t + f''(t)\cos t - f'(t)\sin t$

$= -[f'''(t)+f'(t)]\sin t$

$\therefore \left[\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2\right]^{1/2}$

$= [(f'''(t)+f'(t))^2(\cos^2 t+\sin^2 t)]^{1/2}$

$= f'''(t)+f'(t)$

$\therefore \int\left[\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2\right]^{1/2} dt$

$= f''(t)+f(t)+c$

3. $\lim_{x\to 0}\frac{d}{dx}\int\frac{1-\cos x}{x^2}dx$ equal to ***[Orissa JEE-2007]***

(a) 1
(b) 0
(c) 1/2
(d) None of these

Solution

(c) $\lim_{x\to 0}\frac{d}{dx}\int\left(\frac{1-\cos x}{x^2}\right)dx$

$= \lim_{x\to 0}\frac{1-\cos x}{x^2}$

$= \lim_{x\to 0}\frac{2\sin^2 x/2}{4.x^2/4}$

$= \frac{1}{2}\lim_{x\to 0}\left(\frac{\sin x/2}{x/2}\right)^2 = \frac{1}{2}$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int(\sin^4 x - \cos^4 x)\,dx =$ ***[RPET-2003]***

(a) $-\frac{\cos 2x}{2}+C$ (b) $-\frac{\sin 2x}{2}+C$

(c) $\frac{\sin 2x}{2}+C$ (d) $\frac{\cos 2x}{2}+C$

2. $\int 13^x\,dx$ is ***[Kerala (Engg.)-2002]***

(a) $\frac{13^x}{\log 13}+C$
(b) $13^{x+1}+C$
(c) $14x+C$
(d) $14^{x+1}+C$

3. $\int\frac{ax^3+bx^2+c}{x^4}dx$ equals to ***[RPET-2001]***

(a) $a\log x+\frac{b}{x^2}+\frac{c}{3x^3}+C$

(b) $a\log x+\frac{b}{x^2}-\frac{c}{3x^3}+C$

(c) $a\log x-\frac{b}{x}-\frac{c}{3x^3}+C$

(d) None of these

4. $\int\frac{\cos 2x-1}{\cos 2x+1}dx =$ ***[MPPET-2000]***

(a) $\tan x - x + C$ (b) $x + \tan x + C$

(c) $x - \tan x + C$ (d) $-x - \cot x + C$

5. $\int \frac{x^4+x^2+1}{x^2-x+1}dx$ =

(a) $\frac{1}{3}x^3+\frac{1}{2}x^2+x+c$ (b) $\frac{1}{3}x^3-\frac{1}{2}x^2+x+c$

(c) $\frac{1}{3}x^3+\frac{1}{2}x^2-x+c$ (d) None of these

6. $\int \frac{dx}{\sin^2 x\cos^2 x}$ =

[Roorkee-1976; RPET-1991]

(a) $\tan x+\cot x$
(b) $\cot x-\tan x+c$
(c) $\tan x-\cot x+c$
(d) None of these

7. $\int (e^{a\log x}+e^{x\log a})\,dx$ =

[Kerala PET-2001]

(a) $x^{a-1}+\frac{a^x}{\log a}+c$ (b) $\frac{x^{a+1}}{a+1}+a^x\log a+c$

(c) $\frac{x^{a+1}}{a+1}+\frac{a^x}{\log a}+c$ (d) None of these

8. $\int \frac{3x^3-2\sqrt{x}}{x}dx$ = ***[Roorkee-1976]***

(a) $x^3-\sqrt{x}+c$ (b) $x^3+\sqrt{x}+c$
(c) $x^3-2\sqrt{x}+c$ (d) $x^3-4\sqrt{x}+c$

9. If $f'(x)=\frac{1}{x}+x$ and $f(1)=\frac{5}{2}$, then $f(x)$ =

(a) $\log x+\frac{x^2}{2}+2$

(b) $\log x+\frac{x^2}{2}+1$

(c) $\log x-\frac{x^2}{2}+2$

(d) $\log x-\frac{x^2}{2}+1$

10. If $\int f(x)dx=xe^{-\log|x|}+f(x)$, then $f(x)$ is

(a) 1 (b) ce^x
(c) $\log x+c$ (d) e^x+c

11. $\int f(x)dx=2(f(x))^3+c$, then $f(x)$ is

[BIT Ranchi-1991]

(a) $x/2$ (b) $\sqrt{x/3}$
(c) $\sqrt{\frac{x}{3}}$ (d) $\frac{x^2}{2}$

12. $\int \frac{\cos 2x-\cos 2\alpha}{\cos x-\cos\alpha}dx$ = ***[MPPET-1994]***

(a) $2[\sin x+x\cos\alpha]+c$
(b) $2[\sin x+\sin\alpha]+c$
(c) $2[-\sin x+x\cos\alpha]+c$
(d) $-2[\sin x+\sin\alpha]+c$

13. $\int |x|dx$ is equal to

(a) $\frac{1}{2}x^2+c$ (b) $-\frac{x^2}{2}+c$

(c) $x|x|+c$ (d) $\frac{1}{2}x|x|+c$

14. $\int\{1+2\tan x(\tan x+\sec x)\}^{1/2}dx$ =

[Roorkee-1987]

(a) $\log(\sec x+\tan x)+c$
(b) $\log\sec x(\sec x+\tan x)+c$
(c) $\log(\sec x \quad \tan x)^{1/2}+c$
(d) None of these

15. If $f(0)=f'(0)=0$ and $f''(x)=\tan^2 x$, then $f(x)$ is

(a) $\log\sec x-\frac{1}{2}x^2$ (b) $\log\cos x+\frac{1}{2}x^2$

(c) $\log\sec x+\frac{1}{2}x^2$ (d) None of these

16. $\int \sec x\,dx$

[MPPET-1988, 95; RPET-1996]

(a) $\log\tan\left(\frac{x}{8}+\frac{x}{2}\right)+c$
(b) $-\log(\sec x-\tan x)+c$
(c) $\log(\sec x-\tan x)+c$
(d) None of these

17. $\int\left(x-\frac{1}{x}\right)^2dx$ =

(a) $\frac{1}{3}\left(x-\frac{1}{x}\right)^3+c$ (b) $\frac{x^3}{3}-2x-\frac{1}{x}+c$

(c) $\frac{x^3}{3}+2x+\frac{1}{x}+c$ (d) None of these

18. $\int\sqrt{1+\cos 2x}dx$ =

(a) $\sqrt{2}\sin x+c$
(b) $-\sqrt{2}\sin x+c$
(c) $\pm\sqrt{2}\cos x+c$
(d) $\pm\sqrt{2}\sin x+c$

SOLUTIONS

1. (b) Step 1: $\int\left(\sin^4 x-\cos^4 x\right)dx$

$\int\left[(\sin^2 x)^2-(\cos^2 x)^2\right]dx$

$\int\left[(\sin^2 x-\cos^2 x)\,(\sin^2 x+\cos^2 x)\right]dx$

$\int -\cos 2x\,dx$

($\because \sin^2 x+\cos^2 x=1$ and $\cos^2 x-\sin^2 x=\cos 2x$)

$= -\dfrac{\sin 2x}{2}+C$

NOTE

Formula $\int \cos ax\,dx=\dfrac{\sin ax}{a}+C$

2. (a) Step 1: $\int 13^x dx$

formula: $\int a^x dx=\dfrac{a^x}{\log a}+C$

$= \dfrac{13^x}{\log 13}+C$

3. (c) Step 1: $\int\dfrac{ax^3+bx^2+c}{x^4}dx$

$\int\left[\dfrac{ax^3}{x^4}+\dfrac{bx^2}{x^4}+\dfrac{c}{x^4}\right]dx$

$= \int\dfrac{a}{x}dx+\int\dfrac{b}{x^2}dx+\int\dfrac{c}{x^4}dx$

$= a\log x+\left(-\dfrac{b}{x}\right)-\dfrac{c}{3x^3}+C$

$= a\log x-\dfrac{b}{x}-\dfrac{c}{3x^3}+C$

NOTE

Formula: $\int\dfrac{1}{x}dx=\log x+C$

$\int\dfrac{1}{x^2}dx=-\dfrac{1}{x}+C$

$\int x^n dx=\dfrac{x^{n+1}}{n+1}+C$

4. (c) Step 1: $\int\dfrac{1-2\sin^2 x-1}{2\cos^2 x-1+1}dx=-\int\tan^2 x\,dx$

$= \int-(\sec^2 x-1)\,dx=-[\tan x-x]+C$

$= x-\tan x+C$

5. (a) $\int\dfrac{x^4+x^2+1}{x^2-x+1}dx$

$=\int(x^2+x+1)=\dfrac{x^3}{3}+\dfrac{x^2}{2}+x+c$

6. (c) Step 1: $\int\dfrac{dx}{\sin^2 x\cos^2 x}$

by using formula $\sin^2 x+\cos^2 x=1$

$I=\int\dfrac{\sin^2 x+\cos^2 x}{\sin^2 x\cos^2 x}dx$

$I=\int\dfrac{\sin^2 x}{\sin^2 x\cos^2 x}dx+\int\dfrac{\cos^2 x}{\sin^2 x\cos^2 x}$

$I=\int\sec^2 x dx+\int\text{cosec}^2 x dx$

$= \tan x-\cot x+c$

7. (c) Step 1: by using formula $a^{\log_e N}=N$, we get

$I=\int\left(e^{a\log x}+e^{x\log a}\right)dx$

$= \int(x^a+a^x)dx$

$\Rightarrow \quad \int x^a dx+\int a^x dx$

$= \dfrac{x^{a+1}}{a+1}+\dfrac{a^x}{\log a}+c$

8. (d) Step 1: $\int\dfrac{3x^3-2\sqrt{x}}{x}dx$

$\int\left(\dfrac{3x^3}{x}-\dfrac{2\sqrt{x}}{x}\right)dx$

$I=\int\left(3x^2-\dfrac{2}{\sqrt{x}}\right)dx=\int 3x^2dx-\int\dfrac{2}{\sqrt{x}}dx$

$= \dfrac{3x^3}{3}-2\times 2\sqrt{x}+c$

$= x^3-4\sqrt{x}+c$

NOTE

Formula $\int x^n dx = \frac{x^{n+1}}{n+1} + C$.

9. (a) Step 1: $f'(x) = \frac{1}{x} + x$

Integrating both side, we get

$$f(x) = \log x + \frac{x^2}{2} + C \quad \text{.....(1)}$$

But $f(1) = \frac{5}{2}$ (given)

$$\Rightarrow \quad \frac{5}{2} = \log 1 + \frac{1}{2} + C$$

$$C = \frac{5}{2} - \frac{1}{2} \Rightarrow C = 2$$

Put value of C in equation (1)

$$f(x) = \log x + \frac{x^2}{2} + 2$$

10. (c) Step 1: $e^{-\log|x|} = e^{\log|x|^{-1}} = |x|^{-1} = \frac{1}{|x|}$

Hence $xe^{-\log x} = \frac{x}{|x|}$ $= 1$ if $x > 0$,

$= -1$ if $x < 0$

Hence Integrand $xe^{\log|x|}$ is equal to constant k

Step 2: $\int f(x)dx = k + f(x)$ (here $k = 1$ or -1)

Therefore, function and integral of function both are same.

Hence $f(x) = ce^x$

11. (c) Step 1: $\int f(x)dx = 2(f(x))^3 + c$

Differentiating both side with respect to x

$$\frac{d}{dx}\int f(x)dx = \frac{d}{dx}2(f(x))^3 + \frac{d}{dx}(c)$$

$$f(x) = 6f(x)^2 f'(x) \Rightarrow f'(x)f(x) = \frac{1}{6}$$

integrating we get

$$\frac{(f(x))^2}{2} = \frac{x}{6}$$

$$f(x) = \sqrt{\frac{x}{3}}$$

12. (a) From formula, $\cos 2x = 2\cos^2 x - 1$ and $\cos 2\alpha = 2\cos^2 \alpha - 1$

$$I = 2\int \frac{\cos^2 x - \cos^2 \alpha}{(\cos x - \cos \alpha)} dx$$

$$= 2\int (\cos x + \cos \alpha) dx$$

$$= 2[\sin x + x\cos \alpha] + c$$

13. (b) Step 1: $|x| = +x$ if $x > 0$

$= -x$ if $x < 0$

Step 2: $\int |x| dx = \int -x dx = -\frac{x^2}{2} + c$ if $x < 0$

or $\int |x| dx = \int x dx = \frac{x^2}{2} + c$ if $x \geq 0$

$\therefore$ combining both $\int |x| dx = \frac{x|x|}{2} + c$

14. (b) $\int \{1 + 2\tan^2 x + 2\tan x \sec x\}^{1/2}$

$$= \int (\sec^2 x + \tan^2 x + 2\tan x \sec x)^{1/2} dx$$

$$= \left((\sec x + \tan x)^2\right)^{\frac{1}{2}} dx \quad (\because \sec^2 x - \tan^2 x = 1)$$

$\Rightarrow \quad \int (\sec x + \tan x) dx$

$= \log(\sec x + \tan x) + \log \sec x + c$

$= \log \sec x(\sec x + \tan x) + c$

15. (a) Step 1: $f''(x) = \tan^2 x$

and $f(0) = f'(0) = 0$

Integrating both sides with respect to x

$$\int f''(x)dx = \int \tan^2 x dx$$

$$f'(x) = \int (\sec^2 x - 1) dx$$

$$f'(x) = \tan x - x + c$$

$f'(0) = 0$ (given)

$c = 0$

$f'(x) = \tan x - x$

again, integrating both side with respect to x

$$\int f'(x)dx = \int (\tan x - x) dx$$

$$f(x) = \log \sec x - \frac{x^2}{2} + c$$

$f(0) = 0 \Rightarrow c = 0$

$$f(x) = \log \sec x - \frac{x^2}{2}$$

16. (b) Multiplying numerator and denominator by $\sec x - \tan x$

$$\int \frac{\sec x(\sec x - \tan x)dx}{\sec x - \tan x}$$

$$\int \frac{-(\sec x \tan x - \sec^2 x)dx}{\sec x - \tan x}$$

Step 2: By substitution method

$$t = \sec x - \tan x$$

Hence $dt = (\sec x \tan x - \sec^2 x)dx$

and $I = \int \frac{-dt}{t} = -\log t$

$$= -\log(\sec x - \tan x) + c$$

17. (c) Step 1: $\int \left(x - \frac{1}{x}\right)^2 dx$

$$I = \int \left(x^2 + \frac{1}{x^2} - 2\right)dx$$

$$I = \int x^2 dx + \int \frac{1}{x^2}dx - 2\int dx$$

$$= \frac{x^3}{3} - \frac{1}{x} - 2x + c$$

NOTE

Formula: $\int \frac{1}{x^2}dx = -\frac{1}{x} + c$ and $\int x^n dx = \frac{x^{n+1}}{n+1} + c$

18. (a) Step 1 : $\int \sqrt{1 + \cos 2x}\, dx$

by using formula $\cos 2x = 2\cos^2 x - 1$, we get

$$I = \int \sqrt{2\cos^2 x}dx = \int \sqrt{2}\cos x dx$$

$$= \sqrt{2}\sin x + C$$

NOTE

Formula $\int \cos x\, dx = \sin x + C$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. $\int \tan^2 x\, dx$ is equal to ***[SCRA-1996]***

(a) $\tan x + x + c$ (b) $\tan x - x + c$

(c) $\sec x + x + c$ (d) $\sec x - x + c$

2. $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x}dx =$ ***[MPPET-1996]***

(a) $\tan x + \cot x + c$

(b) $\tan x + \operatorname{cosec} x + c$

(c) $-\tan x + \cot x + c$

(d) $\tan x + \sec x + c$

3. $\int \left(2\sin x + \frac{1}{x}\right)dx$ is equal to ***[MPPET-99]***

(a) $-2\cos x + \log x + C$

(b) $2\cos x + \log x + C$

(c) $-2\sin x - \frac{1}{x^2} + c$

(d) $-2\cos x + \frac{1}{x^2} + C$

4. $\int e^{\log(\sin x)}dx =$ ***[MPPET-1995]***

(a) $\sin x + c$ (b) $-\cos x + c$

(c) $e^{\log \cos x} + c$ (d) None of these

5. The value of $\int \frac{1}{x^4}dx$ is ***[RPET-1995]***

(a) $\frac{1}{-3x^3} + c$ (b) $\frac{1}{3x^3} + c$

(c) $\frac{1}{-4x^3} + c$ (d) $-\frac{1}{3x^2} + c$

6. $\int \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2 dx =$

(a) $x + \cos x + c$

(b) $2\cos^2\frac{x}{2} + c$

(c) $\frac{1}{3}\left(\cos\frac{x}{2} - \frac{x}{2}\right)^3 + c$

(d) $x - \cos x + c$

7. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx =$

(a) $\tan x + \cot x + c$
(b) $\tan x - \cot x + c$
(c) $\operatorname{cosec} x - \cot x + c$
(d) $\sec x - \operatorname{cosec} x + c$

8. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx =$

[MPPET-90]

(a) $\sin x + c$
(b) $\cos x + c$
(c) $x + c$
(d) $x^2 + c$

9. $\int |x| dx; x<0$

(a) $\frac{1}{2}x^2 + c$
(b) $-\frac{1}{2}x^2 + c$
(c) $\frac{x}{2} + c$
(d) None of these

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 7 minutes.
3. The worksheet consists of 7 questions. The maximum marks are 21.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int \frac{1+\cos^2 x}{\sin^2 x}\,dx$ =

[MPPET-1993; BIT Ranchi-1982]

(a) $-\cot x - 2x + c$ (b) $-2\cot x - 2x + c$

(c) $-2\cot x - x + c$ (d) $-2\cot x + x + c$

2. $\int \frac{\cos^4 x - \sin^4 x}{\sin^2 x - 2\sin^4 x}dx$:

(a) $-\cot x$ (b) $\cot x + \tan x$

(c) $-\tan x$ (d) $(\tan x - \cot x)^2$

3. $\int a^x\,da$ = *[MPPET-1994; 96]*

(a) $\frac{a^x}{\log_e a} + c$ (b) $a^x \log_e a + c$

(c) $\frac{a^{x+1}}{x+1} + c$ (d) $x\,a^{x-1} + c$

4. $\int 10^x e^{10x}dx$ =

(a) $\frac{e^{10x}10^x}{10+\log 10}$ (b) $\frac{e^{10x}10^x}{10-\log 10}$

(c) $\frac{10^x}{10+\log 10}$ (d) $\frac{e^{10^x}}{\log 10}$

5. $\int \frac{\sqrt[3]{x^2} + \sqrt[4]{x} + \sqrt[3]{x}}{\sqrt{x}}dx$ =

(a) $\frac{6}{7}x^{7/6} + \frac{4}{3}x^{3/4} + \frac{6}{5}x^{5/6} + c$

(b) $-\frac{6}{5}x^{5/6} + \frac{4}{3}x^{3/4} - \frac{6}{5}x^{7/6} + c$

(c) $\frac{6}{5}x^{1/6} + \frac{4}{5}x^{3/4} - \frac{6}{5}x^{5/6} + c$

(d) None of these

6. $\int e^{-\log x}dx$ is equal to *[Kerala PET-2003]*

(a) $e^{-\log x} + c$ (b) $-x\,e^{-\log x} + c$

(c) $e^{\log x} + c$ (d) $\log|x| + c$

7. $\int \sqrt{1-\sin 2x}dx = \ldots\ldots, x \in (0, \pi/4)$

[MPPET-1987]

(a) $-\sin x + \cos x$ (b) $\sin x - \cos x$

(c) $\tan x + \sec x$ (d) $\sin x + \cos x$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ
6. ⓐ ⓑ ⓒ ⓓ
7. ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

1. (c) $\int \frac{1+\cos^2 x}{\sin^2 x} = \int (\operatorname{cosec}^2 x + \cot^2 x)dx$

$\int (2\operatorname{cosec}^2 x - 1)dx = -2\cot x - x + c$

2. (a) $\int \frac{\cos^4 x - \sin^4 x}{\sin^2 x - 2\sin^4 x}dx$

$= \int \frac{(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)}{\sin^2 x(1 - 2\sin^2 x)}dx$

$= \int \operatorname{cosec}^2 x\,dx = -\cot x + c$

LECTURE

2

Standard Integral Using Substitution

BASIC CONCEPTS

STANDARD INTEGRALS

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$ or $-\cos^{-1}\left(\frac{x}{a}\right)$
2. $\int \frac{dx}{\sqrt{a^2-b^2x^2}} = \frac{1}{b}\sin^{-1}\frac{bx}{a}$
3. $\int \frac{dx}{\sqrt{1-(ax+b)^2}} = \frac{1}{a}\sin^{-1}(ax+b)$
4. $\int \frac{1}{x^2+1}dx = \tan^{-1}x$ or $-\cot^{-1}x$
5. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$ or $\frac{-1}{a}\cot^{-1}\frac{x}{a}$
6. $\int \frac{dx}{a^2+b^2x^2} = \frac{1}{ab}\tan^{-1}\frac{bx}{a}$
7. $\int \frac{dx}{c^2+(ax+b)^2} = \frac{1}{ac}\tan^{-1}\frac{ax+b}{c}$
8. $\int \frac{nx^{n-1}dx}{a^{2n}+x^{2n}} = \frac{1}{a^n}\tan^{-1}\frac{x^n}{a^n}$
9. $\int \frac{dx}{x\sqrt{a^2x^2-1}} = \sec^{-1}(ax)$
10. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right)$
11. $\int \frac{ndx}{x\sqrt{x^{2n}-1}} = \sec^{-1}(x^n)$
12. $\int \frac{ad-bc}{(cx+d)^2}dx = \frac{ax+b}{cx+d}$
13. $\int \frac{ax+b}{cx+d}x = \frac{a}{c}x + \frac{cb-ad}{c^2}\log(cx+d)$
14. $\int \frac{|x|}{x}dx$ or $\int \frac{x}{|x|}dx = |x|, x \neq 0$
15. $\int x^x(1+\log_e x)dx = x^x$

INTEGRATION BY SUBSTITUTION

The following forms are very useful to write integral directly:

(i) $\int f'(ax+b)dx = \frac{f(ax+b)}{a} + c$

(ii) $\int \frac{f'(x)}{f(x)}dx = \log[f(x)] + c$

(iii) $\int \frac{f'(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)} + c$

(iv) $\int \frac{f'(x)}{[f(x)]^2}dx = -\frac{1}{f(x)}$

(v) $\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c$ (provided $n \neq -1$]

(vi) $\int x^{n-1}f'(x^n)dx = \frac{1}{n}f(x^n)$

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)) FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate: $\int \frac{1}{\sqrt{3x+1}-\sqrt{3x-4}}dx$

Solution

$$\int \frac{1}{\sqrt{3x+1}-\sqrt{3x-4}}dx$$

[Multiply N and D by $(\sqrt{3x+1}+\sqrt{3x-4})$] or rationalizing the denominator we get:

$$= \int \frac{\sqrt{3x+1}+\sqrt{3x-4}}{(3x+1)-(3x-4)}dx$$

$$= \frac{1}{5}\int \{(3x+1)^{1/2}+(3x-4)^{1/2}\}dx$$

$$= \frac{1}{5}\left[\frac{(3x+1)^{3/2}}{3\times\frac{3}{2}}+\frac{(3x-4)^{3/2}}{\frac{3}{2}\times 3}\right]+c =$$

$$= \frac{2}{45}[(3x+1)^{3/2}+(3x-4)^{3/2}]+c$$

2. Evaluate: $\int \sin 3x\cos 5x dx$ ***[Foreign-2001]***

Solution

$$\int \sin 3x\cos 5x dx = \frac{1}{2}\int 2\sin 3x\cos 5x dx$$

$$= \frac{1}{2}\int (\sin 8x - \sin 2x)dx$$

$$= \frac{1}{2}\left[\frac{-\cos 8x}{8}+\frac{\cos 2x}{2}\right]+c$$

3. Evaluate: $\int \frac{\sec x\tan x}{\sqrt{4-\sec^2 x}}dx$

Solution

$$\int \frac{\sec x\tan x}{\sqrt{4-\sec^2 x}}dx \mid \text{Put } \sec x = t \Rightarrow \sec x\tan x dx = dt$$

$$= \int \frac{1}{\sqrt{4-t^2}}dt = \sin^{-1}\left(\frac{t}{2}\right)+c$$

$$= \sin^{-1}\left(\frac{\sec x}{2}\right)+c$$

4. Evaluate: $\int \frac{1}{\sqrt{(2-x)^2-3}}dx$

Solution

$$\int \frac{1}{\sqrt{(2-x)^2-3}}dx \mid \text{Put } 2-x = t$$

$$\Rightarrow -dx = dt$$

$$= -\int \frac{1}{\sqrt{t^2-3}}dt = -\log\left|t+\sqrt{t^2-3}\right|+c$$

$$= -\log\left|(2-x)+\sqrt{(2-x)^2-3}\right|+c$$

5. $\int \left\{\frac{\cos x-\cos^3 x}{(1-\cos^3 x)}\right\}^{1/2}dx$

Solution

$$I = \int \frac{\sin x.\sqrt{\cos x}}{\sqrt{\{1-(\cos^{3/2}x)^2\}}}dx$$

Put $\cos^{3/2}x = t$

$$\therefore I = -\frac{2}{3}\sin^{-1}(\cos^{3/2}x)$$

6. Evaluate $\int \frac{1+\cos 4x}{\cot x-\tan x}dx$

[CET-1991]

Solution

$$\int \frac{1+\cos 4x}{\cot x-\tan x}dx$$

$$= \int \frac{2\cos^2(2x)\sin x\cos x}{\cos^2 x-\sin^2 x}dx$$

$$= \int \cos(2x)\sin(2x)dx$$

$$= \frac{1}{2}\int \sin(4x)dx = \frac{1}{2}\left\{\frac{-\cos(4x)}{4}\right\}+C$$

$$= -\frac{1}{8}\cos(4x)+C$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)):
TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Evaluate $\int \frac{dx}{\sqrt{ax+b}-\sqrt{ax+c}}$

2. Evaluate $\int \frac{x\tan^{-1}x^2}{1+x^4}dx$ ***[Roorkee-82]***

3. Evaluate $\int \frac{dx}{\sqrt{1+\sin x}}$

4. Evaluate $\int \frac{dx}{\sqrt{1-e^{2x}}}$

5. Evaluate $\int \frac{e^{\cos^{-1}x}}{\sqrt{1-x^2}}dx$ ***[MP-2000]***

6. Integrate the following with respect to x.

   (i) $\sin^2\frac{x}{2}$ (ii) $\tan^2 x$

   [MP-93, 98, 99]

7. Integrate the following functions with respect to x.

   (i) $\frac{1}{1+\sin x}$ ***[CBSE-92; MP-2000]***

   (ii) $\frac{1}{1-\cos x}$ ***[MP-98]***

8. Evaluate $\int (3x+2)^5 dx$

9. Evaluate $\int \frac{xdx}{\sqrt{4-x^4}}dx$

10. Evaluate $\int \frac{\sin x dx}{1+2\cos x}$

11. Evaluate $\int \frac{\sin(\log x)}{x}dx$ ***[CBSE-82]***

12. Evaluate $\int \frac{\operatorname{cosec}^2 x}{1+\cot x}dx$

    [MP-2001; CBSE-84]

13. Evaluate $\int \frac{\cos 2x}{(\sin x+\cos x)^2}dx$ ***[MP-01, 08]***

14. Evaluate $\int \frac{1}{9+x^2}dx$

15. Evaluate $\int \frac{1+\tan^2\left(\frac{\pi}{4}+x\right)}{1-\tan^2\left(\frac{\pi}{4}+x\right)}dx$

16. Evaluate $\int \frac{1+\tan x}{x+\log\sec x}dx$ ***[CBSE-2000]***

17. Evaluate $\int \sin 4x \cos 3x dx$ ***[CBSE-2007]***

18. Evaluate $\int \frac{3\tan x-\tan^3 x}{1-3\tan^2 x}dx$

ANSWERS

1. $\frac{2}{3a(b-c)}[(ax+b)^{3/2}+(ax+c)^{3/2}]$

2. $\frac{1}{4}(\tan^{-1}x^2)^2$

3. $\sqrt{2}\log\tan\left(\frac{\pi}{8}+\frac{x}{4}\right)$

4. $-\log\left\{e^{-x}+\sqrt{e^{-2x}-1}\right\}$

5. $-e^{\cos^{-1}x}+c$

6. $\frac{1}{2}[x-\sin x]$ (ii) $\tan x - x$

7. (i) $\tan x - \sec x$ (ii) $-\cot x - \operatorname{cosec} x$

8. $\frac{(3x+2)^6}{18}$

9. $\frac{1}{2}\sin^{-1}\left(\frac{x^2}{2}\right)$

10. $\frac{-1}{2}\log(1+2\cos x)$

11. $-\cos(\log x)$

12. $-\log(1+\cot x)$

13. $\log(\cos x+\sin x)$

14. $\frac{1}{3}\tan^{-1}\frac{x}{3}$

15. $\frac{1}{2}\log|\cot x|+c$

16. $\log|x+\log\sec x|+c$

18. $\frac{1}{3}\log\sec(3x)$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\int\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2}\frac{dx}{x}$ ***[IIT-1997]***

Solution

Put $x=t^2 \therefore dx=2t\,dt$

$$I=\int\sqrt{\frac{1-t}{1+t}}\cdot\frac{2t}{t^2}dt=2\int\frac{1-t}{t\sqrt{1-t^2}}dt$$

$$=\int 2\frac{1}{t\sqrt{1-t^2}}dt-2\int\frac{1}{\sqrt{1-t^2}}dt=I_1-I_2$$

For I_1 put $t=\frac{1}{z}$ as above.

$$-2\log\left\{\frac{1+\sqrt{1-x}}{\sqrt{x}}\right\}-2\sin^{-1}\sqrt{x}$$

2. $\int\sqrt{\cos x-\cos^3 x}\,dx$ equals ***[RPET-91]***

(a) $\cos x\sqrt{\cos x}+c$

(b) $\frac{2}{3}\cos x\sqrt{\cos x}+c$

(c) $\cos x\sqrt{\sin x}+c$

(d) $-\frac{2}{3}\cos x\sqrt{\cos x}+c$

Solution

(d) $I=\int\sqrt{\cos x}\sin x\,dx$

$$=-\frac{2}{3}\cos x\sqrt{\cos x}+c$$

(on putting $\cos x=t$)

3. $\int\frac{(x-x^3)^{1/3}}{x^4}dx$ is equal to ***[NDA-2004]***

(a) $\frac{3}{8}\left(\frac{1}{x^2}-1\right)^{4/3}+c$

(b) $-\frac{3}{8}\left(\frac{1}{x^2}-1\right)^{4/3}+c$

(c) $\frac{1}{8}\left(1-\frac{1}{x^2}\right)^{4/3}+c$

(d) $\frac{(x-x^3)^{1/3}}{x^4}+c$

Solution

(b) $I=\int\frac{x\left(\frac{1}{x^2}-1\right)^{1/3}}{x^4}dx$

$$=\int\left(\frac{1}{x^2}-1\right)^{1/3}\left(\frac{1}{x^3}dx\right)=-\frac{3}{8}\left(\frac{1}{x^2}-1\right)^{4/3}+c$$

4. $\int e^{x\log_a a}.e^x dx$ is equal to

(a) $(ae)^x+c$ (b) $\frac{(ae)^x}{\log(ae)}+c$

(c) $\frac{e^x}{1+\log a}+c$ (d) None of these

Solution

(b) $\int e^{x\log a}e^x dx=\int e^{x\log a}e^x dx=\int a^x e^x dx$

$$=\int(ae)^x dx=\frac{(ae)^x}{\log(ae)}+c.$$

5. $\int \frac{x^5 dx}{\sqrt{(1+x^3)}}$ ***[IIT-1985]***

(a) $\frac{2}{3}\sqrt{(1+x^3)}(x^3+2)$

(b) $\frac{2}{9}\sqrt{(1+x^3)}(x^3-4)$

(c) $\frac{2}{9}\sqrt{(1+x^3)}(x^3+4)$

(d) $\frac{2}{9}\sqrt{(1+x^3)}(x^3-2)$

Solution

(d) Here $x^5 = x^3 x^2$ and differential coefficient of x^3 is $3x^2$. In order to remove fractional powers, we put $1 + x^3 = t^2 \Rightarrow 3x^2 dx = 2t.dt$; also $x^3 = t^2 - 1$

$$I = \int \frac{(t^2-1)}{t}\left(\frac{2}{3}tdt\right) = \frac{2}{3}\int (t^2-1)dt$$

$$= \frac{2}{3}\left(\frac{t^3}{3} - t\right) = \frac{2}{9}t(t^2-3)$$

$$= \frac{2}{9}\sqrt{(1+x^3)}(x^3-2) \because (t^2 = 1+x^3)$$

6. $\int \frac{x^2 dx}{(a+bx)^2} =$

(a) $\frac{1}{b^2}\left[x + \frac{2a}{b}\log(a+bx) - \frac{a^2}{b}\frac{1}{a+bx}\right]$

(b) $\frac{1}{b^2}\left[x - \frac{2a}{b}\log(a+bx) + \frac{a^2}{b}\frac{1}{a+bx}\right]$

(c) $\frac{1}{b^2}\left[x + \frac{2a}{b}\log(a+bx) + \frac{a^2}{b}\frac{1}{a+bx}\right]$

(d) $\frac{1}{b^2}\left[x + \frac{a}{b} - \frac{2a}{b}\log(a+bx) - \frac{a^2}{b}\frac{1}{a+bx}\right]$

Solution

(d) Put $a + bx = t$

$$\Rightarrow x = \frac{t-a}{b} \text{ and } dx = \frac{dt}{b}$$

$$\therefore I = \int\left(\frac{t-a}{b}\right)^2 \times \frac{1}{t^2}\frac{dt}{b}$$

$$= \frac{1}{b^3}\int\left(1 - \frac{2a}{t} + a^2 t^{-2}\right)dt$$

$$= \frac{1}{b^3}\left[t - 2a\log t - \frac{a^2}{t}\right]$$

$$= \frac{1}{b^2}\left(x + \frac{a}{b} - \frac{2a}{b}\log(a+bx) - \frac{a^2}{b}\frac{1}{(a+bx)}\right)$$

7. If $\log_e \log_e \log_e$ ..... $\log_e x$ is formed by $l^r(x)$ in which log ..... comes to r times then the value of $\int \frac{1}{xl(x)l^2(x)l^3(x) + ...l^r(x)}dx$ is

(a) $l^{r+1}(x)$ (b) $\frac{l^{r+1}(x)}{r+1}$

(c) $l^r(x) + c$ (d) $l^{r-1}(x) + c$

Solution

(a) Step 1: By taking $r = 2$, given integral is

$$I_2 = \int \frac{dx}{x\log_e x \log_e(\log_e x)}$$

Step 2: Let put $t = \log_e(\log_e x)$

so that $dt = \frac{1}{x\log_e x}dx$

Hence, $I_2 = \int \frac{dt}{t} = \log_e(\log_e(\log_e x))$

$= l^3(x) = l^{2+1}(x)$

In given options by substituting $r = 2$ only option (a) is satisfied.

8. The integral $\int \frac{dx}{(1+\sin x)^{1/2}}$ is

(a) $\sqrt{2}\log\left|\cot\left(\frac{3\pi}{8} - \frac{x}{4}\right)\right| + c$

(b) $\sqrt{2}\log\left|\text{cosec}\left(\frac{\pi}{4} + \frac{x}{2}\right) - \cot\left(\frac{\pi}{4} + \frac{x}{2}\right) + c\right|$

(c) $\sqrt{2}\log\left|\tan\left(\frac{\pi}{8} + \frac{x}{4}\right)\right| + c$

(d) $\sqrt{2}\log\left|\sec\left(\frac{\pi}{4} - \frac{x}{2}\right) + \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)\right| + c$

[IIIT Hyderabad-2002; Roorkee 2000; UPSEAT-99]

Solution

(c) $\int \frac{dx}{(1+\sin x)^{1/2}} = \int \frac{dx}{\cos\frac{x}{2}+\sin\frac{x}{2}}$

$= \frac{1}{r}\int \frac{dx}{\sin\left(\frac{x}{2}+\alpha\right)}$ where, $1 = r\sin\alpha$, $1 = r\cos\alpha$

$= \frac{1}{\sqrt{2}}\int \operatorname{cosec}\left(\frac{x}{2}+\alpha\right)dx$

$= \frac{1}{\sqrt{2}} \frac{\log\left|\tan\left(\frac{x}{4}+\frac{\alpha}{2}\right)\right|}{\frac{1}{2}} + c$

$= \sqrt{2}\log\left|\tan\left(\frac{x}{4}+\frac{\pi}{8}\right)\right| + c.$

$\left[\because \alpha = \pi/4, r = \sqrt{2}\right]$

9. $\int \frac{dx}{\sin x - \cos x + \sqrt{2}}$ is equal to

(a) $-\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$

(b) $\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$

(c) $\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$

(d) $-\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$

[MP PET-2002]

Solution

(d) Step 1: $\frac{1}{\sqrt{2}}\int \frac{dx}{1-\cos\left(x+\frac{\pi}{4}\right)}$

$= \frac{1}{\sqrt{2}}\int \frac{dx}{2\sin^2\left(\frac{x}{2}+\frac{\pi}{8}\right)}$

$= \frac{1}{2\sqrt{2}}\int \operatorname{cosec}^2\left(\frac{x}{2}+\frac{\pi}{8}\right)dx$

$= \frac{1}{2\sqrt{2}} \frac{\left(-\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)\right)}{\frac{1}{2}}$

$= -\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$

NOTES

1. $-\frac{1}{\sqrt{2}}\sin x - \frac{1}{\sqrt{2}}\cos x$

$= -\left\langle \frac{1}{\sqrt{2}}\cos x - (\sin x)\frac{1}{\sqrt{2}} \right\rangle$

$= -\left\langle \cos x\cos\frac{\pi}{4} - \sin\frac{\pi}{4}\sin x \right\rangle = -\cos\left(x+\frac{\pi}{4}\right)$

2. $1-\cos\theta = 2\sin^2\frac{\theta}{2}$

3. $\int \operatorname{cosec}^2(ax+b)dx = -\frac{\cot(ax+b)}{a}+c$

10. $\int \frac{dx}{\cos x + \sqrt{3}\sin x}$ equal to ***[AIEEE-07]***

(a) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$

(b) $\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

(c) $\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$

(d) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

Solution

(a) $\frac{1}{2}\int \frac{dx}{\sin\left(x+\frac{\pi}{6}\right)} = \frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+c$

11. $\int e^{x\log_e 2006}e^x dx$ equal to ***[Gujrat CET-2007]***

(a) $\frac{e^x}{1+\log_e 2006}+C$

(b) $(2006\,e)^x + C$

(c) $\frac{(2006\,e)^x}{\log_e(2006\,e)}+C$

(d) None of these

Solution

(c) $\int e^{x\log_e 2006}.e^x dx$

$= \int e^{\log_e(2006)^x}.e^x dx = \int 2006^x.e^x dx$

$= \int (2006.e)^x dx$

$= \dfrac{(2006.e)^x}{\log_e(2006.e)} + C$

12. The value of $\int \dfrac{1}{1+\cos 8x}dx$ is

[Karnataka CET-2007]

(a) $\dfrac{\tan 2x}{8} + C$ (b) $\dfrac{\tan 8x}{8} + C$

(c) $\dfrac{\tan 4x}{4} + C$ (d) $\dfrac{\tan 4x}{8} + C$

Solution

(d) $\int \dfrac{1}{1+\cos 8x}dx$

$= \int \dfrac{1}{2\cos^2 4x}dx = \dfrac{1}{2}\int \sec^2 4x dx$

$= \dfrac{1}{2}\dfrac{\tan 4x}{4} + C = \dfrac{\tan 4x}{8} + C$

13. The value of $\sqrt{2}\int \dfrac{\sin x dx}{\sin\left(x - \dfrac{\pi}{4}\right)}$ is

[AIEEE-2008]

(a) $x - \log\left|\cos\left(x - \dfrac{\pi}{4}\right)\right| + c$

(b) $x + \log\left|\cos\left(x - \dfrac{\pi}{4}\right)\right| + c$

(c) $x - \log\left|\sin\left(x - \dfrac{\pi}{4}\right)\right| + c$

(d) $x + \log\left|\sin\left(x - \dfrac{\pi}{4}\right)\right| + c$

Solution

(d) $\sqrt{2}\int \dfrac{\sin x dx}{\sin\left(x - \dfrac{\pi}{4}\right)}$

$= \sqrt{2}\int \dfrac{\sin\left(x - \dfrac{\pi}{4} + \dfrac{\pi}{4}\right)}{\sin\left(x - \dfrac{\pi}{4}\right)}dx$

$= \sqrt{2}\int \left[\cos\dfrac{\pi}{4} + \cot\left(x - \dfrac{\pi}{4}\right)\sin\dfrac{\pi}{4}\right]dx$

$= \sqrt{2}.\dfrac{1}{\sqrt{2}}x + \sqrt{2}.\dfrac{1}{\sqrt{2}}\ln\left|\sin\left(x - \dfrac{\pi}{4}\right)\right| + c$

$= x + \ln\left|\sin\left(x - \dfrac{\pi}{4}\right)\right| + c$

c being a constant of integration.

14. $\int x(x^x)^x(2\log x + 1)dx = ...$

[Orissa JEE-2008]

(a) $(x^x)^x + c$
(b) $\log (x)^x + c$
(c) $x^x + c$
(d) None of these

Solution

(a) $\int x(x^x)^x(2\log x + 1)dx$

Let $(x^x)^x = t$

$\Rightarrow$ $x \log x^x = \log t = (x^x)^x + c$

$\Rightarrow$ $x^2 \log x = \log t$

$\Rightarrow$ $2x \log x + x = \dfrac{1}{t}\dfrac{dt}{dx}$

$\Rightarrow$ $(2 \log x + 1)\, x\, .\, (x^x)^x\, dx = dt$

$\therefore$ Integral $\int 1.dt$

$= t + c = (x^x)^x + c$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int\sqrt{1+\sin\left(\frac{x}{4}\right)}\,dx =$ ***[Karnataka CET-03]***

(a) $8\left(\sin\frac{x}{8}-\cos\frac{x}{8}\right)+C$

(b) $\left(\sin\frac{x}{8}+\cos\frac{x}{8}\right)+C$

(c) $\frac{1}{8}\left(\sin\frac{x}{8}-\cos\frac{x}{8}\right)+C$

(d) $8\left(\cos\frac{x}{8}-\sin\frac{x}{8}\right)+C$

2. $\int xe^{x^2}dx =$ ***[RPET-2003]***

(a) $-\frac{e^{x^2}}{2}+C$ (b) $\frac{e^{x^2}}{2}+C$

(c) $\frac{e^x}{2}+C$ (d) $-\frac{e^x}{2}+C$

3. The value of $\int\frac{1}{(x-5)^2}\,dx$ is ***[Karnataka CET-01; Pb. CET-02]***

(a) $\frac{1}{x-5}+C$ (b) $-\frac{1}{x-5}+C$

(c) $\frac{2}{(x-5)^3}+C$ (d) $-2(x-5)^3+C$

4. $\int\frac{e^{\sqrt{x}}}{\sqrt{x}}dx =$ ***[DCE-1999]***

(a) $e^{\sqrt{x}}$ (b) $e^{\sqrt{x}}/2$

(c) $2e^{\sqrt{x}}$ (d) $e^{\sqrt{x}}\sqrt{x}$

5. The value of $\int\frac{x^3}{\sqrt{1+x^4}}\,dx$ is ***[SCRA-1996]***

(a) $(1+x^4)^{1/2}+c$

(b) $-(1+x^4)^{1/4}+c$

(c) $\frac{1}{2}(1+x^4)^{1/2}+c$

(d) $-\frac{1}{2}(1+x^4)^{1/2}+c$

6. $\int\frac{1}{x}\log x\,dx$ is equal to ***[SCRA-1996]***

(a) $\frac{1}{2}\log x+c$ (b) $\frac{1}{2}(\log x)^2+c$

(c) $\frac{1}{2}\log(x^2)+c$ (d) $\log x+c$

7. The value of $\int\left(1+\frac{1}{x^2}\right)e^{\left[x-\frac{1}{x}\right]}dx$ equals ***[Kurukshetra CEE-98]***

(a) $e^{x-\frac{1}{x}}+C$ (b) $e^{x+\frac{1}{x}}+C$

(c) $e^{x^2-\frac{1}{x}}+C$ (d) $e^{x^2+\frac{1}{x^2}}+C$

8. $\int\sqrt{1+\cos x}\,dx$ equals ***[RPET-1996]***

(a) $2\sqrt{2}\sin\frac{x}{2}+c$ (b) $-2\sqrt{2}\sin\frac{x}{2}+c$

(c) $-2\sqrt{2}\cos\frac{x}{2}+c$ (d) $2\sqrt{2}\cos\frac{x}{2}+c$

9. $\int\sqrt{2+\sin 3x}.\cos 3x\,dx =$ ***[IIT-1976]***

(a) $\frac{2}{9}(2+\sin 3x)^{1/2}+c$

(b) $\frac{2}{3}(2+\sin 3x)^{2/3}+c$

(c) $\frac{2}{3}(2+\sin 3x)^{3/2}+c$

(d) $\frac{2}{9}(2+\sin 3x)^{3/2}+c$

10. $\int\frac{dx}{4\cos^3 2x-3\cos 2x} =$

(a) $\frac{1}{3}\log[\sec 6x+\tan 6x]+c$

(b) $\frac{1}{6}\log[\sec 6x+\tan 6x]+c$

(c) $\log[\sec 6x+\tan 6x]+c$

(d) None of these

11. If $\int(\sin 2x - \cos 2x)\,dx \quad \frac{1}{\sqrt{2}}\sin(2x-a)+b$, then ***[Roorkee-1978; MPPET-2001]***

(a) $a=\frac{\pi}{4}, b=0$

(b) $a=-\frac{\pi}{4}, b=0$

(c) $a=\frac{5\pi}{4}$, b = any constant

(d) $a=-\frac{5\pi}{4}$, b = any constant

12. $\int\frac{\sin x}{\sin(x-a)}dx$ is equal to ***[PET(Raj.)91, 99; Kerala(CEE)-03, 08; AIEEE-04; MP PET-08]***

(a) $x\cos a - \sin a \log(x-a)+c$

(b) $x\sin a + c$

(c) $x\sin a + \sin a \,.\, \log\sin(x-a)+c$

(d) $x\cos a + \sin a \,.\, \log\sin(x-a)+c$

13. $\int(1+5x+10x^2+10x^3+5x^4+x^5)\,dx =$ ***[Karnataka CET-89]***

(a) $5+20x+30x^2+20x^3+5x^4+c$

(b) $x+\frac{5x^2}{2}+\frac{10}{3}x^3+\frac{10}{4}x^4+x^5+5x^6+c$

(c) $\left(\frac{(1+x)^6}{3!}\right)+c$

(d) None of these

14. $\int\frac{dx}{\sqrt{x}+\sqrt{x-2}} =$ ***[MPPET-1990]***

(a) $\frac{1}{3}[x^{3/2}-(x-2)^{3/2}]+c$

(b) $\frac{2}{3}[x^{3/2}-(x-2)^{3/2}]+c$

(c) $\frac{1}{3}[(x-2)^{3/2}-x^{3/2}]+c$

(d) $\frac{2}{3}[(x-2)^{3/2}-x^{3/2}]+c$

15. $\int\sin 2x\cos 3x\,dx =$ ***[Roorkee-1976]***

(a) $\frac{1}{2}\left(\cos x+\frac{1}{5}\cos 5x\right)+c$

(b) $\frac{1}{2}\left(\cos x-\frac{1}{5}\cos 5x\right)+c$

(c) $\cos x+\frac{1}{5}\cos 5x+c$

(d) $\cos x-\frac{1}{5}\cos 5x+c$

16. If $d[f(x)] = e^{\tan x}\sec^2 x\,dx$, then $f(x)$ is equal to

(a) $e^{\tan x}+c$ (b) $e^{\sec^2 x}+c$

(c) $e^{\sin x}+c$ (d) None of these

17. $\int e^{3\log x}(x^4+1)^{-1}dx =$ ***[MPPET-2001; Kerala PET-2008]***

(a) $\log(x^4+1)+c$

(b) $1/4\log(x^4+1)+c$

(c) $4\log(x^4+1)+c$

(d) None of these

18. $\int x^2\sec x^3 dx =$ ***[MNR-1986; Roorkee-1975]***

(a) $\log(\sec x^3+\tan x^3)$

(b) $\frac{1}{3}\log(\sec x^3+\tan x^3)$

(c) $\frac{2}{3}\log(\sec x^3+\tan x^3)$

(d)None of these

19. $\int\frac{a^{\sqrt{x}}}{\sqrt{x}}dx =$ ***[Roorkee-1990; MPPET-2001]***

(a) $2a^{\sqrt{x}}\log_e a+c$

(b) $2a^{\sqrt{x}}\log_a e+c$

(c) $2a^{\sqrt{x}}\log_{10} a+c$

(d) $2a^{\sqrt{x}}\log_a 10+c$

20. The value of $\int\frac{2dx}{\sqrt{1-4x^2}}$ is ***[Karnataka CET-01; Pb. CET-01]***

(a) $\tan^{-1}(2x)+C$ (b) $\cot^{-1}(2x)+C$

(c) $\cos^{-1}(2x)+C$ (d) $\sin^{-1}(2x)+C$

21. $\int\frac{e^{m\tan^{-1}x}}{1+x^2}dx$ equals to ***[RPET-2001]***

(a) $e^{\tan^{-1}x}$ (b) $\frac{1}{m}e^{\tan^{-1}x}$

(c) $\frac{1}{m}e^{m\tan^{-1}x}$ (d) None of these

22. $\int\frac{a^x}{\sqrt{1-a^{2x}}}dx =$ ***[MNR-1983, 87]***

(a) $\frac{1}{\log a}\sin^{-1}a^x+c$ (b) $\sin^{-1}a^x+c$

(c) $\frac{1}{\log a}\cos^{-1}a^x+c$ (d) $\cos^{-1}a^x+c$

SOLUTIONS

1. (a) $\int\sqrt{1+\sin\left(\frac{x}{4}\right)}dx$

$$=\int\sqrt{\left(\sin^2\frac{x}{8}+\cos^2\frac{x}{8}+2\sin\frac{x}{8}\cos\frac{x}{8}\right)}dx$$

$$=\int\left(\sin\frac{x}{8}+\cos\frac{x}{8}\right)dx$$

$$=\left(-\cos\frac{x}{8}+\sin\frac{x}{8}\right)8+c$$

$$=8\left(\sin\frac{x}{8}-\cos\frac{x}{8}\right)+c$$

2. (b) Put $x^2=t\Rightarrow dt=2xdx$

$$\frac{1}{2}\int e^t dt=\frac{1}{2}e^{x^2}+c$$

3. (b) $\int\frac{1}{(x-5)^2}dx=-\frac{1}{x-5}+c$ (by formula)

4. (c) Step 1: I.et $\sqrt{x}=t\Rightarrow\frac{1}{2\sqrt{x}}=\frac{dt}{dx}$

$$\Rightarrow\frac{dx}{\sqrt{x}}=2dt$$

Step 2: Put $\frac{dx}{\sqrt{x}}=2dt$ in given integral

$$\therefore\quad\int e^t\times 2dt\Rightarrow 2\int e^t dt$$

$$\Rightarrow\quad 2e^t+c\Rightarrow 2e^{\sqrt{x}}+c$$

5. (c) $\int\frac{x^3}{\sqrt{1+x^4}}dx$

Put $1+x^4=t\Rightarrow dt=4x^3dx$

$$\frac{1}{4}\int\frac{1}{\sqrt{t}}dt=\frac{1}{4}\int t^{-1/2}dt$$

$$=\frac{(1+x^4)^{1/2}}{2}+c$$

6. (b) $\int\frac{1}{x}\log x dx$ Put $\log x=t\Rightarrow dt=\frac{1}{x}dx$

$$\int tdt=\frac{t^2}{2}+c=\frac{(\log x)^2}{2}+c$$

7. (a) Step 1: $t=x-\frac{1}{x}$; $dt=\left(1+\frac{1}{x^2}\right)dx$

Step 2: $\int e^t dt=e^t+C=e^{x-\frac{1}{x}}+C$

8. (a) $\int\sqrt{1+\cos x}dx=\int\sqrt{2}\cos\frac{x}{2}dx$

$(\cos 2x=2\cos^2 x-1)$

$$=2\sqrt{2}\sin\frac{x}{2}+c$$

9. (d) Step 1: $t=2+\sin 3x$, $dt=3\cos 3xdx$

Step 2: $\frac{1}{3}\int\sqrt{t}\,dt=\frac{1}{3}\times\frac{t^{\frac{3}{2}}}{\frac{3}{2}}=\frac{2}{9}t^{3/2}$

$$=\frac{2}{9}(2+\sin 3x)^{3/2}+c$$

10. (b) $\int\frac{dx}{4\cos^3 2x-3\cos 2x}=\int\frac{dx}{\cos 6x}$

$(\cos 3x=4\cos^3 x-3\cos x)$

$$=\int\sec 6x dx=\frac{1}{6}\log(\sec 6x+\tan 6x)+c$$

11. (b) Step 1: $I=\int(\sin 2x-\cos 2x)dx$

$$I=-\frac{\cos 2x}{2}-\frac{\sin 2x}{2}$$

$$=-\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\cos 2x+\frac{1}{\sqrt{2}}\sin 2x\right)$$

$$I=-\frac{1}{\sqrt{2}}\left[\sin\left(2x+\frac{\pi}{4}\right)\right]$$

$$I=\frac{1}{\sqrt{2}}\left[\sin\left(2x+\frac{5\pi}{4}\right)\right]$$

On comparing it with equation, we get

$$a=\frac{-5\pi}{4},b\in R$$

12. (d) Step 1: $\sin x=\sin\big((x-a)+a\big)$

$$=\sin(x-a)\cos a+\cos(x-a)\sin a$$

Step 2: $\int \frac{\sin(x-a)\cos a + \sin a\cos(x-a)}{\sin(x-a)}dx$

$$I = \int \cos a\,dx + \sin a \int \frac{\cos(x-a)}{\sin(x-a)}dx$$

$$= (\cos a)(x) + (\sin a)\log\sin(x-a) + c$$

13. (c) $\int (1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5)dx$

Using binomial given integral

$$I = \int (1+x)^5 dx$$

$$= \frac{(1+x)^6}{6} + c$$

$$= \left(\frac{(1+x)^6}{3!}\right) + c$$

14. (a) Multiplying numerator and denominator by $\sqrt{x} - \sqrt{x-2}$

$$\int \frac{\sqrt{x} - \sqrt{x-2}}{x-(x-2)}dx = \frac{1}{2}\int \sqrt{x}dx - \frac{1}{2}\int \sqrt{x-2}dx$$

$$= \frac{1}{2}\frac{x^{3/2}}{\frac{3}{2}} - \frac{1}{2} \times \frac{(x-2)^{3/2}}{\frac{3}{2}}$$

$$= \frac{1}{3}\left[x^{3/2} - (x-2)^{3/2}\right] + c$$

15. (b) $\int \sin 2x\cos 3x dx = \frac{1}{2}\int 2\sin 2x\cos 3x dx$

$$= \frac{1}{2}\int (\sin(5x) - \sin(x))dx$$

$$= \frac{1}{2}\left\{\frac{-\cos 5x}{5} + \cos x\right\} + c$$

16. (a) $d[f(x)] = e^{\tan x}\sec^2 x dx$

Intigrating both side, we get

$$f(x) = \int e^{\tan x}\sec^2 x dx$$

R.H.S. Put $\tan x = t \Rightarrow dt = \sec^2 x dx$

$$f(x) = \int e^t dt = e^t + c = e^{\tan x} + c$$

17. (b) Step 1: $I = e^{3\log x}(x^4+1)^{-1}$

$$= \int x^3(x^4+1)^{-1}dx \quad \because (e^{\log_e x} = x)$$

Step 2: $I = \int \frac{x^3}{x^4+1}dx$

Let $t = x^4 + 1$. Henced $t = 4x^3 dx$

$$I = \int \frac{dt}{4(t)} = \frac{1}{4}\log t$$

$$= \frac{1}{4}\log(x^4+1) + c$$

NOTE

From formula $\int \frac{f^1(x)dx}{f(x)} = \log\{f(x)\}$

18. (b) Step 1: $t = x^3$, $dt = 3x^2\,dx$

Step 2: $\frac{1}{3}\int \sec t\,dt = \frac{1}{3}\log[\sec t + \tan t]$

19. (b) Step 1: by substitution method

Let $t = \sqrt{x}$ Hence $dt = \frac{1}{2\sqrt{x}}dx$

Step 2: $I = \int 2a^t dt = \frac{2a^t}{\log a} = \frac{2a^{\sqrt{x}}}{\log a} + c$

20. (d) Step 1: $\int \frac{dx}{\sqrt{1-(ax)^2}} = \frac{1}{a}\sin^{-1}(ax) + c$

Step 2: $2\int \frac{dx}{\sqrt{1-4x^2}}$

$$= \frac{2}{2}\sin^{-1}(2x) + c$$

$$= \sin^{-1}(2x) + c$$

21. (c) $\int \frac{e^{m\tan^{-1}x}}{1+x^2}dx$

Put $t = \tan^{-1}x \Rightarrow dt = \frac{1}{1+x^2}dx$

$$\int e^{mt}\,dt = \frac{e^{mt}}{m} + c$$

$$= \frac{e^{m\tan^{-1}x}}{m} + c$$

22. (a) $\int \frac{a^x}{\sqrt{1-a^{2x}}}dx$

Put $a^x = t \Rightarrow dt = a^x \log a dx$

$$\frac{1}{\log a}\int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{\log a}\sin^{-1}(a^x) + c$$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE) FOR IMPROVING SPEED WITH ACCURACY

1. $\int e^{\cos^2 x} \sin 2x dx =$ ***[AICBSE-1980]***

(a) $e^{\cos^2 x} + c$ (b) $-e^{\cos^2 x} + c$

(c) $\frac{1}{2} e^{\cos^2 x} + c$ (d) None of these

2. To find the value of $\int \frac{1+\log x}{x} dx$, the proper subsitution is ***[MPPET-1988]***

(a) $\log x = t$ (b) $1 + \log x = t$

(c) $(1/x) = t$ (d) None of these

3. $\int x^x (1+\log x)\, dx$ is equal to ***[RPET-2000]***

(a) x^x (b) x^{2x}

(c) $x^x \log x$ (d) $\frac{1}{2}(1 + \log x)^2$

4. If $\int \sin 5x \cos 3x dx = -\frac{\cos 8x}{16} + A$, then $A =$ ***[MPPET-1992]***

(a) $-\frac{\cos 2x}{4} + C$ (b) Constant

(c) $\frac{\sin 2x}{4} + C$ (d) None of these

5. If $f(x) = g(x)$, then the value of $\int f'(x) \cdot g(x)\, dx$ is

(a) $\{f(x)\}^2 + c$ (b) $\{g(x)\}^2 + c$

(c) $\frac{1}{2}\{f(x)\}^2 + c$ (d) $\frac{1}{2}\{g(x)\}^2 + c$

6. $\int x \cos x^2\, dx$ is equal to ***[MPPET-1999; Pb. CET-2000]***

(a) $-\frac{1}{2}\sin^2 x + C$ (b) $\frac{1}{2}\sin^2 x + C$

(c) $-\frac{1}{2}\sin x^2 + C$ (d) $\frac{1}{2}\sin x^2 + C$

7. $\int \frac{1}{\sqrt{x}} \sin \sqrt{x} dx =$ ***[MPPET-1989]***

(a) $-\frac{1}{2}\cos\sqrt{x} + c$ (b) $-2\cos\sqrt{x} + c$

(c) $\frac{1}{2}\cos\sqrt{x} + c$ (d) $2\cos\sqrt{x} + c$

8. $\int (x+3)(x^2+6x+10)^9 dx$ equals ***[SCRA-1996]***

(a) $\frac{1}{20}(x^2+6x+10)^{10} + c$

(b) $\frac{1}{20}(x+3)^2 (x^2+6x+10)^{10} + c$

(c) $\frac{1}{16}(x^2+6x+10)^8 + c$

(d) $\frac{1}{38}(x+3)^{19} + \frac{1}{2}(x+3) + c$

9. $\int \frac{1}{x(\log x)^2} dx =$

(a) $\frac{1}{\log x} + c$ (b) $-\frac{1}{\log x} + c$

(c) $\log \log x + c$ (d) $-\log \log x + c$

10. $\int \frac{1}{x\sqrt{1+\log x}} dx =$ ***[Roorkee-1997]***

(a) $\frac{2}{3}(1+\log x)^{3/2} + c$ (b) $(1+\log x)^{3/2} + c$

(c) $2\sqrt{1+\log x} + c$ (d) $\sqrt{1+\log x} + c$

11. $\int \frac{\sin 2x}{1+\sin^2 x} dx =$ ***[Roorkee-76]***

(a) $\log \sin 2x + c$

(b) $\log (1 + \sin^2 x) + c$

(c) $\frac{1}{2}\log (1 + \sin^2 x) + c$

(d) $\tan^{-1}(\sin x) + c$

12. $\int \frac{\sec x \operatorname{cosec} x}{2\cot x - \sec x \operatorname{cosec} x} dx =$ ***[Kerala PET-2008]***

(a) $\log |\sec x + \tan x| + c$

(b) $\log |\sec x + \operatorname{cosec} x| + c$

(c) $\frac{1}{2}\log |\sec 2x + \tan 2x| + c$

(d) $\frac{1}{2}\log |\sec 2x + \operatorname{cosec} 2x| + c$

13. If $\int \frac{dx}{f(x)} = \log\{f(x)\}^2 + c$, then what is $f(x)$ equal to? ***[NDA-2008]***

(a) $2x + \alpha$ (b) $x + \alpha$

(c) $\frac{x}{2} + \alpha$ (d) $x^2 + \alpha$

14. $\int \frac{(\log x)^5}{x} dx$ = ***[Gujrat CET-2007]***

(a) $\log x.(\log x)^5 + \frac{(\log x)^6}{6x} + C$

(b) $\frac{\log x^6}{6} + C$

(c) $\frac{(\log x)^6}{6} + C$

(d) $\frac{(\log x)^6}{3x^2} + C$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 16 minutes.
3. The worksheet consists of 16 questions. The maximum marks are 48.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. If $\int f(x)dx = F(x)$, $f(x)$ is a continuous function, then $\int \frac{f(x)}{F(x)}dx =$

(a) $\log_e f(x)$ (b) $\log_e |F(x)|$
(c) $F(x) + c$ (d) None of these

2. $\int a^{3x-3}dx =$ ***[Roorkee-1977]***

(a) $\frac{a^{3x+3}}{3\log a} + c$ (b) $\frac{a^{3x+3}}{3\log a} + c$

(c) $a^{3x+3}\log a + c$ (d) $3a^{3x-3}\log a + c$

3. $\int e^{-x}\operatorname{cosec}^2(2e^{-x}+5)\,dx$ ***[AISSE-88]***

(a) $\frac{1}{2}\cot(2e^{-x}+5) + c$

(b) $-\frac{1}{2}\cot(2e^{-x}+5) + c$

(c) $2\cot(2e^{-x}+5) + c$

(d) $-2\cot(2e^{-x}+5) + c$

4. $\int\sqrt{1+\sin\frac{x}{2}}dx =$

[IIT-1980; MPPET-1989; Pb. CET-2003]

(a) $\frac{1}{4}\left(\cos\frac{x}{4} - \sin\frac{x}{4}\right) + c$

(b) $4\left(\cos\frac{x}{4} - \sin\frac{x}{4}\right) + c$

(c) $4\left(\sin\frac{x}{4} - \cos\frac{x}{4}\right) + c$

(d) $4\left(\sin\frac{x}{4} + \cos\frac{x}{4}\right) + c$

5. $\int \frac{\sin 2x}{a^2+b^2\sin^2 x}dx =$ ***[Roorkee-1977]***

(a) $\frac{1}{b^2}\log(a^2 + b^2\sin^2 x) + c$

(b) $\frac{1}{b}\log(a^2 + b^2\sin^2 x) + c$

(c) $\log(a^2 + b^2\sin^2 x) + c$

(d) $b^2\log(a^2 + b^2\sin^2 x) + c$

6. If $\int x^{-3}5^{1/x^2}dx = k \times 5^{(1/x^2)}$, then $k =$

(a) $-2\log 5$ (b) $-1/2\log 5$
(c) $\log_2 5$ (d) None of these

7. If $\frac{dI}{dy} = 3^{\cos y}\sin y$, then I is equal to

(a) $\frac{-3^{\cos y}}{\log 3} + c$ (b) $3^{\cos y} + c$

(c) $\frac{3^{\cos y}}{\log 3} + c$ (d) None of these

8. $\int \frac{dx}{\sqrt{x+a}+\sqrt{x+b}} =$ ***[AISSE-1989]***

(a) $\frac{2}{3(a-b)}[(x+a)^{3/2} - (x+b)^{3/2}] + c$

(b) $\frac{2}{3(a-b)}[(x+a)^{3/2} - (x+b)^{3/2}] + c$

(c) $3[(x+a)^{3/2} + (x+b)^{3/2}] + c$

(d) None of these

9. $\int(2\operatorname{cosec}^2 x + 2\sin 3x)\,dx =$ ***[AICBSE-1981]***

(a) $3\cot x + \frac{2}{3}\cos 3x + c$

(b) $\left(-2\cot x - \frac{2}{3}\cos 3x\right) + c$

(c) $3\cot x - \frac{2}{3}\cos 3x + c$

(d) None of these

10. $\int\tan(3x-5)\sec(3x-5)\,dx =$ ***[MPPET-1988]***

(a) $\sec(3x-5) + c$ (b) $\frac{1}{3}\sec(3x-5) + c$

(c) $\tan(3x-5) + c$ (d) None of these

11. $\int \frac{dx}{\tan x + \cot x} =$ ***[MPPET-1991]***

(a) $\frac{\cos 2x}{4} + c$ (b) $\frac{\sin 2x}{4} + c$

(c) $-\frac{\sin 2x}{4} + c$ (d) $-\frac{\cos 2x}{4} + c$

12. $\int \frac{dx}{(ax-b)^4}$

(a) $\frac{1}{3}(ax-b)^3$ (b) $\frac{1}{3a}(ax-b)^3 + k$

(c) $\frac{1}{3a}(b-ax)^{-3} + k$ (d) $-(b-ax)^{-3} + k$

13. If $\int \sin 3x \cos 5x\, dx = 0$ when $x = 0$, then the value of constant of integration ***[Roorkee-1971]***

(a) 1/16 (b) –1/16

(c) 3/16 (d) –3/16

14. $\int x^6 \sin(5x^7)dx = \frac{k}{5}\cos(5x^7), x \neq 0$, then

(a) $k = 7$ (b) $k = -1/7$

(c) $k = -\frac{1}{6}$ (d) $k = 6$

15. $\int \frac{\cos\sqrt{x}}{\sqrt{x}} dx =$

[MPPET-87; IIT-90; SCRA-96; RPET-2001]

(a) $2\cos\sqrt{x} + c$ (b) $2\sin\sqrt{x} + c$

(c) $\sin\sqrt{x} + c$ (d) $\frac{1}{2}\cos\sqrt{x} + c$

16. $\int \frac{e^{\tan^{-1}x}}{1+x^2} dx =$ ***[MPPET-1987]***

(a) $\log(1+x^2) + c$ (b) $\log e^{\tan^{-1}x} + c$

(c) $e^{\tan^{-1}x} + c$ (d) $\tan^{-1} e^{\tan^{-1}x} + c$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	16. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ	
6. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

1. (d) Step 1: $\frac{d}{dx}(F(x)) = \frac{d}{dx}\left(\int f(x)dx\right)$

$= f(x)$

Step 2: $\int \frac{f(x)}{F(x)}dx = \int \frac{d(F(x))}{F(x)} = \int \frac{dt}{t}$

$= \log t = \log_e |F(x)|$

4. (c) Step 1: $1 + \sin\frac{x}{2} = \cos^2\frac{x}{4} + \sin^2\frac{x}{4} + 2\sin\frac{x}{4}\cos\frac{x}{4}$

$= \left(\sin\frac{x}{4} + \cos\frac{\pi}{4}\right)^2$

Hence $I = \int\left(\sin\frac{x}{4} + \cos\frac{\pi}{4}\right)dx$

$= -4\cos\frac{x}{4} + 4\sin\frac{x}{4} + c$

$= 4\left(\sin\frac{x}{4} - \cos\frac{x}{4}\right) + c$

5. (a) Step 1: $t = a^2 + b^2 \sin^2 x$

$dt = b^2(2\sin x \cos x)dx = b^2(\sin 2x)dx$

Step 2: $\frac{1}{b^2}\int \frac{dt}{t} = \frac{1}{b^2}\log t$

$= \frac{1}{b^2}\log(a^2 + b^2 \sin^2 x) + c$

6. (b) Step 1: $t = \frac{1}{x^2} \Rightarrow dt = \frac{-2}{x^3}dx \Rightarrow x^{-3}dx$

$= \frac{dt}{-2}$

Step 2: $-\frac{1}{2}\int 5^t dt = -\frac{1}{2}\left(\frac{5^t}{\log 5}\right)$

$= -\frac{1}{2\log 5} \times 5^{\frac{1}{x^2}}$. Hence $k = -\frac{1}{2\log 5}$

7. (a) Step 1: $dI = 3^{\cos y}\sin y dy$

Step 2: $\int dI = \int 3^{\cos y}\sin y dy$

$I = -\int 3^t dt \quad (t = \cos y, dt = -\sin y dy)$

$= -\frac{3^t}{\log 3} + c = -\frac{3^{\cos x}}{\log 3} + c$

8. Step 1: Rationalization Method

$I = \int \frac{\sqrt{x+a} - \sqrt{x+b}}{a-b}dx$

$= \frac{1}{a-b}\left[\frac{(x+a)^{3/2}}{\frac{3}{2}} - \frac{(x+a)^{3/2}}{\frac{3}{2}}\right] + c$

10. (b) Note: Formula

$\int \sec(ax+b)\tan[ax+b]dx = \frac{\sec(ax+b)}{a}$

11. (d) Step 1: $\frac{1}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} = \frac{\sin 2x}{2}$

Step 2: $\int \frac{\sin 2x}{2}dx = -\frac{\cos 2x}{4} + c$

14. (b) Step 1: $t = 5x^7 \Rightarrow dt = 35x^6 dx$

Step 2: $\frac{1}{35}\int \sin t dt = \frac{-\cos t}{35}$

$= \frac{1}{5}\left(-\frac{1}{7}\right)\cos\left(5x^7\right)$

16. (b) Step 1: Let $t = \tan^{-1} x$ then

$dt = \frac{1}{1+x^2}dx$

Step 2: $\int e^t dt = e^t + c = e^{\tan^{-1} x} + c$

LECTURE



3 Integration of Algebraic Expressions 1

BASIC CONCEPTS

STANDARD INTEGRALS (C.T.M. TO IMPROVE SPEED WITH ACCURACY)

(i) $\int \frac{dt}{\sqrt{(t+\alpha)(t+\beta)}} = 2\log\left[\sqrt{t+\alpha}+\sqrt{t+\beta}\right]$

(ii) $\int \frac{1}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|$

(iii) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|$ or $\frac{1}{a}\tan h^{-1}\left(\frac{x}{a}\right)$

(iv) $\int \frac{dx}{(x-a)(x-b)} = \frac{1}{a-b}\log\frac{x-a}{x-b}$

or $\frac{1}{b-a}\log\frac{x-b}{x-a}$

(v) $\int \frac{ndx}{x^{n-1}+x} = \int \frac{ndx}{x(x^n+1)} = \log\frac{x^n}{x^n+1}$

(vi) $\int \frac{ndx}{x^{n-1}-x} = \int \frac{ndx}{x(x^n-1)} = \log\frac{x^n-1}{x^n}+C$

(vii) $\int \frac{dx}{\sin(x-a)\sin(x-b)}$

$= \frac{1}{\sin(a-b)}\log\frac{\sin(x-a)}{\sin(x-b)}$

(viii) $\int \frac{dx}{\cos(x-a)\cos(x-b)}$

$= \frac{1}{\sin(a-b)}\log\frac{\cos(x-a)}{\cos(x-b)}$

C.T.M: commit to memory.

(ix) $\int \frac{dx}{\sqrt{x^2+a^2}} = \log[x+\sqrt{x^2+a^2}]$ or $\sin h^{-1}(x/a)$

(x) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log[x+\sqrt{x^2-a^2}]$ or $\cos h^{-1}\left(\frac{x}{a}\right)$.

WORKING RULE TO FIND INTEGRALS OF THE FORM

(a) $\int f(a^2-x^2)dx$ (b) $\int f(a^2-x^2)dx$

(c) $\int f(x^2-a^2)dx$ (d) $\int f\left(\frac{a-x}{a+x}\right)dx$

Working Rule Integral	Substitution
$\int f(a^2-x^2)dx$	$x = a\sin\theta$ or $x = a\cos\theta$
$\int f(x^2+a^2)dx$	$x = a\tan 0$ or $x = a\cot 0$
$\int f(x^2-a^2)dx$	$x = a\sec 0$ or $x = a\operatorname{cosec} 0$

$\int f\left(\frac{a-x}{a+x}\right)dx$

or $\int f\left(\frac{a+x}{a-x}\right)dx \qquad x = a\cos 2\,0$

$\int f\left(\frac{x-\alpha}{\beta-x}\right)dx$

or $\int f(x-\alpha)(\beta-x)dx \;\; x = \alpha\cos^2 0 + \beta\sin^2 0$

METHOD OF PARTIAL FRACTIONS FOR RATIONAL FUNCTIONS

Integration by Partial Fractions

Integrals of the type $\int \frac{p(x)}{g(x)}dx$ can be integrated by resolving the integrand into partial functions. We proceed as follows:

Check degree of $p(x)$ and $g(x)$.

If degree of $p(x)$ > degree of $g(x)$, then divide $p(x)$ by $g(x)$ till degree of remainder $f(x)$ be comes lower than degree of denominator $g(x)$.

$$\frac{p(x)}{g(x)} = r(x) + \frac{f(x)}{g(x)}$$

i.e., degree of $f(x)$ < degree of $g(x)$

Case 1:

When the denominator contains non-repeated linear factors. That is, $g(x) = (x-\alpha_1)(x-\alpha_2)\ldots\ldots(x-\alpha_n)$. In such a case, write $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1}{x-\alpha_1} + \frac{A_2}{x-\alpha_2} + \ldots\ldots\ldots + \frac{A_n}{x-\alpha_n}$$

Where, $A_1, A_2, \ldots\ldots A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M. (corresponding to linear factor $x-\alpha_1$ there is a partial fraction of the type $A_1/x-\alpha_1$)

Case 2:

When the denominator contains repeated as well as non-repeated linear factors. That is $g(x) = (x-\alpha_1)^2(x-\alpha_3)\ldots\ldots(x-\alpha_n)$. In such a case express $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1}{x-\alpha_1} + \frac{A_2}{(x-\alpha_1)^2} + \frac{A_3}{x-\alpha_3}\ldots\ldots\ldots + \frac{A_n}{(x-\alpha_n)}$$

Where, $A_1, A_2, \ldots A_n$ are constant to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

NOTE

Corresponding to repeated linear factor $(x-a)^r$ in the denominator, a sum of r partial fractions of the type

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \ldots\ldots\ldots + \frac{A_r}{(x-a)^r} \text{ is taken.}$$

Case 3:

When the denominator contains a non-repeated quadratic factor which cannot be factorized further:

$g(x) = (ax^2 + bx + c)(x-\alpha_3)(x-\alpha_4)\ldots(x-\alpha_n)$.

In such a case express $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1x + A_2}{ax^2+bx+c} + \frac{A_3}{x-\alpha_3} + \ldots\ldots\ldots + \frac{A_n}{x-\alpha_n}$$

Where, $A_1, A_2, \ldots A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

Case 4:

When the denominator contains a repeated quadratic factor which cannot be factorized further.

That is $g(x) = (ax^2 + bx + c)^2(x-\alpha_5)(x-\alpha_6)\ldots(x-\alpha_n)$. In such a case write $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1x + A_2}{ax^2+bx+c} + \frac{A_3x + A_4}{(ax^2+bx+c)^2} + \frac{A_5}{x-\alpha_5} + \ldots + \frac{A_n}{(x-\alpha_n)}$$

Where, $A_1, A_2, \ldots A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

Case 5:

If the integrand contains only even powers of x

(i) Put $x^2 = t$ in the integrand.

(ii) Resolve the resulting rational expression in t into partial fractions.

(iii) Put $t = x^2$ again in the partial fractions and resolve into partial fractions as in case 1 and then integrate both sides.

Example,

$$\frac{1}{(x^2+a^2)(x^2+b^2)}$$

$$=\frac{1}{(b^2-a^2)(x^2+a^2)}+\frac{1}{(a^2-b^2)(x^2+b^2)}$$

THREE IMPORTANT INTEGRALS

(i) $\int\frac{dx}{\sin x(a+b\cos x)}$ multiplying N and D by $\sin x$ we get:

$$=\int\frac{\sin x dx}{(1-\cos x)(1+\cos x)(a+b\cos x)}$$

Put $\cos x = t$ and integrate by Partial fractions.

(ii) $\int\frac{dx}{\cos x(a+b\sin x)}$ multiplying N and D by $\cos x$ we get:

$$\int\frac{\cos x dx}{(1-\sin x)(1+\sin x)(a+b\sin x)}$$

Put $\sin x = t$ and integrate by partial fractions.

(iii) $\int\frac{dx}{(x^2+a^2)(x^2+b^2)}=\left(\frac{1}{b^2-a^2}\right)\left(\frac{1}{a}\right)\tan^{-1}\frac{x}{a}+\left(\frac{1}{a^2-b^2}\right)\left(\frac{1}{b}\right)\tan^{-1}\frac{x}{b}$ ***[NDA-2007]***

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate $\int\frac{(x-1)(x-2)(x-3)}{(x+1)(x+2)(x+3)}dx$

Solution

We have already decomposed the integrand into partial fractions $\frac{(x-1)(x-2)(x-3)}{(x+1)(x+2)(x+3)} = 1+\frac{A}{(x+1)}+\frac{B}{x+2}+\frac{C}{(x+3)}$ .... (1)

$\Rightarrow (x-1)(x-2)(x-3) = (x+1)(x+2)$

$(x+3) + A(x+2)(x+3) + B(x+1)$

$(x+3) + C(x+1)(x+2)$

so by solving we get $A = -12$, $B = 60$ and $C = -60$

using these values the equation (1) can be written as

$$\frac{(x-1)(x-2)(x-3)}{(x+1)(x+2)(x+3)}=1-\frac{12}{x+1}+\frac{60}{x+2}-\frac{60}{x+3}$$

Integrating, we get

$$\int\frac{(x-1)(x-2)(x-3)}{(x+1)(x+2)(x+3)}dx$$

$$=\int\left[1-\frac{12}{x+1}+\frac{60}{x+2}-\frac{60}{x+3}\right]dx$$

$= x - 12\log_e|x+1| + 60\log_e|x+2| - 60\log_e|x+3|$

2. Evaluate $\int\frac{dx}{x(1+4x^3+3x^6)}$

Solution

We put $x^3 = t$, so that $3x^2\,dx = dt$

$$\Rightarrow \frac{1}{x}dx=\frac{dt}{3t}$$

Thus, $\int\frac{dx}{x(1+4x^3+3x^6)}=\frac{1}{3}\int\frac{dt}{t(1+4t+3t^2)}$

$$=\frac{1}{3}\int\frac{dt}{t(1+t)(1+3t)}$$

Resolving $\frac{1}{t(1+t)(1+3t)}$ into partial fractions, we obtain

$$\frac{1}{t(1+t)(1+3t)}=\frac{1}{(1+0)(1+0)}.\frac{1}{t}+\frac{1}{(-1)(1+3(-1))}.\frac{1}{1+t}+\frac{1}{\left(\frac{-1}{3}\right)\left(1-\frac{1}{3}\right)}.\frac{1}{1+3t}$$

$$=\frac{1}{t}+\frac{1}{2}.\frac{1}{1+t}-\frac{9}{2}\frac{1}{1+3t}$$

Thus, $\frac{1}{3}\int\frac{dt}{t(1+t)(1+3t)}$

$$=\frac{1}{3}\int\frac{dt}{t}+\frac{1}{6}\int\frac{dt}{1+t}-\frac{3}{2}\int\frac{dt}{1+3t}$$

$$=\frac{1}{3}\log_e|t|+\frac{1}{6}\log_e|1+t|-\frac{1}{3}\times\frac{3}{2}\log_e|1+3t|+c$$

$$\Rightarrow\int\frac{dx}{x(1+4x^3+3x^6)}$$

$$=\frac{1}{3}\log_e|x^3|+\frac{1}{6}\log_e|1+x^3|-\frac{1}{2}\log_e|1+3x^3|+c$$

3. $\int\frac{x+1}{x(1+xe^x)^2}dx$ is equal to ***[IIT-1996]***

Solution

$$\text{Put } 1+xe^x=t=\int\left(\frac{-1}{t^2}-\frac{1}{t}+\frac{1}{t-1}\right)dt$$

$$=\frac{1}{1+xe^x}+\log\left(\frac{xe^x}{1+xe^x}\right)+c$$

4. Evaluate: $\int\sec x.\log(\sec x+\tan x)dx$

Solution

$$\int\sec x.\log(\sec x+\tan x)dx$$

$$=\int tdt=\frac{t^2}{2}+c$$

$$=\frac{1}{2}[\log(\sec x+\tan x)]^2+c$$

$$\left|\begin{array}{l}\text{Put } \log(\sec x+\tan x)=t\\ \Rightarrow\frac{1}{\sec x+\tan x}(\sec x\tan x+\sec^2 x)dx=dt,\\ \Rightarrow\sec x dx=dt\end{array}\right.$$

5. Evaluate: $\int\frac{1}{\sqrt{(x-a)(x-b)}}dx$

Solution

$$\int\frac{1}{\sqrt{(x-a)(x-b)}}dx$$

$$=\int\frac{1}{\sqrt{x^2-(a+b)x+ab}}dx$$

$$=\int\frac{1}{\sqrt{\left(x-\frac{a+b}{2}\right)^2-\left(\frac{a-b}{2}\right)^2}}dx$$

$$\left(\begin{array}{l}x^2-(a+b)x+ab\\ \left(x-\frac{a+b}{2}\right)^2-\frac{(a+b)^2}{4}+ab\\ \left(x-\frac{a+b}{2}\right)^2-\frac{(a-b)^2}{4}\end{array}\right)$$

$$=\log\left|\left(x-\frac{a+b}{2}\right)+\sqrt{(x-a)(x-b)}\right|+c$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Find $\int\frac{(2x-1)}{(x+1)(x+2)(x+3)}dx$

[CBSE (model question paper)]

2. Evaluate $\int\frac{x^2}{(x^2+1)(x^2+4)}dx$

[CBSE (model question paper)]

3. Evaluate $\int\frac{dx}{x^2-64}$ ***[MP-2000]***

4. Evaluate $\int\frac{dx}{9-4x^2}$ ***[MP-98]***

5. Evaluate $\int\frac{2x-3}{x^2+4}dx$ ***[MP-97]***

6. Evaluate $\int \frac{dx}{\sin x(3+2\cos x)}$

7. Solve $\int \frac{x}{(x-1)^2(x-2)}dx$

8. Evaluate $\int \frac{x+1}{x^2+4x-5}dx$

[AICBSE-1984]

9. Find the value of $\int \frac{x^2}{(x+1)(x-2)(x+3)}\,dx$

[MP-2000]

10. Evaluate $\int \frac{1}{x\{6(\log x)^2+7\log x+2\}}dx$

[PSB-2002]

11. Evaluate $\int \frac{x^2+1}{(x-1)^2(x+3)}$

[CBSE-1993]

12. Evaluate $\int \frac{x^2+1}{x^4+1}\,dx$

[MP-99; CBSE-90, 94, 96 (C)]

13. Evaluate $\int \frac{\sin(x+a)}{\sin(x+b)}dx$

14. Evaluate $\int \frac{\sin 2x}{(1+\sin x)(2+\sin x)}\,dx$

[CBSE-2004]

15. Find the value of $\int \frac{dx}{x(x^n+1)}$ ***[MP-2007]***

ANSWERS

1. $-\frac{1}{6}\log(x-1)-\frac{1}{3}\log(x+2)+\frac{1}{2}\log(x-3)+1$

2. $\frac{1}{3}\tan^{-1}x+\frac{2}{3}\tan^{-1}x+C$

3. $\frac{1}{16}\log\frac{x-8}{x+8}$

4. $\frac{1}{12}\log\left(\frac{3+2x}{3-2x}\right)$

5. $\log(x^2+4)-\frac{3}{2}\tan^{-1}\frac{x}{2}+c$

6. $\frac{1}{10}\log(1-\cos x)-\frac{1}{2}\log(1+\cos x)+\frac{2}{5}\log(3+2\cos x)$

7. $2\log(x-2)-2\log(x-1)+(x-1)^{-1}$

8. $\frac{2}{3}\log(x+5)+\frac{1}{3}\log(x-1)$

9. $\frac{-1}{6}\log(x+1)+\frac{4}{15}\log(x-2)+\frac{9}{10}\log(x+3)$

10. $\log\left|\frac{2\log x+1}{3\log x+2}\right|+c$

11. $\frac{3}{8}\log|x-1|-\frac{1}{2(x-1)}+\frac{5}{8}.\log|x+3|+c$

12. $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)$

13. $x\cos(a-b)+\sin(a-b).\log_e|\sin(x+b)|+c$

14. $\log\left|\frac{(2+\sin x)^4}{(1+\sin x)^2}\right|+C$

Hint: put $\sin x = t$

15. $\frac{1}{n}\log\left(\frac{x^n}{x^n+1}\right)$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. Evaluate $\int \frac{dx}{(x^2-1)\sqrt{x^2+1}}$.

Solution

Put $x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2}dt$

$$\therefore \int \frac{dx}{(x^2-1)\sqrt{x^2+1}}$$

$$= \int \frac{-\frac{1}{t^2}dt}{\left(\frac{1}{t^2}-1\right)\sqrt{\frac{1}{t^2}+1}}$$

$$= -\int \frac{t\,dt}{(1-t^2)\sqrt{1+t^2}}$$

Put $\sqrt{1+t^2} = y$

$\Rightarrow \quad 1 + t^2 = y^2$

$\Rightarrow \quad 2t\,dt = 2y\,dy$

$$= -\int \frac{y\,dy}{(1-(y^2-1)).y} = -\int \frac{dy}{2-y^2}$$

$$= \int \frac{dy}{y^2-2} = \frac{1}{2\sqrt{2}}\log\left|\frac{y-\sqrt{2}}{y+\sqrt{2}}\right| + C$$

$$= \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{1+t^2}-\sqrt{2}}{\sqrt{1+t^2}+\sqrt{2}}\right| + C$$

$$= \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{x^2+1}-\sqrt{2}x}{\sqrt{x^2+1}+\sqrt{2}x}\right| + C$$

2. $\int \frac{dx}{\sin(x-a)\sin(x-b)}$ is

(a) $\frac{1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right| + c$

(b) $\frac{-1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right| + c$

(c) $\log \sin(x-a)\sin(x-b) + c$

(d) $\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|$

Solution

(a) Let $\int \frac{dx}{\sin(x-a)\sin(x-b)}$

$$= \frac{1}{\sin(a-b)}\int \frac{\sin\{(x-b)-(x-a)\}}{\sin(x-a)\sin(x-b)}dx$$

$$= \frac{1}{\sin(a-b)}\int \frac{\sin(x-b)\cos(x-a)}{\sin(x-a)\sin(x-b)} - \frac{\cos(x-b)\sin(x-a)}{\sin(x-a)\sin(x-b)}dx$$

$$= \frac{1}{\sin(a-b)}\left[\int \cot(x-a)dx - \int \cot(x-b)dx\right]$$

$$= \frac{1}{\sin(a-b)}[\log \sin(x-a) - \log \sin(x-b)] + c$$

$$= \frac{1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right| + c$$

3. $\int \frac{dx}{(\sin x + \sin 2x)} =$

(a) $\frac{1}{6}\log(1-\cos x) + \frac{1}{2}\log(1+\cos x) - \frac{2}{3}\log(1+2\cos x)$

(b) $6\log(1-\cos x) + 2\log(1+\cos x) - \frac{2}{3}\log(1+2\cos x)$

(c) $6\log(1-\cos x) + \frac{1}{2}\log(1+\cos x) + \frac{2}{3}\log(1+2\cos x)$

(d) None of these

Solution

(a) $I = \int \frac{dx}{\sin x(1+2\cos x)}$

$$= \int \frac{\sin x dx}{\sin^2 x(1+2\cos x)}$$

$$I = \int \frac{\sin x\,dx}{(1-\cos x)(1+\cos x)(1+2\cos x)}$$

Now differential coefficient of cos x is sin x which is given in numerator and hence we make the substitution

$\cos x = t \quad \Rightarrow -\sin x\,dx \quad dt$

$$\therefore \quad I = -\int \frac{dt}{(1-t)(1+t)(1+2t)}$$

We split the integrand into partial fractions.

$$\therefore \quad I = -\int \left[\frac{1}{6(1-t)} - \frac{1}{2(1+t)} + \frac{4}{3(1+2t)}\right] dt$$

4. If $\int \frac{2x+3}{(x-1)(x^3+1)}dx = \log_e \left\{(x-1)^{\frac{5}{2}}(x^2+1)^a\right\} - \frac{1}{2}\tan^{-1}x + A$, where A is any arbitrary constant, then the value of 'a' is

[MPPET-1998]

(a) 5/4 (b) –5/3
(c) –5/6 (d) –5/4

Solution

(d) $$I = \int \frac{2x+3}{(x-1)(x^2+1)}dx$$

$$= \int \frac{5dx}{2(x-1)} + \int \frac{-\left(\frac{5}{2}x + \frac{1}{2}\right)}{x^2+1}dx,$$

(By partial fraction)

$$I = \frac{5}{2}\log(x-1) - \frac{5}{2}\int \frac{xdx}{1+x^2} - \frac{1}{2}\int \frac{dx}{1+x^2}$$

$$I = \frac{5}{2}\log(x-1) - \frac{5}{4}\log(1+x^2) - \frac{1}{2}\tan^{-1}x + A$$

$$I = \log(x-1)^{5/2}(1+x^2)^{-5/4} - \frac{1}{2}\tan^{-1}x + A$$

on comparing, $a = -5/4$

5. $\int \frac{x^3-1}{x^3+x}dx =$

[Roorkee-88; MPPET-2001]

(a) $x - \log x + \log(x^2+1) - \tan^{-1}x + c$
(b) $x - \log x + 1/2 \log(x^2+1) - \tan^{-1}x + c$
(c) $x - \log x + 1/2 \log(x^2+1) + \tan^{-1}x + c$
(d) None of these

Solution

(b) Step I: $\frac{x^3-1}{x^3+x}$ = Improper fractions Improper fraction is a polynomial function therefore

$$\frac{f(x)}{g(x)} = Q(x) + \frac{R(x)}{\phi(x)};\ \text{where}$$

$R(x)$ [power of $R(x)$] < $\phi(x)$ [power of $\phi(x)$]

Hence $$\frac{x^3-1}{x^3+x} = \frac{x^3+x-x-1}{x^3+x} = 1 - \frac{x+1}{x^3+x}$$

Step 2: $$\frac{x^3-1}{x^3+x}dx = \int dx - \int \frac{x+1}{x(x^2+1)}dx$$

Step 3: $$\frac{x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$(x+1) = A(x^2+1) + (Bx+C)x$$

by putting $x = 0 \Rightarrow A = 1$

by putting $x = 1 \Rightarrow 2 = 2A + B + C$

$$0 = B + C \Rightarrow B = -C$$

by putting $x = -1$

$\Rightarrow \quad 0 = 2A + (C - B)(-1)$

$\Rightarrow \quad 2A = C - B \Rightarrow 2 = C - B$

$\Rightarrow \quad 2 = 2C \Rightarrow C = 1, B = -1$

Step 4: $$I = \int dx - \left[\int \frac{dx}{x} + \int \frac{(1-x)}{x^2+1}dx\right]$$

$$= x - \log x - \tan^{-1}x + \frac{1}{2}\log(x^2+1) + C$$

6. If $\int \frac{e^x-1}{e^x+1}dx = f(x) + c$, then $f(x) =$

[EAMCET-2007]

(a) $2 \log(e^x+1) + C$
(b) $\log(e^{2x}-1) + C$
(c) $2 \log(e^x+1) - x + C$
(d) $\log(e^{2x}+1) + C$

Solution

(c) $$\int \frac{e^x-1}{e^x+1}dx = \int \left(\frac{2e^x}{e^x+1} - 1\right)dx$$

$$= 2 \log(e^x+1) - x + C$$

7. $\int \frac{x}{(x^2-a^2)(x^2-b^2)}dx$ is equal to

[Orissa JEE-2007]

(a) $\frac{1}{a^2-b^2}\log\left|\frac{x^2-a^2}{x^2-b^2}\right|+C$

(b) $\frac{1}{a^2-b^2}\log\left|\frac{x^2-b^2}{x^2-a^2}\right|+C$

(c) $\frac{1}{2(a^2-b^2)}\log\left|\frac{x^2-a^2}{x^2-b^2}\right|+C$

(d) $\frac{1}{2(a^2-b^2)}\log\left|\frac{x^2-b^2}{x^2-a^2}\right|+C$

Solution

(c) Let $I=\int\frac{xdx}{(x^2-a^2)(x^2-b^2)}$,

Put $x^2=t$ then $x\,dx=\frac{1}{2}dt$

$\therefore \quad I=\frac{1}{2}\int\frac{dt}{(t-a^2)(t-b^2)}=$

$$\frac{1}{2}\int\left\{\frac{dt}{(a^2-b^2)(t-a^2)}-\frac{dt}{(a^2-b^2)(t-b^2)}\right\}$$

$$=\frac{1}{2}\frac{1}{(a^2-b^2)}\log\left|\frac{t-a^2}{t-b^2}\right|+C$$

$$\Rightarrow \quad I=\frac{1}{2(a^2-b^2)}\log\left|\frac{x^2-a^2}{x^2-b^2}\right|+C$$

8. The value of $\int\frac{x^2+1}{x^2-1}dx$ is

[Karnataka CET-2007]

(a) $\log\left(\frac{x-1}{x+1}\right)+C$

(b) $\log\left(\frac{x+1}{x-1}\right)+C$

(c) $x+\log\left(\frac{x-1}{x+1}\right)+C$

(d) $\log(x^2-1)+C$

Solution

(c) $I=\int\frac{x^2+1}{x^2-1}dx=\frac{x^2+1-1+1}{x^2-1}dx$

$$\Rightarrow \quad I=\int 1dx+\int\frac{2}{x^2-1}dx$$

$$\Rightarrow \quad I=x+2\int\frac{1}{x^2-1^2}dx$$

$$\Rightarrow \quad I=x+2.\frac{1}{2}\log\left|\frac{x-1}{x+1}\right|+c$$

$$=x+\log\left|\frac{x-1}{x+1}\right|+c$$

9. $\int\frac{x^3+3x+2}{(x^2+1)^2(x+1)}dx$ equals to ***[IIT-1999]***

Solution

$$I=\int\frac{(x^3+x)+2(x+1)}{(x^2+1)^2(x+1)}dx$$

$$=\int\frac{x}{(x^2+1)(x+1)}dx+2\int\frac{1}{(x^2+1)^2}dx=I_1+2I_2$$

$$=2\left[\frac{1}{2}\tan^{-1}x+\frac{1}{2}\frac{x}{x^2+1}\right]$$

by putting $x=-1$ and $x^2=-1$ etc.

$$I_1=\int\left[\frac{1}{2}\frac{x+1}{(x^2+1)}-\frac{1}{2(x+1)}\right]dx$$

$$=\frac{1}{2}\left[\frac{1}{2}\log(x^2+1)+\tan^{-1}x-\log(x+1)\right]$$

$\therefore \quad I=I_1+2I_2$

$$=\frac{1}{4}\log(x^2+1)$$

$$-\frac{1}{2}\log(x+1)+\frac{3}{2}\tan^{-1}x+\frac{x}{x^2+1}$$

10. $\int\frac{x+1}{x(1+xe^x)^2}dx=?$ ***[IIT-1996]***

Solution

Put $1+xe^x=t=\int\left(\frac{-1}{t^2}-\frac{1}{t}+\frac{1}{t-1}\right)dt$

$$=\frac{1}{1+xe^x}+\log\left(\frac{xe^x}{1+xe^x}\right)+c$$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int \frac{1+\tan^2 x}{1-\tan^2 x}\,dx$ equals to *[RPET-2001]*

(a) $\log\left(\frac{1-\tan x}{1+\tan x}\right)+C$

(b) $\log\left(\frac{1+\tan x}{1-\tan x}\right)+C$

(c) $\frac{1}{2}\log\left(\frac{1-\tan x}{1+\tan x}\right)+C$

(d) $\frac{1}{2}\log\left(\frac{1+\tan x}{1-\tan x}\right)+C$

2. $\int \frac{\sin x dx}{3+4\cos^2 x} =$ *[Karnataka CET-2000]*

(a) $\log(3+4\cos^2 x)+C$

(b) $\frac{-1}{2\sqrt{3}}\tan^{-1}\left(\frac{\cos x}{\sqrt{3}}\right)+C$

(c) $\frac{-1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right)+C$

(d) $\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right)+C$

3. $\int \frac{dx}{e^x+e^{-x}} =$ *[Bihar CEE-76; MNR-74]*

(a) $\tan^{-1}(e^{-x})$ (b) $\tan^{-1}(e^x)$

(c) $\log(e^x-e^{-x})$ (d) $\log(e^x+e^{-x})$

4. $\int \frac{x+1}{\sqrt{1+x^2}}dx =$ *[MPPET-1991]*

(a) $\sqrt{1+x^2}+\tan^{-1}x+c$

(b) $\sqrt{1+x^2}-\log\{x+\sqrt{1+x^2}\}+c$

(c) $\sqrt{1+x^2}+\log\{x+\sqrt{1+x^2}\}+c$

(d) $\sqrt{1+x^2}+\log(\sec x+\tan x)+c$

5. $\int \frac{dx}{9x^2-4} =$

(a) $\frac{1}{12}\log\left|\frac{3x+2}{3x-2}\right|+c$

(b) $\frac{1}{6}\log\left|\frac{3x+2}{3x-2}\right|+c$

(c) $\frac{1}{12}\log\left|\frac{3x-2}{3x+2}\right|+c$

(d) $\frac{1}{6}\log\left|\frac{3x-2}{3x+2}\right|+c$

6. $\int \frac{dx}{(x-x^2)} =$ *[Roorkee-82]*

(a) $\log x-\log(1-x)+c$

(b) $\log(1-x^2)+c$

(c) $-\log x+\log(1-x)+c$

(d) $\log(x-x^2)+c$

7. $\int \frac{dx}{(x+1)(x+2)} =$

(a) $\log\frac{x+2}{x+1}+c$

(b) $\log(x+1)+\log(x+2)+c$

(c) $\log\frac{x+1}{x+2}+c$

(d) None of these *[MPPET-1987]*

8. $\int \frac{x^3\tan^{-1}x^4}{1+x^8}dx =$

(a) $\frac{1}{4}(\tan^{-1}x^4)^2+c$

(b) $\frac{1}{2}(\tan^{-1}x^4)^2+c$

(c) $\frac{1}{8}(\tan^{-1}x^2)^2+c$

(d) $\frac{1}{8}(\tan^{-1}x^4)^2+c$

9. $\int \frac{x}{x^4-1}dx =$

(a) $\frac{1}{4}\log\left[\frac{x^2-1}{x^2+1}\right]+c$

(b) $\frac{1}{4}\log\left[\frac{x^2+1}{x^2-1}\right]+c$

(c) $\frac{1}{2}\log\left[\frac{x^2+1}{x^2-1}\right]+c$

(d) $\frac{1}{2}\log\left[\frac{x^2+1}{x^2-1}\right]+c$

10. For $x > 1$, $\int \frac{1}{x(x^4-1)}dx =$ ***[RPET-1997]***

(a) $\log\frac{x^5-1}{x^4}+K$

(b) $\frac{1}{4}\log\frac{x^4-1}{x^4}+K$

(c) $\log\frac{x^4-1}{x}+K$

(d) $\frac{1}{4}\log\frac{x^4-1}{x}+K$

11. $\int \frac{3^x}{\sqrt{9^x-1}}dx =$ ***[EAMCET-02]***

(a) $\frac{1}{\log 3}\log\left|3^x+\sqrt{9^x-1}\right|+C$

(b) $\frac{1}{\log 3}\log\left|9^x+\sqrt{9^x-1}\right|+C$

(c) $\frac{1}{\log 9}\log\left|3^x+\sqrt{9^x-1}\right|+C$

(d) $\frac{1}{\log 9}\log\left|3^x-\sqrt{9^x-1}\right|+C$

12. $\int \frac{x^2+x-1}{x^2+x-6}dx =$ ***[AISSE-88]***

(a) $x+\log(x+3)+\log(x-2)+c$

(b) $x-\log(x+3)+\log(x-2)+c$

(c) $x-\log(x+3)-\log(x-2)+c$

(d) None of these

13. $\int \frac{e^x+1}{e^x-1}dx =$

(a) $2\log(e^{x/2}-e^{-x/2})+c$

(b) $\log(e^{x/2}-e^{-x/2})+c$

(c) $\frac{1}{2}\log(e^{x/2}+e^{-x/2})$

(d) $\log(e^{x/2}+e^{-x/2})+c$

14. $\int \frac{(x+1)^2}{x(x^2+1)}dx =$ ***[MPPET-2003]***

(a) $\log_e x$ (b) $\log_e x+2\tan^{-1}x$

(c) $-\log_a x$ (d) None of these

15. If $\int \frac{2x+3}{(x-1)(x^2+1)}dx = \log_e\{(x-1)^{5/2}(x^2+1)^a\} - \frac{1}{2}\tan^{-1}x + A$, where A is any arbitrary constant, then the function $f(x)$ is ***[MPPET-1998]***

(a) 5/4 (b) – 5/4

(c) 7/4 (d) –7/4

16. $\int \frac{x}{\sqrt{4-x^4}}dx =$ ***[Roorkee-1976]***

(a) $\cos^{-1}\left(\frac{x^2}{2}\right)$ (b) $\frac{1}{2}\cos^{-1}\left(\frac{x^2}{2}\right)$

(c) $\sin^{-1}\left(\frac{x^2}{2}\right)$ (d) $\frac{1}{2}\sin^{-1}\left(\frac{x^2}{2}\right)$

17. If $\int \frac{2x+3}{x^2-5x+6}\,dx = 9\ln(x-3) - 7\ln(x-2) + A$, then $A =$ ***[MPPET-92]***

(a) $5\ln(x-2)+c$

(b) $-4\ln(x-3)+c$

(c) constant

(d) None of these

18. $\int \frac{\cos x}{(1+\sin x)(2+\sin x)}dx =$ ***[Roorkee-1979]***

(a) $\log[(1+\sin x)(2+\sin x)]+c$

(b) $\log\frac{2+\sin x}{1+\sin x}+c$

(c) $\log\frac{1+\sin x}{2+\sin x}+c$

(d) None of these

19. What is $\int \sec x^\circ dx$ equal to? ***[NDA-2009]***

(a) $\ln(\sec x^\circ+\tan x^\circ)+c$

(b) $\frac{\pi\ln\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)}{180}+c$

(c) $\frac{180\ln\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)}{\pi}+c$

(d) $\frac{180\ln\tan\left(\frac{\pi}{4}+\frac{\pi x}{360}\right)}{\pi}+c$

SOLUTIONS

1. (d) $\int \frac{1+\tan^2 x}{1-\tan^2 x}dx = \int \frac{\sec^2 x}{1-\tan^2 x}dx$

Put $t = \tan x \Rightarrow dt = \sec^2 x dx$

$$\int \frac{dt}{1-t^2} = \frac{1}{2}\log\left(\frac{1+t}{1-t}\right)+c$$

$$= \frac{1}{2}\log\left(\frac{1+\tan x}{1-\tan x}\right)+c$$

$$\left(\because \int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left(\frac{a+x}{a-x}\right)\right)$$

2. (c) Step 1: $t = \cos x$. $dt = -\sin dx$

Step 2: $\int \frac{-dt}{3+4t^2} = \frac{-1}{2\sqrt{3}}\tan^{-1}\left(\frac{2t}{\sqrt{3}}\right)+c$

NOTE

Quicker Method: $\int \frac{dx}{a^2+b^2x^2} = \frac{1}{ab}\tan^{-1}\frac{bx}{a}+c$

3. (b) $\int \frac{dx}{e^x+e^{-x}}$

Multiplying by e^x in numerator and denominator we get $\int \frac{e^x}{1+e^{2x}}dx$

Put $e^x = t \Rightarrow e^x dx = dt$

$$\int \frac{dt}{1+t^2} = \tan^{-1}(e^x)+c$$

4. ( ) $\int \frac{x+1}{\sqrt{1+x^2}}dx = \int \frac{x}{\sqrt{1+x^2}}dx + \int \frac{dx}{\sqrt{1+x^2}}$

Put $1+x^2 = t \Rightarrow dt = 2xdx$

$$\frac{1}{2}\int \frac{dt}{\sqrt{t}} + \log\left[x+\sqrt{x^2+1}\right]$$

$$\left(\because \int \frac{dx}{\sqrt{x^2+a^2}} = \log\left[x+\sqrt{x^2+a^2}\right]\right)$$

$$I = \sqrt{1+x^2} + \log\left[x+\sqrt{x^2+1}\right]+c$$

$$\left(\because \int \frac{dx}{\sqrt{x}} = 2\sqrt{x}+c\right)$$

5. (c) $\int \frac{dx}{9x^2-4} = \int \frac{dx}{9\left(x^2-\frac{4}{9}\right)}$

$$\Rightarrow \frac{1}{9}\int \frac{dx}{\left(x-\frac{2}{3}\right)\left(x+\frac{2}{3}\right)} = \frac{1}{12}\log\frac{3x-2}{3x+2}$$

By Using Formula $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-4}{x+4}\right|$

6. (a) $\int \frac{dx}{(x-x^2)} = \int \frac{dx}{x(1-x)}$

$$\Rightarrow \log\frac{x}{1-x}+c = \log x - \log(1-x)+c$$

NOTE

By using formula

$$\int \frac{dx}{(x-a)(x-b)} = \frac{1}{a-b}\log\frac{x-a}{x-b}$$

7. (c) $\int \frac{dx}{(x+1)(x+2)} = \log\frac{x+1}{x+2}+c$

NOTE

By using formula

$$\int \frac{dx}{(x-a)(x-b)} = \frac{1}{a-b}\log\frac{x-a}{x-b}$$

8. (d) Step 1: Given Let $I = \int \frac{x^3\tan^{-1}x^4}{1+x^8}dx$

Step 2: Put,

$\tan^{-1}x^4 = t$, $\frac{4x^3dx}{1+x^8} = dt$, $\frac{x^3dx}{1+x^8} = \frac{dt}{4}$ in the given integral.

We get, $I = \int \frac{tdt}{4}$

Step 3: By using $\int x^n = \frac{x^{n+1}}{n+1}$, we get

$$I = \frac{1}{4}\int tdt = \frac{t^2}{2\times 4}+c$$

$$\Rightarrow I = \frac{(\tan^{-1}x^4)^2}{8}+c$$

9. (a) $\int \frac{x}{x^4-1}dx$

Put $x^2 = t \Rightarrow dt = 2xdx$

$$\frac{1}{2}\int \frac{dt}{t^2-1} = \frac{1}{2}\left[\frac{1}{2}\log\left|\frac{t-1}{t+1}\right|\right]$$

$$= \frac{1}{4}\log\frac{x^2-1}{x^2+1}$$

10. (b) Step 1: $t = x^4$, $dt = 4x^3dx$

Step 2: $\int \frac{4x^3dx}{4x^4(x^4-1)} = \frac{1}{4}\int \frac{dt}{t(t-1)}$

$$= \frac{1}{4}\left(\frac{1}{t-1} - \frac{1}{t}\right)dt =$$

$$\frac{1}{4}\left(\log(t-1) - \log t\right) = \frac{1}{4}\log\frac{t-1}{t}$$

$$= \frac{1}{4}\log\frac{x^4-1}{x^4} + k$$

Quicker Method

Putting $n = 4$ in formula

$$\int \frac{dx}{x(x^n-1)} = \frac{1}{n}\log\left(\frac{x^n-1}{x^n}\right)$$

11. (a) $\int \frac{3^x}{\sqrt{9^x-1}}dx$

Put $3^x = t \Rightarrow dt = 3^x \log 3dx$

$$\frac{1}{\log 3}\int \frac{dt}{\sqrt{t^2-1}} = \frac{1}{\log 3}\left[\log\left|t + \sqrt{t^2-1}\right|\right] + c$$

$$= \frac{1}{\log 3}\left[\log\left|3^x + \sqrt{9^{2x}-1}\right|\right] + c$$

12. (d) $\int \frac{x^2+x-1}{x^2+x-6}dx = \int dx + 5\int \frac{dx}{x^2+x-6}$

$$= x + \int\left(\frac{1}{x-2} - \frac{1}{x+3}\right)dx$$

$$= x + \log\left|\frac{x-2}{x+2}\right| + c$$

13. (a) Step 1:

Dividing numerator and denominator by $e^{x/2}$

$I = \int \frac{e^{x/2} + e^{-x/2}}{e^{x/2} - e^{-x/2}}dx$ is

Step 2: Let $t = e^{x/2} - e^{-x/2}$

$$dt = \frac{1}{2}(e^{x/2} + e^{-x/2})\,dx$$

$$I = \int \frac{2}{t}dt = 2\log(t) = 2\log[e^{x/2} - e^{-x/2}] + c$$

14. (b) Step 1: Let $I = \int \frac{(x+1)^2}{x(x^2+1)}dx$

Step 2: $I = \int \frac{x^2+1+2x}{x(x^2+1)}dx$

$$I = \int \frac{x^2+1}{x(x^2+1)} + \int \frac{2x}{x(x^2+1)}dx$$

$$\Rightarrow \int \frac{1}{x}dx + 2\int \frac{1}{(1+x^2)}dx$$

$$\Rightarrow \log_e x + 2\tan^{-1}x$$

15. (b) Step 1: By partial fraction method

$$\frac{2x+3}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

$$(2x+3) = A(x^2+1) + (Bx+C)(x-1)$$

Now substituting $x = 1, 0, -1$ (because it is true for all values) $A = \frac{5}{2}$

$$3 = A + C(-1) \Rightarrow 3 = \frac{5}{2} - C \Rightarrow C = -\frac{1}{2}$$

$$1 = 2A + (-B+C)(-2)$$

$$1 = 5 + 2B - 2\times\left(-\frac{1}{2}\right) \Rightarrow 2B = -5 \Rightarrow B = -\frac{5}{2}$$

Step 2: $\int \frac{2x+3}{(x-1)(x^2+1)}dx$

$$= \frac{5}{2}\int \frac{dx}{x-1} - \frac{5}{2}\int \frac{x}{x^2+1}dx - \frac{1}{2}\int \frac{dx}{x^2+1}$$

$$= \frac{5}{2}\log(x-1) - \frac{5}{2}\times\frac{1}{2}\log(x^2+1) - \frac{1}{2}\tan^{-1}x$$

$$= \frac{5}{2}\left[\log\frac{(x-1)}{\sqrt{x^2+1}}\right] - \frac{1}{2}\tan^{-1}x + A$$

$$= \log\left\{(x-1)^{\frac{5}{2}}(x^2+1)^{-\frac{5}{4}}\right\} - \frac{1}{2}\tan^{-1}x + A$$

Quicker Method

$$\frac{d}{dx}\left[\left\{\log_e(x-1)^5(x^2+1)^a\right\}-\frac{1}{2}\tan^{-1}x+A\right]$$

$$=\frac{2x+3}{(x-1)(x^2+1)}$$

$$\frac{d}{dx}\left[\frac{5}{2}\log_e(x-1)+a\log_e(x^2+1)-\frac{1}{2}\tan^{-1}x+A\right]$$

$$=\frac{5}{2(x-1)}+\frac{a(2x)}{x^2+1}-\frac{1}{2(1+x^2)}$$

$$=\frac{5(x^2+1)+4a(x-1)(x)-(x-1)}{2(x-1)(x^2+1)}$$

$$=\frac{(5+4a)x^2+.....}{2(x-1)(x^2+1)}$$

Since Numerator of Right hand side does not contain term of x^2. Therefore $4a+5=0\Rightarrow a=\frac{-5}{4}$

16. (d) $\int\frac{x}{\sqrt{4-x^2}}dx=\frac{1}{2}\int\frac{dt}{\sqrt{4-t^2}}$

(Here $t=x^2$)

$\therefore$ Integral $=\frac{1}{2}\sin^{-1}\frac{t}{2}$ or $\frac{1}{2}\sin^{-1}\frac{x^2}{2}$

17. (c) Step 1: Resolving using partial fraction and then integrating

$$\frac{2x+3}{x^2-5x+6}=\frac{2x+3}{(x-2)(x-3)}=\frac{A}{x-2}+\frac{B}{x-3}$$

$$2x+3=A(x-3)+B(x-2)$$

Put $x=2\Rightarrow A=-7$

Put $x=3\Rightarrow B=9$

Step 2: $I=\int\frac{-7\,dx}{x-2}+\int\frac{9}{x-3}dx$

$=-7\log(x-2)+9\log(x-3)+\text{constant}$

18. (c) $\int\frac{\cos x dx}{(1+\sin x)(2+\sin x)}=\int\frac{dt}{(1+t)(2+t)}$;

$t=\sin x$

Here integral $=\int\left(\frac{1}{t+1}-\frac{1}{t+2}\right)dt$

$=\log\left(\frac{t+1}{t+2}\right);t=\sin x$

19. (d) Let $I=\int\sec x^\circ dx$

$=\int\sec\frac{\pi x}{180^\circ}dx$

Put $\frac{180^\circ}{\pi}=t$

$\Rightarrow\quad dx=\frac{180^\circ}{\pi}dt$

$\therefore\quad I=\int\sec t dt.\frac{180^\circ}{\pi}$

$=\frac{180^\circ}{\pi}\log\tan\left(\frac{\pi}{4}+\frac{\pi x}{360^\circ}\right)$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. $\int\frac{e^x}{(1+e^x)(2+e^x)}dx=$

(a) $\log[(1+e^x)/(2+e^x)]+c$

(b) $\log\left[\frac{1+e^x}{2+e^x}\right]+c$

(c) $\log[(1+e^x)\sqrt{2+e^x}]+c$

(d) None of these

2. $\int\frac{dx}{(x^2+1)(x^2+4)}=$ ***[MPPET-1995]***

(a) $\frac{1}{3}\tan^{-1}x-\frac{1}{3}\tan^{-1}\frac{x}{2}+c$

(b) $\frac{1}{3}\tan^{-1}x+\frac{1}{3}\tan^{-1}\frac{x}{2}+c$

(c) $\frac{1}{3}\tan^{-1}x-\frac{1}{6}\tan^{-1}\frac{x}{2}+c$

(d) $\tan^{-1}x-2\tan^{-1}\frac{x}{2}+c$

3. $\int \frac{\log(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}dx =$

(a) $\frac{1}{2}[\log(x+\sqrt{1+x^2})]^2 + c$

(b) $\log(x+\sqrt{1+x^2})^2 + c$

(c) $\log(x+\sqrt{1+x^2}) + c$

(d) None of these

4. $\int \frac{3x^2}{x^6+1}dx =$ ***[MNR-1981; MPPET-1988; RPET-1995]***

(a) $\log(x^6+1) + c$

(b) $\tan^{-1}(x^3) + c$

(c) $3\tan^{-1}(x^3) + c$

(d) $3\tan^{-1}(x^3/3) + c$

5. $\int \frac{1}{x-x^3}dx =$ ***[MPPET-1996]***

(a) $\frac{1}{2}\log\frac{(1-x^2)}{x^2} + C$

(b) $\log\frac{(1-x)}{x(1+x)} + C$

(c) $\log x(1-x^2) + C$

(d) $\frac{1}{2}\log\frac{x^2}{(1-x^2)} + C$

6. $\int \frac{x^3-x-2}{(1-x^2)}dx =$ ***[AICBSE-1985]***

(a) $\log\left(\frac{x+1}{x-1}\right) - \frac{x^2}{2} + c$

(b) $\log\left(\frac{x-1}{x+1}\right) + \frac{x^2}{2} + c$

(c) $\log\left(\frac{x+1}{x-1}\right) + \frac{x^2}{2} + c$

(d) $\log\left(\frac{x-1}{x+1}\right) - \frac{x^2}{2} + c$

7. The value of $\int \frac{dx}{2x^2+x+1}$ ***[RPET-1997]***

(a) $\frac{2}{\sqrt{7}}\tan^{-1}\left(\frac{4x+1}{\sqrt{7}}\right) + c$

(b) $\frac{1}{\sqrt{7}}\tan^{-1}\left(\frac{4x+1}{\sqrt{7}}\right) + c$

(c) $\frac{1}{\sqrt{7}}\tan^{-1}\left(\frac{4x}{\sqrt{7}}\right) + c$

(d) None of these

8. $\int \frac{e^{2x}-1}{e^{2x}+1}dx =$ ***[Roorkee-1976]***

(a) $\frac{e^{2x}-1}{e^{2x}+1} + c$

(b) $\log(e^{2x}+1) - x + c$

(c) $\log(e^{2x}+1)$

(d) None of these

9. $\int \frac{dx}{(e^x-1)^2} =$

(a) $x - \frac{e^x}{e^x-1} - \log(e^x-1)$

(b) $x - \frac{e^x}{e^x-1} + \log(e^x-1)$

(c) $x + \frac{e^{x-1}}{e^x} + \log(e^x-1)$

(d) $x + \frac{e^x}{e^x-1} + \log(e^x-1)$

10. $\int \frac{(\sin^{-1}x)^2}{\sqrt{1-x^2}}dx$ is equal to

(a) $\frac{(\sin^{-1}x)^3}{3} + c$ (b) $(\sin^{-1}x)^3 + c$

(c) $3(\sin^{-1}x)^3 + c$ (d) None of these

11. $\int \frac{x-1}{(x-3)(x-2)}dx =$ ***[Roorkee-1978]***

(a) $\log(x-3) - \log(x-2) + c$

(b) $\log(x-3)^2 - \log(x-2) + c$

(c) $\log(x-3) + \log(x-2) + c$

(d) $\log(x-3)^2 + \log(x-2) + c$

12. $\int \frac{dx}{4x^2+9} =$ ***[MPPET-1991; Roorkee-1977; MNR-1974]***

(a) $\frac{1}{2}\tan^{-1}\left(\frac{2x}{3}\right) + c$ (b) $\frac{3}{2}\tan^{-1}\left(\frac{2x}{3}\right) + c$

(c) $\frac{1}{6}\tan^{-1}\left(\frac{2x}{3}\right) + c$ (d) $\frac{1}{6}\tan^{-1}\left(\frac{3x}{2}\right) + c$

13. $\int \frac{x^2+x-6}{(x-2)(x-1)}dx =$

(a) $x + 2\log(x-1) + c$

(b) $2x + 2\log(x-1) + c$

(c) $x + 4\log(1-x) + c$

(d) $x + 4\log(x-1) + c$

14. $\int \frac{1}{(x-1)(x^2+1)}dx =$ ***[Roorkee-1984]***

(a) $\frac{1}{2}\log(x-1) - \frac{1}{4}\log(x^2+1) - \frac{1}{2}\tan^{-1}x + c$

(b) $\frac{1}{2}\log(x-1) + \frac{1}{4}\log(x^2+1) - \frac{1}{2}\tan^{-1}x + c$

(c) $\frac{1}{2}\log(x-1) - \frac{1}{2}\log(x^2+1) - \frac{1}{2}\tan^{-1}x + c$

(d) None of these

15. $\int \frac{x^2}{(x^2+2)(x^2+3)}dx =$ ***[AISSE-1990]***

(a) $-\sqrt{2}\tan^{-1}x + \sqrt{3}\tan^{-1}x + c$

(b) $-\sqrt{2}\tan^{-1}\frac{x}{\sqrt{2}} + \sqrt{3}\tan^{-1}\frac{x}{\sqrt{3}} + c$

(c) $\sqrt{2}\tan^{-1}\frac{x}{\sqrt{2}} + \sqrt{3}\tan^{-1}\frac{x}{\sqrt{3}} + c$

(d) None of these

16. $\int \frac{x^3dx}{(x^2+1)^3}$ is equal to ***[MPPET-2008]***

(a) $\frac{1}{4}\left\{\frac{2x^2+1}{(x^2+1)^2}\right\} + K$

(b) $-\frac{1}{4}\left\{\frac{2x^2+1}{(x^2+1)^2}\right\} + K$

(c) $\frac{1}{3}\left\{\frac{2x^2-1}{(x^2+1)^2}\right\} + K$

(d) $-\frac{1}{3}\left\{\frac{2x^2-1}{(x^2+1)^2}\right\} + K$

17. The value of $\int \frac{3x+2}{(x-2)^2(x-3)}dx$ is ***[MPPET-2008]***

(a) $11\log\frac{x-3}{x-2} - \frac{8}{x-2} + k$

(b) $11\log\frac{x+3}{x+2} - \frac{8}{x-2} + k$

(c) $11\log\frac{x-3}{x-2} + \frac{8}{x-2} + k$

(d) $11\log\frac{x+3}{x+2} + \frac{8}{x-2} + k$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 15 minutes.
3. The worksheet consists of 15 questions. The maximum marks are 45.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int 2 \sin x \cos x \, dx$ is equal to *[SCRA-1996]*

(a) $\cos 2x + c$ (b) $\sin 2x + c$

(c) $\cos^2 x + c$ (d) $\sin^2 x + c$

2. $\int \frac{2^x}{\sqrt{1-4^x}} dx = K \sin^{-1}(2^x)$, then $K =$

(a) $\frac{1}{2}\log 2$ (b) $\frac{1}{\log 2}$

(c) $-\frac{1}{2}\log 2$ (d) None of these

3. $\int \cos x \{\cot(\sin x)\} dx$ is equal to

(a) $\log |\sin(\sin x)| + c$

(b) $\log |\cos(\sin x)| + c$

(c) $\log |\tan(\sin x)| + c$

(d) None of these

4. What is the value of $\int (\sqrt{x} + x)^{-1} dx$?

[NDA-07]

(a) $\ln(x + \sqrt{x}) + c$

(b) $2 \ln(1 + \sqrt{x}) + c$

(c) $2\ln(x + \sqrt{x}) + c$

(d) $2 \ln(x - \sqrt{x}) + c$

5. $\int \frac{\operatorname{cosec}(\tan^{-1} x)}{1+x^2} dx =$

(a) $\log \tan^{-1} x + c$

(b) $\log \tan(\tan^{-1} x) + c$

(c) $\log \tan(\frac{1}{2}\tan^{-1} x) + c$

(d) $\log \cot(\frac{1}{2}\tan^{-1} x) + c$

6. $\int \frac{2x \tan^{-1} x^2}{1+x^4} dx =$ *[Roorkee-1982]*

(a) $[\tan^{-1} x^2]^2 + c$ (b) $\frac{1}{2}[\tan^{-1} x^2]^2 + c$

(c) $2[\tan^{-1} x^2]^2 + c$ (d) None of these

7. $\int \frac{1}{\sqrt{1-e^{2x}}} dx =$

[MPPET-1993, 2002; RPET-99]

(a) $x - \log[1 + \sqrt{1-e^{2x}}] + c$

(b) $x + \log[1 + \sqrt{1-e^{2x}}] + c$

(c) $\log[1 + \sqrt{1-e^{2x}}] - x + c$

(d) None of these

8. $\int \frac{a^{x/2}}{\sqrt{a^{-x} - a^x}} dx$ is equal to

[Kerala PET-2003; MNR-1983, 87]

(a) $\frac{1}{\log a} \sin^{-1}(a^x) + c$

(b) $\frac{1}{\log a} \tan^{-1}(a^x) + c$

(c) $2\sqrt{a^{-x} - a^x} + c$

(d) $\log(a^x - 1) + c$

9. Correct evaluation of $\int \frac{x}{(x-2)(x-1)} dx$ is

[MPPET-1993]

(a) $\log_e \frac{(x-2)^2}{(x-1)} + p$

(b) $\log_e \left(\frac{x-1}{x-2}\right) + p$

(c) $\frac{x-1}{x-2} + p$

(d) $2 \log_e \left(\frac{x-2}{x-1}\right) + p$

10. Solve $\int \frac{dx}{x + \sqrt{x}}$

(a) $2 \log(\sqrt{x} + 1) + c$

(b) $2 \log(\sqrt{x} - 1) + c$

(c) $\log(\sqrt{x} + 1) + c$

(d) None of these

11. $\int \frac{x^2}{1+x^6}dx$ is equal to

(a) $x^3 + c$

(b) $\frac{1}{3}\tan^{-1}(x^3) + c$

(c) $\log(1 + x^3)$

(d) None of these

12. $\int \frac{e^x}{(2+e^x)(e^x+1)}dx$ is equal to

(a) $\log\left(\frac{e^x+1}{e^x+2}\right)+c$ (b) $\log\left(\frac{e^x+2}{e^x+1}\right)+c$

(c) $\left(\frac{e^x+1}{e^x+2}\right)+c$ (d) $\left(\frac{e^x+2}{e^x+1}\right)+c$

13. The value of $\int \frac{x^2+1}{x(x^2-1)}dx$

[MPPET-1999]

(a) $\log\frac{x^2-1}{x}+C$ (b) $-\log\frac{x}{x^2+1}+C$

(c) $\log\frac{x^2+1}{x^2}+C$ (d) None of these

14. $\int \frac{4x}{x^2+x-2}dx =$

(a) $2\log(x^2+x-2) - \frac{2}{3}\log\left(\frac{x-1}{x+2}\right)+c$

(b) $2\log(x^2+x-2) - 3\log\left(\frac{x-1}{x+2}\right)+c$

(c) $2\log(x^2+x-2)\frac{2}{3}\log\left(\frac{x-1}{x+2}\right)+c$

(d) None of these

15. $\int (\tan x + \cot x)^2\, dx =$

(a) $2x + \tan x - \cot x + c$

(b) $-\tan x + \cot x + c$

(c) $-\tan x - \cot x + c$

(d) $\tan x - \cot x + c$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	6. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

2. (b) Step 1: Substitution Method: $t = 2^x$

Hence $dt = 2^x \log 2 dx$

$$\int \frac{dt}{\log 2\sqrt{1-t^2}} = \frac{1}{\log 2}\sin^{-1}(2^x)$$

7. (a) Step 1: $\int \frac{dx}{e^x\sqrt{e^{-2x}-1}}$

Let $e^{-x} = t$

$$-e^{-x}dx = dt \Rightarrow \frac{dx}{e^x} = -dt$$

Step 2: $\int \frac{-dt}{\sqrt{t^2-1}} = -\log[t + \sqrt{t^2-1}]$

$$= -\log\left[e^{-x} + \sqrt{\frac{1-e^{2x}}{e^x}}\right]+c$$

$$= -\left[\log\left[1+\sqrt{1-e^{2x}}\right] - \log e^x\right]+c$$

$$= x - \log\left[1+\sqrt{1-e^{2x}}\right]+c$$

9. (a) Step 1: $\dfrac{x}{(x-2)(x-1)} = \dfrac{A}{x-2} + \dfrac{B}{x-1}$

Here $A = 2,\ B = -1$

Step 2: Resolving using partial fraction and then integrating

$$\int \frac{x}{(x-2)(x-1)}\,dx = \int \frac{2\,dx}{x-2} - \int \frac{dx}{x-1}$$

$$= 2\log(x-2) - \log(x-1)$$

$$= \log\frac{(x-2)^2}{x-1} + P$$

LECTURE

4 Integration of Algebraic Expressions 2

BASIC CONCEPTS

STANDARD INTEGRALS

(i) $\int \frac{-dx}{(1+x)\sqrt{1-x^2}} = \sqrt{\frac{1-x}{1+x}}$

(ii) $\int \frac{dx}{(1-x)\sqrt{1-x^2}} = \sqrt{\frac{1+x}{1-x}}$

(iii) $\int \frac{dx}{(x+1)\sqrt{x^2-1}} = \sqrt{\frac{x-1}{x+1}}$

(iv) $\int \frac{-dx}{(x-1)\sqrt{x^2-1}} = \sqrt{\frac{x+1}{x-1}}$

(v) $\int \frac{1}{x\sqrt{x^2-1}}dx = \begin{matrix} \sec^{-1}x + c \\ \text{or} \\ -\text{cosec}^{-1}x + c, |x| > 1 \end{matrix}$

(vi) $\int \frac{dx}{x\sqrt{x^2+1}} = \log\left[\frac{\sqrt{x^2+1}-1}{x}\right]$

(vii) $\int \frac{dx}{x^n(1+x^n)^{1/n}} = \frac{1}{1-n}(1+x^{-n})^{1-\frac{1}{n}} + c$

SOME SPECIAL INTEGRALS

(i) $\int \frac{dx}{\sqrt{2ax-x^2}} = \sin^{-1}\left(\frac{x-a}{a}\right)$

(ii) $\int \frac{dx}{2ax-x^2} = \frac{1}{2a}\log\frac{x}{2a-x}$

(iii) $\int \frac{dx}{x^2+2ax} = \frac{1}{2a}\log\frac{x}{x+2a}$

(iv) $\int \frac{dx}{\sqrt{x^2+2ax}} = \log[x+a+\sqrt{x^2+2ax}]$

INTEGRALS OF THE FORM

$$\int \frac{x^2}{x^4+kx^2+1}dx$$

or $\int \frac{x^2-1}{x^4+kx^2+1}dx$,

where, k is a constant positive, negative or zero

Working Rule

(i) Divide the numerator and denominator by x^2.

(ii) Put $x - \frac{1}{x} = t$ or $x + \frac{1}{x} = t$ whichever substitution, on differentiation gives, the numerator of the resulting integrand.

(iii) Evaluate the resulting integral in t.

(iv) Express the result in terms of x.

NOTES

1. $x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2 = \left(x + \frac{1}{x}\right)^2 - 2$

2. $\sin 2x = 1 - (\sin x - \cos x)^2$
$= (\sin x + \cos x)^2 - 1$

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate: $\int \frac{7\cos^3 x + 8\sin^3 x}{3\sin^2 x \cos^2 x} dx$

Solution

We first rewrite the integrand in the integrable form

we have $\frac{7\cos^3 x + 8\sin^3 x}{3\sin^2 x \cos^2 x}$

$$= \frac{7}{3}\frac{\cos^3 x}{\sin^2 x \cos^2 x} + \frac{8}{3}\frac{\sin^3 x}{\sin^2 x \cos^2 x}$$

$$= \frac{7}{3}\frac{\cos x}{\sin x \sin x} + \frac{8}{3}\frac{\sin x}{\cos x . \cos x}$$

Thus, $\int \frac{7\cos^3 x + 8\sin^3 x}{3\sin^2 x \cos^2 x} dx$

$$= \frac{7}{3}\int \cot x \operatorname{cosec} x \, dx + \frac{8}{3}\int \tan x \sec x \, dx$$

$$= \frac{7}{3}(-\operatorname{cosec} x) + \frac{8}{3}\sec x + c$$

$$= \frac{8}{3}\sec x - \frac{7}{3}\operatorname{cosec} x + c$$

2. Show that $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}} = 2\left(\frac{\sqrt{x}-1}{\sqrt{1-x}}\right) + c$

[CBSE-Sample paper-III]

Solution

Let $x = \sin^2 t \ \therefore \ dx = 2 \sin t \cos t \, dt$

$$\therefore \int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$$

$$= \frac{2\sin t \cos t dt}{(1+\sin t)\sqrt{\sin^2 t - \sin^4 t}}$$

$$= \frac{2\sin t \cos t dt}{(1+\sin t)\sin t \cos t}$$

$$= 2\int \frac{dt}{(1+\sin t)} = 2\int \frac{1-\sin t}{1-\sin^2 t} dt = 2\int \frac{1-\sin t}{\cos^2 t} dt$$

$$= 2\left[\int \sec^2 t dt - \int \sec t \tan t dt\right]$$

$$= 2\left[\tan t - \sec t\right] + c$$

$$= 2\left[\frac{\sqrt{x}}{\sqrt{1-x}} - \frac{1}{\sqrt{1-x}}\right] + c = 2\left[\frac{\sqrt{x}-1}{\sqrt{1-x}}\right] + c$$

Proved.

3. Evaluate: $\int x(4+x)^{1/4} dx$

Solution

$$\int x(4+x)^{1/4} dx \ \text{ Put } 4 + x = t$$

$$\Rightarrow x = t - 4, \, dx = dt$$

$$= \int (t-4) t^{1/4} dt = \int (t^{5/4} - 4t^{1/4}) dt$$

$$= \frac{4}{9}t^{9/4} - 4.\frac{4}{5}t^{5/4} + c$$

$$= \frac{4}{9}(4+x)^{9/4} - \frac{16}{5}(4+x)^{5/4} + c$$

4. Evaluate: $\int \frac{1}{1+e^x} dx$

Solution

$$\int \frac{1}{1+e^x} dx = \int \frac{1}{1+\frac{1}{e^{-x}}} dx = \int \frac{e^{-x}}{e^{-x}+1} dx,$$

$$| \text{ Put } e^{-x} + 1 = t, -e^{-x} dx = dt$$

$$= \int \frac{1}{t} dt = -\log|t| + c = -\log|e^{-x} + 1| + c$$

5. Evaluate: $\int \frac{(x^2-1)dx}{x\sqrt{(x^4+3x^2+1)}}$ ***[J.E.E.W.B.-1991]***

Solution

$$I = \int \frac{(x^2-1)dx}{x.x\sqrt{\left(x^2 + \frac{1}{x^2} + 3\right)}}$$

$$= \int \frac{(1-1/x^2)dx}{\sqrt{\{(x+1/x)^2 + 1\}}}$$

$$= \int \frac{dt}{\sqrt{(t^2+1)}} = \log\{t + \sqrt{(t^2+1)}\}$$

where $t = x + \frac{1}{x}$

$$= \log\left[\{x^2 + 1) + \sqrt{(x^4 + 3x^2 + 1)}\}/x\right]$$

6. Evaluate: $\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}}dx$

Solution

$$I=\int\frac{x^4-1}{x^2.x\sqrt{x^2+\frac{1}{x^2}+1}}=\int\frac{\left(x-\frac{1}{x^3}\right)dx}{\sqrt{x^2+\frac{1}{x^2}+1}}$$

Put $x^2+\frac{1}{x^2}+1=t \therefore 2\left(x-\frac{1}{x^3}\right)dx=dt$

$$\therefore \quad I=\frac{1}{2}\int\frac{dt}{\sqrt{t}}=\frac{1}{2}.2\sqrt{t}=\sqrt{t}$$

$$I=\sqrt{x^2+\frac{1}{x^2}+1}=\frac{\sqrt{x^4+x^2+1}}{x}$$

7. Evaluate: $\int\sqrt{\left(\frac{\sin(x-\alpha)}{\sin(x+\alpha)}\right)}dx$

Solution

$$I=\int\sqrt{\left(\frac{\sin(x-\alpha)}{\sin(x+\alpha)}\right)}dx$$

$$=\int\frac{\sin(x-\alpha)}{\sqrt{[\sin(x+\alpha)\sin(x-\alpha)}}dx$$

$$=\int\left\{\frac{\sin x\cos\alpha}{\sqrt{(\sin^2 x-\sin^2\alpha)}}-\frac{\cos x\sin\alpha}{\sqrt{(\sin^2 x-\sin^2\alpha)}}\right\}dx$$

$$=\cos\alpha\int\frac{\sin x dx}{\sqrt{(\cos^2\alpha-\cos^2 x)}}-\sin\alpha\int\frac{\cos x dx}{\sqrt{(\sin^2 x-\sin^2\alpha)}}$$

$\cos x = t$ for 1st and $\sin x = z$ for 2nd

$= \cos\alpha\cos^{-1}(\cos x\sec\alpha)-\sin\alpha\cos h^{-1}(\sin x\operatorname{cosec}\alpha)$

8. Evaluate $\int\frac{dx}{x^4+4}$

Solution

$$I=\frac{1}{4}\int\frac{(x^2+2)-(x^2-2)}{x^4+4}dx$$

$$=\frac{1}{4}\int\frac{x^2+2}{x^4+4}dx-\frac{1}{4}\int\frac{x^2-2}{x^4+4}dx$$

divide num. and deno. by x^2

$$=\frac{1}{4}\int\frac{1+\frac{2}{x^2}}{x^2+\frac{4}{x^2}}dx-\frac{1}{4}\int\frac{1-\frac{2}{x^2}}{x^2+\frac{4}{x^2}}dx=$$

$$\frac{1}{4}I_1-\frac{1}{4}I_2 \text{ (say)} \qquad \ldots(i)$$

To evaluate I_1, put $x-\frac{2}{x}=t\Rightarrow\left(1+\frac{2}{x^2}\right)dx=dt$

and $x^2+\frac{4}{x^2}=t^2+4$,

$$\therefore I_1=\int\frac{dt}{t^2+4}=\frac{1}{2}\tan^{-1}\frac{t}{2}=\frac{1}{2}\tan^{-1}\frac{x^2-2}{2x}$$

To evaluate I_2, put $x+\frac{2}{x}=u$

$$\Rightarrow\left(1-\frac{2}{x^2}\right)dx=du \text{ and } x^2+\frac{4}{x^2}=u^2-4$$

$$\therefore I_2=\int\frac{du}{u^2-(2)^2}=$$

$$\frac{1}{2.2}\log\left|\frac{u-2}{u+2}\right|=\frac{1}{4}\log\left|\frac{x^2-2x+2}{x^2+2x+2}\right|$$

From (i), we have

$$I=\frac{1}{8}\tan^{-1}\frac{x^2-2}{2x}-\frac{1}{16}\log\left|\frac{x^2-2x+2}{x^2+2x+2}\right|+C$$

9. Evaluate $\int\frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}dx$

Solution

Dividing num. and deno by $x\sqrt{x}$, we get

$$\int\frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}dx$$

$$=\int\frac{\frac{1}{\sqrt{x}}-\frac{1}{x\sqrt{x}}}{\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)\sqrt{x+1+\frac{1}{x}}}dx$$

Put $\sqrt{x}+\frac{1}{\sqrt{x}}=t\Rightarrow\left(\frac{1}{2\sqrt{x}}-\frac{1}{2x\sqrt{x}}\right)dx=dt$

$$\Rightarrow\left(\frac{1}{\sqrt{x}}-\frac{1}{x\sqrt{x}}\right)dx=2dt, x+\frac{1}{x}=t^2-2$$

$$=\int\frac{2dt}{t\sqrt{t^2-1}}=2\sec^{-1}t+C$$

$$=2\sec^{-1}\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)+C$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E)/STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Integrate the following with respect to x. $\frac{x^4+x^2-1}{2(x^2+1)}$
2. Integrate $\frac{1}{1+\sqrt{x}}$ with respect to x.
3. Evaluate $\int \sqrt{\tan x}+\sqrt{\cot x}\,dx$ ***[CBSE (model question paper), MPPET-2004]***
4. Evaluate $\int \frac{x^2}{x^2-a^2}dx$.
5. Evaluate $\int \frac{x}{\sqrt{x+2}}dx$ ***[HSB-92]***
6. Solve $\int \frac{x^2dx}{(x^2+1)(3x^2+1)}$
7. Evaluate $\int \frac{x^2+1}{x^4-x^2+1}\,dx$ ***[MP-2000]***
8. Evaluate $\int \sqrt{\sec x-1}dx$. ***[PSB-91 (C)]***

ANSWERS

1. $\frac{x^3}{6}-\frac{1}{2}\tan^{-1}x$
2. $2[\sqrt{x}-\log(\sqrt{x}+1)]$
3. $\sqrt{2}\tan^{-1}\left(\frac{\tan x-1}{\sqrt{2}\tan x}\right)+C$
4. $x+\frac{a}{2}\log\left(\frac{x-a}{x+a}\right)$
5. $\frac{2}{3}(x+2)^{3/2}-4\sqrt{x+2}+c$
6. $\frac{1}{2}\tan^{-1}x-\frac{1}{2\sqrt{3}}\tan^{-1}(x\sqrt{3})$
7. $\tan^{-1}\left(\frac{x^2-1}{x}\right)$
8. $-\log\left|\left(\cos x+\frac{1}{2}\right)+\sqrt{\cos^2 x+\cos x}\right|+c$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\int \sqrt{e^x-1}\,dx$

   (a) $2\left[\sqrt{e^x-1}-\tan^{-1}\sqrt{e^x-1}\right]+c$

   (b) $\sqrt{e^x-1}-\tan^{-1}\sqrt{e^x-1}+c$

   (c) $\sqrt{e^x-1}+\tan^{-1}\sqrt{e^x-1}+c$

   (d) $2\left[\sqrt{e^x-1}+\tan^{-1}\sqrt{e^x-1}\right]+c$

Solution

(a) $A=\int \sqrt{e^x-1}dx$

Let $e^x-1=t^2 \Rightarrow e^x dx=2tdt.$

Hence $dx=\frac{2t}{t^2+1}dt$

$\therefore\ A=\int t\frac{2t}{t^2+1}dt=\int \frac{2t^2}{t^2+1}$

$$= \int \frac{2(t^2+1)-2}{t^2+1}dt = \int 2dt - \int \frac{2dt}{t^2+1}$$

$$= 2t - 2\tan^{-1} t + c$$

$$= 2\sqrt{e^x - 1} - 2\tan^{-1}\sqrt{e^x - 1} + c$$

2. The value of $\int \frac{\sqrt{(x^2-a^2)}}{x}dx$ will be

(a) $\sqrt{(x^2-a^2)} - a\tan^{-1}\left[\frac{\sqrt{(x^2-a^2)}}{a}\right]$

(b) $\sqrt{(x^2-a^2)} + a\tan^{-1}\left[\frac{\sqrt{(x^2-a^2)}}{a}\right]$

(c) $\sqrt{(x^2-a^2)} + a^2\tan^{-1}\left[\sqrt{(x^2-a^2)}\right]$

(d) $\tan^{-1} x/a + c$

Solution

(a) Let $\sqrt{(x^2-a^2)} = t \Rightarrow x^2 - a^2 = t^2$

$$\Rightarrow x^2 = a^2 + t^2$$

$\therefore x\,dx = t\,dt$

$$\therefore \int \frac{\sqrt{(x^2-a^2)}}{x}dx = \int \frac{\sqrt{(x^2-a^2)}x}{x^2}dx$$

$$\Rightarrow I = \int \frac{t}{a^2+t^2}tdt = \int \frac{t^2}{a^2+t^2}dt$$

$$\Rightarrow I = \int \left(1 - \frac{a^2}{a^2+t^2}\right)dt$$

$$= t - a^2\frac{1}{a}\tan^{-1}\left(\frac{t}{a}\right)$$

$$\Rightarrow I = \sqrt{(x^2-a^2)} - a\tan^{-1}\left[\frac{\left\{\sqrt{x^2-a^2}\right\}}{a}\right]$$

3. $\int x\sqrt{\frac{1-x^2}{1+x^2}}dx =$

(a) $\frac{1}{2}\left[\sin^{-1} x^2 + \sqrt{1-x^4}\right] + c$

(b) $\frac{1}{2}\left[\sin^{-1} x^2 + \sqrt{1-x^2}\right] + c$

(c) $\sin^{-1} x^2 + \sqrt{1-x^4} + c$

(d) $\sin^{-1} x^2 + \sqrt{1-x^2} + c$

Solution

(a) $\int x\sqrt{\frac{1-x^2}{1+x^2}}dx = \int \frac{x(1-x^2)}{\sqrt{1-x^4}}dx$

$$\left\langle \begin{matrix} \text{multilpy } N^r \text{ and } D^r \\ \text{by } (1-x^2)^{1/2} \end{matrix} \right\rangle$$

$$\Rightarrow \int \frac{x}{\sqrt{1-x^4}}dx - \int \frac{x^3}{\sqrt{1-x^4}}dx$$

$$\Rightarrow \frac{1}{2}\left[\sin^{-1}(x^2) + \sqrt{1-x^4}\right] + c$$

[By putting $x^2 = t$ and $\sqrt{1-x^4} = \sqrt{t}$ resp.]

4. The value of the integral $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x}dx$ is ***[PUCET-2002; IIT-95]***

(a) $\sin x - 6\tan^{-1}(\sin x) + c$

(b) $\sin x - 2(\sin x)^{-1} + c$

(c) $\sin x - 2(\sin x)^{-1} - 6\tan^{-1}(\sin x) + c$

(d) $\sin x - 2(\sin x)^{-1} + 5\tan^{-1}(\sin x) + c$

Solution

(c) Given integral =

$$\int \frac{[(1-\sin^2 x) + (1-\sin^2 x)^2]\cos x}{\sin^2 x + \sin^4 x}dx$$

Put $\sin x = z \therefore \cos x\,dx = dz$

$\therefore$ given integral $= \int \frac{[(1-z^2 + (1-z^2)^2]dz}{z^2+z^4}$

$$= \int \frac{z^4 - 3z^2 + 2}{z^4+z^2}dz = \int \left[1 + \frac{2-4z^2}{z^4+z^2}\right]dz$$

$$= \int dz + 2\int \left[\frac{1}{z^2} - \frac{3}{z^2+1}\right] + c$$

$$= z - \frac{2}{z} - 6\tan^{-1} z + c$$

$$= \sin x - 2(\sin x)^{-1} - 6\tan^{-1}(\sin x) + c.$$

5. $\int \frac{x^2-1}{x\sqrt{(x^2+\alpha x+1)(x^2+\beta x+1)}}dx$ is equal to ***[PUCET-2002; Roorkee-97]***

(a) $\log\left|\frac{\sqrt{x^2+\alpha x+1} + \sqrt{x^2+\beta x+1}}{\sqrt{x}}\right| + c$

(b) $2\log\left|\frac{\sqrt{x^2+\alpha x+1} - \sqrt{x^2+\beta x+1}}{\sqrt{x}}\right| + c$

(c) $\log\left[\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}\right]+c$

(d) None of these

Solution

(b) Given integral

$$=\int\frac{x^2\left(1-\frac{1}{x^2}\right)dx}{x^2\left[\sqrt{\left(x+\frac{1}{x}+\alpha\right)}\sqrt{x+\frac{1}{x}+\beta}\right]}$$

Put $x+\frac{1}{x}=z \quad \therefore \left(1-\frac{1}{x^2}\right)dx=dz$

$\therefore$ given integral $=\int\frac{dz}{\sqrt{(z+\alpha)(z+\beta)}}$

$$=\int\frac{dz}{\sqrt{z^2+(\alpha+\beta)z+\alpha\beta}}$$

$$=\int\frac{dz}{\sqrt{\left(z+\frac{\alpha+\beta}{2}\right)^2-\left(\frac{\alpha-\beta}{2}\right)^2}}$$

$$=\log\left|z+\frac{\alpha+\beta}{2}-\sqrt{\left(z+\frac{\alpha+\beta}{2}\right)^2+\left(\frac{\alpha-\beta}{2}\right)^2}\right|+c$$

$$=\log\left|x+\frac{1}{x}+\frac{\alpha+\beta}{2}-\sqrt{(z+\alpha)(z+\beta)}\right|+c$$

$$=\log\left|\frac{x+\frac{1}{x}+\frac{\alpha+\beta}{2}-}{\sqrt{\left(x+\frac{1}{x}+\alpha\right)\left(x+\frac{1}{x}+\beta\right)}}\right|+c$$

$$=\log\left|\frac{\frac{2x^2+(\alpha+\beta)x+2}{2x}-}{\frac{\sqrt{(x^2+\alpha x+1)(x^2+\beta x+1)}}{x}}\right|+c$$

$$=\log\frac{1}{2}\left(\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right)^2+c$$

$$=2\log\left(\frac{\sqrt{x^2+\alpha x+1}-\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right)+c$$

where, $\log\frac{1}{2}$ has been merged with c.

6. $\int\frac{2x}{(1-x^2)\sqrt{x^4-1}}dx$ is equal to

[Roorkee-98, 2002]

(a) $\sqrt{\frac{x^2+1}{x^2-1}}+c$ (b) $\sqrt{\frac{x^2-1}{x^2+1}}+c$

(c) $\sqrt{x^4-1}+c$ (d) None of these

Solution

(a) $\int\frac{2x}{(1-x^2)\sqrt{x^4-1}}dx=\int\frac{-2x}{(x^2-1)^{3/2}\sqrt{x^2+1}}dx$

Put $\sqrt{\frac{x^2+1}{x^2-1}}=z$

$$\therefore \quad \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{1/2}\cdot\frac{(x^2-1).2x-(x^2+1)2x}{(x^2-1)^2}dx=dz$$

$$\Rightarrow \left(\frac{\sqrt{x^2-1}}{\sqrt{x^2+1}}\right)\cdot\frac{-2x}{(x^2-1)^2}dx=dz$$

$$\Rightarrow \frac{-2x}{(x^2-1)^{3/2}\sqrt{x^2+1}}dx=dz \quad \therefore \text{ given integral}$$

$$=\int dz=z+c=\sqrt{\frac{x^2+1}{x^2-1}}+c$$

7. $\int\frac{dx}{x^2(x^4+1)^{3/4}}$ is equal to

(a) $\frac{(x^4+1)^{1/4}}{x}+c$ (b) $-\frac{(x^4+1)^{1/4}}{x}+c$

(c) $\frac{3}{4}\frac{(x^4+1)^{3/4}}{x}+c$ (d) $\frac{4}{3}\frac{(x^4+1)^{3/4}}{x}+c$

[IIT-84; RPET-2000; UPSEAT-2002; AMU-2003]

Solution

(b) $I=\int\frac{dx}{x^5\left(1+\frac{1}{x^4}\right)^{3/4}}$

Now put $1+\frac{1}{x^4}=t$, then $-4\frac{1}{x^5}dx=dt$

$$I=-\frac{1}{4}\int\frac{dt}{t^{3/4}}=-t^{1/4}+c \quad =\frac{(x^4+1)^{1/4}}{x}+c$$

8. $\int \frac{dx}{\sqrt[4]{x^4-1}}$ is equal to

[JEE (WB)-2005]

(a) $\frac{1}{2}\log\left|x^2-\sqrt{x^4-1}\right|+c$

(b) $\frac{1}{2}\log\left|x^2+\sqrt{x^4-1}\right|+c$

(c) $-\frac{1}{2}\log\left|x^2+\sqrt{x^4+1}\right|+c$

(d) None of these

Solution

(b) $I=\int \frac{dx}{x^4\sqrt{\left(1-\frac{1}{x^4}\right)}}$

Now let $1/x^2=\sin\theta$, then $-2x^{-3}dx=\cos\theta d\theta$

$$I=-\frac{1}{2}\int\frac{x^2\cos\theta}{\cos\theta}d\theta$$

$$=-\frac{1}{2}\int \text{cosec }\theta d\theta$$

$$=-\frac{1}{2}\log\left|\text{cosec }\theta-\cot\theta\right|+c$$

$$=\frac{1}{2}\log\left|\text{cosec }\theta-\cot\theta\right|+c$$

$$=\frac{1}{2}\log\left|x^2+\sqrt{x^4-1}\right|+c$$

9. $\int \frac{dx}{\sqrt{3-5x-x^2}}$ equals

(a) $\sin^{-1}\left(\frac{2x+5}{\sqrt{37}}\right)+c$

(b) $\cos^{-1}\left(\frac{2x+5}{\sqrt{37}}\right)+c$

(c) $\sin^{-1}(2x+5)+c$

(d) None of these

Solution

(a) $I=\int \frac{dx}{\sqrt{\frac{37}{4}-\left(x+\frac{5}{2}\right)^2}}$

$$=\sin^{-1}\left(\frac{x+5/2}{\sqrt{37}/2}\right)+c=\sin^{-1}\left(\frac{2x+5}{\sqrt{37}}\right)+c$$

10. $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$ is equal to

(a) $\frac{2(\sqrt{x}-1)}{\sqrt{1-x}}+c$ (b) $\frac{2(1-\sqrt{x})}{\sqrt{1-x}}+c$

(c) $\frac{(\sqrt{x}-1)}{\sqrt{1-x}}+c$ (d) None of these

Solution

(a) $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$

Put $x = \sin^2 0$; dx $2\sin 0\cos 0d0$ given integral

$$=\int\frac{2\sin\theta\cos\theta d\theta}{(1+\sin\theta)\sqrt{\sin^2\theta-\sin^4\theta}}$$

$$=\int\frac{2\cos 0\,d0}{(1+\sin 0)\sqrt{1-\sin^2 0}}$$

$$=2\int\frac{d\theta}{(1+\sin\theta)}=2\int\frac{1-\sin\theta}{\cos^2\theta}d0$$

$$=2\int(\sec^2 0-\sec 0\tan 0)d0$$

$$=2(\tan 0-\sec 0)+c$$

$$=2\left(\frac{\sqrt{x}}{\sqrt{1-x}}-\frac{1}{\sqrt{1-x}}\right)+c$$

$$=\frac{2(\sqrt{x}-1)}{\sqrt{1-x}}+c$$

11. $\int \frac{\cos 5x+\cos 4x}{1-2\cos 3x}dx$ is equal to

(a) $\frac{\sin 2x}{2}+\sin x+c$

(b) $-\frac{\sin 2x}{2}-\sin x+c$

(c) $\frac{\sin 2x}{2}-\sin x+c$

(d) $-\frac{\sin 2x}{2}+\sin x+c$

Solution

(b) The given integral is

$$\int\frac{\sin 3x(\cos 5x+\cos 4x)}{\sin 3x(1-2\cos 3x)}dx$$

$$=\int\frac{\sin 3x(\cos 5x+\cos 4x)}{\sin 3x-\sin 6x}dx$$

$$= \int \frac{2\sin\frac{3x}{2}\cos\frac{3x}{2}.2\cos\frac{9x}{2}\cos\frac{x}{2}}{-2\cos\frac{9x}{2}\sin\frac{3x}{2}}dx$$

$$= -\int 2\cos\frac{3x}{2}\cos\frac{x}{2}dx$$

$$= -\int(\cos 2x + \cos x)dx$$

$$= -\left(\frac{\sin 2x}{2} + \sin x\right) + C$$

12. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(fofo.....f)}_{f \text{ occurs } n \text{ times}}(x)$ Then, $\int x^{n-2}g(x)dx$ equals

[IIT-2007]

(a) $\frac{1}{n(n-1)}(1+nx^n)^{1-1/n} + K$

(b) $\frac{1}{(n-1)}(1+nx^n)^{1-1/n} + K$

(c) $\frac{1}{n(n+1)}(1+nx^n)^{1-1/n} + K$

(d) $\frac{1}{(n+1)}(1+nx^n)^{1+1/n} + K$

Solution

(a) Given that $f(x) = \frac{x}{(1+x^n)^{1/n}}$

$\therefore f(f(x)) = \frac{y}{(1+y^n)^{1/n}} = \frac{x}{(1+2x^n)^{1/n}}$

and $f(f(f(x))) = \frac{x}{(1+3x^n)^{1/n}}$

Now, $\frac{(fofofo.....f)(x)}{n \text{ times}} = g(x) = \frac{x}{(1+nx^n)^{1/n}}$

$\int x^{n-2}g(x)dx = \int \frac{x^{n-1}}{(1+nx^n)^{1/n}}dx$

$= \frac{1}{n^2}\int n^2.x^{n-1}.(1+nx^n)^{1/n}dx$

$= \frac{1}{n^2}.\frac{(1+nx^n)^{1-1/n}}{1-\frac{1}{n}} + K = \frac{(1+nx^n)^{1-1/n}}{n(n-1)} + K$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int \frac{x^2}{x^2+4}dx$ equals to ***[RPET-01]***

(a) $x - 2\tan^{1}(x/2) + C$

(b) $x + 2\tan^{1}(x/2) + C$

(c) $x - 4\tan^{-1}(x/2) + C$

(d) $x + 4\tan^{-1}(x/2) + C$

2. $\int(\sec x + \tan x)2\,dx$ ***[MPPET-1987, 92]***

(a) $2(\sec x + \tan x) - x + c$

(b) $\frac{1}{3}(\sec x + \tan x)^3 + c$

(c) $\sec x(\sec x + \tan x) + c$

(d) $2(\sec x + \tan x) + c$

3. $\int \frac{x^2+1}{x^4-x^2+1}dx =$ ***[MPPET-1991]***

(a) $\tan^{-1}\left(\frac{1+x^2}{x}\right) + c$

(b) $\cot^{-1}\left(\frac{1+x^2}{x}\right) + c$

(c) $\tan^{-1}\left(\frac{x^2-1}{x}\right) + c$

(d) $\cot^{-1}\left(\frac{x^2-1}{x}\right) + c$

4. $\int \frac{1}{\cos^2 x(1-\tan x)^2}dx$

(a) $\frac{1}{\tan x - 1} + c$ (b) $\frac{1}{1-\tan x} + c$

(c) $-\frac{1}{3}\frac{1}{(1-\tan x)^3} + c$ (d) None of these

5. $\int \frac{1}{x^2\sqrt{1+x^2}}dx =$

(a) $-\frac{\sqrt{1+x^2}}{x} + c$ (b) $-\frac{\sqrt{1+x^2}}{x} + c$

(c) $-\frac{\sqrt{1-x^2}}{x} + c$ (d) $-\frac{\sqrt{x^2-1}}{x} + c$

6. $\int \frac{dx}{\cos x - \sin x}$ is equal to

(a) $\frac{1}{\sqrt{2}} \log \left| \tan\left(\frac{x}{2} - \frac{3\pi}{8}\right) \right| + C$

(b) $\frac{1}{\sqrt{2}} \log \left| \tan\left(\frac{x}{2} - \frac{\pi}{8}\right) \right| + C$

(c) $\sqrt{2} \log \left| \tan\left(\frac{x}{2} - \frac{3\pi}{8}\right) \right| + C$

(d) None of these

7. $\int \left(\frac{1-x}{1+x}\right)^2 dx =$

(a) $x - 4\log(x+1) + \frac{4}{x+1} + c$

(b) $x - 4\log(x+1) - \frac{4}{x+1} + c$

(c) $x + 4\log(x+1) + \frac{4}{x+1}$

(d) None of these

8. $\int e^{e^{e^x}} . e^{e^x} . e^x$ is equal to

(a) $\frac{1}{2} 2e^{e^x}$ (b) $(e^{e^{e^x}})^2$

(c) $e^{e^{e^x}}$ (d) $\frac{1}{2} e^{e^{e^x}}$

9. $\int \frac{\tan(\log x)}{x} dx$ is equal to

(a) $\log |\cos(\log x)| + c$

(b) $\log |\sec(\log x)| + c$

(c) $\log |\sin(\log x)| + c$

(d) None of these

10. What is the value of $\int \frac{e^x(1+x)}{\sin^2(xe^x)} dx$?

[NDA-2007]

(a) $-e^x \cot x + C$

(b) $\cos^2(xe^x) + C$

(c) $\log \sin(xe^x) + C$

(d) $-\cot(xe^x) + C$

11. $\int \frac{dx}{1+e^x} =$

[MPPET-1991; Roorkee-1977]

(a) $\log(1 + e^x)$ (b) $-\log(1 + e^x)$

(c) $-\log(1 - e^{-x})$ (d) $\log(e^{-x} + e^{-2x})$

12. $\int \frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x} dx =$

[IIT-86]

(a) $\sin 2x + c$ (b) $-\frac{1}{2} \sin 2x + c$

(c) $\frac{1}{2} \sin 2x + c$ (d) $-\sin 2x + c$

13. $\int \frac{\cos x - \sin x}{1 + \sin 2x} dx =$ ***[AISSE-1985]***

(a) $-\frac{1}{\cos x + \sin x} + c$

(b) $\frac{1}{\cos x + \sin x} + c$

(c) $\frac{1}{\cos x - \sin x} + c$

(d) None of these

14. $\int \frac{\sec^2 x}{1 + \tan x} dx =$ ***[MPPET-1987]***

(a) $\log(\cos x + \sin x) + c$

(b) $\log(\sec^2 x)$

(c) $\log(1 + \tan x)$

(d) $-\frac{1}{(1+\tan x)^2}$

15. If $\int \frac{1}{(1+x)\sqrt{x}} dx = f(x) + A$, where A is any arbitrary constant, then the function $f(x)$ is ***[MPPET-1998]***

(a) $2 \tan^{-1} x$ (b) $2\tan^{-1} \sqrt{x}$

(c) $2 \cot^{-1} \sqrt{x}$ (d) $\log_e(1 + x)$

16. $\int \sin x . \cos x . \cos 2x . \cos 4x . \cos 8x dx =$

(a) $\frac{1}{96} \cos 16x + c$ (b) $\frac{-1}{256} \cos 16x + c$

(c) $\frac{-1}{16} \cos 16x + c$ (d) None of these

17. $\int \frac{3x^2}{\sqrt{9 - 16x^6}} dx =$

(a) $\frac{1}{4} \sin^{-1}\left(\frac{4x^3}{3}\right) + c$ (b) $\frac{1}{3} \sin^{-1}\left(\frac{4x^3}{3}\right) + c$

(c) $\frac{1}{4} \sin^{-1} x^3 + c$ (d) $\frac{1}{3} \sin^{-1} x^3 + c$

18. $\int \frac{\sin x \cos x}{\sqrt{1-\sin^4 x}} dx =$

[Karnataka CET-2008]

(a) $\tan^{-1}(\sin^2 x) + C$

(b) $\tan^{-1}(2 \sin x) + C$

(c) $\frac{1}{2} \sin^{-1}(\sin^2 x) + C$

(d) $\frac{1}{2} \cos^{-1}(\sin^2 x) + C$

19. $\int \frac{\sec x}{\sec x + \tan x} dx =$

[Karnataka CET-2008]

(a) $\sec x + \tan x + C$

(b) $\log \sin x + \log \cos x + C$

(c) $\tan x - \sec x + C$

(d) $\log (1 + \sin x) + C$

20. The value of $\int \frac{dx}{x\sqrt{x^4-1}}$ is

(a) $\frac{1}{2} \sec^{-1} x^2 + k$ (b) $\log x \sqrt{x^4-1} + k$

(c) $x \log \sqrt{x^4-1} + k$ (d) $\log \sqrt{x^4-1} + k$

21. $\int \frac{1}{1+\cos x + \sin x} dx$ is equal to

[MPPET-2009]

(a) $\log\left|1 + \tan\frac{x}{2}\right| + c$

(b) $\frac{1}{2}\log\left|1 + \tan\frac{x}{2}\right| + c$

(c) $2\log\left|1 + \tan\frac{x}{2}\right| + c$

(d) $\frac{1}{2}\log\left|1 - \tan\frac{x}{2}\right| + c$

SOLUTIONS

1. (a) $\int \frac{x^2}{x^2+4} dx = \int \frac{x^2+4-4}{x^2+4} dx$

$= \int 1 dx + \int \frac{-4}{x^2+4} dx = x - 2\tan^{-1}\left(\frac{x}{2}\right) + c$

$\left(\because \int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)\right)$

2. (a) $\int (\sec x + \tan x)^2 dx$

$= \int (\sec^2 x + \tan^2 x + 2\tan x \sec x) dx$

$= \int (2\sec^2 x - 1 + 2\sec x \tan x) dx$

$= 2\tan x - x + 2\sec x + c$

$\Rightarrow 2(\sec x + \tan x) - x + c$

3. (c) Step 1: Dividing numerator and denominator by x^2

$$I = \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}-1} dx = \int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2-1} dx$$

Step 2: Let $t = x - \frac{1}{x}$ then $dt = \left(1+\frac{1}{x^2}\right)dx$

$$I = \int \frac{dt}{t^2+1} = \tan^{-1} t = \tan^{-1}\left(x - \frac{1}{x}\right)$$

$$= \tan^{-1}\left(\frac{x^2-1}{x}\right) + c$$

4. (b) $\int \frac{1}{\cos^2 x (1-\tan x)^2} dx = \int \frac{\sec^2 x}{(1-\tan x)^2} dx$

Put $1 - \tan x = t \Rightarrow -\sec^2 x dx = dt$

$$-\int \frac{dt}{t^2} = \frac{1}{1-\tan x} + c$$

5. (a) $\int \frac{1}{x^2\sqrt{1+x^2}} dx = \frac{1}{1-2}(1-x^{-2})^{1-\frac{1}{2}} + c$

$= -(1+x^{-2})^{1/2} + c$

$= \frac{-(x^2+1)^{1/2}}{x} + c$

$\Rightarrow -\frac{\sqrt{1+x^2}}{x} + c$

NOTE

By using formula

$$\int \frac{dx}{x^n(1+x^n)^{1/n}} = \frac{1}{1-n}(1+x^{-n})^{1-\frac{1}{n}} + c$$

6. (b) $\int \frac{dx}{\cos x - \sin x}$

multiplying and dividing by $\sqrt{2}$

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\cos x \sin\frac{\pi}{4} - \sin x \cos\frac{\pi}{4}}$$

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\sin\left(\frac{\pi}{4} - x\right)}$$

$$= \frac{1}{\sqrt{2}} \log\left|\tan\left(\frac{x}{2} - \frac{\pi}{8}\right)\right| + c$$

7. (b) Step 1: Let $t = x + 1$, $dt = dx$ and $x = t - 1$

Step 2: $\int \left\{\frac{1-(t-1)}{t}\right\}^2 dt = \int \left\{\frac{2-t}{t}\right\}^2 dt$

$$= \int \left(\frac{4}{t^2} - \frac{4t}{t^2} + \frac{t^2}{t^2}\right) dt$$

$$= -\frac{4}{t} - 4\log t + t$$

$$= -\frac{4}{x+1} - 4\log(x+1) + (x+1) + c$$

8. (c) $\int e^{e^{e^x}} . e^{e^x} . e^x dx$

Put $e^{e^{e^x}} = t \Rightarrow dt = e^{e^{e^x}} . e^{e^x} . e^x dx$

$$\int dt = t = e^{e^{e^x}}$$

9. (b) Step 1: $t = \log x$; $dt = \frac{dx}{x}$

$$\int \tan(t)\, dt = \log|\sec t| = \log|\sec(\log x)| + c$$

10. (d) Step 1: Let $I = \frac{e^x(1+x)}{\sin^2(xe^x)} dx$

Put $xe^x = t$ and after differentiating both sides, we get $e^x(1+x)dx = dt$ and put this in given integral this in given integral, we get

$$I = \int \frac{dt}{\sin^2 t} = \int \operatorname{cosec}^2 t dt$$

Step 2: Apply: $\int \operatorname{cosec}^2 x dx = -\cot x + c$

We get $\int \operatorname{cosec}^2 t dt = -\cot t + c$

$$= -\cot(xe^x) + c$$

11. (b) Step 1: Multiplying numerator and denominator by e^x

$$I = \int \frac{e^{-x} dx}{1 + e^{-x}}$$

Step 2: Let $t = 1 + e^{-x} \Rightarrow dt = -e^{-x} dx$

$$\int -\frac{dt}{t} = -\log t = -\log(1 + e^{-x}) + c$$

12. (c) Step 1: $\cos^8 x - \sin^8 x$

$$= (\cos^4 x - \sin^4 x)(\cos^4 x + \sin^4 x)$$

$$= (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)$$

$$(1 - 2\sin^2 x \cos^2 x)$$

Step 2: $\int \cos 2x dx = \frac{\sin 2x}{2} + c$

13. (a) $\int \frac{\cos x - \sin x}{1 + \sin 2x} dx = \int \frac{\cos x - \sin x}{(\sin x + \cos x)^2} dx$

Put $t = \sin x + \cos x \Rightarrow dt = (\cos x - \sin x)dx$

$$\int \frac{dt}{t^2} = -\frac{1}{t} = -\frac{1}{\sin x + \cos x} + c$$

14. (c) $\int \frac{\sec^2 x}{1 + \tan x} dx$

Put $t = 1 + \tan x \Rightarrow dt = \sec^2 x dx$

$$\int \frac{dt}{t} = \log(t) = \log(1 + \tan x) + c$$

15. (b) Let $x = t^2$. Hence $dx = 2tdt$

$$\int \frac{2}{1+t^2} dt = 2\tan^{-1} t = 2\tan^{-1}\left(\sqrt{x}\right) + A$$

16. (b) $\int \sin x . \cos x . \cos 2x . \cos 4x . \cos 8x dx$

$$= \frac{1}{2} \int (\sin 2x \cos 2x . \cos 4x \cos 8x) dx$$

$$= \frac{1}{4} \int (\sin 4x . \cos 4x \cos 8x) dx$$

$$= \frac{1}{8} \int (\sin 8x \cos 8x) dx = \frac{1}{16} \int \sin 16x dx$$

$$= \frac{-\cos 16x}{256} + c$$

NOTE

By using formula $\sin 2x = 2\sin x \cos x$

17. (a) $\int \frac{3x^2}{\sqrt{9-16x^6}}dx = \int \frac{x^2}{\sqrt{1-\left(\frac{4x^3}{3}\right)^2}}$

Put $t = \frac{4x^3}{3} \Rightarrow dt = 4x^2 dx$

$\frac{1}{4}\int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{4}\sin^{-1}\left(\frac{4x^3}{3}\right) + c$

18. (c) $\int \frac{\sin x \cos x}{\sqrt{1-\sin^4 x}}dx$

Put $\sin^2 x = t \Rightarrow 2\sin x \cos x dx = dt$

$\frac{1}{2}\int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{2}\sin^{-1}(\sin^2 x) + c$

19. (c) $\int \frac{\sec x}{\sec x + \tan x}dx$

$= \int \frac{\sec x(\sec x - \tan x)}{\sec^2 x - \tan^2 x}dx$

$= \int (\sec^2 x - \tan x \sec x)dx$

$(\because \sec^2 x + \tan^2 x = 1)$

$= \tan x - \sec x + c$

20. (a) $\int \frac{dx}{x\sqrt{x^4-1}} = \frac{1}{2}\sec^{-1} x^2 + k$

By using formula $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + c$

21. (a) Let $I = \int \frac{1}{1+\cos x + \sin x}dx$

$= \int \frac{1}{1 + \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}} + \frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}dx$

$= \int \frac{\sec^2\frac{x}{2}}{2\left(1+\tan\frac{x}{2}\right)}dx$

Put $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

$\therefore I = \int \frac{dt}{1+t} = \log(1+t) + c$

$= \log\left|1+\tan\frac{x}{2}\right| + c$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. $\int \frac{\sec^2 x dx}{\sqrt{\tan^2 x + 4}} =$ ***[MPPET-1991]***

(a) $\log\left[\tan x + \sqrt{\tan^2 x + 4}\right] + c$

(b) $\frac{1}{2}\log\left[\tan x + \sqrt{\tan^2 x + 4}\right] + c$

(c) $\log\left[\frac{1}{2}\tan x + \frac{1}{2}\sqrt{\tan^2 x + 4}\right] + c$

(d) None of these

2. $\int \sqrt{\frac{1+x}{1-x}}dx =$

[RPET-2002]

(a) $-\sin^{-1} x - \sqrt{1-x^2} + C$

(b) $\sin^{-1} x + \sqrt{1-x^2} + C$

(c) $\sin^{-1} x - \sqrt{1-x^2} + C$

(d) $-\sin^{-1} x - \sqrt{x^2-1} + C$

3. $\int \frac{1}{x}\sqrt{\frac{x-1}{x+1}}dx =$

(a) $\log\left[x+\sqrt{x^2-1}\right] - \sec^{-1}x + c$

(b) $\log\left[x-\sqrt{x^2-1}\right] + \sec^{-1}x + c$

(c) $\log\left[x-\sqrt{x^2-1}\right] - \sec^{-1}x + c$

(d) $\log\left[x+\sqrt{x^2-1}\right] + \sec^{-1}x + c$

4. If $\int f(x)\cos x dx = \frac{1}{2}f^2(x)+c$, then $f(x)$ can be ***[Kerala PET-2008]***

(a) x (b) $\sin x$

(c) 1 (d) $\cos x$

5. If $I = \int \frac{x^5}{\sqrt{1+x^3}}dx$, then I is equal to ***[VITEEE-2008]***

(a) $\frac{2}{9}(1+x^3)^{5/2} + \frac{2}{3}(1+x^3)^{3/2} + C$

(b) $\log\left|\sqrt{x}+\sqrt{1+x^3}\right| + C$

(c) $\log\left|\sqrt{x}-\sqrt{1+x^3}\right| + C$

(d) $\frac{2}{9}(1+x^3)^{3/2} - \frac{2}{3}(1+x^3)^{1/2} + C$

6. $\int \frac{\cos 2x + x + 1}{x^2 + \sin 2x + 2x}dx =$ ***[AICBSE-1980]***

(a) $\log(x^2 + \sin 2x + 2x) + c$

(b) $-\log(x^2 + \sin x + 2x) + c$

(c) $\frac{1}{2}\log(x^2 + \sin 2x + 2x) + c$

(d) None of these

7. $\int \frac{1-\tan x}{1+\tan x}dx =$ ***[MPPET-1994]***

(a) $\log\sec\left(\frac{\pi}{4}-x\right)+c$

(b) $\log\cos\left(\frac{\pi}{4}+x\right)+c$

(c) $\log\sin\left(\frac{\pi}{4}+x\right)+c$

(d) None of these

8. $\int \frac{dx}{e^x - 1} =$ ***[MPPET-1989]***

(a) $\ln(1-e^{-x}) + c$ (b) $-\ln(1-e^{-x}) + c$

(c) $\ln(e^x - 1) + c$ (d) None of these

9. $\int \frac{\text{cosec}^2 x}{1+\cot x}dx =$ ***[MNR-1973]***

(a) $\log(1+\cot x) + c$

(b) $-\log(1+\cot x) + c$

(c) $\frac{1}{2(1+\cot x)^2} + c$

(d) None of these

10. If $\int \tan(\sin^{-1}x)dx =$ ***[Kerala PET-2008]***

(a) $\frac{1}{\sqrt{1-x^2}} + c$ (b) $\sqrt{1-x^2} + c$

(c) $\frac{-x}{\sqrt{1-x^2}} + c$ (d) $-\sqrt{1-x^2} + c$

11. $\int \sqrt{(1+\sin x)}dx =$ ***[MPPET-1995]***

(a) $\frac{1}{2}\left(\sin\frac{x}{2} + \cos\frac{x}{2}\right) + C$

(b) $-2\sqrt{(1-\sin x)} + C$

(c) $2\sqrt{1+\sin x} + C$

(d) None of these

12. $\int (e^x + e^{-x})^2 . (e^x - e^{-x})dx$ is equal to ***[MPPET-2008]***

(a) $e^x + C$ (b) $\frac{1}{2}(e^x - e^{-x})^2 + C$

(c) $\frac{1}{2}(e^x + e^{-x})^2 + C$ (d) $\frac{1}{3}(e^x + e^{-x})^3 + C$

13. $\int \frac{\cos x - \sin x}{\sqrt{\sin 2x}}dx$ equals ***[RPET-1996]***

(a) $\cos h^{-1}(\sin x + \cos x) + c$

(b) $\sin^{-1} h(\sin x + \cos x) + c$

(c) $-\cos h^{-1}(\sin x + \cos x) + c$

(d) $-\sin^{-1} h(\sin x + \cos x) + c$

14. $\int \frac{\sec x dx}{\sqrt{\cos 2x}} =$

(a) $\sin^{-1}(\tan x)$ (b) $\tan x$

(c) $\cos^{-1}(\tan x)$ (d) $\frac{\sec x}{\sqrt{\cos x}}$

15. $\int \frac{x^3}{\sqrt{1-x^8}}dx =$

(a) $\frac{1}{2}\sin^{-1}(x^4) + c$ (b) $\frac{1}{3}\sin^{-1}(x^4) + c$

(c) $\frac{1}{4}\sin^{-1}(x^4) + c$ (d) None of these

16. $\int\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+....\right)dx$ =

(a) $-e^x+c$ (b) e^x+c

(c) $e^{-x}+c$ (d) $-e^{-x}+c$

17. Evaluate $\int(1-\cos x)\text{cosec}^2 x dx$.

(a) $\tan\frac{x}{2}+c$ (b) $\cos\frac{x}{2}+c$

(c) $\cot\frac{x}{2}+c$ (d) None of these

18. $\int\frac{10x^9+10^x\log_e 10}{10^x+x^{10}}dx$ will be given by

(a) 10^x-x^{10} (b) 10^x+x^{10}

(c) $(10^x-x^{10})^{-1}$ (d) $\log(10^x+x^{10})$

19. $\int(\sin x-\cos x)^4(\sin x+\cos x)dx$ =

[Kerala PET-2008]

(a) $\frac{\sin x-\cos x}{5}+c$

(b) $\frac{(\sin x-\cos x)^5}{5}+c$

(c) $\frac{\cos x-\sin x}{5}+c$

(d) $\frac{(\cos x-\sin x)^5}{5}+c$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 13 minutes.
3. The worksheet consists of 13 questions. The maximum marks are 39.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int \frac{dx}{1-\cos x-\sin x}$ is equal to *[EAMCET-2002]*

(a) $\log|1+\cot\frac{x}{2}|+c$

(b) $\log|1-\tan\frac{x}{2}|+c$

(c) $\log|1-\cot\frac{x}{2}|+c$

(d) $\log|1+\tan\frac{x}{2}|+c$

2. $\int \frac{dx}{\cos x+\sqrt{3}\sin x}$ equals *[AIEEE-07]*

(a) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$

(b) $\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

(c) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

(d) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

3. $\int \frac{\sec^2 x\tan x dx}{\sec^2 x+\tan^2 x} =$

(a) $-\frac{1}{4}\log[\sec^2 x-\tan^2 x]$

(b) $\frac{1}{4}\log[\sec^2 x+\tan^2 x]$

(c) $-\frac{1}{4}\log[\sec^2 x+\tan^2 x]$

(d) None of these

4. $\int \frac{dx}{\sin x+\sqrt{3}\cos x} =$

(a) $\log\tan\left(\frac{x}{2}+\frac{\pi}{2}\right)+c$

(b) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)+c$

(c) $\log\cot\left(\frac{x}{2}+\frac{\pi}{6}\right)+c$

(d) $\frac{1}{2}\log\cot\left(\frac{x}{2}+\frac{\pi}{6}\right)+c$

5. If $\int\sqrt{2}\sqrt{1+\sin x}dx$ $-4\cos(ax+b)+C$, then the value of (a, b) is *[UPSEAT-2002]*

(a) $1/2, \pi/4$ (b) $1, \pi/2$

(c) $1, 1$ (d) None of these

6. $\int \frac{dx}{x(x^4+1)} =$

(a) $\frac{1}{4}\log\frac{x^4+1}{x^4}+C$

(b) $\frac{1}{4}\log\left(\frac{x^4}{x^4+1}\right)+C$

(c) $\frac{1}{4}\log(x^4+1)+C$

(d) None of these

7. $\int \frac{dx}{\sin^2 x\cos^2 x}$

(a) $\tan x-\cot x+c$ (b) $\tan x+\cot x+c$

(c) $\tan x-\cos x+c$ (d) None of these

8. $\int \frac{\sin x dx}{3+4\cos^2 x}$ is equal to

(a) $\log(3+4\cos^2 x)+c$

(b) $\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{\cos x}{\sqrt{3}}\right)+c$

(c) $-\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right)+c$

(d) $\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{2\cos x}{\sqrt{3}}\right)+c$

9. $\int x^2 e^{x^3}\cos(e^{x^3})\,dx$ is equal to

(a) $\sin(e^{x^3})$ (b) $3\sin(e^{x^3})$

(c) $\frac{1}{3}\sin(e^{x^3})$ (d) $e^{x^3}\sin(e^{x^3})$

10. $\int \frac{e^x(x+1)}{\cos^2(xe^x)}dx$ =

[Roorkee-79; MPPET-95; Pb. CET-2001]

(a) $\tan(xe^x)+c$

(b) $\sec(xe^x)\tan(xe^x)+c$

(c) $-\tan(xe^x)+c$

(d) None of these

11. $\int \frac{\cos x-1}{\cos x+1}dx =$ ***[MPPET-1989, 92]***

(a) $2\tan\frac{x}{2}-x+c$ (b) $\frac{1}{2}\tan\frac{x}{2}-x+c$

(c) $x-\frac{1}{2}\tan\frac{x}{2}+c$ (d) $x-2\tan\frac{x}{2}+c$

12. $\int \frac{(x^3+8)(x-1)}{x^2-2x+4}dx$ equals ***[PET (Raj.)-89]***

(a) $\frac{x^3}{3}+\frac{x^2}{2}-2x+c$ (b) x^3+x^2-2x+c

(c) $\frac{1}{3}(x^3+x^2-x)+c$ (d) None of these

13. $\int \frac{\operatorname{cosec} x}{\log\left(\tan\frac{x}{2}\right)}dx =$

(a) $\log\left(\log\tan\frac{x}{2}\right)+c$

(b) $2\log\left(\log\tan\frac{x}{2}\right)+c$

(c) $\frac{1}{2}\log\left(\log\tan\frac{x}{2}\right)+c$

(d) None of these

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	6. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	
5. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

4. (b) Step 1: $\int \frac{dx}{\sin x+\sqrt{3}\cos x}$

$$\frac{1}{2}\int \frac{dx}{\frac{1}{2}\sin x+\frac{\sqrt{3}}{2}\cos x}$$

$$\frac{1}{2}\int \frac{dx}{\cos\left(x-\frac{\pi}{6}\right)}=\frac{1}{2}\int \sec\left(x-\frac{\pi}{6}\right)dx$$

$$=\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}+\frac{\pi}{4}\right)$$

$$=\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{2\pi}{12}\right)$$

$$=\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{2\pi}{6}\right)+c$$

6. (b) $\int \frac{dx}{x(x^4+1)} = \int \frac{dx}{x^5\left(1+\frac{1}{x^4}\right)}$

Put $1+\frac{1}{x^4} = t \Rightarrow dt = -\frac{4}{x^5}$

$= -\frac{1}{4}\int \frac{dt}{t} = -\frac{1}{4}\log t \quad = -\frac{1}{4}\log\frac{x^4+1}{x^4} + c$

9. (c) Step 1: $t = e^{x^3}$, $dt = e^{x^3}(3x^2)dx$

Step 2: $\frac{1}{3}\int \cos t dt = \frac{1}{3}\sin t + c$

$= \frac{1}{3}\sin(e^{x^3}) + c$

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LECTURE



5 Integration by Parts

BASIC CONCEPTS

STANDARD INTEGRAL

(i) $\int \sin^{-1} x dx = x\sin^{-1} x + \sqrt{1-x^2}$

(ii) $\int \cos^{-1} x dx = x\cos^{-1} x - \sqrt{1-x^2}$

(iii) $\int \tan^{-1} x dx = x\tan^{-1} x - \frac{1}{2}\log(1+x^2)$

(iv) $\int \cot^{-1} x dx = x\cot^{-1} x + \frac{1}{2}\log(1+x^2)$

(v) $\int \sec^{-1} x dx = x\sec^{-1} x - \log[x + \sqrt{x^2-1}]$

(vi) $\int \text{cosec}^{-1} x dx = x\,\text{cosec}^{-1} x + \log[x + \sqrt{x^2-1}]$

(vii) $\int \log_e^x dx = x\log_e^x - x = x\left(\log_e^{x/e}\right)$

INTEGRATION BY PARTS

If u and v are two functions of x, then

$$\int uv\,dx = u\left(\int v\,dx\right) - \int\left(\frac{du}{dx}\right)\left(\int v\,dx\right)dx$$

i.e., integral of the product of two functions

= first × integral of second

$-\int$ ((derivative of first) × (integral of second))dx

Here first and second function is considered by rule of ILATE i.e.,

I Inverse trigonometric function

L Logarithmic

A Algebraic

T Trigonometric

E Exponential

Extension of the Rule of Integration by Parts or Successive Integration or Extended form of Integration by Parts:

$$\int uv\,dx = (uv_1) - (u'v_2) + (u''v_3) - (u'''v_4) + ..$$

$$+ ((-1)^{n-1} u^{n-1} v_n) + 1\,(-1)^n (u^n)(v_{n+1})$$

Where u^n stands for nth diff. coefficient of u and v_n stands for nth integral of v.

Rule

Proceed as usual and successively integrate one function and differentiate the other and attach alternately + and − signs with their products. In the last the integration of only one function will be left.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate $\int \sqrt{x}(\log x)^2 dx$. ***[Roorkee-81]***

Solution

Let $I = \int \sqrt{x}(\log x)^2 dx$

$= (\log x)^2 \int x^{1/2} dx - \int \left[\frac{d}{dx}(\log x)^2 \int x^{1/2} dx \right] dx$

$= (\log x)^2 . \frac{x^{3/2}}{3/2} - \int \left[2\log x . \frac{1}{x} . \frac{x^{3/2}}{3/2} \right] dx$

$\Rightarrow \quad I = \frac{2}{3} x^{3/2} (\log x)^2 - \frac{4}{3} \int x^{1/2} \log x dx$

Integrating by parts again

$$I = \frac{2}{3} x^{3/2} (\log x)^2 - \frac{4}{3} \left[\begin{array}{l} \log x . \int x^{1/2} dx - \\ \int \left\{ \frac{d}{dx} \log x . \int x^{1/2} dx \right\} dx \end{array} \right]$$

$$= \frac{2}{3} x^{3/2} (\log x)^2 - \frac{4}{3} \left[\log x . \frac{x^{3/2}}{3/2} - \int \frac{1}{x} . \frac{x^{3/2}}{3/2} dx \right]$$

$$= \frac{2}{3} x^{3/2} (\log x)^2 - \frac{4}{3} . \frac{2}{3} x^{3/2} . \log x + \frac{4}{3} . \frac{2}{3} \int x^{1/2} dx$$

$$= \frac{2}{3} \left[x^{3/2} (\log x)^2 - \frac{4}{3} x^{3/2} \log x + \frac{4}{3} . \frac{x^{3/2}}{3/2} \right]$$

$$\Rightarrow I = \frac{2}{3} x^{3/2} \left[(\log x)^2 - \frac{4}{3} \log x + \frac{8}{9} \right]$$

2. Evaluate $\int x \sec^2 2x dx$. ***[IIT-75]***

Solution

Let $I = \int x \sec^2 2x dx$

$= x . \int \sec^2 2x dx - \int \left\{ \frac{d}{dx} . x . \int \sec^2 2x dx \right\} dx$

$= x . \frac{\tan 2x}{2} - \int 1 . \frac{\tan 2x}{2} dx$

$= \frac{x \tan 2x}{2} - \frac{1}{2} \int \tan 2x dx$

$= \frac{x \tan 2x}{2} - \frac{1}{2} . \frac{\log(\sec 2x)}{2}$

$\Rightarrow \quad I = \frac{x \tan 2x}{2} - \frac{1}{4} \log(\sec 2x)$

3. Evaluate $\int x^{-2} \sin^{-1} x dx$.

[CBSE-Sample Paper-III]

Solution

Let $I = \int x^{-2} \sin^{-1} x dx$

$= \sin^{-1} x . \int x^{-2} dx - \int \left[\frac{d}{dx} \sin^{-1} x . \int x^{-2} dx \right] dx$

$= -\sin^{-1} x . \frac{1}{x} + \int \frac{1}{x\sqrt{1-x^2}} dx$

$= \frac{-\sin^{-1} x}{x} + \int \frac{x}{x^2 \sqrt{1-x^2}} dx$

$= \frac{-\sin^{-1} x}{x} - \int \frac{t dt}{(1-t^2)t}$ where $1 - x^2 = t^2$

or $-x dx = t dt$

$= \frac{-\sin^{-1} x}{x} - \frac{1}{2} \log \left| \frac{1+t}{1-t} \right| + c$

$= \frac{-\sin^{-1} x}{x} - \frac{1}{2} \log \left| \frac{1+\sqrt{1-x^2}}{1-\sqrt{1-x^2}} \right| + c$

4. If $\int F(x) dx = G(x)$, evaluate $\int F^{-1}(x) dx$.

Hence evaluate $\int \tan^{-1} x dx$ using the fact that $\int \tan x dx = -\log |\cos x|$.

Solution

Let $F^{-1}(x) = t \quad \Rightarrow x = F(t)$

$\Rightarrow \frac{dx}{dt} = F'(t) \Rightarrow dx = F'(t) dt$

Hence $\int F^{-1}(x) dx = \int t F'(t) dt$

Integrate by parts, taking t as the first function

$= t \int F'(t) dt - \int \left(\frac{dt}{dt} \int f'(t) dt \right) dt$

$= tF(t) - \int 1 . F(t) dt \; = tI'(t) - G(t) + c$

$= x F^{-1}(x) - G(F^{-1}(x)) + c$

Since $\int \tan x dx = -\log |\cos x|$ therefore,

$$\int \tan^{-1} x dx = x\tan^{-1}x - \left\{-\log|\cos(\tan^{-1}x)|\right\} + c$$

$= x\tan^{-1}x + \log|\cos\theta| + c$ where $\theta = \tan^{-1}x$

$(\because \tan^{-1}x = \theta) \ \therefore \ x = \tan\theta$

$\Rightarrow \sec^2\theta = 1 + \tan^2\theta$

$$\Rightarrow |\sec\theta| = \sqrt{1+x^2} \quad \Rightarrow |\cos\theta| = \frac{1}{\sqrt{1+x^2}}$$

$$= x\tan^{-1}x - \log\left(\frac{1}{\sqrt{1+x^2}}\right) + c$$

$$= x\tan^{-1}x - \frac{1}{2}\log(1+x^2) + c$$

5. Taking x^2 as or x^3. $\frac{1}{x}$ integrate it by parts and use this to find $\int x^2 \log|x| dx$.

Solution

Now $\int x^2 dx = \int x^3 . \frac{1}{x} dx$

$$\Rightarrow \int x^2 dx = x^3\log|x| - \int (3x^2).\log|x| dx$$

$$\Rightarrow \int 3x^2\log|x| dx = x^3\log|x| - \int x^2 dx$$

$$\Rightarrow 3\int x^2\log|x| dx = x^3\log|x| - \frac{1}{3}x^3 + c$$

$$\Rightarrow \int x^2\log|x| dx = \frac{1}{3}x^3\log|x| - \frac{1}{9}x^3 + c'$$

where $c' = \frac{1}{3}c$ is an arbitrary constant.

6. If $\int \frac{e^x(3-x^2)}{1-2x+x^2}dx = e^x f(x) + c$ then find the value of $f(x)$. ***[EAMCET-2004]***

Solution

$$\frac{3-x^2}{1-2x+x^2} = \frac{(1-x^2)+2}{(1-x)^2} = \frac{1+x}{1-x} + \frac{2}{(1-x)^2}$$

$= f(x) + f'(x)$

$$\therefore \quad I = \int e^x[f(x) + f'(x)]dx = e^x f(x) + c$$

$$\Rightarrow \quad f(x) = \frac{1+x}{1-x}$$

7. Evaluate: $\int \frac{x^3\sin(\tan^{-1}x^4)}{1+x^8}dx$

Solution

$$\int \frac{x^3\sin(\tan^{-1}x^4)}{1+x^8}dx$$

$$\frac{1}{4}\int \sin t dt = -\frac{1}{4}\cos t + c \quad \left| \begin{array}{l} \text{Put } \tan^{-1}x^4 = t \\ \frac{4x^3}{1+x^8}dx = dt \end{array} \right.$$

$$= -\frac{1}{4}\cos(\tan^{-1}x^4) + c$$

8. Evaluate: $\int x^{2n-1}.\sin x^n dx$

Solution

$$\int x^{2n-1}.\sin x^n dx \ , \ | \text{ Put } x^n = t$$

$$\Rightarrow nx^{n-1}dx = dt$$

$$= \int x^{n-1}.x^n.\sin x^n dx = \frac{1}{n}\int t.\sin t dt$$

$$= \frac{1}{n}\left[t(-\cos t) - \int 1.(-\cos t)dt\right]$$

$$= \frac{1}{n}[-t\cos t + \sin t] + c$$

$$= \frac{1}{n}[-x^n\cos x^n + \sin x^n] + c$$

9. $\int u\frac{d^2v}{dx^2}dx - \int v\frac{d^2u}{dx^2}dx = u\frac{dv}{dx} - v\frac{du}{dx}$

Is the above statement true?

Solution

Yes true. Integrate by parts

$$I_1 = u\frac{dv}{dx} - \int \frac{dv}{dx}\frac{du}{dx}dx \quad \ldots\ldots\ldots\ldots (1)$$

$$I_2 = v\frac{du}{dx} - \int \frac{du}{dx}\frac{dv}{dx}dx \quad \ldots\ldots\ldots\ldots :(2)$$

$$\therefore I_1 - I_2 = u\frac{dv}{dx} - v\frac{du}{dx}$$

10. Evaluate $\int x^2(\log x)^2 dx$. ***[Roorkee-1977]***

Solution

$$\int x^2(\log x)^2 dx$$

Integrate by parts, taking $(\log x)^2$ as the first function

$$= (\log x)^2 . \frac{x^3}{3} - \int 2(\log x)^1 . \frac{1}{x} . \frac{x^3}{3}dx$$

$$= \frac{1}{3}x^3(\log x)^2 - \frac{2}{3}\int (\log x).x^2 dx$$

Again integrate by parts, taking $\log x$ as the first function

$$= \frac{1}{3}x^3(\log x)^2 - \frac{2}{3}\left[\log x.\frac{x^3}{3} - \int \frac{1}{x}.\frac{x^3}{3}dx\right]$$

$$= \frac{1}{3}x^3(\log x)^2 - \frac{2}{9}x^3\log x + \frac{2}{9}\int x^2 dx$$

$$= \frac{1}{3}x^3(\log x)^2 - \frac{2}{9}x^3\log x + \frac{2}{9}.\frac{x^3}{3} + C$$

$$= \frac{1}{27}x^3[9(\log x)^2 - 6\log x + 2] + C$$

UNSOLVED SUBJECTIVE PROBLEMS (AT BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Evaluate $\int x\cos^2 x dx$.
2. Evaluate $\int \frac{\log_e x}{x^3}dx$. ***[Roorkee-86; MP-96]***
3. Evaluate $\int \frac{\log_e(\log_e x)}{x}dx$. ***[MP-1996, 2008]***
4. Evaluate $\int x^{2n-1}\sin x^n dx$.
5. Evaluate $\int \sec^{-1} x dx$.
6. Integrate with respect to x $\int x^2 \sin 2x dx$ ***[IIT-74]***
7. Evaluate $\int \sin^{-1}\sqrt{\frac{1-x}{2}}dx$.
8. Evaluate $\int \frac{\sin^{-1}x}{x^2}dx$. ***[CBSE-2004]***
9. Evaluate $\int \log_e x dx$. ***[MP-89, 99, 2000; HPSB-97, 2000(C)]***
10. Evaluate $\int \log(a^2 + x^2)dx$.
11. Evaluate $\int \sqrt{\frac{a-x}{a+x}}dx$. ***[CBSE-88]***
12. Evaluate $\int e^x \sin x dx$.

ANSWERS

1. $\frac{1}{4}\left[x^2 + x\sin 2x + \frac{1}{2}\cos 2x\right]$
2. $\frac{-1}{4x^2}[2\log_e x + 1]$
3. $\log_e x \log_e\left(\frac{\log_e x}{e}\right)$
4. $\frac{1}{n}\left[-x^n\cos x^n + \sin x^n\right]$
5. $x\sec^{-1}x - \log(x + \sqrt{x^2 - 1})$
6. $\left(\frac{1}{4} - \frac{x^2}{2}\right)\cos 2x + \frac{x}{2}\sin 2x$
7. $\frac{1}{2}\left[x\cos^{-1}x - \sqrt{1-x^2}\right]$
8. $-\frac{\sin^{-1}x}{x} + \log\left|\frac{1-\sqrt{1-x^2}}{x}\right| + C$
9. $x\log_e(x/e)$
10. $x\log(a^2 + x^2) - 2x \quad 2a\tan^{-1}\frac{x}{a}$
11. $a\sin^{-1}\frac{x}{a} + \sqrt{a^2 - x^2}$
12. $e^x\frac{(\sin x - \cos x)}{2}$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\int \frac{\sin^{-1}\sqrt{x}-\cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}}dx$ is equal to *[IIT-83; JEE (WB)-92]*

(a) $\frac{2}{\pi}\left[(2x-1)\sin^{-1}\sqrt{x}+\sqrt{x(1-x)}\right]-x+c$

(b) $\frac{2}{\pi}\left[(2x-1)\sin^{-1}\sqrt{x}-\sqrt{x(1-x)}\right]+x+c$

(c) $\frac{\pi}{2}\left[(2x-1)\sin^{-1}\sqrt{x}+\sqrt{x(1-x)}\right]-x+c$

(d) None of these

Solution

(a) $I=\int \frac{\sin^{-1}\sqrt{x}-\left(\frac{\pi}{2}-\sin^{-1}\sqrt{x}\right)}{\pi/2}dx$

$=\frac{4}{\pi}\int \sin^{-1}\sqrt{x}dx-\int dx$

$=\frac{4}{\pi}\left[x\sin^{-1}\sqrt{x}-\int \frac{1}{\sqrt{1-x}}.\frac{xdx}{2\sqrt{x}}\right]-x$

$=\frac{4}{\pi}\left[x\sin^{-1}\sqrt{x}-\frac{1}{2}\int\sqrt{\frac{x}{1-x}}dx\right]-x$

$=\frac{2}{\pi}\left[2x\sin^{-1}\sqrt{x}-\int \sin^2 t dt\right]-x,$

where $x=\sin^2 t$

$=\frac{2}{\pi}\left[2x\sin^{-1}\sqrt{x}-\frac{1}{2}\int(1-\cos 2t)dt\right]-x$

$=\frac{2}{\pi}\left[2x\sin^{-1}\sqrt{x}-\frac{x}{2}+\frac{\sin 2t}{4}\right]-x+c$

put $\sin t=\sqrt{x}$ to get (a)

2. If $\int f(x)dx=g(x)$, then $\int f^{-1}(x)dx$ is equal to

(a) $g^{-1}(x)$ (b) $xf^{-1}(x)-g(f^{-1}(x))$

(c) $xf^{-1}(x)-g^{-1}(x)$ (d) $f^{-1}(x)$

Solution

(b) $\int f(x)dx=g(x)$

$\int f^{-1}(x).1dx=f^{-1}(x)\int dx-\int\left\{\frac{d}{dx}f^{-1}(x)\int dx\right\}dx$

$=xf^{-1}(x)-\int x\frac{d}{dx}f^{-1}(x)dx$

$=xf^{-1}(x)-\int xd\left\{f^{-1}(x)\right\}$

Let $f^{-1}(x)=t\Rightarrow x=f(t).$

and $d\left\{f^{-1}(x)\right\}=dt$

$xf^{-1}(x)-\int f(t).dt=xf^{-1}(x)-g(t)=$

$=xf^{-1}(x)-g\left\{f^{-1}(x)\right\}$

3. $\int \cos\sqrt{x}\,dx=$ *[BIT RANCHI-1990; IIT JEE-1977; RPET-1999]*

(a) $2\left[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}\right]+c$

(b) $2\left[\sqrt{x}\sin\sqrt{x}-\cos\sqrt{x}\right]+c$

(c) $2\left[\cos\sqrt{x}-\sqrt{x}\sin\sqrt{x}\right]+c$

(d) $-2\left[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}\right]+c$

Solution

(a) Put $\sqrt{x}=t\Rightarrow\frac{1}{2\sqrt{x}}dx=dt\Rightarrow dx=2t.dt,$

then it reduces to

$\int 2t.\cos t dt=2\left[t.\sin t-\int\sin t dt\right]$

$=2t\sin t+2\cos t=2[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}]+c$

4. $\int\frac{x^2dx}{(x\sin x+\cos x)^2}=$ *[IIT-1972]*

(a) $\frac{\sin x+\cos x}{x\sin x+\cos x}$ (b) $\frac{\sin x-x\cos x}{x\sin x+\cos x}$

(c) $\frac{\sin x+\cos x}{x\sin x-\cos x}$ (d) None of these

Solution

(b) Differentiation of $x\sin x-\cos x$ is $x\cos x$, then

$I=\int\frac{x^2dx}{(x\sin x+\cos x)^2}$

$=\int\frac{x\cos x}{(x\sin x+\cos x)^2}.\frac{x}{\cos x}dx$

Integrate by parts $\left[\int \frac{1}{t^2}dt = -\frac{1}{t}\right]$

$$\therefore \quad I = \frac{-1}{(x\sin x + \cos x)}$$

$$\frac{x}{\cos x} + \int \frac{1}{(x\sin x + \cos x)} \cdot \frac{\cos x - x(-\sin x)}{\cos^2 x}\, dx$$

$$\Rightarrow \quad -\frac{1}{x\sin x + \cos x} \cdot \frac{x}{\cos x} + \int \sec^2 x dx$$

$$\Rightarrow \quad \frac{-x + x\sin^2 x + \sin x \cos x}{(x\sin x + \cos x)\cos x}$$

$$\Rightarrow \quad \frac{\sin x \cos x - x(1 - \sin^2 x)}{(x\sin x + \cos x)\cos x}$$

$$\Rightarrow \quad \frac{\sin x - x\cos x}{x\sin x + \cos x}$$

5. $\int \cos 2\theta \log\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right) d\theta =$

(a) $(\cos 0 + \sin 0)^2 \log\left(\frac{\cos 0 + \sin 0}{\cos\theta - \sin\theta}\right)$

(b) $\frac{1}{2}\sin 2\theta \log\tan\left(\frac{\pi}{4} + \theta\right) - \frac{1}{2}\log(\sec 2\theta)$

(c) $(\cos\theta - \sin\theta)^2 \log\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right)$

(d) None of these

Solution

(b) We know that

$$\log\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right) = \log\left(\frac{1 + \tan\theta}{1 - \tan\theta}\right)$$

$$= \log\tan\left(\frac{\pi}{4} + \theta\right)$$

$$\int \sec 0\, d0 = \log\tan\left(\frac{\pi}{4} + \frac{0}{2}\right)$$

$$\therefore \quad \int \sec 2\theta\, d\theta = \frac{1}{2}\log\tan\left(\frac{\pi}{4} + \theta\right)$$

$$\therefore \quad 2\sec 2\theta\, d\theta = \frac{d}{d0}\log\tan\left(\frac{\pi}{4} + \theta\right) \quad \text{.....(i)}$$

Integrating the given expression by parts we get,

$$I = \frac{1}{2}\sin 20 \log\tan\left(\frac{\pi}{4} + 0\right) - \frac{1}{2}\int \sin 20.2\sec 0\, d0$$

[By (i)]

$$= \frac{1}{2}\sin 20 \log\tan\left(\frac{\pi}{4} + 0\right) - \int \tan 20\, d0$$

$$= \frac{1}{2}\sin 20 \log\tan\left(\frac{\pi}{4} + 0\right) - \frac{1}{2}\log \sec 2\theta$$

6. $\int \tan^{-1}\sqrt{x}dx =$ ***[A.M.U.-90]***

(a) $x\tan^{-1} x - \frac{1}{2}\log(1 + x^2) + c$

(b) $x\tan^{-1}\sqrt{x} - \sqrt{x} + \log(1 + x)$

(c) $(x \quad 1)\tan^{-1}\sqrt{x} - \sqrt{x} + c$

(d) None of these

Solution

(c) Step 1: Let $\Rightarrow x = \tan^2\theta$

$dx = 2\tan\theta \sec^2\theta\, d\theta$

$$I = \int (\theta)(2\tan\theta)\sec^2\theta d\theta$$

$$= 2\int (\theta\tan\theta)(\sec^2\theta) d\theta$$

Step 2: Integrate by factorization (first function = 0)

$$I = \left[\begin{array}{l}(\theta)\int \tan\theta\sec^2\theta d\theta - \\ \int\left(\frac{d}{d\theta}(\theta)\int \tan\theta\sec^2\theta d\theta\right)d\theta\end{array}\right]$$

$$= 2(\theta)\left(\frac{\tan^2\theta}{2}\right) - \int \frac{\tan^2\theta}{2} d\theta$$

$$= 2\left[(\theta)\left(\frac{\tan^2\theta}{2}\right) - \frac{1}{2}\int (\sec^2\theta - 1) d\theta\right]$$

$$\Rightarrow \quad (\theta)(\tan^2\theta) - (\tan\theta - \theta)$$

$$\Rightarrow \quad x - \tan^{-1}\sqrt{x} + \tan^{-1}\sqrt{x} - \sqrt{x}$$

$$\Rightarrow \quad (x + 1)\tan^{-1}\sqrt{x} - \sqrt{x} + c$$

7. The value of $\int \frac{e^x(2 - x^2)dx}{(1 - x)\sqrt{1 - x^2}}$ is equal to ***[Kerala PET-2001]***

(a) $e^x\sqrt{\frac{1 + x}{1 - x}} + c$ (b) $e^x\sqrt{1 + x} + c$

(c) $e^x\sqrt{1 - x} + c$ (d) $e^x\sqrt{\frac{1 - x}{1 + x}} + c$

Solution

(a) $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}}dx = \int \frac{e^x(1-x^2+1)}{(1-x)\sqrt{1-x^2}}dx$

$= \int \frac{e^x(1-x^2)}{(1-x)\sqrt{1-x^2}}dx + \int \frac{e^x}{(1-x)\sqrt{1-x^2}}dx$

$= \int e^x \sqrt{\frac{1+x}{1-x}}\,dx + \int \frac{e^x}{(1-x)\sqrt{1-x^2}}\,dx$

II I

Integrate by parts (only first part)

$= \sqrt{\frac{1+x}{1-x}}.e^x - \int \frac{1}{2}\left(\frac{1+x}{1-x}\right)^{-1/2}$

$\frac{d}{dx}\left(\frac{1+x}{1-x}\right).e^x dx + \int \frac{e^x dx}{(1-x)\sqrt{1-x^2}}$

$= e^x.\sqrt{\frac{1+x}{1-x}} - \frac{1}{2}\int \sqrt{\frac{1-x}{1+x}}.\frac{(1-x).1-(1+x)(-1)}{(1-x)^2}e^x dx$

$+ \int \frac{e^x}{(1-x)\sqrt{1-x^2}}dx$

$= e^x\sqrt{\frac{1+x}{1-x}} - \frac{1}{2}\int \sqrt{\frac{1-x}{1+x}}.\frac{1-x+1+x}{(1-x)^2}e^x dx$

$+ \int \frac{e^x}{(1-x)\sqrt{1-x^2}}dx$

$= e^x\sqrt{\frac{1+x}{1-x}} - \int \frac{e^x}{(1-x)\sqrt{1-x^2}}$

$dx + \int e^x \frac{}{(1-x)\sqrt{1-x^2}}dx = e^x\sqrt{\frac{1+x}{1-x}} + c$

8. $\int \tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx =$ ***[EAMCET-2007]***

(a) $\frac{1}{2}(x\cos^{-1}x - \sqrt{1-x^2}) + C$

(b) $\frac{1}{2}(x\cos^{-1}x + \sqrt{1-x^2}) + C$

(c) $\frac{1}{2}(x\sin^{-1}x - \sqrt{1-x^2}) + C$

(d) $\frac{1}{2}(x\sin^{-1}x + \sqrt{1-x^2}) + C$

Solution

(a) $\int \tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx$

$= \frac{1}{2}\int \cos^{-1}x\,dx$ (Put $x = \cos 20$)

$= \frac{1}{2}\int 1.\cos^{-1}x\,dx$ (on integrating by parts).

II I

$= \frac{1}{2}\left[x\cos^{-1}x - \int x\cdot\frac{-1}{\sqrt{1-x^2}}\,dx\right]$

$= \frac{1}{2}\left[x\cos^{-1}x - \sqrt{1-x^2}\right] + c$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int \cos(\log_e x)dx$ is equal to ***[MPPET-2003]***

(a) $\frac{1}{2}x\ \{\cos(\log_e x) + \sin(\log_e x)\}$

(b) $x\ \{\cos(\log_e x) + \sin(\log_e x)\}$

(c) $\frac{1}{2}x\{\cos(\log_e x) - \sin(\log_2 x)\}$

(d) $x\ \{\cos(\log_e x) - \sin(\log_e x)\}$

2. $\int x^3 \log x\,dx =$ ***[Karnataka CET-2002]***

(a) $\frac{x^4\log x}{4} + C$

(b) $\frac{1}{16}[4x^4\log x - x^4] + C$

(c) $\frac{1}{8}[x^4\log x - 4x^2] + C$

(d) $\frac{1}{16}[4x^4\log x \quad x^4] + C$

3. $\int e^{\sqrt{x}}\,dx$ is equal to ***[MPPET-1998]***

(a) $e^{\sqrt{x}}+A$ (b) $\frac{1}{2}e^{\sqrt{x}}+A$

(c) $2(\sqrt{x}-1)e^{\sqrt{x}}+A$ (d) $2(\sqrt{x}+1)e^{\sqrt{x}}+A$

4. $\int\frac{1+x^2}{\sqrt{1-x^2}}dx=$ ***[IIT-1977]***

(a) $\frac{3}{2}\sin^{-1}x-\frac{1}{2}x\sqrt{1-x^2}+c$

(b) $\frac{3}{2}\sin^{-1}x+\frac{1}{2}x\sqrt{1-x^2}+c$

(c) $\frac{3}{2}\cos^{-1}x-\frac{1}{2}x\sqrt{1-x^2}+c$

(d) $\frac{3}{2}\cos^{-1}x+\frac{1}{2}x\sqrt{1-x^2}+c$

5. $\int\frac{\sin^{-1}x}{(1-x^2)^{3/2}}dx=$ ***[AISSE-1983, 87]***

(a) $\frac{x}{\sqrt{1-x^2}}\sin^{-1}x+\frac{1}{2}\log(1-x^2)+c$

(b) $\frac{x}{\sqrt{1-x^2}}\sin^{-1}x-\frac{1}{2}\log(1-x^2)+c$

(c) $\frac{1}{\sqrt{1-x^2}}\sin^{-1}x-\frac{1}{2}\log(1-x^2)+c$

(d) $\frac{1}{\sqrt{1-x^2}}\sin^{-1}x+\frac{1}{2}\log(1-x^2)+c$

6. $\int x^2\sin 2x\,dx=$ ***[IIT-1974]***

(a) $\frac{1}{2}x^2\cos 2x+\frac{1}{2}x\sin 2x\quad\frac{1}{4}\cos 2x\quad c$

(b) $-\frac{1}{2}x^2\cos 2x\quad\frac{1}{2}x\sin 2x+\frac{1}{4}\cos 2x+c$

(c) $\frac{1}{2}x^2\cos 2x-\frac{1}{2}x\sin 2x-\frac{1}{4}\cos 2x-c$

(d) None of these

7. $\int(\log x)^2\,dx=$ ***[IIT-1971, 77]***

(a) $x(\log x)^2-2x\log x-2x+c$

(b) $x(\log x)^2-2x\log x-x+c$

(c) $x(\log x)^2-2x\log x\quad 2x\quad c$

(d) $x(\log x)^2-2x\log x-x-c$

8. $\int\tan^{-1}x\,dx=$ ***[Roorkee-1977]***

(a) $x\tan^{-1}x-\frac{1}{2}\log(1+x^2)$

(b) $x\tan^{-1}x-\frac{1}{2}\log(1+x^2)$

(c) $(x-1)\tan^{-1}x$

(d) $x\tan^{-1}x-\log(1+x^2)$

9. $\int[f(x)g''(x)-f''(x)g(x)]dx=$ ***[MPPET-2001]***

(a) $f(x)/g(x)$

(b) $f(x)\,g'(x)-f'(x)\,g(x)$

(c) $f(x)\,g'(x)+f'(x)\,g(x)$

(d) None of these

10. $\int\frac{1}{\log_x e}dx=\int\log_x e\,dx=$ ***[MPPET-1994, 2007]***

(a) $\log\log_x e+c$ (b) $\frac{1}{(\log_x e)^2}+c$

(c) $x\log\left(\frac{x}{e}\right)+c$ (d) None of these

11. If $\frac{d}{dx}f(x)=(x\cos x\quad\sin x)$ and $f(0)=2$, then $f(x)=$ ***[MPPET-1989]***

(a) $x\sin x$ (b) $x\sin x+2$

(c) $x\sin x-2$ (d) None of these

12. $\int x\sqrt{2x+3}dx=$ ***[AISSE-1985]***

(a) $\frac{x}{3}(2x+3)^{3/2}-\frac{1}{15}(2x+3)^{5/2}+c$

(b) $\frac{x}{3}(2x+3)^{3/2}+\frac{1}{15}(2x+3)^{5/2}+c$

(c) $\frac{x}{2}(2x+3)^{3/2}+\frac{1}{6}(2x+3)^{5/2}+c$

(d) None of these

13. $\int\cos\left(2\cot^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)\right)dx=$

(a) $-\frac{x^2}{2}+c$ (b) $\frac{x^2}{2}+c$

(c) x^2+C (d) None of these

14. $\int x\sin^{-1}x\,dx=$ ***[MPPET-1991]***

(a) $\left(\frac{x^2}{2}-\frac{1}{4}\right)\sin^{-1}x+\frac{x}{4}\sqrt{1-x^2}+c$

(b) $\left(\frac{x^2}{2}+\frac{1}{4}\right)\sin^{-1}x-\frac{x}{4}\sqrt{1-x^2}+c$

(c) $\left(\frac{x^2}{2}-\frac{1}{4}\right)\sin^{-1}x-\frac{x}{4}\sqrt{1-x^2}+c$

(d) None of these

SOLUTIONS

1. (a) Let $I = \int \cos(\log x)dx$

Integrating by parts assuming 1 to be second function

$$I = x\cos(\log x) - \int -\frac{\sin(\log x)}{x}.xdx$$

$$I = x\cos(\log x) + \int \sin(\log x)dx$$

$$I = x\cos(\log x) + x\sin(\log x) - \int \frac{\cos(\log x)}{x}.xdx$$

$$2I = x[\cos(\log x) + \sin(\log x)]$$

$$\Rightarrow I = \frac{x}{2}[\cos(\log x) + \sin(\log x)]$$

2. (b) Step 1: Considering log x as first function and then Integration by Parts

$$(\log x)\int x^3dx - \int \frac{d}{dx}(\log x)\int x^3dx\,dx$$

$$= (\log x)\left(\frac{x^4}{4}\right) - \int \frac{1}{x}\times\frac{x^4}{4}dx$$

$$= (\log x)\left(\frac{x^4}{4}\right) - \frac{1}{4}\int x^3dx$$

$$= (\log x)\left(\frac{x^4}{4}\right) - \frac{x^4}{16} + c$$

$$= \frac{1}{16}[4x^4\log x - x^4] + c$$

3. (c) Step 1: Let $x = t^2$ then $dx = 2\,tdt$

Step 2: $I = 2\int te^t dt$

Step 3: Successive Integration by parts

$I = (2)[(t)(e^t) - (1)(e^t)]$ (put $t = \sqrt{x}$)

$= 2[(\sqrt{x} - 1)e^{\sqrt{x}}] + A$

4. (a) $\int \frac{1+x^2}{\sqrt{1-x^2}}dx$

Put $x = \sin\theta \Rightarrow dx = \cos\theta d\theta$, then it reduce to

$$\int (1+\sin^2 0)d0 = 0 + \frac{1}{2}\int (1-\cos 20)d0$$

$$= \frac{30}{2} - \frac{1}{2}\sin 0\sqrt{1-\sin^2 0} + c$$

$$= \frac{3}{2}\sin^{-1}x - \frac{1}{2}x\sqrt{1-x^2} + c$$

5. (d) Step 1: Put $\sin^{-1}x = t \Rightarrow \sin t = x$

$\Rightarrow \cos t dt = dx$

Step 2: $\tan\int \frac{\sin^{-1}x}{(1-x^2)^{3/2}}dx = \int t\sec^2 t\,dt$

$= t\tan t + \log\cos t + c$

$= \sin^{-1}x\tan(\sin^{-1}x) + \log\cos(\sin^{-1}x) + c$

$= \frac{x}{\sqrt{1-x^2}}\sin^{-1}x + \frac{1}{2}\log(1-x^2) + c$

6. (b) Step 1: According to Formula No. 2.1 in Lecture 5 by

Quicker Method

$$I = (x^2)\left(\frac{-\cos 2x}{2}\right) - (2x)\left(\frac{-\sin 2x}{4}\right) + 2\left(\frac{\cos 2x}{8}\right)$$

$$-\frac{1}{2}x^2\cos 2x \quad \frac{1}{2}x\sin 2x \quad \frac{1}{4}\cos 2x \quad c$$

7. (a) Step 1: Integration by parts (Second function is 1)

$$I = (\log x)^2\int 1dx - \int\left[\left\{\frac{d}{dx}(\log x)^2\right\}\int 1dx\right]dx$$

$$= (x)(\log x)^2 - \int \frac{2(\log x)}{x}\times x dx$$

$$= (x)(\log x)^2 - 2[x\log x - x] + c$$

$$= (x)(\log x)^2 - 2x\log x + 2x + c$$

NOTE

$$\int \log x dx = x\log x - x$$

8. (b) $\int \tan^{-1}x dx =$

Integrating by part assuming 1 to be second function

$$= x\tan^{-1}x - \int \frac{x}{1+x^2}dx + c$$

$$= x\tan^{-1}x - \frac{1}{2}\log(1+x^2) + c$$

9. (b) Step-1:

$$I = \int f(x)g''(x)dx - \int f''(x)g(x)dx$$

Integration by parts

$$I=\left[f(x)\int g''(x)dx-\int\left(\frac{d}{dx}(f(x))\int g''(x)dx\right)dx\right]$$

$$-\left[g(x)\int f''(x)dx-\int\left(\frac{d}{dx}\left(g(x)\int f''(x)dx\right)\right)\right]$$

$$=f(x)g'(x)-\int f'(x)g'(x)dx$$

$$-g(x)f'(x)+\int g'(x)f''(x)dx$$

$$=f(x)g'(x)-g(x)f'(x)$$

10. (c) Step-1: $\int \log_e x\,dx$

Integration by parts

Rule: $\int uv\,dx=u\int v dx-\int\left(\frac{du}{dx}\int v dx\right)dx$

$$=(x)(\log_e x)-x=x[\log_e x-\log_e e]$$

$$=x\log_e\left(\frac{x}{e}\right)+c$$

NOTE

Here only one function is given whose integral is not known. Therefore in integration by parts rule second function is defined by 1.

11. (b) Step-1: By formula $\int\frac{d}{dx}f(x)=f(x)$

$$f(x)=\int(x\cos x+\sin x)dx=$$

$$\int x\cos x dx+\int\sin x dx$$

$$=(x)(\sin x)-(1)(-\cos x)-\cos x+c$$

$$=x\sin x+c$$

Step-2: by given condition

$$f(x)\big|_{x=0}=x\sin x+c\big|_{x=0}=2$$

$$f(0)=2=0+c\Rightarrow 2$$

Hence $f(x)=x\sin x+2$

12. (a) $\int x\sqrt{2x+3}dx=\int x(2x+3)^{1/2}dx$

$$=x\frac{(2x+3)^{3/2}}{3/2}.\frac{1}{2}-\int\frac{(2x+3)^{3/2}}{3/2}\frac{1}{2}dx$$

$$=\frac{1}{3}x(2x+3)^{3/2}-\frac{1}{3}\int(2x+3)^{3/2}dx+c$$

$$=\frac{1}{3}x(2x+3)^{3/2}-\frac{1}{15}(2x+3)^{5/2}+c$$

13. (a) Put $x=\cos 0$

$$\therefore\ 2\cot^{-1}\sqrt{\frac{1-\cos 0}{1+\cos\theta}}=2\cot^{-1}\sqrt{\frac{2\sin^2\frac{0}{2}}{2\cos^2\frac{\theta}{2}}}$$

$$=2\cot^{-1}\left(\tan\frac{\theta}{2}\right)=2\left(\frac{\pi}{2}-\tan^{-1}\left(\tan\frac{\theta}{2}\right)\right)$$

$$=\pi-0=\pi-\cos^{-1}x$$

Here

Integral $=\int\cos(\pi-\cos^{-1}x)dx$

$$=-\int\cos(\cos^{-1}x)dx$$

$$=-\int x dx=-\frac{x^2}{2}+c$$

14. (a) Step-1: Integration by parts (taking second function x)

$$I=(\sin^{-1}x)\int x dx-\int\left(\frac{d}{dx}(\sin^{-1}x)\int x dx\right)dx$$

$$=(\sin^{-1}x)\left(\frac{x^2}{2}\right)-\int\frac{1}{\sqrt{1-x^2}}\times\frac{x^2}{2}dx$$

$$=\frac{x^2\sin^{-1}x}{2}-\frac{1}{2}\int\frac{x^2}{\sqrt{1-x^2}}dx$$

$$=\frac{x^2\sin^{-1}x}{2}-\frac{1}{2}\int\frac{(1-1+x^2)dx}{\sqrt{1-x^2}}$$

$$=\frac{x^2}{2}\sin^{-1}x-\frac{1}{2}\int\frac{dx}{\sqrt{1-x^2}}+\frac{1}{2}\int\sqrt{1-x^2}dx$$

Since $\int\sqrt{a^2-x^2}dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}$

$$=\frac{x^2}{2}\sin^{-1}x-\frac{1}{2}\sin^{-1}x+$$

$$\frac{1}{2}\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}x\right]$$

$$=\left(\frac{x^2}{2}-\frac{1}{4}\right)\sin^{-1}x+\frac{x}{4}\sqrt{1-x^2}+c$$

Quicker Method

Inverse Test i.e., obtaining solution by differentiating options

Integral of option (A)

$$\left\{(x)(\sin^{-1}x)+\left(\frac{x^2}{2}-\frac{1}{4}\right)\frac{1}{\sqrt{1-x^2}}\right\}$$

$$+\left\{\frac{1}{4}\sqrt{1-x^2}+\frac{x}{4}\times\frac{1}{2\sqrt{1-x^2}}\times-2x\right\}$$

$$= x\sin^{-1}x+\frac{x^2}{2\sqrt{1-x^2}}-\frac{1}{4\sqrt{1-x^2}}+\frac{\sqrt{1-x^2}}{4}-\frac{x^2}{4\sqrt{1-x^2}}$$

$$= x\sin^{-1}x+\frac{x^2}{4\sqrt{1-x^2}}+\left(\frac{\sqrt{1-x^2}}{4}-\frac{1}{4\sqrt{1-x^2}}\right)$$

$$= x\sin^{-1}x+\frac{x^2}{4\sqrt{1-x^2}}+\left(\frac{-x^2}{4\sqrt{1-x^2}}\right)=x\sin^{-1}x$$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. If $\int \ln(x^2+x).\,dx$ $x\ln(x^2+x)+A$, then $A=$ ***[MPPET-1992]***
   (a) $2x + \ln(x+1)$ + const.
   (b) $2x - \ln(x+1)$ + const.
   (d) constant
   (d) None of these

2. If $\int xe^{2x}\,dx$ is equal to $e^{2x}f(x)+C$ where C is constant of integration, then $f(x)$ is ***[UPSEAT-2001]***
   (a) $(3x-1)/4$ (b) $(2x \quad 1)/2$
   (c) $(2x-1)/4$ (d) $(x-4)/6$

3. If $\int x\sin x\,dx = -x\cos x + A$, then $A=$ ***[MPPET-1992; 2000; RPET-1997]***
   (a) $\sin x$ + constant (b) $\cos x$ − constant
   (c) Constant (d) None of these

4. $\int(1-x^2)\log x\,dx$ ***[DSSE-1982]***
   (a) $\left(x-\frac{x^3}{3}\right)\log x-\left(x-\frac{x^3}{9}\right)+c$
   (b) $\left(x-\frac{x^3}{3}\right)\log x+\left(x-\frac{x^3}{9}\right)+c$
   (c) $\left(x+\frac{x^3}{3}\right)\log x+\left(x+\frac{x^3}{9}\right)+c$
   (d) None of these

5. $\int\tan^{-1}\frac{2x}{1-x^2}dx=$ ***[MPPET-1991]***
   (a) $x\tan^{1}x \quad c$
   (b) $2x\tan^{-1}x-\log(1+x^2)+c$
   (c) $2x\tan^{-1}x-\log(1+x^2)+c$
   (d) None of these

6. $\int\frac{x\tan^{-1}x}{(1+x^2)^{3/2}}dx=$ ***[MP PET-2007]***
   (a) $\frac{x+\tan^{-1}x}{\sqrt{1+x^2}}+c$ (b) $\frac{x-\tan^{-1}x}{\sqrt{1+x^2}}+c$
   (c) $\frac{\tan^{-1}x-x}{\sqrt{1+x^2}}+c$ (d) None of these

7. $\int\frac{\log x\,dx}{x^3}=$ ***[Roorkee-1986]***
   (a) $\frac{1}{4x^2}(2\log x-1)+c$
   (b) $-\frac{1}{4x^2}(2\log x+1)+c$
   (c) $\frac{1}{4x^2}(2\log x+1)+c$
   (d) $\frac{1}{4x^2}(1-2\log x)+c$

8. $\int x^3e^{x^2}dx=$ ***[MNR-1980]***
   (a) $\frac{1}{2}(x^2+1)e^{x^2}+c$ (b) $(x^2+1)e^{x^2}+c$
   (c) $\frac{1}{2}(x^2-1)e^{x^2}+c$ (d) $(x^2-1)e^{x^2}+c$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 10 minutes.
3. The worksheet consists of 10 questions. The maximum marks are 30.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int x\tan^{-1}x\,dx =$ ***[Roorkee-1979]***

(a) $\frac{1}{2}(x^2+1)\tan^{-1}x-\frac{1}{2}x-c$

(b) $\frac{1}{2}(x^2-1)\tan^{-1}x-\frac{1}{2}x+c$

(c) $\frac{1}{2}(x^2+1)\tan^{-1}x+\frac{1}{2}x+c$

(d) $\frac{1}{2}(x^2+1)\tan^{-1}x-x+c$

2. $\int \sin\sqrt{x}\,dx =$ ***[Roorkee-1977]***

(a) $2[\sin\sqrt{x}-\cos\sqrt{x}]+c$

(b) $2[\sin\sqrt{x}-\sqrt{x}\cos\sqrt{x}]+c$

(c) $2[\sin\sqrt{x}+\cos\sqrt{x}]+c$

(d) $2[\sin\sqrt{x}+\sqrt{x}\cos\sqrt{x}]+c$

3. $\int x\cos x\,dx =$ ***[MPPET-1988]***

(a) $x\sin x+\cos x$ (b) $x\sin x-\cos x$

(c) $x\cos x+\sin x$ (d) $x\cos x-\sin x$

4. The value of $\int \frac{\log x}{(x+1)^2}dx$ is ***[UPSEAT-99]***

(a) $\frac{-\log x}{x+1}+\log x-\log(x+1)$

(b) $\frac{\log x}{(x+1)}+\log x-\log(x+1)$

(c) $\frac{\log x}{x+1}-\log x-\log(x+1)$

(d) $\frac{-\log x}{x+1}-\log x-\log(x+1)$

5. $\int x^n\log x\,dx =$

(a) $\frac{x^{n+1}}{n+1}\left\{\log x+\frac{1}{n+1}\right\}+c$

(b) $\frac{x^{n+1}}{n+1}\left\{\log x+\frac{2}{n+1}\right\}+c$

(c) $\frac{x^{n+1}}{n+1}\left\{2\log x-\frac{1}{n+1}\right\}+c$

(d) $\frac{x^{n+1}}{n+1}\left\{\log x-\frac{1}{n+1}\right\}+c$

6. The value of $\int x\sin kx\,dx$ is:

(a) $\frac{\sin kx}{k}+c$

(b) $\frac{\cos kx}{k}+c$

(c) $\frac{\sin x}{k}+c$

(d) $-\frac{x\cos kx}{k}+\frac{\sin kx}{k^2}+c$

7. $\int\log(x+1)dx =$ ***[Roorkee-1974; NDA-2008]***

(a) $(x-1)\log(x+1)-x+c$

(b) $(x-1)\log(x+1)+x-c$

(c) $(x-1)\log(x-1)-x-c$

(d) $(x-1)\log(x-1)+x+c$

8. If $\int f(x)dx=g(x)$, then $\int f(x)g(x)dx =$ ***[Karnataka CET-2008]***

(a) $\frac{1}{2}[g'(x)]^2$

(b) $f'(x)g(x)$

(c) $\frac{1}{2}f^2(x)$

(d) $\frac{1}{2}g^2(x)$

9. $\int x\cos^2 x\,dx =$ ***[IIT-1972]***

(a) $\frac{x^2}{4}-\frac{1}{4}x\sin 2x-\frac{1}{8}\cos 2x+c$

(b) $\frac{x^2}{4}+\frac{1}{4}x\sin 2x+\frac{1}{8}\cos 2x+c$

(c) $\frac{x^2}{4}-\frac{1}{4}x\sin 2x+\frac{1}{8}\cos 2x+c$

(d) $\frac{x^2}{4}+\frac{1}{4}x\sin 2x-\frac{1}{8}\cos 2x+c$

10. $\int\frac{x+\sin x}{1+\cos x}dx =$

(a) $x\tan x-c$ (b) $x\sin x+c$

(c) $x\cos x-c$ (d) $x\tan\frac{x}{2}+c$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ
6. ⓐ ⓑ ⓒ ⓓ
7. ⓐ ⓑ ⓒ ⓓ
8. ⓐ ⓑ ⓒ ⓓ
9. ⓐ ⓑ ⓒ ⓓ
10. ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

4. (a) $\int \frac{\log x}{(x+1)^2} dx = \int \frac{\log(t-1)}{t^2} dt$

Put $\{x+1=t \Rightarrow dx=dt\}$

$$= \log(t-1)\int \frac{1}{t^2} dt - \int \left[\frac{d}{dt}[\log(t-1)] \int \frac{1}{t^2} dt \right] dt$$

$$= -\log(t-1)\left(\frac{1}{t}\right) + \int \frac{1}{t-1}.\frac{1}{t} dt$$

$$= -\log(t-1)\frac{1}{t} + \log\left|\frac{t-1}{t}\right|$$

$$= -\log x \frac{1}{x+1} + \log \frac{x}{x+1}$$

$$\Rightarrow \frac{-1}{x+1}\log x + \log x - \log(x+1)$$

5. (d) $\int x^n \log x dx = \log x.\frac{x^{n+1}}{n+1} - \int \frac{x^{n+1}}{n+1}.\frac{1}{x} dx$

$$= \frac{x^{n+1}}{n+1}\log x - \frac{x^{n+1}}{(n+1)^2} + c$$

$$= \frac{x^{n+1}}{n+1}\left[\log x - \frac{1}{n+1}\right] + c$$

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LECTURE



6 Reduction Formulae

BASIC CONCEPTS

INTEGRALS WHEN: (INITIAL INTEGRAND REAPPEARS AFTER INTEGRATING BY PARTS)

Working Rule

(i) Apply the method of integration by parts twice.

(ii) On integrating by parts second time, we will obtain the given integrand again. Put it equal to I.

(iii) Transpose and collect terms involving I on one side and evaluate I.

(i) $\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + c$

OR $\frac{e^{ax}}{\sqrt{a^2+b^2}}\sin\left(bx - \tan^{-1}\frac{b}{a}\right) + c$

(ii) $\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + c$

OR $\frac{e^{ax}}{\sqrt{a^2+b^2}}\cos\left(bx - \tan^{-1}\frac{b}{a}\right) + c$

THREE STANDARD FORMS OF AN INTEGRAL

Cancellation Method: Split the integral into two integrals. Integrate only the first by integral by parts and second integral will get cancelled.

(i) $\int e^x[f(x) + f'(x)]dx = e^x f(x) + c$

(ii) $\int [xf'(x) + f(x)]dx = xf(x) + c$

(iii) $\int e^{mx}\{mf(x) + f'(x)\}dx = e^{mx}f(x) + c$

SOME SPECIAL INTEGRALS

(i) $\int\sqrt{a^2 - x^2}\,dx =$

$= \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$

(ii) $\int\sqrt{x^2 + a^2}\,dx$

$= \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\log[x + \sqrt{x^2 + a^2}]$

(iii) $\int\sqrt{x^2 - a^2}\,dx$

$= \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log[x + \sqrt{x^2 - a^2}]$

(iv) $\int\sqrt{x^2 + 2ax}\,dx = \left(\frac{x+a}{2}\right)\sqrt{x^2 + 2ax} -$

$\frac{-a^2}{2}\log[x + a + \sqrt{x^2 + 2ax}]$

(v) $\int\sqrt{2ax - x^2}\,dx = \frac{x}{2}\sqrt{2ax - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x-a}{a}\right)$

REDUCTION FORMULAE OF SOME TRIGONOMETRICAL FUNCTIONS

(i) $\int \sin^n x dx = -\frac{1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n}\int \sin^{n-2} x dx$

(ii) $\int \cos^n x dx = \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n}\int \cos^{n-2} x dx$

(iii) $\int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x dx$

(iv) $\int \cot^n x dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x dx$

(v) $\int \sec^n x dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} x dx$

(vi) $\int \operatorname{cosec}^n x dx = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1}\int \operatorname{cosec}^{n-2} x dx$

(vii) $\int \sin^m x . \cos^n x dx = \frac{-\sin^{m+1} x \cos^{m-1} x}{m+n} + \frac{m-1}{m+n}\int \sin^{m-2} x . \cos^n x dx$

Similarly, we can reduce

$\int \sin^m x \cos^n x dx x$

$= \frac{\sin^{m-1} x \cos^{n-1} x}{m+n} + \frac{m-1}{m+n}\int \sin^m x \cos^{n-2} x dx$

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)) FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate $\int \frac{e^x (x^2+1) dx}{(x+1)^2}$

Solution

Let $I = \int \frac{e^x (x^2+1)}{(x+1)^2} dx$

$= \int \frac{e^x [(x+1)^2 - 2x]}{(x+1)^2} dx$

$= \int e^x dx - 2\int \frac{x e^x}{(x+1)^2} dx$

$= e^x - 2\int \frac{(x+1-1)e^x}{(x+1)^2} dx$

$= e^x - 2\int \left[e^x \left\{ \frac{(x+1)}{(x+1)^2} - \frac{1}{(x+1)^2} \right\} \right] dx$

$\Rightarrow \quad I = e^x - 2\int \left[e^x \left\{ \frac{1}{(x+1)} - \frac{1}{(x+1)^2} \right\} \right] dx$

we know that $\int e^x [f(x) + f'(x)] dx = e^x f(x)$

Here $f(x) = \frac{1}{x+1}$

$\therefore \quad f'(x) = -\frac{1}{(x+1)^2}$

$\therefore \quad I = e^x - 2e^x \cdot \frac{1}{(x+1)}$

$= e^x \left[1 - \frac{2}{(x+1)} \right]$

$= e^x \left[\frac{x+1-2}{x+1} \right]$

Hence, $I = e^x \left[\frac{(x-1)}{(x+1)} \right]$

2. Evaluate $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$

[CBSE-Model question paper]

Solution

Let $I = \int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$

Put $x = a \tan^2 \theta$

$\Rightarrow \; dx = 2a \tan \theta \sec^2 \theta \, d\theta$

$\therefore \; I = \int \sin^{-1} \sqrt{\frac{a \tan^2 \theta}{a + a \tan^2 0}} . 2a \tan 0 \sec^2 0 \, d\, 0$

$= \int 2a \sin^{-1} \sqrt{\frac{\tan^2 0}{\sec^2 0}} . \tan \theta \sec^2 \theta d\theta$

$$= \int 2a\sin^{-1}\left(\frac{\sin\theta/\cos\theta}{1/\cos\theta}\right).\tan\theta\sec^2\theta\, d\theta$$

$$= 2a\int \sin^{-1}(\sin\theta)\tan\theta\sec^2\theta\, d\theta$$

$$= 2a\int \theta\tan\theta\sec^2\theta\, d\theta$$

$$= a\int \theta(2\tan\theta\sec^2\theta)d\theta$$

$$= a\left[\int\theta(\tan^2\theta) - \int 1.\tan^2\theta\, d\theta\right]$$

[Integrating by parts and since $\int 2\tan\theta\sec^2\theta\, d\theta = \tan^2\theta$]

$$\therefore \quad I = a\left[\theta\tan^2\theta - \int(\sec^2\theta - 1)d\theta\right]$$

$$= a[\theta\tan^2\theta - \tan\theta + \theta]$$

$$= a[\theta(1+\tan^2\theta) - \tan\theta]$$

$$= a\left[\tan^{-1}\sqrt{\frac{x}{a}}\left\{1+\frac{x}{a}\right\} - \sqrt{\frac{x}{a}}\right]$$

$$= (a+x)\tan^{-1}\sqrt{x/a} - \sqrt{ax}$$

3. Evaluate $\int\sqrt{\frac{a+x}{x}}dx$

Solution

$$\int\sqrt{\frac{a+x}{x}}dx \qquad \text{| Put } x = at^2 \Rightarrow dx = 2at\, dt.$$

$$\int\sqrt{\frac{a+at^2}{at^2}}\times 2atdt = 2a\int\sqrt{1+t^2}dt$$

$$= 2a\left[\frac{t}{2}\sqrt{1+t^2} + \frac{1}{2}\log\left|t+\sqrt{1+t^2}\right|\right] + c$$

$$= 2a\left[\frac{1}{2}.\sqrt{\frac{x}{a}}\sqrt{1+\frac{x}{a}} + \frac{1}{2}\log\left|\sqrt{\frac{x}{a}} + \sqrt{1+\frac{x}{a}}\right|\right] + c$$

$$= \sqrt{ax}\sqrt{1+\frac{x}{a}} + a\log\left|\sqrt{\frac{x}{a}} + \sqrt{1+\frac{x}{a}}\right| + c$$

4. Evaluate $\int e^{3x}.\sin 2x\, dx$

Solution

$$I = \int e^{3x}.\sin 2x\, dx = \sin 2x.\frac{e^{3x}}{3} - \int 2\cos 2x.\frac{e^{3x}}{3}dx$$

II I

$$= \frac{1}{3}\sin 2x.e^{3x} - \frac{2}{3}\left[\cos 2x.\frac{e^{3x}}{3} - \int(-2\sin 2x).\frac{e^{3x}}{3}dx\right]$$

$$I = \frac{1}{3}\sin 2x.e^{3x} - \frac{2}{9}\cos 2x.e^{3x} - \frac{4}{9}I$$

$$\frac{13}{9}I = \frac{(3\sin 2x - 2\cos 2x)e^{3x}}{9}$$

$$\Rightarrow I = \frac{e^{3x}}{13}(3\sin 2x - 2\cos 2x) + c$$

5. Evaluate $\int\frac{x}{\sqrt{8+x-x^2}}dx$

Solution

$$\int\frac{x}{\sqrt{8+x-x^2}}dx$$

$$= \int\frac{-\frac{1}{2}(1-2x)+\frac{1}{2}}{\sqrt{8+x-x^2}}dx$$

$$\left(\begin{array}{l}\text{Let } x = A(1-2x)+B \\ \quad x = -2Ax + (A+B) \\ -2A = 1 : B + A = 0\end{array}\right)$$

$$= -\frac{1}{2}\int\frac{(1-2x)}{\sqrt{8+x-x^2}}dx + \frac{1}{2}\int\frac{1}{\sqrt{8+x-x^2}}dx \quad \ldots(i)$$

Consider $= -\frac{1}{2}\int\frac{(1-2x)}{\sqrt{8+x-x^2}}dx$

| Put $8+x \quad x^2 = t \Rightarrow (1-2x)\, dx = dt$

$$= -\frac{1}{2}\int\frac{1}{\sqrt{t}}dt = -\frac{1}{2}.2\sqrt{t} = -\sqrt{8+x-x^2} \quad \ldots(ii)$$

Consider

$$\frac{1}{2}\int\frac{1}{\sqrt{8+x-x^2}}dx$$

$$= \frac{1}{2}\int\frac{1}{\sqrt{\frac{33}{4} - \left(x-\frac{1}{2}\right)^2}}dx$$

$$\left|\begin{array}{l} -[x^2 - x - 8] = \\ -\left[\left(x-\frac{1}{2}\right)^2 - \frac{1}{4} - 8\right] = \\ \frac{33}{4} - \left(x-\frac{1}{2}\right)^2\end{array}\right.$$

$$= \frac{1}{2}\sin^{-1}\frac{\left(x-\frac{1}{2}\right)}{\frac{\sqrt{33}}{2}} = \frac{1}{2}\sin^{-1}\left(\frac{2x-1}{\sqrt{33}}\right) \quad \ldots\ldots(iii)$$

Substituting from (ii) and (iii) in (i), we get

$$\int \frac{x}{\sqrt{8+x-x^2}}dx$$

$$= -\sqrt{8+x-x^2} + \frac{1}{2}\sin^{-1}\left(\frac{2x-1}{\sqrt{33}}\right) + c$$

6. Evaluate $\int \frac{1}{(2x+3)\sqrt{x+1}}dx$

Solution

$$\int \frac{1}{(2x+3)\sqrt{x+1}}dx$$

$$|\text{ Put } x+1 = t^2 \Rightarrow dx = 2t\,dt$$

$$= \int \frac{1}{[2(t^2-1)+3].t}.2tdt = 2\int \frac{1}{2t^2+1}dt$$

$$= \int \frac{1}{t^2+\frac{1}{2}}dt$$

$$= \sqrt{2}\tan^{-1}\left(\sqrt{2}t\right) + c = \sqrt{2}\tan\left(\sqrt{2}\sqrt{x+1}\right) + c$$

7. Evaluate $\int \frac{1}{(x+1)\sqrt{x^2-1}}dx$

Solution

$$\int \frac{1}{(x+1)\sqrt{x^2-1}}dx$$

$$\left|Put\ \ 1+x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2}dt\right.$$

$$= \int \frac{1}{\frac{1}{t}\sqrt{\left(\frac{1}{t}-1\right)^2-1}} \times \frac{1}{t^2}dt$$

$$= \int \frac{1}{t.\sqrt{\frac{1}{t^2}-\frac{2}{t}+1-1}}dt = -\int \frac{1}{\sqrt{1-2t}}dt$$

$$= -\frac{(1-2t)^{1/2}}{\frac{1}{2}\times -2} + c = \sqrt{1-2t} + c = \sqrt{1-\frac{2}{x+1}} + c$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Evaluate $\int \frac{\sin(\log x)}{x^2}dx$.

2. If $u = \int e^{ax}\cos bx dx$ and $v = \int e^{ax}\sin bx dx$, then prove that

   (i) $(u^2+v^2)\ (a^2+b^2) = e^{2ax}$

   (ii) $\tan^{-1}\frac{v}{u} + \tan^{-1}\frac{b}{a} = bx$

3. Evaluate $\int e^{2x}\left(\frac{1+\sin 2x}{1+\cos 2x}\right)dx$

4. Evaluate $\int \frac{\sin 2x}{(a+b\cos x)^2}$. ***[PSB-1991]***

5. Evaluate $\int \frac{(2+\sin 2x)e^x}{1+\cos 2x}dx$.

   [CBSE (model question paper)]

6. Evaluate $\int e^x \sec x(1+\tan x)dx$.

   [CBSE-82, 85]

7. Evaluate $\int \sin^5 x dx$.

8. Evaluate $\int \cos^4 x dx$.

9. Evaluate $\int \sqrt{x^2+4x+1}dx$. ***[MP-98]***

10. If In = $\{x^n \sin x\ dx$ show that
    $I_n = -x^n \cos x + n x^{n-1}\sin x - n(n-1)I_{n-2}$

11. Evaluate $\int \sqrt{2-3x^2}dx$

12. Evaluate $\int \text{cosec}^3 x dx$

    [CBSE-Practice Sample paper-(VI)]

13. Evaluate $\int \sec^6 x dx$

14. Evaluate $\int \sin^4 x dx$ ***[CBSE-2000, 2004]***

ANSWERS

1. $-\frac{1}{2x}\ [\sin(\log x) + \cos(\log x)]$

3. $\frac{1}{2}e^{2x}\tan x$

4. $\frac{-2}{b^2}\left[\log|a+b\cos x| + \frac{a}{a+b\cos x}\right] + C$

5. $e^x \tan x + C$

6. $e^x \sec x$.

7. $-\left[\cos x - \frac{2}{3}\cos^3 x + \frac{1}{5}\cos^5 x\right]$

8. $\frac{1}{8}\left(3x + 2\sin 2x + \frac{\sin 4x}{4}\right)$

9. $\frac{x+2}{2}\sqrt{x^2+4x+1} - \frac{3}{2}\log$

$\left[(x+2) + \sqrt{x^2+4x+1}\right]$

11. $\sqrt{3}\left[\frac{x}{2}\sqrt{\frac{2}{3}-x^2} + \frac{1}{3}\sin^{-1}\left(x\sqrt{\frac{3}{2}}\right)\right]$

12. $\frac{-\text{cosec}\,x\cot x}{2} + \frac{1}{2}\log|\text{cosec}\,x - \cot x| + c$

13. $\tan x + \frac{2}{3}\tan^3 x + \frac{1}{5}\tan^5 x + c$

14. $\frac{1}{8}\left[3x - 2\sin 2x + \frac{\sin 4x}{4}\right] + c$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\int (x^6 + 7x^5 + 6x^4 + 5x^3 + 4x^2 + 3x + 1)e^x dx = ?$

[Gujrat CET-2007]

(a) $\sum_{i=0}^{6} x^i e^x + C$ (b) $\sum_{i=1}^{7} x^i e^x + C$

(c) $\sum_{i=1}^{6} x^i e^x + C$ (d) None of these

Solution

(c) $\int (x^6 + 7x^5 + 6x^4 + 5x^3 + 4x^2 + 3x + 1)e^x dx$

$\int e^x (x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 +$

$6x^5 + 5x^4 + 4x^3 + 3x^2 + 2x + 1)dx$

$= e^x(x^6 + x^5 + x^4 + x^3 + x^2 + 1) + c$

Since $\int e^x [f(x) + f'(x)]dx = e^x f(x)$

$\Rightarrow \quad \sum_{i=1}^{6} x^i e^x + C$

2. The value of $\int e^x (x^5 + 5x^4 + 1).dx$ is

[Karnataka CET-07]

(a) $e^x . x^5$

(b) $e^x . x^5 + e^x + C$

(c) $e^{x+1} . x^5 + C$

(d) $5x^4 . e^x$

Solution

(b) $\int e^x (x^5 + 5x^4 + 1)dx$;

$\int e^x (x^5 + 5x^4)dx + \int e^x dx$

Now since,

$\int e^x (f(x) + f'(x))dx = e^x f(x) + C$

$\Rightarrow$ required value $e^x x^5 + e^x + C$

$\Rightarrow$ $e^x(x^5 + 1) + C$

3. $\int \left[\log(\log x) + \frac{1}{(\log x)^2}\right]dx =$

(a) $x\log(\log x) + \frac{x}{\log x} + c$

(b) $x\log(\log x) - \frac{x}{\log x} + c$

(c) $x\log(\log x)+\frac{\log x}{x}+c$

(d) $x\log(\log x)-\frac{\log x}{x}+c$

Solution

(b) $\int\left[\log(\log x)+\frac{1}{(\log x)^2}\right]dx=$

$\int\log(\log x)dx+\int\frac{1}{(\log x)^2}dx$

$=x\log(\log x)-\int\frac{x}{x\log x}dx+\int\frac{1}{(\log x)^2}dx$

$=x\log(\log x)-\frac{x}{\log x}-\int\frac{1}{(\log x)^2}dx+$

$\int\frac{1}{(\log x)^2}dx$

$=x\log(\log x)-\frac{x}{\log x}+c$

4. $\int\frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$ is equal to

[IIT-JEE-2006]

(a) $\frac{\sqrt{2x^4-2x^2+1}}{x^3}+c$

(b) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+c$

(c) $\frac{\sqrt{2x^4-2x^2+1}}{x^2}+c$

(d) $\frac{\sqrt{2x^4-2x^2+1}}{x}+c$

Solution

(b) $I=\int\frac{x^2-1}{x^5\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}dx$

$=\int\frac{\left(\frac{1}{x^3}-\frac{1}{x^5}\right)}{\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}dx$

Now Let $2-\frac{2}{x^2}+\frac{1}{x^4}=t$, then

$\left(\frac{1}{x^3}-\frac{1}{x^5}\right)dx=\frac{1}{4}dt$

$\therefore\quad I=\frac{1}{4}\int\frac{dt}{\sqrt{t}}=\frac{1}{2}\sqrt{t}+c$

$=\frac{1}{2}\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+c$

$=\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+c$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int\sin 3x\,dx$ is equal to

[SCRA-1996]

(a) $\sin^3 x+1$ (b) $\sin x^2+x^2+1$

(c) $\frac{\cos^3 x}{3}-\cos x$ (d) $\frac{1}{4}\sin^4 x-\frac{3}{4}\sin^2 x$

2. The value of $\int e^{2x}(2\sin 3x+3\cos 3x)\,dx$ is

[MPPET-2003]

(a) $e^{2x}\sin 3x$ (b) $e^{2x}\cos 3x$

(c) e^{2x} (d) $e^{2x}(2\sin 3x)$

3. $\int\left(\frac{x+2}{x+4}\right)^2 e^x\,dx$ is equal to ***[AMU-2000]***

(a) $e^x\left(\frac{x}{x+4}\right)+C$ (b) $e^x\left(\frac{x+2}{x+4}\right)+C$

(c) $e^x\left(\frac{x-2}{x+4}\right)+C$ (d) $\left(\frac{2xe^x}{x+4}\right)+C$

4. $\int\frac{(x+3)e^x}{(x+4)^2}dx=$ ***[Karnataka CET-2000]***

(a) $\frac{1}{(x+4)^2}+C$ (b) $\frac{e^x}{(x+4)^2}+C$

(c) $\frac{e^x}{x+4}+C$ (d) $\frac{e^x}{x+4}+C$

5. $\int e^{x/2}\sin(x/2+\pi/4)dx=$ ***[Roorkee-1982]***

(a) $e^{x/2}\cos\frac{x}{2}+c$

(b) $\sqrt{2}\ e^{x/2}\cos\frac{x}{2}+c$

(c) $e^{x/2}\sin\frac{x}{2}+c$

(d) $\sqrt{2}\ e^{x/2}\sin\frac{x}{2}+c$

6. $\int e^x \sin x\,dx =$ ***[IIT-78; AICBSE-80; MPPET-89, 99]***

(a) $\frac{1}{2}e^x(\sin x+\cos x)+c$

(b) $\frac{1}{2}e^x(\sin x-\cos x)+c$

(c) $e^x(\sin x+\cos x)+c$

(d) $e^x(\sin x-\cos x)+c$

7. $\int x\sin^2 x\,dx =$ ***[BIT Ranchi-1977; IIT-1972]***

(a) $\frac{x^2}{4}+\frac{x}{4}\sin 2x+\frac{1}{8}\cos 2x+c$

(b) $\frac{x^2}{4}-\frac{x}{4}\sin 2x+\frac{1}{8}\cos 2x+c$

(c) $\frac{x^2}{4}+\frac{x}{4}\sin 2x-\frac{1}{8}\cos 2x+c$

(d) $\frac{x^2}{4}-\frac{x}{4}\sin 2x-\frac{1}{8}\cos 2x+c$

8. $\int x\log x(\log x-1)dx$ is equal to

(a) $2(x\log x-x)^2+c$

(b) $\frac{1}{2}(x\log x-x)^2+c$

(c) $2(x\log x+x)^2+c$

(d) None of these

9. $\int(1+x-x^{-1})e^{x+x^{-1}}dx =$ ***[EAMCET-2003]***

(a) $(x+1)e^{x+x^{-1}}+C$ (b) $(x-1)e^{x+x^{-1}}+C$

(c) $-x\,e^{x+x^{-1}}+C$ (d) $x\,e^{x+x^{-1}}+C$

10. $\int x^2\sin 2x dx$

(a) $\frac{1}{2}x^2\cos 2x+\frac{1}{2}x\sin 2x+\frac{1}{4}\cos 2x+c$

(b) $-\frac{1}{2}x^2\cos 2x+\frac{1}{2}x\sin 2x+\frac{1}{4}\cos 2x+c$

(c) $\frac{1}{2}x^2\cos 2x-\frac{1}{2}x\sin 2x-\frac{1}{4}\cos 2x+c$

(d) None of these

11. $\int\frac{x\sin x^2 e^{\sec x^2}}{\cos^2 x^2}dx =$

(a) $\frac{e^{\sec x^2}}{2}+C$ (b) $\frac{e^{\sec x^3}}{2}+C$

(c) $e^{\sec x^3}+C$ (d) None of these

12. $\int[\sin(\log x)+\cos(\log x)]dx =$ ***[MPPET-1991]***

(a) $x\cos(\log x)+c$

(b) $\sin(\log x)+c$

(c) $\cos(\log x)+c$

(d) $x\sin(\log x)+c$

13. $\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx =$

(a) $xe^{\tan^{-1}x}+c$ (b) $x^2e^{\tan^{-1}x}+c$

(c) $\frac{1}{x}e^{\tan^{-1}x}+c$ (d) $\frac{1}{x^2}e^{\tan^{-1}x}+c$

14. $\int\left(\frac{2+\sin 2x}{1+\cos 2x}\right)ex\,dx$ ***[AISSE-1982]***

(a) $e^x\cot x+c$ (b) $-e^x\cot x+c$

(c) $-e^x\tan x+c$ (d) $e^x\tan x+c$

15. $\int e^x(1+\tan x+\tan^2 x)\,dx$ ***[Karnataka CET-1999]***

(a) $e^x\sin x+C$

(b) $e^x\cos x+C$

(c) $e^x\tan x+C$

(d) $e^x\sec x+C$

16. $\int\frac{xe^x}{(1+x)^2}dx$ is equal to

(a) $-\frac{e^x}{1+x}$ (b) $\frac{1+2xe^x}{1+x}$

(c) $-\frac{1+2xe^x}{1+x}$ (d) $\frac{e^x}{1+x}$

17. The value of $\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx$ ***[AIEEE-2005]***

(a) $\frac{\log x}{(\log x)^2+1}+C$ (b) $\frac{x}{x^2+1}+C$

(c) $\frac{xe^x}{1+x^2}+C$ (d) $\frac{x}{(\log x)^2+1}+C$

SOLUTIONS

1. (c) $\int \sin^3 x dx = \int (1-\cos^2 x)\sin x dx$

$= \int \sin x dx - \int \cos^2 x \sin x dx$

$= -\cos x + \int t^2 dt$

(Put $t = \cos x \Rightarrow dt = -\sin x dx$)

$= \frac{\cos^3 x}{3} - \cos x + c$

2. (a) Let $I = \int e^{2x}(2\sin 3x + 3\cos 3x)dx$

$I = 2\int e^{2x}\sin 3x dx + 3\int e^{2x}\cos 3x dx$

Integrating by part

$I = 2\left\{\frac{e^{2x}}{2}\sin 3x\right\} - 2\int \frac{3e^{2x}}{2}\cos(3x)dx +$

$3\int e^{2x}\cos 3x dx$

$I = e^{2x}\sin 3x - 3\int e^{2x}\cos 3x dx + 3\int e^{2x}\cos 3x dx$

$I = e^{2x}\sin 3x$

3. (a) $I = \int \left(\frac{x+2}{x+4}\right)^2 e^x dx = \int e^x \left[\frac{x^2+4x+4}{(x+4)^2}\right]dx$

$I = \int e^x \left[\frac{x(x+4)}{(x+4)^2} + \frac{4}{(x+4)^2}\right]dx$

$= \int e^x \left[\frac{x}{x+4} + \frac{4}{(x+4)^2}\right]dx = e^x\left(\frac{x}{x+4}\right) + C$

$(\because \int e^x[f(x)+f'(x)]dx = e^x f(x) + C)$

4. (c) $I = \int \frac{(x+3)e^x}{(x+4)^2}dx = \int \frac{(x+4-1)}{(x+4)^2}$

$I = \int e^x\left(\frac{1}{x+4} - \frac{1}{(x+4)^2}\right)dx$

$\therefore \quad I = \frac{e^x}{x+4} + C$

$(\because \int e^x[f(x)+f'(x)]dx = e^x f(x))$

5. (d) Let

$I = \int e^{x/2}\sin\left(\frac{x}{2}+\frac{x}{4}\right)dx = 2\sin\left(\frac{x}{2}+\frac{\pi}{4}\right)e^{x/2}$

$-\int \cos\left(\frac{x}{2}+\frac{\pi}{4}\right)\frac{1}{2}2e^{\frac{x}{2}}dx$

$= 2\sin\left(\frac{x}{2}+\frac{\pi}{4}\right)e^{x/2} - 2e^{x/2}\cos\left(\frac{x}{2}+\frac{\pi}{4}\right) -$

$\int \sin\left(\frac{x}{2}+\frac{\pi}{4}\right)\frac{1}{2}2e^{x/2}$

$\therefore \quad 2I = 2e^{x/2}\left\{\sin\left(\frac{x}{2}+\frac{\pi}{4}\right) - \cos\left(\frac{x}{2}+\frac{\pi}{4}\right)\right\}$

$I = e^{x/2}\left\{\sin\left(\frac{x}{2}+\frac{\pi}{4}\right) - \cos\left(\frac{x}{2}+\frac{\pi}{4}\right)\right\}$

$= \sqrt{2}e^{x/2}\left(\sin\frac{x}{2}\right) = \sqrt{2}e^{x/2}\sin\frac{x}{2} + c$

6. (b) Step-1:

$\int e^{ax}\sin bx dx = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx]$

Step-2: $\frac{e^x}{2}[\sin x - \cos x]$ (Here $a = b = 1$)

7. (d) $I = \int x\sin^2 x dx$

$\Rightarrow \quad 2I = \int x(1-\cos 2x)dx$

$= \frac{x^2}{2} - \int x\cos(2x)dx$

By using formula: Integration by parts

$$\int uv dx = u\int v - \left[\int\left(\frac{du}{dx}\right)\left(\int v dx\right)dx\right]$$

$\Rightarrow \quad 2I = \frac{x^2}{2} - \frac{x\sin(2x)}{2} + \int \frac{\sin(2x)}{2}dx$

$= \frac{x^2}{2} - \frac{x}{2}\sin(2x) - \frac{1}{4}\cos(2x) + c$

$\Rightarrow \quad I = \frac{x^2}{4} - \frac{x}{4}\sin(2x) - \frac{1}{8}\cos(2x) + c$.

8. (b) Step 1: $\int (x\log x - x)\log x dx$

Step 2: $\left(\log x + x\times\frac{1}{x} - 1\right)dx = \log x dx$

$\int t dt = \frac{t^2}{2} = \frac{(x\log x - x)^2}{2}$

9. (d) $\int \left(1 + x - x^{-1}\right)e^{x+x^{-1}}dx$

$= \int\left[xe^{x+x^{-1}}\left(1-\frac{1}{x^2}\right) + e^{x+x^{-1}}\right]dx$

$= xe^{x+x^{-1}} + C$

$(\because \int[xf'(x) + f(x)]dx = xf(x) + c)$

10. (b) Step 1: Quicker Method

$I = (x^2)\left(\frac{-\cos 2x}{2}\right) - (2x)\left(\frac{-\sin 2x}{4}\right) + (2)\left(\frac{\cos 2x}{8}\right)$

$-\frac{1}{2}x^2\cos 2x + \frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x + c$

11. (a)

$\int \frac{x\sin x^2 e^{\sec x^2}}{\cos^2 x^2}dx = \int\left[x\tan x^2\sec x^2 e^{\sec x^2}\right]dx$

Put $\sec x^2 = t \Rightarrow dt = \sec x^2\tan x^2(2x)dx$

$\frac{1}{2}\int e^t dt = \frac{1}{2}e^{\sec^2 x} + C$

12. (d) Step 1: Cancellation method of Integral: Integrating first section by parts

$\left(\sin(\log x)\int 1dx - \int\left(\frac{d}{dx}(\sin(\log x))\right)\int 1dx\right)dx$

$+ \int\cos(\log x)dx$

$= (x)\sin(\log x) - \int\frac{\cos(\log x)}{x}\times x dx + \int\cos(\log x)dx$

$= (x)(\sin(\log(x)) + c$

Last two integral are cancelled.

NOTE

In first section only one function $\sin(\log x)$ is given. Therefore, one is the second function.

13. (a) Putting $\tan^{-1}x = t \Rightarrow \frac{1}{1+x^2}dx = dt$ we get

$\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx = \int e^t\left(\tan t + \sec^2 t\right)dt$

$= e^t\tan t + c = xe^{\tan^{-1}x} + c$

$(\because \int e^x[f(x) + f'(x)]dx = e^x f(x) + C)$

14. (d)

$\int\left(\frac{2+\sin 2x}{1+\cos 2x}\right)e^x dx = \int\frac{2(1+\sin x\cos x)e^x}{2\cos^2 x}dx$

$= \int e^x(\tan x + \sec^2 x)dx = e^x\tan x + c$

15. (c) $I = \int e^x(1 + \tan x + \tan^2 x)dx =$

$\int e^x(\tan x + \sec^2 x)dx$

$I = e^x\tan x + C$

$(\because \int e^x[f(x) + f'(x)]dx = e^x f(x) + C)$

16. (d) $\int\frac{xe^x}{(1+x)^2}dx = \int\frac{x+1-1}{(1+x)^2}dx$

$= \int e^x\left(\frac{1}{1+x} - \frac{1}{(1+x)^2}\right)dx = \frac{e^x}{1+x} + C$

$(\because \int e^x[f(x) + f'(x)]dx = e^x f(x))$

17. (d)

$\int\frac{(\log x - 1)^2}{(1+(\log x)^2)^2}dx = \int\frac{1+(\log x)^2 - 2\log x}{\left[1+(\log x)^2\right]^2}dx$

$= \int\left[\frac{1}{(1+(\log x)^2)} - \frac{2\log x}{(1+(\log x)^2)^2}\right]dx$

Put $\log x = t \Rightarrow dx = e^t dt$

$= \int\left[\frac{e^t}{1+t^2} - \frac{2te^t}{(1+t^2)^2}\right]dt$

$= \int e^t\left[\frac{1}{1+t^2} - \frac{2t}{(1+t^2)^2}\right]dt$

$= \frac{e^t}{1+t^2} + c = \frac{x}{1+(\log x)^2} + c$

$\because \int e^x(f(x) + f'(x))dx = e^x f(x) + c$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE) FOR IMPROVING SPEED WITH ACCURACY

1. $\int e^x (1-\cot x+\cot^2 x)\,dx$ equals

[MPPET-2002]

(a) $e^x \cot x + C$ (b) $e^x \operatorname{cosec} x + C$
(c) $-e^x \cot x + C$ (d) $-e^x \operatorname{cosec} x + C$

2. $\int \frac{e^x}{x+2}\{1+(x+2)\log(x+2)\}\,dx =$

(a) $e^x \log(x+2) + c$
(b) $\frac{e^x}{x+2} + c$
(c) $e^x(x+2) + c$
(d) None of these

3. $\int e^{-2x} \sin 3x\,dx =$

(a) $\frac{1}{13} e^{-2x}[\sin 3x + \cos 3x] + c$
(b) $-\frac{1}{13} e^{2x}[\sin 3x + \cos 3x] + c$
(c) $\frac{1}{13} e^{2x}[2\sin 3x + 3\cos 3x] + c$
(d) $-\frac{1}{13} e^{-2x}[2\sin 3x + 3\cos 3x] + c$

4. If $\int \frac{e^x(1+\sin x)dx}{1+\cos x} = e^x f(x) + c$, then $f(x)$ is equal to

(a) $\sin\frac{x}{2}$ (b) $\cos\frac{x}{2}$
(c) $\tan\frac{x}{2}$ (d) $\log\frac{x}{2}$

5. If $u = \int e^x \cos x\,dx$, $v = \int e^x \sin x\,dx$, then $u + v$ is equal to

(a) du/dx (b) dv/dx
(b) $\frac{du}{dv}$ (d) None of these

6. The value of $\int \sec^3 x\,dx$ will be

[UPSEAT-1999]

(a) $\frac{1}{2}[\sec x \tan x + \log(\sec x + \tan x)]$
(b) $\frac{1}{3}[\sec x \tan x + \log(\sec x + \tan x)]$
(c) $\frac{1}{4}[\sec x \tan x + \log(\sec x + \tan x)]$
(d) $\frac{1}{8}[\sec x \tan x + \log(\sec x + \tan x)]$

7. $\int e^{\sin\theta}[\log\sin\theta + \operatorname{cosec}^2\theta]\cos\theta\,d\theta$ is equal to

[Kerala PET-2008]

(a) $e^{\sin\theta}[\log\sin\theta + \operatorname{cosec}^2\theta] + c$
(b) $e^{\sin\theta}[\log\sin 0 + \operatorname{cosec}\theta] + c$
(c) $e^{\sin\theta}[\log\sin 0 - \operatorname{cosec}\theta] + c$
(d) $e^{\sin\theta}[\log\sin\theta - \operatorname{cosec}^2\theta] + c$

8. The value of $\int e^x \frac{2-\sin 2x}{1-\cos 2x}dx$ is

[MPPET-2008]

(a) $e^x \tan x + C$ (b) $-e^x \tan x + C$
(c) $e^x \cot x + C$ (d) $-e^x \cot x + C$

9. $\int e^x \frac{(x^2+1)}{(x+1)^2}dx$ is equal to

(a) $\left(\frac{x-1}{x+1}\right)e^x + c$ (b) $e^x\left(\frac{x+1}{x-1}\right) + c$
(c) $e^x(x-1)(x-1) + c$ (d) None of these

10. $\int e^x \sin x(\sin x + 2\cos x)\,dx$

[MPPET-1988]

(a) $e^x \sin^2 x + c$ (b) $e^x \sin x + c$
(c) $e^x \sin 2x$ (d) None of these

11. $\int e^x(\log\sin x + \cot x)dx$ is equal to

(a) $e^x \cot x + c$
(b) $e^x \log\sin x + c$
(c) $e^x \tan x + c$
(d) None of these

12. $\int e^x\left[\sin^{-1}\frac{x}{a} + \frac{1}{\sqrt{a^2-x^2}}\right]dx =$

(a) $\frac{1}{a}e^x \sin^{-1}\frac{x}{a} + c$ (b) $ae^x \sin^{-1}\frac{x}{a} + c$
(c) $e^x \sin^{-1}\frac{x}{a} + c$ (d) $\frac{e^x}{\sqrt{a^2-x^2}} + c$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 14 minutes.
3. The worksheet consists of 14 questions. The maximum marks are 42.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int e^{2x}(-\sin x + 2\cos x)dx =$ ***[DSSE-1987]***

(a) $e^{2x}\sin x + c$
(b) $-e^{2x}\sin x + c$
(c) $-e^{2x}\cos x + c$
(d) $e^{2x}\cos x + c$

2. $\int \frac{x-1}{(x+1)^3}e^x dx =$ ***[IIT-1983; MPPET-1990]***

(a) $\frac{-e^x}{(x+1)^2} + c$ (b) $\frac{e^x}{(x+1)^2} + c$
(c) $\frac{e^x}{(x+1)^3} + c$ (d) $\frac{-e^x}{(x+1)^3} + c$

3. $\int g(x)dx = g(x)$, then $\int g(x)(f(x) + f'(x))dx =$

(a) $g(x)f^2(x)$
(b) $g(x)f(x) - g(x)f'(x)$
(c) $g(x)f'(x)$
(d) $g(x)f(x)$

4. $\int xe^x dx$ is equal to ***[SCRA-1996]***

(a) $(x+1)e^x + c$
(b) $(x-1)e^x + c$
(c) $xe^x + 1 + c$
(d) $xe^x - 1 + c$

5. If $u = \int e^{ax}\cos bx\,dx$ and $v = \int e^{ax}\sin bx\,dx$, then $(a^2 + b^2)(u^2 + v^2) =$

(a) $2e^{ax}$ (b) $(a^2 + b^2)e^{2ax}$
(c) e^{2ax} (d) $(a^2 - b^2)e^{2ax}$

6. If $I = \int e^x \sin 2x\,dx$, then for what value of K, $KI = e^x(\sin 2x - 2\cos 2x) + \text{const.}$ ***[MPPET-1992]***

(a) 1 (b) 3
(c) 5 (d) 7

7. $\int \log_{10} x dx =$ ***[Roorkee-1973]***

(a) $x\log_{10} x + c$
(b) $x(\log_{10} x + \log_{10} e) + c$
(c) $\log_{10} x + c$
(d) $x(\log_{10} x - \log_{10} e) + c$

8. $\int e^x \frac{x+1}{(x+2)^2}dx =$

(a) $\frac{e^x}{x+2} + c$ (b) $\frac{e^x}{x+1} + c$
(c) $\frac{e^x}{x} + c$ (d) None of these

9. $\int x^{51}(\tan^{-1}x + \cot^{-1}x)dx =$ ***[MPPET-1991]***

(a) $\frac{x^{52}}{52}(\tan^{-1}x + \cot^{-1}x) + c$
(b) $\frac{\pi x^{52}}{52} + \frac{\pi}{2} + c$
(c) $\frac{\pi x^{52}}{52} + \frac{\pi}{2} + c$
(d) None of these

10. $\int \tan^4 x dx =$

(a) $\tan^3 x - \tan x + x + c$
(b) $\frac{1}{3}\tan^3 x - \tan x + x + c$
(c) $\frac{1}{3}\tan^3 x + \tan x + x + c$
(d) $\frac{1}{3}\tan^3 x + \tan x + 2x + c$

11. $\int \frac{\log x - 1}{(\log x)^2}dx$ is equal to ***[MPPET-1993]***

(a) $x\log x + c$ (b) $\frac{x}{\log x} + c$
(c) $-x\log x + c$ (d) None of these

12. $\int \text{cosec}^4 x\,dx =$ ***[RPET-2002]***

(a) $\cot x + \frac{\cot^3 x}{3} + C$ (b) $\tan x + \frac{\tan^3 x}{3} + C$
(c) $-\cot x - \frac{\cot^3 x}{3} + C$ (d) $-\tan x - \frac{\tan^3 x}{3} + C$

13. $\int (x+1)^2 e^x dx$ = *[EAMCET-2001]*

(a) $xe^x + C$
(b) $x^2 e^x + C$
(c) $(x+1)e^x + C$
(d) $(x^2+1)e^x + C$

14. The value of $\int \frac{dx}{(x-3)\sqrt{x+1}}$ is *[MPPET-2010]*

(a) $\frac{1}{2}\log\left|\frac{\sqrt{x+1}-2}{\sqrt{x+1}+2}\right| + c_1$

(b) $\frac{1}{2}\log\left|\frac{\sqrt{x+1}+2}{\sqrt{x+1}-2}\right| + c_1$

(c) $\frac{1}{2}\log\left|\frac{\sqrt{x+1}-2}{\sqrt{x+1}-2}\right| + c_1$

(d) None of these

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	**6.** ⓐ ⓑ ⓒ ⓓ	**11.** ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	**7.** ⓐ ⓑ ⓒ ⓓ	**12.** ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	**8.** ⓐ ⓑ ⓒ ⓓ	**13.** ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	**9.** ⓐ ⓑ ⓒ ⓓ	**14.** ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	**10.** ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

10. (b) $\int \tan^4 x dx = \int \tan^2 x(\sec^2 x - 1)dx$

$= \int \tan^2 x \sec^2 x dx - \int \tan^2 x dx$

$= \frac{\tan^3 x}{3} - \int (\sec^2 x - 1)dx$

$= \frac{\tan^3 x}{3} - \tan x + x + c$

11. (b) Step-1: $\int \frac{1}{\log x}dx - \int \frac{1}{(\log x)^2}dx =$

Cancellation method of Integral:
Integrating first section by parts

$$I = \left(\frac{1}{\log x}\right)\int dx - \int\left(\frac{d}{dx}\left(\frac{1}{\log x}\right)\int dx\right)dx - \int \frac{dx}{(\log x)^2}$$

$$I = \frac{x}{\log x} - \int \frac{-1}{(\log x)^2} \times \frac{1 \times x}{x} dx - \int \frac{dx}{(\log x)^2}$$

$$= \frac{x}{\log x} + \int \frac{1}{(\log x)^2}dx - \int \frac{dx}{(\log x)^2}$$

$$= \frac{x}{\log x} + c$$

Last two integral are cancelled.

NOTE

In first section only one function $\frac{1}{\log x}$ is given. Therefore, one is the second function.

12. (c) $I = \int \text{cosec}^4 x dx = \int \text{cosec}^2 x . \text{cosec}^2 x dx$

$= \int \text{cosec}^2 x(1 + \cot^2 x)dx$

$$= \int \text{cosec}^2 x(1+\cot^2 x)dx$$

$$= \int \text{cosec}^2 x dx + \int \cot^2 x \, \text{cosec}^2 x dx$$

$$= -\cot x - \frac{\cot^3 x}{3} + C$$

13. (d) $\int (x+1)^2 e^x dx = \int (x^2+2x+1)e^x dx$

$$= \int (x^2+2x)e^x dx + \int e^x dx$$

$$\left[\text{Type } \int [f(x)+f'(x)e^x dx = e^x f(x)\right]$$

$$= x^2 e^x + e^x + c = (x^2+1)\, e^x + c$$

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LECTURE



7 Integration of Different Forms

BASIC CONCEPTS

INTEGRATIONS UNDER DIFFERENT FORM

1. Integrals of the form: $\int \sin^m x \cos^n x\,dx$

Working Rule

(i) If the power of $\sin x$ is an odd positive integer, put $\cos x = t$.

(ii) If the power of $\cos x$ is an odd positive integer, put $\sin x = t$.

(iii) If the power of $\sin x$ and $\cos x$ are both odd positive integers, put $\sin x = t$ or $\cos x = t$. According as $m > n$ or $n > m$.

(iv) If the power of $\sin x$ and $\cos x$ are both even positive integers, use De'Moivre's theorem as follows: Let $\cos x + i \sin x = t$. Then $\cos x - i \sin x = t^{-1}$

Adding these, we get

$$t + \frac{1}{t} = 2\cos x \text{ and } t - \frac{1}{t} = 2i \sin x$$

By De' Moivre's theorem, we have

$$t^n + \frac{1}{t^n} = 2\cos nx \text{ and } t^n - \frac{1}{t^n} = 2i \sin^n x \quad \ldots\ldots\ldots (1)$$

$$\therefore \sin^m x \cos^n x = \frac{1}{(2i)^m}\cdot\frac{1}{2^n}\left(t+\frac{1}{t}\right)^n\left(t-\frac{1}{t}\right)^m$$

$$= \frac{1}{2^{m+n}}\cdot\frac{1}{i^m}\left(t+\frac{1}{t}\right)^n\left(t-\frac{1}{t}\right)^m$$

Now expand each of the factors on the R.H.S. using Binomial theorem. Then group the terms equidistant from the beginning and the end. Thus express all such pairs as the sines or cosines of multiple angles. Further integrate term by term.

(v) If the sum of powers of $\sin x$ and $\cos x$ is an even negative integer, put $\tan x = t$.

2. Integrals of the form: $\int \tan^m x \sec^n x\,dx$

Rule

(i) If m is even or odd integer and n is even positive integer then put $\tan x = t$.

(ii) If m is odd positive integer $n \notin$ even positive integer then put $\sec x = t$

(iii) If $m = 0$ and $n = 2r + 1 \ \forall r \in N$, then

$$\int \sec^{2r+1} x\,dx = \int \sec^{2r-1} x.\sec^2 x\,dx$$

then integrate by parts taking $\sec^2 x$ as second function.

NOTE

If $m \in$ even positive integer and $n \in$ odd positve integer, then integral is non-integrable.

3. Integrals of the form: $\int \cot^m x \operatorname{cosec}^n x\,dx$

Rule

(i) If m is even or odd integer and n is even positive integer then put $\cot x = t$.

(ii) If m is odd positive integer and $n \notin$ even positive integer then put $\operatorname{cosec} x = t$.

(iii) If $m = 0$ and $n = 2r + 1 \ \forall r \in N$, then $\int \text{cosec}^{2r+1} x dx = \int \text{cosec}^{2r-1} x.\text{cosec}^2 x dx$, then integrate by parts taking $\text{cosec}^2 x$ as second function.

NOTE

If $m \in$ even positive integer and $n \in$ odd positive integer then integral is non-integrable.

4. Integrals of the form

(i) $\int \frac{dx}{a^2 + b^2 \sin^2 x}$

$= \frac{1}{a\sqrt{a^2 + b^2}} \tan^{-1} \frac{\sqrt{a^2 + b^2}}{a} \tan x$

(ii) $\int \frac{dx}{(a^2 + b^2 \cos^2 x)}$

$= \frac{1}{a\sqrt{a^2 + b^2}} \tan^{-1} \frac{a}{\sqrt{a^2 + b^2}} \tan x$

(iii) $\int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{1}{ab} \tan^{-1} \frac{b}{a} \tan x$

(iv) $\int \frac{dx}{a^2 + b^2 \sin^2 x + c^2 \cos^2 x} =$

(v) $\int \frac{dx}{(a \sin x + b \cos x)^2} = -\frac{1}{a(a \tan x + b)}$

$= \frac{1}{b(a + b \cot x)}$

$= \int \frac{dx}{r^2(\cos^2(x - \theta))} = \int \frac{dx}{r^2(\sin^2(x + \alpha))}$

(vi) $\int \frac{\phi(\tan x) dx}{a \sin^2 x + b \sin x \cos x + c \cos^2 x}$

We shall always, in such cases, multiply both numerator and denominator by $\sec^2 x$ or $\text{cosec}^2 x$ then put $\tan x = t$ or $\cot x = t$ and solve.

(vii) $\int \frac{dx}{a \cos^2 x + b \sin x \cos x + c \sin^2 x}$

Working Rule

(i) Divide the numerator and denominator by $\sec^2 x$. $\cos^2 x$ or multiply the N and D.

(ii) In the denominator, replace $\sec^2 x$, if any, by $1 + \tan^2 x$.

(iii) Put $\tan x = t \Rightarrow \sec^2 x \, dx = dt$.

(iv) Integrate the resulting rational algebraic function of t.

(v) In the answer, put $t = \tan x$.

5. Integrals of the form: $\int f(x, (ax + b)^{\alpha/n}) dx$, where, α and n are integers.

Working Rule

Put $ax + b = t^n$.

6. Integrals of the form

Integral	Substitution
$\int \frac{x^m}{(a + bx)^p} dx$ m is a + ve integer	Put $a + bx = t$
$\int \frac{dx}{x^m (a + bx)^p}$ where, either (m and p are positive integers) or (m and p are fractions, but $m + p$ = integers > 1)	Put $a + bx = tx$
$\int x^m (a + bx^n)^p dx$, where m, n, p are rationals	
(i) p is a +ve integer	Applying Binomial theorem to $(a + bx^n)^p$.
(ii) p is a –ve integer	Put $x = t^k$ where k = common denominator of m and n.
(iii) $\frac{m+1}{n}$ is an integer	Put $a + bx^n = t^k$, where k = denominator of p.
(iv) $\frac{m+1}{n} + P$ is an integer	Put $a + bx^n = x^n t^k$ where k = denominator of fraction p.

7. $\int \frac{dx}{(x^2 + k)^n} = \frac{x}{k(2n-2)(x^2 + k)^{n-1}} + \frac{2n-3}{k(2n-2)} \int \frac{dx}{(x^2 + k)^{n-1}}$

8. Integrals of the form

(i) $\int \frac{dx}{(a + b \sin x)^2}$

Rule

$$\text{Put } t = \frac{(a\sin x + b)}{(a + b\sin x)} \quad \ldots\ldots\ldots (1)$$

$$\therefore \quad dt = \frac{(a^2 - b^2)\cos x}{(a + b\sin x)^2}dx$$

$$\text{or} \quad \frac{dt}{(a^2 - b^2)\cos x} = \frac{dx}{(a + b\sin x)^2}$$

Now substitute the value of cos x in L.H.S. from (1) in terms of t and then integrate both sides.

(ii) $\int \frac{dx}{(a + b\cos x)^2}$

Rule

$$\text{Put } t = \frac{(a\cos x + b)}{(a + b\cos x)} \quad \ldots\ldots\ldots (1)$$

$$\therefore \quad dt = \frac{(b^2 - a^2)\sin x}{(a + b\cos x)^2}dx$$

$$\text{or} \quad \frac{dt}{(b^2 - a^2)\sin x} = \frac{dx}{(a + b\cos x)^2}$$

Now substitute the value of sin x in L.H.S. from (1) in terms of t and then integrate both sides.

(iii) $\int \sqrt{(\tan^2 x \pm a)}dx$ and

(iv) $\int \sqrt{(\cot^2 x \pm a)}dx$

STANDARD INTEGRALS 5

(i) $\int \sin hx dx = \cos hx + c$

(ii) $\int \cos hx dx = \sin hx + c$

(iii) $\int \sec h^2 x dx = \tan hx + c$

(iv) $\int \text{cosec } h^2 x dx = -\cot hx + c$

(v) $\int \sec hx \tan hx dx = -\sec hx + c$

(vi) $\int \text{cosec } hx \cot hx\, dx = -\text{cosec } hx + c$

INTEGRATION UNDER DIFFERENT FORM 2

9. Integrals of the Form

(a) $\int \frac{dx}{ax^2 + bx + c}$

(b) $\int \frac{dx}{\sqrt{ax^2 + bx + c}}$

(c) $\int \sqrt{ax^2 + bx + c}\, dx$

Working Rule

(i) Make the coefficient of x^2 unity by taking the coefficient of x^2 outside the quadratic.

(ii) Complete the square in the terms involving x, i.e., write $ax^2 + bx + c$ in the form $a[x \pm \alpha)^2 \pm \beta^2]$.

(iii) The integrand is converted to one of the nine special integrals.

(iv) Integrate the function.

10. Integrals of the form

(a) $\int \frac{px + q}{ax^2 + bx + c}dx$

Working Rule

Put $px + q = \lambda\,(2ax + b) + \mu$ or

$px + q = \lambda$ (derivative of quadratic) + μ.

Comparing the coefficient of x and constant term on both sides, we get

$$p = 2a\lambda \text{ and } q = b\lambda + \mu \Rightarrow \lambda = \frac{p}{2a}$$

$$\text{and } \mu = \left(q - \frac{bp}{2a}\right)$$

Then the integral becomes

$$= \frac{p}{2a}\int \frac{2ax + b}{ax^2 + bx + c}dx + \left(q - \frac{bp}{2a}\right)\int \frac{dx}{ax^2 + bx + c}$$

$$= \frac{p}{2a}\log|ax^2 + bx + c| + \left(q - \frac{bp}{2a}\right)\int \frac{dx}{ax^2 + bx + c}$$

10.1 $\int \frac{px + q}{\sqrt{ax^2 + bx + c}}dx$

Working Rule

In this case the integral becomes

$$\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx = \frac{p}{2a}\int \frac{2ax+b}{\sqrt{ax^2+bx+c}}dx +$$

$$\left(q-\frac{bp}{2a}\right)\int \frac{dx}{\sqrt{ax^2+bx+c}}$$

$$= \frac{p}{a}\sqrt{ax^2+bx+c}\left(q-\frac{bp}{2a}\right)\int \frac{dx}{\sqrt{ax^2+bx+c}}$$

10.2 $\int (px+q)\sqrt{ax^2+bx+c}\,dx$

Working Rule

The integral in this case is converted to

$$\int (px+q)\sqrt{ax^2+bx+c}\,dx = \frac{p}{2a}$$

$$\int (2ax+b)\sqrt{ax^2+bx+c}\,dx$$

$$+\left(q-\frac{bp}{2a}\right)\int \sqrt{ax^2+bx+c}\,dx$$

$$= \frac{p}{3a}(ax^2+bx+c)^{3/2} + \left(q-\frac{bp}{2a}\right)$$

$$\int \sqrt{ax^2+bx+c}\,dx$$

10.3. Integrals of the Form $\int \frac{P(x)}{\sqrt{ax^2+bx+c}}dx$,

where, $P(x)$ is a polynomial in x of degree n.

Working Rule

$$\int \frac{P(x)}{\sqrt{ax^2+bx+c}}dx$$

$$= (a_0 + a_1x + a_2x^2 + .. + a_{n-1}x^{n-1})$$

$$\sqrt{ax^2+bx+c} + k\int \frac{dx}{\sqrt{ax^2+bx+c}}$$

where $k, a_0, a_1, ...a_{n-1}$ are constants to be determined by differentiating the above relation and equating the coefficients of various powers of x on both sides.

10.4. Integration of $\frac{px^2+qx+r}{\sqrt{(ax^2+bx+c)}}$

Here write, $px^2 + qx + r = A(ax^2 + bx + c) + B(2ax + b) + C$,

$$= A\int \sqrt{(ax^2+bx+c)}dx + B\int \frac{(2ax+b)dx}{(ax^2+bx+c)}$$

$$+ C\int \frac{dx}{\sqrt{(ax^2+bx+c)}}$$

11. Integral of the Form

(a) $\int \frac{dx}{a+b\cos x}$

(b) $\int \frac{dx}{a+b\sin x}$

(c) $\int \frac{dx}{a+b\cos x+c\sin x}$

Working Rule

(i) Put $\cos x = \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$ and $\sin x = \frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}$ so that the given integrand becomes a function of $\tan\frac{x}{2}$.

(ii) Put $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

(iii) Integrate the resulting rational algebraic function of t.

(iv) In the answer, put $t = \tan\frac{x}{2}$.

12. Integrals of the Form $\int \frac{a\cos x+b\sin x}{c\cos x+d\sin x}dx$

Working Rule

(i) Put Numerator: λ (denominator) + μ (derivative of denominator)
$a \cos x + b \sin x = \lambda (c \cos x + d \sin x) + \mu(-c \sin x + d \cos x)$

(ii) Equate coefficients of sin x and cos x on both sides and find the values of λ and μ.

(iii) Split the given integral into two integrals and evaluate each integral separately, i.e.,

$$\int \frac{a\cos x+b\sin x}{c\cos x+d\sin x}dx = \lambda$$

$$\int 1dx + \mu\int \frac{-c\sin x+d\cos x}{c\cos x+d\sin x}dx$$

$$= \lambda x + \mu \log |a \cos x + b \sin x|.$$

(iv) Substitute the values of λ and μ found in step 2.

13. Integrals of the Form

$$\int \frac{a+b\cos x+c\sin x}{e+f\cos x+g\sin x}dx$$

Working Rule

(i) Put Numerator = l (denominator) + m (derivative of denominator) + n
$a + b\cos x + c\sin x = l(e + f\cos x + g\sin x) + m(-f\sin x + g\cos x) + n$

(ii) Equate coefficients of $\sin x$, $\cos x$ and constant term on both sides and find the values of l, m, n.

(iii) Split the given integral into three integrals and evaluate each integral separately, i.e.,

$$\int \frac{a+b\cos x+c\sin x}{e+f\cos x+g\sin x}dx$$

$$= l\int 1dx + m\int \frac{-f\sin x+g\cos x}{e+f\cos x+g\sin x}dx +$$

$$n\int \frac{dx}{e+f\cos x+g\sin x}$$

$$= lx + m\log|e+f\cos x+g\sin x| + n\int \frac{dx}{e+f\cos x+g\sin x}$$

(iv) Substitute the values of l, m, n found in step (ii).

14. Integrals of the Form: $\int \frac{dx}{P\sqrt{Q}}$, where, P, Q are linear or quadratic functions of x.

	Integral	Substitution
(i)	$\int \frac{1}{(ax+b)\sqrt{cx+d}}dx$	$cx+d=t^2$ or $Q=t^2$
(ii)	$\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$	$px+q=t^2$ or $Q=t^2$
(iii)	$\int \frac{dx}{(px+q)\sqrt{ax^2+bx+c}}$	$px+q=\frac{1}{t}$ or $P=\frac{1}{t}$
(iv)	$\int \frac{dx}{(ax^2+b)\sqrt{cx^2+d}}$	$x=\frac{1}{t}$
(v)	$\int \frac{xdx}{(ax^2+b)\sqrt{cx^2+d}}$	put $cx^2+d=t^2$

(vi) $$\int \frac{dx}{a+b\sin x} = \frac{2}{\sqrt{a^2-b^2}}\tan^{-1}\left(\frac{a\tan\frac{x}{2}+b}{\sqrt{a^2-b^2}}\right), a>b$$

$$= \frac{1}{\sqrt{b^2-a^2}}\log\left(\frac{a\tan\frac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{x}{2}+b+\sqrt{b^2-a^2}}\right), b>a$$

15. $$\int \frac{dx}{a+b\cos x} = \frac{2}{\sqrt{a^2-b^2}}\tan^{-1}\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}, a>b$$

$$= \frac{-1}{\sqrt{b^2-a^2}}\log\frac{\sqrt{b-a}\tan\frac{x}{2}-\sqrt{b+a}}{\sqrt{b-a}\tan\frac{x}{2}+\sqrt{b+a}}, b>a$$

$$= \frac{1}{\sqrt{a^2-b^2}}\cos^{-1}\left(\frac{b+a\cos x}{a+b\cos x}\right), a>b$$

16. $$\int \frac{dx}{a+b\tan x} = \int \frac{\cos x dx}{a\cos x+b\sin x}$$

$$= \frac{a}{a^2+b^2}x + \frac{b}{a^2+b^2}\log(a\cos x+b\sin x)$$

17. $$\int \frac{dx}{a+b\cot x} = \int \frac{\sin x dx}{a\sin x+b\cos x} =$$

$$= \frac{a}{a^2+b^2}x - \frac{b}{a^2+b^2}\log(a\sin x+b\cos x)$$

18. $$\int \frac{c\sin x+d\cos x}{a\sin x+b\cos x}dx$$

$$= \frac{ac+bd}{a^2+b^2}x + \frac{ad-bc}{a^2+b^2}\log[a\sin x+b\cos x]$$

19. $$\int \frac{dx}{a\sin x+b\cos x} =$$

$$\frac{1}{\sqrt{a^2+b^2}}\log\left[\tan\frac{1}{2}\left(x+\tan^{-1}\frac{b}{a}\right)\right]$$

$$\frac{1}{\sqrt{a^2+b^2}}\log\left[\operatorname{cosec}\left(x+\tan^{-1}\frac{b}{a}\right)-\cot\left(x+\tan^{-1}\frac{b}{a}\right)\right]$$

OR

$$\frac{1}{\sqrt{a^2+b^2}}\log\left[\begin{matrix}\sec\left(x-\tan^{-1}\left(\frac{a}{b}\right)\right)+\\ \tan\left(x-\tan^{-1}\left(\frac{a}{b}\right)\right)\end{matrix}\right]$$

$$\frac{1}{\sqrt{a^2+b^2}}\log\left[\tan\left(\frac{\pi}{4}+\frac{1}{2}\left(x-\tan^{-1}\frac{a}{b}\right)\right)\right]$$

Rule

In this case change $\tan^2 x$ into $\sec^2 x - 1$ and $\cot^2 x$ into $\operatorname{cosec}^2 x - 1$ and then proceed as in.

$$\int\sqrt{\sec^2 x\pm\alpha}\ dx \text{ or } \int\sqrt{\operatorname{cosec}^2 x\pm\alpha}\ dx$$

(iv) $$\int\left(\frac{ae^x+be^{-x}}{pe^x+qe^{-x}}\right)dx$$

Rule

Express numerator as l (Denominator) + $m\frac{d}{dx}$ (Denominator) Find l and m by comparing the coefficients of e^x and e^{-x} and split the integral into sum of two integrals as

$$l\int dx+m\int\frac{\text{d.c.(Denominator)}}{\text{(Denominator)}}dx$$
$$= lx + m\,ln\,|\,\text{Denominator}\,| + c$$

20. Some Important Questions with Solution:

(i) Evaluate $\int\sqrt{(\sec^2 x-2)}dx$

Solution

$$\text{Let I} = \int\sqrt{(\sec^2 x-2)}dx = \int\frac{(\sec^2 x-2)}{\sqrt{(\sec^2 x-2)}}dx$$

$$= \int\frac{\sec^2 x dx}{\sqrt{(\sec^2 x-2)}} - 2\int\frac{dx}{\sqrt{(\sec^2 x-2)}}$$

$$= \int\frac{\sec^2 x dx}{\sqrt{(\tan^2 x-1)}} - 2\int\frac{\cos x dx}{\sqrt{1-2(1-\sin^2 x)}}$$

$$= \int\frac{\sec^2 x dx}{\sqrt{(\tan^2 x-1)}} - 2\int\frac{\cos x dx}{\sqrt{(\sqrt{2}\sin x)^2-1}}$$

In first integral on R.H.S. put $\tan x = t$ and in second integral put $\sqrt{2}\ \sin x = u$

i.e., $\sec^2 dx = dt$ and $\cos x\, dx = \frac{du}{\sqrt{2}}$

$$\therefore I = \int\frac{dt}{\sqrt{(t^2-1)}} - \frac{2}{\sqrt{2}}\int\frac{du}{\sqrt{(u^2-1)}}$$

$$= ln\,|\,t+\sqrt{(t^2-1)}\,| - \sqrt{2}\ ln\,|\,u+\sqrt{(u^2-1)}\,| + c$$

$$= ln\,|\,\tan x+\sqrt{(\tan^2 x-1)}\,| - \sqrt{2}\ ln\,|\,\sqrt{2}\ \sin x+\sqrt{(2\sin^2 x-1)}\,| + c$$

(ii) Evaluate $\int\sqrt{(\operatorname{cosec}^2 x\pm a)}dx$

Solution

Let

$$I = \int\sqrt{(\operatorname{cosec}^2 x\pm a)}dx = \int\frac{(\operatorname{cosec}^2 x\pm a)}{\sqrt{(\operatorname{cosec}^2 x\pm a)}}dx$$

$$= \int\frac{\operatorname{cosec}^2 x dx}{\sqrt{(\operatorname{cosec}^2 x\pm a)}} \pm a\int\frac{dx}{\sqrt{(\operatorname{cosec}^2 x\pm a)}}$$

$$= \int\frac{\operatorname{cosec}^2 x dx}{\sqrt{(1\pm a)+\cot^2 x}} \pm a\int\frac{\sin x dx}{\sqrt{1\pm a(1-\cos^2 x)}}$$

In first integral on R.H.S. put $\cot x = t$ and in second integral put $\cos x = u$.

(iii) Evaluate $\int\sqrt{(\operatorname{cosec}^2 x+3)}dx$

Solution

$$\text{Let } I = \int\sqrt{(\operatorname{cosec}^2 x+3)}dx = \int\frac{(\operatorname{cosec}^2 x+3)dx}{\sqrt{(\operatorname{cosec}^2 x+3)}}$$

$$= \int\frac{\operatorname{cosec}^2 x}{\sqrt{(\operatorname{cosec}^2 x+3)}}dx + 3\int\frac{dx}{\sqrt{(\operatorname{cosec}^2 x+3)}}$$

$$= \int\frac{\operatorname{cosec}^2 x dx}{\sqrt{(4+\cot^2 x)}} + 3\int\frac{\sin x dx}{\sqrt{1+3(1-\cos^2 x)}}$$

$$= \int\frac{\operatorname{cosec}^2 x dx}{\sqrt{(4+\cot^2 x)}} + 3\int\frac{\sin x dx}{\sqrt{4-(\sqrt{3}\cos x)^2}}$$

In first integral on R.H.S. put $\cot x = t$ and in second integral put $\sqrt{3}\ \cos x = u$

i.e., $\operatorname{cosec}^2 x\,dx = -dt$, $\sin x\,dx = -\dfrac{du}{\sqrt{3}}$

$$\therefore I = -\int \frac{dt}{\sqrt{(2^2+t^2)}} - \frac{3}{\sqrt{3}}\int \frac{du}{\sqrt{(2^2+u^2)}}$$

$$= -\ln(t+\sqrt{(4+t^2)}) - \sqrt{3}\sin^{-1}(u/2) + c$$

$$= \ln(\cot x + \sqrt{(4+\cot^2 x)}) - \sqrt{3}\sin^{-1}\left(\frac{\sqrt{3}\cos x}{2}\right) + c$$

Example

$$\int \frac{dx}{(\sin x+4)(\sin x-1)}$$

$$= \frac{1}{5}\int \frac{(\sin x+4)-(\sin x-1)dx}{(\sin x+4)(\sin x-1)}$$

$$= \frac{1}{5}\int \frac{dx}{\sin x-1} - \frac{1}{5}\int \frac{dx}{\sin x+4}$$

$$= \frac{2}{5}\left(\frac{1}{\tan\frac{x}{2}-1}\right) - \frac{2}{5\sqrt{15}}\tan^{-1}\left(\frac{4\tan\frac{x}{2}+1}{\sqrt{15}}\right) + C$$

NOTE

$$\frac{2}{5}\int \frac{-1\,dn}{1-\sin x} = \int \frac{dn}{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2} = \int \frac{\sec^2\frac{x}{2}dn}{\left(1-\tan\frac{n}{2}\right)^2}$$

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)) FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate $\int \frac{\tan^5\sqrt{x}\sec^2\sqrt{x}}{\sqrt{x}}dx$

Solution

As a first step, we put $t = \sqrt{x}$, so that $dt = \frac{1}{2\sqrt{x}}dx$.

Since $\frac{1}{\sqrt{x}}dx$ occurs as part of the given integral, we write

$$2dt = \frac{1}{\sqrt{x}}dx.$$

Substituting t and $2\,dt$ in the given integral, we get

$$\int \frac{\tan^5\sqrt{x}\sec^2\sqrt{x}}{\sqrt{x}}dx = \int(\tan^5 t\sec^2 t)(2)dt$$

$$= 2\int \tan^5 t\sec^2 t\,dt$$

Now since $\tan^5(t) = (\tan t)^5$, we let $u = \tan t$. Then

$du = \sec^2 t\,dt$ so that

$$2\int \tan^5 t\sec^2 t\,dt = 2\int u^5 du = 2\frac{u^6}{6} + c$$

$$= \frac{1}{3}u^6 + c = \frac{1}{3}\tan^6 t + c$$

2. Evaluate $\int \sec^3 x\tan x dx$

Solution

$$\int \sec^3 x\tan x dx$$

| Put $\sec x = t \Rightarrow \sec x\tan x\,dx = dt$.

$$= \int t^2 dt = \frac{t^2}{3} + c = \frac{\sec^3 x}{3} + c$$

3. Evaluate $\int \frac{1}{\sin x(2+\cos x)}dx$

$$\int \frac{1}{\sin x(2+\cos x)}dx = \int \frac{\sin x}{\sin^2 x(2+\cos x)}dx$$

[Multiply and divide by $\sin x$]

$$= \int \frac{\sin x}{(1-\cos^2 x)(2+\cos x)}dx$$

| Put $\cos x = t \Rightarrow -\sin dx = dt$

$$= -\int \frac{1}{(1-t)(1+t)(2+t)}dt \quad \text{..... (i)}$$

Considering

$$\frac{1}{(1-t)(1+t)(2+t)} = \frac{A}{1-t} + \frac{B}{1+t} + \frac{C}{2+t} \quad \text{... (ii)}$$

$1 = A(1+t)(2+t) + B(1-t)(2+t) + C(1-t)(1+t)$

$1 = A(t^2 + 3t + 2) + B(2 - t - t^2) + C(1 - t^2)$

Comparing the coefficients, we get

$A - B - C = 0;\ 3A - B = 0;\ 2A + 2B + C = 1$

Solving, we get

$$A = \frac{1}{6}, B = \frac{1}{2}, C = -\frac{1}{3}$$

Substituting in (ii) and integrating, we get

$$\int \frac{1}{(1-t)(1+t)(2+t)}dt = \frac{1}{6}\int \frac{1}{1-t}dt + \frac{1}{2}\int \frac{1}{1+t}dt - \frac{1}{3}\int \frac{1}{2+t}dt$$

$$= -\frac{1}{6}\log|1-t| + \frac{1}{2}\log|1+t| - \frac{1}{3}\log|2+t| + c$$

Substituting in (i), we get

$$\int \frac{1}{\sin x(2+\cos x)}dx$$

$$= \frac{1}{6}\log|1-\cos x| - \frac{1}{2}\log|1+\cos x| + \frac{1}{3}\log|2+\cos x| + c$$

4. $\int \frac{dx}{\sqrt{(\sin^3 x \cos^5 x)}}$

Solution

$\frac{3}{2} + \frac{5}{2} = 4$ = even. Divide above and below by $\cos^4 x$

$$\therefore \quad I = \int \frac{\sec^2 x . \sec^2 x}{\sqrt{(\tan^3 x)}}dx = \int \frac{(1+t^2)}{t\sqrt{t}}dt$$

$$= \frac{2}{3}\frac{(t^2+3)}{\sqrt{t}}, \text{ where } t = \tan x$$

5. Evaluate $\int \frac{\sqrt{\cos 2x}}{\sin x}dx$ ***[IIT-1987]***

Solution

Let $I = \int \frac{\sqrt{\cos 2x}}{\sin x}dx$

$= \int \frac{\sqrt{\cos^2 x - \sin^2 x}}{\sin x}dx$, divide num. and denom. by $\cos x$

$= \int \frac{\sqrt{1-\tan^2 x}}{\tan x}dx$, substitute

$\sqrt{1-\tan^2 x} = t$

$\Rightarrow \quad \tan^2 x = 1 - t^2$

$\Rightarrow \quad 2\tan x \sec^2 x\, dx = -2t\, dt$

$$= \int \frac{t(-t)dt}{\sec^2 x \tan^2 x}$$

$$= -\int \frac{t^2}{(1+1-t^2)(1-t^2)}dt$$

$$= -\int \left(\frac{-1}{2-t^2} + \frac{1}{1-t^2}\right)dt$$

$$= \int \frac{1}{2-t^2}dt - \int \frac{1}{1-t^2}dt$$

$$= \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}+t}{\sqrt{2}-t}\right| - \frac{1}{2}\log\left|\frac{1-t}{1+t}\right| + C$$

$$= \frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}+\sqrt{1-\tan^2 x}}{\sqrt{2}-\sqrt{1-\tan^2 x}}\right| - \frac{1}{2}\log\left|\frac{1-\sqrt{1-\tan^2 x}}{1+\sqrt{1-\tan^2 x}}\right| + C$$

6. Evaluate $\int (\tan x + \sec x)^4 \sec^2 x dx$

Solution

Putting $z = \sec x + \tan x$

$\therefore \quad dz = \sec x\,(\sec x + \tan x)\,dx$

Also $\frac{1}{z} = \sec x - \tan x$ and $z + \frac{1}{z} = 2\sec x$

$$\therefore \quad \int (\tan x + \sec x)^4 (\sec x)(\sec x)dx$$

$$I = \frac{1}{2}\int z^3\left(z + \frac{1}{z}\right)dz = \frac{1}{2}\int (z^4 + z^2)dz$$

$$= \frac{1}{2}\left[\frac{z^5}{5} + \frac{z^3}{3}\right] + C$$

$$= \frac{1}{2}\left[\frac{(\tan x + \sec x)^5}{5} + \frac{(\tan x + \sec x)^3}{3}\right] + C$$

7. Evaluate $\int \frac{2\sin 2\phi - \cos\phi}{6 - \cos^2\phi - 4\sin\phi} d\phi$

Solution

We have $I = \int \frac{2\sin 2\phi - \cos\phi}{6 - \cos^2\phi - 4\sin\phi} d\phi$

$= \int \frac{2.2\sin\phi\cos\phi - \cos\phi}{6 - (1 - \sin^2\phi) - 4\sin\phi} d\phi$

$= \int \frac{(4\sin\phi - 1)\cos\phi}{\sin^2\phi - 4\sin\phi + 5} d\phi$

Putting $\sin\phi = t$ and $\cos\phi\, d\phi = dt$

$I = \int \frac{4t - 1}{t^2 - 4t + 5} dt$

Now let $(4t - 1) = \lambda \frac{d}{dt}(t^2 - 4t + 5) + \mu(4t - 1)$

$= \lambda(2t - 4) + \mu$

comparing coefficients of like powers of t, we get

$2\lambda = 4, -4\lambda + \mu = -1$

$\Rightarrow \quad \lambda = 2, \mu = 7$

$\therefore \quad I = \int \frac{\lambda(2t - 4) + \mu}{t^2 - 4t + 5} dt$

$\Rightarrow \quad I = 2\int \frac{(2t - 4)}{t^2 - 4t + 5} + 7\int \frac{1}{t^2 - 4t + 5} dt$

$I = 2\log|t^2 - 4t + 5| + 7 \int \frac{1}{(t - 2)^2 + (1)^2} dt$

$= 2\log|t^2 - 4t + 5| + 7\tan^{-1}(t - 2) + c$

$= 2\log|\sin^2\phi - 4\sin\phi + 5| + 7\tan^{-1}(\sin\phi - 2) + c$

8. Evaluate $\int \frac{dx}{(x - a)\sqrt{(x - a)(b - x)}}$

Solution

we put $x - a = 1/t$

So that $dx = (-1/t^2)dt$

Therefore, $\int \frac{dx}{(x - a)\sqrt{(x - a)(b - x)}}$

$= \int \frac{(-1/t^2)dt}{(1/t)\sqrt{(1/t)[(b - a) - 1/t]}}$

$= -\int \frac{dt}{\sqrt{(b - a)t - 1}}$

$= \int [(b - a)t - 1]^{-1/2} dt$

$= \frac{(-1)[(b - a)t - 1]^{1/2}}{(b - a)(1/2)} + c$

$= \frac{2}{a - b}\sqrt{(b - a)t - 1} + c$

$= \frac{2}{a - b}\sqrt{\frac{b - x}{x - a}} + c$

9. Evaluate $\int \frac{25 - x - x^2}{\sqrt{(5 + x)(5 - x)}} dx$

Solution

We have $25 - x - x^2 = |(5 + x)(5 - x) - x = (25 - x^2) - x$

Thus $\int \frac{25 - x - x^2}{\sqrt{(5 + x)(5 - x)}} dx$

$= \int \sqrt{25 - x^2} dx - \int \frac{x}{\sqrt{25 - x^2}} dx$

But $\int \sqrt{25 - x^2} dx = \frac{1}{2}x\sqrt{25 - x^2} + \frac{25}{2}\sin^{-1}\left(\frac{x}{5}\right)$

To evaluate $\int \frac{x}{\sqrt{25 - x^2}} dx$

we put $25 - x^2 = t^2$, so that $-2x\, dx = 2t\, dt$ or $x\, dx = -t\, dt$.

Therefore, $\int \frac{x}{\sqrt{25 - x^2}} dx = \int \frac{-t\, dt}{t}$

$= -\int dt = -t = -\sqrt{25 - x^2}$

Hence

$I = \frac{1}{2}x\sqrt{25 - x^2} + \frac{25}{2}\sin^{-1}\left(\frac{x}{5}\right) - \left(-\sqrt{25 - x^2}\right) + c$

$= \frac{1}{2}(x + 2)\sqrt{25 - x^2} + \frac{25}{2}\sin^{-1}\left(\frac{x}{5}\right) + c$

10. If $0 < a < 1$, then find $\int \frac{dx}{1-2a\cos x + a^2}$

[MNR-1997]

Solution

$$I = \int \frac{dx}{(1+a^2) - 2a\left(\frac{1-\tan^{2x/2}}{1+\tan^{2x/2}}\right)}$$

put $\tan(x/2) = t$.

$$\therefore \quad I = \int \frac{2dt}{(1+a^2)(1+t^2) - 2a(1-t^2)}$$

$$= \int \frac{2dt}{(1-a^2) + (1+a)^2 t^2}$$

$$= \frac{2}{(1+a)^2} \int \frac{dt}{\left(\frac{1-a}{1+a}\right)^2 + t^2}$$

$$= \frac{2}{(1+a)^2} \cdot \frac{1}{\frac{(1-a)}{1+a}} \tan^{-1}\left[t\frac{(1+a)}{1-a}\right]$$

$$= \frac{2}{1-a^2} \tan^{-1} t\frac{(1+a)}{(1-a)}$$

11. Evaluate $\int \frac{1}{x(x^7+1)} dx$ ***[AI-2000C]***

$$\int \frac{1}{x(x^7+1)} dx = \int \frac{1}{x^8\left(1+\frac{1}{x^7}\right)} dx,$$

$$\left|\text{Put} \quad 1+\frac{1}{x^7} = t \Rightarrow -\frac{7}{x^8} dx = dt\right.$$

$$= -\frac{1}{7}\int \frac{1}{t} dt = -\frac{1}{7}\log|t| + c = -\frac{1}{7}\log\left|1+\frac{1}{x^7}\right| + c$$

12. Evaluate $\int \frac{11x+8}{x^2(3x+8)} dx$

Solution

$$\int \frac{11x+8}{x^2(3x+8)} dx$$

Considering $\frac{11x+8}{x^2(3x+8)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{3x+8}$ ...(i)

$$\Rightarrow \quad 11x + 8 = Ax(3x+8) + B(3x+8) + Cx^2$$
$$= x^2(3A + C) + x(8A + 3B) + 8A$$

Comparing the coefficients, we get

$3A + C = 0$; $8A + 3B = 11$; $8A = 8$

Solving for A, B and C, we get

$A = 1$, $B = 1$, $C = -3$

Substituting in (i) and integrating, we get

$$\int \frac{11x+8}{x^2(3x+8)} dx = \int \frac{1}{x} dx + \int \frac{1}{x^2} dx - 3\int \frac{1}{3x+8} dx$$

$$= \log|x| - \frac{1}{x} - \log|3x+8| + c$$

13. Evaluate $\int \frac{5}{(x+1)(x^2+4)} dx$

Solution

Consider $\int \frac{5}{(x+1)(x^2+4)} dx$

Considering $\frac{5}{(x+1)(x^2+4)} = \frac{A}{x+1} = \frac{Bx+C}{x^2+4}$ ...(i)

$$\Rightarrow \quad 5 = A(x^2+4) + (Bx + C)(x+1)$$
$$\Rightarrow \quad 5 = x^2(A + B) + x(B + C) + (4A + C)$$

Comparing the coefficients, we get

$A + B = 0$; $B + C = 0$; $4A + C = 5$

Solving for A, B and C, we get

Substituting in (i) and integrating, we get

$$\int \frac{5}{(x+1)(x^2+4)} dx = \int \frac{1}{x+1} dx + \int \frac{-x+1}{x^2+4} dx$$

$$= \log|x+1| - \int \frac{x}{x^2+4} dx + \int \frac{1}{x^2+4} dx$$

$$= \log|x+1| - \frac{1}{2}\log|x^2+4| + \frac{1}{2}\tan^{-1}\frac{x}{2} + c$$

14. Evaluate $\int \frac{5x+3}{\sqrt{x^2+4x+10}} dx$

[CBSE-2010]

Solution

Step 1: $5x + 3 = \lambda \frac{d}{dx}(x^2 + 4x + 10) + \mu$

$$5x + 3 = \lambda(2x + 4) + \mu$$

$$5x + 3 = 2\lambda x + \mu + 4\lambda$$

Equating the coefficients of like powers of x we get

$$2\lambda = 5, \mu + 4\lambda = 3$$

$$\lambda = \frac{5}{2}, \mu + 10 = 3 \Rightarrow \mu = -7$$

Step 2:

$$\int \frac{5x+3}{\sqrt{x^2+4x+10}}dx = \frac{5}{2}\int \frac{2x+4}{\sqrt{x^2+4x+10}} - 7\int \frac{dx}{\sqrt{(x+2)^2+6}}$$

$$\frac{5}{2}\times 2\sqrt{x^2+4x+10} - 7\log\left[x+2+\sqrt{x^2+4x+10}\right]$$

NOTE

$$\int \frac{f'(t)dt}{\sqrt{f(t)}} = 2\sqrt{f(t)}, \int \frac{dt}{\sqrt{t^2+a^2}} = \log\left[t+\sqrt{t^2+a^2}\right]$$

UNSOLVED SUBJECTIVE PROBLEMS (XI BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Evaluate $\int \sin^3 x \cos^2 x dx$.

2. Integrate $\frac{1}{1+\sin^2 x}$ with respect to x.

3. Evaluate $\int \sqrt{\frac{a-x}{a+x}}\,dx$

[CBSE-92]

4. Evaluate $\int \frac{dx}{a\cos x + b\sin x}$

[MP-2001]

5. Evaluate $\int \frac{dx}{a+b\sin^2 x}$.

[MP-98]

6. Evaluate $I = \int \sin^3 x \cos^4 x dx$

[MP-2003]

7. Evaluate $\int \frac{\tan 0 + \tan^3 0}{1+\tan^3 \theta} d0$

8. Evaluate $\int \frac{\sec^2 x}{(1+\tan x)(2+\tan x)}dx$

9. Evaluate $\int \frac{dx}{1+\cot x}$

10. Evaluate $\int e^x \sqrt{e^{2x}+1}dx$

11. Evaluate $\int \frac{\sin x + \cos x}{3\sin x + 2\cos x}dx$

12. Evaluate $\int \frac{dx}{1+3\cos^2 x}$

13. Evaluate $\int \frac{3x+1}{2x^2-2x+3}dx$

[MP-96; 2008]

14. Evaluate $\int (2x+1)\sqrt{x^2+2x+5}dx$

[MP-2000, 2004(B)]

15. Evaluate $\int \frac{dx}{5+4\cos x}$

[MP-98, 99, 2000, 2001; CBSE-2003]

16. Evaluate $\int \frac{dx}{5-4\sin x}$

[MP-2001, 2008; PSB-90(C)]

17. Evaluate $\int x\sqrt{1+x-x^2}dx$

[CBSE-90]

18. Evaluate $\int \frac{1}{(x-1)\sqrt{x^2+4}}dx$

19. Evaluate $\int \frac{1}{1+\cot x}\,dx$.

[CBSE-92; PSB-93, 2002; HPSB-94, 2002(C)]

20. Evaluate $\int \frac{x+2}{2x^2+6x+5}dx$

[CBSE-2007]

21. Evaluate $\int \frac{1}{4+3\cos x}dx$

[CBSE-Sample paper-IV]

22. Evaluate $\int \frac{\sin x\cos x}{a\cos^2 x+b\sin^2 x}dx$

23. Evaluate $\int \frac{x+2}{\sqrt{(x-2)(x-3)}}dx$

[CBSE-2010]

ANSWERS

1. $\frac{\cos^5 x}{5}-\frac{\cos^3 x}{3}$

2. $\frac{1}{\sqrt{2}}\tan^{-1}(\sqrt{2}.\tan x)$

3. $a\sin^{-1}\left(\frac{x}{a}\right)+\sqrt{a^2-x^2}+c$

4. $\frac{1}{\sqrt{a^2+b^2}}\log\tan\frac{x+\tan^{-1}\frac{b}{a}}{2}$

5. $\frac{1}{\sqrt{a(a+b)}}\tan^{-1}\left(\tan x\frac{\sqrt{a+b}}{\sqrt{a}}\right)$

6. $\frac{\cos^7 x}{7}-\frac{\cos^5 x}{5}+c$

7. $\frac{1}{6}\log|\tan^2\theta-\tan\theta+1|-\frac{1}{3}\log|\tan\theta+1|+\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2\tan\theta-1}{\sqrt{3}}\right)+c$

8. $\log\left(\frac{\tan x+1}{\tan x+2}\right)$

9. $\frac{x}{2}-\frac{1}{4}\log\cos 2x-\frac{1}{4}\log(\sec 2x+\tan 2x)$

10. $\frac{1}{2}e^x\sqrt{e^{2x}+1}+\frac{1}{2}\log\left(e^x+\sqrt{e^{2x}+1}\right)$

11. $\frac{1}{13}\log(3\sin x+2\cos x)+\frac{5}{13}x$

12. $\frac{1}{2}\tan^{-1}\left(\frac{\tan x}{2}\right)$

13. $\frac{3}{4}\log(2x^2-2x+3)+\frac{\sqrt{5}}{2}\tan^{-1}\left(\frac{2x-1}{\sqrt{5}}\right)$

14. $\frac{2}{3}(x^2+2x+5)^{\frac{3}{2}}-\frac{1}{2}(x+1)\sqrt{x^2+2x+5}-2\log\left[(x+1)+\sqrt{x^2+2x+5}\right]$

15. $\frac{2}{3}\tan^{-1}\left(\frac{1}{3}\tan\frac{x}{2}\right)$

16. $\frac{2}{3}\times\tan^{-1}\left(\frac{5\tan\frac{x}{2}-4}{3}\right)$

17. $\frac{-1}{3}(1+x-x^2)^{3/2}+\frac{1}{2}\left[\frac{1}{2}\left(x-\frac{1}{2}\right).\sqrt{1+x-x^2}+\frac{5}{8}\sin^{-1}\left(\frac{2x-1}{\sqrt{5}}\right)\right]+c$

18. $\frac{-1}{5}\log\left|\frac{1}{x-1}+\frac{1}{5}+\sqrt{\frac{x^2+4}{5(x-1)^2}}\right|+c$

19. $-\left(\frac{1}{2}\right)\log|\sin x+\cos x|+\left(\frac{1}{2}\right)x+C$

21. $\frac{2}{\sqrt{7}}\tan^{-1}\left(\frac{\tan\frac{x}{2}}{\sqrt{7}}\right)+C$

22. $\frac{1}{2(b-a)}\log(a\cos^2 x+b\sin^2 x)$

23. $\sqrt{x^2-5x+6}+\frac{9}{2}\log\left(x-\frac{5}{2}-\sqrt{x^2-5x+6}\right)+C$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\int\frac{1}{\cos^3 x\sqrt{2\sin 2x}}dx$ is equal to

(a) $\sqrt{\tan x}+\frac{1}{5}\sqrt{\tan^5 x}+c$

(b) $\sqrt{\tan x}+\frac{2}{5}\sqrt{\tan^5 x}+c$

(c) $\frac{1}{2}\sqrt{\tan x}+\frac{2}{5}\sqrt{\tan^5 x}+c$

(d) None of these

Solution

(a) $I=\int\frac{1}{\cos^3 x\sqrt{4\sin x\cos x}}dx=\int\frac{\sec^4 x}{2\sqrt{\tan x}}dx$ ***[AMU-2001]***

$=\int\frac{(1+\tan^2 x)}{2\sqrt{\tan x}}\sec^2 x dx=\sqrt{\tan x}+\frac{1}{5}\sqrt{\tan x}+c$

2. $\int\cos^{-3/7}x\sin^{-11/7}x dx=$

(a) $\log\left|\sin^{4/7}x\right|+c$

(b) $\frac{4}{7}\tan^{4/7}x+c$

(c) $\frac{-7}{4}\tan^{-4/7}x+c$

(d) $\log\left|\cos^{3/7}x\right|+c$

Solution

(c) Let $m+n=-\frac{3}{7}+\left(\frac{-11}{7}\right)=-2$ (–ve integer)

$I=\int\cos^{-3/7}x(\sin^{(-2+3/7)}x)dx$

$=\int\cos^{-3/7}x\sin^{-2}x\sin^{3/7}x dx$

$=\int\frac{\operatorname{cosec}^2 x}{\left(\frac{\cos^{3/7}x}{\sin^{3/7}x}\right)}dx=\int\frac{\operatorname{cosec}^2 x dx}{\cot^{3/7}x}$

Put $\cot x=t\Rightarrow-\operatorname{cosec}^2 x dx=dt$

$=\int\frac{dt}{t^{3/7}}=-\frac{t^{-\frac{3}{7}+1}}{-\frac{3}{7}+1}+c=-\frac{7}{4}t^{4/7}+c.$

$=-\frac{7}{4}\cot^{4/7}x+c=-\frac{7}{4}\tan^{-4/7}x+c$

3. $\int\sec^{2/3}x\operatorname{cosec}^{4/3}dx$ is equal to

(a) $-3(\cot)x^{1/3}+c$ (b) $-3(\tan x)^{-2/3}+c$

(c) $3(\operatorname{cosec}x)^{1/3}+c$ (d) $3(\cos 2x)^{1/3}+c$

Solution

(a) $I=\int\frac{1}{\cos^{2/3}x\sin^{4/3}x}dx$ ***[ICS-2006]***

$=\int\frac{\sin^{2/3}x}{\cos^{2/3}x}\operatorname{cosec}^2 x dx$

$=\int(\cot x)^{-2/3}\operatorname{cosec}^2 x dx$

$=-3(\cot x)^{1/3}+c$

4. $\int\frac{(\sin x)^{98}}{(\cos x)^{100}}dx=$ ***[Gujrat CET-2007]***

(a) $\frac{(\tan x)^{95}}{95}+C$

(b) $\frac{(\tan x)^{97}}{97}+C$

(c) $\frac{(\tan x)^2}{2}+C$

(d) $\frac{(\tan x)^{99}}{99}+C$

Solution

(d) Let

$$I=\int\frac{(\sin x)^{98}}{(\cos x)^{100}}dx=\int\frac{(\sin x)^{98}}{(\cos x)^{98}(\cos x)^2}dx$$

$$\therefore\ \int(\tan)^{98}\left(\frac{1}{\cos x}\right)^2dx=\int(\tan x)^{98}(\sec^2 x)dx$$

Put $\tan x = t$, $\sec^2 x\,dx = dt$

$$\Rightarrow\int t^{98}dt\Rightarrow\frac{t^{99}}{99}+C\therefore I=\frac{(\tan x)^{99}}{99}+C$$

5. $\int\tan^3 2x\sec 2x\,dx$ equals

[IIT-77]

(a) $\sec^3 2x-3\sec 2x+c$

(b) $\sec^3 2x+3\sec 2x+c$

(c) $\frac{1}{6}[\sec^3 2x-3\sec 2x]+c$

(d) $\frac{1}{6}[\sec^3 2x+3\sec 2x]+c$

Solution

(c) $I=\int\tan^2 2x(\tan 2x\sec 2x)dx$

$$=\int(\sec^2 2x-1)(\sec 2x\tan 2x)dx,$$

Now put $\sec 2x = t$ to obtain (c)

6. $\int\frac{dx}{\sin^6 x+\cos^6 x}$ is equal to

(a) $\tan^{-1}(\tan x+\cot x)+c$

(b) $\tan^{-1}(\cot x-\tan x)+c$

(c) $\tan^{-1}(\tan x-\cot x)+c$

(d) None of these

Solution

(c) $\int\frac{dx}{\sin^6 x+\cos^6 x}=\int\frac{\sec^6 x}{1+\tan^6 x}dx$

$$=\int\frac{(1+\tan^2 x)^2\sec^2 x}{1+\tan^6 x}dx$$

Put $x = t$

$\therefore$ $\sec^2 x\,dx = dt$

$$=\int\frac{(1+t^2)^2dt}{1+t^6}$$

$$=\int\frac{(1+t^2)^2dt}{(1+t^2)(1-t^2+t^4)}dt$$

$$=\int\frac{1+t^2}{1-t^2+t^4}dt=\int\frac{1+\frac{1}{t^2}}{t^2-1+\frac{1}{t^2}}dt$$

Put $t-\frac{1}{t}=z$

$\therefore$ $\left(1+\frac{1}{t^2}\right)dt=dz\ t^2+\frac{1}{t^2}=z^2+2$

$\therefore$ given integral $=\int\frac{dz}{z^2+1}=\tan^{-1}z+c$

$$=\tan^{-1}\left(t-\frac{1}{t}\right)+c$$

$= \tan^{-1}(\tan x-\cot x)+c.$

7. $\int\frac{dx}{4\sin^2 x+4\sin x\cos x+5\cos^2 x}=$

[MPPET-1990]

(a) $\tan^{-1}(\tan x+1/2)+c$

(b) $1/4\tan^{-1}(\tan x+1/2)+c$

(c) $4\tan^{-1}(\tan x+1/2)+c$

(d) None of these

Solution

(c) In intergrand multiply both the numerator and denominator with $\sec^2 x$

$$I=\int\frac{\sec^2 x dx}{4\tan^2 x+4\tan x+5}$$

$$=\int\frac{\sec^2 x\,dx}{(2\tan x+1)^2+4}$$

$$=\frac{1}{2\times 2}\tan^{-1}\left(\frac{2\tan x+1}{2}\right)+c$$

$$=\frac{1}{4}\tan^{-1}\left(\frac{2\tan x+1}{2}\right)+c$$

8. $\int \frac{x^3+3x+2}{(x^2+1)^2(x+1)}dx$ is equal to ***[IIT-99]***

(a) $\frac{1}{4}\log\left|\frac{x^2+1}{(x+1)^2}\right|+\frac{3}{2}\tan^{-1}x-\frac{x}{x^2+1}+c$

(b) $\frac{1}{4}\log\left|\frac{(x+1)^2}{x^2+1}\right|+\frac{3}{2}\tan^{-1}x+\frac{x}{x^2+1}+c$

(c) $\frac{1}{4}\log\left|\frac{x^2+1}{(x+1)^2}\right|+\frac{3}{2}\tan^{-1}x+\frac{x}{x^2+1}+c$

(d) None of these

Solution

(c) $\int\frac{x^2+3x+2}{(x^2+1)^2(x+1)}dx$

$=\int\frac{x(x^2+1)+2(x+1)}{(x^2+1)^2(x+1)}dx$

$=\int\left(\frac{x+1}{(x^2+1)(x+1)}\right)dx+2\int\frac{dx}{(x^2+1)^2}$

$=\int\left(\frac{x+1}{2(x^2+1)}-\frac{1}{2(x+1)}\right)dx+\int\frac{2}{(x^2+1)^2}dx$

$=\frac{1}{4}\log|x^2+1|-\frac{1}{2}\log|x+1|+\frac{1}{2}\tan^{-1}x+2I$

where, $I=\int\frac{dx}{(x^2+1)^2}$

Put $x=\tan 0$, $dx=\sec^2 0\,d0$

$=\int\frac{\sec^2\theta d\theta}{(1+\tan^2\theta)^2}=\int\cos^2\theta d\theta$

$=\int\frac{1+\cos 2\theta}{2}d\theta=\frac{1}{2}\left(\theta+\frac{\sin 2\theta}{2}\right)$

$=\frac{1}{2}\left(\theta+\frac{\tan\theta}{1+\tan^2\theta}\right)$

$=\frac{1}{2}\tan^{-1}x+\frac{1}{2}.\frac{x}{1+x^2}$ $\therefore$ given integral

$=\frac{1}{4}\log|x^2+1|-\frac{1}{2}\log|x+1|+\frac{1}{2}\tan^{-1}x+$

$\tan^{-1}x+\frac{x}{1+x^2}+c$

$=\frac{1}{4}\log\left|\frac{x^2+1}{(x+1)^2}\right|+\frac{3}{2}\tan^{-1}x+\frac{x}{1+x^2}+c$

9. If $\int\frac{4e^x+6e^{-x}}{9e^x-4e^{-x}}dx=Ax+B\log(9e^{2x}-4)+c$

then values of A and B are ***[IIT-90]***

(a) $-19/36, -35/36$ (b) $-3/2, -35/36$

(c) $-3/2, -35/36$ (d) $-3/2, 35/36$

Solution

(d)

Let $4e^x+6e^{-x}=A_1(9e^x-4e^x)+B_1(9e^x+4e^{-x})$

$\Rightarrow 9A_1+9B_1=4, -4A_1+4B_1=6$

$\Rightarrow A_1=-\frac{19}{36}, B_1=\frac{35}{36}$

$\therefore I=-\frac{19}{36}x+\frac{35}{36}\log(9e^x-4e^{-x})+c$

$=-\frac{19}{36}x+\frac{35}{36}\log(9e^{2x}-4)-\frac{35}{36}x+c$

$=-\frac{3}{2}x+\frac{35}{36}\log(9e^{2x}-4)+c$

10. If $\int\frac{dx}{x\sqrt{1-x^3}}=a\log\left(\frac{\sqrt{1-x^3}-1}{\sqrt{1-x^3}+1}\right)+b$, then

(a) $a=1/3$ (b) $a=2/3$

(c) $a=-1/3$ (d) $a=-2/3$

Solution

(a) Put $1-x^3=y^2$

11. $\int\frac{dx}{(x+100)\sqrt{x+99}}=f(x)+c\Rightarrow f(x)$ is equal to ***[EAMCET-2004]***

(a) $2\tan^{-1}\left(\sqrt{(x+100)}\right)$ (b) $2\tan^{-1}\left(\sqrt{x+99}\right)$

(c) $2(x+100)^{1/2}$ (d) $3(x+100)^{1/2}$

Solution

(b) Put $x+99=t^2$, then $dx=2t\,dt$ and

$I=\int\frac{2dt}{t^2+1}=2\tan^{-1}(\sqrt{x+99})+c$

12. If

$\int\frac{2x^2+3.dx}{(x^2-1)(x^2-4)}=\log\left[\left(\frac{x-2}{x+2}\right)^a\left(\frac{x+1}{x-1}\right)^b\right]+c$

then the values of a and b respectively are ***[Roorkee (Screening)-2000]***

(a) $\frac{11}{12}, \frac{5}{6}$ (b) $\frac{11}{12}, \frac{-5}{6}$

(c) $-\frac{11}{12}, \frac{5}{6}$ (d) None of these

Solution

$$I = \int \frac{2x^2+3}{(x^2-1)(x^2-4)}$$

$$\frac{2x^2+3}{(x^2-1)(x^2-4)} = \frac{A}{x^2-1} + \frac{B}{x^2-4}$$

$2x^2 + 3 = A(x^2 - 4) + B(x^2 - 1)$

$2x^2 + 3 = x^2(A + B) - 4A - B$

Comparing the coefficient of x^2 and constant term on both sides, $A + B = 2$ .....(i)

$4A + B = -3$ .....(ii)

on solving both the equations $A = -5/3$, $B = 11/3$

$$\int \frac{2x^2+3.dx}{(x^2-1)(x^2-4)} = \int \frac{-\frac{5}{3}dx}{(x^2-1)} + \int \frac{\frac{11}{3}dx}{(x^2-4)}$$

$$= -\frac{5}{3}\int \frac{dx}{(x+1)(x-1)} + \frac{11}{3}\int \frac{dx}{(x+2)(x-2)}$$

$$= -\frac{5}{3}.\frac{1}{2}\int \frac{dx}{x-1} + \frac{5}{6}\int \frac{dx}{x+1} + \frac{11}{3}.\frac{1}{4}\int \frac{dx}{x-2} - \frac{11}{12}\int \frac{dx}{x+2} + c$$

$$= -\frac{5}{6}\log(x-1) + \frac{5}{6}\log(x+1) + \frac{11}{12}\log(x-2) - \frac{11}{12}\log(x+2) + c$$

$$= \frac{5}{6}\log\left(\frac{x+1}{x-1}\right) + \frac{11}{12}\log\left(\frac{x-2}{x+2}\right) + c$$

$$= \log\left(\frac{x+1}{x-1}\right)^{5/6} + \log\left(\frac{x-2}{x+2}\right)^{11/12} + c$$

$\Rightarrow \quad a = \frac{11}{12}$ and $b = \frac{5}{6}$.

13. $\int \frac{\sin x + 8\cos x}{4\sin x + 6\cos x} dx =$ ***[EAMCET-2007]***

(a) $x + \frac{1}{2}\log(4\sin x + 6\cos x) + C$

(b) $2x + \log(2\sin x + 3\cos x) + C$

(c) $x + 2\log(2\sin x + 3\cos x) + C$

(d) $\frac{1}{2}\log(4\sin x + 6\cos x) + C$

Solution

(a) Given function write on this type.

$\sin x + 8\cos x = A(4\sin x + 6\cos x) + B\frac{d}{dx}(4\sin x + 6\cos x)$ comparing the coefficient $\sin x$ and $\cos x$, $A = 1, B = 1/2$

$\therefore$ Given Integral

$$I = \int \frac{(4\sin x + 6\cos x) + \frac{1}{2}(4\cos x - 6\sin x)}{4\sin x + 6\cos x} dx$$

$$= x + \frac{1}{2}\log(4\sin x + 6\cos x) + c$$

14. What is the value of $\int \left(\sqrt{x} + x\right)^{-1} dx$?

[NDA-2007]

(a) $ln(x + \sqrt{x}) + c$ (b) $2\,ln(1 + \sqrt{x}) + c$

(c) $2\,ln(x + \sqrt{x}) + c$ (d) $2\,ln(x - \sqrt{x}) + c$

Solution

(b) $\because$ Let $I = \int \left(\sqrt{x} + x\right)^{-1} dx = \int \frac{1}{\sqrt{x}\left(\sqrt{x}+1\right)} dx$

[put $\sqrt{x} + 1 = t$ then $\frac{dx}{\sqrt{x}} = 2dt$]

$$= \int \frac{2dt}{t} = 2\log t + c$$

$$= 2\,ln(\sqrt{x} + 1) + c$$

15. $\int \frac{dx}{(\sin x + 4)(\sin x - 1)}$

$= \frac{A}{\tan\frac{x}{2} - 1} + B\tan^{-1}(f(x)) + c$, then

(a) $A = \frac{1}{5}, B = -\frac{2}{5\sqrt{15}}, f(x) = \frac{4\tan x + 3}{\sqrt{15}}$

(b) $A = -\frac{1}{5}, B = \frac{1}{\sqrt{15}}, f(x) = \frac{4\tan\left(\frac{x}{2}\right) + 1}{\sqrt{15}}$

(c) $A = \frac{2}{5}, B = -\frac{2}{5\sqrt{5}}, f(x) = \frac{4\tan x + 1}{\sqrt{5}}$

(d) $A = \frac{2}{5}, B = -\frac{2}{5\sqrt{15}}, f(x) = \frac{4\tan\left(\frac{x}{2}\right) + 1}{\sqrt{15}}$

Solution

(d) $\int \frac{dx}{(\sin x+4)(\sin x-1)}$

$= \frac{1}{5}\int \frac{(\sin x+4)-(\sin x-1)}{(\sin x+4)(\sin x-1)}dx$

$= \frac{1}{5}\int \frac{dx}{\sin x-1} - \frac{1}{5}\int \frac{dx}{\sin x+4}$

$= \frac{1}{5}\int \frac{dx}{\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}-1} - \frac{1}{5}\int \frac{dx}{\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}+4}$

put $\tan\frac{x}{2} = t$

$= \frac{1}{5}\int \frac{2dt}{2t-1-t^2} - \frac{1}{5}\int \frac{2dt}{2t+4+4t^2}$

$= \frac{-2}{5}\int \frac{dt}{(t-1)^2} - \frac{1}{10}\int \frac{dt}{t^2+\frac{1}{2}t+1}$

$= -\frac{2}{5}\int \frac{dt}{(t-1)^2} - \frac{1}{10}\int \frac{dt}{\left(t+\frac{1}{4}\right)^2+\left(\frac{\sqrt{15}}{4}\right)^2}$

$= \frac{2}{5}\frac{1}{t-1} - \frac{1}{10}\frac{1}{\frac{\sqrt{15}}{4}}\tan^{-1}\frac{t+\frac{1}{4}}{\frac{\sqrt{15}}{4}} + c$

$-\frac{2}{5}\cdot\frac{1}{\tan\frac{x}{2}-1} - \frac{2}{5\sqrt{15}}\tan^{-1}\left(\frac{4\tan\frac{x}{2}+1}{\sqrt{15}}\right)+c$

$\therefore A = \frac{2}{5}, B = -\frac{2}{5\sqrt{15}}, f(x) = \frac{4\tan\frac{x}{2}+1}{\sqrt{15}}$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int \sec^4 x \tan x\, dx$ ***[AICBSE-1980, 81; SCRA-1996]***

(a) $\frac{1}{4}\sec^4 x + c$ (b) $4\sec^4 x + c$

(c) $\frac{\sec^5 x}{3} + c$ (d) $3\sec^3 x + c$

2. $\int \sin^2 x \cos x\, dx$ is equal to ***[SCRA-1996]***

(a) $\cos^2 x/2$ (b) $\sin^2 x/3$

(c) $\sin^3 x/3$ (d) $-\cos^2 x/2$

3. $\int \frac{\sin^3 x}{\cos^5 x}dx =$

(a) $\frac{1}{4}\tan^3 x + k$ (b) $\frac{1}{4}\tan^4 x + k$

(c) $\tan^4 x + k$ (d) $\tan^3 x \sec^2 x + k$

4. $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x}dx =$ ***[RPET-1995]***

(a) $\cot^{-1}(\tan^2 x) + c$ (b) $\tan^{-1}(\tan^2 x) + c$

(c) $\cot^{-1}(\cot^2 x) + c$ (d) $\tan^{-1}(\cot^2 x) + c$

5. If $I = \int \sec^2 x \operatorname{cosec}^4 x\, dx$ $K\cot^3 x + L\tan x + M\cot x + c$, then

(a) $K = -1/3, L = 1, M = -2$

(b) $K = -1/3, L = 2, M = -2$

(c) $K = 1/3, L = -2, M = 2$

(d) None of these

6. $\int \tan^3 2x \sec 2x\, dx$ ***[IIT-1977]***

(a) $\frac{1}{6}\sec^3 2x - \frac{1}{2}\sec 2x + c$

(b) $\frac{1}{6}\sec^3 2x + \frac{1}{2}\sec 2x + c$

(c) $\frac{1}{9}\sec^2 2x - \frac{1}{3}\sec 2x + c$

(d) None of these

7. $\int \frac{dx}{\sin x \cos x}$ is equal to

(a) $\log|\sin x| + c$

(b) $\log|\tan x| + c$

(c) $\log|\sec x| + c$

(d) None of these

8. $\int \frac{dx}{4\sin^2 x + 4\sin x\cos x + 5\cos^2 x} =$

(a) $\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$

(b) $\frac{1}{4}\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$

(c) $\tan^{-1}\left(\tan x + \frac{1}{4}\right) + c$

(d) $4\tan^{-1}\left(\tan x - \frac{1}{4}\right) + c$

9. $\int\left(\sqrt{\tan x} + \sqrt{\cot x}\right)dx$ is equal to

(a) $\sqrt{2}\tan^{-1}\left[\frac{1}{\sqrt{2}}(\sqrt{\tan x} - \sqrt{\cot x})\right] + c$

(b) $\frac{1}{\sqrt{2}}\tan^{-1}\left[\sqrt{2}(\sqrt{\tan x} - \sqrt{\cot x})\right] + c$

(c) $\sqrt{2}\tan^{-1}\left[\frac{1}{2}(\sqrt{\tan x} - \sqrt{\cot x})\right] + c$

(d) None of these

10. $\int \frac{1-\cos x}{\cos x(1+\cos x)}dx =$

(a) $\log|\sec x + \tan x| - 2\tan\frac{x}{2} + c$

(b) $\log|\cos x(1+\cos x)| + c$

(c) $\log|\cos x| + c$

(d) None of these

11. $\int \frac{\cos x + x\sin x}{x(x+\cos x)}dx =$

(a) $\log x - \log(x + \cos x)$

(b) $\log x + \log(x - \cos x)$

(c) $\log|x + (x + \cos x)| + c$

(d) $\log\left(\frac{x+\cos x}{x}\right)$

12. What is $\int \tan^2 x\sec^4 x dx$ equal to? ***[NDA-2009]***

(a) $\frac{\sec^5 x}{5} + \frac{\sec^3 x}{3} + c$

(b) $\frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + c$

(c) $\frac{\tan^5 x}{5} + \frac{\sec^3 x}{3} + c$

(d) $\frac{\sec^5 x}{5} + \frac{\tan^3 x}{3} + c$

13. $\int \sin^3 x.\cos^2 x dx =$ ***[MPPET-2009]***

(a) $\frac{\sin^5 x}{5} - \frac{\sin^3 x}{3} + c$

(b) $\frac{\sin^5 x}{5} + \frac{\sin^3 x}{3} + c$

(c) $\frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + c$

(d) $\frac{\cos^5 x}{5} + \frac{\cos^3 x}{3} + c$

14. $\int \frac{dx}{1-\sin x} =$ ***[MPPET-1991]***

(a) $x + \cos x + c$

(b) $1 + \sin x + c$

(c) $\sec x - \tan x + c$

(d) $\sec x + \tan x + c$

15. $\int \frac{xdx}{x^2+4x+5} =$ ***[RPET-2002]***

(a) $\frac{1}{2}\log(x^2 + 4x + 5) + 2\tan^{-1}(x) + c$

(b) $\frac{1}{2}\log(x^2 + 4x + 5) - \tan^{-1}(x + 2) + c$

(c) $\frac{1}{2}\log(x^2 + 4x + 5) + \tan^{1}(x + 2) + c$

(d) $\frac{1}{2}\log(x^2 + 4x + 5) - 2\tan^{1}(x + 2) + c$

16. $\int x^3\sqrt{3+5x^4}dx =$

(a) $(3 + 5x^4)^{3/2} + c$ (b) $\frac{1}{5}(3 + 5x^4)^{3/2} + c$

(c) $\frac{1}{30}(3 + 5x^4)^{3/2} + c$ (d) None of these

17. $\int \frac{\sin x}{1+\sin x}dx =$ ***[MPPET-1990, 97]***

(a) $x + 2\left(1 + \tan\frac{x}{2}\right)^{-1} + c$

(b) $x + \left(1 + \tan\frac{x}{2}\right)^{-1} + c$

(c) $x + \frac{1}{2}\left(1 + \tan\frac{x}{2}\right) + c$

(d) $x + \left(1 + \tan\frac{x}{2}\right) + c$

18. $\int \frac{x}{1+\cos x}dx$ is equal to ***[MPPET-1993]***

(a) $x\tan\frac{x}{2}+2\log\cos\left(\frac{x}{2}\right)+c$

(b) $x\tan\frac{x}{2}+2\log\cos\left(\frac{x}{2}\right)+c$

(c) $x\tan\frac{x}{2}-\log\left(\frac{x}{2}\right)+c$

(d) $x\tan\frac{x}{2}-2\log\cos\left(\frac{x}{2}\right)+c$

19. $\int\frac{dx}{1+2\sin x+\cos x}=$ ***[Raj PET-1991]***

(a) $\log\left[1-2\tan\frac{x}{2}\right]+c$

(b) $\frac{1}{2}\log\left[1+2\tan\frac{x}{2}\right]+c$

(c) $2\log\left[1-\tan\frac{x}{2}\right]+c$

(d) None of these

20. $\int\frac{\sin x}{\sin x-\cos x}dx=\int\frac{dx}{1-\cot x}=$ is

[Roorkee-1988]

(a) $\frac{1}{2}[\log(\sin x-\cos x)+2x]+c$

(b) $\frac{1}{2}[\log(\sin x-\cos x)+x]+c$

(c) $\frac{1}{2}[\log(\sin x+\cos x)+x]+c$

(d) None of these

SOLUTIONS

1. (a) $t=\sec x,\ dt=\sec x\tan x dx$

$I=\int t^3dt$

$=\frac{t^4}{4}+c\Rightarrow\frac{\sec^4 x}{4}+c$

2. (c) $\int\sin^2 x\cos x\,dx$

Put $\sin x=t\Rightarrow\cos x dx=dt$

$$I=\int t^2dt=\frac{t^3}{3}+C=\frac{\sin^3 x}{3}+C$$

3. (b) $I=\int\frac{\sin^3 x}{\cos^5 x}dx$

$$I=\int\frac{\sin^3 x}{\cos^3 x\cos^2 x}dx=\int\tan^3 x\sec^2 x dx$$

Putting $\tan x=t\Rightarrow dt=\sec^2 x dx$

$$=\int t^3dt=\frac{t^7}{4}+K=\frac{1}{4}\tan^4 x+K$$

4. (b) Step 1: $I=\int\frac{2\sin x\cos x dx}{\sin^4 x+\cos^4 x}$

Multiplying numerator and denominator by $\sec^4 x$

$$I=\int\frac{2\tan x\sec^2 x dx}{\tan^4 x+1}$$

Step 2: By Substituting method:

$$t=\tan^2 x,\ dt=2\tan x\sec^2 x dx$$

Hence $I=\int\frac{dt}{1+t^2}=\tan^{-1}(t)$

$$=\tan^{-1}(\tan^2 x)+c$$

5. (a) $\int\sec^2 x\,\text{cosec}^4 x dx-K\cot^3 x+L\tan x+M\cot x+c$

Taking L.H.S. of expression

$\int\sec^2 x\,\text{cosec}^4 x dx=\int\sec^2 x(1+\cos^2 x)^2 dx$

$\int\sec^2 x(1+\cot^4 x+2\cot^2 x)dx$

$$= \int (\sec^2 x + \sec^2 x \cot^4 x + 2\sec^2 x \cos^2 x)dx$$

$$= \int \sec^2 x dx + \int \cot^2 x \operatorname{cosec}^2 x dx + 2\int \operatorname{cosec}^2 x dx$$

$$= \tan x + \int -t^2 dt - 2\cot x$$

(Putting $t = \cot x$, $dt = -\operatorname{cosec}^2 x\, dx$)

$$= \tan x - \frac{t^3}{3} - 2\cot x$$

$$\Rightarrow \tan x - \frac{\cot^3 x}{3} - 2\cot x$$

$$\therefore \quad K = -\frac{1}{3};\ L = 1,\ M = -2$$

6. (a) $\int \tan^3 2x \sec 2x dx$

$$= \int (\sec^2 2x - 1)\sec 2x \tan 2x dx$$

$$= \int (\sec^3 2x \tan 2x - \sec 2x \tan 2x)dx$$

$$= \int \sec^3 2x \tan 2x dx - \int \sec 2x \tan 2x dx \quad \ldots(1)$$

Now, we take $\int \sec^3 2x \tan 2x dx$

Put $\sec 2x = t \Rightarrow \sec 2x \tan 2x = \frac{dt}{2}$ then is reduces to $\frac{1}{2}\int t^2 dt = \frac{t^3}{6} = \frac{\sec^3 2x}{6}$

From (1)

$$\int \sec^3 2x \tan 2x dx - \int \sec 2x \tan 2x dx$$

$$= \frac{\sec^3 2x}{6} - \frac{\sec 2x}{2} + c$$

7. (b) $\int \frac{dx}{\sin x \cos x} = 2\int \frac{dx}{\sin 2x} = 2\int \operatorname{cosec} 2x dx$

$$= \frac{2}{2}\log\left(\tan\frac{2x}{2}\right) = \log(\tan x) + c$$

8. (b) Step 1: Multiplying numerator and denominator by $\sec^2 x$

$$I = \int \frac{\sec^2 x dx}{4\tan^2 x + 4\tan x + 5}$$

$$= \int \frac{\sec^2 x dx}{(2\tan x + 1)^2 + 4}$$

$$= \frac{1}{2\times 2}\tan^{-1}\left(\frac{2\tan x + 1}{2}\right) + c$$

NOTE

$$\int \frac{dx}{a^2 + b^2 x^2} = \frac{1}{ab}\tan^{-1}\frac{bx}{a}$$

9. (a) $I = \sqrt{2}\int \frac{\sin x + \cos x}{\sqrt{2\sin x \cos x}} dx$

$$= \sqrt{2}\int \frac{d(\sin x - \cos x)}{\sqrt{1 - (\sin x - \cos x)^2}}$$

$$= \sqrt{2}\sin^{-1}\left(\frac{\sin x - \cos x}{1}\right) = \sqrt{2}\sin^{-1}$$

$$\left(\frac{\text{Perpendicular}}{\text{hypotenous}}\right)$$

$$= \sqrt{2}\tan^{-1}\left(\frac{\sin x - \cos x}{\sqrt{2\sin x \cos x}}\right)$$

$$\left[\sin^{-1}\left(\frac{\text{Perpendicular}}{\text{hypotenous}}\right) = \tan^{-1}\left(\frac{\text{Perpendicular}}{\text{hypotenous}}\right)\right]$$

$$= \sqrt{2}\tan^{-1}\left\{\frac{1}{\sqrt{2}}\left(\sqrt{\tan x} - \sqrt{\cot x}\right)\right\} + c$$

10. (a) Step 1: Integration by using partial fraction

$$\frac{1 - \cos x}{\cos x(1 + \cos x)} = \frac{A}{\cos x} + \frac{B}{1 + \cos x}$$

$$1 - \cos x = A(1 + \cos x) + B(\cos x)$$

$$1 - \cos x = A + (A + B)\cos x$$

$$A = 1,\ A + B = -1,\ B = -2$$

Step 2: $I = \int \frac{dx}{\cos x} - 2\int \frac{dx}{1 + \cos x}$

$$= \int \sec x dx - 2\int \frac{\sec^2 \frac{x}{2}}{2} dx$$

$$= \log|(\sec x + \tan x)| - 2\tan\frac{x}{2} + c$$

11. (a) We have,

$$\frac{\cos x + x\sin x}{x(x + \cos x)} = \frac{x + \cos x - x + x\sin x}{x(x + \cos x)}$$

$$= \frac{1}{x} - \frac{(1 - \sin x)}{x + \cos x}$$

$$= \int \frac{dx}{x} - \int \frac{1-\sin x}{x+\cos x}dx = \log x - \int \frac{1-\sin x}{x+\cos x}dx$$

We take expression $\int \frac{1-\sin x}{x+\cos x}dx$

Putting $x+\cos x = t \Rightarrow dt = 1-\sin x dx$, we get

$$\int \frac{dt}{t} = \log(t) + c = \log(x+\cos x)$$

so required answer will be

$$= \log x - \log(x+\cos x) + c$$

12. (c) Let $I = \int \tan^2 x \sec^4 x dx$

Let $\tan x = t$ and $\sec^2 x\, dx = dt$

$$\therefore \quad I = \int t^2(1+t^2)dt = \int (t^2+t^4)dt$$

$$= \frac{t^5}{5} + \frac{t^3}{3} + c = \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + c$$

13. (c) Let $I = \int (1-\cos^2 x)\cos^2 x.\sin x dx$

Put $\cos x = t$

$$\Rightarrow \quad -\sin x dx = dt$$

$$\therefore \quad I = -\int (1-t^2)t^2 dt = \int (t^4 - t^2)dt$$

$$= \frac{t^5}{5} - \frac{t^3}{3} + c = \frac{(\cos x)^5}{5} - \frac{(\cos x)^3}{3} + c$$

14. (d) $\int \frac{dx}{1-\sin x} = \int \frac{1+\sin x}{1-\sin^2 x}dx$

$$\Rightarrow \int \sec^2 x dx + \int \tan x \sec x dx$$

$$\Rightarrow \int \tan x + \sec x + c$$

15. (d) $\int \frac{x dx}{x^2+4x+5} = \int \frac{x+2-2}{(x+2)^2+1}dx$

$$= \frac{1}{2}\int \frac{2(x+2)}{(x+2)^2+1} - 2\int \frac{dx}{1+(x+2)^2}$$

$$= \frac{1}{2}\int \frac{dt}{t} - 2\int \frac{dx}{1+(x+2)^2}$$

Putting $1 + (x + 2)^2 = t$ in first expression

$$\Rightarrow \quad 2(x+2)\,dx = dt$$

$$= \frac{1}{2}\log t - 2\tan^{-1}(x+2) + c$$

$$= \frac{1}{2}\log(x^2+4x+5) - 2\tan^{-1}(x+2) + c$$

16. (c) $\int x^3\sqrt{3+5x^4}dx$

Putting $3+5x^4 = t^2 \Rightarrow 2tdt = 20x^3$

$$\frac{1}{10}\int t^2 dt = \frac{1}{10}\frac{t^3}{3} + c$$

$$= \frac{1}{30}(3+5x^4)^{3/2} + c$$

17. (a) Step 1: $\frac{\sin x}{1+\sin x} = 1 - \frac{1}{1+\sin x}$

Step 2: $I = \int dx - \int \frac{dx}{1+\sin x}$

$$= x - \int \frac{dx}{\left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)^2}$$

$$= x - \int \frac{\sec^2\frac{x}{2}dx}{\left(\tan\frac{x}{2}+1\right)^2}$$

$$= x - \int \frac{2dt}{t^2}; t = 1+\tan\frac{x}{2} = dt = \frac{1}{2}\sec^2\frac{x}{2}dx$$

$$= x + \frac{2}{t} = x + 2\left(1+\tan\frac{x}{2}\right)^{-1} + c$$

18. (a) $I = \frac{1}{2}\int x\sec^2\frac{x}{2}dx$

Successive Integration by parts:

$$\int uvdx = (u)(v_1) - (u^1)(v_2) + (u^{11})(v_3).....$$

$$+(-1)^{n-1}(u^{n-1})(v_n) + ...$$

Here u^n = differentiation of u, n times

v_n = Integration of v, n times

19. (b) Step 1: dividendo = $1+\cos x + 2\sin x$

$$= 2\cos^2\frac{x}{2} + 2\left(2\sin\frac{x}{2}\cos\frac{x}{2}\right)$$

$$I = \int \frac{dx}{2\cos^2\frac{x}{2} + 4\sin\frac{x}{2}\cos\frac{x}{2}}$$

Multiplying numerator and denominator by $\sec^2\frac{x}{2}$

$$I=\frac{1}{2}\int\frac{\sec^2\frac{x}{2}dx}{1+2\tan\frac{x}{2}}$$

Step 2: $t=1+2\tan\frac{x}{2}$

$$dt=\sec^2\frac{x}{2}dx$$

$$I=\frac{1}{2}\int\frac{dt}{t}=\frac{1}{2}\log t$$

$$=\frac{1}{2}\log\left[1+2\tan\frac{x}{2}\right]+c$$

20. (b) Step 1: $\frac{1}{2}\int\frac{2\sin x}{\sin x-\cos x}dx$

Step 2: $\frac{1}{2}\int\left[\frac{\sin x+\cos x+\sin x-\cos x}{\sin x-\cos x}\right]dx$

$$=\frac{1}{2}\left[\int\frac{\sin x+\cos x}{\sin x-\cos x}dx+\int dx\right]$$

$$=\frac{1}{2}[\log(\sin x-\cos x)+x]+c$$

Quicker Method: (Substituting $a=1, b=-1$ in following formula)

$$\int\frac{dx}{a+b\cot x}=\int\frac{\sin x dx}{a\sin x+b\cos x}$$

$$=\frac{a}{a^2+b^2}\log[a\sin x+b\cos x]$$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. $\int\frac{dx}{(1+x^2)\sqrt{1-x^2}}=$ ***[MNR-1985]***

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{\sqrt{1-x^2}}{x\sqrt{2}}\right]+c$

(b) $\frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{x\sqrt{2}}{\sqrt{1-x^2}}\right]+c$

(c) $\sqrt{2}\tan^{-1}\left[\frac{\sqrt{1-x^2}}{x\sqrt{2}}\right]$

(d) $\sqrt{2}\tan^{-1}\left[\frac{x\sqrt{2}}{\sqrt{1-x^2}}\right]$

2. The value of $\int\sqrt{1+\sec x}\,dx$ is ***[RPET-1998]***

(a) $\frac{1}{2}\sin^{-1}(\sqrt{2}\sin x)+c$

(b) $-2\sin^{-1}(\sqrt{2}\sin\frac{x}{2})+c$

(c) $2\sin^{-1}(\sqrt{2}\sin x)+c$

(d) $2\sin^{-1}(\sqrt{2}\sin\frac{x}{2})+c$

3. $\int\frac{1}{\cos x(1+\cos x)}dx=$

(a) $\log(\sec x+\tan x)+2\tan\frac{x}{2}+c$

(b) $\log(\sec x+\tan x)-2\tan\frac{x}{2}+c$

(c) $\log(\sec x+\tan x)+\tan\frac{x}{2}+c$

(d) $\log(\sec x+\tan x)-\tan\frac{x}{2}+c$

4. $\int 2x\cos^3 x^2\sin(x^2)\,dx$

(a) $-\frac{1}{4}\cos^4 x^2+c$ (b) $\frac{1}{4}\cos^4 x^2+c$

(c) $\cos^4 x^2+c$ (d) None of these

5. $\int\frac{dx}{1+3\sin^2 x}=$ ***[Roorkee-1989; DCE-2001]***

(a) $\frac{1}{3}\tan^{-1}(3\tan^2 x)+c$

(b) $\frac{1}{2}\tan^{-1}(2\tan x)+c$

(c) $\tan^{-1}(\tan x)+c$

(d) None of these

6. $\int\sqrt{\frac{x}{1-x}}dx$ =

(a) $\sin^{-1}\sqrt{x}+c$

(b) $\sin^{-1}\sqrt{x}-\sqrt{x(1-x)}+c$

(c) $\sin^{-1}\sqrt{x}+\sqrt{x(1-x)}+c$

(d) None of these

7. $\int\sqrt{\frac{\cos x-\cos^3 x}{1-\cos^3 x}}dx$ =

(a) $-\frac{2}{3}\sin^{-1}(\cos^{3/2}x)+c$

(b) $\frac{2}{3}\cos^{-1}(\cos^{3/2}x)+c$

(c) $-\frac{3}{2}\sin^{-1}(\cos^{3/2}x)+c$

(d) $\frac{3}{2}\cos^{-1}(\cos^{3/2}x)+c$

8. $\int\frac{\sin x\cos x}{a\cos^2 x+b\sin^2 x}dx$ =

[AICBSE-1988; 89]

(a) $\frac{1}{2}(b-a)\log(a\cos^2 x+b\sin^2 x)+c$

(b) $\frac{1}{b-a}\log(a\cos^2 x+b\sin^2 x)+c$

(c) $\frac{1}{2}\log(a\cos^2 x+b\sin^2 x)+c$

(d) None of these

9. $\int\frac{3\cos x+3\sin x}{4\sin x+5\cos x}dx$ =

[EAMCET-1991]

(a) $\frac{27}{41}x-\frac{3}{41}\log(4\sin x+5\cos x)+c$

(b) $\frac{27}{41}x+\frac{3}{41}\log(4\sin x+5\cos x)+c$

(c) $\frac{3}{41}x-\frac{27}{41}\log(4\sin x+5\cos x)+c$

(d) $\frac{3}{41}x+\frac{27}{41}\log(4\sin x+5\cos x)+c$

10. $\int\frac{3\sin x+2\cos x}{3\cos x+2\sin x}dx$ =

(a) $\frac{12}{13}x-\frac{5}{13}\log(3\cos x+2\sin x)$

(b) $\frac{12}{13}x+\frac{5}{13}\log(3\cos x+2\sin x)$

(c) $\frac{13}{12}x+\frac{5}{13}\log(3\cos x+2\sin x)$

(d) None of these

11. $\int\frac{dx}{4\sin^2 x+5\cos^2 x}$ = ***[AISSE-1986]***

(a) $\frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{2\tan x}{\sqrt{5}}\right)+c$

(b) $\frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{\tan x}{\sqrt{5}}\right)+c$

(c) $\frac{1}{2\sqrt{5}}\tan^{-1}\left(\frac{2\tan x}{\sqrt{5}}\right)+c$

(d) None of these

12. If $\int\sec^4 x\ \operatorname{cosec}^2 x\,dx = l\tan^3 x+m\tan x+n\cot x+K$, then $l+m+n$ =

(a) 1/3 (b) 2

(c) −1 (d) 4/3

13. $\int\sqrt{e^x-1}\,dx$ = ***[Kerala (Engg.)-2005]***

(a) $2\left[\sqrt{e^x-1}-\tan^{-1}\sqrt{e^x-1}\right]+c$

(b) $\sqrt{e^x-1}-\tan^{-1}\sqrt{e^x-1}+c$

(c) $2\left[\sqrt{e^x-1}+\tan^{-1}\sqrt{e^x-1}\right]+c$

(d) $2\left[\sqrt{e^x-1}-\tan^{-1}\sqrt{e^x+1}\right]+c$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 14 minutes.
3. The worksheet consists of 14 questions. The maximum marks are 42.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int \frac{1+x\sin x+\cos x}{x(1+\cos x)}dx =$

(a) $\log x + 2\log \sec \frac{x}{2}$

(b) $\log x - 2\log \sec \frac{x}{2}$

(c) $\log \frac{x}{2} + \log \sec \frac{x}{2}$

(d) $\log x - \log \sec \frac{x}{2}$

2. $2\int \frac{\cos x-\sin x}{\cos x+\sin x}(1+\sin 2x)\,dx$ is equal to

(a) $\sin 2x + c$ (b) $\cos 2x + c$

(c) $\tan 2x + c$ (d) None of these

3. $\int \frac{e^{5\log x}-e^{4\log x}}{e^{3\log x}-e^{2\log x}}dx =$ ***[MNR-1985]***

(a) $e.3^{3x} + c$ (b) $\frac{x^3}{3} + c$

(c) $3x^2 + c$ (d) None of these

4. If $\int \frac{e^{\log(1+1/x^2)}}{x^2+1/x^2}dx$ is

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(x-\frac{1}{x}\right)$

(b) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(x-\frac{1}{x}\right)$

(c) $\frac{1}{\sqrt{2}}\log\left(\frac{x^2+1}{x\sqrt{2}}\right)$

(d) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{x\sqrt{2}}\right)$

5. $\int \sqrt{\sec x - 1}\,dx$ is equal to

(a) $2\log\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

(b) $\log\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

(c) $-2\log\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

(d) None of these

6. $\int \frac{dx}{(1+\sin x)^{1/2}} =$ ***[Roorkee-2000]***

(a) $\frac{1}{\sqrt{2}}\log\left|\cot\left(\frac{3\pi}{8}-\frac{x}{4}\right)\right|+C$

(b) $\frac{1}{\sqrt{2}}\log\left|\operatorname{cosec}\left(\frac{\pi}{4}+\frac{x}{2}\right)+\cot\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|$

(c) $\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{\pi}{8}+\frac{x}{4}\right)\right|+C$

(d) $\frac{1}{\sqrt{2}}\log\left|\sec\left(\frac{\pi}{4}-\frac{x}{2}\right)+\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|+C$

7. If $\int \frac{dx}{\sin^4 x+\cos^4 x} = \frac{1}{\sqrt{2}}\tan^{-1}(f(x))+c$, then

(a) $f(x) = \tan x - \cot x$

(b) $f\left(\frac{\pi}{4}\right) = 0$

(c) $f(x)$ is continuous on R

(d) $f(x) = \frac{1}{2}(\tan x - \cot x)$

8. $\int \frac{e^{2x}-2e^x}{e^{2x}+1}dx$ is equal to

[Roorkee-1999; Him.U.CET-2002]

(a) $\log(e^{2x}+1) - \tan^{-1}(e^x) + c$

(b) $\frac{1}{2}\log(e^{2x}+1) - \tan^{-1}(e^x) + c$

(c) $\frac{1}{2}\log(e^{2x}+1) - 2\tan^{-1}(e^x) + c$

(d) None of these

9. $\int \sec^{2/3} x \cdot \operatorname{cosec}^{4/3} x\, dx =$

(a) $3\tan^{1/2} x + c$ (b) $-3\tan^{1/3} x + c$

(c) $-3\tan^{-1} x + c$ (d) $3\tan^{-1} x + c$

10. $\int \cos^2 x \sin^3 x dx =$

(a) $\frac{1}{5}\cos^3 x - \frac{1}{3}\cos^2 x + c$

(b) $\frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + c$

(c) $\frac{1}{5}\cos^2 x - \frac{1}{3}\cos^4 x + c$

(d) $-\frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + c$

11. $\int \frac{x \quad dx}{(x^2+4)\sqrt{x^2+9}} =$

(a) $\frac{1}{2\sqrt{5}}\log\left|\frac{\sqrt{x^2+9}-5}{\sqrt{x^2+9}+5}\right|$

(b) $\frac{1}{2\sqrt{5}}\log\left|\frac{\sqrt{x^2+9}-\sqrt{5}}{\sqrt{x^2+9}+\sqrt{5}}\right|$

(c) $\frac{1}{\sqrt{5}}\tan^{-1}\frac{\sqrt{x^2+9}}{\sqrt{5}}$

(d) $\frac{1}{2\sqrt{5}}\log\left|\frac{\sqrt{x^2+9}+5}{\sqrt{x^2+9}-5}\right|$

12. $\int \frac{\sin x dx}{\sqrt{\sin^2 x - \sin^2 \alpha}} =$ ***[NDA-05]***

(a) $-\sin^{-1}(\sec\alpha \cos x)$ (b) $\sin^{-1}(\sec\alpha \sec x)$

(c) $\sin^{-1}\left(\frac{\cos x}{\cos\alpha}\right)$ (d) None of these

13. The value of $\int \frac{dx}{2+\cos x}$ is ***[MPPET-2010]***

(a) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\tan(x/2)}{\sqrt{3}}\right)+c$

(b) $-\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\tan(x/2)}{\sqrt{3}}\right)+c$

(c) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\tan(x/2)}{\sqrt{5}}\right)+c$

(d) None of these

14. $\int \frac{x-1}{(x+1)(x-2)}dx$ is ***[MPPET-2010]***

(a) $\frac{2}{3}\log|x+1|+\frac{1}{3}\log|x-2|+c$

(b) $\frac{2}{3}\log(x+1)+\frac{1}{3}\log(x-2)+c$

(c) $\frac{2}{3}\log|x+1|+c$

(d) None of these

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	6. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

1. (a)

$$\int \frac{1+x\sin x+\cos x}{x(1+\cos x)}dx = \int \frac{1}{x}dx + \int \frac{\sin x}{1+\cos x}dx$$

$$= \log x + \int \frac{\sin x}{1+\cos x}dx$$

$$= \log x + \int \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}dx$$

$$= \log x + \int \tan\frac{x}{2}dx$$

$$= \log x + 2\log\left(\sec\frac{x}{2}\right) + c$$

3. (b) Step-1: $e^{n\log x} = e^{\log x^n} = x^n$

Step-2: $I = \int \frac{x^5 - x^4}{x^3 - x^2}dx = \int \frac{x^4}{x^2}dx = \frac{x^3}{3} + c$

5. (c) $\int \sqrt{\sec x - 1}dx$

$$\int \sqrt{\frac{1-\cos x}{\cos x}}dx = \int \frac{\sqrt{2}\sin\frac{x}{2}}{\sqrt{2\cos^2\frac{x}{2}-1}}dx$$

$$= \int \frac{\sin\frac{x}{2}}{\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}}dx$$

put $\cos\frac{x}{2} = t$; $-\frac{1}{2}\sin\frac{x}{2}dx = dt$

$$= -\int \frac{2dt}{\sqrt{t^2-\frac{1}{2}}} = -2\log\left(t + \sqrt{t^2-\frac{1}{2}}\right)$$

$$= -2\log\left(\cos\frac{x}{2} + \sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right) + c$$

LECTURE



8

Test Your Skills

ASSERTION/REASONING

ASSERTION AND REASONING TYPE QUESTIONS

Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.

(a) **Assertion** is True, **Reason** is True and **Reason** is a correct explanation for **Assertion**

(b) **Assertion** is True, **Reason** is True and **Reason** is NOT a correct explanation for **Assertion**

(c) **Assertion** is True and **Reason** is False

(d) **Assertion** is False and **Reason** is True

1. **Assertion:** $\int e^x(\tan x + \sec^2 x)dx$

$= e^x.\tan x + c$

Reason: $\int e^x\{f(x) + f'(x)\}dx$

$= e^x f(x) + c$

2. **Assertion:** If $I_n = \int \cot^n x dx$, then $5(I_6 + I_4)$ $= -\cot^5 x$

Reason: If $I_n = \int \cot^n x dx$

then $I_n = -\frac{\cot^{n-1} x}{n} - I_{n-2}$,

where $n \geq 2$

3. **Assertion:** $\int \frac{dx}{\sqrt{1-(x^2)^2}} = \sin^{-1} x^2 + c$

Reason: $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c$

4. **Assertion:** $\int_{-1}^{1} |x| dx$ can be found while $\int |x| dx$ cannot be found.

Reason: $|x|$ is non differentiable at $x = 0$.

5. **Assertion:** $\int \frac{(3-2x)}{\sqrt{(4+2x-x^2)}}dx$

$= 2\sqrt{(4+2x-x^2)} + \sin^{-1}\left(\frac{x-1}{\sqrt{5}}\right) + c$

Reason: $\int \frac{dx}{\sqrt{(a^2-x^2)}} = \frac{x}{2}\sqrt{(a^2-x^2)} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$

6. **Assertion:** $\int \tan 5x \tan 3x \tan 2x \, dx$

$= \frac{\ln|\sec 5x|}{5} - \frac{\ln|\sec 3x|}{3} - \frac{\ln|\sec 2x|}{2} + c$

Reason: $\tan 5x - \tan 3x - \tan 2x = \tan 5x \tan 3x \tan 2x$

7. **Assertion:** If $y(x-y)^2 = x$, then $\int \frac{dx}{(x-3y)} = \frac{1}{2}\ln\{(x-y)^2 - 1\}$

Reason: $\int \frac{dx}{(x-3y)} = \ln(x-3y) + c$

8. **Assertion:** $\int e^{x^2} dx = e^{x^2} + c$

Reason: $\int e^x dx = e^x + c$

9. **Assertion:** If $\int \frac{1}{f(x)} dx = 2\ln|f(x)| + c$, then $f(x) = \frac{x}{2}$

Reason: When $f(x) = \frac{x}{2}$ then $\int \frac{1}{f(x)} dx = \int \frac{2}{x} dx = 2\ln|x| + c$

10. **Assertion:** The function $F(x) = \int \sin^2 x \, dx$ satisfies $F(x+\pi) = F(x) \forall x \in R$

Reason: $\sin^2(x+\pi) = \sin^2 x$

11. **Assertion:** $\int \frac{e^x}{e^x+1} dx = \log(e^x+1) + C$

Reason: $\int \frac{f'(x)}{f(x)} dx = \log|f(x)| + C$

12. **Assertion:** $\int e^x \left(\sqrt{x} + \frac{1}{2\sqrt{x}} \right) dx = e^x \sqrt{x} + C$

Reason: $\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$

13. **Assertion:** $\int \frac{dx}{x(1+\log x)^2} = -\frac{1}{1+\log x} + C$

Reason: $\int (f(x))^n f'(x) dx = \frac{(f(x))^{n+1}}{n+1} + C \ (n+1 \neq 0)$

14. **Assertion:** $\int \frac{1+\log x}{\sqrt{(x^x)^2 - 1}} dx = \sec^{-1}(x^2) + C$

Reason: $\int \frac{dx}{x\sqrt{x^2-1}} dx = \sec^{-1} x + C$

15. **Assertion:** $\int \frac{10x^9 + 10^x \log_e 10}{10^x + x^{10}} dx = \log|10^x + x^{10}| + c$

Reason: $\int \frac{f'(x)}{f(x)} = \log|f(x)| + c$

16. **Assertion:** $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx = \tan(xe^x) + c$

Reason: $\int \sec^2 x dx = \tan x + c$

17. Let $F(x)$ be an indefinite integral of $\sin 2x$ ***[IIT-2007]***

Assertion: The function $F(x)$ satisfies $F(x+\pi) = F(x)$ for all real x because

Reason: $\sin^2(x+\pi) = \sin^2 x$ for all real x.

ASSERTION/REASONING: SOLUTIONS

1. (a) $\because \int e^x \{f(x) + f'(x)\} dx = \int e^x f(x) dx + \int e^x f'(x) dx$

$= f(x).e^x - \int f'(x) e^x dx + \int e^x f'(x) dx$

$= e^x f(x)$

For $f(x) = \tan x$

$\int e^x (\tan x + \sec^2 x) dx = e^x . \tan x + c$

2. (c) $\because I_n = \int \cot^n x dx$

$= \int \cot^{n-2} x . (\operatorname{cosec}^2 x - 1) dx$

$= \int \cot^{n-2} x \operatorname{cosec}^2 x dx - I_{n-2}$

$= -\frac{\cot^{n-1} x}{(n-1)} - I_{n-2}$

For $n = 6$ $\quad 5(I_6 + I_4) = -\cot^5 x$

3. (d) The formula $\int \frac{dx}{\sqrt{1-x^{2n}}} = \sin^{-1} x^n$ is valid only for $n = 1$.

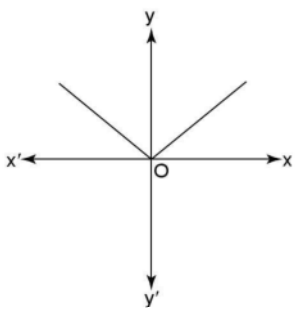
4. (b) $\because \int_{-1}^{1} |x|\,dx = \int_{-1}^{0} |x|\,dx + \int_{0}^{1} |x|\,dx$

$$= \int_{-1}^{0} (-x)dx + \int_{0}^{1} (x)dx$$

$$= -\left\{\frac{x^2}{2}\right\}_{-1}^{0} + \left\{\frac{x^2}{2}\right\}_{0}^{1}$$

$$= -\left(0 - \frac{1}{2}\right) + \left(\frac{1}{2} - 0\right) = 1$$

and $\int |x|\,dx$ cannot be found, since condition on x is not given.
Also, $|x|$ is non differentiable at $x = 0$



5. (c) Let $I = \int \frac{(3-2x)}{\sqrt{(4+2x-x^2)}}dx$

$$= \int \frac{(2-2x)}{\sqrt{(4+2x-x^2)}}dx + \int \frac{dx}{\sqrt{(4+2x-x^2)}}$$

$$= 2\sqrt{(4+2x-x^2)} + \int \frac{dx}{\sqrt{5-(x-1)^2}}$$

$$= 2\sqrt{(4+2x-x^2)} + \sin^{-1}\left(\frac{x-1}{\sqrt{5}}\right) + c$$

and $\int \sqrt{(a^2-x^2)}dx = \frac{x}{2}\sqrt{(a^2-x^2)} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{2}\right) + c$

6. (a) $\because 5x = 3x + 2x$

$\therefore \quad \tan 5x = \dfrac{\tan 3x + \tan 2x}{1 - \tan 3x \tan 2x}$

$\therefore \quad \tan 5x - \tan 3x - \tan 2x$

$= \tan 5x \tan 3x \tan 2x$

7. (c) Let $P = \int \frac{dx}{(x-3y)} = \frac{1}{2}\ln\{(x-y)^2 - 1\}$

$\because \quad P = \int \frac{dx}{(x-3y)}$

$\Rightarrow \quad \frac{dP}{dx} = \frac{1}{(x-3y)}$ ..... (i)

Also, $P = \frac{1}{2}\ln\{(x-y)^2 - 1\}$

$\therefore \quad \frac{dP}{dx} = \frac{2(x-y)\left(1 - \frac{dy}{dx}\right)}{2\{(x-y)^2 - 1\}}$

$= \frac{(x-y)\left(1 - \frac{dy}{dx}\right)}{(x-y)^2 - 1}$ ..... (ii)

Given $y(x-y)^2 = x$

$\ln y + 2\ln(x-y) = \ln x$

$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} + \frac{2}{(x-y)}\left(1 - \frac{dy}{dx}\right) = \frac{1}{x}$

$\Rightarrow \quad \frac{dy}{dx}\left(\frac{1}{y} - \frac{2}{x-y}\right) = \frac{1}{x} - \frac{2}{x-y}$

$= \frac{x-y-2x}{x(x-y)}$

$\Rightarrow \quad \frac{dy}{dx}\left(\frac{x-3y}{y(x-y)}\right) = -\frac{(x+y)}{x(x-y)}$

or $\quad \frac{dy}{dx} = -\frac{y(x+y)}{x(x-3y)}$

Now from equation (ii)

$$\frac{dP}{dx} = \frac{(x-y)\left\{1 + \frac{y(x+y)}{x(x-3y)}\right\}}{(x-y)^2 - 1}$$

$$= \frac{(x-y)\left\{\frac{x^2 - 2xy + y^2}{x(x-3y)}\right\}}{\left(\frac{x}{y} - 1\right)}$$

$$= \frac{y(x-y)^2}{x(x-3y)} = \frac{1}{x-3y}$$ ..... (iii)

$\therefore$ It is true from (i)

Hence, $\int \frac{dx}{x-3y}$

$= \frac{1}{2}\ln\{(x-y)^2-1\}$

$\because$ y is variable.

$\therefore$ $\int \frac{dx}{x-2y} \neq \ln(x-3y)$

8. (d) $\because$ $\int e^{x^2}dx$ cannot expressed in terms of elementary function, then integral is known as inexpressible or that is "cannot be found".

9. (a) $\because$ $\int \frac{1}{f(x)}dx = 2\ln|f(x)| + c$

Differentiating both sides with respect to x, then

$\because$ $\frac{1}{f(x)} = \frac{2}{f(x)}f'(x)$

or $f'(x) = \frac{1}{2}$

$\Rightarrow$ $\therefore f(x) = \frac{x}{2} + c$

If $f(0) = 0$,

then $f(x) = \frac{x}{2}$

10. (a) $\because$ $I'(x+\pi) = \int \sin^2(x+\pi)dx$

$= \int \sin^2 x dx$ $\{\because \sin^2(x+\pi) = \sin^2 x\} = I'(x)$

11. (a) $\left[\because \frac{d}{dx}(e^x+1) = e^x\right]$

$\therefore$ $\int \frac{e^x}{e^x+1}dx = \log(e^x+1) + C$

12. (a) $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

13. (a) $\frac{d}{dx}(1+\log x) = \frac{1}{x}$

14. (a) Put $x^x = z$

$\therefore$ $x^x(1+\log x)\,dx$ $\quad dz$

15. (a) $\frac{d}{dx}(10^x + x^{10})$

$= 10^x \log_x 10 + 10x^{10-1}$

$= 10^x \log_e 10 + 10x^9$

16. (a) Put $xe^x = t$.

17. (d) $F(x) = \int \sin^2 x dx = \frac{1}{2}\int(1-\cos 2x)dx =$

$= \frac{1}{2}\left(1 - \frac{1}{2}\sin 2x\right) + C$

$F(x+\pi) \neq F(x)$

$\therefore$ **Statement 1**: Statement to be written is False

Statement 2: $\sin^2(\pi + x) = \sin^2 x$, is True

$\forall x \in R$

MENTAL PREPARATION TEST

1. Evaluate $\int \frac{\cos x}{\sqrt{\sin^2 x - 2\sin x - 3}}dx$. ***[MP-2000]***

2. Evaluate $\int \frac{1}{x^2+4x+8}dx$. ***[CBSE-96, 2002]***

3. Evaluate $\int \frac{e^{3x}}{4e^{6x}-9}dx$. ***[HPSEB-96]***

4. Evaluate $\int \sqrt{1 \mp \sin x}\ dx$. Given $0 < x < \frac{\pi}{2}$ ***[PSB-94]***

5. Evaluate $\int \frac{1}{e^x+1}dx$. ***[CBSE-2003]***

6. Evaluate $\int \frac{1-\sin 2x}{x+\cos^2 x}dxI$. ***[CBSE-2000]***

7. Evaluate $\int \frac{1}{\sqrt{8+3x-x^2}}dx$. ***[CBSE-92]***

8. Evaluate $\int \sqrt{\frac{1-x}{1+x}}dx$. ***[PSB-93, 99(C)]***

9. Evaluate $\int \frac{x+\sin x}{1+\cos x}dx$.

[MP-92, 98, 99; PSB-92]

10. Evaluate $\int \frac{x^2 \tan^{-1} x}{1+x^2}dx$. ***[MP-88, 92, 99]***

11. Evaluate $\int \tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}dx$. Given $-\frac{\pi}{2} < x < \frac{\pi}{2}$ ***[CBSE-2003]***

12. Evaluate $\int \frac{2x}{\sqrt{1-x^2-x^4}}dx$.

[CBSE-2005(Foreign-I)]

13. Integrate the following with respect to x.
$\frac{\sin x + \text{cosec } x}{\tan x}$ ***[MP-93, 98, 99]***

14. Integrate with respect to x $\int (\log x)^2 dx$.

[CBSE-84, 88; PSB-91, 93]

15. Evaluate $\int \frac{e^x(1-\sin x)}{1-\cos x}dx$. ***[MP-99]***

16. Evaluate $\int \frac{dx}{2x^2+4x+6}$

[MP-2000, 2005 (A)]

17. Evaluate $\int \frac{1}{x(x^6+1)}dx$. ***[CBSE-96]***

18. Evaluate $\int x^2 \sin^{-1} x dx$. ***[MP-2001]***

19. Evaluate $\int \sec x . \log(\sec x + \tan x)dx$.

[CBSE-82]

20. Evaluate $\int \frac{4\sin x + 5\cos x}{5\sin x + 4\cos x}dx$.

[PSB-2001(C)]

21. Evaluate $\int \frac{2\cos 2x + \sec^2 x}{\sin 2x + \tan x - 5}dx$.

[HPSB-91]

22. Evaluate $\int \frac{1}{\sqrt{x+3}-\sqrt{x+2}}dx$.

[CBSE-2002]

23. Evaluate $\int \sin^{-1}\left(\frac{2x}{1+x^2}\right)dx$.

[MP-98, 99, 08; CBSE-83, 90; HSB-2002]

24. Evaluate $\int \frac{dx}{5+4\sin x}$.

[MP-99, 2000, 2001, 2002, 2003]

25. Evaluate $\int \frac{\sin 2x}{(a+b\cos x)^2}dx$.

[CBSE-2005 (O.D.)-I]

26. Evaluate $\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx$. ***[MP-93, 99]***

27. Evaluate $\int \sqrt{\frac{a+x}{a-x}}dx$. ***[MP-95, 99, 2008]***

28. Evaluate $\int \frac{xe^x}{(1+x)^2}dx$.

[MP-90, 95, 96, 97, 2000]

29. Evaluate $\int \sec^3 x dx$

[CBSE-96(C); PSB-91(C), 92(C); HSB-2001]

30. Evaluate $\int \sqrt{x^2+2x+5}dx$. ***[MP-98]***

31. Evaluate $\int \frac{10x^9 + 10^x \log_e 10}{10^x + x^{10}}dx$.

[MLNR-79]

32. Evaluate $\int \sqrt{2ax-x^2}dx$.

[PSB-90 (C)]

33. Find the value of $\int \frac{3x+1}{\sqrt{5-2x-x^2}}dx$.

[MP-2004 (C)]

34. Evaluate $\int \frac{x}{(x^2+1)(x+1)}dx$.

[CBSE-2002, 2005 (Foreign)-I]

35. Evaluate $\int \frac{2\cos x}{3\sin^2 x}dx$. ***[CBSE-2008]***

36. Evaluate $\int \sqrt{2x^2+3x+4}\ dx$. ***[MP-2008]***

TOPIC WISE WARMUP TEST 1

1. If $\int \frac{2x^2+3}{(x^2-1)(x^2+4)}dx = a\log\left(\frac{x+1}{x-1}\right) + b\tan^{-1}x/2 + c$, then (a, b) is equal to ***[PET (Raj.)-2000]***
   (a) $(-1/2, 1/2)$ (b) $(1/2, 1/2)$
   (c) $(-1, 1)$ (d) $(1, -1)$

2. $\int [f(x)g''(x) - f''(x)g(x)]dx$ is equal to ***[MP;-01; ICS (Pre)-04]***
   (a) $f(x)\,g'(x) - f'(x)\,g(x) + c$
   (b) $f(x)\,g'(x) + f'(x)\,g(x) + c$
   (c) $f'(x)\,g(x) - f(x)\,g'(x) + c$
   (d) None of these

3. $\int \frac{dx}{\sqrt{e^x+1}}$ equals ***[IIT (Allahabad)-2001]***
   (a) $2\sin h^{-1}(e^{x/2}) + c$
   (b) $-2\sin h^{-1}(e^{x/2}) + c$
   (c) $2\sin h^{-1}(e^{-x}) + c$
   (d) $-2\sin h^{-1}(e^{-x}) + c$

4. $\int \tan^3 2x \sec 2x\,dx$ equals ***[IIT-1977]***
   (a) $\sec^3 2x - 3\sec 2x + c$
   (b) $\sec^3 2x + 3\sec 2x + c$
   (c) $\frac{1}{6}[\sec^3 2x - 3\sec 2x] + c$
   (d) $\frac{1}{6}[\sec^3 2x + 3\sec 2x] + c$

5. $\int \frac{dx}{\sqrt{1-e^{2x}}}$ equals ***[MP 93; Roorkee 75]***
   (a) $\cos h^{-1}(e^{-x}) + c$
   (b) $-\cos h^{-1}(e^{x}) + c$
   (c) $\sin h^{-1}(e^{x}) + c$
   (d) $-\sin h^{-1}(e^{-x}) + c$

6. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}}dx = Ax + B\log(9e^{2x} - 4) + C$, then values of A and B are ***[IIT-1990]***
   (a) $-19/36, -35/36$
   (b) $-3/2, -35/36$
   (c) $-3/2, -35/36$
   (d) $-3/2, 35/36$

7. $\int \frac{dx}{\sqrt{2-3x-x^2}}$ is equal to ***[EAMCET-96]***
   (a) $\tan^{-1}\left(\frac{2x+3}{\sqrt{17}}\right) + c$
   (b) $\sec^{-1}\left(\frac{2x+3}{\sqrt{17}}\right) + c$
   (c) $\sin^{-1}\left(\frac{2x+3}{\sqrt{17}}\right) + c$
   (d) $\cos^{-1}\left(\frac{2x+3}{\sqrt{17}}\right) + c$

8. If $\int \frac{dx}{x\sqrt{1-x^3}} = a\log\left[\frac{\sqrt{1-x^3}-1}{\sqrt{1-x^3}+1}\right] + b$, then
   (a) $a = 1/3$ (b) $1 = 2/3$
   (c) $a = -1/3$ (d) $a = -2/3$

9. $\int \frac{\sin^{-1}\sqrt{x} - \cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x} + \cos^{-1}\sqrt{x}}dx$ is equal to ***[IIT-83; JEE(WB)-92]***
   (a) $\frac{2}{\pi}[(2x-1)\sin^{-1}\sqrt{x} + \sqrt{x(1-x)}] - x + c$
   (b) $\frac{2}{\pi}[(2x-1)\sin^{-1}\sqrt{x} - \sqrt{x(1-x)}] + x + c$
   (c) $\frac{2}{\pi}[(2x-1)\sin^{-1}\sqrt{x} + \sqrt{x(1-x)}] - x + c$
   (d) None of these

10. $\int \frac{x^2}{(x\sin x + \cos x)^2}dx$ is equal to ***[MNR-89; RPET-2000]***
    (a) $\frac{\sin x + x\cos x}{x\sin x + \cos x} + c$ (b) $\frac{\sin x - x\cos x}{x\sin x + \cos x} + c$
    (c) $\frac{\sin x - x\cos x}{x\sin x - \cos x} + c$ (d) None of these

11. $\int \frac{x-1}{(x+1)^3}e^x dx$ is equal to ***[MPPET-90; IIT-83]***
    (a) $\frac{e^x}{(x+1)^2} + c$ (b) $-\frac{e^x}{(x+1)^2} + c$
    (c) $\frac{e^x}{(x+1)^3} + c$ (d) $-\frac{e^x}{(x+1)^3} + c$

12. If $f(x) = \int \frac{2\sin x - \sin 2x}{x^3} dx, x \neq 0$, then $\lim_{x\to\infty} f'(x)$ is equal to

(a) 0 (b) 1
(c) 2 (d) 1/2

13. $\int \frac{e^x}{(2+e^x)(e^x+1)} dx$ is equal to ***[Kerala CEE-2003]***

(a) $\log\left(\frac{e^x+1}{e^x+2}\right)+c$ (b) $\log\left(\frac{e^x+2}{e^x+1}\right)+c$
(c) $\left(\frac{e^x+1}{e^x+2}\right)+c$ (d) $\left(\frac{e^x+2}{e^x+1}\right)+c$

14. $\int \frac{f'(x)}{f(x)\log[f(x)]} dx$ is equal to ***[Kerala(CEE)-2003]***

(a) $\frac{f(x)}{\log f(x)}+c$ (b) $f(x).\log f(x)+c$
(c) $\log[\log f(x)]+c$ (d) $\frac{1}{\log[\log f(x)]}+c$

15. $\int x^2(ax+b)^{-2} dx$ is equal to ***[Kerala PET-2003]***

(a) $\frac{2}{a^2}\left(x-\frac{b}{a}\log(ax+b)\right)+c$
(b) $\frac{2}{a^2}\left(x-\frac{b}{a}\log(ax+b)\right)-\frac{x^2}{a(ax+b)}+c$
(c) $\frac{2}{a^2}\left(x+\frac{b}{a}\log(ax+b)\right)+\frac{x^2}{a(ax+b)}+c$
(d) $\frac{2}{a^2}\left(x+\frac{b}{a}\log(ax+b)\right)-\frac{x^2}{a(ax+b)}+c$

16. $\int \frac{\left[\sqrt{1+x^2}+x\right]^n}{\sqrt{1+x^2}} dx$ is equal to ***[Pb. CET-2002; Roorkee-1995]***

(a) $(\sqrt{1+x^2}+x)^n+c$
(b) $\frac{1}{n}(\sqrt{1+x^2}+x)^n+c$
(c) $\frac{(\sqrt{1+x^2}+x)^{n+1}}{n+1}+c$
(d) None of these

17. $\int \frac{1}{x^2(x^4+1)^{3/4}} dx =$ ***[AMU-2003]***

(a) $\frac{(x^4+1)^{1/4}}{x}+C$ (b) $-\frac{(x^4+1)^{1/4}}{x}+C$
(c) $\frac{3}{4}\frac{(x^4+1)^{3/4}}{x}+C$ (d) $\frac{4}{3}\frac{(x^4+1)^{3/4}}{x}+C$

18. $\int \frac{x^5 dx}{\sqrt{(1+x^3)}} =$ ***[IIT-1975]***

(a) $\frac{2}{3}\sqrt{(1+x^3)}(x^3+2)$
(b) $\frac{2}{9}\sqrt{(1+x^3)}(x^3-4)$
(c) $\frac{2}{9}\sqrt{(1+x^3)}(x^3+4)$
(d) $\frac{2}{9}\sqrt{(1+x^3)}(x^3-2)$

19. $\int \frac{dx}{\sin(x-a)\sin(x-b)}$ is ***[Kerala (Engg.)-2005]***

(a) $\frac{1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|+c$
(b) $\frac{-1}{\sin(a-b)}\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|+c$
(c) $\log \sin(x-a)\sin(x-b)+c$
(d) $\log\left|\frac{\sin(x-a)}{\sin(x-b)}\right|$

20. $\int \frac{(\sin\theta+\cos\theta)}{\sqrt{\sin 2\theta}} d\theta$ is equal to ***[Kerala (Engg.)-2005]***

(a) $\log|\cos 0 - \sin 0 + \sqrt{\sin 2\theta}|$
(b) $\log|\sin 0 - \cos 0 + \sqrt{\sin 2\theta}|$
(c) $\sin^{-1}(\sin 0 - \cos 0)+c$
(d) $\sin^{-1}(\sin 0 + \cos 0)+c$

21. $\int \cos\sqrt{x} dx$ is equal to ***[PET (Raj.)-88, 99, 2001; IIT-77; Ranchi-90]***

(a) $2[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}]+c$
(b) $2[\sqrt{x}\sin\sqrt{x}-\cos\sqrt{x}]+c$
(c) $2[\cos\sqrt{x}-\sqrt{x}\sin\sqrt{x}]+c$
(d) $-2[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}]+c$

22. The value of $\int \sec^3 x dx$ will be ***[UPSEAT-99; PET (Raj.)-90, 92]***

(a) $\frac{1}{2}[\sec x \tan x + \log(\sec x + \tan x)]$

(b) $\frac{1}{3}[\sec x \tan x + \log(\sec x + \tan x)]$

(c) $\frac{1}{4}[\sec x \tan x + \log(\sec x + \tan x)]$

(d) $\frac{1}{8}[\sec x \tan x + \log(\sec x + \tan x)]$

23. $\int (1 + x - x^{-1}) e^{x - x^{-1}} dx$ is equal to ***[EAMCET-2003]***

(a) $(x+1)e^{x+x^{-1}} + c$ (b) $(x-1)e^{x+x^{-1}} + c$

(c) $-x\ e^{x+x^{-1}} + c$ (d) $x\ e^{x+x^{-1}} + c$

24. $\int \frac{dx}{\sqrt{x}(x+9)}$ is equal to ***[EAMCET-2001]***

(a) $\frac{2}{3}\tan^{-1}\sqrt{x} + c$ (b) $\frac{2}{3}\tan^{-1}\frac{\sqrt{x}}{3} + c$

(c) $\tan^{1}\sqrt{x} + c$ (d) $\tan^{1}(\sqrt{x}/3) + c$

25. $\int (x+1)^2 e^x dx$ is equal to ***[EAMCET-2001]***

(a) $x\,e^x + C$ (b) $x^2 e^x + C$

(c) $(x+1)e^x + C$ (d) $(x^2+1)e^x + C$

26. $\int \frac{2x}{(1-x^2)\sqrt{x^4-1}} dx$ is equal to ***[Roorkee-98]***

(a) $\sqrt{\frac{x^2+1}{x^2-1}} + C$ (b) $\sqrt{\frac{x^2-1}{x^2+1}} + C$

(c) $\sqrt{x^4-1} + C$ (d) None of these

TOPIC WISE WARMUP TEST 2

1. $\int \tan x \tan 2x \tan 3x\, dx$ is equal to ***[IIT (Hydrabad-2000)]***

(a) $\frac{1}{3}\log|\sec 3x| - \frac{1}{2}\log|\sec 2x| + \log|\sec x| + C$

(b) $\frac{1}{3}\log|\sec 3x| - \frac{1}{2}\log|\sec 2x| - \log|\sec x| + C$

(c) $\frac{1}{3}\log|\sec 3x| + \frac{1}{2}\log|\sec 2x| + \log|\sec x| + C$

(d) None of these

2. $\int \frac{dx}{\sin x - \cos x + \sqrt{2}}$ equals ***[MPPET-2002]***

(a) $-\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

(b) $\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

(c) $\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

(d) $-\frac{1}{\sqrt{2}}\cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + c$

3. $\int u\frac{d^2v}{dx^2}dx - \int v\frac{d^2u}{dx^2}dx$ is equal to ***[MP-1990]***

(a) $u\frac{dv}{dx} - v\frac{du}{dx} + c_1$ (b) $2\frac{du}{dx}\frac{dv}{dx} - c_1$

(c) $uv + c$ (d) c_1

4. $\int \frac{dx}{\sqrt{1-x-x^2}}$ is equal to

(a) $\sin^{1}\left(\frac{2x+1}{5}\right) + c$

(b) $\sin^{-1}\left(\frac{2x+1}{3}\right) + c$

(c) $-\sin^{-1}\left(\frac{2x+1}{3}\right) + c$

(d) $\sin^{-1}\left(\frac{2x+1}{\sqrt{5}}\right) + c$

5. $\int \frac{dx}{1+x+x^2+x^3} =$ ***[MP-91]***

(a) $\log\sqrt{1+x} - \frac{1}{2}\log\sqrt{1+x^2} + \frac{1}{2}\tan^{-1}x + c$

(b) $\log\sqrt{1+x} - \log\sqrt{1+x^2} + \tan^{-1}x + c$

(c) $\log\sqrt{1+x^2} - \log\sqrt{1+x} + \frac{1}{2}\tan^{-1}x + c$

(d) $\log\sqrt{1+x} + \tan^{-1}x + \log\sqrt{1+x^2} + c$

6. $\int x \sin^{-1} x\, dx$ is equal to

[MP-2000; MP-91]

(a) $\left(\frac{x^2}{2} - \frac{1}{4}\right)\sin^{-1}x + \frac{x}{4}\sqrt{1-x^2} + c$

(b) $\left(\frac{x^2}{2} + \frac{1}{4}\right)\sin^{-1}x + \frac{x}{4}\sqrt{1-x^2} + c$

(c) $\left(\frac{x^2}{2} - \frac{1}{4}\right)\sin^{-1}x - \frac{x}{4}\sqrt{1-x^2} + c$

(d) $\left(\frac{x^2}{2} + \frac{1}{4}\right)\sin^{-1}x - \frac{x}{4}\sqrt{1-x^2} + c$

7. $\int \frac{dx}{(x^2+1)(x^2+4)} = K\tan^{-1}x + L\tan^{-1}\frac{x}{2} + c$

[Roorkee-98; MPPET-1995]

(a) $K = 1/3$

(b) $L = 2/3$

(c) $-K = 1/3$

(d) $L = 1/6$

8. $\int x\cos^3 x^2 \sin x^2\, dx$ is equal to

(a) $-\frac{1}{8}\sin^4 x^2$ (b) $\frac{1}{8}\cos^4 x^2$

(c) $-\frac{1}{8}\cos^4 x^2$ (d) None of these

9. $\int \frac{x^2+1}{x(x^2-1)}dx$ is equal to

[MPCET-1999]

(a) $\log(x^2+1) - \log x$

(b) $\log(x^2+1) + \log x$

(c) $\log x - \log(x^2+1)$

(d) $\log(x^2-1) - \log x$

10. The two possible values $\frac{1}{2a}\log\frac{x-a}{x+a}$ and $\frac{1}{2a}\log\frac{a-x}{a+x}$ of the integral $\int\frac{dx}{x^2-a^2}$ differ by

(a) $\frac{i\pi}{2a}$ (b) $\frac{\pi}{2a}$

(c) $\frac{i\pi}{4a}$ (d) None of these

11. If $\int\frac{(2x^2+1)dx}{(x^2-4)(x^2-1)} = \log\left[\left(\frac{x+1}{x-1}\right)^a\left(\frac{x-2}{x+2}\right)^b\right] + c$, then values of a and b respectively are

[Roorkee (Screening)-2000]

(a) 1/2, 3/4 (b) –1, 3/2

(c) 1, 3/2 (d) –1/2, 3/4

12. The value of $\int\frac{e^x(2-x^2)dx}{(1-x)\sqrt{1-x^2}}$ is equal to

(a) $e^x\sqrt{\frac{1+x}{1-x}} + c$ (b) $e^x\sqrt{1+x} + c$

(c) $e^x\sqrt{1-x} + c$ (d) $e^x\sqrt{\frac{1-x}{1+x}} + c$

13. $\int\frac{x^2-1}{x\sqrt{(x^2+\alpha\ x+1)(x^2+\beta\ x+1)}}\,dx$ is equal to

(a) $2\log\left[\frac{\sqrt{x^2+\alpha\ x+1}+\sqrt{(x^2+\beta\ x+1)}}{\sqrt{x}}\right]+C$

(b) $2\log\left[\frac{\sqrt{x^2+\alpha\ x+1}-\sqrt{(x^2+\beta\ x+1)}}{\sqrt{x}}\right]+C$

(c) $\log\left[\sqrt{(x^2+\alpha\ x+1)}-\sqrt{(x^2+\beta\ x+1)}\right]+C$

(d) None of these

[Roorkee-1997; PUCET-2002]

14. $\int\frac{xe^x dx}{\sqrt{1+e^x}} = f(x)\ \sqrt{1+e^x} - 2\log(g(x)) + c$, then ***[MNR-1993]***

(a) $f(x) \neq 2x - 4$

(b) $g(x) = \frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}$

(c) $f(x) = x - 2$

(d) $g(x) = \frac{\sqrt{1+e^x}+1}{\sqrt{1+e^x}-1}$

15. $\int\frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$ is equal to ***[AMU-2001]***

(a) $\sqrt{\tan x} + \frac{\tan^{5/2}x}{5} + C$

(b) $\sqrt{\tan x} + \frac{2}{5}\tan^{5/2}x + C$

(c) $2\sqrt{\tan x} + \frac{2}{5}\tan^{5/2}x + C$

(d) None of these

16. $\int\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2}\frac{dx}{x}$ is equal to *[IIT(Screening)-97]*

(a) $2\left[\log\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right|+\cos^{-1}\sqrt{x}\right]+C$

(b) $-2\left[\log\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right|-\cos^{-1}\sqrt{x}\right]+C$

(c) $-2\left[\log\left|\frac{\sqrt{x}}{1+\sqrt{1-x}}\right|-\cos^{-1}\sqrt{x}\right]+C$

(d) None of these

17. If $\int f(x)\,dx \quad x\cos\pi x + c$, then $f(1/2) =$

(a) 0 (b) π

(c) $-\pi/2$ (d) $\pi/2$

18. $\int \sin x^\circ\,dx =$

(a) $\cos x^\circ$

(b) $-\cos x^\circ$

(c) $-(\cos x^\circ)\,\frac{180}{\pi}$

(d) None of these

19. $\int\frac{e^{\log\left(1+\frac{1}{x^2}\right)}dx}{x^2+\frac{1}{x^2}}dx =$

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(x-\frac{1}{x}\right)$

(b) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(x-\frac{1}{x}\right)$

(c) $\frac{1}{\sqrt{2}}\log\left(\frac{x^2-1}{x\sqrt{2}}\right)$

(d) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{x\sqrt{2}}\right)$

20. $\int\frac{dx}{(x+1)^{3/2}(x-1)^{1/2}}$

(a) $\sqrt{\frac{x+1}{x-1}}$ (b) $\sqrt{\frac{x-1}{x+1}}$

(c) $\sqrt{(x+1)(x-1)}$ (d) None of these

21. If $\int\frac{4x^3+\lambda 4^x}{(1+x^2)\sqrt{1-x^2}}=dx \quad \log(4^x+x^4)$, then $\lambda =$ *[MPPET-95]*

(a) 1 (b) $\log_e 4$

(c) $\log_4 e$ (d) 4

22. $\int\frac{dx}{(1+x^2)\sqrt{1-x^2}} =$

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{\sqrt{1-x^2}}{x\sqrt{2}}\right]+C$

(b) $\frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{x\sqrt{2}}{\sqrt{1-x^2}}\right]+C$

(c) $\sqrt{2}\tan^{-1}\left[\frac{\sqrt{1-x^2}}{x\sqrt{2}}\right]+C$

(d) $-\sqrt{2}\tan^{-1}\left[\frac{\sqrt{1-x^2}}{x\sqrt{2}}\right]+C$

23. If $\int\sqrt{2}\sqrt{1+\sin x}\;dx = -4\cos(ax+b)+c$, then (a, b) is equal to *[UPSEAT-2002]*

(a) $(1, \pi/2)$ (b) $(1/2, \pi/4)$

(c) $(1, 1)$ (d) $(-1/2, \pi/4)$

24. $\int\frac{(x+1)dx}{x^4+4x^3+2x^2-4x+1}$

(a) $\frac{-2}{x^2+2x-1}+c$

(b) $-\frac{1}{2(x^2+2x-1)}$

(c) $\frac{1}{2(x^2+2x-1)}$

(d) None of these

25. If $f'(x)=\frac{x}{\sqrt{1+x}}$ and $f(0)=0$, then $f(x) =$

(a) $\frac{2}{3}\{(1+x)^{3/2}-6(1+x)^{1/2}+1\}$

(b) $\frac{2}{3}\{(1+x)^{3/2}-3(1+x)^{1/2}+2\}$

(c) $\frac{2}{3}\{(1+x)^{3/2}-4(1+x)^{1/2}+2\}$

(d) $\frac{2}{3}\{(1+x)^{3/2}-3(1+x)^{1/2}+1\}$

26. The acceleration of a particle moving along the x-axis is given by $\frac{d^2x}{dt^2} = -4x$. If the particle starts from rest at $x = 5$, the velocity when it reaches $x = 3$ is

(a) 0 (b) 4

(c) 8 (d) None of these

27. $\int \frac{dx}{\sqrt{2-3x-x^2}}$ is equal to *[EAMCET-96]*

(a) $\tan^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+c$

(b) $\sec^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+c$

(c) $\sin^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+c$

(d) $\cos^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+c$

TOPIC WISE WARMUP TEST 1: SOLUTIONS

1. (b) $I = \int \frac{1}{x^2-1}dx + \int \frac{dx}{x^2+4}$

$= \frac{1}{2}\log\frac{x-1}{x+1} + \frac{1}{2}\tan^{-1}\frac{x}{2}$

hence $a = 1/2$, $b = 1/2$

2. (a) $f(x)g'(x) - \int f'(x)g'(x)dx -$

$\{g(x)f'(x) - \int g'(x)f'(x)dx\} + c$

$= f(x)\,g'(x) - g(x)f'(x) + c$

3. (b) $\int \frac{e^{-x/2}dx}{\sqrt{1+e^{-x}}} \Rightarrow \int \frac{-2dt}{\sqrt{1+t^2}}$

$= -2\sin h^{-1}(t) = -2\sin h^{-1}(e^{-x/2})$

4. (c) $\int(\sec^2 2x - 1)(\sec 2x \tan 2x)\,dx$;

$t = \sec 2x$

5. (b) $\int \frac{e^{-x}dx}{(e^{-x})^2 - 1}$; $t = e{-}x$;

$\int \frac{-dt}{\sqrt{t^2-1}} = -\cos h^{-1}(e^{-x}) + c$

6. (d) $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} = \frac{4e^{2x}+6}{9e^{2x}-4}$

$= \frac{A(9e^{2x}-4) + B(18e^{2x})}{9e^{2x}-4}$,

$4 = 9A + 18B$, $6 = -4A$

7. (c) $\int \frac{dx}{\sqrt{\frac{17}{4} - \left(x+\frac{3}{2}\right)^2}}$

$= \sin^{-1}\left(\frac{x+\frac{3}{2}}{\sqrt{17/2}}\right) = \sin^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)$

8. (c) $1 - x^3 = y^3 \Rightarrow -3x^2\,dx \quad 2y\,dy$

$\int \frac{2y\ dy}{-3x^3\sqrt{1-x^3}} = \frac{-2}{3}\int \frac{ydy}{(1-y^2)\sqrt{y^2}}$

$= \frac{-2}{3}\int \frac{dy}{1-y^2}$

$= \frac{-2}{3} \times \frac{1}{2\times 1}\log\frac{1+y}{1-y}$

$= \frac{-1}{3}\log\frac{1+y}{1-y}$

9. (a) $\frac{4}{\pi}\int \sin^{-1}\sqrt{x}dx - \int dx$

$= \frac{4}{\pi}[\int 0\sin 20d0] - x \quad (x = \sin^2\theta)$

$= \frac{4}{\pi}\left[\theta\left(\frac{-\cos 2\theta}{2}\right) - (1)\left(\frac{-\sin 2\theta}{4}\right)\right] - x$;

$\sin 20 = 2\sqrt{x}\sqrt{1-x}$, $\cos 20 = 1 - 2x^2$.

10. (b) $\int \underset{\text{I}}{\left(\frac{x}{\cos x}\right)} \underset{\text{II}}{\left(\frac{x\cos x}{(x\sin x+\cos x)^2}\right)} dx$

$$= \frac{\sin x - x\cos x}{x\sin x + \cos x}$$

11. (a) $\int e^x \left\{\frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}\right\} dx = \frac{e^x}{(x+1)^2}$

12. (b) $\lim_{x\to 0} \frac{2\sin x - \sin 2x}{x^3}$

$$= \lim_{x\to 0} \frac{2-2\cos x}{x^2} = \lim_{x\to 0} \frac{2\times 2\sin^2\frac{x}{2}}{x^2} = 1$$

13. (a) Let $e^x = t \Rightarrow ex\,dx \quad dt$

$$\therefore \quad I = \int \frac{t}{(2+t)(1+t)} dt$$

$$= \int \left[\left(\frac{1}{1+t}\right) - \left(\frac{1}{2+t}\right)\right] dt$$

$$= \log(1+t) - \log(2+t) + C$$

$$= \log\left(\frac{1+e^x}{2+e^x}\right) + C$$

14. (c) Let $\log[f(x)] = t = \log t + c = \log[\log f(x)] + c$

$$\frac{1}{f(x)} . f'(x)dx = dt \quad \therefore I = \int \frac{1}{t} dt$$

15. (b) Let $ax + b = t$

$$\Rightarrow \quad dx = \frac{1}{a} dt \text{ and } x = \left(\frac{t-b}{a}\right)$$

$$\therefore \quad I = \frac{1}{a^3} \int \frac{(t-b)^2}{t^2} dt$$

$$= \frac{1}{a^3} \int \left(1 + \frac{b^2}{t^2} - \frac{2b}{t}\right) dt$$

$$= \frac{1}{a^3}\left(t - \frac{b^2}{t} - 2b\log t\right) + c$$

$$= \frac{1}{a^3}\left(ax + b - \frac{b^2}{ax+b} - 2b\log(ax+b)\right) + c$$

$$= \frac{2}{a^2}\left(x - \frac{b}{a}\log(ax+b)\right) - \frac{x^2}{a(ax+b)} + c$$

16. (b) Put $(\sqrt{1+x^2} + x))^n = z$

$$\therefore n(\sqrt{1+x^2} + x))^{n-1} . \left(\frac{2x}{2\sqrt{1+x^2}} + 1\right) dx = dz$$

$$\Rightarrow \frac{(\sqrt{1+x^2} + x)^n}{\sqrt{1+x^2}} dx = \frac{dz}{n}$$

$$\therefore \quad \text{given integral} = \int \frac{dz}{n} = \frac{1}{n} z + c$$

$$= \frac{1}{n}(\sqrt{1+x^2} + x)^n + c$$

17. (b) $\int \frac{dx}{x^2(1+x^4)^{3/4}} = \int \frac{dx}{x^5\left(1+\frac{1}{x^4}\right)^{3/4}}$

$$= -\frac{1}{4}\int \left(1+\frac{1}{x^4}\right)^{-3/4} . \left(\frac{-4}{x^5}\right) dx$$

$$= -\frac{1}{4} \frac{\left(1+\frac{1}{x^4}\right)^{1/4}}{1/4} + c$$

$$\left[\because \frac{d}{dx}\left(1+\frac{1}{x^4}\right) = -\frac{4}{x^5}\right]$$

$$= -\left(1+\frac{1}{x^4}\right)^{1/4} + c = \frac{-(x^4+1)^{1/4}}{x} + c$$

18. (d) Putting $1 + x^3 = t^2$

$\Rightarrow \quad 3x^2\,dx \quad 2t\,dt.$

Also $\quad x^3 = t^2 - 1$

$$I = \int \frac{t^2-1}{t}\left(\frac{2t}{3} dt\right) = \frac{2}{3}\int (t^2-1) dt$$

$$= \frac{2}{3}\left(\frac{t^3}{3} - t\right) = \frac{2}{9} t(t^3 - 3)$$

$$= \frac{2}{9}\sqrt{1+x^3}(x^3 - 2)$$

19. (a) $\int \frac{dx}{\sin(x-a)\sin(x-b)} = \frac{1}{\sin(a-b)}$

$$\int \frac{\sin\{(x-b)-(x-a)\}}{\sin(x-a)\sin(x-b)} dx$$

$$= \frac{1}{\sin(a-b)}\left[\int \cot(x-a)dx - \int \cot(x-b)dx\right]$$

$$= \frac{1}{\sin(a-b)} \left[\log\sin(x-a) - \log\sin(x-b)\right] + c$$

$$= \frac{1}{\sin(a-b)} \log\left|\frac{\sin(x-a)}{\sin(x-b)}\right| + c$$

20. (c) Let $I = \int \frac{\sin\theta + \cos\theta}{\sqrt{2\sin 0\cos 0}} d\theta$

$$= \int \frac{\sin\theta + \cos\theta}{\sqrt{1-(\sin 0 - \cos 0)^2}} d\theta$$

Let $\sin\theta - \cos\theta = t$

$\Rightarrow$ $(\cos\theta + \sin\theta)\, d\theta = dt$

$\therefore$ $I = \int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + c$

$= \sin^{-1}(\sin 0 - \cos 0) + c$

21. (a) Putting $\sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$

$\Rightarrow$ $dx = 2t\ dt$

$\therefore$ $\int 2t.\cos t dt = 2\left[t.\sin t - \int \sin t dt\right]$

$= 2t\sin t + 2\cos t$

$= 2\left[\sqrt{x}\sin\sqrt{x} + \cos\sqrt{x}\right] + c$

22. (a) $\int \sec^3 x dx = \int \sec x.\sec^2 x dx = I$ (say)

$\Rightarrow$ $I = \sec x\tan x - \int \sec x\tan^2 x dx$

$\Rightarrow$ $I = \sec x\tan x - \int \sec x(\sec^2 x - 1) dx$

$\Rightarrow$ $I = \sec x\tan x - \int \sec^3 x dx + \int \sec x dx$

$\Rightarrow$ $2I = \sec x\tan x + \log(\sec x + \tan x)$

$\Rightarrow$ $I = \frac{1}{2}\left[\sec x\tan x + \log(\sec x + \tan x)\right]$

23. (d) $I = \int \left[e^{x+x^{-1}}(x - x^{-1}) + e^{x-x^{-1}}\right] dx$

$= \int \left[x\left\{e^{x-x^{-1}}\left(1 - \frac{1}{x^2}\right)\right\} + e^{x+x^{-1}}\right] dx$

$= \int \left[xf'(x) + f(x)\right] dx = xf(x) + c$

24. (b) $\int \frac{dx}{\sqrt{x}(x+9)}$ Put $\sqrt{x} = 3\tan 0$

$\therefore$ $\frac{2}{2\sqrt{x}} dx = 3\sec^2\theta d\theta$

$= \int \frac{2(3\sec^2\theta d\theta)}{9\tan^2\theta + 9} = \frac{2}{3}\int \frac{\sec^2\theta}{\sec^2\theta} d\theta$

$= \frac{2}{3}\int d\theta = \frac{2}{3}\theta + c$

$= \frac{2}{3}\tan^{-1}\frac{\sqrt{x}}{3} + c$

25. (d) $\int (x+1)^2 e^x dx = \int (x^2 + 2x + 1) e^x dx$

$= \int (x^2 + 2x) e^x dx + \int e^x dx$

[Type $\int [f(x) + f'(x)] e^x dx = e^x f(x)$]

$= x^2 e^x + e^x + c = (x^2 + 1) e^x + c$

26. (a) $\int \frac{2x}{(1-x^2)\sqrt{x^4 - 1}} dx$

$= \int \frac{-2x}{(x^2-1)^{3/2}\sqrt{x^2+1}} dx$

Put $\sqrt{\frac{x^2+1}{x^2-1}} = z$

$\therefore \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{1/2} \cdot \frac{(x^2-1).2x - (x^2+1)2x}{(x^2-1)^2} dx = dz$

$\Rightarrow$ $\frac{\sqrt{x^2-1}}{\sqrt{x^2+1}} \cdot \frac{-2x}{(x^2-1)^2} dx = dz$

$\Rightarrow$ $\frac{-2x}{(x^2-1)^{3/2}\sqrt{x^2+1}} dx = dz$

$\therefore$ given integral $= \int dz = z + c$

$= \sqrt{\frac{x^2+1}{x^2-1}} + c$

TOPIC WISE WARMUP TEST 2: SOLUTIONS

1. (b) $\tan 3x = \tan(2x + x)$

$$= \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x}$$

$\therefore \quad \tan 3x - \tan 3x \tan 2x \tan x$

$= \tan 2x + \tan x$

$\therefore \quad \tan 3x \cdot \tan 2x \tan x$

$= \tan 3x - \tan 2x - \tan x$

$\therefore \quad \int \tan 3x \tan 2x \tan x dx = \frac{1}{3}$

$\log|\sec 3x| - \frac{1}{2}\log|\sec 2x| - \log|\sec x| + c$

2. (d) We have $\int \frac{1}{\sin x - \cos x + \sqrt{2}} dx$

$$= \frac{1}{\sqrt{2}} \int \frac{1}{\left(\frac{1}{\sqrt{2}}\sin x - \frac{1}{\sqrt{2}}\cos x\right) + 1} dx$$

$$= \frac{1}{\sqrt{2}} \int \frac{1}{1 - \cos\left(x + \frac{\pi}{4}\right)} dx$$

$$= \frac{1}{2\sqrt{2}} \int \operatorname{cosec}^2\left(\frac{x}{2} + \frac{\pi}{8}\right) dx$$

$$= \frac{1}{\sqrt{2}} \int \operatorname{cosec}^2\left(\frac{x}{2} + \frac{\pi}{8}\right) d\left(\frac{x}{2} + \frac{\pi}{8}\right)$$

$$= -\frac{1}{\sqrt{2}} \cot\left(\frac{x}{2} + \frac{\pi}{8}\right) + C$$

3. (a) Integrate by parts taking u and v as first and second parts in given integrals.

4. (d) $\int \frac{dx}{\sqrt{1 - x - x^2}} = \int \frac{dx}{\sqrt{1 - x - x^2 + \frac{1}{4} - \frac{1}{4}}}$

$$= \int \frac{dx}{\sqrt{1 + \frac{1}{4} - \left(x^2 + x + \frac{1}{4}\right)}}$$

$$= \int \frac{dx}{\sqrt{\frac{5}{4} - \left(x + \frac{1}{2}\right)^2}}$$

$$= \int \frac{dx}{\sqrt{\left(\frac{\sqrt{5}}{4}\right)^2 - \left(x + \frac{1}{2}\right)^2}}$$

$$= \sin^{-1}\left[\frac{x + \frac{1}{2}}{\sqrt{5}/2}\right] + C = \sin^{-1}\left(\frac{2x + 1}{\sqrt{5}}\right) + C$$

5. (a) $I = \int \frac{dx}{(1 + x)(x^2 + 1)} = \frac{1}{2}\int \frac{1}{1 + x^2} dx +$

$$\frac{1}{2}\int \frac{1}{1 + x} dx - \frac{1}{2}\int \frac{x}{1 + x^2} dx$$

$$= \log\sqrt{1 + x} - \frac{1}{2}\log\sqrt{1 + x^2} + \frac{1}{2}\tan^{-1} x + c$$

6. (a) Integrating by parts

$$I = \frac{x^2}{2}\sin^{-1} x - \int \frac{1}{\sqrt{1 - x^2}} \cdot \frac{x^2}{2} dx$$

$$= \frac{x^2}{2}\sin^{-1} x + \frac{1}{2}\int \sqrt{1 - x^2} dx - \frac{1}{2}\int \frac{dx}{\sqrt{1 - x^2}}$$

$$= \left(\frac{x^2}{2} - \frac{1}{4}\right)\sin^{-1} x + \frac{x}{4}\sqrt{1 - x^2} + C$$

7. (a) $\frac{1}{(x^2 + 1)(x^2 + 4)} = \frac{1}{3}\left(\frac{1}{x^2 + 1} - \frac{1}{x^2 + 4}\right)$

$\therefore \quad \int \frac{dx}{(x^2 + 1)(x^2 + 4)}$

$$= \frac{1}{3}\int \frac{dx}{x^2 + 1} - \frac{1}{3}\int \frac{dx}{x^2 + 2^2}$$

$$= \frac{1}{3}\tan^{-1} x - \frac{1}{3} \cdot \frac{1}{2}\tan^{-1}\frac{x}{2} + c$$

$$= \frac{1}{3}\tan^{-1} x - \frac{1}{6}\tan^{-1}\frac{x}{2} + c$$

$$= K\tan^{-1} x + L\tan^{-1}\frac{x}{2} + c$$

$\therefore \quad K = 1/3, L = -1/6$

8. (c) Put $t = \cos x^2 \Rightarrow dt = -2x \sin x^2\, dx$

$$\Rightarrow x \sin x^2\, dx = -\frac{dt}{2}$$

Then $\int x\cos^3 x^2 \sin x^2 dx = -\frac{1}{2}\int t^3 dt$

$$= \frac{-1}{2}\cdot\frac{t^4}{4} + C = -\frac{1}{8}\cos^4 x^2 + C$$

9. (d) $\int \frac{x^2+1}{x(x^2-1)}dx = \int \frac{x^2+1}{x(x-1)(x+1)}dx$

$$= \int\left(\frac{-1}{x} + \frac{1}{x-1} + \frac{1}{x+1}\right)dx$$

$= \log(x-1) + \log(x+1) - \log x + c = \log\left(\frac{x^2-1}{x}\right) + c$

$= \log(x^2-1) - \log x + c$

10. (a) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left(\frac{x-a}{x+a}\right) + c$ .....(1)

$$= \frac{1}{2a}\log\left(-\frac{a-x}{a+x}\right) + c$$

$$= \frac{1}{2a}\log(-1) + \frac{1}{2a}\log\left(\frac{a-x}{a+x}\right) + c$$

$$= \frac{1}{2a}\log(\pi i) + \frac{1}{2a}\log\left(\frac{a-x}{a+x}\right) + c$$

..... (2)

(1) and (2) shows that required difference $= i\,\pi/2a$

11. (a) $I = \int \frac{3}{x^2-4}dx - \int \frac{1}{x^2-1}dx$ (resolving into partial fraction)

$$= \frac{3}{4}\log\left(\frac{x-2}{x+2}\right) - \frac{1}{2}\log\left(\frac{x-1}{x+1}\right) + C$$

$\Rightarrow \quad a = 1/2, \quad b = 3/4.$

12. (a) $\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}}dx = \int e^x \frac{1+(1-x^2)}{(1-x)\sqrt{1-x^2}}dx$

$$= \int e^x\cdot\left[\frac{1}{(1-x)\sqrt{1-x^2}} + \sqrt{\frac{1+x}{1-x}}\right]dx$$

Now $\frac{d}{dx}\left(\sqrt{\frac{1+x}{1-x}}\right)$

$$= \frac{1}{2}\cdot\left(\frac{1+x}{1-x}\right)^{-1/2}\cdot\frac{(1-x).1-(1+x)(-1)}{(1-x)^2}$$

$$= \frac{1}{2}\cdot\sqrt{\frac{1-x}{1+x}}\cdot\frac{1-x+1+x}{(1-x)^2}$$

$$= \frac{1}{\sqrt{1+x}(1-x)^{3/2}} = \frac{1}{(-x)\sqrt{1-x^2}}$$

$\therefore \quad \int \frac{e^{x(2-x^2)}}{(1-x)\sqrt{1-x^2}} = \int e^x(f(x) + f'(x))dx$

where $f(x) = \sqrt{\frac{1+x}{1-x}}$

$= e^x f(x) + C = e^x \cdot \sqrt{\frac{1+x}{1-x}} + C$

13. (a) We have $\int \frac{x^2-1}{x\sqrt{x^2+\alpha x+1)(x^2+\beta x+1)}}dx$

$$= \int \frac{1-\frac{1}{x^2}}{\sqrt{\left(x+\frac{1}{x}+\alpha\right)\left(x+\frac{1}{x}+\beta\right)}}dx$$

$$= \int \frac{1}{\sqrt{\left(x+\frac{1}{x}\right)^2 + (\alpha+\beta)\left(x+\frac{1}{x}\right) + \alpha\beta}}d\left(x+\frac{1}{x}\right)$$

$= \int \frac{1}{\sqrt{t^2+(\alpha+\beta)t+\alpha\beta}}dt$, where, $t = x + \frac{1}{x}$

$$= \int \frac{1}{\sqrt{\left(t+\frac{\alpha+\beta}{2}\right)^2 - \left(\frac{\alpha-\beta}{2}\right)^2}}dt$$

$$= \log\left|\left(t+\frac{\alpha-\beta}{2}\right) + \sqrt{\left(t+\frac{\alpha+\beta}{2}\right)^2 - \left(\frac{\alpha-\beta}{2}\right)^2}\right| + C$$

$$= \log\left|t+\frac{\alpha+\beta}{2} + \sqrt{t^2+(\alpha+\beta)t+\alpha\beta}\right| + C$$

$= \log\left|(t+\alpha)+(t+\beta)+2\sqrt{(t+\alpha)(t+\beta)}\right| + C - \log 2$

$= \log\{\sqrt{t+\alpha}+\sqrt{t+\beta}\}^2 + C_1$

$= 2\log\left\{\frac{\sqrt{x^2+\alpha x+1}+\sqrt{x^2+\beta x+1}}{\sqrt{x}}\right\} + C_1$

14. (b) Put $1 + e^x = t^2 \quad \therefore e^x\,dx = 2t\,dt$

Also $x = \log(t^2 - 1) \quad \therefore I = \int \frac{\log(t^2-1).2t\,dt}{t}$

$= 2\int \log(t^2-1)dt = 2\int \log(t^2-1).1\,dt$

$= 2\left[\log(t^2-1).t - \int \frac{2t}{t^2-1}t\,dt\right] = 2t\log(t^2-1)$

$-4\int \frac{t^2-1+1}{t^2-1}dt$

$= 2t\log(t^2-1) - 4t - 4\log\left(\frac{t-1}{t+1}\right) + C$

$= 2\sqrt{1+e^x}\,,x - 4\sqrt{1+e^x} - 4\,.\,\frac{1}{2}\log$

$\left(\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right) + C$

$= 2(x-2)\sqrt{1+e^x} - 2\log\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C$

$\therefore f(x) = 2(x-2)\,,\, g(x) = \frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}$

15. (a) $I = \int \frac{1}{\cos^3 x\sqrt{4\sin x\cos x}}dx$

$= \int \frac{\sec^4 x}{2\sqrt{\tan x}}dx = \int \frac{(1+\tan^2 x)}{2\sqrt{\tan x}}\sec^2 x\,dx$

$= \sqrt{\tan x} + \frac{1}{5}\sqrt{\tan^5 x} + C$

16. (b) Put $x = \cos^2 0 \Rightarrow dx \quad -2\cos 0\sin 0\,d0$

$\therefore$ given integral

$= \int\left(\frac{1-\cos\theta}{1+\cos\theta}\right)^{1/2}\left(\frac{-2\cos\theta\sin\theta\,d\theta}{\cos^2\theta}\right)$

$= -2\int \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cdot\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\cos\theta}d\theta$

$= -2\int \frac{2\sin^2\frac{\theta}{2}}{\cos\theta}d\theta = -2\int\frac{1-\cos\theta}{\cos\theta}d\theta$

$= -2\int(\sec\theta - 1)d\theta$

$= -2\int \frac{2\sin^2\frac{\theta}{2}}{\cos\theta}d\theta = -2\int\frac{1-\cos\theta}{\cos\theta}d\theta$

$= -2\int(\sec\theta - 1)d\theta$

$= -2\,[\log|\sec\theta + \tan\theta| - \theta]$

$= -2\left[\log\left|\frac{1}{\sqrt{x}} + \frac{\sqrt{1-x}}{\sqrt{x}}\right| - \cos^{-1}\sqrt{x}\right] + C$

$= -2\left[\log\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right| - \cos^{-1}\sqrt{x}\right] + C$

17. (c) $\int f'(x)dx = x\cos\pi x + C$

Differentiate both sides with respect to x

$f(x) = x\ (-\sin\pi x \times \pi) + \cos\pi x = -\pi x\sin\pi x + \cos\pi x$

$\Rightarrow \quad f(1/2) = -\pi\,.\,\frac{1}{2}\sin\frac{\pi}{2} + \cos\frac{\pi}{2} = -\frac{\pi}{2}$

18. (c) $\int \sin x^\circ dx = \int \sin\frac{\pi}{180^\circ}x\,dx$

$= \frac{-\cos\frac{\pi}{180}x}{\frac{\pi}{180}} = -(\cos x^\circ)\frac{180}{\pi}$

19. (d) Since $e^{\log x} = x$ therefore,

$e^{\log\left(1-\frac{1}{x^2}\right)} = 1 + \frac{1}{x^2}$

Put $\quad x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2}\right)dx = dt$

and $\quad x^2 + \frac{1}{x^2} - 2 = t^2$

$\Rightarrow \quad x^2 + \frac{1}{x^2} = t^2 + 2$

$\therefore \quad \int \frac{e^{\log\left(1+\frac{1}{x^2}\right)}}{x^2 + \frac{1}{x^2}}dx = \int \frac{1+\frac{1}{x^2}}{x^2 + \frac{1}{x^2} + 2 - 2}dx$

$= \int \frac{1+\frac{1}{x^2}}{\left(x^2 + \frac{1}{x^2} - 2\right) + 2}dx$

$= \int \frac{1+\frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + (\sqrt{2})^2}dx$

$= \int \frac{dt}{t^2 + (\sqrt{2})^2} = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right) + C$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{x-\frac{1}{x}}{\sqrt{2}}\right]+C$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{x\sqrt{2}}\right)+C$$

20. (b) Since $\frac{3}{2}+\frac{1}{2}=2=$ An even integer.

Put $\frac{x+1}{x-1}=z$

$$\therefore \quad \frac{(x-1).1-(x+1).1}{(x-1)^2}dx=dz$$

$$\Rightarrow \quad -\frac{2dx}{(x-1)^2}=dz \quad \therefore \int\frac{dx}{(x+1)^{3/2}(x-1)^{1/2}}$$

$$= \frac{1}{2}\int\frac{-(x-1)^2\ dz}{(x-1)^{3/2}\ Z^{3/2}\ (x-1)^{1/2}}$$

$$= \frac{1}{2}\int -Z^{-3/2}dz = -\frac{1}{2}\frac{Z^{-1/2}}{-\frac{1}{2}}+C$$

$$= \frac{1}{\sqrt{z}}+C=\sqrt{\frac{x-1}{x+1}}+C$$

21. (b) $\int\frac{4x^3+\lambda 4^x}{4^x+x^4}dx=\log(4^x+x^4)$ diff. both sides with respect to x, we get

$$\frac{4x^3+\lambda 4^x}{4^x+x^4}+\frac{1}{4^x+x^4}\ (4^x\log_e 4+4x^3)$$

$$\Rightarrow \quad \lambda 4^x=4^x\log_e 4 \Rightarrow \lambda=\log_e 4$$

22. (b) Put $x=\frac{1}{t}\ \therefore\ 4x=-\frac{1}{t^2}\ dt$

$\therefore$ given integral

$$= \int\frac{-\frac{1}{t^2}dt}{\left(1+\frac{1}{t^2}\right)\sqrt{1-\frac{1}{t^2}}}=\int\frac{-tdt}{(t^2+1)\sqrt{t^2-1}}$$

Put $t^2-1=u^2\ \therefore\ 2t\,dt=2u\,du \Rightarrow t\,dt=u\,du$

$\therefore$ given integral

$$=-\int\frac{udu}{(u^2+2)u}=-\int\frac{du}{u^2+2}=-\frac{1}{\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}}$$

$$= -\frac{1}{\sqrt{2}}\tan^{-1}\frac{\sqrt{t^2-1}}{\sqrt{2}}+c_1$$

$$= -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\sqrt{\frac{1}{x^2}-1}\right)+c_1$$

$$= -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{2}x}\right)+c_1$$

$$= -\frac{1}{\sqrt{2}}\left[\frac{\pi}{2}-\cot^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{2}x}\right)\right]+$$

$c_1\left[\tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}\right]$

$$= \frac{1}{\sqrt{2}}\cot^{-1}\left(\frac{\sqrt{1-x^2}}{\sqrt{2}x}\right)+c_1-\frac{\pi}{2\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{2}x}{\sqrt{1-x^2}}\right)+c \text{ where } c=$$

$$c_1-\frac{\pi}{2\sqrt{2}}$$

23. (b) $\int\sqrt{2}\sqrt{1+\sin x}dx=\sqrt{2}\int\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)dx$

$$= 2\int\left(\cos\frac{\pi}{4}\cos\frac{x}{2}+\sin\frac{\pi}{4}\sin\frac{x}{2}\right)dx.$$

$$= 2\int\left[\cos\left(\frac{\pi}{4}-\frac{x}{2}\right)\right]dx=-4\sin\left(\frac{\pi}{4}-\frac{x}{2}\right)+c$$

$$= -4\cos\left(\frac{\pi}{2}-\frac{\pi}{4}+\frac{x}{2}\right)+c$$

$$= -4\cos\left(\frac{\pi}{4}+\frac{x}{2}\right)+c \quad \therefore \quad a=\frac{1}{2}, b=\frac{\pi}{4}$$

$\therefore$ answer is (b).

24. (b) $\int\frac{(x+1)dx}{x^4+4x^3+2x^2-4x+1}$

$$= \int\frac{(x+1)dx}{(x^2+2x-1)^2}$$

Let $x^2+2x-1=t$

$\Rightarrow$ $(2x+2)\,dx \quad dt$

$\Rightarrow$ $2\,(x+1)\,dx \quad dt$

$\Rightarrow$ $(x+1)\,dx \quad dt/2$

$$= \int\frac{dt}{2t^2}=-\frac{1}{2t}+C=-\frac{1}{2(x^2+2x-1)}$$

25. (b) $f'(x) = \frac{x}{\sqrt{1+x}} \Rightarrow \int f'(x)dx = \int \frac{x}{\sqrt{1+x}}dx$

$\Rightarrow \quad f(x) = \int \sqrt{1+x}dx - \int \frac{1}{\sqrt{1+x}}dx$

$= \frac{2}{3}(1+x)^{3/2} - 2(1+x)^{1/2} + c$

$\Rightarrow f(0 = \frac{2}{3} \times 1 - 2 \times 1 + c \Rightarrow 0$

$= -\frac{4}{3} + c \Rightarrow c = \frac{4}{3}$

hence $f(x) = \frac{2}{3} \times (1+x)^{3/2} - 2(1+x)^{1/2} + \frac{4}{3}$

$\Rightarrow f(x) = \frac{2}{3}[(1+x)^{3/2} - 3(1+x)^{1/2} + 2]$

26. (d) $\frac{d^2x}{dt^2} = -4x$ on integration

$\Rightarrow \frac{dx}{dt} = -\frac{4x^2}{2} + C = -2x^2 + C$

at $x = 5$, then $\frac{dy}{dt} = 0$, So $O = -2.(5)^2 + C$

$\Rightarrow C = 50$

Hence $\frac{dx}{dt} = -2x^2 + 50$, Put $x = 3$, we get

$\frac{dx}{dt} = -2.(3)^2 + 50 \Rightarrow = -2 \times 9 + 50 = 32.$

27. (c) $\int \frac{dx}{\sqrt{\frac{17}{4} - (x + 3/2)^2}}$

$= \sin^{-1} \frac{2}{\sqrt{17}} (x + 3/2) + c = \sin^{-1}\left(\frac{2x+3}{\sqrt{17}}\right) + c$

ANSWERS

LECTURE 1

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (b)	**4.** (b)	**7.** (d)
2. (a)	**5.** (a)	**8.** (c)
3. (a)	**6.** (a)	**9.** (b)

Worksheet: To Check the Preparation Level

1. (c)	**4.** (a)	**7.** (d)
2. (a)	**5.** (a)	
3. (c)	**6.** (d)	

LECTURE 2

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (b)	**5.** (c, d)	**9.** (b)	**13.** (c)
2. (b)	**6.** (d)	**10.** (c)	**14.** (b)
3. (a)	**7.** (b)	**11.** (b)	
4. (a)	**8.** (a)	**12.** (c)	

Worksheet: To Check the Preparation Level

1. (b)	**5.** (a)	**9.** (b)	**13.** (d)
2. (b)	**6.** (b)	**10.** (b)	**14.** (b)
3. (a)	**7.** (a)	**11.** (d)	**15.** (b)
4. (c)	**8.** (b)	**12.** (c)	**16.** (c)

LECTURE 3

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (b)	**6.** (d)	**11.** (b)	**16.** (b)
2. (c)	**7.** (a)	**12.** (c)	**17.** (c)
3. (a)	**8.** (c)	**13.** (a)	
4. (b)	**9.** (a)	**14.** (a)	
5. (d)	**10.** (a)	**15.** (b)	

Worksheet: To Check the Preparation Level

1. (d)	**5.** (c)	**9.** (b)	**13.** (a)
2. (b)	**6.** (b)	**10.** (a)	**14.** (a)
3. (a)	**7.** (a)	**11.** (b)	**15.** (d)
4. (b)	**8.** (a)	**12.** (a)	

LECTURE 4

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (a)	**6.** (c)	**11.** (b)	**16.** (b)
2. (c)	**7.** (c)	**12.** (d)	**17.** (a)
3. (a)	**8.** (a)	**13.** (a)	**18.** (d)
4. (b)	**9.** (b)	**14.** (a)	**19.** (b)
5. (d)	**10.** (d)	**15.** (c)	

Worksheet: To Check the Preparation Level

1. (c)	**5.** (a)	**9.** (c)	**13.** (a)
2. (a)	**6.** (b)	**10.** (a)	
3. (b)	**7.** (a)	**11.** (d)	
4. (b)	**8.** (c)	**12.** (a)	

LECTURE 5

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (d)	**3.** (c)	**5.** (d)	**7.** (b)
2. (c)	**4.** (a)	**6.** (b)	**8.** (c)

Worksheet: To Check the Preparation Level

1. (a)	**4.** (a)	**7.** (a)	**10.** (d)
2. (b)	**5.** (d)	**8.** (d)	
3. (a)	**6.** (d)	**9.** (b)	

LECTURE 6

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (c)	**4.** (c)	**7.** (c)	**10.** (a)
2. (a)	**5.** (b)	**8.** (d)	**11.** (b)
3. (d)	**6.** (a)	**9.** (a)	**12.** (c)

Worksheet: To Check the Preparation Level

1. (d)	**5.** (c)	**9.** (a)	**13.** (d)
2. (b)	**6.** (c)	**10.** (b)	**14.** (a)
3. (d)	**7.** (d)	**11.** (b)	
4. (b)	**8.** (a)	**12.** (c)	

LECTURE 7

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (b)	**5.** (b)	**9.** (a)	**13.** (a)
2. (d)	**6.** (b)	**10.** (a)	
3. (d)	**7.** (a)	**11.** (c)	
4. (a)	**8.** (a)	**12.** (d)	

Worksheet: To Check the Preparation Level

1. (a)	**5.** (c)	**9.** (b)	**13.** (a)
2. (a)	**6.** (c)	**10.** (b)	**14.** (a)
3. (b)	**7.** (b)	**11.** (b)	
4. (d)	**8.** (c)	**12.** (a)	

LECTURE 8

Mental Preparation Test

1. $\log\left[\sin x - 1 + \sqrt{\sin^2 x - 2\sin x - 3}\right]$
2. $\frac{1}{2}\tan^{-1}\left(\frac{x+2}{2}\right) + C$
3. $\frac{1}{36}\log\left|\frac{2e^{3x}-3}{2e^{3x}+3}\right|$
4. $2\left(\sin\frac{x}{2} \pm \cos\frac{x}{2}\right) + C$
5. $-\log|1 + e^{-x}| + C$
6. $\log|x + \cos^2 x| + C$
7. $\sin^{-1}\left(\frac{2x-3}{\sqrt{41}}\right) + C$
8. $\sin^{-1} x + \sqrt{1-x^2} + c$
9. $x \tan\frac{x}{2}$
10. $x\tan^{-1} x - \log\sqrt{1+x^2} - \frac{1}{2}(\tan^{-1} x)^2$
11. $\frac{\pi}{4}x - \frac{x^2}{2} + C$
12. $\sin^{-1}\left(\frac{2x^2+1}{\sqrt{5}}\right) + C$
13. $\sin x - \operatorname{cosec} x$

14. $x(\log x)^2 - 2x(\log x - 1)$

15. $-e^x \cot \frac{x}{2} + c$

16. $\frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right)$

17. $\frac{1}{6}\log\left|\frac{x^6}{x^6+1}\right| + C$

18. $\frac{1}{3}x^3 \sin^{1}x + \frac{1}{3}\sqrt{1-x^2} - \frac{1}{9}(1-x^2)3/2$

19. $\frac{1}{2}\{\log|\sec x + \tan x|\}^2 + C$

20. $\frac{40}{41}x + \frac{9}{41}\log|5\sin x + 4\cos x| + C$

21. $\log|\sin 2x + \tan x - 5| + C$

22. $\frac{2}{3}\left[(x+3)^{3/2} + (x+2)^{3/2}\right] + C$

23. $2x \tan^{-1} x - \log(1+x^2) + C$

24. $\frac{2}{3}\tan^{-1}\left(\frac{5\tan\frac{x}{2}+4}{3}\right)$

25. $\frac{-2}{b^2}\left[\log(a+b\cos x) + \frac{a}{(a+b\cos x)}\right] + C$

26. $-2\cos\sqrt{x}\ (4)\ a \sin^{1}(x/a) - \sqrt{a^2-x^2}$

27. $e^x.\frac{1}{1+x}$

28. $\frac{1}{2}\left[-\frac{\cos 7x}{7} + \cos 7\right] + C$

29. $\frac{1}{2}\sec x \tan x + \frac{1}{2}\log|\sec x + \tan x| + C$

30. $\frac{1}{2}(x+2)\sqrt{x^2+4x+5} + \frac{1}{2}\log\left[(x+2) + \sqrt{x^2+4x+5}\right]$

31. $\log(10^x + x^{10})$

32. $\frac{1}{2}(x-a)\sqrt{2ax-x^2} + \frac{a^2}{2}.\sin^{-1}\left(\frac{x-a}{a}\right) + C$

33. 2

34. $\frac{1}{4}\log(x^2+1) + \frac{1}{2}\tan^{1}x - \frac{1}{2}\log(x+1) + c$

35. $-\frac{2}{3}\operatorname{cosec} x + c$

36. $\left[\frac{(4x+3)\sqrt{2x^2+3x+4}}{8} + \frac{23\sqrt{2}}{32}\right.$
$\left.\log\left(\frac{4x+3}{4} + \frac{\sqrt{2x^3+3x+4}}{\sqrt{2}}\right)\right]$

PART B

Definite Integration and Areas Under Curve

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LECTURE



1 Basics of Definite Integral

BASIC CONCEPTS

1. If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$ then $\int_a^b f(x)dx = F(b) - F(a)$ is read as the definite integral of $f(x)$ with respect to x between the limits a and b.
2. $\int_a^b f(x)dx = \int_a^b f(t)dt \Rightarrow$ i.e., the value of the definite integral does not change with the change of the variable if integration of the limits and form of the integral are same in both cases.
3. $\int_a^a f(x)dx = 0 \Rightarrow$ A definite integral with equal limits of integration is zero.
4. $\int_a^b f(x)dx = -\int_b^a f(x)dx$: i.e., interchange of the limits of the definite integral does not change it absolute value but only the sign.
5. Whenever there is a change of variable in a definite integral the limits must be changed accordingly.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate: $\int_0^4 \frac{1}{\sqrt{x^2+2x+3}}dx$

Solution

We have $\int_0^4 \frac{1}{\sqrt{x^2+2x+3}}dx$

$= \int_0^4 \frac{1}{\sqrt{(x+1)^2+(\sqrt{2})^2}}dx$

$= \left[\log|x+1+\sqrt{(x+1)^2+(\sqrt{2})^2}|\right]_0^4$

$= \left[\log|x+1+\sqrt{x^2+2x+3}|\right]_0^4$

$= \log(5+\sqrt{16+8+3} - \log(1+\sqrt{3}))$

$= \log\left(\frac{5+3\sqrt{3}}{1+\sqrt{3}}\right)$

2. Evaluate $\int_2^4 \frac{x}{x^2+1}dx$ ***[CBSE-1993]***

Solution

We have $\int_2^4 \frac{x}{x^2+1}dx = \frac{1}{2}\int_2^4 \frac{2x}{x^2+1}dx$

$= \frac{1}{2}\left[\log(x^2+1)\right]_2^4$

$= \frac{1}{2}$ [log (4^2+1) – log (2^2+1)]

$= \frac{1}{2}$ log 17 – log 5 $= \frac{1}{2}$ log $\left(\frac{17}{5}\right)$

3. Evaluate $\int_0^{\pi/2} \sqrt{\cos 0}\sin^3 0 d0$ ***[CBSE-90]***

Solution

Let $\cos\theta = t$ Then $d(\cos\theta) = dt$

$\Rightarrow -\sin\theta\, d\theta = dt$

when $\theta = 0$, $t = \cos 0 = 1$ when $\theta = \pi/2$, $t = \cos\frac{\pi}{2} = 0$

$\therefore \int_0^{\pi/2} \sqrt{\cos\theta}\sin^3\theta d\theta$

$= \int_1^0 \sqrt{t}\sin^3\theta\left(\frac{-dt}{\sin\theta}\right) = -\int_1^0 \sqrt{t}\sin^2\theta dt$

$= -\int_1^0 \sqrt{t}(1-t^2)dt$

$= -\int_1^0 (\sqrt{t} - t^{5/2})dt = -\left[\frac{2}{3}t^{3/2} - \frac{2}{7}t^{7/2}\right]_1^0$

$= -\left[0 - \left(\frac{2}{3} - \frac{2}{7}\right)\right] = \frac{8}{21}$

4. Evaluate $\int_0^{\pi/2} \frac{1}{3+2\cos x}dx$ ***[CBSE-1993]***

Solution

We have $I = \int_0^{\pi/2} \frac{1}{3+2\cos x}dx$

$= \int_0^{\pi/2} \frac{1}{3+2\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)}dx$

$= \int_0^{\pi/2} \frac{1+\tan^2\frac{x}{2}}{3\left(1+\tan^2\frac{x}{2}\right)+2\left(1-\tan^2\frac{x}{2}\right)}dx$

$= \int_0^{\pi/2} \frac{\sec^2\frac{x}{2}}{5+\tan^2\frac{x}{2}}dx$ Let $\tan\frac{x}{2} = t$ then

$d\left(\tan\frac{x}{2}\right) = dt$

$\Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

$\Rightarrow dx = \frac{2dt}{\sec^2\frac{x}{2}}$

Now $x = 0 \Rightarrow t = \tan 0 = 0$ and $x = \frac{\pi}{2}$

$\Rightarrow t = \tan\frac{\pi}{4} = 1$

$\therefore I = \int_0^1 \frac{\sec^2\frac{x}{2}}{5+t^2}.\frac{2dt}{\sec^2\frac{x}{2}} = 2\int_0^1 \frac{dt}{(\sqrt{5})^2+t^2}$

$= 2.\frac{1}{\sqrt{5}}\left[\tan^{-1}\left[\frac{t}{\sqrt{5}}\right]\right]_0^1$

$= \frac{2}{\sqrt{5}}\left[\tan^{-1}\frac{1}{\sqrt{5}} - \tan^{-1}0\right]$

$= \frac{2}{\sqrt{5}}\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$

5. Evaluate $\int_0^{\pi/2} \frac{\cos x}{(3\cos x + \sin x)}dx$

[CBSE-91 (C)]

Solution

Let $\cos x = K(3\cos x + \sin x) + L\frac{d}{dx}(3\cos x + \sin x)$

Then $\cos x = K(3\cos x + \sin x) + L(-3\sin x + \cos x)$ comparing coefficients of $\cos x$ and $\sin x$, we get $3K + L = 1$ and $K - 3L = 0$

Solving these two equations, we have

$$K = \frac{3}{10} \text{ and } L = \frac{1}{10}$$

$$\therefore \cos x = \frac{3}{10}(3\cos x + \sin x) + \frac{1}{10}(-3\sin x + \cos x)$$

$$\therefore \int_0^{\pi/2} \frac{\cos x}{(3\cos x + \sin x)} dx$$

$$= \int_0^{\pi/2} \frac{\frac{3}{10}(3\cos x + \sin x) + \frac{1}{10}(-3\sin x + \cos x)}{3\cos x + \sin x} dx$$

$$= \frac{3}{10}\int_0^{\pi/2} \frac{3\cos x + \sin x}{3\cos x + \sin x} dx$$

$$+ \frac{1}{10}\int_0^{\pi/2} \frac{-3\sin x + \cos x}{3\cos x + \sin x} dx$$

$$= \frac{3}{10}\int_0^{\pi/2} 1.dx + \frac{1}{10}\int_0^{\pi/2} \frac{-3\sin x + \cos x}{3\cos x + \sin x} dx$$

$$= \frac{3}{10}[x]_0^{\pi/2} + \frac{1}{10}[\log|3\cos x + \sin x|]_0^{\pi/2}$$

$$= \frac{3}{10}\left(\frac{\pi}{2} - 0\right) + \frac{1}{10}(\log 1 - \log 3) = \frac{3\pi}{20} - \frac{1}{10}\log 3$$

6. Evaluate $\int_0^1 \frac{e^x}{1+e^{2x}} dx$

[CBSE-1997]

Solution

Let $I = \int_0^1 \frac{e^x}{1+e^{2x}} dx$

Put $e^x = t$

$\Rightarrow e^x dx = dt$

when $x = 0$

$\Rightarrow e^0 = t \Rightarrow t = 1$

when $x = 1$

$\Rightarrow e^1 = t \Rightarrow t = e$

$$\therefore \int_0^1 \frac{e^x}{1+e^{2x}} dx = \int_1^e \frac{dt}{1+t^2} = \left[\tan^{-1}(t)\right]_1^e$$

$$\left(\therefore \int \frac{1}{1+t^2} dt = \tan^{-1} t\right)$$

$$= [\tan^{-1}(e) - \tan^{-1}(1)] = \tan^{-1} e - \frac{\pi}{4}$$

7. Evaluate $\int_0^1 xe^{2x} dx$ ***[MP-99]***

Solution

$$\int xe^{2x} dx = x.\frac{e^{2x}}{2} - \int 1.\frac{e^{2x}}{2} dx$$

$$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} = \frac{1}{2}e^{2x}\left[x - \frac{1}{2}\right]$$

$$\therefore \int_0^1 xe^{2x} dx = \frac{1}{2}\left[e^{2x}\left(x - \frac{1}{2}\right)\right]_0^1$$

$$= \frac{1}{2}\left[e^2\left(1 - \frac{1}{2}\right) - e^0\left(0 - \frac{1}{2}\right)\right]$$

$$= \frac{1}{4}e^2 + \frac{1}{4} = \frac{1}{4}(e^2 + 1)$$

8. If for non-zero x, $af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5$ where $a \neq b$, then $\int_1^2 f(x)dx$ = ...

[IIT 1996]

Solution

$$af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5 \quad \text{..... (1)}$$

Integrating both sides

$$a\int_1^2 f(x)dx + b\int_1^2 f\left(\frac{1}{x}\right)dx = [\log x - 5x]_1^2$$

$= \log 2 - 5$ ..... (2)

Replacing x by $\frac{1}{x}$ in (1), we get $af\left(\frac{1}{x}\right) +$ $bf(x) = x \quad 5$

Integrating both sides, we get

$$a\int_1^2 f\left(\frac{1}{x}\right)dx + b\int_1^2 f(x)dx = \left[\frac{x^2}{2} - 5x\right]_1^2 = -\frac{7}{2}$$

..... (3)

Eliminate $\int_1^2 f\left(\frac{1}{x}\right)$ between (2) and (3) by multiplying (2) by a and (3) by b and subtracting

$$\therefore \quad (a^2-b^2)\int_1^2 f(x)dx = a(\log 2-5)+b.\frac{7}{2}$$

$$\therefore \quad \int_1^2 f(x)dx = \frac{1}{(a^2-b^2)}\left[a(\log 2-5)+\frac{7b}{2}\right]$$

9. Evaluate $\int_0^1 \log\left[\sqrt{1-x}+\sqrt{1+x}\right]dx$

Solution

$$I = \left[x\log\left\{\sqrt{(1-x)}+\sqrt{(1+x)}\right\}\right]_0^1$$

$$-\int_0^1 x.\frac{1}{\sqrt{(1-x)}+\sqrt{(1+x)}}.$$

$$\left\{\frac{-1}{2\sqrt{(1-x)}}+\frac{-1}{2\sqrt{(1+x)}}\right\}dx$$

$$= \log\sqrt{2}-\int_0^1 \frac{x}{2\sqrt{(1-x^2)}}\frac{\sqrt{(1-x)}-\sqrt{(1+x)}}{\sqrt{(1-x)}+\sqrt{(1+x)}}dx$$

Rationalize

$$= \log\sqrt{2}-\int_0^1 \frac{x}{2\sqrt{(1-x^2)}}\frac{\left[\sqrt{(1-x)}-\sqrt{(1+x)}\right]^2}{(1-x)-(1+x)}$$

$$= \log\sqrt{2}-\int_0^1 \frac{x}{2\sqrt{(1-x^2)}}.$$

$$\frac{(1-x)+(1+x)-2\sqrt{(1-x^2)}}{-2x}dx$$

$$= \log\sqrt{2}+\int_0^1 \frac{1}{4\sqrt{(1-x^2)}}\{2-2\sqrt{(1-x^2)}\}dx$$

$$= \log\sqrt{2}+\frac{1}{2}\int_0^1\left\{\frac{1}{\sqrt{(1-x^2)}}-1\right\}dx$$

$$= \log\sqrt{2}+\frac{1}{2}\left[\sin^{-1}x-x\right]_0^1$$

$$= \log\sqrt{2}+\frac{1}{2}\left[\frac{\pi}{2}-1\right] = \frac{1}{2}\log 2+\frac{\pi}{4}-\frac{1}{2}$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{5\cos x+3\sin x}dx$ ***[PSB-2001(S)]***

2. Evaluate the following:

(i) $\int_1^2 x^3dx$

(ii) $\int_1^4 \sqrt{x}dx$

3. Evaluate the following definite integrals on the basis of fundamental theorem of integral calculus.

(i) $\int_0^{\pi/2} \sin^2 dx$

(ii) $\int_0^{\pi/4} \tan^2 dx$

4. Evaluate $\int_{\pi/4}^{\pi/2} e^x(\log\sin x+\cot x)dx$

5. Evaluate $\int_{-\infty}^{\infty} xe^{-x^2}dx$

[AISSE-2000]

6. Evaluate the following definite integral on the basis of fundamental theorem of integral calculus:

(i) $\int_0^a \frac{dx}{\sqrt{a^2-x^2}}$

7. Prove that $\int_2^3 \frac{dx}{x^2-1} = \frac{1}{2}\log\frac{3}{2}$

[MP-98, 99]

8. Prove that $\int_0^1 \tan^{-1}xdx = \frac{\pi}{4}-\frac{1}{2}\log 2$

[MP-2000, 2002]

9. Evaluate $\int_2^5 (3x^2 + 6x + 2)\,dx$

10. Evaluate the integral $\int_0^1 \frac{1-x^2}{1+x^2+x^4}dx$

11. Evaluate $\int_1^2 \frac{1}{x(1+\log x)^2}dx$

[CBSE-97, 2003]

12. Evaluate $\int_{\pi/6}^{\pi/2} \frac{\operatorname{cosec} x \cot x}{1+\operatorname{cosec}^2 x}dx$ ***[CBSE-97]***

13. Evaluate $\int_0^1 \frac{2x}{5x^2+1}dx$ ***[CBSE-93]***

14. Evaluate $\int_0^{\pi} \frac{1}{5+3\cos x}dx$

[CBSE-90 (C); HSB-93]

15. Evaluate $\int_0^{\pi/2} \frac{dx}{2+3\cos x}$

[MP-99, 2005-(A)]

16. Evaluate $\int_0^1 xe^x dx$ ***[HPSB-2001 (C)]***

17. Prove that $\int_0^1 \sin^{-1} x dx = \frac{\pi}{2} - 1$

[AISSE-85]

ANSWERS

1. $\frac{1}{\sqrt{34}}\log\left(\frac{8+\sqrt{34}}{8-\sqrt{34}}\right)$

2. (i) 15/4 (ii) 14/3

3. (i) $\pi/4$ (ii) $1 - \pi/4$

4. $e^{\pi/4}\log\sqrt{2}$

5. 0

6. (i) $\frac{\pi}{2}$

9. 186

10. $\frac{1}{2}\log 3$

11. $\frac{\log 2}{1+\log 2}$

12. $\tan^{-1}(1/3)$

13. $\frac{1}{5}\log 6$

14. $\frac{\pi}{4}$

15. $\frac{1}{\sqrt{5}}\log\frac{\sqrt{5}+1}{\sqrt{5}-1}$

16. 1

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. If $f(y) = e^y$, $g(y) = y$, $y > 0$ and $I'(t) = \int_0^t f(t-y)g(y)dy$, then $F(t)$ equals:

[AIEEE-2003]

(a) te^{-t} (b) $1 - e^{-t}(1 + t)$

(c) $e^t - (1 + t)$ (d) te^t

Solution

(c) $I'(t) = \int_0^t e^{t-y}ydy = e^t\int_0^t e^{-y}ydy$

$= e^t[-ye^{-y} - e^{-y}]_0^t$

$= e^t[-te^{-t} - e^{-t} + 1] = e^t - (1 + t)$

2. $\int_0^{\pi} \sin mx \sin nx\,dx$ equals $(m, n \in Z, m \neq n)$:

[NDA-2006]

(a) $m + n$ (b) 0

(c) $m - n$ (d) 1

Solution

(b) $I = \frac{1}{2}\int_0^{\pi}[\cos(m-n)x - \cos(m+n)x]dx$

$= \frac{1}{2}\left[\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n}\right]_0^{\pi} = 0$

3. If $f(x) = A \sin\left(\frac{\pi x}{2}\right) + B, f'(1/2) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2A}{\pi}$, then values of A and B are:

[IIT (Screening)-95; NDA-2006]

(a) $\pi/2, \pi/2$ (b) $2/\pi, 3/\pi$
(c) $4/\pi, 0$ (d) $0, -4/\pi$

Solution

(c) $\int_0^1 f(x)dx = \frac{2A}{\pi}$

$\Rightarrow \int_0^1 \left\{A\sin\left(\frac{\pi x}{2}\right) + B\right\}dx = \frac{2A}{\pi}$

$\Rightarrow B - \left(\frac{-2A}{\pi}\right) = \frac{2A}{\pi} \Rightarrow B = 0$

$\therefore f(x) = A \sin \frac{\pi x}{2}$

$\Rightarrow f'(x) = \frac{\pi}{2} A \cos\frac{\pi x}{2}$

$f'\left(\frac{1}{2}\right) = \frac{\pi}{2} A\left(\frac{1}{\sqrt{2}}\right) = \sqrt{2} \Rightarrow A = \frac{4}{\pi}$

4. If $I\ (m, n) = \int_0^1 t^m (1+t)^n dt$, then $I\ (m, n)$ equals: ***[IIT (Screening)-2003]***

(a) $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$

(b) $\frac{n}{m+1} I(m+1, n-1)$

(c) $\frac{2^n}{m+1} + \frac{n}{m+1} I(m+1, n-1)$

(d) $\frac{m}{n+1} I(m+1, n-1)$

Solution

(a) $I_{(m, n)} = \int_0^1 t^m (1+t)^n dt$

$= \left[(1+t)^n \frac{t^{m+1}}{m+1}\right]_0^1 - \int_0^1 n(1+t)^{n-1} \frac{t^{m+1}}{m+1} dt$

$= \frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$

5. If $I_m = \int_1^x (\ln x)^m dx$ and $I_m = x\ (ln\ x)^m - k\, I_{m-1}$, then k is equal to: ***[NDA-2004]***

(a) $m - 1$ (b) m
(c) $m + 1$ (d) $m^2 + 1$

Solution

(b) $I_m = \int_1^x (\log x)^m dx = x(\log x)^m$

$-\int_1^x m(\log x)^{m-1} dx$

(integrating by parts)

$= x\ (ln)^m - m\ I_{m\ \ 1}$

$\therefore \quad k = m$

6. $\int_0^{\pi/2} \frac{1 - \sin 3x}{1 + \sin x} dx$ is equal to ***[ICS-2006]***

(a) 2 (b) $\frac{2\pi}{3}$
(c) $\frac{3\pi}{2} - 4$ (d) $\frac{3\pi}{2} + 4$

Solution

(c) $I = \int_0^{\pi/2} \frac{1 - 3\sin x + 4\sin^3 x}{1 + \sin x} dx$

$= \int_0^{\pi/2} \frac{(1+\sin x)(1 - 4\sin x + 4\sin^2 x)}{1 + \sin x} dx$

$= \int_0^{\pi/2} (1 - 4\sin x + 4\sin^2 x) dx$

$= \frac{\pi}{4} - 4 + 4.\frac{1}{2}.\frac{\pi}{2} = \frac{3\pi}{2} - 4$

7. If $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then $\int_0^1 f(x)g(x)dx$ is equal to: ***[AIEEE-2003]***

(a) $e + \frac{e^2}{2} + \frac{5}{2}$ (b) $e - \frac{e^2}{2} - \frac{5}{2}$
(c) $e + \frac{e^2}{2} - \frac{3}{2}$ (d) $e - \frac{e^2}{2} - \frac{3}{2}$

Solution

(d) $\because\ f'(x) = f(x) \Rightarrow \dfrac{f'(x)}{f(x)} = 1$

$\Rightarrow \log f(x) = x$

(on integration)

$\Rightarrow\ f(x) = e^x$

Now $\int_0^1 f(x)g(x)dx = \int_0^1 f(x)(x^2 - f(x))dx$

$= \int_0^1 e^x(x^2 - e^x)dx = \int_0^1 (x^2e^x - e^{2x})dx$

$= \left[x^2e^x - 2xe^x + 2e^x - e^{2x}/2\right]_0^1$

$= e - \dfrac{e^2}{2} - \dfrac{3}{2}$

8. $\int_0^{\pi/4} (\cos x - \sin x)dx + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)dx + \int_{2\pi}^{\pi/4} (\cos x - \sin x)dx$ is equal to:

[RPET-2000]

(a) $\sqrt{2} - 2$ (b) $2\sqrt{2} - 2$

(c) $3\sqrt{2} - 2$ (d) $4\sqrt{2} - 2$

Solution

(d) $I = \int_0^{\pi/4} (\cos x - \sin x)dx$

$+ \int_{\pi/4}^{5\pi/4} (\sin x - \cos)dx + \int_{2\pi}^{\pi/4} (\cos x - \sin x)dx$

$= [\sin x + \cos x]_0^{\pi}{}_4 - [\sin x + \cos x]_{\frac{\pi}{4}}^{5\pi}{}_4$

$+ [\sin x + \cos x]_{2\pi}^{\frac{\pi}{4}}$

$I = \left[\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}} - 1\right]$

$-\left[-\dfrac{1}{\sqrt{2}} - \dfrac{1}{\sqrt{2}} - \left(\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}}\right)\right]$

$+\left[\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}} - 1\right]$

$I = [\sqrt{2} - 1] - [-\sqrt{2} - \sqrt{2}]$

$+ [\sqrt{2} - 1]$

$I = [\sqrt{2} - 1 + 2\sqrt{2} + \sqrt{2} - 1]$

$= 4\sqrt{2} - 2.$

9. $\int_{\pi/3}^{\pi/2} \dfrac{\sqrt{1+\cos x}}{(1-\cos x)^{\frac{5}{2}}} dx =$

[AI CBSE-1980]

(a) 5/2 (b) 3/2

(c) 1/2 (d) 2/5

Solution

(b) $I = \int_{\pi/3}^{\pi/2} \dfrac{\sqrt{1+\cos x}}{(1-\cos x)^{5/2}}$

$\times \sqrt{\dfrac{1-\cos x}{1-\cos x}}dx \int_{\pi/3}^{\pi/2} \dfrac{\sin x}{(1-\cos x)^3} dx$

Now, put $1 - \cos x = t$

Also, when $x = \pi/3$, $t = 1/2$

and $x = \pi/2$, $t = 1$

Therefore, $I = \int_{1/2}^1 \dfrac{dt}{t^3} = \left|\dfrac{t^{-2}}{-2}\right|_{1/2}^1 = \dfrac{3}{2}$

10. $\int_0^1 \dfrac{e^{-x}}{1+e^{-x}} dx =$ ***[Roorkee-1996]***

(a) $\log\left(\dfrac{1+e}{e}\right) - \dfrac{1}{e} + 1$

(b) $\log\left(\dfrac{1+e}{2e}\right) - \dfrac{1}{e} + 1$

(c) $\log\left(\dfrac{1+e}{2e}\right) + \dfrac{1}{e} - 1$

(d) None of these

Solution

(b) Put $1 + e^{-x} = t \Rightarrow -e^{-x}\,dx = dt$, then we have

$I = \int_2^{1-1/e} \dfrac{(t-1)(-dt)}{t} = \int_2^{1+1/e} \left(\dfrac{1}{t} - 1\right)dt$

$= \left[\log_e t - t\right]_2^{1+1/e} = \log_e\left(1 + \dfrac{1}{e}\right)$

$-\left(1 + \dfrac{1}{e}\right) - \log_e 2 + 2$

$= \log_e\left(\dfrac{e+1}{2e}\right) - \dfrac{1}{e} + 1$

11. If $\int_0^1 x\log\left(1+\frac{x}{2}\right)dx = a + b\log\frac{2}{3}$, then

[SCRA-1986]

(a) $a = 3/2,\ b = 3/2$
(b) $a = 3/4,\ b = -3/4$
(c) $a = 3/4,\ b = 3/2$
(d) $a = b$

Solution

(c) Integrate it by parts taking $\log(1 + x/2)$ as first function

$$= \left[\log\left(1+\frac{x}{2}\right)\frac{x^2}{2}\right]_0^2 - \int_0^1 \frac{1}{1+x/2}\frac{1}{2}\frac{x^2}{2}dx$$

$$= \frac{1}{2}\log\frac{3}{2} - \frac{1}{2}\int_0^1 \frac{x^2}{x+2}dx$$

$$= \frac{1}{2}\log\frac{3}{2} - \frac{1}{2}\left[\frac{1}{2} - 2 + 4\log 3 - 4\log 2\right]$$

$$= \frac{3}{4} + \frac{3}{2}\log\frac{2}{3}$$

On comparing with the given value $a = 3/4$, $b = 3/2$.

12. Value of $\int_0^a \sqrt{a^2 - x^2}dx$ is

[MP PET-2007]

(a) $a^2\pi$ (b) $\frac{a^2\pi}{4}$

(c) $\frac{a^2\pi}{2}$ (d) $\frac{\pi}{2}$

Solution

(c) $\int_0^a \sqrt{a^2 - x^2}dx$

$$= \left[\frac{1}{2}x\sqrt{a^2 - x^2} + \frac{1}{2}a^2\sin^{-1}\frac{x}{a}\right]_0^a$$

$$= \frac{1}{2}a\sqrt{a^2 - a^2} + \frac{1}{2}a^2\sin^{-1}\frac{a}{a} - 0$$

$$= \frac{a^2}{2}\sin^{-1}1 = \frac{a^2\pi}{2}$$

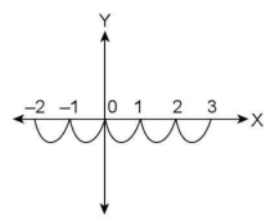
13. If $f(x) = x(x - 1),\ 0 \le x \le 1$, and $f(x + 1)$ $f(x)\ \forall\ x \in R$, Then $\left|\int_2^4 f(x)dx\right|$ is:

(a) 1 sq. unit (b) $\frac{1}{3}$ sq. unit

(c) $\frac{2}{3}$ sq. unit (d) $\frac{1}{6}$ sq. unit

Solution

(b) Graph of $y = f(x)$ will be of the following form. From the graph we can observe that period of $f(x)$ is 1 unit.



$$\therefore \quad \left|\int_2^4 f(x)dx\right| = 2\left|\int_0^1 x(x-1)dx\right| = \frac{1}{3} \text{ sq. unit}$$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int_2^3 \frac{dx}{x^2 - x} =$

[EAMCET-2002]

(a) log (2/3) (b) log (1/4)
(c) log (4/3) (d) log (8/3)

2. $\int_1^3 (x - 1)(x - 2)(x - 3)\,dx =$

[Karnataka CET-2002]

(a) 3 (b) 2
(c) 1 (d) 0

3. The value of $\int_0^1 \frac{\tan^{-1}x}{1+x^2}dx$ is

[RPET-2001; SCRA-87; MNR-90]

(a) $\pi/4$ (b) $\pi^2/32$
(c) 1 (d) None of these

4. $\int_{-1}^{0} \frac{dx}{x^2+2x+2} =$ ***[MPPET-2000]***

(a) 0 (b) $\pi/4$
(c) $\pi/2$ (d) $-\pi/4$

5. $\int_{0}^{2/3} \frac{dx}{4+9x^2} =$ ***[MPPET-1997]***

(a) $\pi/12$ (b) $\pi/24$
(c) $\pi/4$ (d) 0

6. Given that $\int_{0}^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)(x^2+c^2)}$

$= \frac{\pi}{2(a+b)(b+c)(c+a)}$, then the value of

$\int_{0}^{\infty} \frac{dx}{(x^2+4)(x^2+9)}$ is ***[Karnataka CET-93]***

(a) $\pi/60$ (b) $\pi/20$
(c) $\pi/40$ (d) $\pi/80$

7. The value of $\int_{1}^{2} \log x \, dx$ is ***[Roorkee-1995]***

(a) $\log \frac{2}{e}$ (b) $\log 4$
(c) $\log \frac{4}{e}$ (d) $\log 2$

8. $\int_{0}^{2\pi} (\sin x \quad \cos x) \, dx =$ ***[SCRA-1991]***

(a) 0 (b) 2
(c) -2 (d) 1

9. $\int_{0}^{\pi/6} \frac{\sin x}{\cos^3 x} dx =$ ***[SCRA-79]***

(a) 2/3 (b) 1/6
(c) 2 (d) 1/3

10. $\int_{0}^{\pi/2} \frac{\sin x \cos x}{1+\sin^4 x} dx =$ ***[AISSE-1988]***

(a) $\pi/2$ (b) $\pi/4$
(c) $\pi/6$ (d) $\pi/8$

11. $\int_{0}^{2\pi} \sqrt{1+\sin\frac{x}{2}} dx =$

[MNR-1987; UPSEAT-2000]

(a) 0 (b) 2
(c) 8 (d) 4

12. $\int_{\pi/4}^{\pi/2} \cos\theta \operatorname{cosec}^2\theta d\theta =$ ***[Roorkee-1978]***

(a) $\sqrt{2}-1$
(b) $1-\sqrt{2}$
(c) $\sqrt{2}+1$
(d) None of these

13. $\int_{0}^{1} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx =$ ***[Karnataka CET-99]***

(a) $\frac{\pi}{2} - 2\log\sqrt{2}$ (b) $\frac{\pi}{2} + 2\log\sqrt{2}$
(c) $\frac{\pi}{4} - \log\sqrt{2}$ (d) $\frac{\pi}{4} - \log\sqrt{2}$

14. $\int_{0}^{\pi/4} \tan^2 x \, dx =$

[Roorkee-1983; Pb. CET-2000]

(a) $1-\frac{\pi}{4}$ (b) $1+\frac{\pi}{4}$
(c) $\frac{\pi}{4}-1$ (d) $\frac{\pi}{4}$

15. The value of the integral $\int_{0}^{\pi} e^x \sin x \, dx$ is

[MPPET-2007]

(a) $1+e^{\pi}$ (b) $(1+e^{\pi})/2$
(c) $e^{\pi}/2$ (d) None of these

16. If $x(x^4+1)\phi(x) = 1$, then $\int_{1}^{2} \phi(x) dx =$

[SCRA-1986]

(a) $\frac{1}{4}\log\frac{32}{17}$ (b) $\frac{1}{2}\log\frac{32}{17}$
(c) $\frac{1}{4}\log\frac{16}{17}$ (d) None of these

17. If $I = \int_{0}^{\pi/4} \sin^2 x dx$ and $J = \int_{0}^{\pi/4} \cos^2 x dx$, then $I =$ ***[SCRA-1989]***

(a) $\frac{\pi}{4} - J$ (b) $2J$
(c) J (d) $J/2$

18. $\int_0^{b-c} f''(x+a)dx$ = ***[SCRA-1990]***

(a) $f'(a)-f'(b)$

(b) $f'(b-c+a)-f'(a)$

(c) $f'(b+c-a)-f'(a)$

(d) None of these

19. $\int_1^2 e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)dx =$

[MNR-1990; AMU-1999; UPSEAT-2000; Pb. CET-2004]

(a) $\frac{e^2}{2}+e$ (b) $e-\frac{e^2}{2}$

(c) $\frac{e^2}{2}-e$ (d) None of these

20. $\int_0^{\pi/2}\frac{\cos x}{(1+\sin x)(2+\sin x)}dx =$

[UPSEAT-1999]

(a) $\log\frac{4}{3}$ (b) $\log\frac{1}{3}$

(c) $\log\frac{3}{4}$ (d) None of these

21. What is $\int_a^b \frac{\log x}{x}dx$ = ***[NDA-2007]***

(a) (1/2) log (ab) . log (b/a)

(b) log b/log a

(c) log (b/a)

(d) (1/2) log $[(a + b)/ab]$

22. The integral $\int_0^1 \frac{dx}{1-x+x^2}$ has value

[MPPET-2007]

(a) $2\pi/3$ (b) $2\pi/\sqrt{3}$

(c) $2\pi/3\sqrt{3}$ (d) None of these

23. $\int_0^{\pi/2}\frac{dx}{2+\cos x}$ = ***[BIT Ranchi-1992]***

(a) $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(b) $\sqrt{3}\tan^{-1}(\sqrt{3})$

(c) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(d) $2\sqrt{3}\tan^{-1}(\sqrt{3})$

24. $\int_0^{\pi/2}\frac{\cos x}{1+\cos x+\sin x}dx =$ ***[Roorkee-89]***

(a) $\frac{\pi}{4}+\frac{1}{2}\log 2$ (b) $\frac{\pi}{4}+\log 2$

(c) $\frac{\pi}{4}-\frac{1}{2}\log 2$ (d) $\frac{\pi}{4}-\log 2$

25. $\int_0^{\pi/2}\frac{\sin x\cos x dx}{\cos^2 x+3\cos x+2} =$ ***[MNR-1981]***

(a) log (8/9) (b) log (9/8)

(c) log (8 × 9) (d) None of these

26. $\int_0^{\pi/2}(\sin x-\cos x)\log(\sin x \quad \cos x)\,dx =$

[SCRA-1986]

(a) –1 (b) 1

(c) 0 (d) None of these

27. $\int_3^5 \frac{x^2}{x^2-4}dx =$ ***[Roorkee-92]***

(a) $2-\log_e(15/7)$

(b) $2+\log_e(15/7)$

(c) $2+4\log_e 3-4\log_e 7+4\log_e 5$

(d) $2-\tan^{-1}(15/7)$

28. $\int_0^{\pi/3}\frac{\cos x}{3+4\sin x}dx =$

(a) $\frac{1}{4}\log\left(\frac{3+2\sqrt{3}}{2}\right)$

(b) $\frac{1}{2}\log\left(\frac{3+2\sqrt{3}}{2}\right)$

(c) $\frac{1}{3}\log\left(\frac{3+2\sqrt{3}}{3}\right)$

(d) None of these

SOLUTIONS

1. (c) $I=\int_2^3 \frac{dx}{x^2-x}=\int_2^3 \frac{dx}{x(x-1)}=\int_2^3\left[\frac{1}{x-1}-\frac{1}{x}\right]dx$

$=\int_2^3 \frac{1}{x-1}dx-\int_2^3 \frac{1}{x}dx=\left[\log(x-1)\right]_2^3-\left[\log x\right]_2^3$

$=[\log 2-\log 1]-[\log 3-\log 2]$

$=2\log 2-\log 3=\log\frac{4}{3}$

2. (d) $I=\int_1^3 (x-1)(x-2)(x-3)dx$

$=\int_1^3 \left(x^3-6x^2+11x-6\right)dx$

$=\left[\frac{x^4}{4}-\frac{6x^3}{3}+\frac{11x^2}{2}-6x\right]_1^3=0$

3. (b) Step 1: $\int \frac{\tan^{-1}x}{1+x^2}dx=\frac{(\tan^{-1}x)^2}{2}$

Step 2: $\int_0^1 \frac{\tan^{-1}x}{1+x^2}dx=\frac{(\tan^{-1}x)^2}{2}\Big|_0^1=\frac{(\tan^{-1}1)^2-0}{2}$

$=\pi^2/32$

4. (b) Step 1: $I=\int_{-1}^0 \frac{dx}{(x+1)^2+1}=\tan^{-1}(x+1)\Big|_{-1}^0$

$=\tan^{-1}1-\tan^{-1}0=\frac{\pi}{4}$

5. (b) $\int_0^{2/3} \frac{dx}{4+9x^2}=\frac{1}{9}\int_0^{2/3} \frac{dx}{(2/3)^2+x^2}$

$=\frac{1}{9}\times\frac{1}{2/3}\left(\tan^{-1}\frac{x}{2/3}\right)\Big|_0^{2/3}$

$=\frac{\pi}{4}\times\frac{1}{6}=\frac{\pi}{24}$

6. (a) Step 1: put $a = 2$, $b = 3$, $c = 0$ in given condition

we get ans $=\frac{\pi}{60}$.

7. (c) $\int_1^2 \log x dx=\left[x\log x-x\right]_1^2=2\log 2-2+1$

$=\log 4-1=\log 4-\log e=\log\frac{4}{e}$.

8. (a) $\int_0^{2\pi}(\sin x+\cos x)dx=\left[-\cos x+\sin x\right]_0^{2\pi}=0$

9. (b) Let $I=\int_0^{\pi/6}\frac{\sin x}{\cos^3 x}dx=\int_0^{\pi/6}\tan x\sec^2 x dx$

Put $t \quad \tan x \Rightarrow dt \quad \sec^2 x\,dx$, then we have

$I=\int_0^{1/\sqrt{3}} t dt=\left[\frac{t^2}{2}\right]_0^{1/\sqrt{3}}=\frac{1}{6}$

10. (d) Put $\sin^2 x = t \Rightarrow dt \quad 2\sin x\cos x\,dx$

Now $\int_0^{\pi/2}\frac{\sin x\cos x}{1+\sin^4 x}dx=\frac{1}{2}\int_0^1\frac{1}{1+t^2}dt$

$=\frac{1}{2}\left[\tan^{-1}t\right]_0^1=\frac{\pi}{8}$

11. (c) $\int_0^{2\pi}\sqrt{1+\sin\frac{x}{2}}dx=\int_0^{2\pi}\left|\sin\frac{x}{4}+\cos\frac{x}{4}\right|dx$

$=4\left[\sin\frac{x}{4}-\cos\frac{x}{4}\right]_0^{2\pi}$

$=4\,[1+1]=8$

13. (a) $\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right)dx$

Putting $x=\tan\theta$, $\therefore\ dx=\sec^2\theta d\theta$

As $x=1\Rightarrow 0=\frac{\pi}{4}$

and $x=0\Rightarrow 0=0$, then

$I=2\int_0^{\pi/4} 0\sec^2 0 d0$

$=2\left[\theta\tan\theta\right]_0^{\pi/4}-2\int_0^{\pi/4}\tan\theta d\theta$

$=\frac{\pi}{2}+2\left[\log\cos 0\right]_0^{\pi/4}=\frac{\pi}{2}-2\log\sqrt{2}$

14. (a) $\int_0^{\frac{\pi}{4}} \tan^2 x\,dx = \int_0^{\frac{\pi}{4}} (\sec^2 x - 1)\,dx$

$= \int_0^{\frac{\pi}{4}} \sec^2 x\,dx - \int_0^{\frac{\pi}{4}} 1\,dx$

$= [\tan x]_0^{\frac{\pi}{4}} - [x]_0^{\frac{\pi}{4}}$

$= \left(\tan\frac{\pi}{4} - \frac{\pi}{4}\right) = 1 - \frac{\pi}{4}.$

15. (a) By $\int e^{ax}\sin bx\ dx$

$= \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$

$\int_0^{\pi} e^x \sin x\,dx = \left[\frac{e^x}{1+1}(\sin x - \cos x)\right]_0^{\pi}$

$= \left[\frac{e^x}{2}(\sin x - \cos x)\right]_0^{\pi}$

$= \frac{e^{\pi}}{2}(\sin\pi - \cos\pi) - \frac{e^0}{2}(\sin 0 - \cos 0)$

$= \frac{e^{\pi}}{2}(1) - \frac{1}{2}(-1)$

$= \frac{e^{\pi}}{2} + \frac{1}{2} = \frac{e^{\pi}+1}{2}$

16. (a) $\phi(x) = \frac{1}{x(x^4+1)} = \frac{1}{x} - \frac{x^3}{x^4+1}$

$= \int_1^2 \phi(x)dx = \int_1^2 \left(\frac{1}{x} - \frac{x^3}{x^4+1}\right)dx$

$= [\log x]_1^2 - \left[\frac{1}{4}\log(x^4+1)\right]_1^2$

$= \frac{1}{4}\log\frac{32}{17}$

17. (a) $I = \int_0^{\pi/4} \sin^2 x\,dx$ and $J = \int_0^{\pi/4} \cos^2 x\,dx$ then

Adding $I + J$, we get

$= \int_0^{\pi/4} dx = \frac{\pi}{4} \Rightarrow I = \frac{\pi}{4} - J$

18. (b) $\int_0^{b-c} f''(x+a)dx = f'(x+a)\Big|_0^{b-c}$

$= f'(b-c+a) - f'(a)$

19. (c) $\int_1^2 e^x\left(\frac{1}{x} - \frac{1}{x^2}\right)dx = \frac{1}{x}e^x\Big|_1^2 = \frac{e^2}{2} - e$

20. (a) $\int_0^{\pi/2} \frac{\cos x}{(1+\sin x)(2+\sin x)}dx$

Putting $\sin x = t \Rightarrow \cos x dx = dt$

so that reduced integral is

$\int_0^1 \left(\frac{1}{1+t} - \frac{1}{2+t}\right)dt = [\log(1+t) - \log(2+t)]_0^1$

$= \log\frac{2}{3} - \log\frac{1}{2} = \log\frac{4}{3}$

21. (c) Step 1: Let $t = \log x$ then $dt = \frac{1}{x}dx$

and limits are

$t \quad \log a, t \quad \log b$

Step 2: $\int_a^b \frac{\log x}{x}dx = \int_{\log a}^{\log b} t\,dt = \frac{t^2}{2}\Big|_{\log a}^{\log b}$

$= \frac{1}{2}\left[(\log b)^2 - (\log a)^2\right] = \frac{1}{2}\log(ab)\log\frac{b}{a} + c$

Note that $\int f(x)f'(x)dx = \frac{[f(x)]^2}{2} + c$

22. (c) $\int_0^1 \frac{1}{x^2 - x + 1}dx = \int_0^1 \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}dx$

$= \left[\frac{2}{\sqrt{3}}\tan^{-1}\frac{\left(x - \frac{1}{2}\right)}{\frac{\sqrt{3}}{2}}\right]_0^1 = \frac{2}{\sqrt{3}}\left[\tan\left(\frac{2x-1}{\sqrt{3}}\right)\right]_0^1$

$= \frac{2}{\sqrt{3}}\left\{\tan^{-1}\frac{1}{\sqrt{3}} - \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right\}$

$= \frac{2}{\sqrt{3}}\left(\frac{\pi}{6} + \frac{\pi}{6}\right) = \frac{2}{\sqrt{3}}\frac{2\pi}{6} = \frac{2\pi}{3\sqrt{3}}$

23. (c) $I=\int_0^{\pi/2}\frac{dx}{2+\cos x}=\int_0^{\pi/2}\frac{dx}{2+\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$

$$=\int_0^{\pi/2}\frac{\sec^2\frac{x}{2}}{3+\tan^2\frac{x}{2}}$$

Putting $t=\tan^2\frac{x}{2}\Rightarrow dt=\frac{1}{2}\sec^2\frac{x}{2}dx$, then

$$I=2\int_0^1\frac{dt}{3+t^2}=\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Note: By using formula $\int\frac{dx}{a^2+x^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}$

24. (c) $\int_0^{\pi/2}\frac{\cos x}{1+\cos x+\sin x}dx$

$$=\int_0^{\pi/2}\frac{\cos^2\left(\frac{x}{2}\right)-\sin^2\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)+2\sin\frac{x}{2}\cos\frac{x}{2}}dx$$

$$=\frac{1}{2}\int_0^{\pi/2}\frac{1-\tan^2\left(\frac{x}{2}\right)}{1+\tan\left(\frac{x}{2}\right)}dx$$

$$=\frac{1}{2}\int_0^{\pi/2}\left(1-\tan\left(\frac{x}{2}\right)\right)dx$$

$$=\frac{1}{2}\left[x+2\log\cos\frac{x}{2}\right]_0^{\pi/2}$$

$$=\frac{\pi}{4}+\log\frac{1}{\sqrt{2}}$$

$$=\frac{\pi}{4}-\frac{1}{2}\log 2$$

25. (b) $I=\int_0^{\pi/2}\frac{\sin x\cos x dx}{\cos^2 x+3\cos x+2}$

Putting $\cos x=t\Rightarrow -\sin x dx=dt$, then

$$I=-\int_0^1\frac{tdt}{t^2+3t+2}=\int_0^1\left[\frac{2}{t+2}-\frac{1}{t+1}\right]dt$$

$$=[2\log(t+2)-\log(t+1)]_0^1$$

$=[2\log 3-\log 2-2\log 2]=[2\log 3-3\log 2]$

$=[\log 9-\log 8]$

$=\log(9/8)$

26. (c) $\int_0^{\pi/2}(\sin x-\cos x)\log(\sin x+\cos x)$

Putting $\sin x+\cos x=t$

$\Rightarrow -(\sin x-\cos x)dx=dt$

Also as $x=0$ to $\frac{\pi}{2}$, t 1 to 1

Since here limit is 1 to 1, therefore the value of integral will be zero.

Note that $\because\int_a^a f(x)dx=0$

27. (a) $I=\int_3^5\frac{x^2}{x^2-4}dx=\int_3^5\left(1+\frac{4}{x^2-4}\right)dx$

$$=\left[x+4.\frac{1}{4}\log\frac{x+2}{x-2}\right]_3^5$$

$$=2+\log\frac{7}{3}-\log 5=2+\log\frac{7}{15}$$

28. (d) $\int_0^{\pi/3}\frac{\cos x}{3+4\sin x}dx=$

Putting $3+4\sin x=t\Rightarrow dt=4\cos x\,dx$

As $x=\frac{\pi}{3}\Rightarrow t=3+2\sqrt{3}$

and $x=0\Rightarrow t=3$, then

$$I=\frac{1}{4}\int_3^{3+2\sqrt{3}}\frac{dt}{t}=\frac{1}{4}\log t\int_3^{3-2\sqrt{3}}$$

$$=\frac{1}{4}\left[\log(3+2\sqrt{3})-\log 3\right]$$

$$=\frac{1}{4}\log\frac{3+2\sqrt{3}}{3}$$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. $\int_{\pi/6}^{\pi/4} \operatorname{cosec} 2x\, dx =$ ***[MNR-1980]***

(a) log 3 (b) log $\sqrt{3}$
(c) log 9 (d) None of these

2. $\int_{1}^{\sqrt{3}} \frac{1}{1+x^2} dx =$ ***[DCE-2002]***

(a) $\pi/12$ (b) $\pi/6$
(c) $\pi/4$ (d) $\pi/3$

3. $\int_{1/4}^{1/2} \frac{dx}{\sqrt{x-x^2}} =$ ***[SCRA-86]***

(a) π (b) $\pi/2$
(c) $\pi/3$ (d) $\pi/6$

4. $\int_{1}^{e} \frac{1+\log x}{x} dx =$ ***[SCRA-86]***

(a) 3/2 (b) 1/2
(c) $1/e$ (d) None of these

5. The value of $\int_{0}^{1} \frac{dx}{e^x + e^{-x}}$ is ***[SCRA-1980]***

(a) $\tan^{-1}\left(\frac{1-e}{1+e}\right)$ (b) $\tan^{-1}\left(\frac{e-1}{e+1}\right)$
(c) $\frac{\pi}{4}$ (d) $\tan^{-1} e - \frac{\pi}{4}$

6. $\int_{0}^{\pi/3} \cos 3x\, dx =$ ***[SCRA-80]***

(a) π (b) 0
(c) $\pi/2$ (d) $\pi/4$

7. $\int_{0}^{\pi/4} \tan^6 x \sec^2 x\, dx =$ ***[BIT (Ranchi)-1981]***

(a) 1/7 (b) 2/7
(c) 1 (d) None of these

8. $\int_{0}^{1} \cos^{-1} x\, dx =$ ***[DSSE-1988]***

(a) 0 (b) 1
(c) 2 (d) None of these

9. $\int_{0}^{\pi/8} \frac{\sec^2 2x}{2} dx =$

(a) 1/4 (b) 1/3
(c) 1/2 (d) None of these

10. $\int_{0}^{1} \tan^{-1} x\, dx =$ ***[Karnataka CET-93; RPET-1997]***

(a) $\frac{\pi}{4} - \frac{1}{2}\log 2$ (b) $\pi - \frac{1}{2}\log 2$
(c) $\frac{\pi}{4} - \log 2$ (d) $\pi - \log 2$

11. $\int_{0}^{1} e^{2\ln x} dx =$ ***[MPPET-1990]***

(a) 0 (b) 1/2
(c) 1/3 (d) 1/4

12. The integral $\int_{-1}^{3} \left(\tan^{-1}\frac{x}{x^2+1} + \tan^{-1}\frac{x^2+1}{x} \right) dx =$ ***[Karnataka CET-2000]***

(a) π (b) 2π
(c) 3π (d) None of these

13. $\int_{0}^{a} x^2 \sin x^3\, dx$ equals ***[RPET-1996]***

(a) $(1 - \cos a^3) + c$
(b) $3(1 - \cos a^3) \quad c$
(c) $-\frac{1}{3}(1 - \cos a^3) - c$
(d) $\frac{1}{3}(1 - \cos a^3) - c$

14. $\int_{0}^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx =$ ***[UPSEAT-2003]***

(a) –log 2 (b) log 2
(c) $\pi/2$ (d) 0

15. $\int_{0}^{\pi/2} e^x \sin x\, dx =$ ***[Roorkee-1978]***

(a) $\frac{1}{2}(e^{\pi/2} - 1)$ (b) $\frac{1}{2}(e^{\pi/2} + 1)$
(c) $\frac{1}{2}(1 - e^{\pi/2})$ (d) $2(e^{\pi/2} + 1)$

16. $\int_0^1 \frac{dx}{[ax+b(1-x)]^2} =$ ***[SCRA-1986]***

(a) a/b (b) b/a
(c) ab (d) $1/ab$

17. $\int_0^{\pi} \frac{dx}{1+\sin x} =$ ***[CEE-93]***

(a) 0 (b) 1/2
(c) 2 (d) 3/2

18. $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} =$

(a) $\frac{\pi}{2(a-b)}$ (b) $\frac{\pi}{2(b-a)}$
(c) $\frac{\pi}{a+b}$ (d) $\frac{\pi}{2(a+b)}$

19. $\int_0^{\pi/6} (2+3x^2)\cos 3x\, dx =$ ***[DSSE-1985]***

(a) $\frac{1}{36}(\pi+16)$ (b) $\frac{1}{36}(\pi-16)$
(c) $\frac{1}{36}(\pi^2-16)$ (d) $\frac{1}{36}(\pi^2+16)$

20. $\int_{-\pi/4}^{\pi/2} e^{-x}\sin x dx =$ ***[CEE-1993]***

(a) $-\frac{1}{2}e^{-\pi/2}$ (b) $-\frac{\sqrt{2}}{2}e^{-\pi/4}$
(c) $-\sqrt{2}(e^{-\pi/4}+e^{-\pi/4})$ (d) 0

21. $\int_0^{\pi/4} \frac{4\sin 2\theta d\theta}{\sin^4\theta+\cos^4\theta} =$ ***[SCRA-86]***

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) π (d) None of these

22. $\int_0^1 \frac{e^x(x-1)}{(x+1)^3}dx =$ ***[SCRA-86]***

(a) $\frac{e}{4}$ (b) $\frac{e}{4}-1$
(c) $\frac{e}{4}+1$ (d) None of these

23. $\int_0^{\pi/4} \sec x \log(\sec x+\tan x)\, dx =$

(a) $\frac{1}{2}[\log(1+\sqrt{2})]^2$ (b) $[\log(1+\sqrt{2})]^2$
(c) $\frac{1}{2}[\log(\sqrt{2}-1)]^2$ (d) $[\log(\sqrt{2}-1)]^2$

24. If $\frac{d[f(x)]}{dx} = g(x)$ for $a \le x \le b$, then $\int_a^b f(x)\, g(x)\, dx$ equals ***[CEE-93]***

(a) $f(b)-f(a)$
(b) $g(b)-g(a)$
(c) $\frac{[f(b)]^2-[f(a)]^2}{2}$
(d) $\frac{[g(b)]^2-[g(a)]^2}{2}$

25. Value of integral $\int_0^{\pi} \frac{1}{3+2\sin x+\cos x}dx$ is ***[MP PET-2007]***

(a) $\pi/2$ (b) $2\pi/3$
(c) $\pi/4$ (d) $-\pi/4$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 18 minutes.
3. The worksheet consists of 18 questions. The maximum marks are 54
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int_1^2 \frac{1}{x^2} e^{-1/x}\,dx =$

[DCE-2001]

(a) $\sqrt{e}+1$ (b) $\sqrt{e}-1$

(c) $\frac{\sqrt{e}+1}{e}$ (d) $\frac{\sqrt{e}-1}{e}$

2. $\int_1^x \frac{\log(x^2)}{x}dx =$

[DCE-1999]

(a) $(\log x)^2$ (b) $\frac{1}{2}(\log x)^2$

(c) $\log x^2/2$ (d) None of these

3. $\int_0^{\pi/2} \frac{d\theta}{5+3\cos\theta} =$

(a) $\tan^{-1}\frac{1}{2}$ (b) $\tan^{-1}\frac{1}{3}$

(c) $\frac{1}{2}\tan^{-1}\frac{1}{2}$ (d) $\frac{1}{3}\tan^{-1}\frac{1}{3}$

4. $\int_0^1 \frac{\sin^2 x+\cos^4 x}{\cos^2 x+\sin^4 x}.x^5\,dx =$

(a) 1/3 (b) 1/4

(c) 1/5 (d) 1/6

5. $\int_0^{\pi/2} \frac{\cos 0}{\sqrt{4-\sin^2 0}}d\theta$ is equal to

(a) $\pi/2$ (b) $\pi/6$

(c) $\pi/3$ (d) $\pi/5$

6. $\int_0^3 \frac{3x+1}{x^2+9}dx$ is equal to

(a) $\log(2\sqrt{2})+\frac{\pi}{12}$ (b) $\log(2\sqrt{2})+\frac{\pi}{2}$

(c) $\log(2\sqrt{2})+\frac{\pi}{6}$ (d) $\log(2\sqrt{2})+\frac{\pi}{3}$

7. $\int_0^1 \frac{x^7}{\sqrt{1-x^4}}dx$ is equal to

(a) 1 (b) 1/3

(c) 2/3 (d) $\pi/3$

8. Value of $\int_0^{1/2} \frac{dx}{\sqrt{x-x^2}}$ is *[Assam JEE-1998]*

(a) 1/2 (b) π

(c) $\pi/2$ (d) $\pi/4$

9. The value of

$$\int_0^1 \left(1-\frac{x}{1!}+\frac{x^2}{2!}-\frac{x^3}{3!}+\ldots\frac{(-1)^n x^n}{n!}+\ldots\right)e^{2x}dx \text{ is:}$$

[Kerala P.E.T.-2001]

(a) 0 (b) $e-1$

(c) 1 (d) e

10. $\int_0^{\pi/2} \frac{\sin x\cos x}{1+\sin x}dx$ is

(a) $\frac{3}{2}-\log 2$ (b) $1-\log 2$

(c) $3-\log 2$ (d) $3+\log 2$

11. $(\int_0^a x dx) \le (a+4)$, then

(a) $0 \le a \le 4$ (b) $-2 \le a \le 4$

(c) $-2 \le a \le 0$ (d) $a \le -2$ or $a \le 4$

12. $\int_0^{(\pi/2)^{1/3}} 0^5 \sin 0^3\,d0$ is equal to *[VIT-2005]*

(a) 1 (b) 1/2

(c) 1/3 (d) 1/4

13. If $2f(x)-3f\left(\frac{1}{x}\right)=x$, then $\int_1^2 f(x)dx$ is equal to *[J & K-2005]*

(a) $\frac{3}{5}\ln 2$ (b) $\frac{-3}{5}(1+\ln 2)$

(c) $\frac{-3}{5}\ln 2$ (d) None of these

14. If $\int_a^b x^3 dx = 0$ and $\int_a^b x^2 dx = \frac{2}{3}$, then the value of a and b respectively are ***[Kerala PET-2008]***

(a) 1, –1 (b) –1, 1

(c) 1, 1 (d) –1, –1

15. What is the value of $\int_0^1 xe^{x^2} dx$?

[NDA-2008]

(a) $\frac{(e-1)}{2}$ (b) $e^2 - 1$

(c) $2(e-1)$ (d) $e-1$

16. If $f(t) = \int_{-t}^{t} \frac{dx}{1+x^2}$, then $f'(1)$ is

[Roorkee-2000]

(a) Zero (b) 2/3

(c) –1 (d) 1

17. $\int_0^{\frac{\pi^2}{4}} \sin\sqrt{x}\,dx$ is ***[MPPET-2010]***

(a) 0 (b) 1

(c) 2 (d) 4

18. Value of the integral $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx$ is

[MPPET-2010]

(a) 0 (b) 1/2

(c) 1 (d) 2

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	16. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	17. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ	18. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ	
6. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ	
7. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

2. (a) $\int_1^x \frac{\log(x^2)}{x} dx = \int_1^x \frac{2\log x}{x} dx$

Let $\log x = t \Rightarrow \frac{dx}{x} = dt$

$$I = 2\int_0^{\log x} t\,dt = 2\left[\frac{t^2}{2}\right]_0^{\log x} = (\log x)^2$$

9. (b)

$$\int_0^1 \left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots + \frac{(-1)^n (x^n)}{n!} + \ldots\right) e^{2x} dx$$

$$\left(\because 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots + \frac{(-1)^n (x^n)}{n!} + \ldots = e^{-x}\right)$$

$$= \int_0^1 e^{-x} e^{2x} dx$$

$$= \int_0^1 e^x dx = e^x \Big|_0^1$$

$$= e - 1$$

11. (b) $\int_0^a x dx \le a + 4$

Taking expression $\int_0^a x dx = \frac{x^2}{2}\Big|_0^a = \frac{a^2}{2}$

$$\Rightarrow \frac{a^2}{2} \le a + 4 \Rightarrow a^2 \le 2a + 8 \Rightarrow a^2 - 2a - 8 \le 0$$

$$\Rightarrow (a+2)(a-4) \le 0 \Rightarrow -2 \le a \le 4$$

12. (c) $\int_0^{(\pi/2)^{1/3}} \theta^5 \sin\theta^3 d\theta = \int_0^{(\pi/2)^{1/2}} \theta^2 \theta^3 \sin\theta^3 d\theta$

Let $t = \theta^3 \Rightarrow dt = 3\theta^2 d\theta$, then

$$\Rightarrow \frac{1}{3}\int_0^{\pi/2} t \sin t dt = \frac{1}{3}\left[-t\cos t + \sin t\right]_0^{\pi/2} = \frac{1}{3}$$

13. (b) $2f(x) - 3f\left(\frac{1}{x}\right) = x$ ..... (1)

replace x by $\frac{1}{x}$

$$2f\left(\frac{1}{x}\right) - 3f(x) = \frac{1}{x} \quad \text{..... (2)}$$

Equation (1) and (2), we get adding

$$4f(x) - 6f\left(\frac{1}{x}\right) = 2x$$

$$-9f(x) + 6f\left(\frac{1}{x}\right) = \frac{3}{x}$$

$f(x) = -\frac{1}{5}\left\{2x + \frac{3}{x}\right\}$, then put value of $f(x)$ in integration

$$-\frac{1}{5}\int_1^2 \left[2x + \frac{3}{x}\right] dx = -\frac{1}{5}\left[x^2 + 3\log x\right]_1^2$$

$$= -\frac{1}{5}\left[3 + 3\log 2\right]$$

$$= \frac{-3}{5}\left[1 + \log 2\right]$$

14. (b) $\int_a^b x^3 dx = 0$, (given)

$$\left|\frac{1}{4}x^4\right|_a^b = 0 \Rightarrow \frac{1}{4}(b^4 - a^4) = 0 \Rightarrow b^4 - a^4 = 0$$

$$\int_a^b x^2 dx = \frac{2}{3} \Rightarrow \left|\frac{x^3}{3}\right|_a^b = \frac{2}{3} \Rightarrow b^3 - a^3 = 2$$

$(b^4 - a^4) = 0 \Rightarrow (b^2 - a^2)(b^2 + a^2) = 0$

or $(b-a)(b+a) = 0 \Rightarrow b = \pm a$

but $b = a$ does not satisfy the equation,

$\therefore \quad b = -a$

$b^3 - a^3 = 2 \Rightarrow (-a)^3 - a^3 = 2$

$-2a^3 = 2$ or $a^3 = -1 \Rightarrow a \quad -1 \Rightarrow b \quad -a$

Hence $b \quad -(-1) = 1 \Rightarrow a \quad -1, b \quad 1$

15. (a) $\int_0^1 x e^{x^2} dx$

Let $x^2 = t \Rightarrow 2x dx = dt$

$$\frac{1}{2}\int_0^1 e^t dt = \frac{1}{2}\left[e^t\right]_0^1 \quad = \frac{1}{2}\left[e - 1\right]$$

16. (d) $f(t) = \int_{-t}^{t} \frac{dx}{1+x^2}$

$\left(\because f(-x) = f(x)\right)$

so $f(t) = 2\int_0^t \frac{dx}{1+x^2}$

$f(t) = 2\tan^{-1}x\Big|_0^t$

$f(t) = 2\tan^{-1}t$

Differentiating both side w.r.t. we get

$f'(t) = \frac{2}{1+t^2}$ Put $t = 1$, $f'(1) = 1$

17. (c) $\int_0^{\pi^2/4} \sin\sqrt{x} dx$

$\sqrt{x} = t$

$x = t^2 \Rightarrow dx = 2t dt$

$2\int_0^{\pi/2} t\sin t = -t\cos t - \int -\cos t dt$

$= 2\left[-t\cos t + \sin t\right]_0^{\pi/2} = 2$

18. (c) $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2}\Big|_0^1 = 1$

LECTURE



2 Properties of Definite Integral

BASIC CONCEPTS

1. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$: i.e., If the interval of integration is split up into a finite number of sub-interval, the value of the definite integral over the given interval is equal to the sum of the definite integrals over all the sub-intervals.

2. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$: i.e., the integral remains unchanged if x is replaced by $a - x$ in the integrand.

3. $$\int_0^{2a} \frac{f(x)dx}{f(x)+f(2a-x)} = a$$

$$(10)\ \int_0^{\pi/2} \frac{f(x)dx}{f(x)+f\left(\frac{\pi}{2}-x\right)} = \frac{\pi}{4}$$

4. $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

5. $$\int_a^b \frac{f(x)dx}{f(x)+f(a+b-x)} = \frac{b-a}{2}$$

$$(5.1)\ \int_{\pi/6}^{\pi/3} \frac{f(x)dx}{f(x)+f\left(\frac{\pi}{2}-x\right)} = \frac{\pi}{12}$$

6. $\int_{-a}^{0} f(x)dx = \int_0^a f(-x)dx$

7. $$\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx \quad \text{if } f(-x) = f(x)$$

$$0 \quad \text{if } f(-x) = -f(x)$$

i.e., the value of the definite integral between the limits-a to a is twice the integral of the same integrand with limits 0 to a if $f(x)$ is an even function and is zero if $f(x)$ is an odd function.

NOTE

This property is useful for changing the limits from $-a$ to a to 0 to a.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Prove that $\int_0^{\pi} |\cos x|\,dx = 2$

[MP-99; CBSE-90, 92, 94]

Solution

Let $I = \int_0^{\pi} |\cos x|\,dx$

Clearly $|\cos x| = \cos x$, if $\cos x > 0$
i.e., x lies between 0 and $\pi/2$ and $|\cos x| = -\cos x$ if $\cos x < 0$
i.e., x lies between $\pi/2$ and π.

$\therefore\ I = \int_0^{\pi/2} |\cos x|\,dx + \int_{\pi/2}^{\pi} |\cos x|\,dx$

$$= \int_0^{\pi/2} \cos x\,dx + \int_{\pi/2}^{\pi} (-\cos x)\,dx$$

$$= [\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi}$$

$$= \left[\sin\frac{\pi}{4} - \sin 0\right] - \left[\sin\pi - \sin\frac{\pi}{2}\right]$$

$$= (1-0) - (0-1) = 1 + 1 = 2$$

Proved.

2. Prove that $\int_{\pi/6}^{\pi/3} \frac{1}{1+\sqrt{\tan x}}\,dx = \frac{\pi}{12}$

[MP-99]

Solution

$$\text{Let } I = \int_{\pi/6}^{\pi/3} \frac{1}{1+\sqrt{\tan x}}\,dx = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\frac{\sqrt{\sin x}}{\sqrt{\cos x}}}$$

$$\Rightarrow I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx \quad \ldots\ldots\ldots (1)$$

Again

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)}+\sqrt{\cos\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right)}}\,dx$$

$$\left[\because \int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx\right]$$

$$\Rightarrow I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}\,dx$$

$$\Rightarrow I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \quad \ldots\ldots\ldots (2)$$

Adding equations (1) and (2), we get

$$I + I = = \int_{\pi/6}^{\pi/3} dx = [x]_{\pi/6}^{\pi/3}$$

$$= \int_{\pi/6}^{\pi/3} dx = [x]_{\pi/6}^{\pi/3}$$

$$\Rightarrow 2I = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$\Rightarrow I = \frac{\pi}{12}$$

Proved.

3. Prove that $\int_0^{2\pi} \frac{\sin 2\theta}{a - b\cos\theta}\,d\theta = 0, (a > b > 0)$

[Roorkee-88]

Solution

$$\text{Let } I = \int_0^{2\pi} \frac{\sin 2\theta}{a - b\cos\theta}\,d\theta$$

$$= \int_0^{2\pi} \frac{\sin 2(2\pi-\theta)}{a - b\cos(2\pi-\theta)}\,d\theta$$

$$\Rightarrow I = -\int_0^{2\pi} \frac{\sin 2\theta}{a - b\cos\theta}\,d\theta$$

$$\Rightarrow I = -I \Rightarrow I + I = 0$$

$$\Rightarrow 2I = 0 \Rightarrow I = 0$$

Proved.

4. Evaluate $\int_{-1}^{1} f(x)\,dx$ where

$$f(x) = \begin{cases} 1-2x, & x \le 0 \\ 1+2x, & x \ge 0 \end{cases}$$

[CBSE-95 (C)]

Solution

We have $\int_{-1}^{1} f(x)\,dx = \int_{-1}^{0} f(x)\,dx + \int_0^1 f(x)\,dx$

$$= \int_{-1}^{0} (1-2x)\,dx + \int_0^1 (1+2x)\,dx$$

[by definition of $f(x)$]

$$= \left[x - x^2\right]_{-1}^{0} + \left[x + x^2\right]_0^1$$

$$= [0 - (-1-1)] + [(1+1) - (0)] = 4$$

5. Evaluate $\int_{-\pi/2}^{\pi/2} \sin^2 x\,dx$

[CBSE-91 (C); HPSB-1996]

Solution

Let $f(x) = \sin^2 x$. Then $f(-x) = \sin^2(-x)$

$= [\sin(-x)]^2 = [-\sin x]^2$

$= \sin^2 x = f(x)$

Therefore $f(x)$ is an even function

So, $\int_{-\pi/2}^{\pi/2} \sin^2 x dx = 2\int_0^{\pi/2} \sin^2 x dx$

[using property]

$= 2I$, say, where $I = \int_0^{\pi/2} \sin^2 x dx$

Now $I = \int_0^{\pi/2} \sin^2 x dx$ ..... (i)

$\Rightarrow I = \int_0^{\pi/2} \sin^2\left(\frac{\pi}{2} - x\right) dx$ [using property]

$\Rightarrow I = \int_0^{\pi/2} \cos^2 x dx$ ..... (ii)

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} (\sin^2 x + \cos^2 x) dx$$

$$= \int_0^{\pi/2} 1.dx = [x]_0^{\pi/2} = \frac{\pi}{2}$$

Hence $\int_{-\pi/2}^{\pi/2} \sin^2 x dx = \frac{\pi}{2}$

6. Evaluate $\int_1^4 f(x)dx$ where $f(x)$

$$= \begin{cases} 4x+3 & \text{if} \quad 1 \le x \le 2 \\ 3x+5 & \text{if} \quad 2 \le x \le 4 \end{cases}$$ ***[CBSE-1992]***

Solution

$$I = \int_1^4 f(x)dx = \int_1^2 f(x)dx + \int_2^4 f(x)dx$$

$$= \int_1^2 (4x+3)dx + \int_2^4 (3x+5)dx$$

$$= \left[\frac{4x^2}{2} + 3x\right]_1^2 + \left[\frac{3x^2}{2} + 5x\right]_2^4$$

$$= [(2\ (2)^2 + 3(2)) - (2+3)] +$$

$$\left[\left(\frac{3(4)^2}{2} + 5(4)\right) - \left(\frac{3(2)^2}{2} + 5(2)\right)\right]$$

$$= [8 + 6 - 5] + [24 + 20 - 6 - 10]$$

$$= 9 + 28 = 37$$

7. Evaluate $\int_{\pi/4}^{\pi/2} \cos 2x \log \sin x dx$

[CBSE-2003]

Solution

We have $I = \int_{\pi/4}^{\pi/2} \underset{II}{\cos 2x} \underset{I}{\log \sin x}\ dx$

$$= \left[\frac{1}{2}(\log \sin x)\sin 2x\right]_{\pi/2}^{\pi/2}$$

$$- \int_{\pi/4}^{\pi/2} \frac{1}{2} \cot x \sin 2x dx$$

$$= \left[0 - \frac{1}{2}\log\left(\frac{1}{\sqrt{2}}\right)\right] - \int_{\pi/4}^{\pi/2} \cos^2 x dx$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\int_{\pi/4}^{\pi/2} (1 + \cos 2x) dx$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left[\left(\frac{\pi}{2} + 0\right) - \left(\frac{\pi}{4} + \frac{1}{2}\right)\right]$$

$$= \frac{1}{4}\log 2 - \frac{\pi}{8} + \frac{1}{4}$$

8. Prove that $\int_0^{\pi} \frac{x}{1 + \sin\alpha \sin x} dx = \frac{\pi}{\cos\alpha}\left(\frac{\pi}{2} - \alpha\right)$

[Roorkee-1994]

Solution

Let $I = \int_0^{\pi} \frac{x}{1 + \sin\alpha \sin} dx$ ..... (i)

then $I = \int_0^{\pi} \frac{\pi - x}{1 + \sin\alpha \sin(\pi - x)} dx$

$$\left[\because \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$$

$\Rightarrow I = \int_0^{\pi} \frac{\pi - x}{1 + \sin\alpha \sin x} dx$ .. (ii)

On adding (i) and (ii), we get

$$2I = \int_0^{\pi} \frac{\pi}{1 + \sin\alpha \sin x} dx$$

$$= \pi \int_0^{\pi} \frac{dx}{1 + \sin\alpha \sin x}$$

Put $\tan\frac{x}{2}=t\Rightarrow \sec^2\frac{x}{2}.\frac{1}{2}dx=dt\Rightarrow dx=\frac{2dt}{1+t^2}$

when $x=0,\ t=0$

when $x=\pi/2,\ t\to\infty$

$$= \pi\lim_{t\to\infty}\int_0^t \frac{\frac{2dt}{1+t^2}}{1+\sin\alpha.\frac{2t}{1+t^2}} \qquad \int$$

$$= 2\pi\lim_{t\to\infty}\int_0^t \frac{dt}{1+t^2+2t\sin\alpha}$$

$$= 2\pi\lim_{t\to\infty}\int_0^t \frac{dt}{(1+\sin\alpha)^2+\cos^2\alpha}$$

$$= 2\pi\lim_{t\to\infty}\frac{1}{\cos\alpha}\left[\tan^{-1}\left(\frac{t+\sin\alpha}{\cos\alpha}\right)\right]_0^t$$

$$= \frac{2\pi}{\cos\alpha}\left[\lim_{t\to\infty}\tan^{-1}\frac{t+\sin\alpha}{\cos\alpha}-\tan^{-1}(\tan\alpha)\right]$$

$$= \frac{2\pi}{\cos\alpha}\left(\frac{\pi}{2}-\alpha\right)$$

$\Rightarrow I=\frac{\pi}{\cos\alpha}\left(\frac{\pi}{2}-\alpha\right)$ Proved.

9. Evaluate $\int_0^a \frac{x^4}{\sqrt{a^2-x^2}}dx$

[CBSE-1994]

Solution

Let $x = a\sin 0$ Then $dx = d(a\sin 0) = a\cos 0\, d0$

Also $x = 0 \Rightarrow a\sin\theta = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$

And, $x = a \Rightarrow a\sin\theta = a$

$\Rightarrow \sin 0 = 1 \Rightarrow 0 = \frac{\pi}{2}$

$$\therefore \int_0^a \frac{x^4}{\sqrt{a^2-x^2}}dx$$

$$= \int_0^{\pi/2}\frac{(a\sin 0)^4}{\sqrt{a^2-a^2\sin^2 0}}a\cos\theta d\theta$$

$$= a^4\int_0^{\pi/2}\sin^4 0 d0 = a^4.\frac{1}{4}\int_0^{\pi/2}(2\sin^2 x)^2 dx$$

$$= \frac{1}{4}a^4\int_0^{\pi/2}(1-\cos 2x)^2 dx$$

$$= \frac{a^4}{4}\int_0^{\pi/2}(1-2\cos 2x+\cos^2 2x)dx$$

$$= \frac{a^4}{4}\int_0^{\pi/2}1-2\cos 2x+\frac{1+\cos 4x}{2}dx$$

$$= \frac{a^4}{8}\int_0^{\pi/2}(3-4\cos 2x+\cos 4x)dx$$

$$= \frac{a^4}{8}\left[3x-\frac{4}{2}\sin 2x+\frac{\sin 4x}{4}\right]_0^{\pi/2}$$

$$= \frac{a^4}{8}\left[\left\{\frac{3\pi}{2}-2\sin\pi+\frac{1}{4}\sin 2\pi\right\}-\{0-0+0\}\right]$$

$$= \frac{a^4}{8}\left[\frac{3\pi}{2}-0+0\right]=\frac{a^4}{8}\times\frac{3\pi}{2}=\frac{3\pi a^4}{16}$$

10. Find the value of $\int_0^\infty \frac{dx}{(x+\sqrt{x^2+1})^3}$

[EAMCET-1992]

Solution

By putting $x = \tan 0$, we get $\int_0^\infty \frac{dx}{(x+\sqrt{x^2+1})^3}$

$$= \int_0^{\pi/2}\frac{\sec^2 0 d0}{(\tan 0+\sec 0)^3}=\int_0^{\pi/2}\frac{\cos 0}{(1+\sin 0)^3}d\theta$$

$$= \left[-\frac{1}{2(1+\sin 0)^2}\right]_0^{\pi/2}=-\frac{1}{8}+\frac{1}{2}=\frac{3}{8}$$

11. Evaluate: ***[CBSE-2010]***

$\int_{-\pi/2}^{\pi/2}\sin^5 x dx$

Solution

$\int_{-\pi/2}^{\pi/2}\sin^5 x dx = 0$

NOTE

($\because \sin^5 x$ is an odd function)

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Prove that: $\int_{-\pi/2}^{\pi/2} x\sin x\,dx = 2\int_0^{\pi/2} x\sin x\,dx$

2. Prove that: $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx = \frac{b-a}{2}$

3. Prove that: $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)} = \frac{\pi}{4}$

4. Evaluate $\int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}}dx$ ***[PSB-89 (C)]***

5. Prove that: $\int_0^{\pi} \frac{\sin(2nx)}{\sin x}dx = 0$, n is any integer

6. Prove that $\int_{-1}^{1} \sin^3 x.\cos^2 x\,dx = 0$

[MP-97; MLNR-91]

7. Evaluate $\int_0^1 \frac{\log_e(1+x)}{1+x^x}dx$

[MP-97, 98, 99, 2000; IIT-97]

8. Prove that: $\int_0^{\pi/4} \log(1+\tan x)dx = \frac{\pi}{8}\log 2$

[MP-2007; CBSE-2007]

9. Evaluate $\int_0^{\infty} \frac{x^2dx}{(x^2+a^2)(x^2+b^2)}$

10. Evaluate $\int_0^1 x(1-x)^5dx$ ***[CBSE-96 (C)]***

11. Evaluate $\int_0^{\pi/2} \frac{1}{5+4\sin x}dx$

[AISSE-90; CBSE-90 (C)]

12. Evaluate $\int_{-\pi/4}^{\pi/4} x^3\sin^4 x\,dx$ ***[CBSE-95 (C)]***

13. Evaluate $\int_0^{\pi/2} \frac{1}{a^2\sin^2 x+b^2\cos^2 x}dx$

[AISSE-86; HSB-2001]

14. Evaluate $\int_0^1 \frac{1-x}{1+x}dx$ ***[HSB-1987]***

15. Prove that $\int_0^{\pi/2} \frac{dx}{1+\tan x} = \int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}dx = \frac{\pi}{4}$

[MP-97; CBSE-94 (C); Roorkee-80; UP-85; CBSE-92; PSB-92, MLNR-90; IIT-83]

16. Prove that $\int_0^a \sqrt{\frac{a-x}{a+x}}dx = a\left(\frac{\pi}{2}-1\right)$

[MP-2003]

17. Evaluate $\int_0^1 \frac{5x^3}{\sqrt{1-x^8}}dx$ ***[MP-99]***

18. Evaluate $\int_1^3 \frac{\sqrt{4-x}}{\sqrt{x}+\sqrt{4-x}}dx$

[CBSE-Sample Paper-I]

19. Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

[DSB-2002 (S)]

20. Evaluate $\int_{-1}^1 \log\frac{(2-x)}{2+x}dx$

21. Evaluate $\int_1^2 \frac{\log x}{x^2}dx$

[MP-94, 95, 99, 2000]

22. Prove that $\int_0^{2a} f(x)dx = \int_0^{2a} f(2a-x)dx$

[CBSE-93, 2002 (C)]

23. Evaluate $\int_0^2 x\sqrt{2-x}dx$ ***[MP-2000]***

24. Evaluate $\int_0^{\pi/2} \frac{\sin x}{1+\cos^2 x}dx$

[CBSE-97; HSB-99]

25. Evaluate $\int_0^1 \sqrt{\frac{1-x}{1+x}}dx$

[CBSE-2004]

26. Evaluate $\int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x}dx$

[PSB-90; HSB-99]

27. Evaluate $\int_{-1}^{1} e^{|x|}dx$ ***[CBSE-91 (C)]***

28. Evaluate $\int_0^{\pi/2} \frac{\sin 2x}{\sin^4 x+\cos^4 x}dx$ ***[CBSE-2003; HPSB-97 (C)]***

29. Evaluate $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x+\cos^n x}dx$ ***[CBSE-2004]***

30. Prove that $\int_0^{\pi} e^{\sec\theta}\sec^3\theta\,(\sin^2\theta+\cos\theta+\sin\theta+\sin\theta\cos\theta)d\theta = -(e+e^{-1})$

ANSWERS

4. $\frac{1}{2}a$
7. $\frac{\pi}{8}\log_e 2$
9. $\frac{\pi}{2(a+b)}$
10. $\frac{1}{42}$
11. $\frac{2}{3}\tan^{-1}\left(\frac{1}{3}\right)$
12. 0
13. $\frac{\pi}{2ab}$
14. $-1+2\log 2$
17. $\frac{5\pi}{8}$
20. 0
21. $\frac{1}{2}-\frac{1}{2}\log 2$
23. $\frac{16\sqrt{2}}{15}$
24. $\frac{\pi}{4}$
25. $\frac{\pi}{2}-1$
26. $\frac{\pi}{4}$
27. $2e-2$
28. $\frac{\pi}{2}$
29. $\frac{\pi}{4}$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\int_{\pi/4}^{3\pi/4} \frac{dx}{1+\cos x}$ is equal to ***[IIT-1999]***

(a) 2 (b) -2

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

Solution

(a) Use property of integration

$$I = \int_{\pi/4}^{3\pi/4} \frac{dx}{1+\cos x} \quad \because \cos(\pi - x) = -\cos x$$

$$\therefore \quad 2I = \int_{\pi/4}^{3\pi/4} \frac{2}{1-\cos^2 x}dx$$

$$= 2\int_{\pi/4}^{3\pi/4} \operatorname{cosec}^2 x\,dx$$

2. $\int_0^{\pi/2} \log\sin x dx$ equals

[PET (Raj.)-88, 89, 95, 96, 96; MP-94, 2000; UPSEAT-84, 2001]

(a) $\frac{\pi}{2}\log 2$ (b) $-\frac{\pi}{2}\log 2$

(c) $\frac{\pi}{2}\log_{10}2$ (d) $-\frac{\pi}{2}\log_{10}2$

Solution

(b) $I = \int_0^{\pi/2} \log\sin x dx,$

$$I = \int_0^{\pi/2} \log\cos x dx \text{ (by property)}$$

$$\therefore \quad 2I = \int_0^{\pi/2} \log(\sin x \cos x)\,dx$$

$$= \int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right)dx$$

$$= \int_0^{\pi/2} \log\sin 2x\,dx - \frac{\pi}{2}\log 2$$

$$= \frac{1}{2}\int_0^{\pi} \log\sin t\,dt - \frac{\pi}{2}\log 2\,, \text{ where } t = 2x$$

$$= 2.\ \frac{1}{2}\int_0^{\pi/2} \log\sin t\,dt - \frac{\pi}{2}\log 2 = I - \frac{\pi}{2}\log 2$$

$$\Rightarrow I = -\frac{\pi}{2}\log 2$$

3. If $f(x)$ is an even function, then $\int_0^{\pi/2} f(\cos 2x)\cos x\,dx$ is equal to:

[IIT (Main)-2003]

(a) $\sqrt{2}\int_0^{\pi/4} f(\sin 2x)\sin x\,dx$

(b) $\sqrt{2}\int_0^{\pi/4} f(\sin 2x)\cos x\,dx$

(c) $\frac{1}{\sqrt{2}}\int_0^{\pi/4} f(\sin 2x)\sin x\,dx$

(d) $\frac{1}{\sqrt{2}}\int_0^{\pi/4} f(\sin 2x)\cos x\,dx$

Solution

(b) $I = \int_0^{\pi/2} f(\cos 2x)\cos x\,dx$ ..... (1)

$$I = \int_0^{\pi/2} f(\cos(\pi - 2x))\cos\left(\frac{\pi}{2} - x\right)dx \quad \text{[by Property]}$$

$$= \int_0^{\pi/2} f(-\cos 2x)\sin x\,dx$$

$$= \int_0^{\pi/2} f(\cos 2x)\sin x\,dx \qquad \text{..... (2)}$$

($\because f(x)$ is even function)

$$\therefore \quad 2I = \int_0^{\pi/2} f(\cos 2x)(\cos x + \sin x)\,dx$$

$$= \sqrt{2}\int_0^{\pi/2} f(\cos 2x)\sin(x + \pi/4)\,dx$$

$$= \sqrt{2}\int_{-\pi/4}^{\pi/4} f(\sin 2t)\cos t\ dt\,,$$

where $x = \frac{\pi}{4} + t$

$$= 2\sqrt{2}\int_0^{\pi/4} f(\sin 2t)\cos t\ dt$$

$$\therefore \quad I = \sqrt{2}\int_0^{\pi/4} f(\sin 2x)\cos x\ dx$$

4. $\int_{-2}^{0} \{x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)\}dx$ is equal to ***[IIT (Screening)-2005]***

(a) 4 (b) 1

(c) 0 (d) 2

Solution

(a) $I = \int_{-2}^{0} \{(x+1)^3 + 2 + (x+1)\cos(x+1)\}dx$

$$= \int_{-1}^{1} (t^3 + 2 + t\cos t)dt\,, \text{ where, } t = x + 1$$

$$= \int_{-1}^{1} (t^3 + t\cos t)dt + 2\int_{-1}^{1} dt$$

$$= 0 + 2[t]_{-1}^{1} = 4$$

5. $\int_0^{\pi} \frac{\cos x}{x^4 + (\pi - x)^4}dx$ is equal to

[NDA-2004]

(a) 0 (b) π

(c) $\pi/4$ (d) $\pi/2$

Solution

(a) $I = \int_0^{\pi} \frac{\cos x}{x^4(\pi - x)^4}dx = \int_0^{\pi} \frac{\cos(\pi - x)}{(\pi - x)^4 + x^4}dx\,,$

by property

$= -I \quad \Rightarrow I = 0$

6. Let $I_1 = \int_0^{\infty} \frac{\log(x+1/x)}{1+x^2}dx$ and $I_2 = \int_0^{\pi/2} \log\sin 2t\ dt$, then $I_1 + I_2$ is equal to

[ICS-2006]

(a) $(\pi/2)\log 2$ (b) $-(\pi/2)\log 2$
(c) $\pi \log 2$ (d) $-\pi \log 2$

Solution

(a) $I_1 = \int_0^{\pi/2} .\log.(\tan t + \cot t)dt$,

where $x = \tan t$

$= \int_0^{\pi/2} \log(2/\sin 2t)dt = (\pi/2)\log 2 - I_2$

$\Rightarrow I_1 + I_2 = (\pi/2)\log 2$

7. $\int_0^{\infty} \frac{dx}{(x+\sqrt{x^2+1})^3} =$ ***[EAMCET-1992]***

(a) 3/8 (b) 1/8
(c) – 3/8 (d) None of these

Solution

(a) Putting $x = \tan\theta$, we get $\int_0^{\infty} \frac{dx}{(x+\sqrt{x^2+1})^3}$

$= \int_0^{\pi/2} \frac{\sec^2 0\, d0}{(\tan\theta+\sec\theta)^3} = \int_0^{\pi/2} \frac{\cos 0}{(1+\sin\theta)^3}d\theta$

$= \left[-\frac{1}{2(1+\sin 0+\sec 0)^2}\right]_0^{\pi/2} = -\frac{1}{8}+\frac{1}{2} = \frac{3}{8}$

8. The value of $\int_0^{\pi/2} \frac{5\sin x + 3\cos x}{\sin x + \cos x}dx$ is

[UPSC-2007]

(a) 0 (b) $\pi/2$
(c) 4π (d) 2π

Solution

(d) $I = \int_0^{\pi/2} \frac{5\sin x + 3\cos x}{\sin x + \cos x}dx$ ..... (1)

$I = \int_0^{\pi/2} \frac{5\sin\left(\frac{\pi}{2}-x\right)+3\cos\left(\frac{\pi}{2}-x\right)}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}dx$

$= \int_0^{\pi/2} \frac{5\cos x + 3\sin x}{\cos x + \sin x}dx$ .... (2)

Adding equation (1) and (2),

$2I = \int_0^{\pi/2} \frac{(5\sin x + 3\cos x) + (5\cos x + 3\sin x)}{\cos x + \sin x}dx$

$= \int_0^{\pi/2} \frac{8(\sin x + \cos x)}{\cos x + \sin x}dx$

$= \int_0^{\pi/2} 8dx = \frac{8\pi}{2} = 4\pi \therefore I = 2\pi$

9. The value of $\int_0^{\pi/2} (\sin^{100} x - \cos^{100} x)dx$ is:

[Karnataka CET-2007]

(a) $\frac{1}{100}$ (b) $\frac{100!}{(100)^{100}}$

(c) $\frac{\pi}{100}$ (d) 0

Solution

(d) $\int_0^{\pi/2} (\sin^{100} x - \cos^{100} x)dx$

(replace x by $\frac{\pi}{2} - x$)

$= \int_0^{\pi/2} \left(\sin\left(\frac{\pi}{2}-x\right)\right)^{100} - \left(\cos\left(\frac{\pi}{2}-x\right)\right)^{100} dx$

$= \int_0^{\pi/2} (\cos^{100} x - \sin^{100} x)dx$

$I = -I \Rightarrow 2I = 0 \Rightarrow I = 0$

10. If $k\int_0^1 x.f(3x)dx = \int_0^3 t.f(t)dt$, then the value of k is

[Karnataka CET-2007]

(a) 9 (b) 3
(c) 1/9 (d) 1/3

Solution

(a) $k\int_0^1 x.f(3x)dx = \int_0^3 t.f(t)dt$ Think that

$k\int_0^1 xf(3x)dx$

Let $3x = t \Rightarrow dx = dt/3$

$x = 0 \Rightarrow t = 0$; $x = 1 \Rightarrow t = 3$

$$\Rightarrow \quad k\int_0^3 \frac{1}{3}\frac{t}{3}f(t)dt$$

$$\Rightarrow \quad \frac{k}{9}\int_0^3 tf(t)=\int_0^3 t.f(t)dt \Rightarrow \frac{k}{9}=1\Rightarrow k=9$$

11. $\int_0^3 x(3-x)^{3/2}dx$ = ***[Gujarat CET-2007]***

(a) $\frac{108\sqrt{3}}{35}$ (b) $\frac{-108\sqrt{3}}{35}$

(c) $\frac{54\sqrt{3}}{35}$ (d) None of these

Solution

$$\text{(a)}\int_0^3 (3-x)(3-(3-x))^{3/2}dx=\int_0^3 (3-x)(x^{3/2})dx$$

$$=3\int_0^3 x^{3/2}dx-\int_0^3 x^{5/2}dx=\frac{108\sqrt{3}}{35}$$

12. $\int_0^{\pi/4}\frac{\sec x dx}{1+2\sin^2 x}$ is equal to ***[MNR-94]***

(a) $\frac{1}{3}\left[\log(\sqrt{2}+1)+\frac{\pi}{2\sqrt{2}}\right]$

(b) $\frac{1}{3}\left[\log(\sqrt{2}+1)-\frac{\pi}{2\sqrt{2}}\right]$

(c) $3\left[\log(\sqrt{2}+1)-\frac{\pi}{2\sqrt{2}}\right]$

(d) $3\left[\log(\sqrt{2}+1)+\frac{\pi}{2\sqrt{2}}\right]$

Solution

(a) Let $I=\int_0^{\pi/4}\frac{\cos x}{\cos^2 x(1+2\sin^2 x)}dx$

$$=\int_0^{\pi/4}\frac{\cos x dx}{(1-\sin^2 x)(1+2\sin^2 x)}$$

$$=\frac{1}{3}\int_0^{\frac{1}{\sqrt{2}}}\left(\frac{1}{1-t^2}+\frac{2}{1+2t^2}\right)dt$$ By partial fractions, where, $t = \sin x$

$$=\frac{1}{3}\left[\frac{1}{2.1}\log\frac{1+t}{1-t}+\frac{2}{\sqrt{2}}\tan^{-1}t\sqrt{2}\right]_0^{1/\sqrt{2}}$$

$$=\frac{1}{3}\left[\frac{1}{2}\log\frac{(\sqrt{2}+1)}{(\sqrt{2}-1)}+\sqrt{2}\tan^{-1}1\right]$$

$$=\frac{1}{3}\left[\frac{1}{2}\log(\sqrt{2}+1)^2+\sqrt{2}.\frac{\pi}{4}\right]$$

$$=\frac{1}{3}\left[\log(\sqrt{2}+1)+\frac{\pi}{2\sqrt{2}}\right]$$

13. Find the value of $\int_{-1}^{3/2}|x\sin\pi x|dx$

[M.N.R.-96; IIT-82]

(a) $\frac{3}{\pi}+\frac{1}{\pi^2}$ (b) $\frac{3}{\pi}-\frac{1}{\pi^2}$

(c) $\frac{1}{\pi}(\pi+1)$ (d) None of these

Solution

(a) We know that sin θ is +ive when $0 \le \theta < \pi$ and sin θ is – ive when $\pi \le \theta \le 3\pi/2$. Hence we write

$$\int_{-1}^{3/2}|x\sin\pi x|dx=\int_{-1}^{0}x\sin\pi x dx$$

$$+\int_0^1 x\sin\pi x dx+\int_1^{3/2}-x(\sin\pi x)dx$$

$$=\int_{-1}^{1}x\sin\pi x dx+\int_1^{3/2}(-x\sin\pi x)dx$$

$$=2\int_0^1 x\sin\pi x dx-\int_1^{3/2}x\sin\pi x dx$$

by prop. as $f(-x)$ $f(x)$. Integrate by parts

$$I=2\left[-\frac{x}{\pi}\cos\pi x+\frac{1}{\pi^2}\sin\pi x\right]_0^1-$$

$$\left[-\frac{x}{\pi}\cos\pi x+\frac{1}{\pi^2}\sin\pi x\right]_1^{3/2}$$

$$=2\left(-\frac{\cos\pi}{\pi}\right)-\left(0+\frac{\cos\pi}{\pi}+\frac{1}{\pi^2}\sin\frac{3\pi}{2}\right)$$

$$=-\frac{3}{\pi}(-1)-\frac{1}{\pi^2}(-1)=\frac{3}{\pi}+\frac{1}{\pi^2}$$

14. $\int_0^{\pi} \frac{x\,dx}{1+\cos\alpha\sin x}(0<\alpha<\pi) =$ ***[Roorkee-2001]***

(a) $\frac{\pi\alpha}{\sin\alpha}$ (b) $\frac{2\pi\alpha}{\sin\alpha}$

(c) $\frac{\pi\alpha}{\cos\alpha}$ (d) $\frac{\pi}{\cos\alpha}$

Solution

(a) $I=\int_0^{\pi}\frac{x}{1+\cos\alpha\sin x}dx \;\therefore\; \int_0^{\pi}\frac{(\pi-x)dx}{1+\cos\alpha\sin(\pi-x)}$

$=\int_0^{\pi}\frac{(\pi-x)dx}{1+\cos\alpha\sin x}$,

Adding $2I=\int_0^{\pi}\frac{\pi dx}{1+\cos\alpha\sin x}$

$=2\pi\int_0^{\pi/2}\frac{dx}{1+\cos\alpha\sin x}$

or $I=\pi\int_0^{\pi/2}\frac{dx}{1+2\cos\alpha\sin\frac{1}{2}x\cos\frac{1}{2}x}$

$=\pi\int_0^{\pi/2}\frac{\sec^2\frac{1}{2}dx}{1+\tan^2\frac{1}{2}x+2\cos\alpha+\tan\frac{1}{2}x}$

Now put $\tan\frac{1}{2}x=t$. Then $\frac{1}{2}\sec^2\frac{1}{2}xdx=dt$.

$\therefore\; I=\pi\int_0^{1}\frac{2dt}{1+t^2+2t\cos\alpha}$

$=2\pi\int_0^{1}\frac{dt}{(t+\cos\alpha)^2+\sin^2\alpha}$

$=\frac{2\pi}{\sin\alpha}\left[\tan^{-1}\left(\frac{t+\cos\alpha}{\sin\alpha}\right)\right]_0^1$

$=\frac{2\pi}{\sin\alpha}\left[\tan^{-1}\left(\frac{1+\cos\alpha}{\sin\alpha}\right)-\tan^{-1}\left(\frac{0+\cos\alpha}{\sin\alpha}\right)\right]$

$=\frac{2\pi}{\sin\alpha}\left[\tan^{-1}\cot\left(\frac{1}{2}\alpha\right)-\tan^{-1}\cot\alpha\right]$

$=\frac{2\pi}{\sin\alpha}\left[\begin{matrix}\tan^{-1}\tan\left(\frac{1}{2}\pi-\frac{1}{2}\alpha\right)-\\ \tan^{-1}\tan\left(\frac{1}{2}\pi-\alpha\right)\end{matrix}\right]$

$=\frac{2\pi}{\sin\alpha}.\frac{1}{2}\alpha=\frac{\pi\alpha}{\sin\alpha}$.

15. Match the integrals in Column I with the values in Column II.

Column I		Column II	
(a)	$\int_{-1}^{1}\frac{dx}{1+x^2}$	(p)	$\int_2^3\frac{dx}{1-x^2}$
(b)	$\int_1^2\frac{dx}{x\sqrt{x^2-1}}$	(q)	2 log (2/3)
(c)	$\frac{1}{2}\log\left(\frac{2}{3}\right)$	(r)	$\frac{\pi}{3}$
(d)	$\int_0^{1}\frac{dx}{\sqrt{1-x^2}}$	(s)	$\frac{\pi}{2}$

[IIT-2007]

Solution

(a) $I=\int_{-1}^{1}\frac{dx}{1+x^2}\Rightarrow f(x)=\frac{1}{1+x^2}$ is an even function

$I=2\int_0^{1}\frac{dx}{1+x^2}=\left|2(\tan^{-1}x)\right|_0^1=\frac{\pi}{2}\;(a)\to(s)$

(b) $I=\int_0^{1}\frac{dx}{\sqrt{1-x^2}}=(\sin^{-1}x)\Big|_0^1=\frac{\pi}{2}\quad(b)\to(s)$

(c) $\int_2^3\frac{dx}{1-x^2}=-\int_2^3\frac{dx}{x^2-1}=-\left[\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right|\right]_2^3$

$=-\frac{1}{2}\ln\frac{2}{4}\times\frac{3}{1}=\frac{1}{2}\ln\left(\frac{2}{3}\right)\therefore(c)\to(p)$

(d) $I=\int_1^2\frac{dx}{x\sqrt{x^2-1}}$ Put $x=\sec\theta$

$dx=\sec 0\tan 0d0$

$=\int_0^{\pi/3}\frac{\sec 0\tan 0d0}{\sec 0\tan 0}=\int_0^{\pi/3}d\theta=\frac{\pi}{3}$

$\therefore(d)\to(r)$

16. Suppose we define definite integral with the formula $\int_a^b f(x)dx=\frac{b-a}{2}\{f(a)+f(b)\}$, for more accurate result for $c\in(a,b)$, $F(c)=\frac{c-a}{2}\{f(a)+f(c)\}+\frac{b-c}{2}\{f(b)+f(c)\}$ where $c=\frac{a+b}{2}$, $\int_a^b f(x)dx=\frac{b-a}{4}\{f(a)+f(b)+2f(c)\}$ Then $\int_0^{\pi/2}\sin x\,dx$ is equal to

[IIT-JEE-2006]

(a) $\frac{\pi}{8\sqrt{2}}$ (b) $\frac{\pi}{4\sqrt{2}}$

(c) $\frac{\pi}{4}(1+\sqrt{2})$ (d) $\frac{\pi}{8}(1+\sqrt{2})$

Solution

(d) $\int_0^{\pi/2} \sin x dx = \frac{\pi/2-0}{4}\{\sin 0+\sin \pi/2\}+ 2\sin\left(\frac{0+\pi/2}{2}\right)$

$= \frac{\pi}{8}(1+\sqrt{2})$

17. $\int_{-1}^{1} g'(x)dx =$ ***[IIT JEE-2008]***

(a) $2g(-1)$ (b) 0

(c) $-2g(1)$ (d) $2g(1)$

Solution

(d) As obtained earlier

$y' = \frac{1}{3[1-\{f(x)\}^2]}$ which is an even function

$\int_{-1}^{1} g'(x) = g(1)-g(-1) = 2g(1)$

18. Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0,1], p(0) = 1$ and $p(1) = 41$. Then $\int_0^1 p(x)dx$ equals

[AIEEE-2010]

(a) 21 (b) 41

(c) 42 (d) $\sqrt{41}$

Solution

(a) $p'(x) = p'(1-x)$

$\Rightarrow p(x) = -p(1-x)+c$

put $x = 0$

$p(0) = -p(1)+c$

$\Rightarrow c$ 42

$I = \int_0^1 p(x);\quad I = \int_0^1 p(1-x)dx$

$2I = \int_0^1 (p(x)+p(1-x))dx$

$= \int_0^1 42dx$

$21 = 42$

$\Rightarrow I = 21$

19. What is the value of $\int_0^{\pi/2} \ln(\tan x)dx$?

[NDA-2009]

(a) 0 (b) 1

(c) -1 (d) $\pi/4$

Solution

Let $I = \int_0^{\pi/2} \log(\tan x)dx$ .....(i)

and $I = \int_0^{\pi/2} \log\left\{\tan\left(\frac{\pi}{2}-x\right)\right\}dx$

$\Rightarrow I = \int_0^{\pi/2} \log(\cot x)dx$ .....(ii)

From Equ. (i) and (ii), we get

$2I = \int_0^{\frac{\pi}{2}} \log(\tan x \cot x)dx = 0 \Rightarrow I = 0$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. The value of the integral $I = \int_0^1 x(1-x)^n dx$ is

[AIEEE-2003; NDA-2009]

(a) $\frac{1}{n+1}$ (b) $\frac{1}{n+2}$

(c) $\frac{1}{n+1}-\frac{1}{n+2}$ (d) $\frac{1}{n+1}+\frac{1}{n+2}$

2. The value of $\int_{-\pi/2}^{\pi/2} (3\sin x + \sin^3 x)\,dx$ is

[MPPET-2003]

(a) 3

(b) 2

(c) 0

(d) 10/3

3. $\int_0^{\pi/2} \sin 2x \log \tan x\, dx$ is equal to

[Kerala (Engg.)-2002; AICBSE-90; Karnataka CET-96, 98]

(a) π (b) $\pi/2$

(c) 0 (d) 2π

4. $\int_{-1/2}^{1/2} (\cos x)\left[\log\left(\frac{1-x}{1+x}\right)\right]dx =$

[Karnataka CET-2002]

(a) 0 (b) 1
(c) $e^{1/2}$ (d) $2e^{1/2}$

5. The value of $\int_0^{\pi/2} \frac{\sin^{2/3} x}{\sin^{2/3} x + \cos^{2/3} x}dx$ is

[RPET-2001]

(a) $\pi/4$ (b) $\pi/2$
(c) $3\pi/4$ (d) π

6. $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}}$ is *[DCE-2001]*

(a) $\pi/3$ (b) $\pi/6$
(c) $\pi/12$ (d) $\pi/2$

7. If m and n are integers and $m = n$, then $\int_0^{\pi/2} \sin mx \sin nx dx =$

(a) $\pi/4$ (b) 0
(c) $\pi/2$ (d) π

8. $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is equal to

[IIT-93; DCE-2000, 01; HPSB-92]

(a) 0 (b) 1
(c) $\pi/2$ (d) $\pi/4$

9. $\int_{-3}^{3} \frac{x^2 \sin 2x}{x^2+1}dx =$

(a) 0 (b) 1
(c) $2\log_e 3$ (d) None of these

10. If $\int_{-3}^{2} f(x)dx = \frac{7}{3}$ and $\int_{-3}^{9} f(x)dx = -\frac{5}{6}$ then what is the value of $\int_{2}^{9} f(x)dx$?

[NDA-2007]

(a) $-\frac{19}{6}$ (b) $\frac{19}{6}$
(c) $\frac{3}{2}$ (d) $-\frac{3}{2}$

11. The value of $\int_{-1}^{1} \frac{\sin x - x^2}{3-|x|}dx$ is

[Roorkee-1995]

(a) 0 (b) $2\int_0^1 \frac{\sin x}{3-|x|}dx$
(c) $2\int_0^1 \frac{-x^2}{3-|x|}dx$ (d) $2\int_0^1 \frac{\sin x - x^2}{3-|x|}dx$

12. $\int_0^{\pi/2} \frac{\cos x - \sin x}{1+\sin x \cos x}dx =$ *[Karnataka CET-2004]*

(a) 2 (b) −2
(c) 0 (d) None of these

13. $\int_0^{2\pi} \frac{\sin 2\theta}{a - b\cos\theta}d\theta =$ *[Roorkee-88]*

(a) 1 (b) 2
(c) $\pi/4$ (d) 0

14. Let $\int_0^1 f(x)\,dx = 1$, $\int_0^1 x f(x)\,dx = a$ and $\int_0^1 x^2 f(x)\,dx = a^2$, then the value of $\int_0^1 (x-a)^2 f(x)\,dx =$ *[IIT-1990]*

(a) 0 (b) a^2
(c) $a^2 - 1$ (d) $a^2 - 2a + 2$

15. To find the numerical value of $\int_{-2}^{2} (px^2 + qx + s)\,dx$, it is necessary to know the values of constants *[IIT-92]*

(a) p (b) q
(c) s (d) p and s

16. $\int_{-\pi}^{\pi} (\cos px - \sin qx)^2\,dx$ is equal to (where p and q are integers)

[IIT-92; UPSEAT-2001]

(a) $-\pi$ (b) 0
(c) π (d) 2π

17. $\int_{-3}^{3} \frac{x^2 \sin x}{1+x^6}dx =$

(a) 4 (b) 2
(c) 0 (d) None of these

18. The value of $\int_0^1 \frac{dx}{x+\sqrt{1-x^2}}$ is

[MPPET-2003]

(a) $\pi/3$ (b) $\pi/2$
(c) 1/2 (d) $\pi/4$

19. $\int_0^{\infty} \frac{x\,dx}{(1+x)(1+x^2)} =$ *[Karnataka CET-2003]*

(a) 0 (b) $\pi/2$
(c) $\pi/4$ (d) 1

20. The value of $\int_0^1 \frac{x^4+1}{x^2+1}dx$ is **[MPPET-1998]**

(a) $\frac{1}{6}(3\pi-4)$ (b) $\frac{1}{6}(3-4\pi)$

(c) $\frac{1}{6}(3\pi+4)$ (d) $\frac{1}{6}(3+4\pi)$

21. If $\int_{\ln 2}^{x}(e^x-1)^{-1}dx=\ln\frac{3}{2}$, then what is the value of x? **[NDA-2007]**

(a) e^2 (b) $1/e$

(c) $ln\ 4$ (d) 1

22. $\int_8^{15}\frac{dx}{(x-3)\sqrt{x+1}}=$ **[UPSEAT-2003]**

(a) $\frac{1}{2}\log\frac{5}{3}$ (b) $\frac{1}{3}\log\frac{5}{3}$

(c) $\frac{1}{2}\log\frac{3}{5}$ (d) $\frac{1}{5}\log\frac{3}{5}$

23. $\int_0^{1/\sqrt{2}}\frac{\sin^{-1}x}{(1-x^2)^{3/2}}dx=$ **[Roorkee-84]**

(a) $\frac{\pi}{4}+\frac{1}{2}\log 2$ (b) $\frac{\pi}{4}-\frac{1}{2}\log 2$

(c) $\frac{\pi}{2}+\log 2$ (d) $\frac{\pi}{2}-\log 2$

24. If $I_1=\int_x^1\frac{dt}{1+t^2}$ and $I_2=\int_1^{1/x}\frac{dt}{1+t^2}$ for $x>0$, then

(a) $I_1=I_2$ (b) $I_1>I_2$

(c) $I_2>I_1$ (d) None of these

SOLUTIONS

1. (c) $I=\int_0^1 x(1-x)^n dx=\int_0^1(1-x)\{1-(1-x)\}^n$

$=\int_0^1 x^n(1-x)dx=\int_0^1 x^n dx-\int_0^1 x^{n+1}dx$

$I=\frac{1}{n+1}-\frac{1}{n+2}$

2. (c) $I=\int_{-\pi/2}^{\pi/2}(3\sin x+\sin^3 x)dx=0$

NOTE

Function $(3\sin x + \sin^3 x)$ is an odd function $\int_{-a}^{a} f(odd)=0$

3. (c) $I=\int_0^{\pi/2}\sin 2x\log\tan x dx$

$I=\int_0^{\pi/2}\sin 2\left(\frac{\pi}{2}-x\right)\log\tan\left(\frac{\pi}{2}-x\right)dx$

$\left[\because\int_0^a f(x)dx=\int_0^a f(a-x)dx\right]$

$=\int_0^{\pi/2}\sin 2x\log\cot x dx$

$=-\int_0^{\pi/2}\sin 2x\log\tan x dx$

$\therefore I=-I\Rightarrow 2I=0\Rightarrow I=0$

4.

$\cos x$ is odd function, $\log\left(\frac{1-x}{1+x}\right)$ is even function $\cos x\ \log\left(\frac{1-x}{1+x}\right)$ is an odd function.

Hence $\int_{-1/2}^{1/2}\cos x\log\left(\frac{1-x}{1+x}\right)dx=0$.

5.

Let $I=\int_0^{\pi/2}\frac{\sin^{2/3}x}{\sin^{2/3}x+\cos^{2/3}x}dx$ .....(1)

using property $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$

Also

$$I=\int_0^{\pi/2}\frac{\sin^{2/3}(\pi/2-x)}{\sin^{2/3}(\pi/2-x)+\cos^{2/3}(\pi/2-x)}dx$$

$$\Rightarrow\quad I=\int_0^{\pi/2}\frac{\cos^{2/3}x}{\cos^{2/3}x+\sin^{2/3}x}dx \qquad \text{.....(2)}$$

(1) + (2), we get

$$2I=\int_0^{\pi/2}dx\ ,\ I=\frac{\pi}{4}$$

6. (c) $I=\int_{\pi/6}^{\pi/3}\frac{dx}{1+\sqrt{\cot x}}=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cot x}}dx$

..... (1)

$$I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}$$

$$=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx \qquad \text{..... (2)}$$

Adding (1) and (2),

$$2I=\int_{\pi/6}^{\pi/3}dx;\ I=\frac{1}{2}\left(\frac{\pi}{3}-\frac{\pi}{6}\right)=\frac{\pi}{12}$$

7. (a) $I=\int_0^{\pi/2}\sin^2 mxdx \quad (\because m=n)$

Now $I=\int_0^{\pi/2}\cos^2 mxdx$

$$2I=x\big|_0^{\pi/2}\quad I=\frac{\pi}{4}$$

8. (d) $I=\int_0^{\pi/2}\frac{dx}{1+\tan^3 x}=\int_0^{\pi/2}\frac{\cos^3 x}{\sin^3 x+\cos^3 x}dx$

.....(i)

$$=\int_0^{\pi/2}\frac{\sin^3 x}{\cos^3 x+\sin^3 x}dx \qquad \text{.....(ii)}$$

Adding both, we get $2I=\int_0^{\pi/2}dx\Rightarrow I=\frac{\pi}{4}$

9. (a) $\int_{-3}^{3}\frac{x^2\sin 2x}{x^2+1}dx$

$\because$ The above function is odd, so

$$\int_{-3}^{3}\frac{x^2\sin 2x}{x^2+1}dx=0$$

10. (a) We know that

$$\int_{-3}^{9}f(x)dx=\int_{-3}^{2}f(x)dx+\int_{2}^{9}f(x)dx \qquad \text{..... (1)}$$

But $\int_{-3}^{9}f(x)dx=\frac{-5}{6}$ and $\int_{-3}^{2}f(x)dx=\frac{7}{3}$

From equation (1)

$$\frac{-5}{6}=\frac{7}{5}+\int_2^9 f(x)dx$$

$$\Rightarrow\int_2^9 f(x)dx=\frac{-5-14}{6}=\frac{-19}{6}$$

11. (c) $\int_{-1}^{1}\frac{\sin x-x^2}{3-|x|}dx=\int_{-1}^{1}\frac{\sin x}{3-|x|}dx-\int_{-1}^{1}\frac{x^2}{3-|x|}dx$

Here, $f(x)=\frac{\sin x}{3-|x|}$ is an odd function but

$f(x)=\frac{x^2}{3-|x|}$ is an even function

$$\therefore I=-\int_{-1}^{1}\frac{x^2}{3-|x|}dx=-2\int_0^1\frac{x^2}{3-|x|}dx$$

$$=2\int_0^1\frac{-x^2}{3-|x|}dx$$

12. (c) $\int_0^{\pi/2}\frac{\cos x-\sin x}{1+\sin x\cos x}dx=I$ ..... (1)

Now, $I=\int_0^{\pi/2}\frac{\cos\left(\frac{\pi}{2}-x\right)-\sin\left(\frac{\pi}{2}-x\right)}{1+\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)}$

$$=\int_0^{\pi/2}\frac{\sin x-\cos x}{1+\sin x\cos x} \qquad \text{..... (2)}$$

On adding $2I=0\Rightarrow I=0$

13. (d) $\int_0^{2\pi}\frac{\sin 20}{a-b\cos 0}d\theta=\int_0^{2\pi}\frac{\sin(4\pi-20)}{a-b\cos(2\pi-0)}d\theta$

$$\Rightarrow I=-\int_0^{2\pi}\frac{\sin 20}{a-b\cos 0}d\theta$$

$$\Rightarrow 2I=0\Rightarrow\int_0^{2\pi}\frac{\sin 20}{a-b\cos\theta}=0$$

14. (a) $\int_0^1 (x^2 - 2ax + a^2) f(x)dx = \int_0^1 x^2 f(x)dx - 2a$

$\int_0^1 xf(x) + a^2 \int_0^1 f(x)dx$

$= a^2 - 2a \cdot a + a^2 = 0$

15. (d) Given integral $I = \int_{-2}^{2} (px^2 + s)dx + q\int_{-2}^{2} xdx$

$= 2\int_0^2 (px^2 + s)dx + 0 = \frac{4}{3}(4p + 3s)$

Thus, to find the numerical value of I, it is necessary to know the value of p and s.

16. (d) $I = \int_{-\pi}^{\pi} (\cos px - \sin qx)^2 dx$

$= \int_{-\pi}^{\pi} (\cos^2 px + \sin^2 qx - 2\sin qx \cos px)dx$

$= \int_{-\pi}^{\pi} (\cos^2 px + \sin^2 qx)dx - 2\int_{-\pi}^{\pi} \sin qx \cos pxdx$

$= 2\int_0^{\pi} (\cos^2 px + \sin^2 qx)dx - 0$

$= \int_0^{\pi} (2 + \cos 2px - \cos 2qx)dx = 2\pi$

17. (c) $\int_{-3}^{3} \frac{x^2 \sin x}{1 + x^6}$

by prop. of D.I. $\int_{-a}^{a} f(x)dx = 0$

Since function $\sin x$ is odd, then

when $f(x) = f(-x)$

$\int_{-3}^{3} \frac{x^2 \sin x}{1 + x^6} = 0$

18. (d) $\int_0^1 \frac{dx}{x + \sqrt{1 - x^2}} = \int_0^{\frac{\pi}{2}} \frac{\cos 0 d0}{\sin 0 + \cos 0} = \frac{\pi}{4}$

19. (c) $\int_0^{\infty} \frac{xdx}{(1 + x)(1 + x^2)} =$

put $x = \tan\theta; dx = \sec^2\theta$;

$I = \int_0^{\pi/2} \frac{\tan\theta \sec^2\theta d\theta}{(1 + \tan\theta)(1 + \tan^2\theta)}$

$\int_0^{\pi/2} \frac{\tan\theta}{1 + \tan\theta} d0 = \int_0^{\pi/2} \frac{\sin\theta}{\cos\theta + \sin\theta}$,

Now, $I = \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - 0\right)}{\cos\left(\frac{\pi}{2} - \theta\right) + \sin\left(\frac{\pi}{2} - \theta\right)}$

$= \int_0^{\pi/2} \frac{\cos\theta}{\sin\theta + \cos\theta} d0$

$2I = \int_0^{\pi/2} \frac{(\sin\theta + \cos\theta)d\theta}{\sin\theta + \cos\theta}$

$= \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$

20. (a) Step 1: $\frac{x^4 + 1}{x^2 + 1}$ is a improper fraction (because degree of numerator is > degree of denominator)

Step 2: Inproper fraction can be expressed as sum of polynomial and proper fraction.

i.e., $\frac{f(x)}{g(x)} = \phi(x) + \frac{R(x)}{\phi(x)}$

where degree of $R(x)$ < degree of $\phi(x)$

Hence $\frac{x^4 + 1}{x^2 + 1} = \frac{x^4 - 1 + 2}{x^2 + 1} = x^2 - 1 + \frac{2}{x^2 + 1}$

Step-3: $\int_0^1 \frac{x^4 + 1}{x^2 + 1} dx = \int_0^1 \left(x^2 - 1 + \frac{2}{x^2 + 1}\right) dx$

$= \left(\frac{x^3}{3} - x + 2\tan^{-1} x\right)\Big|_0^1$

$= \left(\frac{1}{3} - 1 + 2\tan^{-1} 1\right) - (0 - 0 + 0)$

$= -\frac{2}{3} + \frac{\pi}{2} = \frac{1}{6}(3\pi - 4)$

21. (c) Given that $\int_{\ln 2}^{x} \frac{1}{e^x - 1} dx = \ln\frac{3}{2}$

Let $I = \int_{\ln 2}^{x} \frac{1}{e^x - 1} dx$

Put $e^x - 1 = t$ and $dx = \frac{dt}{1 + t}$

$\therefore \quad I = \int_{\ln 2}^{x} \frac{1}{t(1 + t)} dt = \int_{\ln 2}^{x} \left[\frac{1}{t} - \frac{1}{1 + t}\right] dt$

$= [\ln t - \ln(1 + t)]_{\ln 2}^{x}$

$= \left[\ln(e^x - 1) - \ln e^x\right]_{\ln 2}^{x} = \left[\ln\frac{e^x - 1}{e^x}\right]_{\ln 2}^{x}$

$$= \ln\left(\frac{e^x-1}{e^x}\right) - \ln\frac{1}{2} = \ln 2\left(\frac{e^x-1}{e^x}\right)$$

But $I = \ln\frac{3}{2}$

$\therefore\ 2\left(\frac{e^x-1}{e^x}\right) = \frac{3}{2} \Rightarrow 4e^x - 4 = 3e^x$

$\Rightarrow\ e^x = 4 \Rightarrow x = \ln 4$

22. (a) $I = \int_8^{15} \frac{dx}{(x-3)\sqrt{x+1}}$

Put $x = \tan^2\theta \Rightarrow \theta = \tan^{-1}\sqrt{x}$

$dx = 2\tan\theta\sec^2\theta d\theta$

$$I = \int_{\tan^{-1}\sqrt{8}}^{\tan^{-1}\sqrt{15}} \frac{2\tan 0\sec^2 0}{(\tan^2\theta - 3)\sqrt{\tan^2\theta+1}} d\theta$$

$$= \int_{\tan^{-1}\sqrt{8}}^{\tan^{-1}\sqrt{15}} \frac{2\tan 0\sec^2 0}{(\sec^2 0 - 4)\sec 0} d\theta$$

$$= \int_{\tan^{-1}\sqrt{8}}^{\tan^{-1}\sqrt{15}} \frac{2\tan\theta\sec\theta}{(\sec^2\theta - 4)} d0$$

Put $t = \sec 0 \Rightarrow dt = \sec 0\tan 0 d0$

$$\therefore\ I = \int \frac{1}{t^2-4} dt = \frac{1}{4}\log\left(\frac{t-2}{t+2}\right)$$

$$\therefore\ I = \left[\frac{1}{2}\log\frac{(\sec 0 - 2)}{(\sec\theta + 2)}\right]_{\tan^{-1}\sqrt{8}}^{\tan^{-1}\sqrt{15}}$$

$$= \frac{1}{2}\left[\log\frac{2}{6} - \log\frac{1}{5}\right] = \frac{1}{2}\log\frac{5}{3}$$

23. (b) $I = \int_0^{1/\sqrt{2}} \frac{\sin^{-1}x}{(1-x^2)^{3/2}} dx$

Putting $\sin^{-1}x = t \Rightarrow \frac{1}{\sqrt{1-x^2}} dx = dt$

and $x = \sin t$

Also $t = 0$ to $\frac{\pi}{4}$ as $x = 0$ to $\frac{1}{\sqrt{2}}$

$$I = \int_0^{\pi/4} t.\sec^2 t dt = \frac{\pi}{4} - \frac{1}{2}\log 2$$

24. (a) $I_1 = \int_x^1 \frac{dt}{1+t^2}$ and $I_2 = \int_1^{1/x} \frac{dt}{1+t^2}$

taking $I_1 = \int_x^1 \frac{dt}{1+t^2} = \tan^{-1}t\Big|_x^1 = \frac{\pi}{4} - \tan^{-1}x$

and $I_2 = \int_1^{1/x} \frac{dt}{1+t^2} = \tan^{-1}t\Big|_1^{1/x}$

$$= \tan^{-1}\frac{1}{x} - \frac{\pi}{4} = \tan^{-1}x$$

Clearly we can observe that for $x > 0$ $I_1 = I_2$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. If $I_1 = \int_e^{e^2} \frac{dx}{\log x}$ and $I_2 = \int_1^2 \frac{e^x}{x} dx$, then

(a) $I_1 = I_2$ (b) $2I_1 = I_2$
(c) $I_1 = 2I_2$ (b) None of these

2. $\int_{-1}^{1} \log(x + \sqrt{x^2+1}) dx =$

[MPPET-2001]

(a) 0 (b) log 2
(c) $\log\frac{1}{2}$ (d) None of these

3. The value of $\int_0^{\pi/2} \frac{e^{x^2}}{e^{x^2} + e^{\left(\frac{\pi}{2}-x\right)^2}} dx$ is

[AMU-1999]

(a) $\pi/4$ (b) $\pi/2$
(c) $e^{\pi^2/16}$ (d) $e^{\pi^2/4}$

4. The value of $\int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x} + 2^{\cos x}} dx$ is

[Karnataka CET-1999; Kerala (Engg.)-2005]

(a) $\pi/4$ (b) $\pi/2$
(c) π (d) 2π

5. The value of $\int_{-1}^{1}(\sqrt{1+x+x^2}-\sqrt{1-x+x^2})dx$ is

(a) 0 (b) 1
(c) −1 (d) None of these

6. $\int_{-1}^{1}\log\left(\frac{1+x}{1-x}\right)dx =$ ***[MPPET-1995; Pb. CET-2000]***

(a) 2 (b) 1
(c) 0 (d) π

7. $\int_{0}^{\pi/2}\frac{\sqrt{\cos x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx =$ ***[MNR-1989; UPSEAT-2002; MPPET-2009]***

(a) 0 (b) $\pi/2$
(c) $\pi/4$ (d) None of these

8. $\int_{0}^{\pi/2}\frac{\sin^{3/2}x dx}{\cos^{3/2}x+\sin^{3/2}x} =$ ***[Roorkee 1989; BIT Ranchi-1989]***

(a) 0 (b) π
(c) $\pi/2$ (d) $\pi/4$

9. $\int_{-1}^{1} x^{17}\cos^4 x\, dx =$ ***[MPPET-1990]***

(a) −2 (b) −1
(c) 0 (d) 2

10. What is the value of integral $\int_{0}^{1}x(1-x)^9 dx$? ***[NDA-2007]***

(a) 1/110 (b) 1/111
(c) 1/112 (d) 1/119

11. If $\int_{-1}^{4} f(x)dx = 4$ and $\int_{2}^{4}(3-f(x))\,dx = 7$, then the value of $\int_{2}^{-1} f(x)\,dx$ is

(a) 2 (b) −3
(c) −5 (d) None of these

12. $\int_{0}^{\infty}\frac{\log(1+x^2)}{1+x^2}dx =$

(a) $\pi\log\frac{1}{2}$ (b) $\pi\log 2$
(c) $2\pi\log\frac{1}{2}$ (d) $2\,\pi\log 2$

13. $\int_{0}^{a}\frac{x dx}{\sqrt{a^2+x^2}} =$

(a) $a(\sqrt{2}-1)$ (b) $a(1-\sqrt{2})$
(c) $a(1+\sqrt{2})$ (d) $2a\sqrt{3}$

14. $\int_{0}^{1}\frac{dx}{\sqrt{1+x}-\sqrt{x}} =$ ***[SCRA-1986]***

(a) $\frac{2\sqrt{2}}{3}$ (b) $\frac{4\sqrt{2}}{3}$
(c) $\frac{8\sqrt{2}}{2}$ (d) None of these

15. The value of $\int_{0}^{\pi/2}\frac{\sin x}{1+\cos^2 x}dx$ is ***[RPET-1995]***

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

16. $\int_{0}^{1}(1-x)^9\,dx =$ ***[SCRA-79]***

(a) 1 (b) 1/10
(c) 11/10 (d) 2

17. $\int_{0}^{2}\sqrt{\frac{2+x}{2-x}}dx =$ ***[MNR-1984; CEE-1993]***

(a) $\pi+2$ (b) $\pi+\frac{3}{2}$
(c) $\pi+1$ (d) None of these

18. The value of $\int_{2}^{3}\frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$ is ***[IIT-1994; Kurukshetra CEE-1998]***

(a) 1 (b) 0
(c) −1 (d) 1/2

19. If $\int_{-a}^{a}\sqrt{\frac{a-x}{a+x}}dx = k\pi$, then $k =$ ***[AISSE-1986; SCRA-1986]***

(a) $-a$ (b) $-2a$
(c) $2a$ (d) a

20. The value of $\int_{1/\pi}^{2/\pi}\frac{\sin(1/x)}{x^2}dx =$ ***[IIT-1990]***

(a) 2 (b) −1
(c) 0 (d) 1

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 17 minutes.
3. The worksheet consists of 17 questions. The maximum marks are 51.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. Let $I = \int_0^1 \frac{e^x dx}{x+1}$, then the value of the integral $\int_0^1 \frac{xe^{x^2}}{x^2+1} dx$ is

(a) I^2 (b) $\frac{1}{2}I$

(c) $2I$ (d) $\frac{1}{2}I^2$

2. The value of $\int_0^2 \frac{3^{\sqrt{x}}}{\sqrt{x}} dx$ is ***[SCRA-1992]***

(a) $\frac{2}{\log 3}(3^{\sqrt{2}}-1)$ (b) 0

(c) $\frac{2.\sqrt{2}}{\log 3}$ (d) $\frac{3^{\sqrt{2}}}{\sqrt{2}}$

3. The solution for x of the equation $\int_{\sqrt{2}}^{x} \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{2}$ is ***[AIEEE-2007]***

(a) $-\sqrt{2}$ (b) $\sqrt{3}/2$

(c) $2\sqrt{2}$ (d) 2

4. Find the value of integral $\int_0^{\pi} \frac{\sin kx}{\sin x} dx$, where, k is an even integer.

(a) 3 (b) 2

(c) 1 (d) 0

5. If $f(x) = x + \phi(x)$, where $\phi(x)$ is an even function, then the value of $\int_{-1}^{1} xf(x)dx$ is equal to

(a) 0

(b) 2/3

(c) $2[\phi(1) - \phi(0)]$

(d) $\frac{2}{3} + 2[\phi(1) + \phi(0)]$

6. $\int_{-\pi/4}^{\pi/4} \frac{e^x \sec^2 x dx}{e^{2\pi}-1}$ is equal to

(a) 0 (b) 2

(c) e (d) None of these

7. $\int_{-1}^{1} \log \frac{1-x}{1+x} dx =$

(a) 2 (b) 1

(c) –1 (d) 0

8. If $\int_{-1}^{1} f(x)\,dx = 0$, then ***[SCRA-1990]***

(a) $f(x) = f(-x)$ (b) $f(-x) = -f(x)$

(c) $f(x) = 2f(x)$ (d) None of these

9. The value

$$\int_{-2}^{2} \left[p \ln\left(\frac{1+x}{1-x}\right) + q \ln\left(\frac{1-x}{1+x}\right)^{-2} + r \right] dx$$

depends on ***[Orissa JEE-2003]***

(a) The value of p

(b) The value of q

(c) The value of r

(d) The value of p and q

10. The value of $\int_{-2}^{2} (ax^3 + bx + c)dx$ depends on the: ***[Karnataka CET-2008]***

(a) value of a (b) values of a and b

(c) value of b (d) value of c

11. The value of $\int_0^{12a} \frac{f(x)}{f(x) + f(12a - x)} dx$ ***[MP PET-2008]***

(a) a (b) $2a$

(c) $3a$ (d) $6a$

12. The value of $\int_0^{\pi/2} \frac{dx}{1+\cot x}$ is ***[MP PET-2008]***

(a) π (b) $\pi/2$

(c) $\pi/3$ (d) $\pi/4$

13. If $\int_0^1 x^m(1-x)^n dx = \int_0^1 x^n(1-x)^p dx$, then p is equal to *[UPSC-2007]*

(a) $2n$ (b) m
(c) $m+n$ (d) m/n

14. $\int_0^{4014} \frac{2^x}{2^x + 2^{4014-x}} dx =$ *[Gujarat CET-2007]*

(a) 2^{2007} (b) 2^{4014}
(c) 4014 (d) 2007

15. $\int_0^{\pi/2} \frac{2\sqrt{\cos\theta}}{3(\sqrt{\sin\theta}+\sqrt{\cos\theta})} d\theta$ is equal to *[Orissa-JEE-2007]*

(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/2$ (d) None of these

16. The value of $\int_1^4 e^{\sqrt{x}} dx$ is *[Orissa-JEE-2007]*

(a) e^2 (b) $2e^2$
(c) $4e^2$ (d) $3e^2$

17. $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$ is *[MPPET-2010]*

(a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{4}$
(c) $\frac{\pi}{2}$ (d) None of these

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	16. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ	17. ⓐ ⓑ ⓒ ⓓ
6. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

2. (a) $\int_0^2 \frac{3\sqrt{x}}{\sqrt{x}} dx$

Put $\sqrt{x} = t \Rightarrow dt = \frac{1}{2\sqrt{x}}$, then

$$= 2\int_0^2 3^t dt = 2.\frac{3^t}{\log 3}\Big|_0^{\sqrt{2}}$$

$$= 2\left[\frac{3\sqrt{2}}{\log 3} - \frac{1}{\log 3}\right] = \frac{2}{\log 3}\left[3^{\sqrt{2}} - 1\right]$$

3. (d) $\int_{\sqrt{2}}^{x} \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{12}$

$$\Rightarrow \left[\sec^{-1} t\right]_{\sqrt{2}}^{x} = \frac{\pi}{12}$$

$$\Rightarrow \sec^{-1} x - \frac{\pi}{4} = \frac{\pi}{12}$$

$$\sec^{-1} x = \frac{\pi}{3} \Rightarrow x = 2$$

4. (d) $I = \int_0^{\pi} \frac{\sin kx}{\sin x} = \int_0^{\pi} \frac{\sin(k\pi - kx)}{\sin(\pi - x)} = -\int_0^{\pi} \frac{\sin kx}{\sin x} dx$

($\because \sin(k\pi - kx) = -\sin kx$, k is even integer)

$I = -I \Rightarrow I = 0$

9. (c) Since $\log\left(\frac{1+x}{1-x}\right)$ is odd function

$$\therefore \int_{-2}^{2} \left\{ p\log\left(\frac{1+x}{1-x}\right) + q\log\left(\frac{1-x}{1+x}\right)^2 + r \right\} dx$$

$$= r\int_{-2}^{2} dx = 4r$$

(Hence depends on the value of r)

11. (a) $I = \int_0^{2a} \frac{f(x)}{f(x) + f(2a-x)} dx$ ..... (1)

$$I = \int_0^{2a} \frac{f(2a-x)}{f(2a-x) + f(x)} dx \quad \text{..... (2)}$$

Adding (1) and (2), we get

$$2I = \int_0^{2a} dx = 2a \Rightarrow I = a$$

14. (d) $I = \int_0^{4014} \frac{2^x}{2^x + 2^{4014-x}} dx$ ..... (1)

Now $I = \int_0^{4014} \frac{2^{4014-x}}{2^{4014-x} + 2^x}$ ..... (2)

On adding equation (1) and (2)

$$2I = \int_0^{4014} [x]_0^{4014} \Rightarrow I = \frac{4014}{2} = 2007$$

17. (a) $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}$ ..... (1) then

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin\left(\frac{\pi}{2} - x\right)}}{\sqrt{\sin\left(\frac{\pi}{2} - x\right)} + \sqrt{\cos\left(\frac{\pi}{2} - x\right)}}$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} \quad \text{..... (2)}$$

Adding equation (1) and (2), we get

$$2I = \int_0^{\pi/2} dx = x\Big|_0^{\pi/2} = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

LECTURE



3 Some more Properties of Definite Integral

BASIC CONCEPTS

1. $\int_a^{2a} f(x)dx = \int_0^a f(2a-x)dx$

2. $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$

   $= 2\int_0^a f(x)dx$ if $f(2a-x) = f(x) = 0$

   if $f(2a-x) = -f(x)$

NOTE

Apply this property for making the upper limit half of the given limit.

3. $\int_a^b xf(x)dx = \frac{a+b}{2}\int_a^b f(x)dx$ if $f(a+b-x) = f(x)$

4. If $f(x)$ is a periodic function with period T i.e., $f(x+T) = f(x)$, then

   (i) $\int_a^{a+T} f(x)dx$ is independent of a i.e.,

   $$\int_0^{a+t} f(x)dx = \int_0^t f(x)dx$$

   (ii) $\int_0^{nT} f(x)dx = n\int_0^T f(x)dx$

   if $f(T+x) = f(x)$

   (iii) If $f(x)$ is periodic with period T, then

   $$\int_{a+2T}^{b+2T} f(x)dx = \int_a^b f(x)dx = \int_{a+nT}^{b+nT} f(x)dx$$

   (iv) $\int_{aT}^{bT} f(x)dx = (b-a)\int_0^T f(x)dx$

5. Integration of Modulus Functions and the Greatest integer function: $\int_a^b |f(x)|dx$ and $\int_a^b [x]dx$

   While integrating the modulus function and the greatest integer function divide the range of integration into suitable parts and then discuss the sign of these functions in these parts.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)) FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate $\int_{-3}^{3} |x+1|dx$ *[CBSE-1992]*

Solution

$$I = \int_{-3}^{3} |x+1|dx$$

Let $f(x) = |x+1| = \begin{cases} x+1 & \text{if } x+1 \geq 0 \\ -(x+1) & \text{if } x+1 < 0 \end{cases}$

$= \begin{cases} x+1 & \text{if } x \geq -1 \\ -(x+1) & \text{if } x < -1 \end{cases}$

$I = \int_{-3}^{3} f(x)dx = \int_{-3}^{-1} f(x)dx + \int_{-1}^{3} f(x)dx$

$= -\int_{-3}^{-1} (x+1)dx + \int_{-1}^{3} (x+1)dx$

$= -\left[\frac{(x+1)^2}{2}\right]_{-3}^{-1} + \left[\frac{(x+1)^2}{2}\right]_{-1}^{3}$

$= -\left[\frac{(-1+1)^2}{2} - \frac{(-3+1)^2}{2}\right] +$

$\left[\frac{(3+1)^2}{2} - \frac{(-1+1)^2}{2}\right]$

$= 2 + 8 = 10$

2. Evaluate $\int_{-\pi/4}^{\pi/4} |\sin x| dx$ ***[CBSE-91]***

Solution

$I = \int_{-\pi/4}^{\pi/4} |\sin x| dx$

Let $f(x) = |\sin x| = \begin{cases} \sin x & \text{if } x \geq 0 \\ -\sin x & \text{if } x < 0 \end{cases}$

$I = \int_{-\pi/4}^{\pi/4} f(x)dx = \int_{-\pi/4}^{0} -\sin x dx + \int_{0}^{\pi/4} \sin x dx$

$= -[-\cos x]_{-\pi/4}^{0} + [-\cos x]_{0}^{\pi/4}$

$= \left[\cos 0 - \cos(-\pi/4)\right] - \left[\cos\frac{\pi}{4} - \cos 0\right]$

$= \left[1 - \frac{1}{\sqrt{2}}\right] - \left[\frac{1}{\sqrt{2}} - 1\right] = 1 - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1$

$= 2 - \frac{1+1}{\sqrt{2}} = 2 - \frac{2}{\sqrt{2}} = (2 - \sqrt{2})$

3. Evaluate $\int_{0}^{8} |x-5| dx$ ***[CBSE-1998]***

Solution

Let $I = \int_{0}^{8} |x-5| dx$

Let $f(x) = |x-5| = \begin{cases} x-5 & \text{if } x-5 \geq 0 \\ -(x-5) & \text{if } x-5 < 0 \end{cases}$

$= \begin{cases} x-5 & \text{if } x \geq 5 \\ 5-x & \text{if } x < 5 \end{cases}$

$I = \int_{0}^{8} f(x)dx = \int_{0}^{5} f(x)dx + \int_{5}^{8} f(x)dx$

$= \int_{0}^{5} (5-x)dx + \int_{5}^{8} (x-5)dx$

$= \left[\frac{(5-x)^2}{-2}\right]_{0}^{5} + \left[\frac{(x-5)^2}{2}\right]_{5}^{8}$

$= \left[\frac{(5-5)^2}{-2} - \frac{(5-0)^2}{-2}\right] + \left[\frac{(8-5)^2}{2} - \frac{(5-5)^2}{2}\right]$

$= \frac{25}{2} + \frac{9}{2} = \frac{34}{2} = 17$

4. Evaluate $\int_{0}^{\pi} |\cos x| dx$ ***[CBSE-92, 94]***

Solution

Clearly $[\cos x] = \begin{cases} \cos x & \text{when } 0 \leq x \leq \frac{\pi}{2} \\ -\cos x & \text{when } \frac{\pi}{2} \leq x \leq \pi \end{cases}$

$\int_{0}^{\pi} [\cos x]dx = \int_{0}^{\pi/2} [\cos x]dx + \int_{\pi/2}^{\pi} [\cos x]dx$

$= \int_{0}^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} -\cos x dx$

$= [\sin x]_{0}^{\pi/2} - [\sin x]_{\pi/2}^{\pi}$

$= (1 + 1) = 2$

5. Evaluate the definite integrals $\int_{4}^{12} x.\sqrt[3]{x-4} dx$

Solution

We let $t = (x-4)^{1/3}$

$\Rightarrow t^3 = x - 4$

$\Rightarrow x = t^3 + 4$ Thus $\frac{dx}{dt} = 3t^2$

$\Rightarrow dx = 3t^2\, dt$

when $x = 4$, $t = 0$ and

when $x = 12$, $t = (12-4)^{1/3}$

$= 8^{1/3} = 2$ therefore, $\int_{4}^{12} x.\sqrt[3]{x-4} dx$

$= \int_{0}^{2} (t^3+4)t.3t^2 dt = 3\int_{0}^{2} (t^6 + 4t^3)dt$

$= 3\left(\frac{t^7}{7} + \frac{4}{4}t^4\right)_{0}^{2}$

$= 3\left[\frac{2^7}{7} + 2^4 - 0 - 0\right] = 3\left[\frac{128}{7} + 16\right] = \frac{3(240)}{7}$

$= 720/7$

6. For $x > 0$, let $f(x) = \int_1^x \frac{\ln t}{1+t}dt$. Find the function $f(x) + f\left(\frac{1}{x}\right)$ and show that $f(e) + f\left(\frac{1}{e}\right)$. Here $\ln t = \log_e t$. ***[IIT-2000]***

Solution

$$f(x) = \int_1^x \frac{\log t}{1+t}dt \quad \ldots\ldots\ldots\ldots (1)$$

$$f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\log t}{1+t}dt$$

Put $t = \frac{1}{z}$ $\therefore$ $dt = -\frac{1}{z^2}\,dz$ and adjust the limits.

$$\therefore \quad f\left(\frac{1}{x}\right) = \int_1^x -\frac{\log z}{(z+1)/z}\left(-\frac{1}{z^2}dz\right)$$

$$= \int_1^x \frac{\log z}{z(z+1)}dz = \int_1^x \frac{\log t}{t(1+t)}dt$$

$$f(x) + f\left(\frac{1}{x}\right) = \int_1^x \frac{\log t}{1+t}\left(1+\frac{1}{t}\right)dt$$

$$= \int_1^x \frac{\log t}{t} = \left[\frac{1}{2}(\log t)^2\right]_1^x = \frac{1}{2}(\log x)^2$$

Now put $x = e$ in both sides

$$\therefore \quad f(e) + f\left(\frac{1}{e}\right) = \frac{1}{2}[(\log e)^2] = \frac{1}{2}$$

7. Consider the integrals $I_1 = \int_0^1 e^{-y}\cos^2 y\,dy$ $I_2 = \int_0^1 e^{-y^2}\cos^2 y\,dy$, $I_3 = \int_0^1 e^{-y^2}dy$, $I_4 = \int_0^1 e^{-\frac{1}{2}y^2}dy$, where, $0 < y < 1$. If I be the greatest amongst I_1, I_2, I_3, I_4 then prove that $I = I_4$.

Solution

Since $0 < y < 1$ i.e., y is + ive but less than 1

Then $y > y^2 > \frac{1}{2}y^2$

$\Rightarrow -y < -y^2 < -\frac{1}{2}y^2$. Also $e > 1$

$$\therefore \quad e^{-y} < e^{-y^2} < e^{-\frac{1}{2}y^2} \quad \ldots\ldots\ldots (1)$$

Also $\cos^2 y \le 1$ ..... (2)

$$\therefore \quad e^{-y}\cos^2 y \le e^{-y^2}\cos^2 y < e^{-y^2} < e^{-\frac{1}{2}y^2}$$

by (1) and (2)

$\therefore$ $I_1 < I_2 < I_3 < I_4$ Hence I_4 is greatest.

8. Determine a positive integer $n \le 5$ such that $\int_0^1 e^x(x-1)^n dx = 16 - 6e$ ***[IIT-1992]***

Solution

$$I_1 = \int_0^1 e^x(x-1)dx = [e^x(x-1) - e^x.1]_0^1$$

$$= 1 - (e-1) = 2 - e \quad \ldots\ldots\ldots\ldots (1)$$

$$I_n = [e^x(x-1)^n]_0^1 - \int_0^1 e^x.n(x-1)^{n-1}dx$$

$$= 0 - (-1)^n - nI_{n-1} = (-1)^{n+1} - nI_{n-1}$$

Putting $n = 2, 3, 4, \ldots\ldots$ in the above, we get

$I_2 = (-1)^3 - 2I_1 = -1 - 2(2-e) = 2e - 5$

$I_3 = (-1)^4 - 3I_2 = 1 - 3(2e-5)$

$= 16 - 6e$ $\therefore$ $n = 3$

$= -2[\cot x]_{\pi/4}^{3\pi/4} = 4$ $\therefore$ $I = 2$

9. Prove that $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx = \pi^2$ ***[IIT Re-ex-1997]***

Solution

$$I = 0 + 2\int_0^{\pi} \frac{2x\sin x}{1+\cos^2 x}dx \text{ [By property]}$$

$$= 4\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\int_0^{\pi} dx = 4.\frac{\pi^2}{4} = \pi^2$$

10. Evaluate $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)}dx$ ***[IIT-2004]***

Solution

$I = I_1 + I_2 = I_1$ as $I_2 = 0$ by Property

and $I_1 = 2\int_0^{\pi/3} \frac{\pi dx}{2 - \cos\left(|x| + \frac{\pi}{3}\right)}$

$$= 2\pi\int_0^{\pi/3} \frac{dx}{2 - \cos\left(x + \frac{\pi}{3}\right)}$$

because $|x| = x$ as x is +ive in the range 0 to $\pi/3$

Now put $x + \dfrac{\pi}{3} = t$ and adjust the limits.

$$\therefore \quad I = I_1 = 2\pi \int_{\pi/3}^{2\pi/3} \frac{dt}{2 - \cos t}$$

$$= 2\pi \int_{\pi/3}^{2\pi/3} \frac{\sec^2(t/2)dt}{2\left(1 + \tan^2 \frac{t}{2}\right) - \left(1 - \tan^2 \frac{t}{2}\right)}$$

Now put $\tan(t/2) = z$

$$\therefore \quad \frac{1}{2}\sec^2 \frac{t}{2} dt = dz \text{ and adjust the limits.}$$

$$\therefore \quad I = 4\pi \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{dz}{1 + 3z^2} = \frac{4\pi}{3} \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{dz}{z^2 + \left(\frac{1}{\sqrt{3}}\right)^2}$$

$$= \frac{4\pi}{3.\sqrt{3}} \left[\tan^{-1} z\sqrt{3}\right]_{1/\sqrt{3}}^{\sqrt{3}}$$

$$= \frac{4\pi}{\sqrt{3}} \left[\tan^{-1} 3 - \tan^{-1} 1\right]$$

$$= \frac{4\pi}{\sqrt{3}} \tan^{-1} \frac{3-1}{1+3.1} = \frac{4\pi}{\sqrt{3}} \tan^{-1} \frac{1}{2}$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Evaluate $\int_0^{\pi} x \sin^2 x dx$

2. Evaluate $\int_0^{\infty} \frac{x \tan^{-1} x}{(1+x^2)^2} dx$

[MP-99, 2000]

3. Evaluate $\int_{-2}^{2} |2x + 3| dx$

[CBSE-92]

4. Evaluate $\int_{-6}^{6} |x + 2| dx$

[CBSE-91]

5. Evaluate $\int_0^{\pi/4} \sqrt{1 - \sin 2x} dx$

[AISSE-82; CBSE-2004]

6. Evaluate $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$

7. Evaluate $\int_{-5}^{0} f(x)dx$ where $f(x) = |x| + |x + 2| + |x + 5|$

[CBSE-2005 (outside Delhi)-I]

8. Prove that $\int_0^{\pi/2} x \cot x dx = \frac{\pi}{2} \log 2$

9. Evaluate $\int_0^{\pi/4} \sqrt{1 + \sin 2x} dx$

[PSB-90 (C)]

10. Evaluate $\int_0^{\pi} \frac{\sin x}{\sin x + \cos x} dx$

[AISSE-86 (C)]

11. Evaluate $\int_{-\pi/2}^{\pi/2} |\sin x| dx$

[CBSE-92 (C); PSB-91 (C)]

12. Evaluate $\int_{-1}^{1} |2x + 1| dx$ ***[CBSE-92]***

13. Evaluate $\int_0^{1} |2x - 1| dx$

[PSB-96]

14. Prove that $\int_0^{\pi/2} (2 \log_e \sin x - \log_e \sin 2x) dx = \frac{-\pi}{2} \log_e 2$ ***[MP-96, CBSE-91 (C)]***

ANSWERS

1. $\pi^2/4$

2. $\frac{\pi}{8}$

3. 25/2

4. 40

5. $\sqrt{2}-1$

6. $\frac{\pi^2}{2}-\pi$

7. 31.5

9. 1

10. $\frac{\pi}{2}$

11. 2

12. 5/2

13. 1/2

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. Let $g(x) = \int_0^x f(t)dt$, where f is such that $\frac{1}{2} \le f(t) \le 1$ for $t \in [0, 1]$ and $0 \le f(t) \le \frac{1}{2}$ for $t \in (1, 2]$. Then $g(2)$ satisfies the inequality. ***[IIT Sc.-2000]***

(a) $-\frac{3}{2} \le g(2) < \frac{1}{2}$ (b) $0 \le g(2) < 2$

(c) $\frac{3}{2} < g(2) \le \frac{5}{2}$ (d) $2 < g(2) < 4$

Solution

$$g(2) = \int_0^2 f(t)dt = \int_0^1 f(t)dt + \int_1^2 f(t)dt = I_1 + I_2$$

Now we know from property that

If $f(x)$ is defined in $[a, b]$ then $\int_a^b f(x)dx$ lies between $L(b-a)$ and $G(b-a)$, where L and G are the least and greatest values of the function $f(x)$ in the interval $[a, b]$.

Now $\frac{1}{2} \le f(t) \le 1, t \in [0, 1]$

$L = \frac{1}{2}$ and $G = 1, b - a = 1 - 0 = 1$

$$\frac{1}{2} \cdot 1 \le I_1 \le 1 \qquad \ldots\ldots\ldots (1)$$

Again $0 \le f(t) \le \frac{1}{2}, t \in [1, 2]$

$L = 0, G = \frac{1}{2}$ and $b - a = 2 - 1 = 1$

$$\therefore \quad 0.1 \le I_2 \le \frac{1}{2} \cdot 1 \text{ or } 0 \le I_2 \le \frac{1}{2} \qquad \ldots (2)$$

Adding the inequalities in (1) and (2), we get

$$\frac{1}{2} + 0 \le I_1 + I_2 \le 1 + \frac{1}{2} \text{ or } \frac{1}{2} \le g(2) \le \frac{3}{2}$$

2. If $f(x) = \begin{cases} e^{\cos x}\sin x & \text{for } |x| \le 2 \\ 2 & \text{otherwise} \end{cases}$, then

$\int_{-2}^{3} f(x)dx =$ ***[IIT Sc.-2000]***

(a) 0 (b) 1

(c) 2 (d) 3

Solution

(c) $|x| \le 2 \Rightarrow -2 \le x \le 2$ and $f(x) = e^{\cos x}\sin x$ is an odd function.

$$I = \int_{-2}^{3} f(x)dx = \int_{-2}^{2} f(x)dx + \int_{2}^{3} f(x)dx$$

Since $f(x)$ is an odd function in $(-2, 2)$ so that by Prop. Ist integral is zero and $f(x) = 2$ in $(2, 3)$

$$\therefore \quad I = 0 + \int_2^3 2dx = [2x]_2^3 = 2$$

3. Let $T > 0$ be a fixed real number. Suppose f is a continuous function such that for all $x \in R$, $f(x + T) = f(x)$.

If $I = \int_0^T f(x)dx$, then the value of $\int_3^{3+3T} f(2x)dx$ is: ***[IIT Sc.-2002]***

(a) $\frac{3}{2}I$ (b) $2I$

(c) $3I$ (d) $6I$

Solution

Put $2x = z$

$\therefore \int_3^{3+3T} f(2x)dx = \int_6^{6+6T} f(z)\frac{1}{2}dz$

$= \frac{1}{2}\int_0^{6T} f(z)dz = 3I$ (using property $\int_a^{a+T} f(x)dx = \int_0^T f(x)dx$)

$f(z)$ is periodic function of period T.

4. Let for all real numbers x, $f(x) = x - [x]$, where $[x]$ is equal to the integral part of x. Then value of $\int_{-1}^{1} f(x)dx$ is equal to: ***[IIT-98]***

(a) 1 (b) 2

(c) 0 (d) 1/2

Solution

(a) Obviously in the interval $[-1, 1]$

$$f(x) = \begin{cases} x+1, & -1 \le x < 0 \\ x, & 0 \le x < 1 \end{cases}$$

$\therefore \int_{-1}^{1} f(x)dx = \int_{-1}^{0} f(x)dx + \int_0^1 f(x)dx$

$= \int_{-1}^{0}(x+1)dx + \int_0^1 xdx = \left[\frac{x^2}{2}+x\right]_{-1}^{0} + \left[\frac{x^2}{2}\right]_0^1$

$= 0 - \left(\frac{1}{2}-1\right) + \left(\frac{1}{2}-0\right) = 1$

5. The value of $\int_1^a [x]f'(x)dx$, $a > 1$, where $[x]$ denotes greatest integer not exceeding x, is: ***[AIEEE-2006]***

(a) $[a]f([a]) - \{f(1) + f(2) + \ldots\ldots + f(a)\}$

(b) $af([a] - \{f(1) + f(2) + \ldots\ldots + f(a)\}$

(c) $af(a) - \{f(1) + f(2) + \ldots\ldots + f([a])\}$

(d) $[a]f(a) - \{f(1) + f(2) + \ldots\ldots + f([a])\}$

Solution

(d) $I = \int_1^2 [x]f'(x)dx + \int_2^3 [x]f'(x)dx$

$+\ldots. + \int_{[a]}^{a} [x]f'(x)dx$

$= \int_1^2 1f'(x)dx + \int_2^3 2f'(x)dx$

$+\ldots. + \int_{[a]}^{a} [a]f'(x)dx$

$= \{f(2) - f(1)\} + 2\{f(3) - f(2)\} + \ldots\ldots + [a]\{f(a) - f[a]\}$

$= -\{f(1) + f(2) + \ldots\ldots + f([a])\} + [a]f(a)$

6. If $[\ ]$ denotes the greatest integer function, then $\int_0^{3/2} [x^2]dx$ is equal to: ***[Aligarh, 98; Haryana (CEET) 95; DCE 97, 2001, 2003; IIT 88; ICS (Pre) 2004]***

(a) $2 - \sqrt{2}$ (b) $2 + \sqrt{2}$

(c) $1 - \sqrt{2}$ (d) $1 + \sqrt{2}$

Solution

(a) $I = \int_0^1 [x^2]dx + \int_1^{\sqrt{2}} [x^2]dx + \int_{\sqrt{2}}^{3/2} [x^2]dx$

$= \int_0^1 0dx + \int_1^{\sqrt{2}} 1dx + \int_{\sqrt{2}}^{3/2} 2dx$

$= 0 + (\sqrt{2} - 1) + 2(3/2) - \sqrt{2}) = 2 - \sqrt{2}$

7. If $I_1 = \int_0^1 2^{x^2}dx$, $I_2 = \int_0^1 2^{x^3}dx$, $I_3 = \int_1^2 2^{x^3}dx$ and $I_4 = \int_1^2 2^{x^4}dx$, then: ***[AIEEE-2005]***

(a) $I_3 > I_4$ (b) $I_3 = I_4$

(c) $I_1 > I_2$ (d) $I_2 > I_1$

Solution

(c) $0 < x < 1 \Rightarrow x^2 > x^3 = 2^{x^2} > 2^{x^3}$

$\Rightarrow \int_0^1 2^{x^2}dx > \int_0^1 2^{x^3}dx \Rightarrow I_1 > I_2$

Also $1 < x < 2$

$\Rightarrow x^2 < x^3 \Rightarrow 2^{x^2} < 2^{x^3}$

$\Rightarrow \int_1^2 2x^2 dx < \int_1^2 2^{x^3} dx$

$\Rightarrow I_3 < I_4$

8. If $[x]$ denotes the greatest integer less than or equal to x, then the value of the integral $\int_0^2 x^2[x]dx$ equals:

[Kurukshetra CEE-1996, Pb. CET-2001]

(a) 5/3 (b) 7/3
(c) 8/3 (d) 4/3

Solution

(b) $\int_0^2 x^2[x]dx = \int_0^1 x^2[0]dx$

$+\int_1^2 x^2(1)dx = 0 + \left[\frac{x^3}{3}\right]_1^2 = \frac{7}{3}$

9. If $\int_0^1 e^{x^2}(x-\alpha)dx = 0$, then:

[MNR 94; UPSEAT-2000; Bihar (CEE)-2003]

(a) $1 < \alpha < 2$ (b) $\alpha < 0$
(c) $0 < \alpha < 1$ (d) $\alpha = 0$

Solution

(c) $I = 0 \Rightarrow \int_0^1 e^{x^2} x dx = \int_0^1 e^{x^2} \alpha\, dx$

But $\int_0^1 e^{x^2} x dx = \frac{1}{2}(e-1)$ and $\int_0^1 \alpha e^{x^2} dx$ has max. value ae and min. value α, so its value = $\frac{1}{2}(e-1)$ only when $0 < \alpha < 1$.

10. $\lim_{x \to \frac{\pi}{2}} \frac{\int_{\pi/2}^{x} (2^{-\cos t} - 1)dt}{\int_{\pi^2/4}^{x^2} \left(\sqrt{t} - \frac{\pi}{2}\right)dt}$ is equal to:

(a) $\frac{\log_e 2}{2\pi}$ (b) $\frac{\log_e 2}{\pi}$
(c) $\frac{4\log_e 2}{\pi}$ (d) None of these

Solution

(b) $\lim_{x \to \frac{\pi}{2}} \frac{\int_{\pi/2}^{x} (2^{-\cos t} - 1)dt}{\int_{\pi^2/4}^{x^2} \left(\sqrt{t} - \frac{\pi}{2}\right)dt} = \lim_{x \to \frac{\pi}{2}} \left(\frac{2^{-\cos x} - 1}{\left(x - \frac{\pi}{2}\right).2x} \right)$

Applying L' Hospital's rule

$= \lim_{x \to \frac{\pi}{2}} \frac{2^{-\cos x}(\log_e 2)\sin x}{2x + 2\left(x - \frac{\pi}{2}\right)} = \frac{\log_e 2}{\pi}$

11. $\int_{-1/7}^{20/7} \sin(x - [x])dx$ (where [.] denotes greatest integer) is:

(a) 0 (b) 3 (cos 1 – 1)
(c) 3 (1 – cos 1) (d) None of these

Solution

(c) $I = \int_{-1/7}^{20/7} \sin(x - [x])dx = \int_{-1/7}^{1-\frac{1}{7}} \sin(\{x\})dx$

$= 3\int_0^{-1} \sin x dx = 3[-\cos x]_0^1 = 3(1 - \cos 1)$

12. The value of $\int_{\pi/4}^{3\pi/4} \frac{\phi}{1 + \sin\phi} d\phi$ is $= \pi(\sqrt{2} - 1)$

[IIT-1993]

(a) $\pi \tan\frac{\pi}{8}$ (b) $\log \tan\frac{\pi}{8}$
(c) $\tan\frac{\pi}{8}$ (d) None of these

Solution

(a) $I = \int_{\pi/4}^{3\pi/4} \frac{\phi}{1 + \sin\phi} d\phi$

$= \int_{\pi/4}^{3\pi/4} \frac{\pi - \phi}{1 + \sin(\pi - \phi)} d\phi$

$\left\{\because \frac{\pi}{4} + \frac{3\pi}{4} = \pi\right\}$

$\Rightarrow 2I = \int_{\pi/4}^{3\pi/4} \frac{\pi}{1 + \sin\phi} d\phi$

On solving $I = \pi(\sqrt{2-1}) = \pi \tan\frac{\pi}{8}$

13. $\int_{-2}^{2} \frac{dx}{1+|x-1|}$ is equal to

[DCE-96]

(a) log 2 (b) 2 log 2
(c) 3 log 2 (d) 4 log 2

Solution

(c) $I = \int_{-2}^{1} \frac{dx}{1-(x-1)} + \int_{1}^{2} \frac{dx}{1+(x-1)}$

$= \int_{-2}^{1} \frac{dx}{2-x} + \int_{1}^{2} \frac{dx}{x}$

$= -[\log(2-x)]_{-2}^{1} + [\log x]_{1}^{2}$

$= \log 4 + \log 2 = 3 \log 2$

14. The value of integral $\int_{e^{-1}}^{e^2} \left|\frac{\log_e x}{x}\right| dx$ is

[IIT Sc.-2000]

(a) $\frac{3}{2}$ (b) $\frac{5}{2}$
(c) 3 (d) 5

Solution

(b) Put $\log_e x = z \quad \therefore x = e^z$

$\therefore$ $dx = e^z\, dz$ and limits are adjusted as -1 to 2.

$\therefore \quad I = \int_{-1}^{2} \left|\frac{z}{e^z}\right| e^z dz = \int_{-1}^{2} |z|\, dz$, e^z is always +ive

$= \int_{-1}^{0} -z dz + \int_{0}^{2} z dz = \left[-\frac{z^2}{2}\right]_{-1}^{0} + \left[\frac{z^2}{2}\right]_{0}^{2}$

$= \frac{1}{2} + 2 = \frac{5}{2}$

15. $\left|\int_0^1 (1-y^2)dy\right| + \left|\int_0^1 (y^2-1)dy\right|$ equals:

[IIT-JEE-2006]

(a) 2/3 (b) 1/3
(c) 4/3 (d) 1

Solution

(c) Exp. $= \left|\int_0^1 (1-y^2)dy\right| + \left|\int_0^1 (1-y^2)dy\right|$

$= 2\int_0^1 (1-y^2)dy \quad [\because 1-y^2 > 0 \ \forall\, y]$

$= 2\left[y - \frac{y^3}{3}\right]_0^1 = \frac{4}{3}$

16. $\int_0^{n\pi+v} |\sin x|\, dx$ is equal to ***[IIT-94; AMU-2003]***

(a) $2n + 1$ (b) $2n + 1 + \cos v$
(c) $2n + 1 - \cos v$ (d) $2n + \cos v$

Solution

(c) $I = \int_0^{n\pi} |\sin x|\, dx + \int_{n\pi}^{n\pi+v} |\sin x|\, dx$

$= n\int_0^{\pi} |\sin x|\, dx + \int_0^{v} \sin(n\pi+\theta)|\, d\theta$

where $x = n\pi + \theta$

$= n\int_0^{\pi} \sin x\, dx + \int_0^{v} \sin\theta\, d\theta$

$= 2n + 1 - \cos v$

17. If $I = \int_0^{3\pi} f(\cos^2 x)dx$ & $J = \int_0^{\pi} f(\cos^2 x)dx$.

Then: ***[Orissa JEE-2008]***

(a) $I = 5J$ (b) $I = J$
(c) $I = 3J$ (d) none of these

Solution

(c) Now period of $f(\cos^2 x)$ is π,

$\therefore \quad I = \int_0^{3\pi} f(\cos^2 x)dx$

$= 3\int_0^{\pi} f(\cos^2 x)dx = 3J$

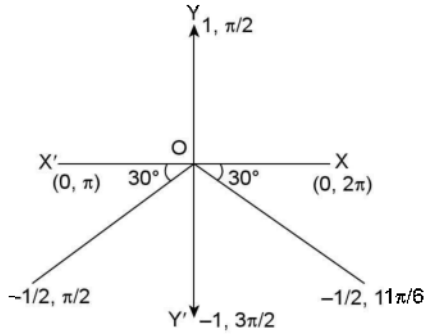
18. The value of $\int_{\pi}^{2\pi} [2\sin x]dx$, where, [ ] represents the greatest integer function, is

[IIT 1995, AMU 2003]

(a) $-\frac{5\pi}{3}$ (b) $\frac{5\pi}{3}$
(c) $\frac{3}{5\pi}$ (d) $\frac{2\pi}{3}$

Solution

(a) It is a question of greatest integer function. We subdivide the interval π to 2π as under keeping in view that we have to evaluate $[2 \sin x]$



We know that $\sin\frac{\pi}{6}=\frac{1}{2}$,

$$\sin\left(\pi+\frac{\pi}{6}\right)=\sin\frac{7\pi}{6}=-\frac{1}{2}$$

$$\sin\frac{11\pi}{6}=\sin\left(2\pi-\frac{\pi}{6}\right)=-\sin\frac{\pi}{6}=-\frac{1}{2}$$

$$\sin\frac{9\pi}{6}=\sin\frac{3\pi}{2}=-1$$

Hence we divide the interval π to 2π as

$$\left(\pi,\frac{7\pi}{6}\right),\left(\frac{7\pi}{6},\frac{11\pi}{6}\right),\left(\frac{11\pi}{6},2\pi\right)$$

$$\sin x=\left(0,-\frac{1}{2}\right),\left(-1,-\frac{1}{2}\right),\left(0,-\frac{1}{2}\right)$$

$2\sin x = (0,-1),(-2,-1),(0,-1)$

$[2\sin x] = -1\ -2\ -1$

$\therefore\ I=I_1+I_2+I_3=\int -1dx+\int -2dx+\int -1dx$

between proper limits

$$=-\frac{\pi}{6}-2\left(\frac{4\pi}{6}\right)-\frac{\pi}{6}=-\frac{10\pi}{6}=-\frac{5\pi}{3}$$

19. $\int_0^1 \frac{1}{(5+2x-2x^2)\{1+e^{(2-4x)}\}}dx=$

[Roorkee-1998]

(a) $\frac{1}{\sqrt{11}}\log\frac{\sqrt{11}+1}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{11}}\log\frac{10}{\sqrt{11}+1}$

(c) $\frac{10}{\sqrt{11}}\log\frac{1}{\sqrt{11}+1}$ (d) $\frac{\sqrt{11}}{1}\log\frac{10}{\sqrt{11}+1}$

Solution

(a) Apply property

$$\therefore\ I=\int_0^1 \frac{dx}{[5+2(1-x)-2(1-x)^2][1+e^{2-4(1-x)}]}$$

$$=\int_0^0 \frac{dx}{[5+2x-2x^2](1+e^{-2(2-4)})}$$

$$=\int_0^1 \frac{e^{(2-4x)}dx}{(5+2x-2x^2)(1+e^{(2-4x)})}\ \text{as}$$

$e^{y}=\frac{1}{e^{y}}$, adding

$$2I=\int_0^1 \frac{(1+e^{(2-4x)})dx}{(5+2x-2x^2)(1+e^{(2-4x)})}$$

$$\text{or } I=\frac{1}{2}\int_0^1 \frac{dx}{2\left[\left(\frac{\sqrt{11}}{2}\right)^2-\left(x-\frac{1}{2}\right)^2\right]}$$

$$I=\frac{1}{4}.\frac{1}{2.\frac{\sqrt{11}}{2}}\left[\log\frac{\frac{\sqrt{11}}{2}+x-\frac{1}{2}}{\frac{\sqrt{11}}{2}-\left(x-\frac{1}{2}\right)}\right]_0^1 \text{ Formula}$$

$$=\frac{1}{4\sqrt{11}}\left[\log\frac{\sqrt{11}+1}{\sqrt{11}-1}-\log\frac{\sqrt{11}-1}{\sqrt{11}+1}\right]$$

$$=\frac{1}{4\sqrt{11}}\log\left(\frac{\sqrt{11}+1}{\sqrt{11}-1}\right)^2$$

$\because\ \log m\ \ \log n=\log(m/n)$

$$=2.\frac{1}{4\sqrt{11}}\log\frac{(\sqrt{11}+1)^2}{(\sqrt{10})^2}\ \text{(conj. of } D^r)$$

$$=\frac{1}{\sqrt{11}}\log\frac{\sqrt{11}+1}{\sqrt{10}}$$

20. The value of $5050\frac{\int_0^1(1-x^{50})^{100}dx}{\int_0^1(1-x^{50})^{101}dx}$ is

[IIT-JEE-2006]

(a) 5050 (b) 5051

(c) 5052 (d) 1

Solution

Let given integral be $5050\ \frac{I_{100}}{I_{101}}$

Now $I_{101}=\int_0^1(1-x^{50})(1-x^{50})^{100}dx$

$$= I_{100} - \int_0^1 x^{50}(1-x^{50})^{100}\,dx$$

$$= I_{100} - \int x.x^{49}(1-x^{50})^{100}\,dx$$

$$= I_{100} - \left[\frac{-x(1-x^{50})^{101}}{101}\right]_0^1 - \int_0^1 \frac{(1-x^{50})^{101}}{5050}\,dx$$

$$= I_{100} - 0 - \frac{1}{5050} I_{101} \Rightarrow \frac{5051}{5050} I_{101} = I_{100}$$

$$\therefore \quad 5050\frac{I_{100}}{I_{101}} = 5051$$

21. $\int_{-1}^{1}\left[x+\left[x+[x]\right]\right]dx$, where, $[\cdot]$ denote greatest integer function, is equal to

(a) –2 (b) –1

(c) –3 (d) –4

Solution

(c)

$$I = \int_{-1}^{0}\left[x+\left[x+[x]\right]\right]dx + \int_{0}^{1}\left[x+\left[x+[x]\right]\right]dx$$

$I = I_1 + I_2$ For $I_1 \rightarrow [x] = -1$ and $-1 < x < 0$

$\Rightarrow -2 < x + [x] < -1 \Rightarrow [x + [x]] = -2$

and now, $-3 < x + [x + [x]] < -2$

$\Rightarrow [x + [x + [x]]] = -3;$

$$I_1 = \int_{-1}^{0}(-3)dx = -3[x]_{-1}^{0} = -3$$

For $I_2 \rightarrow [x] = 0$ and $0 < x < 1$

$\Rightarrow 0 < x + [x] < 1 \Rightarrow [x + [x]] = 0$

Again $0 \le x + [x + [x]] < 1$

$\Rightarrow [x + [x + [x]]] = 0$ So, $I_2 = 0$

$\therefore \quad I_1 + I_2 = -3$

22. $\int_0^{\pi} e^{\cos x}\left\{2\sin\left(\frac{1}{2}\cos x\right)+3\cos\left(\frac{1}{2}\cos x\right)\right\}\sin x dx$ is equal to ***[IIT (Main)-2005]***

(a) $\frac{24}{5}e\left(\cos\frac{1}{2}+\frac{1}{2}\sin\frac{1}{2}-1\right)$

(b) $\frac{24}{5}e\left(\cos\frac{1}{2}+\frac{1}{2}\sin\frac{1}{2}\right)$

(c) $\frac{6}{5}e\left(\cos\frac{1}{2}-\frac{1}{2}\sin\frac{1}{2}+1\right)$

(d) None of these

Solution

(a) $I = \int_0^{\pi} e^{|\cos x|}\left\{2\sin\left(\frac{1}{2}\cos x\right)+3\cos\left(\frac{1}{2}\cos x\right)\right\}$

$\sin x dx$

$$= 0 + 2.3\int_0^{\pi/2} e^{\cos x}\cos\left(\frac{1}{2}\cos x\right)\sin x\,dx$$

[By property]

$$= 6\int_0^{\pi/2} e^{\cos x}\cos\left(\frac{1}{2}\cos x\right)\sin x\,dx$$

$$= 6\int_0^1 e^{t}\cos\left(\frac{t}{2}\right)dt, \text{ where } t = \cos x$$

$$= 6\left[\frac{4}{5}e^{t}\left(\cos\frac{t}{2}+\frac{1}{2}\sin\frac{t}{2}\right)\right]$$

$$= \frac{24}{5}e\left(\cos\frac{1}{2}+\frac{1}{2}\sin\frac{1}{2}-1\right)$$

23. If $f(\pi) = 2$ and $\int_0^{\pi}[f(x)+f''(x)]\sin x dx = 5$, then $f(0)$ is equal to, (it is given that $f(x)$ is continuous in $[0, \pi]$)

(a) 7 (b) 3

(c) 5 (d) 1

Solution

(b) $\int_0^{\pi}[f(x)\sin x dx + \int_0^{\pi} f''(x)\sin x dx = 5$

$\Rightarrow [f(x)(-\cos x)]_0^{\pi} + \int_0^{\pi} f'(x)\cos x dx +$

$\int_0^{\pi} f''(x)\sin x dx = 5$

$\Rightarrow [-f(x)\cos x]_0^{\pi} + [f'(x)\sin x dx]_0^{\pi} -$

$\int_0^{\pi} f''(x)\sin x dx + \int_0^{\pi} f''(x)\sin x dx = 5$

$\Rightarrow f(\pi) + f(0) = 5$

$\Rightarrow f(0) = 5 - f(\pi) = 3$

24. $\int_0^1(\{x\}[x+1])dx$ (where $\{.\}$ denotes fractional part of x and $[.]$ denotes greatest integer of x) is equal to

(a) 1 (b) 0

(c) 1/2 (d) None of these

Solution

(c) $\{x\} = x$ when $0 < x < 1$ and $[x + 1] = 1$

$$\therefore \quad \int_0^1(\{x\}[x+1])dx = \int_0^1 x dx = \frac{1}{2}$$

25. $\int_0^{\pi} [\cot x]dx$, where $[.]$ denotes the greatest integer function, is equal to *[AIEEE-2009]*

(a) $\frac{\pi}{2}$ (b) 1

(c) -1 (d) $-\frac{\pi}{2}$

Solution

(d) Let $I = \int_0^{\pi} (\cot x)dx$ .....(1)

$= \int_0^{\pi} [\cot(\pi - x)dx]$

$= \int_0^{\pi} [-\cot x]dx$ .....(2)

Adding equations (1) and (2)

$2I = \int_0^{\pi} [\cot x]dx + \int_0^{\pi} [-\cot x]dx = \int_0^{\pi} (-1)dx$

$[\because [x] + [-x] = -1 \text{ if } x \notin Z = 0 \text{ if } x \in Z]$

$= [-x]_0^{\pi} = -\pi \qquad \therefore I = -\frac{\pi}{2}$

26. $\int_0^{\pi/2} \log \sin x \, dx =$

(a) $-\pi \log 2$ (b) $\pi \log 2$

(c) $-\frac{\pi}{2}\log 2$ (d) $\frac{\pi}{2}\log 2$

Solution

(c) Let $I = \int_0^{\pi/2} \log \sin x \, dx$ .....(i)

$I = \int_0^{\pi/2} \log \sin\left(\frac{\pi}{2} - x\right)dx$

$= \int_0^{\pi/2} \log \cos x \, dx$ .....(ii)

On adding Equations (i) and (ii), we get

$2I = \int_0^{\pi/2} (\log \sin x + \log \cos x)dx$

$= \int_0^{\pi/2} (\log \sin 2x - \log 2)dx$

27. If $I_n = \int_{\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x}dx$, $n = 0, 1, 2, \ldots$ then *[IIT-2009]*

(a) $I_n = I_{n+2}$ (b) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$

(c) $\sum_{m=1}^{10} I_{2m} = 0$ (d) $I_n = I_{n+1}$

Solution

(a, b, c) $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x}dx$

$= \int_0^{\pi} \left(\frac{\sin nx}{(1+\pi^x)\sin x} + \frac{\pi^x \sin nx}{(1+\pi^x)\sin x} \right)dx$

$= \int_0^{\pi} \frac{\sin nx}{\sin x}$

Now, $I_{n+2} - I_n = \int_0^{\pi} \frac{\sin(n+2)x - \sin nx}{\sin x}dx$

$= \int_0^{\pi} \frac{2\cos(n+1)x \cdot \sin x}{\sin x}dx = 0$

$\Rightarrow I_1 = \pi, I_2 = \int_0^{\pi} 2\cos x \, dx = 0$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\int_{-2}^{2} |[x]| \, dx =$ *[EAMCET-2003]*

(a) 1 (b) 2

(c) 3 (d) 4

2. $\int_0^{2\pi} (\sin x + |\sin x|) \, dx =$ *[Karnataka CET-2003]*

(a) 0 (b) 4

(c) 8 (d) 1

3. $\int_0^{\pi/2} |\sin x - \cos x| \, dx =$ *[Roorkee-90; MPPET-01; UPSEAT-01]*

(a) 0 (b) $2(\sqrt{2} - 1)$

(c) $\sqrt{2} - 1$ (d) $2(\sqrt{2} + 1)$

4. If $I = \sqrt{2}\int_0^{100\pi}\sqrt{(1-\cos 2x)}dx$, then the value of I is:
(a) $100\sqrt{2}$ (b) $200\sqrt{2}$
(c) $50\sqrt{2}$ (d) None of these

5. $\int_0^{\pi} \log \sin^2 x\, dx =$ ***[MPPET-1997]***
(a) $2\pi\log_e(1/2)$ (b) $\pi\log_e 2 - c$
(c) $\frac{\pi}{2}\log_e\left(\frac{1}{2}\right)+c$ (d) None of these

6. $\int_{-\pi/2}^{\pi/2}\frac{\sin x}{1+\cos^2 x}e^{-\cos^2 x}dx$ is equal to:
[AMU-1999]
(a) $2e^{-1}$ (b) 1
(c) 0 (d) None of these

7. $f(x) = f(2-x)$, then $\int_{0.5}^{1.5} x f(x)\, dx$ equals:
[AMU-1999]
(a) $\int_0^1 f(x)\, dx$ (b) $\int_{0.5}^{1.5} f(x)dx$
(c) $2\int_{0.5}^{1.5} f(x)\, dx$ (d) 0

8. If $f: R \to R$ and $g: R \to R$ are one to one, real valued functions, then the value of the integral $\int_{-\pi}^{\pi}[f(x)+f(-x)]\,[g(x)-g(-x)]\,dx$ is:
[DCE-01; MPPET-04]
(a) 0 (b) π
(c) 1 (d) None of these

9. The value of $\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}\,dx$, $a > 0$ is:
[IIT Screening 2001; AIEEE-2005; Kerala PET-2008]
(a) π (b) $a\pi$
(c) $\pi/2$ (d) 2π

10. $\int_0^1 \log\sin\left(\frac{\pi}{2}x\right)dx =$ ***[RPET-1997]***
(a) $-\log 2$ (b) $\log 2$
(c) $\frac{\pi}{2}\log 2$ (d) $-\frac{\pi}{2}\log 2$

11. $\int_0^{\pi/4}\log(1+\tan\theta)d\theta =$
[SCRA-86; Karnataka CET-2000, 05]
(a) $\frac{\pi}{4}\log 2$ (b) $\frac{\pi}{4}\log\frac{1}{2}$
(c) $\frac{\pi}{8}\log 2$ (d) $\frac{\pi}{8}\log\frac{1}{2}$

12. $\int_0^{\pi/2}\frac{x\sin x\cos x}{\cos^4 x+\sin^4 x}dx =$ ***[IIT-1985]***
(a) 0 (b) $\pi/8$
(c) $\pi^2/8$ (d) $\pi^2/16$

13. The value of $\int_{1/e}^{\tan x}\frac{tdt}{1+t^2}+\int_{1/e}^{\cot x}\frac{dt}{t(1+t^2)} =$
[IIT Screening]
(a) -1 (b) 1
(c) 0 (d) None

14. $\int_0^{1/2}\frac{x\sin^{-1}x}{\sqrt{1-x^2}}dx =$ ***[IIT-1984]***
(a) $\frac{1}{2}+\frac{\sqrt{3}\pi}{12}$ (b) $\frac{1}{2}-\frac{\sqrt{3}\pi}{12}$
(c) $\frac{1}{2}-\frac{\sqrt{3\pi}}{12}$ (d) None of these

15. $\int_0^{\pi}\frac{x\tan x}{\sec x+\tan x}dx =$ ***[MNR-1984]***
(a) $\frac{\pi}{2}-1$ (b) $\pi\left(\frac{\pi}{2}+1\right)$
(c) $\frac{\pi}{2}+1$ (d) $\pi\left(\frac{\pi}{2}-1\right)$

16. The value of $\int_1^5 (|x-3|+|1-x|)\,dx$ is:
[IIT-Screening]
(a) 10 (b) 5/6
(c) 21 (d) 12

17. $\int_0^{\pi/4}\left[\sqrt{\tan x}+\sqrt{\cot x}\right]dx$ equals:
[RPET-1997]
(a) $\sqrt{2}\,\pi$ (b) $\pi/2$
(c) $\pi/\sqrt{2}$ (d) 2π

18. The integral of $\log_e \tan x$ under the limits from 0 to $\pi/2$ is: ***[MPPET-2007]***
(a) 0 (b) 1
(c) -1 (d) None of these

19. Suppose f is such that $f(-x) = -f(x)$ for every real x and $\int_0^1 f(x)\,dx = 5$, then $\int_{-1}^0 f(t)\,dt =$
[MPPET-2000]

(a) 10 (b) 5
(c) 0 (d) –5

20. The value of the integral $\int_{1/n}^{an-1/n} \frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}$ dx is: ***[AMU-2002]***
(a) $a/2$ (b) $(na+2)/2n$
(c) $(na-2)/2n$ (d) None of these

21. If n is a positive integer and $[x]$ is the greatest integer not exceeding x, then $\int_0^n \{x-[x]\}\,dx$ equals:
(a) $\frac{n^2}{2}$ (b) $\frac{n(n-1)}{2}$
(c) $\frac{n}{2}$ (d) $\frac{n^2}{2}-n$

22. Find the value of $\int_0^9 [\sqrt{x}+2]dx$, where [.] is the greatest integer function: ***[UPSEAT-2002]***
(a) 31 (b) 22
(c) 23 (d) None of these

23. If n is any integer, then $\int_0^{\pi} e^{\cos^2 x}\cos^3(2n+1)x\,dx =$ ***[MNR-1982]***
(a) x (b) 1
(c) 0 (d) None or these

24. $\int_{-2}^{2}|1-x^2|\,dx =$ ***[IIT-89; BIT Mesra-96; Kurukshetra CEE-98; MPPET-02; Kerala (Engg.)-02; AIEEE-2004]***
(a) 2 (b) 4
(c) 6 (d) 8

25. For $n>0$, $\int_0^{2\pi} \frac{x\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}dx =$ ***[IIT-1996]***
(a) π^1 (b) π
(c) π^{-2} (d) π^2

26. $\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx$ equals ***[RPET-1997; IIT Screening-2004]***
(a) $\left(\frac{\pi}{2}-1\right)$ (b) $\left(\frac{\pi}{2}+1\right)$
(c) $\pi/2$ (d) $(\pi+1)$

27. $\int_0^{\pi/2}\{x-|\sin x|\}\,dx =$ ***[AMU-1999]***
(a) $\pi^2/8$ (b) $\frac{\pi^2}{8}-1$
(c) $\frac{\pi^2}{8}-2$ (d) None of these

28. The value of $\int_0^{\sqrt{2}}[x^2]\,dx$, where [.] is the greatest integer function: ***[AIEEE-2002]***
(a) $2-\sqrt{2}$ (b) $2+\sqrt{2}$
(c) $\sqrt{2}-1$ (d) $\sqrt{2}-2$

29. $\int_{-3\pi/2}^{-\pi/2}\{(x+\pi)^3+\cos^2(x+3\pi)\}dx$ equals: ***[AIEEE-2006]***
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}-1$
(c) $\frac{\pi^2}{32}$ (d) $\frac{\pi^4}{32}+\frac{\pi}{2}$

30. The value of $\int_a^b \frac{x}{|x|}dx$, $a<b<0$ is: ***[Orissa JEE-2003]***
(a) $-(|a|+|b|)$ (b) $|b|-|a|$
(c) $|a|-|b|$ (d) $|a|+|b|$

31. The value of $I=\int_0^1 x\left|x-\frac{1}{2}\right|\,dx$ is: ***[UPSEAT-2003]***
(a) 1/3 (b) 1/4
(c) 1/8 (d) None of these

32. The value of $\int_0^1 |3x^2-1|\,dx$ is: ***[AMU-1999]***
(a) 0 (b) $4/3\sqrt{2}$
(c) 3/7 (d) 5/6

33. The expression $\frac{\int_0^n [x]dx}{\int_0^n \{x\}dx}$, where $[x]$ and $\{x\}$ are integral and fractional parts of x and $n\in N$, is equal to:
(a) $\frac{1}{n-1}$ (b) $\frac{1}{n}$
(c) n (d) $n-1$

34. Let f be a real-valued function defined on the interval $(-1,1)$ such that $e^{-x}f(x)=2+\int_0^x\sqrt{t^4+1}\,dt$, for all $x\in(-1,1)$, and let f^{-1} be the inverse function of f. Then $(f^{-1})'(2)$ is equal to: ***[IIT (2)-2010]***

(a) 1 (b) 1/3
(c) 1/2 (d) 1/*e*

SOLUTIONS

1. (d) $\int_{-2}^{2}|[x]|\,dx=\int_{-2}^{-1}|[x]|\,dx+\int_{-1}^{0}|[x]|\,dx+\int_{0}^{1}|[x]|\,dx+\int_{1}^{2}|[x]|\,dx$

$=\int_{-2}^{-1}2\,dx+\int_{-1}^{0}1\,dx+\int_{0}^{1}0\,dx+\int_{1}^{2}1\,dx$

$=2[x]_{-2}^{-1}+[x]_{-1}^{0}+0+[x]_{1}^{2}$

$=2(-1+2)+(0+1)+(2-1)=2+1+1=4$

2. (b) $\int_0^{2\pi}(\sin x+|\sin x|)dx$

$=\int_0^{\pi}2\sin x\,dx+\int_{\pi}^{2\pi}0.dx=2[-\cos x]_0^{\pi}+0$

$=-2(\cos\pi-\cos 0)=-2(-1-1)=4$

3. (b) Step 1: $\sin x-\cos x<0$ if $0<x<\frac{\pi}{4}$ and $\sin x-\cos x>0$ if $\frac{\pi}{4}<x<\frac{\pi}{2}$ and sign of y and $|y|$ is same

Step 2:

$I=\int_0^{\pi/4}-(\sin x-\cos x)dx+\int_{\pi/4}^{\pi/2}(\sin x-\cos x)dx.$

$=(\cos x-\sin x)\Big|_0^{\frac{\pi}{4}}+(-\cos x-\sin x)\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

$=\left\{\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)-(1+0)\right\}+\left\{(-0-1)-\left(-\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2}}\right)\right\}$

$=\sqrt{2}-1-1+\sqrt{2}=2(\sqrt{2}-1)$

NOTE

considering modulus function and limits, range of limits $(0,\pi/4)$ is divided into various interval on the basis of sign of modulus function and then value of $\int_0^{\pi/2}|\sin x-\cos x|dx$ is obtained.

4. (b)

$I=\int_0^{100\pi}\sqrt{1-\cos 2x}\,dx=\int_0^{\pi}(\sqrt{1-\cos 2x})dx+\int_{\pi}^{2\pi}(\sqrt{1-\cos 2x})dx+.....+\int_{(r-1)\pi}^{\pi}\sqrt{1-\cos 2x}dx+.....+\int_{99\pi}^{100\pi}(\sqrt{1-\cos 2x})dx$

$\because\ \int_0^{na}f(x)dx=n\int_0^{a}f(x)dx,$

if $f(a+x)=f(x)$

$\therefore\ I=100\int_0^{\pi}\sqrt{(1-\cos 2x)}dx=100\sqrt{2}$

$\int_0^{\pi}\sin x\,dx=200\sqrt{2}\int_0^{\pi/2}\sin x\,dx$

$=200\sqrt{2}\left[-\cos x\right]_0^{\pi/2}=200\sqrt{2}$

5. (a) $\int_0^{\pi}\log(\sin x)^2dx=2\int_0^{\pi}\log(\sin x)dx$

$=2\times 2\int_0^{\frac{\pi}{2}}\log(\sin x)dx$

$=4\times\left(-\frac{\pi}{2}\log 2\right)=-2\pi\log 2=2\pi\log_e\left(\frac{1}{2}\right)$

6. (c) $I=\int_{-\pi/2}^{\pi/2}\frac{\sin x}{1+\cos^2 x}e^{-\cos^2 x}dx$

$\because\ \frac{\sin x}{1+\cos^2 x}e^{-\cos^2 x}$ is an odd function

$\therefore\ I=0$

7. (b) $I=\int_{0.5}^{1.5}xf(x)dx=\int_{0.5}^{1.5}(2-x)f(2-x)dx$

$\left[\because\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$

$=\int_{0.5}^{1.5}(2-x)f(x)dx=2$

$\int_{0.5}^{1.5}f(x)dx-I\Rightarrow I=\int_{0.5}^{1.5}f(x)dx$

8. (a) Let $\phi(x)=[f(x)+f(-x)][g(x)-g(-x)]$, then

$\phi(-x)=[f(-x)+f(x)][g(-x)-g(x)]=-\phi(x)$

$\therefore \int_{-\pi}^{\pi}\phi(x)dx=0$

$\Rightarrow \int_{-\pi}^{\pi}[f(x)+f(-x)][g(x)-g(-x)]dx=0$

9. (c) Step 1: From formula

$\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$

$I=\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^{-x}}dx=\int_{-\pi}^{\pi}\frac{a^x\cos^2 x}{a^x+1}dx$

Step 2:

$I+I=2I=\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx+\int_{-\pi}^{\pi}\frac{a^x\cos^2 x}{a^x+1}dx$

$2I=\int_{-\pi}^{\pi}\left(\frac{a^x+1}{a^x+1}\right)\cos^2 x dx=\int_{-\pi}^{\pi}\cos^2 x dx$

$=2\int_0^{\pi}\cos^2 x dx$; because $\cos^2 x$ is a even function

$(f(-x)=f(x))$

$=2\times 2\int_0^{\pi/2}\cos^2 x dx$; as $f(\pi-x)=f(x)$

$=4\times\frac{1}{2}\times\frac{\pi}{2}=\pi$

Hence $I=\frac{\pi}{2}$

Quicker Method: This question All for $a > 0$ is true

Hence take $a = 1$

$I=\int_{-\pi}^{\pi}\frac{\cos^2 x}{2}dx=\frac{1}{2}\int_{-\pi}^{\pi}\cos^2 x dx=\frac{\pi}{2}$

10. (a) $\int_0^1 \log\sin\left(\frac{\pi}{2}x\right)dx$

Put $\frac{\pi}{2}x=\theta\Rightarrow dx=\frac{2}{\pi}d\theta$, then

As $x = 0$ to 1, $0 = 0$ to $\frac{\pi}{2}$, then

it reduces to

$\frac{2}{\pi}\int_0^{\pi/2}\log\sin 0 d0=\frac{2}{\pi}\left[-\frac{\pi}{2}\log 2\right]=-\log 2$

11. (c)

$I=\int_0^{\pi/4}\log(1+\tan\theta)d\theta$

$\Rightarrow I=\int_0^{\pi/4}\log\left\{1+\tan\left(\frac{\pi}{4}-\theta\right)\right\}d\theta$

$I=\int_0^{\pi/4}\log\left(1+\frac{1-\tan 0}{1+\tan\theta}\right)d\theta$

$=\int_0^{\pi/4}\log 2d0-\int_0^{\pi/4}\log(1+\tan 0)d0$

$I=\frac{1}{2}\int_0^{\pi/4}\log\ 2\ d\theta=\frac{\log 2}{2}|\theta|_0^{\pi/4}$

$=\frac{\pi}{8}\log 2$

12. (d) $I=\int_0^{\pi/2}\frac{x\sin x\cos x}{\cos^4 x+\sin^4 x}$ ..... (i)

$=\int_0^{\pi/2}\frac{\left(\frac{\pi}{2}-x\right)\cos x\sin x}{\sin^4 x+\cos^4 x}$ ..... (ii)

By adding (i) and (ii), we get

$2I=\frac{\pi}{2}\int_0^{\pi/2}\frac{\cos x\sin x}{\cos^4 x+\sin^4 x}dx$

$\Rightarrow I=\frac{\pi}{4}\int_0^{\pi/2}\frac{\cos x\sin x}{1+\tan^4 x}dx=\frac{\pi}{4}\int_0^{\pi/2}\frac{\tan x\sec^2 x}{1+\tan^4 x}$

Now, put $\tan^2 x=t$, $\Rightarrow dt=2\tan x\sec^2 x dx$, we get

$I=\frac{\pi}{8}\int_0^{\infty}\frac{dt}{1+t^2}=\frac{\pi}{8}\left[\tan^{-1}t\right]_0^{\infty}=\frac{\pi^2}{16}$

13. (b) $\int_{1/e}^{\tan x}\frac{tdt}{1+t^2}+\int_{1/e}^{\cot x}\frac{dt}{t(1+t^2)}$

Consider $\int_{1/e}^{\cot x}\frac{dt}{t(1+t^2)}=\int_e^{\tan x}\frac{-\frac{1}{u^2}du}{\frac{1}{u}\left(1+\frac{1}{u^2}\right)}$

(substituting $u = 1/t$)

$=\int_e^{\tan x}\frac{-udu}{u^2+1}=\int_{\tan x}^{e}\frac{udu}{u^2+1}=\int_{\tan x}^{e}\frac{tdt}{t^2+1}$

$\therefore$ Given Integral

$=\int_{1/e}^{\tan x}\frac{tdt}{1+t^2}+\int_{\tan x}^{e}\frac{tdt}{t^2+1}$

$=\int_{1/e}^{e}\frac{tdt}{t^2+1}=\frac{1}{2}\log_e|t^2+1|_{1/e}^{e}$

$$=\frac{1}{2}\left[\log_e(e^2+1)-\log_e\left(\frac{1}{e^2}+1\right)\right]$$

$$=\frac{1}{2}(\log_e e^2)=1$$

14. (b) $I=\int_0^{1/2}\frac{x\sin^{-1}x}{\sqrt{1-x^2}}dx$

Let, $\sin^{-1}x=0\Rightarrow x=\sin 0$

$\therefore\quad dx=\cos 0 d0$

also, $x=0\Rightarrow 0=0$ and $x=\frac{1}{2}\Rightarrow 0=\frac{\pi}{6}$

$$I=\int_0^{\pi/6}\frac{\theta\sin\theta}{\sqrt{1-\sin^2 0}}.\cos 0 d0$$

$$=(-0\cos 0)_0^{\pi/6}+\int_0^{\pi/6}\cos 0 d0$$

$$=\left(-\frac{\pi}{6}\cos\frac{\pi}{6}+0\right)+\left(\sin\frac{\pi}{6}-\sin 0\right)$$

$$=\frac{-\sqrt{3}\pi}{12}+\frac{1}{2}$$

15. (d) $\int_0^{\pi}\frac{x\tan x}{\sec x+\tan x}dx$ ..... (i)

$$\Rightarrow\int_0^{\pi}\frac{(\pi-x)\tan(\pi-x)}{\sec(\pi-x)+\cos(\pi-x)}dx \quad \text{..... (ii)}$$

By adding (i) and (ii), we get

$$2I=\frac{\pi}{2}\int_0^{\pi}\frac{\tan x}{\sec x+\tan x}dx=\frac{\pi}{2}\int_0^{\pi}\frac{\sin x}{1+\sin x}dx$$

$$=\frac{\pi}{2}\left[\int_0^{\pi}1dx-\int_0^{\pi}\frac{dx}{1+\sin x}\right]$$

On Solving, we get

$$I=\frac{\pi^2}{2}-\pi=\pi\left(\frac{\pi}{2}-1\right)$$

16. (d)

$$\int_1^5(|x-3|+|1-x|)dx=\int_1^5|x-3|dx+\int_1^5|1-x|dx$$

$$=\int_1^3-(x-3)dx+\int_3^5(x-3)dx+\int_1^5-(1-x)dx$$

$$=-\left[\frac{x^2}{2}-3x\right]_1^3+\left[\frac{x^2}{2}-3x\right]_3^5-\left[x-\frac{x^2}{2}\right]_1^5$$

$$=12$$

17. (c) $I=\int_0^{\pi/4}\left[\sqrt{\tan x}+\sqrt{\cot x}\right]dx$

$$=\int_0^{\pi/4}\frac{\sin x+\cos x}{\sqrt{\sin x\cos x}}dx$$

$$=\sqrt{2}\int_0^{\pi/4}\frac{\sin x+\cos x}{\sqrt{1-(\sin x-\cos x)^2}}dx$$

Put $\sin x-\cos x=t;(\cos x+\sin x)dx=dt$

$$I=\sqrt{2}\int_{-1}^{0}\frac{dt}{\sqrt{1-t^2}}\Rightarrow I=\sqrt{2}\left[\sin^{-1}t\right]_{-1}^{0}$$

$$=\sqrt{2}\left[0-\left(-\frac{\pi}{2}\right)\right]=\frac{\pi}{\sqrt{2}}$$

18. (a) $I=\int_0^{\pi/2}\log_e\tan x dx$ ..... (1), then

$$I=\int_0^{\pi/2}\log_e\tan\left(\frac{\pi}{2}-x\right)dx=\int_0^{\pi/2}\log\cot x \quad \text{.. (2)}$$

Adding equation (1) and (2), we get

$$2I=\int_0^{\pi/2}\log_e\tan x dx+\int_0^{\pi/2}\log_e\cot x$$

$$=\int_0^{\pi/2}(\log_e\tan x+\log_e\cot x)dx$$

$$=\int_0^{\pi/2}(\log_e\tan x.\cot x)dx$$

$$(\because\log_e a+\log_e b=\log_e ab)$$

$$=\int_0^{\pi/2}\log 1\,dx=0$$

19. (d) Step 1: $\int_0^1 f(x)dx=\int_{-1}^0 f(x)dx+\int_0^1 f(x)dx$

Step 2: For ($f(-x)=-f(x)$) odd function,

$\int_{-1}^{1}f(x)dx=0$ is defined

Hence For Step 1

$$0=\int_{-1}^0 f(x)dx+\int_0^1 f(x)dx$$

Hence $\int_{-1}^0 f(x)dx=-\int_0^1 f(x)dx=-5$

NOTE

Only change in variable does not make any difference.

20. (c)

$$I=\int_{1/n}^{n-1/n}\frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}dx=\int_{1/n}^{n-1/n}\frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}dx$$

....(i)

$$= \int_{1/n}^{a-\frac{1}{n}} \frac{\sqrt{\frac{1}{n}+a-\frac{1}{n}-x}}{\sqrt{a-\left(\frac{1}{n}+a-\frac{1}{n}-k\right)}+\sqrt{\frac{1}{n}+a-\frac{1}{n}-x}}dx$$

$\because \quad \int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

$$I = \int_{1/n}^{a-\frac{1}{n}} \frac{\sqrt{a-x}}{\sqrt{x}+\sqrt{a-x}}dx \qquad \text{.....(ii)}$$

Adding (i) and (ii), we get $2I = \int_{1/n}^{a-\frac{1}{n}} 1dx = [x]_{1/n}^{a-\frac{1}{n}}$

$$\Rightarrow 2I = a-\frac{1}{n}-\frac{1}{n} = \frac{na-2}{n} \Rightarrow I = \frac{na-2}{2n}$$

21. (c) $x-[x]$ is a periodic function with period 1

$$\therefore \quad \int_0^1 \{x-[x]\}dx = n\int_0^1 (x-[x])dx$$

$$n\left[\int_0^1 xdx - \int_0^1 [x]dx\right] = n\left[\left(\frac{x^2}{2}\right)_0^1 - 0\right] = \frac{n}{2}$$

22. (a) $\int_0^9 \left[\sqrt{x}+2\right]dx = \int_0^1 2dx + \int_1^4 3dx + \int_4^9 4dx$

$= 2+(12-3)+(36-16) = 2+9+20 = 31$

23. (c) $f(x) = \int_0^{\pi} e^{\cos^2 x}\cos^3(2n+1)x dx$

$\because \quad \cos(2n+1)(\pi-x)$

$= \cos[(2n+1)\pi-(2n+1)x]$

$= -\cos(2n+1)x$ and $\cos^2(\pi-x) = \cos^2 x$

Hence by the property of definite integral

$\int_0^{\pi} e^{\cos^2 x}\cos^3(2n+1)x dx = 0$,

$[\because f(2a-x) = -f(x)]$

24. (b) $I = \int_{-2}^{2}|1-x^2|dx$ or $I = 2\int_0^2|(1-x^2)|dx$

$\left(\because \int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx\right)$ when

$f(-x) = f(x)$

$$I = 2\left\{\int_0^1 (1-x^2)dx + \int_1^2 (x^2-1)dx\right\}$$

$$= 2\left\{\left[x-\frac{x^3}{3}\right]_0^1 + \left[\frac{x^3}{3}-x\right]_1^2\right\}$$

$$= 2\left(1-\frac{1}{3}+\frac{8}{3}-2-\frac{1}{3}+1\right) = 4$$

25. (d) Let $I = \int_0^{2\pi} \frac{x\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}dx$ ..... (1)

$$I = \int_0^{2\pi} \frac{(2\pi-x)[\sin(2\pi-x)]^{2n}}{[\sin(2\pi-x)]^{2n}+[\cos(2\pi-x)]^{2n}}dx$$

$\left[\because \int_0^a f(x)dx = \int_0^a [f(a-x)]dx\right]$

$$= \int_0^{2\pi} \frac{(2\pi-x).\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x} = \int_0^{2\pi} \frac{2\pi\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}dx -$$

$$\int_0^{2\pi} \frac{x\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}dx$$

$$I = \int_0^{2\pi} \frac{2\pi\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x} - I \quad [\text{From (1)}]$$

$$2I = \int_0^{2\pi} \frac{2\pi\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}$$

$$\Rightarrow \quad I = \int_0^{2\pi} \frac{\pi\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}$$

By using property $\int_0^{2a}[f(x)+f(2a-x)]dx$

$$= \pi\left[\int_0^{\pi} \frac{\sin^{2n}x dx}{\sin^{2n}x+\cos^{2n}x}dx + \int_0^{\pi} \frac{\sin^{2n}x dx}{\sin^{2n}x+\cos^{2n}x}\right], \text{ we get}$$

$$I = \pi\left[\int_0^{\pi} \frac{\pi\sin^{2n}x dx}{\sin^{2n}x+\cos^{2n}x} + \int_0^{\pi} \frac{\sin^{2n}(2\pi-x)}{\sin^{2n}(2\pi-x)+\cos^{2n}(2\pi-x)}\right]$$

$$I = 2\pi\int_0^{\pi} \frac{\sin^{2n}x dx}{\sin^{2n}x+\cos^{2n}x}dx$$

$$I = 4\pi\left[\int_0^{\pi/2} \frac{\sin^{2n}x dx}{\sin^{2n}x+\cos^{2n}x}\right] \qquad \text{..... (2)}$$

Applying the property again,

$$\Rightarrow I = 4\pi\int_0^{\pi/2} \frac{\sin^{2n}\left(\frac{\pi}{2}-x\right)dx}{\sin^{2n}\left(\frac{\pi}{2}-x\right)+\cos^{2n}\left(\frac{\pi}{2}-x\right)}$$

$$\Rightarrow I = 4\pi\int_0^{\pi/2} \frac{\cos^{2n}x}{\cos^{2n}x+\sin^{2n}x}dx \qquad \text{..... (3)}$$

Adding equation (2) and (3)

$$2I = 4\pi\int_0^{\pi/2} \frac{\sin^{2n}x+\cos^{2n}x}{\sin^{2n}x+\cos^{2n}x}dx$$

$$2I = 4\pi\int_0^{\pi/2} dx$$

$= 4\pi[x]_0^{\pi/2}$

$= 4\pi\frac{\pi}{2} = 2\pi^2$

$I = \pi^2$

26. (a) Step-1: Rationalisation method:

$$I = \int_0^1 \frac{1-x}{\sqrt{1-x^2}}dx = \int_0^1 \frac{dx}{\sqrt{1-x^2}} - \int_0^1 \frac{xdx}{\sqrt{1-x^2}}$$

$$= (\sin^{-1}x + \sqrt{1-x^2})_0^1$$

$$= (\sin^{-1}1 + \sqrt{1-1}) - (\sin^{-1}0 + \sqrt{1-0}) = \frac{\pi}{2} - 1$$

27. (a) $\int_0^{\pi/2}(x - [\sin x])dx = \int_0^{\pi/2} xdx - \int_0^{\pi/2}[\sin x]dx$

$$= \left(\frac{x^2}{2}\right)_0^{\pi/2} = \frac{\pi^2}{8}$$

NOTE

$\because \int_0^{\pi/2}[\sin x]dx = 0$

28. (c) Step 1: Greatest integer function $x = [x] \le x < [x]+1$ and $n \le x < n+1$ for $[x] = n$

Step 2: $0 \le x < 1 \Rightarrow 0 \le x^2 < 1 \Rightarrow [x^2] = 0$

$1 \le x < \sqrt{2} \Rightarrow 1 \le x^2 < 2 \Rightarrow [x^2] = 1$

Hence $\int_0^{\sqrt{2}}[x^2]dx = \int_0^1[x^2]dx + \int_1^{\sqrt{2}}[x^2]dx$

$\int_0^1 0dx + \int_1^{\sqrt{2}}dx = [x]_1^{\sqrt{2}} = \sqrt{2} - 1$.

NOTE

For all values of x, sign of x, $[x]$ are equal. Hence x, $[x]$ are of same sign.

29. (b) Let $I = \int_{-3\pi/2}^{-\pi/2}[(x+\pi)^3 + \cos^2(x+3\pi)]dx$

and

$$I = \int_{-3\pi/2}^{-\pi/2}\left[\left(-\frac{\pi}{2} - \frac{3\pi}{2} - x + \pi\right)^3 + \cos^2\left(-\frac{\pi}{2} - \frac{3\pi}{2} - x + 3\pi\right)\right]dx$$

$$\Rightarrow I = \int_{-3\pi/2}^{-\pi/2}[-(x+\pi)^3 + \cos^2(\pi - x)]dx \quad \text{....(ii)}$$

On adding Eqs. (i) and (ii), we get

$$2I = \int_{-3\pi/2}^{-\pi/2} 2\cos^2 x\, dx$$

$$= \int_{-3\pi/2}^{-\pi/2}(1+\cos 2x)\, dx = \left[x + \frac{\sin 2x}{2}\right]_{-3\pi/2}^{-\pi/2}$$

$$= -\frac{\pi}{2} + \frac{3\pi}{2} = \pi \Rightarrow I = \frac{\pi}{2}$$

30. (b) $\int_a^b \frac{x}{|x|}dx = -\int_a^b dx \quad (\because a < b < 0)$

$= -(b-a) = |b| - |a|$

31. (c)

$$I = \int_0^1 x\left|x - \frac{1}{2}\right|dx$$

$$= -\int_0^{1/2} x\left(x - \frac{1}{2}\right) + \int_{1/2}^1 x\left(x - \frac{1}{2}\right)dx$$

$$= \int_0^{1/2}\left(\frac{1}{2}x - x^2\right)dx + \int_{1/2}^1\left(x^2 - \frac{1}{2}x\right)dx$$

$$= \left[\frac{x^2}{4} - \frac{x^3}{3}\right]_0^{1/2} + \left[\frac{x^3}{3} - \frac{x^2}{4}\right]_{1/2}^1$$

$$= \left(\frac{1}{16} - \frac{1}{24}\right) + \left(\frac{1}{3} - \frac{1}{4} + \frac{1}{16} - \frac{1}{24}\right) = \frac{1}{8}$$

32. (b)

$$\int_0^1 |3x^2 - 1|dx$$

$$= \int_0^{1/\sqrt{3}}(1-3x^2)dx + \int_{1/\sqrt{3}}^1 -(1-3x^2)dx$$

$$= \left[x - x^3\right]_0^{1/\sqrt{3}} + \left[x^3 - x\right]_{1/\sqrt{3}}^1$$

$$= \frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{3}} + \frac{-1}{3\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{4}{3\sqrt{3}}$$

33. (d)

$$\int_0^n (x)dx = \int_0^1 0dx + \int_1^2 1dx +$$

$$\int_2^3 2dx + \ldots\ldots\int_{n-1}^n (n-1)dx$$

$$= 1(2-1) + 2(3-2) + \ldots\ldots\ldots(n-1)(n-(n-1))$$

$$= 1 + 2 + 3 + \ldots\ldots\ldots n - 1 = \frac{n(n-1)}{2}$$

$$\int_0^n \{x\}dx = n\int_0^1 \{x\}dx = n\int_0^1 x dx = n\frac{x^2}{2}\Big|_0^1 = \frac{n}{2}$$

Expression $\dfrac{\int_0^n [x]dx}{\int_0^n \{x\}dx} = \dfrac{\frac{n(n-1)}{2}}{\frac{n}{2}} = n-1$

34. (b) $f(x) = e^x\left(2 + \int_0^x \sqrt{t^4+1}dt\right)$

Let $g(x) = f^{-1}(x)$

$\Rightarrow g(f(x)) = x$

$\Rightarrow g'(f(x))f'(x) = 1$

$\Rightarrow g'(2) = \dfrac{1}{f'(0)}$ $(\because f(0) = 2)$

Now $f'(x) = e^x\left(2 + \int_0^x \sqrt{t^4+1}dt\right) + e^x\sqrt{x^4+1}$

(Applying Leibinitz Rule)

$\Rightarrow f'(0) = 2+1 = 3$

$\Rightarrow g'(2) = \dfrac{1}{3}$

$\Rightarrow (f^{-1})'(2) = \dfrac{1}{3}$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The correct evaluation of $\int_0^{\pi/2} \sin x \sin 2x$ is

[MPPET-1993, 03]

(a) 4/3 (b) 1/3
(c) 3/4 (d) 2/3

2. If $[x]$ denotes the greatest integer less than or equal to x, then the value of $\int_1^5 [|x-3|]\,dx$ is

[Roorkee-1999]

(a) 1 (b) 2
(c) 4 (d) 8

3. $\int_0^{\pi} \sqrt{\dfrac{1+\cos 2x}{2}}dx =$ ***[AMU-2001]***

(a) 0 (b) 2
(c) 1 (d) -1

4. $\int_0^{10\pi} e^{x-[x]}\,dx =$ ***[AMU-2002]***

(a) $e^{1000}-1$ (b) $\dfrac{e^{1000}-1}{e-1}$

(c) $1000(e-1)$ (d) $\dfrac{e-1}{1000}$

5. The integral $\int_{-1/2}^{1/2}\left\{[x] + \log\left(\dfrac{1+x}{1-x}\right)\right\}dx$ equals

(where, [.] is the greatest integer function)

[IIT Screening-2002]

(a) $-1/2$ (b) 0
(c) 1 (d) $2\log\dfrac{1}{2}$

6. $\int_0^{\pi} \dfrac{x\,dx}{a^2\cos^2 x + b^2\sin^2 x} =$

[Karnataka CET-2003]

(a) $\dfrac{\pi}{ab}$ (b) $\dfrac{\pi}{2ab}$

(c) $\dfrac{\pi^2}{ab}$ (d) $\dfrac{\pi^2}{2ab}$

7. $\int_0^1 \sin\left(2\tan^{-1}\sqrt{\dfrac{1+x}{1-x}}\right)dx =$

[EAMCET-2003]

(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/2$ (d) π

8. If $n \in N$, then $\int_{-n}^{n} (-1)^{[x]}\,dx$ equals

(a) $2n$ (b) n
(c) n^2 (d) 0

9. If $[x]$ denotes the greatest integer function, then the value of $\int_{0.5}^{4.5} [x]dx + \int_{-1}^{1} |x|\,dx$ is

(a) 9 (b) 8
(c) 7 (d) 6

10. $\int_0^{\pi} x \sin x \, dx =$ ***[SCRA-80]***

(a) π (b) 0
(c) 1 (d) π^2

11. $\int_0^{\pi} |\cos x| \, dx =$

[MPPET-98;Pb. CET-2001]

(a) π (b) 0
(c) 2 (d) 1

12. $\int_0^{\pi/2} x \cot x \, dx$ equals

[RPET-1997]

(a) $-\frac{\pi}{2}\log 2$ (b) $\frac{\pi}{2}\log 2$
(c) $\pi\log 2$ (d) $-\pi \log 2$

13. $\int_{-\pi/2}^{\pi/2} \log\left(\frac{2-\sin\theta}{2+\sin\theta}\right) d\theta =$

(a) 0 (b) 1
(c) 2 (d) None of these

14. If $\int_0^{\pi} x f(\sin x)\, dx = k\int_0^{\pi} f(\sin x)dx$, then the value of k will be

(a) π (b) $\pi/2$
(c) $\pi/4$ (d) 1

15. $\int_0^{\pi} \frac{x\tan x}{\sec x + \cos x} dx =$

[BIT Ranchi-1986; MNR-1985, UPSEAT-02]

(a) $\frac{\pi^2}{4}$ (b) $\frac{\pi^2}{2}$
(c) $\frac{3\pi^2}{2}$ (d) $\frac{\pi^2}{2}$

16. $\int_{-1}^{1} x \tan^{-1}x \, dx$ equals ***[RPET-1997]***

(a) $\left(\frac{\pi}{2}-1\right)$ (b) $\left(\frac{\pi}{2}+1\right)$
(c) $(\pi-1)$ (d) 0

17. $\int_0^3 |2-x| \, dx =$

[RPET-99]

(a) 2/7 (b) 5/2
(c) 3/2 (d) –3/2

18. Let $I_1 = \int_a^{\pi-a} xf(\sin x)\, dx$, $I_2 = \int_a^{\pi-a} f(\sin x)\, dx$, then I_2 is equal to ***[AMU-2000]***

(a) $\frac{\pi}{2} I_1$ (b) πI_1
(c) $\frac{2}{\pi} I_1$ (d) $2I_1$

19. The value of $\int_0^{\pi/2} \log|\tan x + \cot x| \, dx$ is

(a) $\pi \log 2$ (b) $-\pi\log 2$
(c) $\frac{\pi}{2}\log 2$ (d) $-\frac{\pi}{2}\log 2$

20. $\int_{a-c}^{b+c} f(x-c)\, dx$ is equal to

(a) $\int_a^b f(x-c)\, dx$
(b) $\int_a^b f(x)\, dx$
(c) $\int_a^b f(a+b-c+x)\, dx$
(d) None of these

21. The value of $\int_0^{\pi/2} \frac{\left(\frac{\pi}{4}-x\right)dx}{\sqrt{\sin x + \cos x}}$ is

(a) 0 (b) $\pi/4\sqrt{2}$
(c) $\pi\sqrt{2}/4$ (d) None of these

22. The function $F(x) = \int_0^x \log(t+\sqrt{1+t^2})dt$ is

(a) an even function (b) an odd function
(c) a periodic function (d) None of these

23. $\int_{-1}^{1} |1-x| \, dx$ is equal to

(a) –2 (b) 0
(c) 2 (d) 4

24. $\int_0^a \frac{dx}{x+\sqrt{a^2-x^2}}$ is

[Kerala P.E.T.-2001]

(a) $a^2/4$ (b) $\pi/4$
(c) $\pi/2$ (d) π

25. $\int_1^9 e^{\sqrt{t}}\,dt$ is equal to

[VIT-2006]

(a) e^3 (b) $4e^3$
(c) $4(e^3 - e)$ (d) $4e^3 - 2e$

26. The value of $\int_{\log\frac{1}{2}}^{\log 2} \log(x + \sqrt{1+x^2})\,dx$ is

[Kerala PET-2008]

(a) $2\log 2$ (b) $1 - \log 2$
(c) $\log 2$ (d) 0

27. The value of $\int_1^2 \frac{dx}{(x+1)\sqrt{x^2-1}}$ is

[MP PET-2008]

(a) 1 (b) $\frac{1}{\sqrt{3}}$
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{-2}{\sqrt{3}}$

28. If $f(t) = \int_{-t}^{t} \frac{e^{-|x|}}{2}\,dx$, then $\lim_{t\to\infty} f(t) =$

[EAMCET-2007]

(a) 1 (b) 1/2
(c) 0 (d) –1

29. $\int_0^2 |x - 1|\,dx =$

[SCRA-1990; RPET-2001; UPSEAT-2003]

(a) 0 (b) 2
(c) 1/2 (d) 1

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 20 minutes.
3. The worksheet consists of 20 questions. The maximum marks are 60.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int_4^{10} \frac{[x^2]dx}{[x^2-28x+196]+[x^2]}$ is

(a) 0 (b) 1
(c) 3 (d) None of these

2. The value of $\int_{\pi/6}^{5\pi/6} \sqrt{4-4\sin^2 t}\, dt$ is

(a) 0 (b) 2
(c) 1 (d) None of these

3. $\int_{-\pi/2}^{\pi/2} \sqrt{\frac{1}{2}(1-\cos 2x)}dx =$

(a) 0 (b) 2
(c) 1/2 (d) None of these

4. Given function $f(x) = \begin{cases} x^2, & \text{for } 0 \le x < 1 \\ \sqrt{x}, & \text{for } 1 \le x \le 2 \end{cases}$,

evaluate $\int_0^2 f(x)dx$

(a) $\frac{1}{3}(4\sqrt{2}-1)$ (b) $\frac{1}{3}(4\sqrt{2}+1)$
(c) $\frac{1}{3}(4\sqrt{3}+1)$ (d) $\frac{1}{3}(4\sqrt{3}-1)$

5. If $\int_0^a f(2a-x)dx = \mu$ and $\int_0^{2a} f(x)dx$ then $\int_0^{2a} f(x)dx$ equals

(a) $2\lambda - \mu$ (b) $\lambda + \mu$
(c) $\mu - \lambda$ (d) $\lambda - 2\mu$

6. The value of $\int_{-1}^{1} \left[2|x| - |x|^3\right]dx$ is

(a) 0 (b) 3/4
(c) 3/2 (d) 3

7. $\int_{-1}^{2} |x|\, dx =$

(a) 5/2 (b) 2
(c) 3/2 (d) 1

8. If $f(x) = \begin{cases} |x| & -1 \le x \le 1 \\ |x-2| & 1 < x \le 3 \end{cases}$, then $\int_{-1}^{3} f(x)dx$ is equal to

(a) 0 (b) 1
(c) 2 (d) 4

9. $\int_0^{2\gamma} 0.\sin^2 0\cos 0 d0$ is equal to

(a) $\pi/16$ (b) $3\pi/16$
(c) $16\pi/3$ (d) 0

10. The value of the integral $\int_{-\pi/3}^{\pi/3} \frac{x\sin x}{\cos^2 x}dx$ is

(a) $\left(\frac{\pi}{3} - \log\tan\frac{3\pi}{2}\right)$
(b) $2\left(\frac{2\pi}{3} - \log\tan\frac{5\pi}{12}\right)$
(c) $3\left(\frac{\pi}{2} - \log\tan\frac{\pi}{12}\right)$
(d) None of these

11. The value of $\int_{-\pi}^{\pi} (1-x^2)\sin x\cos^2 x dx$ is

(a) 0 (b) $\pi - \frac{\pi^2}{3}$
(c) $2\pi - \pi^3$ (d) $\frac{7}{2} - 2\pi^3$

12. $\int_{-2}^{2} \sin^3 x\, |x|\, dx$ is equal to

(a) 0 (b) 1
(c) 4 (d) None of these

13. $\int_0^{\pi} \frac{xdx}{1+\sin x}$ is equal to ***[MPPET-2009]***

(a) $-\pi$ (b) $\pi/2$
(c) π (d) None of these

14. Let $I_1 = \int_a^{\pi-a} xf(\sin x)dx, I_2 = \int_a^{\pi-a} f(\sin x)dx$, then I_2 is equal to

(a) $\frac{\pi}{2}I_1$ (b) πI_1

(c) $\frac{2}{\pi}I_1$ (d) $2I_1$

15. What is the value of $\int_{-1}^{1} x|x|dx$?

[NDA-2007]

(a) 2 (b) 1

(c) 1/4 (d) 0

16. $\int_0^1 f(1-x)\,dx$ has the same value as the integral ***[SCRA-90]***

(a) $\int_0^1 f(x)\,dx$ (b) $\int_0^1 f(-x)\,dx$

(c) $\int_0^1 f(x-1)\,dx$ (d) $\int_{-1}^1 f(x)dx$

17. $\int_0^2 |(1-x)|\,dx =$

(a) 1 (b) 2

(c) 3 (d) 0

18. If $\int_2^e \left[\frac{1}{\log x} - \frac{1}{(\log x)^2}\right] dx = \alpha + \frac{\beta}{\log 2}$, then

[SCRA-1986]

(a) $\alpha = e, \beta = -2$ (b) $\alpha = e, \beta = 2$

(c) $\alpha = -e, \beta = 2$ (d) $\alpha = -e, \beta = -2$

19. $\int_0^1 \frac{\log x}{\sqrt{1-x^2}}dx =$

[BIT Ranchi-1984]

(a) $\frac{\pi}{2}\log 2$ (b) $\pi\log 2$

(c) $-\frac{\pi}{2}\log 2$ (d) $-\pi\log 2$

20. $\int_{-2}^1 \frac{|x|}{x}dx$ is equal to

(a) 1 (b) 3

(c) –1 (d) –3

21. The value of $\int_{-1}^{1}|x+1|dx$ is

[MPPET-2010]

(a) 1 (b) 0

(c) 2 (d) None of these

22. The value of $\int_0^3 [x]dx$ is

[MPPET-2010]

(a) –3 (b) 3

(c) 2 (d) 1

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	17. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	18. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ	19. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ	20. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ	21. ⓐ ⓑ ⓒ ⓓ
6. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ	22. ⓐ ⓑ ⓒ ⓓ
7. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ	
8. ⓐ ⓑ ⓒ ⓓ	16. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

1. (c)

$$I=\int_4^{10}\frac{dx}{(14-x)^2+[x^2]}$$

$$=\int_4^{10}\frac{[(14-x)^2]dx}{[14-(14-x)]^2+[(14-x)^2]}$$

$$\left(\because \int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right)$$

$$\therefore\quad I=\int_4^{10}\frac{[(14-x)^2]dx}{[x^2]+[14-x]^2}$$

Adding $2I=\int_4^{10}\frac{[x^2]+(14-x)^2}{[x^2]+(14-x)^2}dx=\int_4^{10}dx$

$I=3$

2. (b) $\int_{\pi/6}^{5\pi/6}\sqrt{4(1-\sin^2 t)}dt=\int_{\pi/6}^{5\pi/6}2|\cos t|dt$

$$=2\int_{\pi/6}^{\pi/2}-2\int_{\pi/2}^{5\pi/6}\cos t dt=2\sin t\Big|_{\pi/6}^{\pi/2}-2\sin t\Big|_{\pi/2}^{\pi/6}dt$$

$$=2\left(1-\frac{1}{2}\right)-2\left(\frac{1}{2}-1\right)=1+1=2$$

3. (b)

$$I=\int_{-\pi/2}^{\pi/2}\sqrt{\frac{1}{2}(2\sin^2 x)}dx$$

$$=\int_{-\pi/2}^{\pi/2}|\sin x|\,dx=2\int_0^{\pi/2}\sin x dx$$

$$=2(-\cos x)\Big|_0^{\pi/2}=2$$

4. (a) $\int_0^2 f(x)dx=\int_0^1 x^2dx+\int_1^2\sqrt{x}dx\frac{x^3}{3}\Big|_0^1+\frac{x^{3/2}}{3/2}\Big|_1^2$

$$=\frac{1}{3}+\frac{2}{3}(2\sqrt{2}-1)=\frac{4\sqrt{2}-1}{3}$$

5. (b) $I=\int_0^{2a}f(x)dx=\int_0^a f(x)dx+\int_a^{2a}f(x)dx$

consider $\int_a^{2a}f(x)dx=\int_a^0 f(2a-t)(-dt)$

Put $2a\quad x=t$

$$=\int_0^a f(2a-t)dt=\int_0^a f(2a-x)dx$$

$$\therefore\quad \int_0^{2a}f(x)dx=\int_0^a f(x)dx+\int_0^a f(2a-x)dx$$

$$=\lambda+\mu$$

8. (c) $\int_{-1}^3 f(x)dx=\int_{-1}^1|x|dx+\int_1^3|x-2|dx$

$$=2\int_0^1 xdx-\int_1^2(x-2)dx+\int_2^3(x-2)dx=2$$

9. (d)

$$\int_0^{2\pi}0\sin^2 0\cos 0d0=\frac{0+2\pi}{2}\int_0^{2\pi}\sin^2 0\cos 0d0$$

$$\left(\because \int_a^b xf(x)dx=\frac{a+b}{2}\int_a^b f(x)dx\right)$$

$$=\pi\left(\frac{\sin^2\theta}{3}\right)\Big|_0^{2\pi}=0$$

10. (b) $\int_{-\pi/3}^{\pi/3}\frac{x\sin x}{\cos^2 x}dx=2\int_0^{\pi/3}\frac{x\sin x}{\cos^2 x}dx$

$\left\{\frac{x\sin x}{\cos^2 x}\right.$ is an even function $\}$

$$=2\int_0^{\pi/3}x\tan x\sec x dx$$

$$=2\left[x\int\tan x\sec x-\int\sec x dx\right]_0^{\pi/3}$$

$$=2[x\sec x-\log(\sec x+\tan x)]_0^{\pi/3}$$

$$=2\left(\frac{\pi}{3}\times 2-\log(2+\sqrt{3})\right)=2\left(\frac{2\pi}{3}-\log\tan\frac{5\pi}{12}\right)$$

18. (a) $\int_2^e\left[\frac{1}{\log x}-\frac{1}{(\log x)^2}\right]dx=\int_{\ln 2}^1\left[\frac{1}{t}-\frac{1}{t^2}\right]e^t dt$

$[\log x=t\Rightarrow x=e^t,\ dx=e^t dt]$

$$=\frac{e^t}{t}\Big|_{\ln 2}^1=\left[e-\frac{2}{\ln 2}\right]$$

20. (c) $\int_{-2}^1\frac{|x|}{x}dx=-\int_{-2}^0\frac{x}{x}dx+\int_0^1 dx=-2+1=-1$

21. (c) $\int_{-1}^1|x+1|dx=\int_{-1}^0(x+1)dx+\int_0^1(x+1)^2$

$$=\frac{x^2}{2}+x\Big|_{-1}^0+\frac{x^2}{2}+x\Big|_0^1=\frac{1}{2}+\frac{3}{2}=2$$

22. (b) $\int_0^3[x]dx=\int_0^1 0dx+\int_1^2 1dx+\int_2^3 2dx$

$$=x\Big|_1^2+2x\Big|_2^3=1+2=3$$

LECTURE

4

Estimation of Integral and other Definite Integrals

BASIC CONCEPTS

1. If $f(x)$ is continuous on $[a, b]$, m is the least and M is the greatest value of $f(x)$ on $[a, b]$, then $m(b-a) \le \int_a^b f(x)dx \le M(b-a)$

2. If a function $f(x)$ is continuous on the interval $[a, b]$, then there exists a point $c \in (a, b)$ such that $\int_a^b f(x)dx = f(c)\ (b-a)$, where $a < c < b$.

3. $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx$

$= \frac{n-1}{n} \times \frac{n-3}{n-2} \times \frac{n-5}{n-4} \ldots\ldots \frac{1}{2} \times \frac{\pi}{2}$, if n is even.

$= \frac{n-1}{n} \times \frac{n-3}{n-2} \times \frac{n-5}{n-4} \times \ldots\ldots\ldots \times \frac{2}{3}$ if n is odd.

4. gamma $(n) = \Gamma n = \int_0^\infty e^{-x} x^{n-1} dx : \Gamma n+1 = n\Gamma n$

gamma $\left(\frac{1}{2}\right) = \Gamma\frac{1}{2} = \int_0^\infty e^{-x} x^{-1/2} dx = \sqrt{\pi}, \Gamma\frac{9}{2} = \frac{7}{2}\Gamma\frac{7}{2}$

$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}, \Gamma\frac{7}{2} = \frac{5}{2}\Gamma\frac{5}{2}, \Gamma\frac{5}{2} = \frac{3}{2}\Gamma\frac{3}{2}$

1. $\int_{-\pi/2}^{\pi/2} \sin^m x \cos^n x\,dx = 0$ if m is odd

$= 2\int_0^{\pi/2} \sin^m x \cos^n x\,dx$ if m is even

2. $\int_0^{\pi} \sin^m x \cos^n x\,dx = 0$ if n is odd.

$= 2\int_0^{\pi/2} \sin^m x \cos^n x\,dx$ if n is even

3. $\int_0^{2\pi} \sin^m x \cos^n x\,dx = 0$ if m or n is odd

$= 4\int_0^{\pi/2} \sin^m x \cos^n x\,dx$, if m and n both are even

5. $\int_0^{\pi/2} \sin^m x \cos^n x\,dx = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$

OR $\int_0^{\pi/2} \sin^m x \cos^n x\,dx$

$= \frac{(m-1)(m-3).....(2 \text{ or } 1)(n-1)(n-3).....(2 \text{ or } 1)}{(m+n)(m+n-2).....(2 \text{ or } 1)} \times \left(1 \text{ or } \frac{\pi}{2}\right)$

If m and n are both even then only multiply by $\pi/2$.

6. **Leibnitz Rule for Differentiation Under Integral Sign:**
In this function $g(x)$ and $h(x)$ are defined on $[a, b]$ and differentiable at all points $x \in (a, b)$ and $f(t)$ is continuous on $[a, b]$, then $\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(t)dt\right]$

$= \frac{d}{dx}[h(x)].f[h(x)] - \frac{d}{dx}[g(x)].f[g(x)]$

$= f(h(x))\frac{d}{dx}(h(x)) - f(g(x))\frac{d}{dx}(g(x))$

7. $\int_0^{\pi/2} \log\sin x\,dx = \int_0^{\pi/2} \log\cos x\,dx = -\frac{\pi}{2}\log 2$

   $\int_0^{\pi/2} \log\operatorname{cosec} x\,dx = \int_0^{\pi/2} \log\sec x\,dx = \frac{\pi}{2}\log 2$

8. If $f(x) \geq 0\ \forall\ x \in [a, b]$ then $\int_a^b f(x)dx \geq 0$

9. If $f(x) \leq g(x) \forall\ x \in [a, b]$ then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$

10. $\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|\,dx$

11. $\int_a^b |f(x)|dx = \frac{(f(a))^2 + (f(b))^2}{2f'(x)}$

    If $f(x)$ is linear expression in x.

12. $\int_0^n [x]dx = \frac{n(n-1)}{2}$

13. $\int_a^b [f(x)]dx = \left(\frac{b-a}{2}\right)[f(a) + f(b) - 1]$

    $f(x)$ is linear expression in x.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)) FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Prove that $\int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\,dt + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\,dt = \frac{\pi}{4}$

Solution

$I = f(x)$ after integration and putting the limits

Now $f'(x) = \sin^{-1}\sqrt{(\sin^2 x)}\ .\ (2\sin x\cos x) - 0 + \cos^{-1}\sqrt{\cos^2 x}\ (-2\cos x\sin x) - 0$

$\therefore\ f'(x) = 0$. Hence $f(x) = c$ (constant)

In order to find c, the constant of integration, we evaluate $f(x)$ at $x = \pi/4$.

$$\therefore\ I = \int_0^{1/2} \sin^{-1}\sqrt{t}\,dt + \int_0^{1/2} \cos^{-1}\sqrt{t}\,dt$$

$$= \int_0^{1/2} (\sin^{-1}\sqrt{t} + \cos^{-1}\sqrt{t})dt$$

$$= \int_0^{1/2} \frac{\pi}{2}dt = \frac{\pi}{4} = c\ \ \therefore f(x) = \frac{\pi}{4}.$$

2. Prove that $\int_0^1 \tan^{-1}\left(\frac{1}{1-x+x^2}\right)dx = 2\int_0^1 \tan^{-1}x\,dx$.

   Hence or otherwise, evaluate the integral $\int_0^1 \tan^{-1}(1-x+x^2)dx$ ***[IIT-1998]***

Solution

$$I = \frac{\pi}{2} - \log 2 \qquad \ldots(1)$$

Now $\int_0^1 \tan^{-1}(1-x+x^2)dx$

$$= \int_0^1 \cot^{-1}\frac{1}{1-x+x^2}dx$$

$$= \int_0^1 \left(\frac{\pi}{2} - \tan^{-1}\frac{1}{1-x+x^2}\right)dx$$

$$= \left[\frac{\pi}{2}x\right]_0^1 - I$$

$$= \frac{\pi}{2} - \left(\frac{\pi}{2} - \log 2\right) = \log 2 \text{ by (1)}$$

3. Let $\frac{d}{dx}F(x) = \frac{e^{\sin x}}{x}, x > 0$. If $\int_1^4 \frac{2e^{\sin x^2}}{x}dx =$ $F(k) - F(1)$ then one of the possible values of k is ..... ***[IIT Re-ex.-1997]***

Solution

Ans $k = 16$.

$$\int_1^4 \frac{2e^{\sin x^2}}{x}dx = F(k) - F(1)$$

$$= [F(x)]_1^k$$

Put $x^2 = t\ \therefore\ 2x\,dx = dt$

$$\therefore\quad I = \int_1^{16} \frac{e^{\sin t}}{t}dt = [F(t)]_1^{16}\ \ \therefore\quad k = 16.$$

4. Evaluate $\int_0^{\pi/2} \frac{\cos^9 x}{\cos^3 x + \sin^3 x}dx$

[Roorkee-2001]

Solution

By properties of integral

$$2I = \int_0^{\pi/2} \frac{\cos^9 x + \sin^9 x}{\cos^3 x + \sin^3 x} dx$$

$$\text{or } I = \frac{1}{2}\int_0^{\pi/2} [\cos^6 x + \sin^6 x - \cos^3 x \sin^3 x]\, dx$$

$$= \frac{1}{2}\left[\frac{5.3.1}{6.4.2}\frac{\pi}{2} + \frac{5.3.1}{6.4.2}\frac{\pi}{2} - \frac{2.2}{6.4.2}\right]$$

$$= \frac{1}{2}\left[\frac{5}{16}\pi - \frac{1}{12}\right]$$

5. Evaluate: $\int_1^2 \frac{5x^2}{x^2+4x+3}$ ***[CBSE-2010]***

Solution

Now $\frac{5x^2}{x^2+4x+3} = 5 + \frac{-20x-15}{x^2+4x+3}$ (By long division)

$$= 5 - \frac{20x+15}{(x+1)(x+3)}$$

$$5 - \left[\frac{-5}{2(x+1)} + \frac{45}{2(x+3)}\right]$$

[By partial fraction]

$$5 + \frac{5}{2}.\frac{1}{x+1} - \frac{45}{2}.\frac{1}{x+3}$$

Hence

$$\int_1^2 \frac{5x^2}{x^2+4x+3} dx = \int_1^2 \left(5 + \frac{5}{2}.\frac{1}{x+1} - \frac{45}{2}.\frac{1}{x+3}\right) dx$$

$$= 5[x]_1^2 + \frac{5}{2}[\log|x+1|]_1^2 - \frac{45}{2}[\log|x+3|]_1^2$$

$$= 5(2-1) - \frac{5}{2}(\log 3 - \log 2) - \frac{45}{2}(\log 5 - \log 4)$$

$$= 5 + \frac{5}{2}\log\frac{3}{2} - \frac{45}{2}\log\frac{5}{4}$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Prove that: $\int_0^{\pi} \cos^3 x dx = 0$
2. Prove that:
$$\int_0^{\pi/2} (\sin x - \cos x)\log(\sin x + \cos x).dx = 0$$
3. Evaluate the integral $\int_0^{\pi/2} \sqrt{\sin x}\cos^5 x dx$.
4. Evaluate $\int_1^4 |x-1| + |x-2| + |x-3|\, dx$
[CBSE-Model question paper-I]
5. Evaluate the integral by using properties of definite integral, $\int_0^{1.5} [x^2]\, dx$
6. Evaluate $\int_{-\pi/2}^{\pi/2} (\sin|x| + + \cos|x|) dx$
[CBSE-2000]
7. Evaluate $\int_0^{\pi/2} \sin^4 x dx$
[PSB-88 C]

ANSWERS

3. $\frac{64}{231}$
4. 19/2
5. $2 - \sqrt{2}$
6. 4
7. $\frac{3\pi}{16}$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. Let $f(x) = \int_1^x \sqrt{2-t^2}\ dt$. Then the real roots of the equation $x^2 - f'(x) = 0$ are

[IIT Screening-2002]

(a) ± 1 (b) $\pm\frac{1}{\sqrt{2}}$

(c) $\pm\frac{1}{2}$ (d) 0 and 1

Solution

(a) $f(x) = \int_1^x \sqrt{2-t^2}\ dt$, $f'(x) = \sqrt{2-x^2} \cdot 1 - \sqrt{2-1} \cdot 0 = \sqrt{2-x^2}$

$\therefore\ x^2 - f'(x) = \sqrt{2-x^2}$

or $x^4 + x^2 - 2 = 0$

or $(x^2+2)(x^2-1) = 0$

$\therefore\ x = \pm 1$ (only real)

2. Let $f: (0, \infty) \to R$ and $F(x) = \int_0^x f(t)dt$. If $F(x^2) = x^2(1+x)$, then $f(4)$ equals

[IIT Sc.-2001]

(a) 5/4 (b) 7

(c) 4 (d) 2

Solution

(c) By definition of $F(x)$ we have

$$F(x^2) = \int_0^{x^2} f(t)\,dt = x^2 + x^3 \quad \text{(given)}$$

differentiate both sides $f(x^2) \cdot 2x + 0 = 2x + 3x^2$

Put $x = 2$

$4f(4) = 16$ or $f(4) = 4$

3. The slope of the tangent to the curve $y = \int_0^{x^2} \frac{dt}{1+t^2}$ at point $x = 1$ is

[ICS (Pre.)-2004]

(a) 2 (b) 1

(c) 1/2 (d) 1/4

Solution

(b) $y = \int_0^{x^2} \frac{dt}{1+t^2} \Rightarrow \frac{dy}{dx} = \frac{1}{1+x^4}(2x) - 0 = \frac{2x}{1+x^4}$

$\therefore$ required slope $= \left(\frac{dy}{dx}\right)_{x=1} = \frac{2}{1+1} = 1$

4. If $y(x) = \int_{\pi^2/16}^{x^2} \frac{(\cos x)\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta$, then $y(\pi) =$

[IIT-2005]

(a) π (b) 2π

(c) 0 (d) None of these

Solution

(b) We have $y(x) = \int_{\pi^2/16}^{x^2} \frac{(\cos x)\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta$

$= \cos x \int_{\pi^2/16}^{x^2} \frac{\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta$

[$\because \cos x$ is independent of θ]

$$\Rightarrow \frac{dy}{dx} = -\sin x \int_{\pi^2/16}^{x^2} \frac{\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta + \cos x \frac{d}{dx}\left[\int_{\pi^2/16}^{x^2} \frac{\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta\right]$$

$$= -\sin x \int_{\pi^2/16}^{x^2} \frac{\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta + \cos x \left[\frac{\cos x}{1+\sin^2 x} \cdot 2x - 0\right]$$

(using Leibnitz theorem)

$$\Rightarrow \left|\frac{dy}{dx}\right|_{x=\pi}$$

$$= -\sin x \int_{\pi^2/16}^{x^2} \frac{\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}} d\theta + \frac{\cos^2\pi \cdot 2\pi}{1+\sin^2\pi}$$

$= 0 \times (\ ) + 2\pi = 2\pi$

5. $f(x)$ is a continuous function for all real values of x and satisfies $\int_0^x f(t)dt = \int_x^1 t^2 f(t)dt + \frac{x^{16}}{8} + \frac{x^6}{3} + a$, then the value of '$a$' is equal to

(a) –1/24 (b) 17/168
(c) 1/7 (d) None of these

Solution

(d) $\int_0^x f(t)dt = \int_x^1 t^2 f(t)dt + \frac{x^{16}}{8} + \frac{x^6}{3} + a$ .. (i)

For $x = 1$

$\int_0^1 f(t)dt = \frac{1}{8} + \frac{1}{3} + a$ ..... (ii)

On differentiating equation (i) with respect to x

$f(x) = -x^2 f(x) + 2x^{15} + 2x^5$

$\Rightarrow f(x) = \frac{2(x^{15} + x^5)}{1 + x^2}$

Now, from equation (ii)

$2\int_0^1 \left(\frac{x^{15} + x^5}{1 + x^2}\right)dx = \frac{1}{8} + \frac{1}{3} + a \Rightarrow a = -\frac{167}{840}$

6. If $I_n = \int_0^{\pi/2} x^n \sin x dx$, then $I_4 + 12 I_2$ is equal to

[DCE-97]

(a) 4π (b) $(\pi/2)^2$
(c) $3(\pi/2)^3$ (d) $4(\pi/2)^3$

Solution

(d) $I_n = \int_0^{\pi/2} x^n \sin x dx$

$= \left[x^n(-\cos x)\right]_0^{\pi/2} - n\int_0^{\pi/2} x^{n-1}(-\cos x)dx$

$= n\left[(x^{n-1}\sin x)_0^{\pi/2} - (n-1)\int_0^{\pi/2} x^{n-2}\sin x dx\right]$

$= n\left[\left(\frac{\pi}{2}\right)^{n-1} - (n-1)I_{n-2}\right]$

$\therefore \quad I_4 = 4\left(\frac{\pi}{2}\right)^3 - 12I_2$

7. $\int_0^{\pi/2} (\sin x)^{\cos x}(\cos x \cot x - \log(\sin x)^{\sin x})dx$ equals

[IIT-JEE-2006]

(a) 0 (b) 1
(c) 1/2 (d) π

Solution

(b) $I = \int_0^{\pi/2} \frac{d}{dx}(\sin x)^{\cos x} dx = 1$

8. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}}dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}}dx$. Then which one of the following is true?

[AIEEE-2008]

(a) $I > \frac{2}{3}$ and $J < 2$ (b) $I > \frac{2}{3}$ and $J > 2$
(c) $I < \frac{2}{3}$ and $J < 2$ (d) $I < \frac{2}{3}$ and $J > 2$

Solution

(c) In the interval of integration $\sin x < x$

$I = \int_0^1 \frac{\sin x}{\sqrt{x}}dx < \int_0^1 \frac{x}{\sqrt{x}}dx = \int_0^1 \sqrt{x}dx$

$= \left[\frac{2}{3}x^{3/2}\right]_0^1 = \frac{2}{3} \therefore I < \frac{2}{3}$

Also $J = \int_0^1 \frac{\cos x}{\sqrt{x}}dx < \int_0^1 \frac{1}{\sqrt{x}}dx = \left[2\sqrt{x}\right]_0^1 = 2$

$\therefore \quad J < 2$

9. $\int_2^4 \log[x]dx$ is .....

[Orissa JEE-2008]

(a) log 2 (b) log 3
(c) log 5 (d) None of these

Solution

(d) $\int_2^4 \log[x]dx = \int_2^3 \log[x]dx + \int_3^4 \log[x]dx$

$= \int_2^3 \log 2dx + \int_3^4 \log[3]dx \quad \left[\because [x] = \begin{matrix} 2, 2 \le x < 3 \\ 3, 3 \le x < 4 \end{matrix}\right]$

$= (\log 2)(3-2) + (\log 3)(4-3) = \log 3 + \log 2$
$= \log 6.$

10. If $I_{10} = \int_0^{\pi/2} x^{10} \sin x \, dx$. Then the value of $I_{10} + 90 I_8$ is: ***[Orissa JEE-2008]***

(a) $10(\pi/2)^6$ (b) $10(\pi/2)^9$
(c) $10(\pi/2)^7$ (d) None of these

Solution

(b) $I_{10} = \int_0^{\pi/2} x^{10} \sin x \, dx$

$= \left[\left[x^{10}\int \sin x dx\right]_0^{\pi/2} - \int_0^{\pi/2}\left[10x^9 \int \sin x dx\right]\right] dx$

$= \left[-x^{10}\cos x\right]_0^{\pi/2} - 10\int_0^{\pi/2} x^9(-\cos x)dx$

$= \left(-\left(\frac{\pi}{2}\right)^{10}\cos\frac{\pi}{2}\right) - (-0) + 10$

$\left[\left[x^9 \int \cos x dx\right]_0^{\pi/2} - \int_0^{\pi/2} 9x^8(\sin x)\right]$

$= 10\left[x^9 \sin x\right]_0^{\pi/2} - 90\int_0^{\pi/2} x^8 \sin x dx$

$\Rightarrow I_{10} = 10\left(\frac{\pi}{2}\right)^9 \sin\frac{\pi}{2} - 90I_8$

$\Rightarrow I_{10} + 90I_8 = 10\left(\frac{\pi}{2}\right)^9$

11. If $I = \int_0^{\pi/2} \cos(\sin x)dx$,

$J = \int_0^{\pi/2} \sin(\cos x)dx$, $K = \int_0^{\pi/2} \cos x \, dx$ then:

[Orissa JEE-2008]

(a) $I > J > K$ (b) $J > K > I$

(c) $K > J > I$ (d) $I > K > J$

Solution

(d) $\because x > \sin x$ for $x \in (0, \pi/2]$

or $\cos x < \cos(\sin x)$

Integrating both sides,

$\Rightarrow \int_0^{\pi/2} \cos x dx < \int_0^{\pi/2} \cos(\sin x)dx$

$\Rightarrow K < 1$ ..... (i)

$\because x < \frac{\pi}{2} - \cos x$ for $x \in \left[0, \frac{\pi}{2}\right)$

$\cos x > \cos\left(\frac{\pi}{2} - \cos x\right) = \sin(\cos x)$

or $\cos x > \sin(\cos x)$

$\Rightarrow \int_0^{\pi/2} \cos x dx > \int_0^{\pi/2} \sin(\cos x)dx$ Integrating both sides,

$\Rightarrow K > J$ ..... (ii)

from (i) and (ii), we get $I > K > J$.

12. The maximum value of $f(x)$, where $f(x) = \int_0^x \sin\{x(1-x)\}dx$ occurs at which one of the following points? ***[UPSC-2007]***

(a) $x = 0$ (b) $x = 1$

(c) $x = -1$ (d) None of these

Solution

(b) $f(x) = \int_0^x \sin\{t(1-t)\}dt$

$f'(x) = \sin x (1 - x)$

$\Rightarrow f'(x) = 0$

$\Rightarrow \sin x (1 - x) = 0$

$\Rightarrow x(1 - x) = 0, \pm\pi, \pm 2\pi$

Thus $x - x^2 = n\pi$

$\Rightarrow x^2 - x + n\pi = 0$

$\therefore x = \frac{1 \pm \sqrt{1 - 4n\pi}}{2}$

For real roots to exist $1 \geq 4n\pi \Rightarrow n \leq \frac{1}{4\pi}$.

Thus $n = 0$ or negative values.

$f''(x) = \cos\{x(1-x)\} \cdot (1 - 2x)$

$f''(0) = 1 > 0$, $f''(1) = -1 < 0$

But $f(x)$ is maximum at $x = 1$.

13. Least value of the function $f(x) = \int_0^x (1 - \cos t)dt$ on the interval $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is:

(a) $\frac{3\pi}{2} + 1$ (b) $\frac{\pi}{2} - 1$

(c) $\frac{\pi}{2} + 1$ (d) None of these

Solution

(b) $f'(x) = 1 - \cos x$ always positive, so $f(x)$ is an increasing function so least value of $f(x)$ on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is $f\left(\frac{\pi}{2}\right)$.

$$\therefore \quad f\left(\frac{\pi}{2}\right)=\int_0^{\pi/2}(1-\cos t)dt$$

$$=[t-\sin t]_0^{\pi/2}=\left(\frac{\pi}{2}-1\right)$$

14. Prove that $\int_0^{\pi}\frac{x\sin 2x\sin(\pi/2\cos x)}{2x-\pi}dx=\frac{8}{\pi^2}$

[IIT-1991]

Solution

Putting $x=\frac{\pi}{2}+t \therefore dx=dt$ and $2x-\pi=2t$

Adjust the limits $\therefore$ sin $2x$ = sin $(\pi + 2t)$ = $-$ sin $2t$ cos x = $-$sin t

$$\therefore \quad I=\int_{-\pi/2}^{\pi/2}\frac{\left(\frac{\pi}{2}+t\right)(-\sin 2t)\sin\left(-\frac{\pi}{2}\sin t\right)}{2t}\text{ ..(a)}$$

$$=\int_{-\pi/2}^{\pi/2}\left(\frac{\pi}{4}\right)\frac{1}{t}\sin 2t\sin\left(\frac{\pi}{2}\sin t\right)dt+$$

$$\frac{1}{2}\int_{-\pi/2}^{\pi/2}\sin 2t\sin\left(\frac{\pi}{2}\sin t\right)dt$$

$$=0+2\,.\,\frac{1}{2}\int_0^{\pi/2}2\sin t\cos t\sin\left(\frac{\pi}{2}\sin t\right)dt$$

[using the property of definite integrals]

$$=2\int_0^{\pi/2}\left(\frac{2}{\pi}z\right)\sin z.\frac{2}{\pi}dz,$$

Putting $\frac{\pi}{2}\sin t=z \therefore \frac{\pi}{2}\cos t dt=dz$

$$=\frac{8}{\pi^2}\int_0^{\pi/2}z\sin z dz$$

$$=\frac{8}{\pi^2}[z(-\cos z)+\sin z]_0^{\pi/2}$$

$$=\frac{8}{\pi^2}[(0+1)-0]=\frac{8}{\pi^2}$$

15. $f(x)$ = min $\{x + 2, 1, 2 - x\}$, then $\int_{-2}^{2}f(x)dx$ is equal to

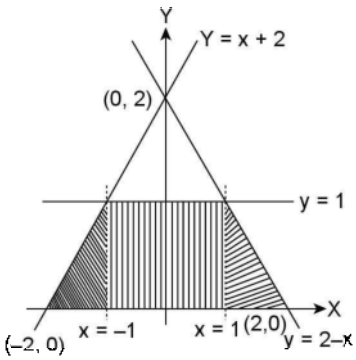
(a) 1 (b) 2

(c) 3 (d) 0

Solution

(c) $\therefore \quad \int_{-2}^{4}f(x)dx$

$$=\int_{-2}^{-1}(x+2)dx+\int_{-1}^{1}dx+\int_1^2(2-x)dx$$



$$=\frac{1}{2}+2+\frac{1}{2}=3$$

16. Let $f(x)$ = Maximum $[x^2, (1 - x)^2, 2x(1 - x)]$ where $0 \le 1 \le 1$. The area of the region bounded by the curves $y = f(x)$, x – axis $x = 0$ and $x = 1$ is

(a) 17/27 (b) 14/27

(c) 19/27 (d) None of these

Solution

(a) $f(x)$ = Max $[x^2, (1 - x)^2, 2x (1 - x)]$

$\therefore \quad f(x) = x^2$

If $x^2 > (1 - x)^2, x^2 > 2x (1 - x)$

If $x > \frac{2}{3} = (1 - x)^2$

if $(1 - x)^2 > x^2, (1 - x)^2 > 2x (1 - x)$ i.e.,

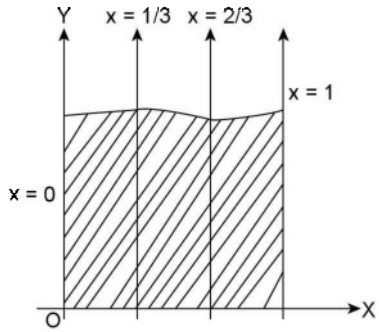
if $x < \frac{1}{3}$

and $f(x) = 2x (1 - x)$

if $2x (1 - x) > x^2, 2x (1 - x) > (1 - x)^2$ i.e.,

if $\frac{1}{3} < x < \frac{2}{3}$

$$\text{Thus } f(x)=\begin{cases}(1-x)^2 & , \quad 0\le x<\frac{1}{3}\\ 2x(1-x) & , \quad \frac{1}{3}\le x<\frac{2}{3}\\ x^2 & , \quad \frac{2}{3}\le x\le 1\end{cases}$$



The required area

$$= \int_0^{1/3} (1-x)^2\,dx + \int_{1/3}^{2/3} 2x(1-x)\,dx + \int_{2/3}^{1} x^2\,dx$$

$$= \left|\frac{(1-x)^2}{-3}\right|_0^{1/3} + \left(x^2 - \frac{2x^3}{3}\right)_{1/3}^{2/3} + \left|\frac{x^3}{3}\right|_{2/3}^{1}$$

$$= \frac{1}{3} - \frac{8}{81} + \frac{3}{9} - \frac{2}{3}\left(\frac{9}{27} - \frac{1}{27}\right) + \frac{1}{3}\left(1 - \frac{8}{27}\right)$$

$$= \frac{19}{81} + \frac{13}{81} + \frac{19}{81} = \frac{51}{81} = \frac{17}{27}$$

17. $\int_0^{\infty} \left[\frac{[x+1]}{e^x}\right] dx$ (where [.] denotes greatest integer function) is equal to

(a) 0 (b) 1

(c) ∞ (d) None of these

Solution

(a) $\frac{x+1}{e^x} < 1$ ..... (i)

$\Rightarrow \frac{[x+1]}{e^x} < 1$ ..... (ii)

(as $[x+1] < x+1$)

$\therefore \left[\frac{[x+1]}{e^x}\right] = 0$ [from equation (i) and (ii)]

18. The value of $\int_{-1}^{1} [x[1+\sin \pi x]+1]\,dx$ is, ([.] denotes the greatest integer)

(a) 2 (b) 0

(c) 1 (d) None of these

Solution

(a) $\int_{-1}^{1} [x[1+\sin \pi x]+1]\,dx$

$= \int_{-1}^{0} [x[1+\sin \pi x]+1]\,dx + \int_{0}^{1} [x[1+\sin \pi x]+1]\,dx$

Now, $-1 < x < 0 \Rightarrow [1 + \sin \pi x] = 0$

$0 < x < 1 \Rightarrow [1 + \sin \pi x] = 1$

$\Rightarrow [x\,[1 + \sin \pi x] + 1] = 1$

$\therefore \int_{-1}^{1} [x[1+\sin \pi x]+1]\,dx = 2$

19. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'\left(\frac{1}{4}\right) = 0$. Then,

[IIT JEE-2008]

(a) $f''(x)$ vanishes at least twice on [0, 1]

(b) $f'\left(\frac{1}{2}\right) = 0$

(c) $\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right) \sin x\,dx = 0$

(d) $\int_0^{1/2} f(t) e^{\sin \pi t}\,dt = \int_{1/2}^{1} f(1-t) e^{\sin \pi t}\,dt$

Solution

(a,b,c,d) $f(x) = f(1-x)$

changing x to $x + \frac{1}{2}$

we have $f\left(x + \frac{1}{2}\right) = f\left(\frac{1}{2} - x\right)$

giving immediately that $f\left(\frac{1}{2} + x\right)$ is an even function. The product of an even function and an odd function is odd. Then

$$\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right) \sin x\,dx = 0$$

(from properties of definite integral)

Thus choice (c) is correct.

Now

$$\int_{1/2}^{1} f(1-t) e^{\sin \pi t}\,dt$$

$$= -\int_{1/2}^{0} f(u) e^{\sin \pi u}\,du \quad (put\ 1-t = u)$$

$$= \int_{0}^{1/2} f(u) e^{\sin \pi u}\,du$$ Thus choice (a) is correct.

Again $f(x) = f(1-x)$. On differentiation yields

$f'(x) = -f'(1-x)$

We have $f'\left(\frac{1}{2}\right) = -f'\left(\frac{1}{2}\right) \Rightarrow f'\left(\frac{1}{2}\right) = 0$

Thus choice (b) is correct.

$f'\left(\frac{1}{4}\right)=0$ (given)

$f'(-x)=-f'(x)$ gives

$f'\left(-\frac{1}{4}\right)=-f'\left(\frac{1}{4}\right)\Rightarrow f'\left(\frac{3}{4}\right)=0$

Thus $f'(x)$ vanishes at x =1/4, 1/2 and 3/4.

Applying Rolle's theorem to $f'(x)$ we have $f''(x)$ vanishes at least once in each of the intervals, $\left(\frac{1}{4},\frac{1}{2}\right)$ and $\left(\frac{1}{2},\frac{3}{4}\right)$

As $\left(\frac{1}{4},\frac{1}{2}\right)$ and $\left(\frac{1}{2},\frac{3}{4}\right)\subseteq[0,1]$

We can concluded that $f''(x)$ vanishes at least twice on [0, 1]. Thus choice (a) is also correct.

20. Consider the function f: $(-\infty, \infty)\to(-\infty, \infty)$ defined by $f(x)=\frac{x^2-ax+1}{x^2+ax+1}, 0<a<2$.

Let $g(x)=\int_0^{e^x}\frac{f'(t)}{1+t^2}dt$. Which of the following is true? ***[IIT JEE-2008]***

(a) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$

(b) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$

(c) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$

(d) $g'(x)$ is does not change sign on $(-\infty, \infty)$.

Solution

(b) We have $g'(x)=\frac{f'(e^x)e^x}{1+e^{2x}}$

Now $f'(e^x)=\frac{2a.(e^{2x}-1)}{(e^{2x}+ae^x+1)^2}$

It is seen from the above that $f'(e^x)$ and so $g'(x)$ is positive on $(0, \infty)$ and negative on $(-\infty, 0)$.

21. Let f be a non-negative function defined on the interval [0, 1]. If $\int_0^x\sqrt{1-(f'(t))^2}\,dt=\int_0^x f(t)dt$, $0\le x\le 1$ and $f(0)=0$, then ***[IIT-2009]***

(a) $f\left(\frac{1}{2}\right)<\frac{1}{2}$ and $f\left(\frac{1}{3}\right)>\frac{1}{3}$

(b) $f\left(\frac{1}{2}\right)>\frac{1}{2}$ and $f\left(\frac{1}{3}\right)>\frac{1}{3}$

(c) $f\left(\frac{1}{2}\right)<\frac{1}{2}$ and $f\left(\frac{1}{3}\right)<\frac{1}{3}$

(d) $f\left(\frac{1}{2}\right)>\frac{1}{2}$ and $f\left(\frac{1}{3}\right)<\frac{1}{3}$

Solution

(c) $f'=\pm\sqrt{1-f^2}$

$\Rightarrow f(x)=\sin x$ or $f(x)=-\sin x$ (not possible)

$\Rightarrow f(x)=\sin x$. Also $x>\sin x\ \forall\ x>0$

22. Let f: $R\to R$ be a continuous function which satisfies $f(x)=\int_0^x f(t)dt$. Then the value of $f(\ln 5)$ is ***[IIT-2009]***

(a) 0 (b) 1

(c) 2 (d) None of these

Solution

(a) $f(x)=\int_0^x f(t)dt\Rightarrow f(0)=0$

also, $f'(x)=f(x), x>0\Rightarrow f(x)=ke^x, x>0$

$\because$ $f(0)=0$ and $f(x)$ is continuous

$\Rightarrow f(x)=0\ \forall\ x>0$

$\therefore$ $f(\ln 5)=0$.

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. The value of $\int_0^{2\pi}|\sin^3\theta|\,d\theta$ is ***[Roorkee qualifying-1998]***

(a) 0

(b) 3/8

(c) 8/3

(d) π

2. $\int_{-1}^{1} \sin^3 x \cos^2 x \, dx =$ ***[MNR-1991; UPSEAT-2000]***

(a) 0 (b) 1
(c) 1/2 (d) 2

3. The value of the definite integral $\int_0^1 \frac{x dx}{x^3+16}$ lies in the interval $[a, b]$. The smallest such interval is:

(a) [0, 1/17] (b) [0, 1]
(c) [0, 1/27] (d) None of these

4. Let a, b, c be non - zero real numbers such that $\int_0^1 (1 + \cos^8 x)(ax^2 + bx \quad c)\, dx = \int_0^2 (1 + \cos^8 x)(ax^2 + bx \quad c)\, dx$. Then the quadratic equation $ax^2 + bx \quad c = 0$ has ***[IIT-81; CEE-93]***

(a) No root in (0, 2)
(b) At least one root in (0, 2)
(c) A double root in (0, 2)
(d) None of these

5. $\int_0^{\pi} \frac{dx}{1-2a\cos x + a^2} =$ ***[CEE-1993]***

(a) $\frac{\pi}{2(1-a^2)}$ (b) $\pi(1-a^2)$

(c) $\frac{\pi}{1-a^2}$ (d) None of these

6. What is the value of $\int_0^{\pi/2} \cos^8 x dx$? ***[NDA-2007]***

(a) $\frac{35\pi}{256}$ (b) $\frac{70}{256}$

(c) $\frac{16}{35}$ (d) $\frac{8\pi}{35}$

7. The correct evaluation of $\int_0^{\pi/2} \left|\sin\left(x - \frac{\pi}{4}\right)\right| dx$ is ***[MPPET-93]***

(a) $2 + \sqrt{2}$ (b) $2 - \sqrt{2}$
(c) $-2 + \sqrt{2}$ (d) 0

8. The points of extremum of $\int_0^{x^2} \frac{t^2-5t+4}{2+e^t}$ at are ***[IIT-Screening]***

(a) $x = -2$ (b) $x = 1$
(c) $x = 0$ (d) All of the above

9. The value of the integral $\int_{-\pi/4}^{\pi/4} \sin^4 x dx$ is ***[IIT Screening; MPPET-03]***

(a) 3/2 (b) –8/3
(c) 3/8 (d) 8/3

10. $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16\sin 2x} dx =$ ***[IIT-1983]***

(a) $\frac{1}{20}\log 3$ (b) $\log 3$

(c) $\frac{1}{20}\log 5$ (d) None of these

11. $\int_{-\pi/2}^{\pi/2} \sin^2 x \cos^2 x(\sin x - \cos x)\, dx =$ ***[EAMCET-1992]***

(a) 2/15 (b) 4/15
(c) 6/15 (d) 8/15

12. $\int_0^{\pi} \sin^5(x/2) dx$ equals ***[Kurukshetra CEE-1996]***

(a) 16/15 (b) 32/15
(c) 8/15 (d) 5/6

13. $\int_0^{\infty} \frac{x \ln x dx}{(1+x^2)^2} =$ ***[AMU-2000]***

(a) 0 (b) 1
(c) ∞ (d) None of these

14. If $I_n = \int_1^e (\log x)^n dx$, then $I_n + n I_{n-1}$ is equal to

(a) $1/e$ (b) e
(c) $e - 1$ (d) None of these

15. $\int_{-1}^{1/2} \frac{e^x(2-x^2)dx}{(1-x)\sqrt{1-x^2}}$ is equal to

(a) $\frac{\sqrt{e}}{2}(\sqrt{3}+1)$ (b) $\sqrt{3e}/2$

(c) $\sqrt{3e}$ (d) $\sqrt{e/3}$

16. If $I_{m,n} = \int_0^1 x^m (\log x)^n dx$, then $I_{m,n}$ is equal to

(a) $\frac{n}{n+1} I_{m,n-1}$ (b) $-\frac{m}{n+1} I_{m,n-1}$

(c) $-\frac{n}{m+1} I_{m,n-1}$ (d) $\frac{m}{n+1} I_{m,n-1}$

17. The integral $\int_0^{\pi} \frac{dx}{3+2\sin x+\cos x}$ evaluates to **[MPPET-07]**

(a) π (b) $\pi/4$
(c) 2π (d) None of these

18. If $\int_0^{t^2} xf(x)dx = \frac{2t^5}{5}, t>0$, then $f\left(\frac{4}{25}\right)$ = **[IIT (Screening)-04]**

(a) 2/5 (b) 5/2
(c) –2/5 (d) 1

19. $\lim_{x\to 0} \frac{\int_0^{x^2} \tan\sqrt{t}\,dt}{x^3}$ is equal to

(a) 2/3 (b) 3/2
(c) 1 (d) None of these

20. $\int_{-1}^{1} \sin^{11} x\,dx$ is equal to **[MNR-1995]**

(a) $\frac{10}{11}.\frac{8}{9}.\frac{6}{7}.\frac{4}{5}.\frac{2}{3}$ (b) $\frac{10}{11}.\frac{8}{9}.\frac{6}{7}.\frac{4}{5}.\frac{2}{3}.\frac{\pi}{2}$
(c) 1 (d) 0

21. $\int_0^{\pi} x\sin^3 x\,dx$ = **[CEE-1993]**

(a) $4\pi/3$ (b) $2\pi/3$
(c) 0 (d) None of these

22. The value of $\int_1^3 \sqrt{3+x^3}\,dx$ lies in the interval **[CEE-1993]**

(a) (1, 3) (b) (2, 30)
(c) $(4, 2\sqrt{30})$ (d) None of these

23. $\int_0^1 \frac{d}{dx}\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right]dx$ is equal to

(a) 0 (b) π
(c) $\pi/2$ (d) $\pi/4$

24. $I_n = \int_0^{\pi/4} \tan^n x\,dx$, then $\lim_{n\to\infty} n\,[I_n + I_{n-2}]$ equals **[AIEEE-2002]**

(a) 1/2 (b) 1
(c) ∞ (d) 0

25. If $\int_0^x f(t)dt = x + \int_x^1 tf(t)dt$, then the value of $f(1)$ is **[IIT-98]**

(a) 1/2 (b) 0
(c) 1 (d) –1/2

26. If vlauc(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2}dx$ is (arc)

[IIT (1)-2010]

(a) $\frac{22}{7}-\pi$ (b) $\frac{2}{105}$
(c) 0 (d) $\frac{71}{15}-\frac{3\pi}{2}$

27. Let f be a real-valued function defined on the interval (0, ∞) by $f(x) = \ell nx + \int_0^x \sqrt{1+\sin t}\,dt$.

Then which of the following statement(s) is (are) true? **[IIT (1)-2010]**

(a) $f''(x)$ exists for all $x\in(0,\infty)$
(b) $f''(x)$ exists for all $x\in(0,\infty)$ and f' is continuous on $(0,\infty)$, but not differentiable on $(0,\infty)$
(c) there exists $\alpha>1$ such that $|f'(x)|<|f(x)|$ for all $x\in(\alpha,\infty)$
(d) there exists $\beta>0$ such that $|f(x)|+|f'(x)|\le\beta$ for all $x\in(0,\infty)$

SOLUTIONS

1. (c) $\int_0^{2\pi}|\sin^3 0|d0 = 2\int_0^{\pi}|\sin^3 0|d0$

$\left(\int_0^{2a} f(x) = \int_0^a f(x)dx; f(2a-x)=f(x)\right)$

$= 4\int_0^{\pi/4}\sin^3 0 d0 = 4.\frac{2}{3} = \frac{8}{3}$

(using walli's formula)

2. (a) $\because$ $\sin^3 x\cos^2 x$ is odd function

$\int_{-1}^{1}\sin^3 x\cos^2 x dx = 0$

3. (a) Consider

$f(x) = \frac{x}{x^3+16}; f'(x) = \frac{16-2x^2}{(x^3+16)^2}$

$\because$ $\forall x\in(0,1), f'(x)>0$, $f(x)$ is increasing.

$$f(0)(1-0) \le \int_0^1 f(x)dx \le f(1)[1-0]$$

$$0 \le \int_0^1 f(x)dx \le \frac{1}{17}$$

4. (b) $\int_0^1 (1+\cos^8 x)(ax^2+bx+c)dx$

$$= \int_0^2 (1+\cos^8 x)(ax^2+bx+c)dx$$

Let us consider a function

$$F(x) = \int_0^x (1+\cos^8 x)(ax^2+bx+c)$$

Here $F(1) = F(2)$ (given)

$\therefore$ $F'(x)$ will have atleast one root in (1, 2),

$\therefore$ $ax^2 + bx + c$ has a root in (1, 2) or (0, 20 (By rolli's theorem)

5. (c) $I = \int_0^{\pi} \frac{dx}{1+a^2-2a\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)}$

$$= \int_0^{\pi} \frac{\left(1+\tan^2\frac{x}{2}\right)}{(1+a^2)\left(1+\tan^2\frac{x}{2}\right)-2a+2a\tan^2\frac{x}{2}}$$

put $\tan\frac{x}{2} = t$,

$$I = \int_0^{\infty} \frac{2dt}{(1+a^2-2a)+t^2(1+a^2+2a)}$$

$$\frac{1}{(1+a)^2}\int_0^{\infty} \frac{2dt}{\left(\frac{1-a}{1+a}\right)^2+t^2}$$

$$= \frac{2}{(1+a)^2}\left[\tan^{-1}\left(\frac{t(1+a)}{1-a}\right)\right]_0^{\infty} \cdot \frac{1}{\left(\frac{1-a}{1+a}\right)}$$

$$= \frac{2}{1-a^2}\left[\tan^{-1}\infty - \tan^{-1}0\right] = \frac{2}{1-a^2}\cdot\frac{\pi}{2} = \frac{\pi}{1-a^2}$$

6. (a) $\int_0^{\pi/2} \cos^8 x dx$

$$= \frac{8-1}{8}\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}$$

$$= \frac{105\pi}{768} = \frac{35\pi}{256}$$

7. (b) $\int_0^{\pi/2} \left|\sin\left(x-\frac{\pi}{4}\right)\right| dx =$

$$-\int_0^{\pi/4} \sin\left(x-\frac{\pi}{4}\right)dx + \int_{\pi/4}^{\pi/2} \sin\left(x-\frac{\pi}{4}\right)dx$$

$$= +\cos\left(x-\frac{\pi}{4}\right)\Big|_0^{\pi/4} - \left(\cos\left(x-\frac{\pi}{4}\right)\right)\Big|_{\pi/4}^{\pi/2}$$

$$= 1-\frac{1}{\sqrt{2}} - \left(\frac{1}{\sqrt{2}}-1\right)$$

$$= 2-\sqrt{2}$$

8. (d) $F(x) = \int_0^{x^2} \left(\frac{t^2-5t+4}{2+e^t}\right)dt$; for point of extremum, $F'(x) = 0$

$$\frac{(x^2)^2-5x^2+4}{2+e^{x^2}} \times 2x - 0 = 0,$$

$x = 0, 1, -1, 2, -2$

9. (b) $\int_{\pi/4}^{\pi/4} [\text{cosec}^2 x(1+\cos^2 x)dx]$

$$= 2\left(\int_0^{\pi/4} \text{cosec}^2 x dx + \int_0^{\pi/4} \text{cosec}^2 x \cot^2 x dx\right)$$

$$= \left[-\cot x - \frac{\cot^3 x}{3}\right]_{-\pi/4}^{\pi/4} = \left[-1-\frac{1}{3}-\left(1+\frac{1}{3}\right)\right] = \frac{-8}{3}$$

10. (a) $\int_0^{\pi/4} \frac{\sin x+\cos x}{9+16\sin 2x}dx$;

$d(\sin x - \cos x) = (\cos x + \sin x)dx$

also $(\sin x - \cos x)^2 = 1 - \sin 2x$

$$I = \int_{-1}^0 \frac{dt}{9+16(1-t^2)}$$

$$= \int_{-1}^0 \frac{dt}{25-16t^2}$$

$$= \frac{1}{16}\int_{-1}^0 \frac{dt}{\frac{25}{16}-t^2}$$

$$= \frac{1}{16}\times\frac{1}{2.\frac{5}{4}}\log\left(\frac{5+4t}{5-4t}\right)\Bigg|_{-1}^0$$

$$= \frac{1}{40}\log 9 = \frac{1}{40}\log 3^2 = \frac{1}{20}\log 3$$

11. (a) $I = \int_{-\pi/2}^{\pi/2} \sin^2 x\cos^2 x(\sin x - \cos x)dx$

$$= \int_{-\pi/2}^{\pi/2} \sin^3 x\cos^2 x dx + \int_{-\pi/2}^{\pi/2} \sin^2 x\cos^3 x dx$$

$$= 0 + \frac{1.2}{5.3.1} = \frac{2}{15}$$

($\because$ $\sin^3 x \cos^2 x$ is odd function)

12. (a) $\int_0^{\pi}\sin^5\left(\frac{x}{2}\right)dx = 2\int_0^{\pi}\sin^5 t\,dt \quad (t = x/2)$

$= \frac{24}{5}.\frac{2}{3} = \frac{16}{15}$

13. (a) $I = \int_0^{\infty}\frac{x\ln x\,dx}{(1+x^2)^2}dx = \int_{\infty}^{1}\frac{\frac{1}{t}\ln\left(\frac{1}{t}\right)}{\left(1+\frac{1}{t^2}\right)^2}\left(\frac{-1}{t^2}\right)dt$

$I = \int_{\infty}^{0}\frac{t\ln(t)}{(1+t^2)^2} = -I \Rightarrow I = 0$

14. (b) $I_n = \int_1^e(\log x)^n dx$

$= (\log x)^n\int_1^e dx - \int_1^e n(\log x)^{n-1}.\frac{1}{x}x\,dx$

$= \left[(\log x)^n x\right]_1^e - n\int_1^e(\log x)^{n-1}dx$

$I_n = e - nI_{n-1} \Rightarrow I_n + nI_{n-1} = e$

15. (c) $\int_{-1}^{1/2}\frac{e^x(2-x^2)dx}{(1-x)\sqrt{1-x^2}}$

$= \int_{-1}^{1/2}e^x\left(\frac{1}{(1-x)\sqrt{1-x^2}} + \frac{1-x^2}{(1-x)\sqrt{1-x^2}}\right)dx$

$I = \int_{-1}^{1/2}e^x\left(\sqrt{\frac{1+x}{1-x}} + \frac{1}{(1-x)\sqrt{1-x^2}}\right)dx$

Also

$\frac{d}{dx}\left(\sqrt{\frac{1+x}{1-x}}\right) = \frac{1}{(1-x)\sqrt{1-x^2}};\ I = e^x\sqrt{\frac{1+x}{1-x}}\Big|_{-1}^{1/2}$

$= e^{1/2}\sqrt{\frac{1+\frac{1}{2}}{1-\frac{1}{2}}} = e^{-1}.0 = \sqrt{3e}$

16. (c) $I_{m,n} = \int_0^1 x^m(\log x)^n dx$

$= (\log x)^n\frac{x^{m+1}}{m+1}\Big|_0^1 - \int n(\log x)^{n-1}\frac{x^m}{m+1}dx$

$= 0 - 0 - \frac{n}{m+1}I_{(m,n-1)}$

$\left(\because \lim_{x\to 0}(\log x)^n.x^{m+1} = 0\right) = \frac{-n}{m+1}I_{m,n-1}$

17. (b) $I = \int_0^{\pi}\frac{1}{3+\frac{4\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} + \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}dx$

$I = \int_0^{\pi}\frac{\sec^2\frac{x}{2}dx}{3+3\tan^2\frac{x}{2}+4\tan\frac{x}{2}+1-\tan^2\frac{x}{2}}$

$I = \int_0^{\pi}\frac{\sec^2\frac{x}{2}dx}{2\tan^2\frac{x}{2}+4\tan\frac{x}{2}+4}dx$

Put $\tan\frac{x}{2} = t$; $\frac{1}{2}\sec^2 x dx = dt$

$I = \int_0^{\infty}\frac{dt}{t^2+2t+2}$;

$I = \int_0^{\infty}\frac{dt}{(t+1)^2+1} = \left[\tan^{-1}\frac{(t+1)}{1}\right]_0^{\infty}$

$\tan^{-1}\infty - \tan^{-1}1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

18. (a) $\int_0^{t^2}xf(x)dx = \frac{2t^5}{5}$; differentiating $t^2f(t^2)\,2t$

$= \frac{2}{5}.5t^4$

$f(t^2) = t \Rightarrow f\left(\frac{4}{25}\right) = \sqrt{\frac{4}{25}} = \frac{2}{5}$

19. (a) $\lim_{x\to 0}\frac{\int_0^{x^2}\tan\sqrt{t}dt}{x^3}\left(\frac{0}{0}\right) = \lim_{x\to 0}\frac{\tan x.2x}{3x^2}$

$= \frac{2}{3}\lim_{x\to 0}\frac{\tan x}{x} = \frac{2}{3}.1 = \frac{2}{3}$

20. (d) $\sin^{11}x$ is odd function

$\therefore \int_{-1}^{1}\sin^{11}x\,dx = 0$

21. (b) $I = \int_0^{\pi}x\sin^3 x\,dx = \int_0^{\pi}(\pi - x)\sin^3(\pi - x)dx$

$I = \int_0^{\pi}(\pi - x)\sin^3 x\,dx$; Adding both, we get

$2I = \pi\int_0^{\pi}\sin^3 x\,dx = 2\pi\int_0^{\pi/2}\sin^3 x\,dx$

$I = \pi\times\frac{3-1}{3} = \frac{2\pi}{3}$

22. (c) $I=\int_1^3\sqrt{3+x^3}$;

$\because$ $\sqrt{3+x^3}$ is an decreasing function in $(1, 3)$

$(3-1)f(1)<I<(3-1)f(3)\Rightarrow 4<I<2\sqrt{30}$

23. (c) $\int_0^1\frac{d}{dx}\left[\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right]dx=\sin^{-1}\left(\frac{2x}{1+x^2}\right)\Big|_0^1$

$=\sin^{-1}1=\frac{\pi}{2}$

24. (b) Step 1:

$\int\tan^n 0d0=\int\tan^{n-2}0(\sec^2 0-1)d0$

$=\int\tan^{n-2}\theta\sec^2\theta d\theta-\int\tan^{n-2}\theta d\theta$

$I_n=\frac{\tan^{n-1}\theta}{n-1}\Big|_0^{\pi/4}-\int_0^{\pi/4}\tan^{n-2}0d0$

$I_n=\frac{1}{n-1}-I_{n-2}$

$=\left[\frac{\tan^{n-1}x}{n-1}\right]_0^{\pi/4}-I_{n-2}\Rightarrow I_n+I_{n-2}=\frac{1}{n-1}$

Hence $I_n+I_{n-2}=\frac{1}{n-1}$ (Note)

Step 2:

$\lim_{x\to\infty}n(I_n+I_{n-2})=\lim_{x\to\infty}\frac{n}{n-1}=\lim_{x\to\infty}\frac{1}{1-\frac{1}{n}}$

$=\frac{1}{1-0}=1.$

25. (a) $\int_0^x f(t)dt=x+\int_x^1 tf'(t)dt$

differentiating w.r.t. x, $f(x)$. $1 = 1 + 0 - x f(x)$

$f(x)=\frac{1}{1+x}\Rightarrow x=1$ or $f(1)=\frac{1}{2}$

26. (a) $\int_0^1\frac{x^4(1-x)^4}{1+x^2}=\int_0^1\frac{x^4[(1+x^2)-2x]^2}{1+x^2}$

$=\int_0^1\frac{x^4[(1+x^2)^2-4x(1+x^2)+4x^2]}{1+x^2}$

$=\int_0^1 x^4\left[(1+x^2)-4x+\frac{4x^2}{1+x^2}\right]dx$

$=\int\left[x^6+x^4-4x^5+\frac{4x^6}{1+x^2}\right]dx$

Now on polynomial division of x^6 by $1 + x^2$, we obtain

$=\int\left[x^6+x^4-4x^5+4\left[(x^4-x^2+1)-\frac{1}{1+x^2}\right]\right]dx$

$=\int\left[\left(x^6-4x^5+5x^4-4x^2+4\right)-\frac{4}{1+x^2}\right]dx$

$=\left[\frac{x^7}{7}-\frac{4x^6}{6}+\frac{5x^5}{5}-\frac{4x^3}{3}+4x\right]_0^1-4\left[\tan^{-x}x\right]_0^1$

$=\left(\frac{1}{7}-\frac{4}{6}+1-\frac{4}{3}+4\right)-4\left(\frac{\pi}{4}\right)$

$=\left[\frac{1}{7}-\frac{12}{6}+5\right]-\pi=\left(\frac{1}{7}+3\right)-\pi=\frac{22}{7}-\pi$

27. (b,c) $f(x)=\ln x+\int_0^2\sqrt{1+\sin t}dt$

$f'(x)=\frac{1}{x}+\sqrt{1+\sin x}$

$f''(x)=-\frac{1}{x^2}+\frac{\cos x}{2\sqrt{1+\sin x}}$

(a) f'' is not defined for $x=-\frac{\pi}{2}+2n\pi, n\in I$,

So (a) is wrong

(b) $f'(x)$ always exist for $x > 0$

(c) $|f'| < |f|$

Since $f' > 0$ and $f > 0$

$f' < f$

$\frac{1}{x}+\sqrt{1+\sin x}<\ln x+\int_0^x\sqrt{1+\sin x}dx$

LHS is bounded *RHS* is increasing with range ∞

So there exist some α beyond which *RHS* is greater than *LHS*

(d) $|f| + |f'| \le b$ is wrong as f is *MI* & its range is not bound while β is finite.

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The value of $\int_0^{\pi} |\sin^3\theta| \, d\theta$ is

[UPSEAT-2003]

(a) 0 (b) 3/8
(c) 4/3 (d) π

2. $\int_0^{\pi} \sin^2 x \, dx$ is equal to

[MPPET-1999]

(a) π (b) $\pi/2$
(c) 0 (d) None of these

3. The value of the integral $\int_0^1 e^{x^2} dx$ lies in interval

[CEE-1993]

(a) (0, 1) (b) (–1, 0)
(c) (1, e) (d) None of these

4. The correct evaluation of $\int_0^{\pi} |\sin^4 x| \, dx$ is

[MPPET-1993]

(a) $8\pi/3$ (b) $2\pi/3$
(c) $4\pi/3$ (d) $3\pi/8$

5. Let a,b,c be non-zero real numbers such that $\int_0^3 (3ax^2 + 2bx + c)\, dx = \int_1^3 (3ax^2 + 2bx + c)\, dx$, then

[BIT Ranchi-1991]

(a) $a + b - c = 3$ (b) $a + b - c = 1$
(c) $a + b - c = 0$ (d) $a + b + c = 2$

6. $\int_0^{\pi/2} \sin^4 x \cos^6 x \, dx =$

[RPET-1999; EAMCET-2002]

(a) $5\pi/512$ (b) $3\pi/512$
(c) $\pi/512$ (d) None of these

7. If $u_n = \int_0^{\pi/4} \tan^n x dx$, then $u_n + u_{n-2} =$

[UPSEAT-2002]

(a) $1/n - 1$ (b) $1/n + 1$
(c) $1/2n - 1$ (d) $1/2n + 1$

8. $\int_0^1 \frac{2 - x^2}{(1+x)\sqrt{1-x^2}} \, dx$ is equal to

(a) $\pi/4$ (b) $\pi/3$
(c) $\pi/2$ (d) None of these

9. Let $f: R \to R$ be a differentiable function and $f(1) = 4$, then the value of $\lim_{x \to 1} \int_4^{f(x)} \frac{2tdt}{x-1}$ is

(a) $8f'(1)$ (b) $4f'(1)$
(c) $2f'(1)$ (d) $f'(1)$

10. $\int_0^1 \tan^{-1}(1 - x + x^2) dx =$ ***[IIT-98]***

(a) log 2 (b) $\log \frac{1}{2}$

(c) $\pi \log 2$ (d) $\frac{\pi}{2} \log \frac{1}{2}$

11. If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x, x \in \left(0, \frac{\pi}{2}\right)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ equals to

[IIT (Screening)-2005]

(a) 3 (b) 1/3
(c) $1/\sqrt{3}$ (d) $\sqrt{3}$

12. If $f(x) = \int_{x^2}^{x^2-1} e^{-t^2} dt$, find the interval in which $f(x)$ is increasing

[IIT (Screening) 2003]

(a) $(0, \infty)$ (b) $(-\infty, 0)$
(c) $[-2, 2]$ (d) No where

13. The value of the integral $\int_0^{\pi/4} \frac{\sqrt{\tan x}}{\sin x \cos x} dx$ equals ***[Kurukshetra CEE-1996]***

(a) 1 (b) 2
(c) 0 (d) 4

14. If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$, then for any positive integer n, the value of $n(I_{n-1} + I_{n+1})$ is

[RPET-99; Kurukshetra CET-2000]

(a) 1 (b) 2
(c) $\pi/4$ (d) π

15. Let $a_n = \int_0^{\pi/4} \tan^n x \, dx$. Then $a_2 + a_4, a_3 + a_5, a_4 + a_6$ are in

(a) A.P. (b) G.P.
(c) H.P. (d) None of these

16. $\int_0^{\lambda} \frac{ydy}{\sqrt{y+\lambda}}$ is equal to

[MPPET-2008]

(a) $\frac{2}{3}(2-\sqrt{2})\lambda\sqrt{\lambda}$ (b) $\frac{2}{3}(2+\sqrt{2})\lambda\sqrt{\lambda}$

(c) $\frac{1}{3}(2-\sqrt{2})\lambda\sqrt{\lambda}$ (d) $\frac{1}{3}(2+\sqrt{2})\lambda\sqrt{\lambda}$

17. $\lim_{x\to\pi/4} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2-\frac{\pi^2}{16}}$ equals

(a) $\frac{8}{\pi}f(2)$ (b) $\frac{2}{\pi}f(2)$

(c) $\frac{2}{\pi}f\left(\frac{1}{2}\right)$ (d) $4f(2)$

18. $\int_0^{2\pi} x\sin^6 x\cos^5 x\, dx$ ***[EAMCET-2007]***

(a) 2π (b) $\pi/2$

(c) 0 (d) $-\pi$

19. $\int_0^{\pi/2} \sin^5 x\, dx =$

(a) 8/15 (b) 4/15

(c) $8\sqrt{\pi}/15$ (d) $8\pi/15$

20. $\int_{-\pi/2}^{\pi/2} \sin^4 x\cos^6 x\, dx =$

[EAMCET-2002]

(a) $\frac{3\pi}{64}$ (b) $\frac{3\pi}{572}$

(c) $\frac{3\pi}{256}$ (d) $\frac{3\pi}{128}$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 16 minutes.
3. The worksheet consists of 16 questions. The maximum marks are 48.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\int_0^{\pi/2} \frac{\cos 2x dx}{(\sin x + \cos x)^2}$ is equal to

(a) 0 (b) $\pi/4$
(c) $\pi/2$ (d) None of these

2. $\int_0^2 ([x]^2 - [x^2])dx$ is equal to

(a) $-4 + \sqrt{3} + \sqrt{2}$ (b) $4 + \sqrt{3} - \sqrt{2}$
(c) $4 + 2\sqrt{3} + \sqrt{2}$ (d) None of these

3. $\int_0^{\pi/8} \cos^3 4\theta d\theta$ is equal to

(a) 5/3 (b) 5/4
(c) 1/3 (d) 1/6

4. If $I_n = \int_0^{\pi/4} \tan^n 0 d0$, then $I_8 + I_6$ equals

[Kurukshetra CEE-96]

(a) 1/4 (b) 1/5
(c) 1/6 (d) 1/7

5. $\int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx =$

(a) $2\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$ (b) $\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$
(c) $2\sin^{-1}\left(\frac{\sqrt{3}+1}{2}\right)$ (d) $\sin^{-1}\left(\frac{\sqrt{3}+1}{2}\right)$

6. Consider the following statements

1. $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x} dx = \frac{3}{4}$
2. $\int_0^4 (|x-1| + |x-3|)dx = 10$

Which of these is/are correct?

(a) Only (1) (b) Onle (2)
(c) Both of these (d) None of these

7. If $f(x) = \operatorname{cosec}(x - \pi/3) \operatorname{cosec}(x - \pi/6)$, then the value of $\int_0^{\pi/2} f(x)dx$ is

(a) $4 \log \sqrt{3}$ (b) $-4 \log \sqrt{3}$
(c) $2 \log \sqrt{3}$ (d) $-1/4$

8. $\int_0^1 \frac{xdx}{\left[x + \sqrt{1-x^2}\right]\sqrt{1-x^2}}$ is equal to

(a) 0 (b) 1
(c) $\pi/4$ (d) $\pi^2/2$

9. The value of the integral $\int_1^\infty \frac{x \log x}{(1-x^2)x} dx$ is

(a) 0 (b) log 7
(c) 5 log 13 (d) None of these

10. $\frac{d}{dx}\int_2^{x^2} (t-1)dt$ is equal to

(a) $x^2 - 1$ (b) $x(x^2 - 1)$
(c) x 1 (d) $2x(x^2 - 1)$

11. $\int_0^1 \tan^{-1}\left(\frac{1}{x^2 - x + 1}\right) dx$ is

(a) $ln\ 2$ (b) $-ln\ 2$
(c) $\frac{\pi}{2} + ln\ 2$ (d) $\frac{\pi}{2} - ln\ 2$

12. If $P = \int_0^{3\pi} f(\cos^2 x)dx$ and $Q = \int_0^{\pi} f(\cos^2 x)dx$, then

(a) $P - Q = 0$ (b) $P - 2Q = 0$
(c) $P - 3Q = 0$ (d) $P - 5Q = 0$

13. If $\int_{-3}^{2} f(x)dx = \frac{7}{3}$ and $\int_{-3}^{9} f(x)dx = -\frac{5}{6}$, then what is the value of $\int_2^9 f(x)dx$?

[NDA-2007]

(a) $-\frac{19}{6}$ (b) $\frac{19}{6}$
(c) $\frac{3}{2}$ (d) $-\frac{3}{2}$

14. The value of $\int_0^{\pi/2} (\sqrt{\sin\theta}\cos\theta)^3 d\theta$ is:

[AMU-1999]

(a) 2/9 (b) 2/15
(c) 8/45 (d) 5/2

15. $\int_0^{\pi} \cos^3 x\, dx =$

[MPPET-96; Pb.CET-02]

(a) -1 (b) 0
(c) 1 (d) π

16. The value of the integral $\int_0^{\pi} \sin^6 x dx$ is:

[MPPET-2010]

(a) $\frac{3}{4}\pi$ (b) $\frac{5}{32}\pi$
(c) $\frac{3}{16}\pi$ (d) None of these

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	16. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ	
6. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

2. (a) $\int_0^2 \left([x]^2 - [x^2]\right) dx =$

$\int_0^1 0^2 dx + \int_1^2 1^2 dx -$

$\left[\int_0^1 0 dx + \int_1^{\sqrt{2}} dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx + \int_{\sqrt{3}}^2 3dx\right]$

$1(2-1) - \left[\sqrt{2} - 1 + 2(\sqrt{3} - \sqrt{2}) + 3(2 - \sqrt{3})\right]$

$= 1 - (\sqrt{2} - 1 + 2\sqrt{3} - 2\sqrt{2} + 6 - 3\sqrt{3})$

$= 1 - (-\sqrt{2} - \sqrt{3} + 5) = \sqrt{2} + \sqrt{3} - 4$

3. (d) $\int_0^{\pi/8} \cos^3 40 d0 = \int_0^{\pi/2} \cos^3 t \frac{dt}{4}$

Put $40 = t; d0 = \frac{dt}{4} = \frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$

4. (d) We know that $I_n + I_{n-2} = \frac{1}{n-1}$

$n = 8;\ I_8 + I_6 = \frac{1}{7}$

5. (a)

$$\int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos)^2}} dx$$

$$= \int_{\frac{1}{2} - \frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2} - \frac{1}{2}} \frac{dt}{\sqrt{1 - t^2}}$$

Put $\sin x - \cos x = t$

$$= \sin^{-1} t \Big|_{\frac{1}{2} - \frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2} - \frac{1}{2}} = 2\sin^{-1}\left(\frac{\sqrt{3} - 1}{2}\right)$$

7. (b) $f(x) = \operatorname{cosec}\left(x - \frac{\pi}{3}\right)\operatorname{cosec}\left(x - \frac{\pi}{6}\right) =$

$$= \frac{\sin\left(x - \frac{\pi}{6} - \left(x - \frac{\pi}{3}\right)\right)}{\sin\left(x - \frac{\pi}{3}\right)\sin\left(x - \frac{\pi}{6}\right)} \times \frac{1}{\sin\frac{\pi}{6}}$$

$$= \left[\cot\left(x - \frac{\pi}{3}\right) - \cot\left(x - \frac{\pi}{6}\right)\right].2$$

$$2.\int_0^{\pi/2}\left[\cot\left(x-\frac{\pi}{3}\right)-\cot\left(x-\frac{\pi}{6}\right)\right]dx$$

$$=2.\left[\log\sin\left(x-\frac{\pi}{3}\right)\right]\Bigg|_0^{\pi/2}$$

$$\left[\log\sin\left(x-\frac{\pi}{6}\right)\right]\Bigg|_0^{\pi/2}$$

$$=2.\left[\log\left(\frac{1/2}{\sqrt{3}/2}\right)-\log\left(\frac{\sqrt{3}/2}{1/2}\right)\right]$$

$$=2\left[\log\frac{1}{\sqrt{3}}-\log\sqrt{3}\right]=-4\log\sqrt{3}$$

11. (d) $\int_0^1\tan^{-1}\left(\frac{1}{x^2-x+1}\right)dx$

$$=\int_0^1\tan^{-1}\left(\frac{x-(x-1)}{1+x(x-1)}\right)dx$$

$$=\int_0^1\left[\tan^{-1}x-\tan^{-1}(x-1)\right]dx$$

$$=\int_0^1\tan^{-1}x+\int_0^1\tan^{-1}(1-x)dx$$

$$=2\int_0^1\tan^{-1}xdx$$

$$\left[\because\int_0^1\tan^{-1}(1-x)dx=\int_0^1\tan^{-1}(1-(1-x))dx=\int_0^1\tan^{-1}xdx\right]$$

$$=2.\left[\tan^{-1}x.x-\int\frac{x}{1+x^2}dx\right]_0^1$$

$$=2.\left[x\tan^{-1}x-\frac{1}{2}\log(1+x^2)\right]_0^1$$

$$=2\left[\frac{\pi}{4}-\frac{1}{2}\log 2\right]=\frac{\pi}{2}-\log 2$$

13. (a) $\int_{-3}^{9}f(x)dx=\int_{-3}^{2}f(x)dx+\int_2^9f(x)dx$

$$\frac{-5}{6}=\frac{7}{3}+\int_2^9f(x)dx$$

$$\Rightarrow\int_2^9f(x)dx=\frac{-5}{6}-\frac{7}{3}=\frac{-19}{6}$$

14. (c)

$$\int_0^{\pi/2}(\sqrt{\sin\theta}\cos\theta)^3d\theta$$

$$=\int_0^{\pi/2}(\sin 0\sqrt{\sin 0}\cos^3 0)d0$$

$$=\int_0^1(t\sqrt{t})\cos^2\theta dt\ \text{ put }\ \sin\theta=t\ ;\ \cos\theta d\theta=dt$$

$$=\int_0^1t^{3/2}(1-t^2)dt=\int_0^1\left(t^{3/2}-t^{7/2}\right)dt$$

$$=\frac{t^{5/2}}{5/2}\Bigg|_0^1-\frac{t^{9/2}}{9/2}\Bigg|_0^1=\frac{2}{5}-\frac{2}{9}=\frac{8}{45}$$

16. (b) $\int_0^{\pi/2}\sin^6xdx=\frac{6-1}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}=\frac{5\pi}{32}$

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LECTURE



5 Definite Integral as a Limit of Sum

BASIC CONCEPTS

The definite integral $\int_a^b f(x)dx$ can also be expressed as a limiting case of summation of an infinite series, provided $f(x)$ is continuous on $[a, b]$. We define

$$\int_a^b f(x)dx = (b-a)\lim_{n\to\infty}\frac{1}{n}$$

$$[f(a)+f(a+h)+....+f\{a+(n-1)h\}]$$

$$h=\frac{b-a}{n}=(b-a)\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{n-1}f\left\{a+(b-a)\frac{r}{n}\right\}$$

in particular if $a = 0$ and $b = 1$,

we get $\int_0^1 f(x)dx=\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{n-1}f\left(\frac{r}{n}\right)$

Working Rule

If a given series is expressible in the form $\frac{1}{n}\sum f\left(\frac{r}{n}\right)$, then its limit as $n \to \infty$ can be expressed as a definite integral by writing x for $\frac{r}{n}$, dx for $\frac{1}{n}$ and integral sign $\int$ for Σ. The lower and upper limits of integration are the values of $\lim_{n\to\infty}\left(\frac{r}{n}\right)$ for the least and the greatest values of r respectively. In general

$$\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{nk-1}f\left(\frac{r}{n}\right)=\int_0^k f(x)dx, m,k\in N$$

Example

Sum of the series

$$\lim_{n\to\infty}\left[\frac{1}{\sqrt{n^2-1^2}}+\frac{1}{\sqrt{n^2-2^2}}+.....+\frac{1}{\sqrt{n^2-(n-1)^2}}\right]$$ is

$[n \in N]$

(a) $\pi/2$ (b) $\pi/4$

(c) $\pi/6$ (d) none of these

Solution

(a) Let

$$S=\lim_{n\to\infty}\left[\frac{1}{\sqrt{n^2-1^2}}+\frac{1}{\sqrt{n^2-2^2}}+.....+\frac{1}{\sqrt{n^2-(n-1)^2}}\right]$$

Step 1: Converting each term are a function of $\frac{r}{n}$

$$S=\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{\sqrt{1-\left(\frac{1}{n}\right)^2}}+\frac{1}{\sqrt{1-\left(\frac{2}{n}\right)^2}}+.....+\frac{1}{\sqrt{n-\left(\frac{n-1}{n}\right)^2}}\right]$$

Step 2: Writing the series as $\frac{1}{n}\sum f\left(\frac{r}{n}\right)$

$$S = \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n-1} \frac{1}{\sqrt{1-\left(\frac{r}{n}\right)^2}}$$

Step 3: Replace $\frac{r}{n}$ by x, $\frac{1}{n}$ by dx and Σ by $\int$

For the limits of integration

Lower limit $= \lim_{n\to\infty} \frac{r}{n}$ (for $r = 1$) $\lim_{n\to\infty} \frac{1}{n} = 0$

Upper limit $= \lim_{n\to\infty} \frac{r}{n}$ (for $r = n - 1$)

$$= \lim_{n\to\infty} \frac{n-1}{n} = 1$$

$$\therefore \quad S = \int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx = \left[\sin^{-1} x\right]_0^1$$

$$= \sin^{-1} 1 - \sin^{-1} 0 = \frac{\pi}{2}$$

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Evaluate the integral as limit of a sum $\int_0^2 (2x+1)dx$ ***[CBSE-96]***

Solution

We have

$$\int_a^b f(x)dx = \lim_{h\to 0} h[f(a) + f(a+h) + f(a+2h) + ... + f(a+(n-1)h)]$$

where $h = \frac{b-a}{n}$

Here $a = 0, b = 2, f(x) = 2x + 1$

and $h = \frac{2-0}{n} = \frac{2}{n}$

$$\therefore \quad \int_0^2 (2x+1)dx$$

$$= \lim_{h\to 0} h[f(0) + f(0+h) + f(0+2h) + f(0+3h) + .. + f(0+(n-1)h)]$$

$$= \lim_{h\to 0} h[f(0) + f(h) + f(2h) + f(3h) + .. + f(n-1)h]$$

$$= \lim_{h\to 0} h[(0+1) + (2h+1) + (2.2h+1) + (2.3h+1) + .. + 2(n-1)h + 1]$$

$$= \lim_{h\to 0} h[n + 2h(1 + 2 + 3 + ... + (n-1))]$$

$$= \lim_{h\to 0} h[n + nh(n-1)] = \lim_{n\to\infty} \frac{2}{n}\left[n + n.\frac{2}{n}(n-1)\right]$$

$[\because h = \frac{2}{n}$ and $h \to 0 \Rightarrow n \to \infty]$

$$= \lim_{n\to\infty} \left[2 + 4\left(\frac{n-1}{n}\right)\right] = \lim_{n\to\infty} \left[2 + 4\left(1 - \frac{1}{n}\right)\right]$$

$= 2 + 4 = 6$

2. Evaluate the integral as limit of sums: $\int_0^2 (x^2+3)dx$

[CBSE-1994, 2001 (C)]

Solution

We have

$$\int_a^b f(x)\,dx = \lim_{h\to 0} h[f(a) + f(a+h) + f(a+2h) + ... + f(a+(n-1)h)]$$

where $h = \frac{b-a}{n}$

Here $a = 0, b = 2,$

$f(x) = x^2 + 3$ and $h = \frac{2-0}{n} = \frac{2}{n}$

$$\therefore \quad \int_0^2 (x^2+3)dx$$

$$= \lim_{h\to 0} h[f(0) + f(0+h) + f(0+2h) + .. + f(0+(n-1)h)]$$

$$= \lim_{h\to 0} h[f(0) + f(h) + f(2h) + ... + f(n-1)h]$$

3. Evaluate the definite integral as the limit of a sum $\int_0^{\pi/2} \sin 2x dx$

[AISSE-95]

Solution

Let $f(x) = \sin 2x$, $a = 0$, $b = \frac{\pi}{2}$

and $nh = \frac{\pi}{2} - 0 = \frac{\pi}{2}$

$\therefore \quad \int_0^{\pi/2} \sin 2x dx$

$= \lim_{h\to 0} h[f(0) + f(0+h) +$

$.....+ f\{0+(n-1)h\}]$

$= \lim_{h\to 0} h[\sin 0 + \sin(2h) + \sin 4h +$

$.....+ \sin\{0+(n-1)h\}]$

$= \lim_{h\to 0} h\left[\frac{\sin\left\{0+\left(\frac{n-1}{2}\right)h\right\}\sin\frac{nh}{2}}{\sin\frac{h}{2}}\right]$

$= \lim_{h\to 0}\left[\frac{\sin\left(0+\frac{nh}{2}-\frac{h}{2}\right)\sin\frac{nh}{2}}{\sin\frac{h}{2}}.h\right]$

$= \lim_{h\to 0}\left[2\sin\left(\frac{nh}{2}-\frac{h}{2}\right)\sin\frac{nh}{2}.\frac{\frac{h}{2}}{\sin\frac{h}{2}}\right]$

$= \lim_{h\to 0} 2\sin\left(\frac{\pi}{4}-\frac{h}{2}\right)\sin\frac{\pi}{4}\times 1$

$= 2\sin\frac{\pi}{4}.\sin\frac{\pi}{4} = 2\times\frac{1}{2} = 1.$

$= \lim_{h\to 0} h[(0+3)+(h^2+3)+(2^2h^2+3)+$

$(3^2h^2+3)+...+((n-1)^2h^2+3)]$

$= \lim_{h\to 0} h[3n + h^2(1^2+2^2+....+(n-1)^2]$

$= \lim_{h\to 0}\left[3n + h^2\frac{n(n-1)(2n-1)}{6}\right]$

$= \lim_{n\to\infty}\left[3n + \frac{4}{n^2}.\frac{n(n-1)(2n-1)}{6}\right]\frac{2}{n}$

$= \lim_{n\to\infty}\left[6+\frac{8}{6}\frac{(n-1)(2n-1)}{n^2}\right]$

$= \lim_{n\to\infty}\left[6+\frac{8}{6}\left(1-\frac{1}{n}\right)\left(2-\frac{1}{n}\right)\right]$

$= 6+\frac{8}{6}(1-0)(2-0)$

$= 6+\frac{8}{6} = \frac{26}{3}$

4. Evaluate the integral as limit of sums $\int_0^2 e^x dx$ ***[CBSE-91]***

Solution

We have $\int_a^b f(x)dx = \lim_{h\to 0} h[f(a) + f(a+h) + f(a+2h) + ... + f(a+(n-1)h]$ where $h = \frac{b-a}{n}$

Here $a = 0$, $b = 2$, $f(x) = e^x$ and

$h = \frac{2-0}{n} = \frac{2}{n}$

$\therefore \quad \int_0^2 e^x dx$

$= \lim_{h\to 0} h\ [f(0) + f(h) + f(2h) + ... + f((n-1)h)]$

$= \lim_{h\to 0} h\ [e^0 + e^h + e^{2h} + ... + e^{(n-1)h}]$

$= \lim_{h\to 0} h\left[e^0\left\{\frac{(e^h)^n - 1}{e^h - 1}\right\}\right]$

[using $a + ar + ... + ar^{n-1} = a\left(\frac{r^n-1}{r-1}\right)$]

$= \lim_{h\to 0} h\left[\frac{e^{nh}-1}{e^h-1}\right]$

$= \lim_{h\to 0}\frac{h}{h}\left[\frac{e^2-1}{\left(\frac{e^h-1}{h}\right)}\right]$

$\left[\because h = \frac{2}{n} \Rightarrow nh = 2\right]$

$= \lim_{h\to 0}\frac{e^2-1}{\left(\frac{e^h-1}{h}\right)} = \frac{e^2-1}{1}$

$\left[\because \lim_{h\to 0}\frac{e^h-1}{h} = 1\right] = e^2 - 1$

UNSOLVED SUBJECTIVE PROBLEMS (XI BOARD/C.B.S.E./STATE): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Find the value of the integral as the limit of a sum $\int_1^2 x dx$.
2. Evaluate the integral as limit of sums $\int_0^3 (x+4)dx$ ***[CBSE-95]***
3. Evaluate the integral as limit of sums $\int_0^2 (x+3)dx$ ***[CBSE-95]***
4. Evaluate the integral as limit of sums: $\int_0^2 (x^2+2)dx$ ***[CBSE-2000]***
5. Evaluate the integral as limit of sums: $\int_1^3 (2x^2+5x)$ ***[CBSE-97]***
6. Evaluate as the limit of sum $\int_a^b x dx$. ***[PSB-1988]***
7. Evaluate as the limit of sum $\int_a^b \sin x dx$.
8. Evaluate: $\int_a^b (2x^2+3x+5)dx$ as limit of a sum. ***[CBSE-2007]***
9. Evaluate the integral as limit of sums: $\int_0^2 (x^2+1)dx$ ***[CBSE-91(c), 92, 2005 (Delhi)-I]***

ANSWERS

1. $\frac{3}{2}$
2. $\frac{33}{2}$
3. 8
4. $\frac{20}{3}$
5. $\frac{112}{3}$
6. $\frac{b^2-a^2}{2}$
7. $\cos a - \cos b$
9. 14/3

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. $\lim_{n\to\infty}\left\{\left(1+\frac{1^2}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)....\left(1+\frac{n^2}{n^2}\right)\right\}^{1/n}$ is equal to ***[JEE (WB)-2005]***

(a) $2e^{\pi-4/2}$ (b) $2e^{\pi-2/2}$

(c) $e^{\pi-4/2}$ (d) $2e^{\pi-1/2}$

Solution

(a) Let

$$A=\lim_{n\to\infty}\left\{\left(1+\frac{1^2}{n^2}\right)\left(1+\frac{2^2}{n^2}\right).....\left(1+\frac{n^2}{n^2}\right)\right\}^{1/n}$$

$$\Rightarrow \log A=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\log\left(1+\frac{r^2}{n^2}\right)$$

$$=\int_0^1 \log(1+x^2)dx=[x\log(1+x^2)]_0^1$$

$$-\int_0^1 \frac{2x^2}{1+x^2}dx$$

$$=\log 2-2[x-\tan^{-1}x]_0^1=\log 2-2+\frac{\pi}{4}$$

$$\therefore \quad A=2e^{\pi-4/2}$$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\lim_{n\to\infty}\left[\frac{1^k+2^k+3^k+....+n^k}{n^{k+1}}\right]$ is equal to

[AIEEE-2002]

(a) $\frac{1}{k}$ (b) $\frac{1}{k+1}$

(c) $\frac{1}{k+2}$ (d) None of these

2. $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}+....+\frac{1}{n}\sec^2 1\right]$ equals ***[AIEEE-05]***

(a) $\frac{1}{2}\operatorname{cosec} 1$ (b) $\frac{1}{2}\sec 1$

(c) $\frac{1}{2}\tan 1$ (d) $\tan 1$

3. $\lim_{n\to\infty}\frac{1}{n}\left[\sec^2\frac{\pi}{4n}+\sec^2\frac{2\pi}{4n}+....+\sec^2\frac{n\pi}{4n}\right]=$

(a) $\frac{2}{\pi}$ (b) $\frac{4}{\pi}$

(c) $\frac{6}{\pi}$ (d) $\frac{8}{\pi}$

4. $\lim_{n\to\infty}\left[\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+...+\sqrt{n}}{n\sqrt{n}}\right]=$

(a) 1/3 (b) 2/3

(c) 3/4 (d) 2/4

5. $\lim_{n\to\infty}\left[\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+.....+\frac{1}{n}\right]=$

(a) $\frac{\pi}{4}+\frac{1}{2}\log 2$ (b) $\frac{\pi}{4}-\frac{1}{2}\log 2$

(c) $\frac{\pi}{4}-\frac{1}{3}\log 2$ (d) None of these

6. $\lim_{n\to\infty}\sum_{r=n}^{n-1}\frac{1}{\sqrt{4n^2-r^2}}=\frac{1}{2n}+\frac{1}{\sqrt{4n^2-1}}$

$+....+\frac{1}{\sqrt{3n^2+2n-1}}=$

[Roorkee-1997]

(a) π/3 (b) π/4

(c) π/6 (d) None of these

7. Show that $\lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{n}{n^2+r^2}=$

(a) π/3 (b) π/4

(c) π/6 (d) None of these

8. $\lim_{n\to\infty}\left[\frac{1}{1+n^3}+\frac{4}{8+n^3}+\frac{9}{27+n^3}+.....+\frac{1}{2n}\right]=$

[RPET-97]

(a) $\frac{1}{3}\log 2$ (b) $\frac{1}{2}\log 2$

(c) $\frac{1}{3}\log 3$ (d) None of these

9. $\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)^{2\cdot n^2}\left(1+\frac{2^2}{n^2}\right)^{4/n^2}...\left(1+\frac{n^2}{n^2}\right)^{2n/n^2}\right]=$

(a) 2/e (b) 4/e

(c) 6/e (d) None of these

10. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\sqrt{\frac{n+r}{n-r}}=$

(a) $\frac{\pi}{2}+1$ (b) $\frac{\pi}{3}+1$

(c) $\frac{\pi}{2}+3$ (d) None of these

11. $\lim_{n\to\infty}\left(\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+.....+\frac{1}{na}\right)=$

[Karnataka CET-1999]

(a) $\log_e a$ (b) $\log_a e$

(c) $2\log_e a$ (d) None of these

12. $\lim_{n\to\infty}\frac{n}{(n+1)^2}+\frac{n}{(n+2)^2}+...+\frac{n}{(n+n)^2}=$

(a) 1/2 (b) 1

(c) 2 (d) 3

13. $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$ equals

[IIT-1997; Re-Exam]

(a) $1+\sqrt{5}$ (b) $-1+\sqrt{5}$

(c) $-1+\sqrt{2}$ (d) $1+\sqrt{2}$

14. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}e^{\frac{r}{n}}$ is

[AIEEE-2004]

(a) e (b) $e-1$
(c) $1-e$ (d) $e+1$

15. The value of $\lim_{x\to\infty}\frac{1}{x^3}\int_0^x \frac{t\,\ell n(1+t)}{t^4+4}dt$ is

[IIT (I)-2010]

(a) 0 (b) 1/12
(c) 1/24 (d) 1/64

SOLUTIONS

1. (b) $Lim\left[\frac{1^k+2^k+3^k+.....n^k}{n^{k+1}}\right]$

$=Lim\frac{1}{n}\left[\left(\frac{1}{n}\right)^k+\left(\frac{2}{k}\right)^k+\left(\frac{3}{n}\right)^k+.....\left(\frac{n}{n}\right)^k\right]$

$=\int_0^1 x^k dx=\frac{x^{k+1}}{k+1}\Big|_0^1=\frac{1}{k+1}$

2. (c)

$\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{n}\sec^2\frac{1}{n^2}+\frac{2}{n}\sec^2\frac{4}{n^2}+ .....+\frac{n}{n}\sec^2\left(\frac{n}{n}\right)^2\right]$

$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{r}{n}\sec^2\left(\frac{r}{n}\right)^2=\frac{1}{2}\int_0^1 2x\sec^2 x^2 dx$

$=\frac{1}{2}\tan x^2\Big|_0^1=\frac{\tan 1}{2}$

3. (b) $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left[\sec^2\frac{r\pi}{4n}\right]=\int_0^1\sec^2\frac{\pi}{4}x dx$

$=\frac{\tan\frac{\pi}{4}x}{\frac{\pi}{4}}\Big|_0^1=\frac{4}{\pi}$

4. (b) $\lim_{n\to\infty}\left[\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+.....+\sqrt{n}}{n\sqrt{n}}\right]$

$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\sqrt{\frac{r}{n}}$

$=\int_0^1\sqrt{x}dx=\frac{x^{3/2}}{3/2}\Big|_0^1=\frac{2}{3}$

5. (a) $\lim_{n\to\infty}\left[\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+.....+\frac{n+n}{n^2+n^2}\right]$

dividing by n^2 in numerator and denominator

$\lim_{n\to\infty}\frac{1}{n}\left[\sum_{r=1}^{n}\frac{1+\frac{r}{n}}{1+\left(\frac{r}{n}\right)^2}\right]=\int_0^1\frac{1+x}{1+x^2}dx$

$=\tan^{-1}x+\frac{1}{2}\log(1+x^2)\Big|_0^1=\frac{\pi}{4}+\frac{1}{2}\log 2$

6. (c) $\lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n}\frac{1}{\sqrt{4-\left(\frac{r}{n}\right)^2}}=\int_0^1\frac{1}{\sqrt{4-x^2}}dx$

$=\sin^{-1}\frac{x}{2}\Big|_0^1=\frac{\pi}{6}$

7. (b) $\lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n}\frac{1}{1+\left(\frac{r}{n}\right)^2}$

$=\int_0^1\frac{1}{1+x^2}dx=\tan^{-1}x\Big|_0^1=\frac{\pi}{4}$

8. (a)

$\lim_{n\to\infty}\left[\frac{1^2}{1+n^3}+\frac{2^2}{2^3+n^3}+\frac{3^2}{3^3+n^3}+.....\frac{n^2}{n^3+n^3}\right]$

$=\lim_{n\to\infty}\frac{1}{n}\frac{(r/n)^2}{(r/n)^3+1}=\int_0^1\frac{x^2}{x^3+1}dx$

$=\frac{1}{3}\log(x^3+1)\Big|_0^1$

$=\frac{1}{3}\log 2$

9. (a) Let

$$A=\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)^{2/n^2}\left(1+\frac{2^2}{n^2}\right)^{4/n^2}\right.$$
$$\left.\ldots\ldots\left(1+\frac{n^2}{n^2}\right)^{2n/n^2}\right]$$

$$\log A=\lim_{n\to\infty}\left[\frac{2}{n^2}\log\left(1+\frac{1}{n^2}\right)+\frac{4}{n^2}\log\left(1+\frac{2^2}{n^2}\right)+\ldots\ldots\frac{2n}{n^2}\log\left(1+\frac{n^2}{n^2}\right)\right]$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{2r}{n}\log\left(1+\left(\frac{r}{n}\right)^2\right)$$

$$=\int_0^1 2x\log(1+x^2)\,dx$$

put $t=x^2$

$$\log A=\int_0^1\log(1+t)\,dt$$

$$\log A=t\log(1+t)\Big|_0^1-\int_0^1\frac{t}{1+t}dt$$ [Integration by parts]

$$=\log 2-[t-\log(1+t)]_0^1=\log 2-[1]$$

$$\log A=\log\frac{2}{e}\Rightarrow A=\frac{2}{e}$$

10. (a) $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\sqrt{\frac{1+\frac{r}{n}}{1-\frac{r}{n}}}=\int_0^1\sqrt{\frac{1+x}{1-x}}dx$

$$=\int_0^1\frac{1+x}{\sqrt{1-x^2}}dx=\sin^{-1}x\Big|_0^1-\sqrt{1-x^2}\Big|_0^1$$

$$=\frac{\pi}{2}-[0-1]=\frac{\pi}{2}+1$$

11. (a) $\lim_{n\to\infty}\left(\frac{1}{n+0}+\frac{1}{n+1}+\frac{1}{n+2}+\ldots\ldots\right.$
$$\left.\ldots\ldots+\frac{1}{n+(a-1)n}\right)$$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{(a-1)}\frac{1}{1+\frac{r}{n}}dx=\int_0^{(a-1)}\frac{1}{1+x}dx=\log(1+x)\Big|_0^{a-1}$$

$$=\log_e a$$

12. (a) $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{1}{\left(1+\frac{r}{n}\right)^2}=\int_0^1\frac{1}{(1+x)^2}dx$

$$=\frac{-1}{1+x}\Big|_0^1=-\frac{1}{2}+1=\frac{1}{2}$$

13. (b) $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r/n}{\sqrt{1+\left(\frac{r}{n}\right)^2}}=\int_0^2\frac{x}{\sqrt{1+x^2}}dx$

$$=\sqrt{1+x^2}\Big|_0^2=\sqrt{5}-1$$

14. (b) $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{n}e^{r/n}=\int_0^1 e^x dx=e^x\Big|_0^1=e^1-1$

15. (b) Applying L-Hospital's, we get

$$\lim_{x\to 0}\frac{x\ln(1+x)}{(x^4+4)\times 3x^2}=\lim_{x\to 0}\frac{1}{4}\times\frac{1}{3}=\frac{1}{12}$$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. Show that $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\sin^2\frac{r\pi}{2n}=$

(a) 1/2 (b) 3/4
(c) 2/3 (d) None of these

2. $\lim_{n\to\infty}\frac{1}{n\sqrt{n}}\left[\sqrt{n+1}+\sqrt{n+2}+.....+\sqrt{2n}\right]=$

(a) $\frac{2}{3}(2\sqrt{2}-1)$ (b) $\frac{3}{2}(2\sqrt{2}-1)$

(c) $\frac{2}{3}(2\sqrt{2}+1)$ (d) $\frac{3}{2}(2\sqrt{2}+1)$

3. $\lim_{n\to\infty}\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\left(1+\frac{3}{n}\right).....\left(1+\frac{n}{n}\right)\right]^{1/n}=$

(a) $2/e$ (b) $4/e$

(c) $6/e$ (d) None of these

4. $\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+.....+\frac{1}{n+5n}\right)=$

(a) log 6 (b) log 8

(c) log 9 (d) log 10

5. $\lim_{n\to\infty}\frac{1}{n}\left[\left(\frac{1}{n}\right)^m+\left(\frac{2}{n}\right)^m+...+\left(\frac{n-1}{n}\right)^m\right]=$

(a) $1/m+1$ (b) $1/m$

(c) $1/m-1$ (d) None of these

6. $\lim_{n\to\infty}\left[\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+n}}+\frac{1}{\sqrt{n^2+2n}}+.....+\frac{1}{\sqrt{n^2+(n-1)n}}\right]$

is equal to

[RPET-2000]

(a) 2 (b) $2\sqrt{2}$

(c) $2-2\sqrt{2}$ (d) $2\sqrt{2}-2$

7. $\lim_{n\to\infty}\frac{n}{n^2}+\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+...+\frac{n}{n^2+(n-1)^2}=$

(a) $\pi/4$ (b) $\pi/5$

(c) $\pi/3$ (d) $\pi/6$

8. $\lim_{n\to\infty}\left[\frac{1}{na}+\frac{1}{na+1}+\frac{1}{na+2}+.....+\frac{1}{nb}\right]$ is equal to

(a) log (b/a) (b) log (a/b)

(c) log a (d) log b

9. $\lim_{n\to\infty}\frac{1^{99}+2^{99}+.....+n^{99}}{n^{100}}$ is equal to

[EAMCET-94]

(a) $\frac{9}{100}$ (b) $\frac{1}{100}$

(c) $\frac{1}{99}$ (d) $\frac{1}{101}$

10. $\lim_{n\to\infty}\left[\frac{1}{\sqrt{(n^2-1^2)}}+\frac{1}{\sqrt{(n^2-2^2)}}+.....+\frac{1}{\sqrt{\{n^2-(n-1)^2\}}}\right]=$

(a) $\pi/3$ (b) $\pi/4$

(c) $\pi/2$ (d) None of these

11. $\lim_{n\to\infty}\frac{1}{n+1}+\frac{1}{n+2}+\frac{1}{n+3}+...+\frac{1}{n+n}=$

(a) log 2 (b) 2 log 2

(c) 2 log 4 (d) 3 log 4

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 7 minutes.
3. The worksheet consists of 7 questions. The maximum marks are 21.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. $\lim_{n\to\infty}\frac{1}{n+1}+\frac{1}{n+2}+.....+\frac{1}{3n}=$

(a) log 2 (b) log 3
(c) log 4 (d) log 5

2. $\lim_{n\to\infty}\frac{1}{n+m}+\frac{1}{n+2m}+.....+\frac{1}{n+nm}=$

(a) $\frac{1}{m}\log(1+m)$ (b) $\frac{1}{m}\log(1-m)$

(c) $\frac{1}{n}\log(1-m)$ (d) $\frac{1}{n}\log(1+m)$

3. $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{n^2+k^2}$ is equal to ***[Roorkee-1999]***

(a) $\frac{1}{2}$ log 2 (b) log 2

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

4. $\lim_{n\to\infty}\frac{1+2^4+3^4+.....+n^4}{n^5}-$

$\lim_{n\to\infty}\frac{1+2^3+3^3+.....+n^3}{n^5}=$

(a) 1/30 (b) 0
(c) 1/4 (d) 1/5

5. If $f(n)=\frac{1}{n}\left[(n+1)(n+2).....(n+n)\right]^{1/n}$, then $\lim_{n\to\infty} f(n)$ equals

(a) e
(b) $1/e$
(c) $2/e$
(d) $4/e$

6. Determine the limit of the sum of the following series is $\lim_{n\to\infty}\frac{1}{n}\left(\frac{1}{n+1}+\frac{2}{n+2}+.....+\frac{3n}{4n}\right)$

(a) 3 – log 4
(b) 2 – log 4
(c) 3 – log 2
(d) None of these

7. $\lim_{n\to\infty}\left[\frac{1}{n}+\frac{n^2}{(n+1)^3}+\frac{n^2}{(n+2)^3}+.....+\frac{1}{8n}\right]$

(a) 3/8
(b) 4/8
(c) 2/8
(d) 1/8

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ
6. ⓐ ⓑ ⓒ ⓓ
7. ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

1. (b)

$$\lim_{n\to\infty}\frac{1}{n+1}+\frac{1}{n+2}+.....+\frac{1}{n+2n}=\lim_{n\to\infty}\sum_{r=1}^{2n}\frac{1}{n+r}$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{1}{1+\frac{r}{n}}=\int_0^2\frac{1}{1+x}dx=\log(1+x)\Big|_0^2$$

$$=\log 3$$

4. (d) $\lim_{n\to\infty}\frac{1^4+2^4+3^4+.....,n^4}{n^4}.\frac{1}{n}-$

$$\lim_{n\to\infty}\frac{1^3+2^3+3^3+.....n^3}{n^5}$$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^4-0$$

$$=\int_0^1 x^4dx=\frac{x^5}{5}\Big|_0^1=\frac{1}{5}$$

5. (d) $f(n)=\left[\frac{(n+1)(n+2).....(n+n)}{n^n}\right]^{1/n}$

$$f(n)=\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right).....\left(1+\frac{n}{n}\right)\right]^{1/n}$$

$$\log f(n)=\frac{1}{n}\left[\log\left(1+\frac{1}{n}\right)+\log\left(1+\frac{2}{n}\right)\right.$$

$$\left.+....\log\left(1+\frac{n}{n}\right)\right]$$

$$=\lim_{n\to\infty}\log f(n)=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\log\left(1+\frac{r}{n}\right)$$

$$=\log\left(\lim_{n\to\infty}f(n)\right)=\int_0^1\log(1+x)dx$$

$$=\left[(1+x)\log(1+x)-(1+x)\right]_0^1$$

$$\log\left(\lim_{n\to\infty}f(n)\right)=2\log 2-1=\log(4/e)$$

$$\lim_{n\to\infty}f(n)=\frac{4}{e}$$

6. (a) $\lim_{n\to\infty}\frac{1}{n}\left(\frac{1}{n+1}+\frac{2}{n+2}+.....\frac{3n}{n+3n}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{3n}\frac{r}{n+r}=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{3n}\frac{r/n}{1+\frac{r}{n}}$$

$$=\int_0^3\frac{x}{1+x}dx=\int_0^3\left(1-\frac{1}{1+x}\right)dx$$

$$=\left[x-\log(1+x)\right]_0^3$$

$$=3-\log 4$$

7. (a) $\lim_{n\to\infty}\left[\frac{n^2}{n^3}+\frac{n^2}{(n+1)^3}+.....\frac{n^2}{(n+n)^3}\right]$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{n}\frac{n^3}{(n+r)^3}=\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{n}\frac{1}{\left(1+\frac{r}{n}\right)^3}$$

$$=\int_0^1\frac{1}{(1+x)^3}dx=\frac{1}{-2(1+x)^2}\Big|_0^1=\frac{-1}{2}\left[\frac{1}{4}-1\right]$$

$$=\frac{3}{8}$$

LECTURE



6 Area Under Curve 1

BASIC CONCEPTS

AREAS OF BOUNDED REGIONS

1. **Curve tracing**
In order to find the area bounded by several curves, it is important to have rough sketch of the required portion. The following steps are very useful in tracing a cartesian curve $f(x, y) = 0$.

Step 1: Symmetry

(i) The curve is symmetrical about x-axis if all powers of y in the equation of the given curve are even.

(ii) The curve is symmetrical about y-axis if all powers of x in the equation of the given curve are even.

(iii) The curve is symmetrical about the line $y = x$, if the equation of the given curve remains unchanged on intercharging x and y.

(iv) The curve is symmetrical in opposite quadrants, if the equation of the given curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively.

Step 2: Origin

If there is no constant term in the equation of the given curve, then the curve passes through the origin.

In that case, the tangents at the origin are given by equating to zero the lowest degree terms in the equation of the given curve.

For example, the curve $y^3 = x^3 + axy$ passes through the origin and the tangents at the origin are given by $axy = 0$ i.e., $x = 0$ and $y = 0$.

Step 3: Intersection with the Co-ordinate Axes

(i) To find the points of intersection of the curve with x-axis, put $y = 0$ in the equation of the given curve and get the corresponding values of x.

(ii) To find the points of intersection of the curve with y-axis, put $x = 0$ in the equation of the given curve and get the corresponding values of y.

Step 4: Asymptotes

Find out the asymptotes of the curve.

(i) The vertical asymptotes or the asymptotes parallel to y-axis of the given curve are obtained by equating to zero the coefficient of the highest power of y in the equation of the given curve.

(ii) The horizontal asympototes or the asympotes parallel to x-axis of the given curve are obtained by equating to zero the coefficient of the highest power of x in the equation of the given curve.

Step 5: Region

Find out the regions of the plane in which no part of the curve lies. To determine such regions we solve the given equation for y in terms of x or

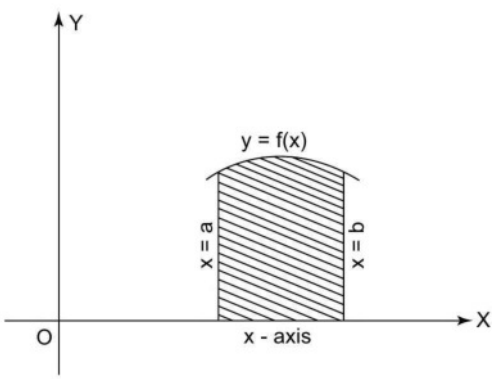
vice-versa. Suppose that y becomes imaginary for $x > a$, the curve does not lie in the region $x > a$.

Step 6: Critical Points

Find out the values of x at which $\frac{dy}{dx} = 0$. At such points y generally changes its character from an increasing function of x to a decreasing function of x or vice-versa.

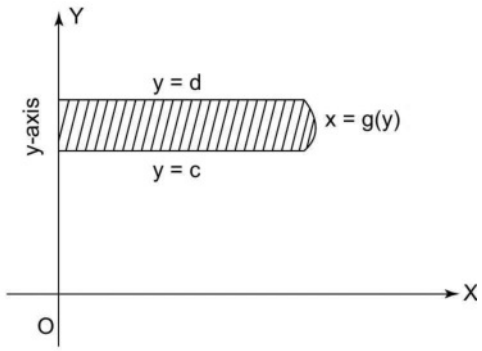
2. **Area of Plane Regions**

(i) The area bounded by the curve $y = f(x)$, x-axis and



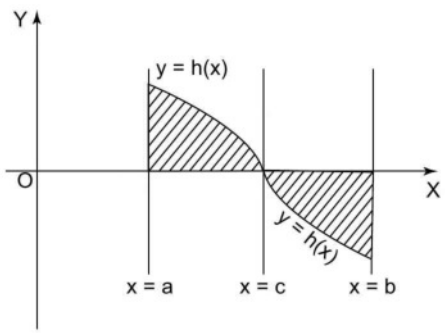
the ordinate $x = a$ and $x = b$ (where $b > a$) is given by $A = \int_a^b y\,dx = \int_a^b f(x)\,dx$

(ii) The area bounded by the curve $x = g(y)$, y-axis and the abscissae $y = c$ and $y = d$ (where $d > c$) is given by



$$A = \int_c^d x\,dy = \int_c^d g(y)\,dy$$

(iii) The area bounded by the curve $y = h(x)$, x-axis and the two ordinate $x = a$ and $x = b$ is given by



$$A = \left|\int_a^c y\,dx\right| + \left|\int_c^b y\,dx\right| = \int_a^c h(x)\,dx - \int_c^b h(x)\,dx$$

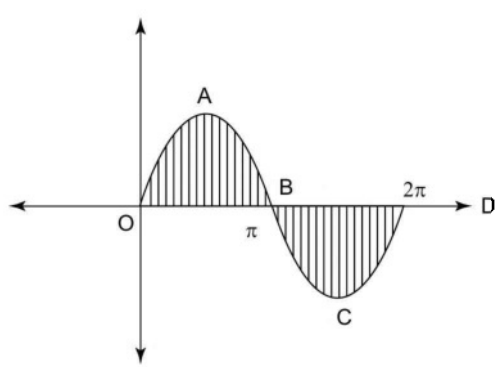
where c is a point in between a and b.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)) FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Find the area bounded by the curve $y = x \sin x$, x-axis from $x = 0$ to $x = 2\pi$.

[MP-99; Roorkee-81; RPET-1985]

Solution



Hence, required area = Area OAB + Area BCD

$$= \int_0^{\pi} y\,dx + \int_{\pi}^{2\pi} -y\,dx$$

$$= \int_0^{\pi} x\sin x\,dx - \int_{\pi}^{2\pi} x\sin x\,dx$$

$$= \left[-x\cos x + \sin x\right]_0^{\pi} - \left[-x\cos x + \sin x\right]_{\pi}^{2\pi}$$

$= [-\pi\cos\pi + \sin\pi - 0] - [-2\pi\cos 2\pi + \sin 2\pi + \pi\cos\pi - \sin\pi]$

$= (\pi) - (-2\pi - \pi)$

$= \pi + 3\pi = 4\pi$ sq. units.

2. Sketch the region bounded by $y = 2x - x^2$ and x-axis and find its area using integration.

[CBSE-1993; KCET-2008]

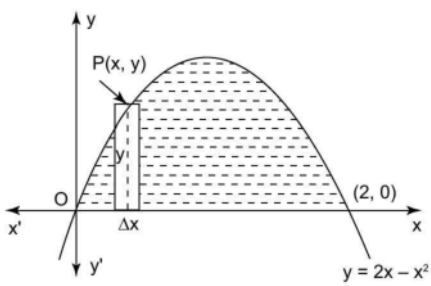
Solution

We have $y = 2x - x^2$

Clearly, it represents a parabola opening downward which cuts x-axis at (0, 0) and (2, 0). The rough sketch of the curve is as shown in fig. The required region is the shaded region in fig.

Here, we slice this region into verticle strips. For the approximating rectangle shown in fig., we have

Length = y, width = Δx and area = $y\Delta x$



The approximating rectangle can move from $x = 0$ to $x = 2$

$\therefore$ Required area $= \int_0^2 y\,dx = \int_0^2 (2x - x^2)\,dx$

[$\because$ $P(x, y)$ lies on $y = 2x - x^2$]

$= \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$ sq. units.

3. Using integration, find the area of the region bounded by the following curves after making a rough sketch. $y = 1 + |x + 1|$, $x = -3$, $x = 3$, $y = 0$. ***[CBSE-95]***

Solution

We have $y = 1 + |x + 1|$

$$\Rightarrow y = \begin{cases} 1 + x + 1 & \text{if} \quad x + 1 \geq 0 \\ 1 - (x + 1) & \text{if} \quad x + 1 < 0 \end{cases}$$

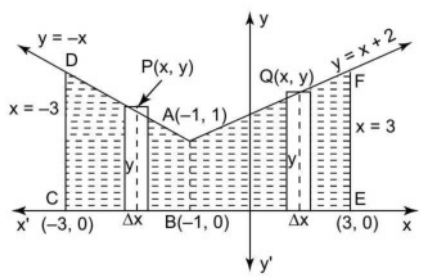
$$= \begin{cases} x + 2 & \text{if} \quad x \geq -1 \\ -x & \text{if} \quad x < -1 \end{cases}$$

Thus the equations of the given curves are $y = x + 2$ for $x \geq -1$, $y = -x$ for $x < -1$, $x = -3$, $x = 3$ and $y = 0$

Clearly, $y = x + 2$ is a straight line cutting x-axis and y-axis at (–2, 0) and (0, 2) respectively. So, $y = x + 2$, for $x > -1$ represents the part of the line which is on the right side of $x = -1$.

Clearly, $x = -3$ and $x = 3$ are lines parallel to y-axis. The region bounded by the given curves is shaded portion shown in figure.

We slice the shaded region into verticle strips. We observe that verticle strips change their character at A.



So, required area = Area $CDAB$ + Area $ABEF$

Area *CDAB*: In area $CDAB$, each verticle strip has its upper end on $y = -x$ and lower end on x-axis. So, the approximating rectangle shown in figure has length = y, width = Δx and area = $y\Delta x$. Since the approximating rectangle can move from $x = -3$ to $x = -1$. $\therefore$ Area $CDAB = \int_{-3}^{-1} y\,dx = \int_{-3}^{-1} -x\,dx$ [$\because$ $P(x, y)$ lies on $y = -x$]

Area *ABEF*: In area $ABEF$, each verticle strip has its lower end on x-axis and the upper end on $y = x + 2$. So, the approximating rectangle has length = y, width = Δx and area = $y\Delta x$. As it can move from $x = -1$ to $x = 3$.

$\therefore$ Area $ABEF = \int_{-1}^{3} y\,dx = \int_{-1}^{3} (x + 2)\,dx$

[$\because$ $Q(x, y)$ lies on $y = x + 2$]

Hence required area = Area $CDAB$ + Area $ABEF = \int_{-3}^{-1} -x\,dx + \int_{-1}^{3} (x + 2)\,dx$

$$= -\left[\frac{x^2}{2}\right]_{-3}^{-1} + \left[\frac{x^2}{2} + 2x\right]_{-1}^{3}$$

$$= -\left[\frac{1}{2} - \frac{9}{2}\right] + \left[\left(\frac{9}{2} + 6\right) - \left(\frac{1}{2} - 2\right)\right]$$

= 16 sq. units.

4. Using integration, find the area of the region bounded by the line $2y = 5x + 7$, x-axis and the lines $x = 2$ and $x = 8$.

[CBSE-1995]

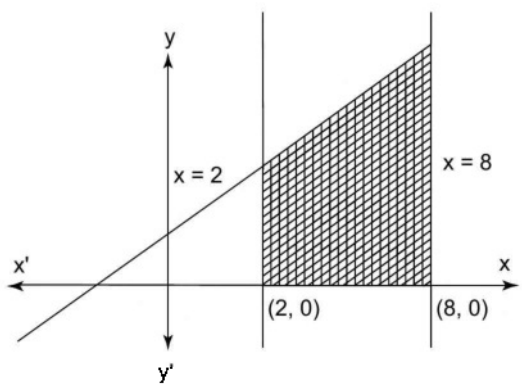
Solution

Given equations in the curves are

$2y = 5x + 7$ ..... (1)

$x = 2$ ..... (2)

$x = 8$ ..... (3)



Equation (1) represents the line passing through the points $\left(0, \frac{7}{2}\right)$ and $\left(-\frac{7}{5}, 0\right)$. equation (2) represents the line parallel to y-axis and passing through (2, 0). Equation (3) represents the line parallel to y-axis and passes through (8, 0).

$(1) \Rightarrow y = \frac{5x+7}{2}$

Required area $= \int_2^8 y\,dx = \int_2^8 \left(\frac{5x+7}{2}\right) dx$

$= \frac{1}{2}\left[\frac{5x^2}{2} + 7x\right]_2^8$

$= \frac{1}{2}\left[\left(\frac{5(64)}{2} + 56\right) - \left(\frac{5(4)}{2} + 14\right)\right]$

$= \frac{1}{2}[(160 + 56) - (10 + 14)]$

$= \frac{1}{2}[216 - 24] = \frac{192}{2} = 96$ sq. units.

5. Let A_n be the area bounded by the curve $y = (\tan x)^n$ and the lines $x = 0$, $y = 0$ and $x = \frac{\pi}{4}$. Prove that for $n \geq 2$, $A_n + A_{n-2} = \frac{1}{n-1}$ and deduce $\frac{1}{2n+2} < A_n < \frac{1}{2n-2}$ ***[IIT-1996]***

Solution

$$A_n = \int_0^{\pi/4} \tan^n x\,dx = \int_0^{\pi/4} \tan^{n-2}(\sec^2 x - 1)\,dx$$

or $A_n = \left[\frac{\tan^{n-1} x}{n-1}\right]_0^{\pi/4} - A_{n-2}$

$\therefore A_n + A_{n-2} = \frac{1}{n-1}$ ..... (1)

Again in the interval 0 to $\pi/4$, $\tan x$ is +ive and < 1.

$\therefore \tan^n x \leq \tan^{n-2} x$

$\therefore A_n < A_{n-2}$

$\therefore A_n + A_n < A_n + A_{n-2} = \frac{1}{n-1}$, by (1)

$\therefore A_n < \frac{1}{2(n-1)}$ ..... (2)

$A_{n-2} < A_n \therefore A_{n-2} + A_n < A_n + A_n$

or $\frac{1}{(n+2)-1} < 2A_n$ by (1)

$\therefore \frac{1}{2n+2} < A_n$ ..... (3)

$\therefore \frac{1}{2n+2} < A_n < \frac{1}{2n-2}$ by (2) and (3)

UNSOLVED SUBJECTIVE PROBLEMS (XI BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Find the area bounded by the y-axis, the curve $x = \log y$ and the abscissae $y = a$, $y = b$ $(b > a > 1)$.

2. Find the area bounded by the y-axis, the curve $y = \log x$ and the abscissae $y = 1$, $y = 2$.

3. Find the area of the curve $y = \sin x$, the x-axis and $x = 0, x = \frac{\pi}{3}$. Also find the area of the curve $y = \sin 2x$, the x-axis and $x = 0$, $x = \frac{\pi}{3}$. Show that the ratio of the above two areas 2 : 3.
4. Find the area of the region bounded by the curve $xy - 3x - 2y - 10 = 0$, x-axis and the lines $x = 3$, $x = 4$ ***[CBSE-91(C)]***
5. Find the area bounded by the curve $y = \cos x$, the x-axis and $x = 0$, $x = 2\pi$ ***[MP-2001]***
6. Find the area bounded by the line $\frac{x}{a} + \frac{y}{b} = 1$ and the coordinate axes. ***[MP-99]***
7. Find the area of the region between the curve $y^2 = 4x$ and $x = 3$. ***[MP-98; CBSE-92(C)]***
8. Find the area bounded by the curve $y^2 = 4ax$ and the lines $y = 2a$ and y-axis. ***[CBSE-94]***
9. Find the area bounded by the curve $y^2 = 4a^2(x - 1)$ and the lines $x = 1$ and $y = 4a$. ***[CBSE-94]***
10. Draw a rough sketch of the curve $y = \cos^2 x$ in $[0, \pi]$ and find the area enclosed by the curve, the line $x = 0$, $x = \pi$ and the x-axis. ***[CBSE-90]***
11. Make a rough sketch of the graph of the function $y = \cos 3x$, $0 \le x \le \frac{\pi}{6}$, and determine the area enclosed between the curve and the axes. ***[DSB-84]***
12. Find the complete area of the circle $x^2 + y^2 = 25$. ***[MP-2005 (B)]***
13. Find the area of quadrant of a circle $x^2 + y^2 = 1$. ***[MP-99, 2005 (C)]***

ANSWERS

1. $b\log\frac{b}{e} - a\log\frac{a}{e}$ sq. units.
2. $e(e - 1)$ sq. units.
4. $3 + 16 \log 2$ sq. units.
5. 4 sq. units
6. $\frac{ab}{2}$ sq. units
7. $8\sqrt{3}$ sq. units
8. $\frac{2}{3}a^2$ sq. units
9. $\frac{16}{3}a$ sq. units
10. $\frac{\pi}{2}$ sq. units
11. 1/3 sq. units
12. 25π sq. unit
13. $\frac{\pi}{4}$ sq. units

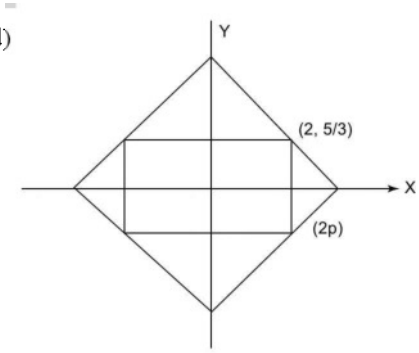
SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. The area of the quadrilateral formed by the tangent at the end points of latus recta to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$, is ***[IIT-2003]***

(a) 27/4 sq. units
(b) 9 sq. units
(c) 27/2 sq. units
(d) 27 sq. units

Solution

(d)



Equation of tangent at end of latus rectum

$$= \frac{x}{9/2} + \frac{y}{3} = 1$$

$$\text{Area} = 4 \times \left(\frac{1}{2} \left(\frac{9}{2} \times 3 \right) \right) = 27 \text{ sq. units.}$$

2. The area bounded by the curve $x = a\cos^3 t$, $y = a\sin^3 t$ is ***[IIT-91; ICS-2000]***

(a) $3\pi a^2/8$ (b) $3\pi a^2/16$

(c) $3\pi a^2/32$ (d) $3\pi a^2$

Solution

Eliminating t, cartesian equation of the curve is $x^{2/3} + y^{2/3} = a^{2/3}$ which shows that curve is symmetrical about both axes. Also putting $y = 0$, we get $x = a$. So the required area $= 4\int_0^a y\,dx$

But when $x = 0$ then $t = \pi/2$ and when, $x = a$ then $t = 0$. So area $= 4\int_{\pi/2}^{0} a\sin^3 t.(-3a\cos^2 t\sin t\,dt)$

$$= 12a^2 \int_0^{\pi/2} \sin^4 t\cos^2 t\,dt$$

$$= 12a^2.\frac{3.1.1\,\pi}{6.4.2\,2} = \frac{3\pi a^2}{8}$$

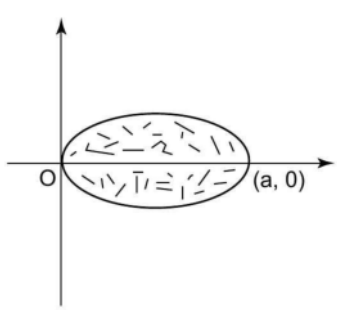
3. The area of a loop of the curve $ay^2 = x^2(a - x)$ is ***[Roorkee-85; PET (Raj.)-96, 97]***

(a) $8a^2/15$ (b) $4a^2/15$

(c) $a^2/15$ (d) $16a^2/15$

Solution

(a) The curve is symmetrical about x-axis. also on putting $x = 0$, we get $y = 0$ and on putting $y = 0$ we get $x = 0, a$. So one loop of the curve lies between $(0, 0)$ and $(a, 0)$. The area of this loop $= 2\int_0^a y\,dx = 2\int_0^a x\sqrt{\frac{a-x}{a}}\,dx$



$$= 4a^2 \int_0^{\pi/2} \sin^3 t\cos^2 t\,dt \text{ (on putting } x = a\sin^2 t)$$

$$= 4a^2.\frac{2.1}{5.3.1}\frac{8a^2}{15}$$

4. The area of the region bounded by the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ $(x, y > 0)$ and the coordinate axes is ***[CET-86]***

(a) a^2 (b) $a^2/2$

(c) $a^2/3$ (d) $a^2/6$

Solution

(d) area $= \int_0^a (\sqrt{a} - \sqrt{x})^2 dx.$

5. If the area between the parabola $y^2 = 4ax$ and its any double ordinate is k times the area of the rectangle formed by this double ordinate and the distance from the vertex, then value of k will be ***[DCE-98]***

(a) 2/3 (b) 3/2

(c) 1/3 (d) 3

Solution

(a) Let ends of a double ordinate be $(at^2, 2at)$ and $(at^2, -2at)$

Then area between parabola and this double ordinate $= 2\int_0^{at^2} 2\sqrt{a}\sqrt{x}dx$

$$= \frac{8}{3}\sqrt{a}[x^{3/2}]_0^{at^2} = \frac{8}{3}a^2t^3 \quad \ldots\ldots\ldots\ldots (1)$$

Area of the rectangle $= at^2 \times 4at = 4a^2t^3$

As given $\frac{8}{3}a^2t^3 = k(4a^2t^3) \Rightarrow k = \frac{2}{3}$

6. The area of the figure bounded by the parabola $(y - 2)^2 = x - 1$, the tangent to it at the point with ordinate 3 and x-axis is ***[UPSEAT-96; Roorkee-2000]***

(a) 9 (b) 6

(c) 3 (d) 1

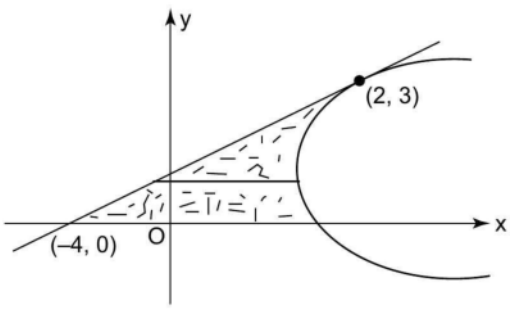
Solution

(a) Given point with ordinate 3 is (2, 3) tangent at this point is $x - 2y + 4 = 0$. It meets x-axis at $(-4, 0)$. The area is as shown in the diagram and area $= \int_{y=0}^{3} \{(y^2 - 4y + 5) - (2y - 4)\}dy$

$$= \int_0^3 (y^2 - 6y + 9)dy$$

$$= \left[\frac{y^3}{3} - 3y^2 + 9y\right]_0^3$$

$= 9 - 27 + 27 = 9.$



7. Area of the region bounded by the curve $y = \tan x$, tangent drawn to the curve at $x = \frac{\pi}{4}$ and the x-axis is

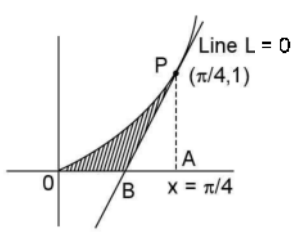
[DCE-2001]

(a) $\frac{1}{4}$ (b) $\log\sqrt{2} + \frac{1}{4}$

(c) $\log\sqrt{2} - \frac{1}{4}$ (d) None of these

Solution

(c) Curve is $y = \tan x$ BC is the tangent at P (or at $x = \pi/4$) $\frac{dy}{dx}$ at $P\left(\frac{\pi}{4}, 1\right) = \sec^2\frac{\pi}{4} = 2$



Equation of tangent at point P $(\pi/4, 1)$

$$y - 1 = 2\left(x - \frac{\pi}{4}\right), y - 1 = 2x - \frac{\pi}{2},$$

$$y = 2x - \frac{\pi}{4} + 1$$

The line meets at x-axis then $y = 0$

$$0 = 2x - \frac{\pi}{2} + 1 \Rightarrow x = \frac{\pi}{4} - \frac{1}{2} \text{ Point}$$

$$B = \left(\frac{\pi}{4} - \frac{1}{2}, 0\right) \text{ and } A = \left(\frac{\pi}{4}, 0\right)$$

$\Rightarrow AB = 1/2$, $PA = 1$; B area of

$$\Delta PBA = \frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$$

Required shaded area $A = \int_0^{\pi/4} \tan x dx$ – Area of ΔPBA

$$\Rightarrow -[\log(\cos x)]_0^{\pi/4} - \frac{1}{4} \Rightarrow \log\sqrt{2} - \frac{1}{4}$$

8. Area inside the parabola $y^2 = 4ax$, between the lines $x = a$ and $x = 4a$ is equal to

[Pb. CET-2002; Karnataka CET-2005]

(a) $4a^2$ (b) $8a^2$

(c) $28\frac{a^2}{3}$ (d) $35\frac{a^2}{3}$

Solution

(c) We have $y^2 = 4ax \Rightarrow y = 2\sqrt{ax}$

We know the equations of lines $x = a$ and $x = 4a$ ∴ The area inside the parabola between the lines

$$A = \int_a^{4a} y dx = \int_a^{4a} 2\sqrt{ax} dx = 2\sqrt{a}$$

$$= \int_a^{4a} x^{\frac{1}{2}} dx = 2\sqrt{a}\left[\frac{x^{3/2}}{\frac{3}{2}}\right]_a^{4a}$$

$$\frac{4}{3}a^{\frac{1}{2}}\left[(4a)^{\frac{3}{2}} - (a)^{\frac{3}{2}}\right]$$

$$= \frac{4}{3}a^{\frac{1}{2}}a^{\frac{3}{2}}[8 - 1] = \frac{28}{3}a^2$$

9. The area of the region between the curves $y = \sqrt{\frac{1 + \sin x}{\cos x}}$ and $y = \sqrt{\frac{1 - \sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is

[IIT JEE-2008]

(a) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

(b) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

(c) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

(d) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

Solution

(b) The desired area

$$= \int_0^{\pi/4} \left(\sqrt{\frac{1+\sin x}{\cos x}} - \sqrt{\frac{1-\sin x}{\cos x}} \right) dx$$

Now $$\frac{1+\sin x}{\cos x} = \frac{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)^2}{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}$$

$$= \frac{\cos\frac{x}{2}+\sin\frac{x}{2}}{\cos\frac{x}{2}-\sin\frac{x}{2}} = \frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}$$

The expression for area reduces to

$$\int_0^{\pi/4} \left(\sqrt{\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}} - \sqrt{\frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}}} \right) dx$$

$$= \int_0^{\pi/4} \frac{2\tan\frac{x}{2}}{\sqrt{1-\tan^2\frac{x}{2}}} dx$$

Let $\tan\frac{x}{2} = t$

$\Rightarrow$ the expression $= \int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

10. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is ***[MPPET-2009]***

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$

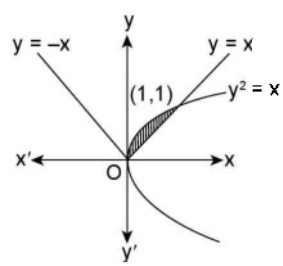
(c) $\frac{2}{3}$ (d) 1

Solution

(a)

Given curves are $y^2 = x$ and $y = |x|$

$\therefore$ Required area $= \int_0^1 \left(\sqrt{x} - x\right) dx$



$$= \left[\frac{2x^{3/2}}{3} - \frac{x^2}{2}\right]_0^1 = \left[\frac{3}{2} - \frac{1}{2}\right] = \frac{1}{6} \text{ sq. unit}$$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. Area bounded by the parabola $y^2 = 4ax$ and its latus rectum is ***[RPET-1997, 2000,2002]***

(a) $\frac{2}{3}a^2$ sq. unit (b) $\frac{4}{3}a^2$ sq. u.

(c) $\frac{8}{3}a^2$ sq. u. (d) $\frac{3}{8}a^2$ sq. u.

2. Area bounded by the curve $y = \log x$, x-axis and the ordinates $x = 1$, $x = 2$ is ***[MPPET-2004]***

(a) log 4 sq. units (b) log 4 + 1 sq. units

(c) log 4 – 1 sq. units (d) None of these

3. Area bounded by the curve $y = \sin x$ between $x = 0$ and $x = 2\pi$ is

(a) 2 sq. units (b) 4 sq. units

(c) 8 sq. units (d) None of these

4. Area enclosed by the parabola $ay = 3(a^2 - x^2)$ and x-axis is

(a) $4a^2$ sq. units (b) $12a^2$ sq. units

(c) $4a^3$ sq. units (d) None of these

5. Area under the curve $y = \sin 2x + \cos 2x$ between $x = 0$ and $x = \frac{\pi}{4}$ is

[AICBSE-1979]

(a) 2 sq. units (b) 1 sq. units
(c) 3 sq. units (d) 4 sq. units

6. The area bounded by curve $y^2 = x$, line $y = 4$ and y-axis is

[Roorkee-1995; RPET-2003]

(a) 16/3 (b) 64/3
(c) $7\sqrt{2}$ (d) None of these

7. The measurement of the area bounded by the coordinate axes and the curve $y = \log_e x$ is

[MPPET-1998]

(a) 1 (b) 2
(c) 3 (d) ∞

8. The area between the curve $y = \sin^2 x$, x-axis and the ordinates $x = 0$ and $x = \frac{\pi}{2}$ is

[RPET-1996; MP PET-07]

(a) $\pi/2$ (b) $\pi/4$
(c) $\pi/8$ (d) π

9. The area of the region bounded by the x-axis and the curve defined by $y = \tan x$ $(-\pi/3 \le x \le \pi/3)$ is

[Kurukshetra CEE-1998]

(a) $\log \sqrt{2}$ (b) $-\log \sqrt{2}$
(c) $2 \log 2$ (d) 0

10. The area between the curve $y = 4 + 3x - x^2$ and x-axis is ***[RPET-2001]***

(a) 125/6 (b) 125/3
(c) 125/2 (d) None of these

11. If the area above the x-axis, bounded by the curve $y = 2^{kx}$ and $x = 0$ and $x = 2$ is $\frac{3}{\ln 2}$, then the value of k is ***[Orissa JEE-2003]***

(a) 1/2 (b) 1
(c) -1 (d) 2

12. If a curve $y = a\sqrt{x} + bx$ passses through the point (1, 2) and the area bounded by the curve, line $x = 4$ and x-axis is 8 square unit, then

[MPPET-2002]

(a) $a = 3, b = -1$ (b) $a = 3, b = 1$
(c) $a = -3, b = 1$ (d) $a = -3, b = -1$

13. Area enclosed between the curve $y^2(2a - x) = x^3$ and the line $x = 2a$ above x-axis is

[MPPET-2001]

(a) πa^2 (b) $3\pi a^2/2$
(c) $2\pi a^2$ (d) $3\pi a^2$

14. The area bounded by the x-axis, the curve $y = f(x)$ and the lines $x = 1$, $x = b$ is equal to $\sqrt{b^2 + 1} - \sqrt{2}$ for all $b > 1$, then $f(x)$ is

[MPPET-2000; AMU-2000]

(a) $\sqrt{x - 1}$ (b) $\sqrt{x + 1}$
(c) $\sqrt{x^2 + 1}$ (d) $x/\sqrt{1 + x^2}$

15. If the ordinate $x = a$ divides the area bounded by the curve $y = \left(1 + \frac{8}{x^2}\right)$ x-axis and the ordinates $x = 2$, $x = 4$ into two equal parts, then $a =$ ***[IIT-1983]***

(a) 8 (b) $2\sqrt{2}$
(c) 2 (d) $\sqrt{2}$

16. Area ofthe region bounded by the curves $x = at^2$, $y = 2at$ and x-axis is (between the limits $1 \le t \le 3$)

(a) $26a^2$ (b) $8a^2$
(c) $26a^2/3$ (d) $104a^2/3$

17. The area of the region bounded by $y = -1, y = 2$, $x = y^3$, $x = 0$ is

(a) 11/4 sq. units (b) 17/4 sq.units
(c) 23/4 sq. units (d) None of these

18. Draw a rough graph of $y^2 + 1 = x$, $x \le 2$ and Find the area enclosed by the curve and the ordinate $x = 2$

(a) 1/3 sq.units (b) 2/3 sq. units
(c) 4/3 sq. units (d) None of these

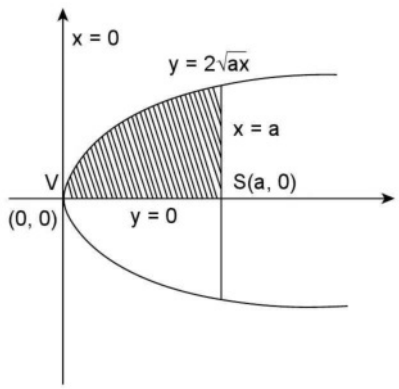
19. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval ***[IIT (2)-2010]***

(a) $\left(\frac{3}{4}, 3\right)$ (b) $\left(\frac{21}{64}, \frac{11}{16}\right)$

(c) (9, 10) (d) $\left(0, \frac{21}{64}\right)$

SOLUTIONS

1. (c) Step 1:



Required Area $\Rightarrow$ curve $y = 2\sqrt{ax}$, $y = 0$ is twice of the area bounded by the curve ordinate, $x = a$ and x-axis is

$$= 2\int_0^a 2\sqrt{ax}\,dx = 4\sqrt{a} \times \frac{x^{3/2}}{\frac{3}{2}}\Bigg|_0^a$$

$$= \frac{8\sqrt{a}}{3}\left[a^{3/2} - 0\right] = \frac{8a^2}{3}$$

2. (c) Given curve $y = \log x$ and $x = 1$, $x = 2$

Hence, required area $= \int_1^2 \log x\,dx = (x \log x - x)_1^2$

$= 2\log 2 - 1 = (\log 4 - 1)$ sq. unit

3. (b) Area is in squre unit Hence it values is always positive

Hence from the defination of bounded area

$$= \int_0^{2\pi} |x\sin x|\,dx + \int_0^{\pi} (x\sin x)\,dx + \int_{\pi}^{2\pi} -(x\sin x)\,dx$$

$$[-x\cos x + \sin x]_0^{\pi} - [-x\cos x + \sin x]_{\pi}^{2\pi} = 4\pi$$

NOTES

1. Curve $y = f(x)$, x-axis ($y = 0$) and area bounded by ordinates $x = 0$ and $x = b$ ($b > a$) is

$$= \int_a^b f(x)\,dx \text{ or } \int_a^b |f(x)|\,dx$$

2. If some part of required area is above x-axis and some part is below x-axis then we divide the interval $[a, b]$ in sub interval in which area is either above x-axis or below x-axis
3. If we neglect the note 2 then for finding area we can take only modulus of the function and the range of the limit is change by taking modulus of the function in different some intervals for which we can find value of the $\int_a^b |f(x)|\,dx$

4. (a) Area $= \int_{-a}^{a} \frac{3}{a}(a^2 - x^2)\,dx = \frac{6}{a}\int_0^a (a^2 - x^2)\,dx$

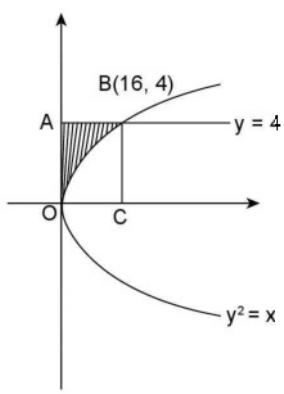
$$= \frac{6}{a}\left(a^2 x\Big|_0^a - \frac{x^3}{3}\right) = \frac{6}{a}\left(a^2 - \frac{a^3}{3}\right) = 4a^2$$

5. (b) Required area $= \int_0^{\pi/4} (\sin 2x + \cos 2x)\,dx$

$$= \left[-\frac{\cos 2x}{2} + \frac{\sin 2x}{2}\right]_0^{\pi/4}$$

$$= \frac{1}{2}\left[-\cos\frac{\pi}{2} + \sin\frac{\pi}{2} + \cos 0 - \sin 0\right] = 1 \text{ sq. unit}$$

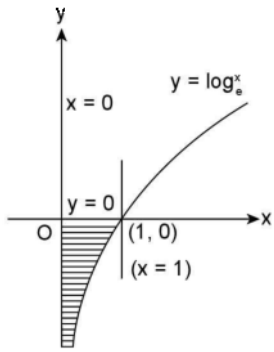
6. (b)



Required area = area of $OABC$ – area of OBC

$$= 16 \times 4 - \int_0^{16} \sqrt{x}\,dx = 64 - \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^{16} = \frac{64}{3}$$

7. (d) Step 1:



Step 2: Required Area

$$= \int_0^1 |\log x|\, dx = \int_0^1 -\log x dx = -(x\log x - x)|_0^1$$

$$= -(\log 1 - 1)\left(\lim_{x>0} x\log x - 0\right)$$

$$= (0-1) - (0-0) = 1$$

8. (b) Required area

$$A = \int_0^{\pi/2} \sin^2 x dx = \int_0^{\pi/2}\left(\frac{1-\cos 2x}{2}\right)dx$$

$$= \frac{1}{2}[x]_0^{\pi/2} - \frac{1}{4}[\sin 2x]_0^{\pi/2} = \frac{\pi}{4}$$

9. (c) Required area $= 2\int_0^{\pi/3} \tan x dx$

$$= 2[\log \sec x]_0^{\pi/3} = 2\log(2)$$

10. (a) Solving $y = 0$ and $y = 4 + 3x - x^2$, we get $x = -1$ and $x = 4$

$\because$ curve does not intersect x-axis between $x = -1$ and $x = 4$

$$\therefore \text{Area} = \int_{-1}^{4}(4+3x-x^2)dx = 4x + \frac{3x^2}{2} - \frac{x^3}{3}\Big|_{-1}^{4}$$

$$= 16 + 24 - \frac{64}{3} - \left[-4 + \frac{3}{2} + \frac{2}{6}\right]$$

$$= \frac{112+13}{6} = \frac{125}{6}$$

11. (b) $\int_0^2 2^{kx} dx = \frac{3}{\log 2} \Rightarrow \frac{2^{kx}}{\log 2}\cdot\frac{1}{k}\Big|_0^2 = \frac{3}{\log 2}$

$$= \frac{1}{k\log 2}\left[2^{2k} - 1\right] = \frac{3}{\log 2}$$

$$= 2^{2k} - 1 = 3k$$

options, only (b) satisfies the above condition.

12. (a) Step 1: Limit of a definite integral is defined from the point of intersection of the curve and x-axis

Step 2: From defination required area

$$= \int_0^4 \left(a\sqrt{x} + bx\right)dx$$

$$= \left(\frac{ax^{3/2}}{3/2} + \frac{bx^2}{2}\right)\Big|_0^4 = \frac{16a}{3} + 8b$$

Step 3: According to question $\frac{16a}{3} + 8b = 8$

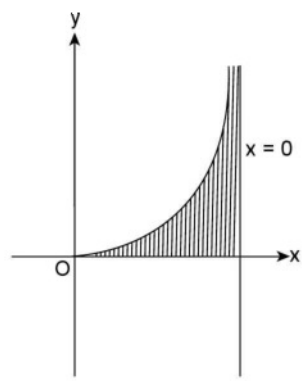
and $2 = a + b \Rightarrow b = 2 - a$

Step 4: $\frac{2a}{3} + b = 1$

$$\frac{2a}{3} + (2-a) = 1 \Rightarrow \frac{-a}{3} = -1$$

$\Rightarrow$ $a = 3$; Hence $b = -1$

13. (b) Curve $y^2(2a - x) = x^3$ is symmetrical about x-axis and passes through origin.



Also $\frac{x^3}{2a-x} < 0$ for $x > 2a$ or $x < 0$. So does not lie in $x > 2a$ and $x < 0$.

Hence curve lies completely on $0 \le x \le 2a$

$$\therefore \text{Area} = \int_0^{2a} \frac{x^{3/2}}{\sqrt{2a-x}} dx$$

Put $x = 2a \sin^2\theta$

As $x = 2a$ to $0 = \frac{\pi}{2}$ and $x = 0$ to $0 = 0$

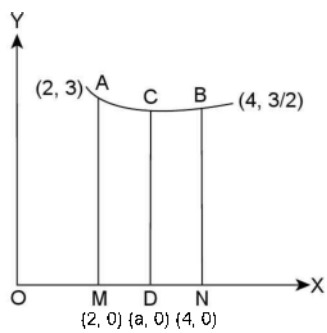
$$= \int_0^{\pi/2} 8a^2 \sin^4\theta d\theta = 8a^2\left[\frac{3}{4}, \frac{1}{2}, \frac{\pi}{2}\right] = \frac{3}{2}\pi a^2$$

Note: Applying Gamma function.

14. (d) $\int_1^b f(x)dx = \sqrt{b^2+1} - \sqrt{2} = \sqrt{b^2+1} - \sqrt{1+1} = \left[\sqrt{x^2+1}\right]_1^b$

$\therefore \quad f(x) = \dfrac{d}{dx}\sqrt{x^2+1} = \dfrac{2x}{2\sqrt{x^2+1}} = \dfrac{x}{\sqrt{x^2+1}}$

15. (b) Step 1: Area bounded by curve $y = 1 + \dfrac{8}{x^2}$, x-axis and ordinates $x = 2$ and $x = 4$ is $\int_2^4 \left(1 + \dfrac{8}{x^2}\right)dx$



$= \left(x - \dfrac{8}{x}\right)\Big|_2^4$

$= \left(4 - \dfrac{8}{4}\right) - \left(2 - \dfrac{8}{2}\right)$

$= 2 - (-2) = 4$

Step 2: According to question straight line $x = a$ divide the given area in two equal parts.

Hence $\int_2^a \left(1 + \dfrac{8}{x^2}\right)dx = 2$

$\left(x - \dfrac{8}{x}\right)\Big|_2^a = 2$

$\left(a - \dfrac{8}{a}\right) - (2-4) = 2$

$a^2 - 8 = 0 \Rightarrow a = 2\sqrt{2}$

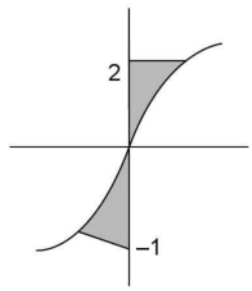
16. (d) Area $= \int_{x_1}^{x_2} y\,dx = \int_1^3 2at\,d(at^2)$

$= 2a^2\int_1^3 t(2t\,dt)$

$= 4a^2\int_1^3 t^2 dt = 4a^2 \times \dfrac{26}{3}$

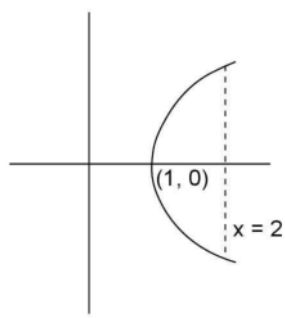
$= \dfrac{104a^2}{3}$

17. (d) Area $= \int_{-1}^{2} y^3 dx = \dfrac{y^4}{4}\Big|_{-1}^{2}$



$= \dfrac{16-(-1)}{4} = \dfrac{15}{4}$ sq. units

18. (c) Curve $y^2 + 1 = x$ or $y^2 = x - 1$; $x \le 2$



Area $= 2\int_1^2 \sqrt{x-1}dx = 2\dfrac{(x-1)^{3/2}}{3/2}\Big|_1^2$

$= \dfrac{4}{3}(1-0) = \dfrac{4}{3}$ sq. units.

19. (a) By estimation of integration

$$\int_0^{1/2} f(x)dx < \int_0^1 f(x)dx < \int_0^{3/4} f(x)dx$$

$$\Rightarrow \quad \frac{15}{16} < \int_0^1 f(x)dx < \frac{525}{256}$$

Hence option (a) is correct.

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The area bounded by the curve $y = \log x$, the x-axis and ordinate $x = e$ is
[MPPET-94; Pb. CET-2000]
(a) e (b) 1
(c) ∞ (d) None of these

2. The area enclosed by the curve $y = \sin x$, $y = 0$, $x = 0$ and $x = \frac{\pi}{2}$ is
(a) π (b) 2π
(c) 1 (d) 2

3. The area bounded by the curve $y = x^3$, x-axis and two ordinates $x = 1$ to $x = 2$ equal to
[MPPET-1994]
(a) 15/2 sq. unit (b) 15/4 sq. unit
(c) 17/2 sq. unit (d) 17/4 sq. unit

4. The area between the curve $y^2 = 4ax$, x-axis and the ordinates $x = 0$ and $x = a$ is
[RPET-1996]
(a) $\frac{4}{3}a^2$ (b) $\frac{8}{3}a^2$
(c) $\frac{2}{3}a^2$ (d) $\frac{5}{3}a^2$

5. The area bounded by the curve $y = x$, x-axis and ordinates $x = -1$ to $x = 2$ is
[RPET-2001]
(a) 0 (b) 1/2
(c) 3/2 (d) 5/2

6. The area of the region (in the square units) bounded by the curve $x^2 = 4y$, line $x = 2$ and x-axis is *[MPPET-2002]*
(a) 1 (b) 2/3
(c) 4/3 (d) 8/3

7. The area of the curve $xy^2 = a^2(a - x)$ bounded by y-axis is *[RPET-1996]*
(a) πa^2 (b) $2\pi a^2$
(c) $3\pi a^2$ (d) $4\pi a^2$

8. The area bounded by the curve $y = f(x)$, x-axis and ordinates $x = 1$ and $x = b$ is $(b - 1)\sin(3b + 4)$, then $f(x)$ is *[RPET-2000]*
(a) $3(x - 1)\cos(3x + 4) + \sin(3x + 4)$
(b) $(b - 1)\sin(3x + 4) + 3\cos(3x + 4)$
(c) $(b - 1)\cos(3x + 4) + 3\sin(3x + 4)$
(d) None of these

9. The area bounded by the x-axis and the curve $y = 4x - x^2 - 3$ is
(a) 4/3 (b) 3/4
(c) 7 (d) 3/2

10. The slope of the tangent to a curve $y = f(x)$ at $\{x, f(x)\}$ is $2x + 1$. If the curve passes through the point (1, 2), then area bounded by the curve x-axis and the line $x = 1$ is
(a) 5/6 (b) 6/5
(c) 1/6 (d) 6

11. The area bounded by the curve $y = \frac{3}{x^2}$, x-axis and the ordinates $x = 1$ to $x = 2$ is
(a) 5/2 (b) 3/2
(c) 3 (d) 1

12. Area bounded by the curve $y = x(x - 1)(x - 2)$ and x-axis, is
(a) 1/2 (b) 1
(c) 0 (d) 3/2

13. Caculate the area of the region bounded by the curve $xy = x^2 + 1$, the x-axis and the lines $x = 1$ and $x = 2$
(a) $\frac{3}{2} - \log 2$ (b) $\frac{3}{2} + \log 2$
(c) $\frac{3}{2}\log 2$ (d) $3 \log 2$

14. Find the area enclosed between the curve $y^2 = x^2(4 - x^2)$, the coordinate axes and the ordinate $x = 2$
(a) 2/3 sq. units (b) 4/3 sq. units
(c) 8/3 sq. units (d) 32/3 sq. units

15. A plane area is bounded by the curve $y = x(4 - x)$ and the x-axis. Then its area will be
[MPPET-99, 2003]
(a) 16/3 sq.units (b) 32/3 sq. units
(c) 64/3 sq. units (d) 8/3 sq. units

16. Find the area bounded by the curve $y = 4x(x - 1)(x - 2)$ and the x-axis
(a) 1/2 sq. units (b) 3/2 sq. units
(c) 2 sq. units (d) None of these

17. The area bounded by $y = [x]$ and the two ordinates $x = 1$ and $x = 1.7$ is
(a) 17/10 (b) 1
(c) 17/5 (d) 7/10

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 13 minutes.
3. The worksheet consists of 13 questions. The maximum marks are 39.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. Area under the curve $y = \sqrt{3x+4}$ between $x = 0$ and $x = 4$, is ***[AICBSE -1979, 80]***

(a) 56/9 sq. units (b) 64/9 sq. units
(c) 8 sq. units (d) None of these

2. Area under the curve $y = x^2 - 4x$ within the x-axis and the line $x = 2$, is ***[SCRA-1991]***

(a) 16/3 sq. units
(b) –16/3 sq. units
(c) 4/7 sq. units
(d) Cannot be calculated

3. The area bounded by the st. line $y = 3x$, $y = 0$, $x = 2$, $x = 5$

(a) 13.5 sq. units (b) 26.5 sq. units
(c) 28 sq. units (d) 31.5 sq. units

4. The area bounded by the parabola $y^2 = 4ax$, and two ordinates $x = 4$, $x = 9$ is

(a) $4a^2$ (b) $4a^2.4$
(c) $4a^2(9-4)$ (d) $152\sqrt{a}/3$

5. The area bounded by the curve $y = 2\cos x$ and the x-axis from $x = 0$ to $x = 2\pi$ is

(a) 2 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 8 sq. units

6. The area enclosed by the curve $y = \sin 2x$, x-axis and the lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ is

(a) $\frac{1}{2}$ sq. units
(b) 1 sq. units
(c) 2 sq. units
(d) None of these

7. The area of the region bounded by $y = \tan x$, $x = 0$ and $x = \frac{\pi}{4}$ is

(a) $\frac{1}{2}\log 2$ (b) $\log 2$
(c) $\frac{3}{2}\log 2$ (d) $2\log 2$

8. The area of the region bounded by the curve $y = x - x^2$ between $x = 0$ and $x = 1$ is

(a) 1/6 (b) 1/3
(c) 1/2 (d) 5/6

9. The area of the region bounded by $a^2y^2 = x^2(a^2 - x^2)$

(a) $\frac{a^2}{2}$ (b) $\frac{2a^2}{3}$
(c) $\frac{4a^2}{3}$ (d) None of these

10. Area bounded by the curve $y = |x - 1|$, $y = 0$, $x = 0$ and $x = 1/2$ is

(a) $\frac{3}{4}$ sq. units (b) $\frac{3}{16}$ sq. units
(c) $\frac{3}{8}$ sq. units (d) 1 sq. units

11. Area bounded by the curve $y = \sin^2 x$ and lines $x = \frac{\pi}{2}$, $x = \pi$ and x-axis is

(a) $\frac{\pi}{2}$ sq. units (b) $\frac{\pi}{4}$ sq. units
(c) $\frac{\pi}{8}$ sq. units (d) None of these

12. The area bounded by the curve $y^2 = 9x$ and the line $x = a$, $x = 4$ and x-axis is

(a) 7 (b) 14
(c) 28 (d) 14/3

13. What is the area of the region bounded by the line $3x - 5y = 15$, $x = 1$, $x = 3$ and x-axis in sq. units? ***[NDA-2008]***

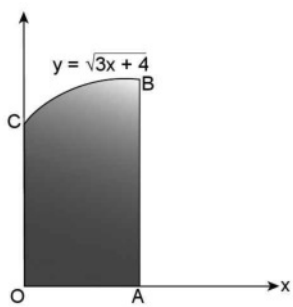
(a) $\frac{36}{5}$ (b) $\frac{18}{5}$
(c) $\frac{9}{5}$ (d) $\frac{3}{5}$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ
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13. ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

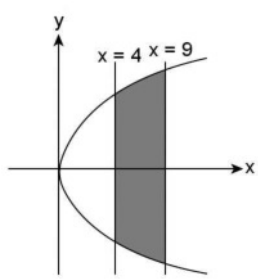
1. (d) Area $= \int_0^4 \sqrt{3x+4}\,dx = \left|\frac{(3x+4)^{3/2}}{3^{3/2}}\right|_0^4$



$= \frac{2}{9} \times 56 = \frac{112}{9}$ sq. unit

2. (a) $\int_0^2 (x^2 - 4x)dx = \left[\frac{x^3}{3} - \frac{4x^2}{2}\right]_0^2 = \frac{16}{3}$ sq. unit

4. (d) Shaded area $A = 2\int_4^9 \sqrt{4ax}$

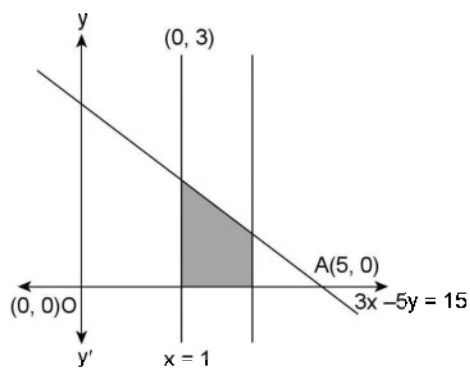


$A = 4\sqrt{a} \times \frac{2}{3}\left[x^{3/2}\right]_4^9 = \frac{152\sqrt{a}}{3}$

13. (b) The given equation of line can be rewritten as

$$\frac{x}{5} - \frac{y}{3} = 1$$

and $y = \frac{3x-15}{5}$



$\therefore$ Required area $= \int_1^3 y\,dx = \int_1^3 \left(\frac{3x-15}{5}\right)dx$

$= \frac{1}{5}\int_1^3 (3x-15)dx$

$= \frac{1}{5}\left[\frac{3x^2}{2} - 15x\right]_1^3 = \frac{1}{5}\left[\frac{27}{2} - 45 - \frac{3}{2} + 15\right]$

$= \frac{1}{5}\left[\frac{24}{2} - 30\right] = \frac{1}{5}[12-30]$

$= \frac{-18}{5} = \frac{18}{5}$ sq. unit (neglecting negative sign)

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LECTURE

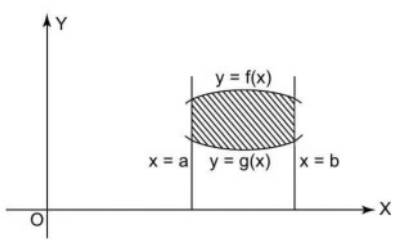


7 Area Under Curve 2

BASIC CONCEPTS

POINTS TO REMEMBER

1. If we have two curve $y = f(x)$ and $y = g(x)$, such that $y = f(x)$ lies above the curve $y = g(x)$ and both are above the axis of x then the area bounded between them and the ordinates $x = a$ and $x = b(b > a)$, is given by

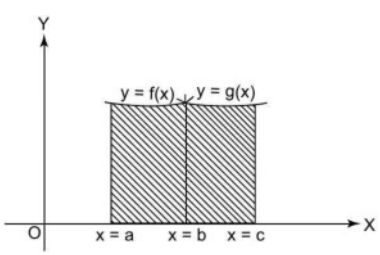


$$A = \int_a^b f(x)dx - \int_a^b g(x)dx$$

i.e., upper curve area-lower curve area.

2. The area bounded by the curves $y = f(x)$ and $y = g(x)$ between the ordinates $x = a$ and $x = b$ is given by

$$A = \int_a^c f(x)dx + \int_c^b g(x)dx$$



where $x = c$ is the point of intersection of the two curves.

3. Trace the curve with the help of the above points.

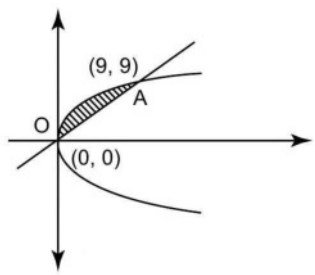
NOTES

1. The area of the region bounded by $y^2 = 4ax$ and $x^2 = 4$ by is $\frac{16ab}{3}$ sq. units.
2. The area of the region bounded by $y^2 = 4ax$, $y = mx$ is: $\frac{8a^2}{3m^3}$ sq. units
3. The area of the region bounded by $x^2 = 4ay$, $x = my$ is $\frac{8a^2}{3m^3}$ sq. units
4. The area of the region bounded by $y^2 = 4ax$ and its latus rectum is $\frac{2}{a}$ sq. units.
5. The area of the region bounded by one arch of $\sin(ax)$ or $\cos ax$ and x-axis is $\frac{2}{a}$ sq. units
6. Area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab sq. units.
7. Area of the region bounded by the curves
   (i) $|x-k| + |y-k| = a$ is $2a^2$
   (ii) $|x| + |y| = a$ is $2a^2$
   (iii) $\left.\begin{matrix} y = |x| + a \text{ and} \\ y = +|x| - a \end{matrix}\right]$ is $2a^2$
   (iv) $y = |x-a|$ and $y = a$ Ans a^2

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Find the area enclosed between the parabola $y^2 = 9x$ and the line $y = x$. ***[MP-99]***

Solution



Given $y^2 = 9x$ and $y = x$

$\Rightarrow x^2 - 9x = 0 \quad \Rightarrow x = 0, x = 9$

$\therefore$ Required area $= \int_0^9 (y_1 - y_2)dx$

$$= \int_0^9 \left(\sqrt{9x} - x\right)dx$$

$$= \left[3\frac{x^{3/2}}{3/2} - \frac{x^2}{2}\right]_0^9 = \left[2\times 9\times 3 - \frac{81}{2}\right] = \frac{27}{2}$$

sq. units.

2. Find the area of triangle by integration, whose vertices are the points (–1, 1), (0, 5) and (3, 2).

[AISSE-1994]

Solution

Equation of AB, $y - 1 = \frac{5-1}{0+1}(x+1)$

$\Rightarrow y = 4x + 5$ ..... (1)

Equation of BC, $y - 5 = \frac{2-5}{3-0}(x-0)$

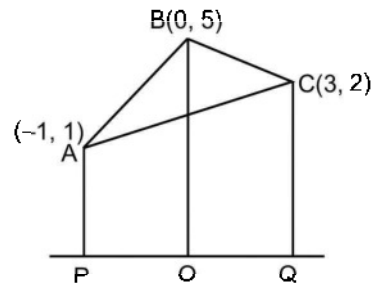
$\Rightarrow y = 5 - x$ ..... (2)

and equation of AC $y - 1 = \frac{2-1}{3+1}(x+1)$

$\Rightarrow y = \frac{1}{4}x + \frac{5}{4}$ ..... (3)

$\therefore$ Required area = Area $PABO$ + Area $OBCQ$ – Area $PACQ$

$$= \int_{-1}^{0}(4x+5)dx + \int_0^3 (5-x)dx - \int_{-1}^{3}\left(\frac{x}{4} + \frac{5}{4}\right)dx$$



$$= [2x^2 + 5x]_{-1}^{0} + \left[5x - \frac{x^2}{2}\right]_0^3 - \left[\frac{x^2}{8} + \frac{5}{4}x\right]_{-1}^{3}$$

$$= 3 + \frac{21}{2} - 6 = \frac{15}{2} \text{ sq. units}$$

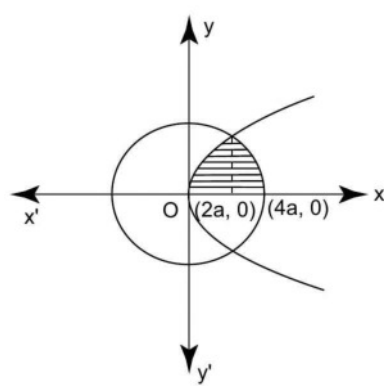
3. Draw a rough sketch of the region $\{(x, y) : y^2 \le 3x, 3x^2 + 3y^2 \le 12\}$ and find the area enclosed by the region using method of integration. ***[CBSE-1997]***

Solution

Given equation to the curves are $y^2 = 3x$ ..... (1)

$3x^2 + 3y^2 = 12$ ..... (2)

$\Rightarrow x^2 + y^2 = 4$ ..... (3)



Equation (1) represents a parabola whose vertex is (0, 0) and axis is x-axis.

Equation (2) represents a circle whose centre is (0, 0) and radius is 2.

Put $y^2 = 3x$ in (3) $x^2 + 3x = 4 \Rightarrow x = 1$ or $x = -4$ neglecting $x = -4$ [$\because$ for $x = -4$, $y^2 = 3(-4) < 0$]

$(1) \Rightarrow y = \sqrt{3}\sqrt{x}$

$(2) \Rightarrow y = \sqrt{4 - x^2}$

Required area

$$= 2\left[\int_0^1 (y \text{ of parabola})\, dx + \int_1^2 (y \text{ of circle})\, dx\right]$$

$$= 2\left[\int_0^1 \sqrt{3}\sqrt{x}dx + \int_1^2 \sqrt{4 - x^2}dx\right]$$

$$= 2\left[\sqrt{3}\left\{\frac{2x^{3/2}}{3}\right\}_0^1 + \int_1^2 \sqrt{2^2 - x^2}dx\right]$$

$$= 2\left[\frac{2}{\sqrt{3}}[1 - 0] + \left\{\frac{x}{2}\sqrt{4 - x^2} + \frac{2^2}{2}\sin^{-1}\frac{x}{2}\right\}_1^2\right]$$

$$= 2\left[\begin{array}{l}\frac{2}{\sqrt{3}} + \left\{\frac{2}{2}\sqrt{4-4} + 2\sin^{-1}\frac{2}{2}\right\} \\ -\left\{\frac{1}{2}\sqrt{4-1} + 2\sin^{-1}\frac{1}{2}\right\}\end{array}\right]$$

$$= 2\left[\frac{2}{\sqrt{3}} + \left[0 + 2\times\frac{\pi}{2}\right] - \frac{\sqrt{3}}{2} - 2\left(\frac{\pi}{6}\right)\right]$$

$$= 2\left[\frac{2}{\sqrt{3}} + \pi - \frac{\sqrt{3}}{2} - \frac{\pi}{3}\right]$$

$$= \left[\frac{4}{\sqrt{3}} - \sqrt{3}\right] + \left[2\pi - 2\frac{\pi}{3}\right]$$

$$= \left(\frac{1}{\sqrt{3}} + \frac{4\pi}{3}\right) \text{ sq. units.}$$

4. Find the area common to the circle $x^2 + y^2 = 16a^2$ and the parabola $y^2 = 6ax$.

[CBSE-2004]

Solution

Given equation to the curves are

$y^2 = 6ax$ ..... (1)

$x^2 + y^2 = 16a^2$ ..... (2)

Equation (1) represents a parabola whose vertex is (0, 0) and axis is x-axis.

Equation (2) represents a circle whose centre is (0, 0) and radius is $4a$.

Put $y^2 = 6ax$ in (2) $x^2 + 6ax = 16a^2$

$\Rightarrow x^2 + 6ax - 16a^2 = 0$

$\Rightarrow x^2 + 8ax - 2ax - 16a^2 = 0$

$\Rightarrow x(x + 8a) - 2a(x + 8a) = 0$

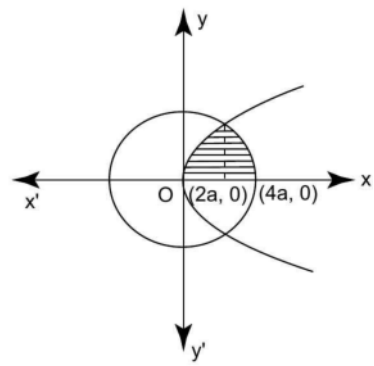
$\Rightarrow (x - 2a)(x + 8a) = 0$

$\Rightarrow x = 2a, x = -8a$ neglecting $x = -8a$

$[\because$ for $x = -8a$, $y^2 = 6a.(-8a) < 0]$

$(1) \Rightarrow y = \sqrt{6ax}$

$(2) \Rightarrow y = \sqrt{16a^2 - x^2}$



Required area $= 2\left[\int_0^{2a} (y \text{ of parabola})dx + \int_{2a}^{4a} (y \text{ of circle})dx\right]$

$$= 2\left[\int_0^{2a} \sqrt{6a}x^{1/2}dx + \int_{2a}^{4a} \sqrt{(4a)^2 - x^2}dx\right]$$

$$= 2\left\{\left[\frac{2\sqrt{6a}.x^{3/2}}{3}\right]_0^{2a} + \left[\frac{x}{2}\sqrt{16a^2 - x^2} + 16a^2\sin^{-1}\left(\frac{x}{4a}\right)\right]_{2a}^{4a}\right\}$$

on solving this, we get result $\frac{4a^2}{3}(4\pi + \sqrt{3})$ sq. units.

5. Find the area of the region defined by $\{(x, y): x + 1 \le y \le x^2 + 3, 0 \le x \le 3\}$

[CBSE-Sample paper-III]

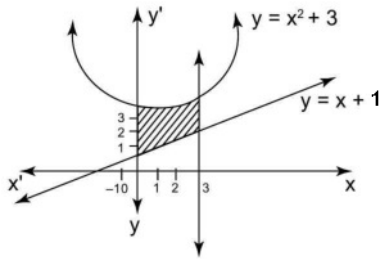
Solution

$y = x^2 + 3$ and $y = x + 1$ are respectively the equation of the parabola and straight line $y = x^2 + 3$ opens upwards and its vertex is at (0, 3)

The straight line $y = x + 1$ passes through the points (0, 1) and (–1, 0).

$\therefore$ The required area $= \int_0^3 (x^2 + 3)dx$

$$-\int_0^3 (x + 1)dx = \int_0^3 (x^2 - x + 2)dx$$



$$= \left[\frac{x^3}{3} - \frac{x^2}{2} + 2x\right]_0^3 = \frac{27}{3} - \frac{9}{2} + 6$$

$= 15 - 4.5 = 10.5$ sq. units.

6. Find the area of the region $\{(x, y) : x^2 \le y \le |x|\}$ ***[CBSE-02]***

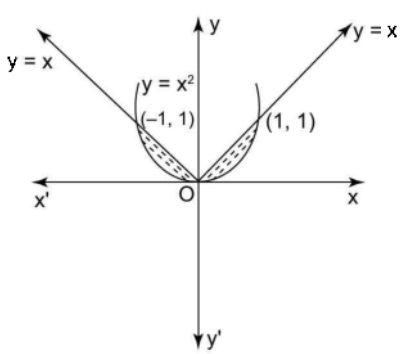
Solution

Let $R = \{(x, y) : x^2 \le y \le |x|\}$.

Then, $R = \{(x, y): x^2 \le y\} \cap \{(x, y) : y \le |x|\}$
$= \{(x, y) : x^2 \le y\} \cap [\{(x, y) : y \le x, x \ge 0\}$
$\cup \{(x, y) : y \le -x, x < 0\}]$
$= R_1 \cap (R_2 \cup R_3)$, where $= \{(x, y) : x^2 \le y\} = R_1$, $R_2 = \{(x, y) : y \le x, x \ge 0\}$ and $R_3 = \{(x, y) : y \le -x, x < 0\}$

Region R_1: Clearly, $x^2 = y$ represents a parabola with vertex at (0, 0), positive direction of y-axis as its axis and it opens upwards. The interior of the parabola is the region R_1.

Region R_2: Clearly, $y = x, x \ge 0$ is a line passing through the origin and making an angle of 45° with the positive direction of x-axis. So, R_2 is the region lying below the line $y = x$.



Region R_3: Clearly, $y = x$, $x < 0$ is a line passing through the origin and making an angle of 135° with the positive direction of x-axis. So, R_3 is the region lying below the line $y = -x$. The required region R is the shaded region shown in figure

Since both the curves are symmetrical about y-axis. So

Required area = 2 (shaded area in first quadrant)

$$= 2\int_0^1 (x - x^2)dx = 2 \times \frac{1}{6} = \frac{1}{3} \text{ sq. units}$$

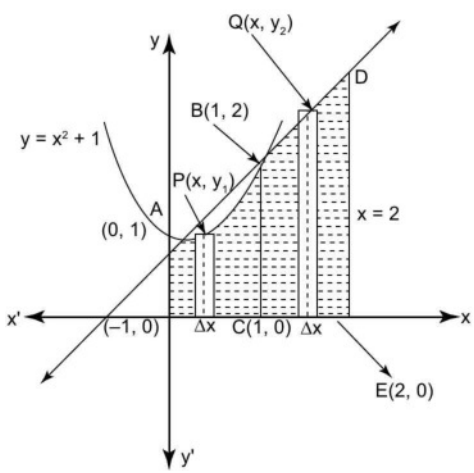
7. Find the area of the region $\{(x, y): 0 \le y \le x^2 + 1, 0 \le y \le x + 1, 0 \le x \le 2\}$ ***[CBSE-2001]***

Solution

Let $R = \{(x, y) : 0 \le y \le x^2 + 1, 0 \le y \le x + 1, 0 \le x \le 2\}$. Then

$R = \{(x, y) : 0 \le y \le x^2 + 1\} \cap \{(x, y): 0 \le y \le x + 1\} \cap \{(x, y) : 0 \le x \le 2\} = R_1 \cap R_2 \cap R_3$, where $R_1 = \{(x, y) : 0 \le y \le x^2 + 1\}$

$R_2 = \{(x, y) : 0 \le y \le x + 1\}$ and $R_3 = \{(x, y): 0 \le x \le 2\}$



Region R_1: Clearly $y = x^2 + 1$ represents a parabola with vertex at (0, 1), axis along the positive direction of y -axis and it opens upwards. Since, $0 \le y \le x^2 + 1 \Leftrightarrow y \ge 0$ and $y \le x^2 + 1$. So, R_1 is the region lying above x-axis and outside the parabola $y = x^2 + 1$.

Region R_2: Clearly $y = x + 1$ represents a straight line cutting x-axis at (–1, 0) and y-axis at (0, 1). Since $0 \le y \le x + 1 \Leftrightarrow y \ge 0$ and $y \le x + 1$. Therefore, R_2 is the region lying above x-axis and below the line $y = x + 1$.

Region R_3: Clearly, R_3 is the region lying between the lines $x = 0$ i.e., y-axis and $x = 2$.

Hence, the required region R is the shaded region as show in figure. Solving $y = x^2 + 1$ and $y = x + 1$, we find that these two intersect at (0, 1) and (1, 2). Now, we slice the shaded region into verticle strips. we observe that verticle strips change their character at the point B. Draw a line BC parallel to the y-axis which devides the area $OABDEO$ into two portions $OABCO$ and $BDECB$. For the area $OABCO$, the approximating rectangle has

Length $= y_1$, width $= \Delta x$ and Area $= y_1 \Delta x$

As it can move from $x = 0$ to $x = 1$

So, area $OABCO = \int_0^1 y_1 dx = \int_0^1 (x^2 + 1) dx$

[$\because P(x, y_1)$ lies on $y = x^2 + 1 \ \because y_1 = x^2 + 1$]

For the area $BDECB$, the approximating rectangle shown in figure has: Length $= y_2$, width $= \Delta x$ and Area $= y_2 \Delta x$

Sinec it can move from $x = 1$ to $x = 2$

So, area $BDECB = \int_1^2 y_2 dx = \int_1^2 (x + 1) dx$

[$\because Q(x, y_2)$ lies on $y = x + 1 \ \therefore y_2 = x + 1$]

Hence Required area = Area $OABCO$ + Area $BDECB$

$$= \int_0^1 (x^2 + 1) dx + \int_1^2 (x + 1) dx$$

$$= \left[\frac{x^3}{3} + x\right]_0^1 + \left[\frac{x^2}{2} + x\right]_1^2$$

$$= \frac{4}{3} + \frac{5}{2} = \frac{23}{6} \text{ sq. units}$$

8. Find the area of the region enclosed between the two circles $x^2 + y^2 = 1$ and $(x - 1)^2 + y^2 = 1$

[CBSE-91, 96(C), 97(C), 99; HPB-96, 99, 2002; HPSB-2002(C)]

Solution

The equations of the given curves are $x^2 + y^2 = 1$ ..... (i)

and $(x - 1)^2 + (y - 0)^2 = 1$ ..... (ii)

Clearly, $x^2 + y^2 = 1$ represents a circle with centre at (0, 0) and radius unity. Also $(x - 1)^2 + (y - 0)^2 = 1$ represents a circle with centre at (1, 0) and radius unity. To find the points of intersection of the given curves, we solve (i) and (ii) simultaneously. Solving (i) and (ii) simultaneously we find that the two curves intersect at $A(1/2, \sqrt{3}/2)$ and $D(1/2, -\sqrt{3}/2)$

Since both the curves are symmetrical about x-axis. So, required area = 2 (Area $OABCO$)

Now, we slice the area $OABCO$ into verticle strips. We observe that the verticle strips change their character at $A(1/2, \sqrt{3}/2)$. So, Area $OABCO$ = Area $OACO$ + Area $CABC$

When area $OACO$ is sliced into verticle strips, we find that each strip has its upper end on the circle $(x - 1)^2 + (y - 0)^2 = 1$ and the lower end on x-axis. So the approximating rectangle shown in figure has, Length $= y_1$, width $= \Delta x$ and area $= y_1 \Delta x$. As it can move from $x = 0$ to $x = 1/2$

$$\therefore \text{Area } CACO = \int_0^{1/2} y_1 dx$$

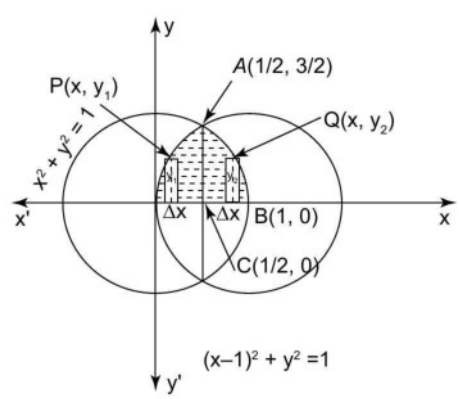
$$= \int_0^{1/2} \sqrt{1 - (x - 1)^2} dx$$

[$\because P(x, y_1)$ lies on $(x - 1)^2 + y^2 = 1 \ \therefore (x - 1)^2 + y_1^2 = 1$

$\Rightarrow y_1 = \sqrt{1 - (x - 1)^2}$]

Similarly, approximating rectangle in the region $CABC$ has,

Length $= y_2$, width $= \Delta x$ and area $= y_2 \Delta x$. As it can move from $x = 1/2$ to $x = 1$.



$$\therefore \text{Area } CABC = \int_{1/2}^1 y_2 dx = \int_{1/2}^1 \sqrt{1 - x^2} dx$$

[$\because Q(x, y_2)$ lies on $x^2 + y^2 = 1 \ \therefore x^2 + y_2^2 = 1$

$\Rightarrow y_2 = \sqrt{1 - x^2}$]

Hence required area $= 2\left[\int_0^{1/2} \sqrt{1 - (x - 1)^2} dx + \int_{1/2}^1 \sqrt{1 - x^2} dx\right]$

$$= 2\left[\left[\frac{1}{2}(x - 1)\sqrt{1 - (x - 1)^2} + \frac{1}{2}\sin^{-1}\left(\frac{x - 1}{1}\right)\right]_0^{1/2}\right.$$

$$+\left[\frac{1}{2}x\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}\left(\frac{x}{1}\right)\right]_{1/2}^{1}\Bigg]$$

$$=\left[\left\{-\frac{\sqrt{3}}{4}+\sin^{-1}\left(\frac{-1}{2}\right)-\sin^{-1}(-1)\right\}+\left\{\sin^{-1}(1)-\frac{\sqrt{3}}{4}-\sin^{-1}\left(\frac{-1}{2}\right)\right\}\right]$$

$$=-\frac{\sqrt{3}}{4}-\frac{\pi}{6}+\frac{\pi}{2}+\frac{\pi}{2}-\frac{\sqrt{3}}{4}-\frac{\pi}{6}$$

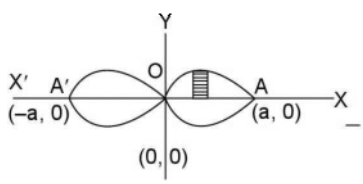
$$=\left(\frac{2\pi}{3}-\frac{\sqrt{3}}{2}\right)\text{sq. units.}$$

9. Find the area of a loop as well as the whole area of the curve $a^2y^2 \quad x^2(a^2-x^2)$.

Solution

The curve is symmetrical about both the axes. It cuts x-axis

$y \quad 0$ at $(0, 0)$, $(a, 0)$ and $(-a, 0)$.



$$\text{Area of a loop} = 2\int_0^a y\,dx = 2\int_0^a \frac{x}{a}\sqrt{a^2-x^2}\,dx$$

$$=-\frac{1}{a}\int_0^A \sqrt{a^2-x^2}(-2x)\,dx$$

$$=-\frac{1}{a}\left[\frac{2}{3}(a^2-x^2)^{3/2}\right]_0^A=\frac{2}{3}a^2$$

$$\text{Total area} = 2\times\frac{2}{3}a^2=\frac{4}{3}a^2$$

NOTE

You can put $x = a\sin\theta$ and adjust the limits and evaluate the integral by Gamma funtion as in next example.

10. The line $y \quad mx$ bisects the area enclosed by the lines $x \quad 0, y = 0, x = 3/2$ and the curve $y = 1 + 4x - x^2$. Find the value of m.

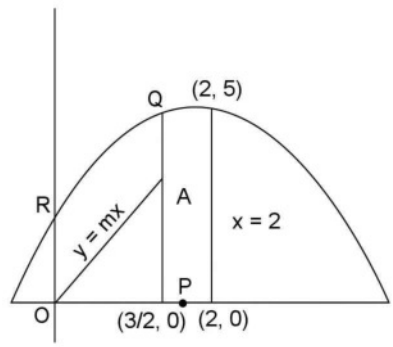
[Roorkee-1991]

Solution

The equation $y \quad 1 + 4x - x^2$ can be written as

$$y = -(x-2)^2 + 5 \text{ or } (x-2)^2 = -(y-5)$$

which is a parabola with vertex $(2, 5)$ and symmetrical about the line $x = 2$ and is as shown in the figure.



Let OA represent the line $y = mx$. Then as given this line biscets the area $OPQR$ so that Area $OPQR$ = 2 Area OPA.

$$\text{Now area } OPQR = \int_0^{3/2} y\,dx$$

$$=\int_0^{3/2}(1+4x-x^2)\,dx$$

$$=\left[x+2x^2-\frac{1}{3}x^3\right]_0^{3/2}$$

$$=\left(\frac{3}{2}+2.\frac{9}{4}-\frac{1}{3}.\frac{27}{8}\right)-0$$

$$=6-\frac{9}{8}=\frac{39}{8}$$

$$\text{And area } OPA = \int_0^{3/2} y\,dx=\int_0^{3/2} mx\,dx$$

$$=\frac{1}{2}m\left[x^2\right]_0^{3/2}=\frac{1}{2}m\left[\frac{9}{4}-0\right]=\frac{9}{8}\,m$$

$\therefore$ From the given condition we have

$$\text{Area } OPA = \frac{1}{2}\text{ Area } OPQR \text{ or } \frac{9m}{8}=\frac{39}{16}$$

$$\text{or } m \quad \frac{13}{6}$$

11. Find the ratio in which the area bounded by the curves $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$. ***[Roorkee-1994]***

Solution

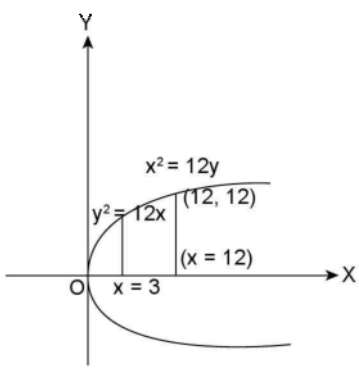
The two curves meet at $P(12, 12)$

$$\therefore A = \int_0^{12} (y_1 - y_1)dx = \int_0^{12} \left(2\sqrt{3}x^{1/2} - \frac{x^2}{12} \right) dx$$

$$= \left[2\sqrt{3}.\frac{2}{3}x^{3/2} - \frac{x^3}{12.3} \right]_0^{12}$$

$$= \left(\frac{4}{\sqrt{3}}.12.\sqrt{12} - 48 \right) = 96 - 48 = 4$$

Again Area $A_1 = \int_0^3 (y_1 - y_2)dx$



$$= \left[2\sqrt{3}\frac{2}{3}x^{3/2} - \frac{x^3}{12.3} \right]_0^3$$

$$= \left[\frac{4}{3}\sqrt{3}3\sqrt{3} - \frac{3}{4} \right] = \frac{45}{4}$$

$$\therefore \quad A_2 \quad A - A_1 = 48 - \frac{45}{4} = \frac{3}{4} \times 49$$

$$\therefore \quad \frac{A_1}{A_2} = \frac{15}{49}$$

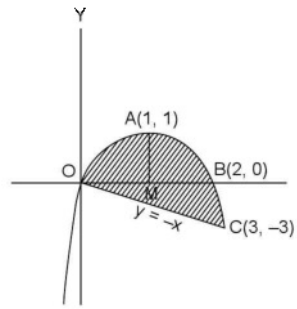
12. Find the area bounded by the curve $y \quad 2x - x^2$ and the straight line $y = -x$. ***[Roorkee-1992]***

Solution

$y = 2x - x^2$ is $(x - 1)^2 = -(y - 1)$

It represents a parabola with vertex at (1, 1).

$$A = \left| \int_0^3 (y_1 - y_2)dx \right|$$



$$= \left| \int_0^3 (2x - x^2) + x dx \right|$$

$$= \left| \left(\frac{3x^2}{2} - x^3 \right) \right| = \frac{9}{2}$$

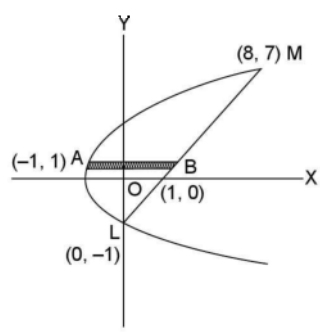
13. Find the area in the plane bounded by the curves $y \quad x - 1$ and $(y - 1)^2 = 4(x + 1)$. ***[JEE Bihar-1995]***

Solution

Eliminating y we get the points of intersection as

$(x - 2)^2 = 4(x + 1)$ or $x^2 - 8x = 0$

$\therefore x = 0, 8 \quad \therefore y = -1, 7$



Hence the points of intersection are $(0, -1)$ and $(8, 7)$. The vertex of the parabola is at the point $(-1, 1)$. The line cut axes at $(1, 0)$ and $(0, -1)$.

$$\therefore \quad A = \int_{y=-1}^{7} (x_1 - x_2)dy$$

$$= \int_{-1}^{7} \left[(y+1) - \left\{ \frac{(y-1)^2}{4} - 1 \right\} \right] dy$$

$$= \left\{ \frac{1}{2}y^2 + y - \frac{(y-1)^3}{12} + y \right\}_{-1}^{7} = \frac{64}{3} \text{ sq. units}$$

14. Compute the area of the region bounded by the curves $y = ex \ln x$ and $y = \frac{\ln x}{ex}$, where, $\ln e = 1$. ***[IIT-1990]***

Solution

The two curves are $y \quad ex \log x$ ..... (1)

and $y = \frac{\log x}{ex}$ [Here ln stands for natural log)

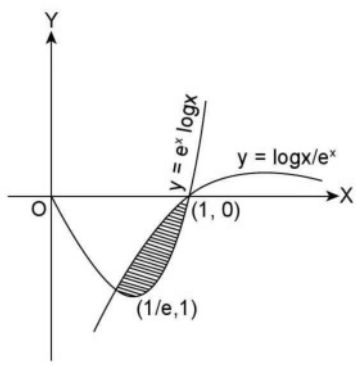
The two curves meet, where

$$ex \log x = \frac{\log x}{ex} \Rightarrow \left(ex - \frac{1}{ex}\right)\log x = 0$$

$\Rightarrow \log x = 0$ or $ex - \frac{1}{ex} = 0$

$\Rightarrow x \quad 1$ or $x^2 = \frac{1}{e^2}$

$\Rightarrow x = 1$ or $x = 1/e \quad (\because x > 0)$



The required area is shown shaded in the figure

Required area $= \int_{1/e}^{1} \left(\frac{\log x}{ex} - ex \log x\right) dx$

$$= \frac{1}{e}\int_{1/e}^{1} \frac{\log x}{x} dx - e\int_{1/e}^{1} x \log x dx$$

$$= \frac{1}{e} I_1 - eI_2 \qquad \text{..... (1)}$$

Now $I_1 = \int_{1/e}^{1} \log x . \frac{1}{x} dx = \left[\frac{(\log x)^2}{2}\right]_{1/e}^{1}$

$$= \frac{1}{2}\left[(\log 1)^2 - \left(\log \frac{1}{e}\right)^2\right]$$

$$= \frac{1}{2}\left[0 - (0 - \log e)^2\right]$$

$$= \frac{1}{2}\left[-(-1)^2\right] = -\frac{1}{2} \qquad \text{..... (2)}$$

and $I_2 = \int_{1/e}^{1} x \log x dx$

$$= \left[\log \frac{x^2}{2}\right]_{1/e}^{1} - \int_{1/e}^{1} \frac{1}{x} \cdot \frac{x^2}{2} dx$$

$$= \left[\frac{x^2}{2}\log x - \frac{x^2}{4}\right]_{1/e}^{1}$$

$$= \left(0 - \frac{1}{4}\right) - \left(\frac{1}{2e^2}(-1) - \frac{1}{4e^2}\right)$$

$$= -\frac{1}{4} + \frac{3}{4e^2} \qquad \text{..... (3)}$$

From (1), (2) and (3), we get required area

$$= -\frac{1}{2e} - e\left[-\frac{1}{4} + \frac{3}{4e^2}\right]$$

$$= \frac{e}{4} - \frac{5}{4e} \text{ sq. units.}$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Using integration find the area enclosed by the curves $y = |x - 1|$ and $y = 1 - |x - 1|$.
[CBSE-Practice Sample paper-II]

2. Using integration find the area of the triangular region bounded by the lines $2x + y = 4$, $3x - 2y = 6$ and $x - 3y + 5 = 0$.
[CBSE-Practice Sample paper-III]

3. Find the area of the region $\{(x, y): x^2 + y^2 \leq 2ax, y^2 \leq ax, x \geq 0, y \geq 0\}$
[CBSE-1992]

4. Find the area of the region between the circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$.
[AICBSE-99C; PSB-92]

5. Draw a rough sketch of the region $\{(x, y): y^2 \le 5x, 5x^2 + 5y^2 \le 36\}$ and find the area enclosed by the region using method of integration. ***[CBSE-97]***

6. Find the area bounded by the curves $x^2 = 8y$ and $y^2 = 8x$. ***[MP-2000]***

7. Find the area enclosed between the parabola $y^2 = 16x$ and the line $y = x$. ***[MP-99; CBSE-96, 97]***

8. Find the area bounded by the curves $y^2 = 9x$ and $x^2 = 9y$. ***[MP-97, 2000, 2005(A)]***

9. Find the area of the region bounded by the curve $y^2 = 2y - x$ and the y-axis. ***[CBSE-93]***

10. Find the area of the region $\{(x, y): x^2 \le y \le x\}$ ***[CBSE-91, 92, 2005(O.D.)-I]***

11. Using integration, find the area of the triangle ABC whose vertices have coordinates $A(2, 5)$, $B(4, 7)$ and $C(6, 2)$. ***[CBSE-94, 2001; HPSB-2000(C)]***

12. Using integration, find the area of the region bounded by the triangle whose vertices are (2, 1), (3, 4) and (5, 2) ***[CBSE-2001]***

13. Sketch the graph $y = |x - 5|$. Evaluate $\int_0^1 |x - 5|\, dx$. What does this value of the integral represent on the graph. ***[CBSE-94]***

14. Find the area of the circle $x^2 + y^2 = 16$ which is exterior to the parabola $y^2 = 6x$, by using integration. ***[CBSE-2007]***

15. Find the area enclosed by the parabola $y^2 = x$ and the line $x + y = 2$. ***[CBSE-2005 (O.D.)-I]***

16. Draw a rough sketch of the graph of the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and evaluate the area of the region under the curve and above the x-axis. ***[HSB-89, 96; PSB-91, 96; HPSB-96]***

ANSWERS

1. 1/2 sq. units
2. 3.5 sq. units
3. $\left(\frac{\pi}{4} - \frac{2}{3}\right)a^2$ sq. units.
4. $\left(\frac{8\pi}{3} - 2\sqrt{3}\right)$ sq. units
5. $\frac{4\sqrt{5}}{3}a^{3/2} + \frac{18}{\pi} - a\sqrt{\frac{36}{5} - a^2}$
   $-\frac{36}{5}\sin^{-1}\left(\frac{a\sqrt{5}}{6}\right), a = \frac{-25 + \sqrt{1345}}{10}$
6. $\frac{64}{3}$ sq. units
7. $\frac{128}{3}$ sq. units
8. 27 sq. units
9. $\frac{4}{3}$ sq. units
10. $\frac{1}{6}$ sq. units
11. 7 sq. units
12. 4 sq. units
13. 9/2 sq. units.
15. 9/2
16. 3π sq. units.

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

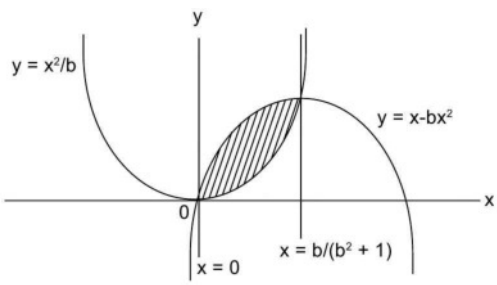
1. If the area bounded by the curves $y = x - bx^2$ and $y = \frac{1}{b}x^2$, where $b > 0$ is maximum, then $b =$

(a) 0 (b) 1

(c) 2 (d) None of these

Solution

(b) Area bounded by the given curves is



$$A = \int_0^{\frac{b}{b^2+1}} \left(x - bx^2 - \frac{x^2}{b} \right) dx$$

$$= \left| \frac{x^2}{2} - \left(\frac{b^2+1}{b} \right) \frac{x^3}{3} \right|_0^{\frac{b}{b^2+1}} = \frac{1}{6} \left(\frac{b}{b^2+1} \right)^2$$

$$\therefore \log A = \frac{1}{6} [2 \log b - 2 \log (b^2 + 1)]$$

$$\Rightarrow \frac{1}{A}\frac{dA}{db} = \frac{1}{3}\left(\frac{1}{b} - \frac{2b}{b^2+1} \right)$$

i.e., $$\frac{dA}{db} = \frac{A}{3}\left[\frac{b^2+1-2b^2}{b(b^2+1)} \right]$$

$$= \frac{A}{3}\frac{1-b^2}{b(b^2+1)} = -\frac{(b-1)(b+1)}{b(b^2+1)} \cdot \frac{A}{3}$$

$$\frac{dA}{db} = 0 \Rightarrow b = 1 \quad [\because b > 0]$$

Also $\frac{dA}{db} > 0$ for $0 < b < 1$ and < 0 for $b > 1$

$\therefore A$ is max. when $b = 1$

2. The area of the region ling inside $x^2 + (y - 1)^2 = 1$ and outside $c^2 x^2 + y^2 = c^2$ where $c = (\sqrt{2} - 1)$ is ***[Roorkee-1999]***

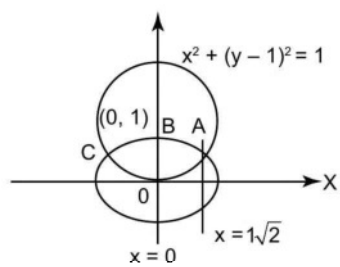
(a) $(4 - \sqrt{2})\frac{\pi}{4} + \frac{1}{\sqrt{2}}$ (b) $(4 + \sqrt{2})\frac{\pi}{4} - \frac{1}{\sqrt{2}}$

(c) $(4 + \sqrt{2})\frac{\pi}{4} + \frac{1}{\sqrt{2}}$ (d) None of these

Solution

(a) Area of $OABCO$

$$= 2\left[\int_0^{1/\sqrt{2}} C\sqrt{1-x^2}\,dx - \int_0^{1/\sqrt{2}} (1 - \sqrt{1-x^2})\,dx \right]$$



$$= 2\left[\int_0^{1/\sqrt{2}} (C+1)\sqrt{1-x^2}\,dx - \int_0^{1/\sqrt{2}} 1.dx \right]$$

$$= 2\left[\sqrt{2} \int_0^{1/\sqrt{2}} \sqrt{1-x^2} - \frac{1}{\sqrt{2}} \right]$$

$$= 2\left[\frac{\sqrt{2}}{2}\left(x\sqrt{1-x^2} + \sin^{-1} x \right) \right]_0^{\frac{1}{\sqrt{2}}} - \frac{1}{\sqrt{2}}$$

$$= 2\left(\frac{\sqrt{2}}{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{2}}{2} \cdot \frac{\pi}{4} - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{\sqrt{2}\pi}{4} - \frac{\sqrt{2}}{2}$$

$\therefore$ area inside the circle and outside the ellipse

$$= \pi - \left(\frac{\sqrt{2}\pi}{4} - \frac{\sqrt{2}}{2} \right)$$

$$= (4 - \sqrt{2})\frac{\pi}{4} + \frac{1}{\sqrt{2}}$$

3. The area enclosed between the curves $y = \log_e (x + e)$, $x = \log_e (1/y)$ and the x-axis is ***[Roorkee-90]***

(a) 2 (b) 1

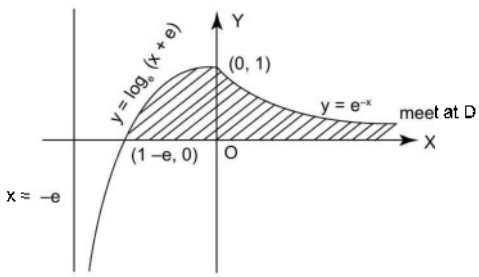
(c) 4 (d) None of these

Solution

(a) Required area $= \int_{1-e}^{0} \log_e(x+e) + \int_{0}^{\infty} e^{-x}$

$= |x\log_e(x+e)|_{1-e}^{1} - \int_{1-e}^{0} \frac{x}{x+e}dx - |e^{-x}|_0^{\infty}$

$= \int_0^{1-e}\left(1 - \frac{e}{x+e}\right)dx - e^{-\infty} + e^0$



$= |x - e\log(x+e)|_0^{1-e} - 0 + 1,$

$= 1 - e + e\log e + 1 = 2$

4. The area of the region formed by $x^2 + y^2 - 6x - 4y + 12 \le 0$, $y \le x$ and $x \le 5/2$ is

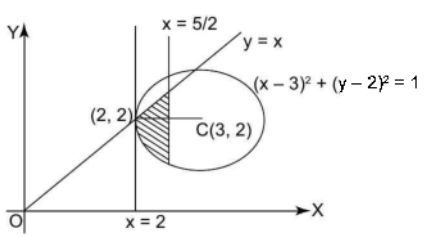
(a) $\frac{\pi}{6} - \frac{\sqrt{3}+1}{8}$ (b) $\frac{\pi}{6} + \frac{\sqrt{3}-1}{8}$

(c) $\frac{\pi}{6} - \frac{\sqrt{3}-1}{8}$ (d) None of these

Solution

(c) The required area

$= \int_2^{5/2} x dx - \int_2^{5/2} [2 - \sqrt{1-(x-3)^2}]dx$



$= \left|\frac{x^2}{2}\right|_2^{5/2} - |2x|_2^{5/2} + \left|\frac{x-3}{2}\sqrt{1-(x-3)^2} + \frac{1}{2}\sin^{-1}(x-3)\right|_2^{5/2}$

$= \left(\frac{28}{8} - 2\right) - (5-4) + \left(-\frac{1}{4}\right)\sqrt{1-\frac{1}{4}}$

$+ \frac{1}{2}\sin^{-1}\left(-\frac{1}{2}\right) + \frac{1}{2}\sqrt{1-1} - \frac{1}{2}\sin^{-1}(-1)$

$= \frac{9}{8} - 1 - \frac{\sqrt{3}}{8} - \frac{\pi}{12} - \frac{1}{2}\left(\frac{-\pi}{2}\right)$

$= \frac{1}{8} - \frac{\sqrt{3}}{8} + \frac{\pi}{6} = \frac{\pi}{6} - \frac{\sqrt{3}-1}{8}$

5. Area bounded by the curves $y = \sqrt{x}$, $x = 2y + 3$ in first quadrant and x-axis is

[IIT (Screening)-2003]

(a) $2\sqrt{3}$ (b) 18

(c) 9 (d) 34/3

Solution

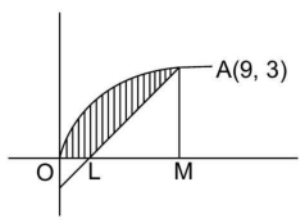
(c) $y^2 = x$, $x = 2y + 3$

The line and parabola meet where $y^2 - 2y - 3 = 0$

$\therefore y = 3, -1$

$\therefore x = 9, 1$ $\therefore$ (9, 3) and (1, –1)

We have to find the area between the two curves and x-axis in first quadrant.



$A = \int_0^9 y dx - \Delta ALM$

$= \frac{2}{3}[x^{3/2}]_0^9 - \frac{1}{2}.6.3 = \frac{2}{3}.27 - 9$

$= 18 - 9 = 9$

6. For which of the following values of m, is the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals 9/2?

[IIT-1999]

(a) –4 (b) –2

(c) 2 (d) 4

Solution

(b, d) The two curves meet at $mx = x - x^2$ or $x^2 = x(1 - m)$

$\therefore x = 0, 1 - m.$

$\int (y_1 - y_2)dx = \int (x - x^2 - mx)dx$

$= \left[(1-m)\frac{x^2}{2} - \frac{x^3}{3}\right]_0^{1-m} = \frac{9}{2}$ if $m < 1$

or $(1-m)^3\left[\frac{1}{2}-\frac{1}{3}\right]=\frac{9}{2}$ or $(1-m)^3=27$

$\therefore\ m=-2$

But if $m>1$ then $1-m$ is –ive, then

$$\left[(1-m)\frac{x^2}{2}-\frac{x^3}{3}\right]_{1-m}^{0}=\frac{9}{2}-(1-m)^3$$

$$\left(\frac{1}{2}-\frac{1}{3}\right)=\frac{9}{2}$$

$\therefore\ (1-m)^3=-27$ or $1-m=-3\ \therefore\ m=4.$

7. The area of the smaller portion between the circle $x^2+y^2=9$ and the line $x=1$ is ***[PET (Raj.)-91, 2002]***

(a) $9\sec^{-1}3-\sqrt{8}$

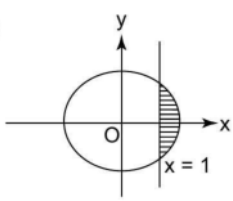
(b) $9\,\text{cosec}^{-1}3-\sqrt{8}$

(c) $\sec^{-1}3-\sqrt{8}$

(d) None of these

Solution

(a)



Since the area is symmetrical about x-axis, so

$$\text{area}=2\int_1^3\sqrt{9-x^2}\,dx$$

$$=[x\sqrt{9-x^2}+9\sin^{-1}x/3]_1^3$$

$$=9\,\pi/2-\sqrt{8}-9\sin^{-1}(1/3)$$

$$=9\,(\pi/2-\sin^{-1}1/3-\sqrt{8}=9\sin^{-1}3-\sqrt{8}.$$

8. The area of the smaller portion between curves $x^2+y^2=8$ and $y^2=2x$ is ***[PET (Raj.)-98; DCE-99]***

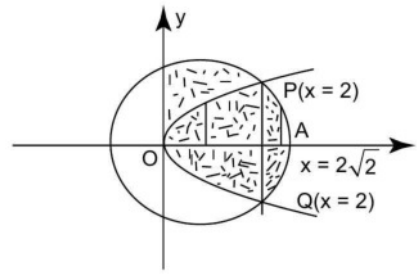
(a) $\pi+2/3$ (b) $2\pi+2/3$

(c) $2\pi+4/3$ (d) $\pi+4/3$

Solution

(c) Two curves meet at P and Q where $x=2$. Obviously the required area lies between $x=0$ and $x=2\sqrt{2}$. It is symmetrical about x-axis and bounded by two given curves.

So required area



$$=2\left[\int_0^2\sqrt{2}\sqrt{x}\,dx+\int_0^{2\sqrt{2}}\sqrt{8-x^2}\,dx\right]$$

$$=2\left[\left(\frac{2\sqrt{2}}{3}x^{3/2}\right)_0^2+\left(\frac{x}{2}\sqrt{8-x^2}+4\sin^{-1}\frac{x}{2\sqrt{2}}\right)_0^{2\sqrt{2}}\right]$$

$$=2\left[\left(\frac{8}{3}-0\right)+(2\pi-2-\pi)\right]=2\pi+4/3$$

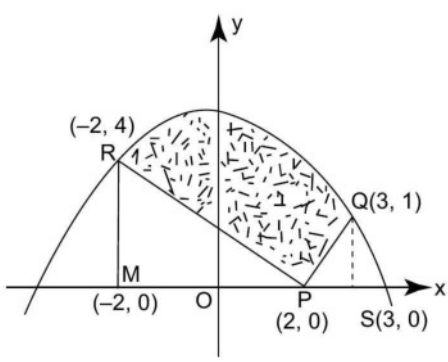
9. The area bounded by curves $3x^2+5y=32$ and $y=|x-2|$ is ***[MNR-94]***

(a) 25 (b) 17/2

(c) 33/2 (d) 33

Solution

(c) First curve is a parabola $x^2=-\frac{5}{3}\left(y-\frac{32}{5}\right)$ whose vertex is on y-axis. Second curve is given by $y=\begin{cases}x-2, & x\ge 2\\ -(x-2), & x<2\end{cases}$ which consists of two perpendicular lines PQ and PR as shown in the figure.



These lines meet the parabola at $Q(3,1)$ and $R(-2,4)$. Hence required area

$$= \int_{-2}^{3} y\,dx - \Delta PRM - \Delta PQS \text{ (parabola)}$$

$$= \int_{-2}^{3} \frac{1}{5}(32 - 3x^2)dx - \frac{1}{2}(4,4) - \frac{1}{2}(1,1)$$

$$= \frac{1}{5}[32x - x^3]_{-2}^{3} - \frac{17}{2} = \frac{1}{5}[125] - \frac{17}{2} = \frac{33}{2}.$$

10. The area bounded by curves $y = ex \log x$ and $y = \frac{\log x}{ex}$ is ***[IIT-90]***

(a) $\frac{e^2 - 5}{e}$ (b) $\frac{e}{4} - \frac{5}{4e}$

(c) $\frac{e^2 + 5}{4e}$ (d) None of these

Solution

(b) Two curves meet at points where $ex \log x = \frac{\log x}{ex}$

$\Rightarrow e^2 x^2 \log x - \log x = 0$

$\Rightarrow \log x\,(e^2 x^2 - 1) = 0.$

$\Rightarrow \log x = 0,\ x^2 = \frac{1}{e^2}$

$\Rightarrow \quad x = 1, \frac{1}{e} \quad (\because\ x \neq -\frac{1}{e})$

$\therefore$ required area $= \int_{1/e}^{1} \left(\frac{\log x}{ex} - ex \log x \right) dx$

$$= \left[\frac{(\log x)^2}{2e} \right]_{1/e}^{1} - e\left[\frac{1}{2}x^2 \log x - \frac{x^2}{4} \right]_{1/e}^{1}$$

$$= -\frac{1}{2e} - e\left[-\frac{1}{4} + \frac{3}{4e^2} \right]$$

$$= \frac{e}{4} - \frac{4}{4e}.$$

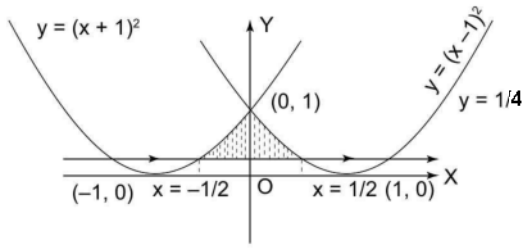
11. The area bounded by the parabolas $y = (x + 1)^2$ and $y = (x - 1)^2$ and the line $y = 1/4$ is ***[IIT-(Screening) 2005]***

(a) 4 (b) 1/6

(c) 4/3 (d) 1/3

Solution

(d) Given parabolas intersect at (0, 1). Also $y = 1/4$ meets $y = (x - 1)^2$ at $x = 1/2$, $x = 3/2$ and it meets $y = (x + 1)^2$ at $x = -1/2$, $x = -3/2$.



As shown required area is symmetrical about y-axis. So

$$\text{Area} = 2\int_{0}^{1/2} \left\{ (x - 1)^2 - \frac{1}{4} \right\} dx$$

$$= 2\int_{0}^{1/2} (x^2 - 2x + 3/4)dx$$

$$= 2\left[\frac{x^3}{3} - x^2 + \frac{3}{4}x \right]_{0}^{1/2}$$

$$= 2\left[\frac{1}{24} - \frac{1}{4} + \frac{3}{8} \right] = \frac{1}{3}.$$

12. The area bounded by $x = -4y^2$ and $x - 1 = -5y^2$ is ***[IIT-JEE-2006]***

(a) 4/3 (b) 3/4

(c) 2/3 (d) 3/2

Solution

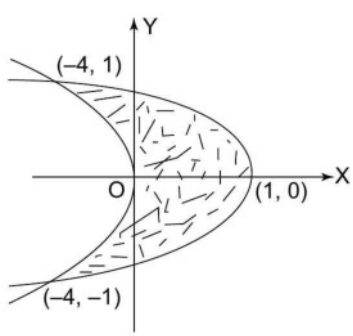
(a) Given curves meet at points (–4, –1) and (–4, 1).

Hence required area

$$= 2\int_{0}^{1} \{(1 - 5y^2) - (-4y^2)\}dy$$

$$= 2\int_{0}^{1} (1 - y^2)dy$$

$$= 2\left[y - \frac{y^3}{3} \right]_{0}^{1} = \frac{4}{3}$$



13. The area between the parabola $4y = 3x^2$ and the line $2y = 3x + 12$ is given by the integral ***[NDA-2004]***

(a) $\int_{-2}^{4} \frac{3x^2}{4} dx$

(b) $\int_{0}^{4} \left(\frac{3x+12}{2} - \frac{3x^2}{4} \right) dx$

(c) $\int_{-2}^{4} \left(\frac{3x+12}{2} - \frac{3x^2}{4} \right) dx$

(d) $\int_{-2}^{2} \left(\frac{3x+12}{2} - \frac{3x^2}{4} \right) dx$

Solution

(c) Two curves meet at $x = -2$ and $x = 4$. So required area $= \int_{-2}^{4} \left(\frac{3x+12}{2} - \frac{3x^2}{4} \right) dx$

14. Area of a loop of the curve $r = a \sin 3\theta$ is ***[VIT-2006]***

(a) $\pi a^2/6$ (b) $\pi a^2/8$

(c) $\pi a^2/12$ (d) $\pi a^2/24$

Solution

(c) Area of one loop $= \int_{0=0}^{\pi/3} \frac{1}{2} r^2 d\theta$

$= \frac{1}{2} \int_{0}^{\pi/3} a^2 \sin^2 3\theta d\theta$

$= \frac{1}{2} \cdot \frac{1}{3} a^2 \int_{0}^{\pi} \sin^2 t \, dt$, where $t = 3\theta$

$= \frac{1}{3} a^2 \int_{0}^{\pi/2} \sin^2 t \, dt$

$= \frac{1}{3} a^2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi a^2}{12}$

15. If $f(x) = \begin{cases} x & , \ 0 \le x < 1/2 \\ 1/2 & , \ x = 1/2 \\ 1-x & , \ 1/2 < x \le 1 \end{cases}$ and $g(x) = (x - 1/2)^2$, $x \in R$. Then the area of the portion bounded between $g(x)$ and $f(x)$ in the interval $[1/2, \sqrt{3}/2]$ is ***[ICS-2006]***

(a) $\sqrt{3}/4 - 1/3$ (b) $\sqrt{3}/4 + 1/3$

(c) 0 (d) $\sqrt{3}/12$

Solution

(b) In the given interval given curves meet at $x = \sqrt{3}/2$

So, required area

$= \int_{1/2}^{\sqrt{3}/2} \{f(x) - g(x)\} dx$

$= \int_{1/2}^{\sqrt{3}/2} [(1 - x) - (x - 1/2)^2] dx$

$= \left[x \frac{x^2}{2} - \frac{(x - 1/2)^3}{3} \right]_{1/2}^{\sqrt{3}/2}$

$= \frac{\sqrt{3}}{4} - \frac{1}{3}$.

16. The area of the region enclosed by curves $y = x \log x$ and $y = 2x - 2x^2$ is ***[Roorkee-98]***

(a) 1/2 (b) 5/12

(c) 7/12 (d) 1

Solution

(c) Two curves meet at (1, 0). Also $y = 2x - 2x^2$ passes through origin and $y = x \log x$ approches to origin. Hence required area is as shown in the diagram, and area

$= \int_{0}^{1} \{(2x - 2x^2) - x \log x\} dx$

$= \left[x^2 - \frac{2x^3}{3} - \frac{x^2}{2} \log x + \frac{x^2}{4} \right]_0^1$

$= 1 - 2/3 + \frac{1}{4} = \frac{7}{12}$.

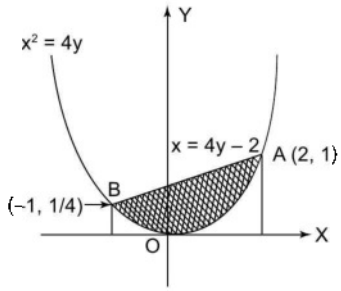
17. Area bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is ***[SCRA-1986; IIT-1981; Pb. CET-2003; MP-97, 2000, 2001; CBSE-2004, 05; PSB-91(C); HPSB-2001(C)]***

(a) $\frac{8}{9}$ sq. unit (b) $\frac{9}{8}$ sq. unit

(c) $\frac{4}{3}$ sq. unit (d) None of these

Solution

(b) Solving the equations $x^2 = 4y$ and $x = 4y - 2$ simultaneously. The points of intersection of the parabola and the line are $A(2, 1)$ and $B(-1, 1/4)$.



$\therefore$ The required area = shaded area

$$= \left[\int_{-1}^{2} ydx_{(\text{for } x=4y-2)}\right] - \left[\int_{-1}^{2} ydx_{(\text{for } x^2=4y)}\right]$$

$$= \int_{-1}^{2} \frac{1}{4}(x+2)dx - \int_{-1}^{2} \frac{1}{4}x^2 dx$$

$$= \frac{1}{4}\left[\frac{x^2}{2}+2x\right]_{-1}^{2} - \frac{1}{4}\left[\frac{x^3}{3}\right]_{-1}^{2} = \frac{9}{8} \text{ sq. unit}$$

18. Let y be the function which passes through $(1, 2)$ having slope $(2x + 1)$. The area bounded between the curve and x-axis is ***[DCE-2005]***

(a) 6 sq. unit (b) 5/6 sq. unit

(c) 1/6 sq. unit (d) None of these

Solution

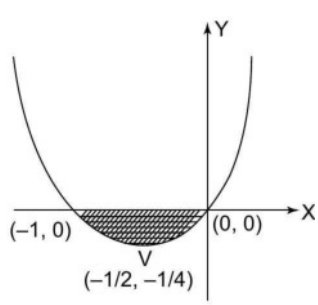
(c) $\frac{dy}{dx} = 2x + 1$

$\Rightarrow y = x^2 + x + c$

$\Rightarrow y = x^2 + x,$

$[\because\ c = 0$ by putting $x = 1, y = 2)$

$\Rightarrow \left(x+\frac{1}{2}\right)^2 = y + \frac{1}{4},$ which is a equation of parabola, whose vertices is, $V\left(\frac{-1}{2}, \frac{-1}{4}\right)$



$\therefore$ Required area $= \left|\int_{-1}^{0} (x^2 + x)dx\right|$

$$= \left(\frac{x^3}{3} + \frac{x^2}{2}\right)_{-1}^{0}$$

$$= \left|\frac{-1}{3} + \frac{1}{2}\right| = \frac{1}{6} \text{ sq. unit}$$

19. Curve $2y = 2 - 3x - 2x^2$ and x-axis area bounded by the region? ***[UPSC-2007]***

(a) $\frac{125}{48}$ square unit (b) 4 square unit

(c) 3 square unit (d) $\frac{125}{24}$ square unit

Solution

(a) $y = \frac{1}{2}(2 - 3x - 2x^2) = -\frac{1}{2}(2x - 1)(x + 2)$

Area of required curve

$$= \int_{-2}^{1/2} ydx = \int_{-2}^{1/2} -\frac{1}{2}(2x-1)(x+2)dx$$

$$= \frac{1}{2}\int_{-2}^{1/2} (2 - 3x - 2x^2)dx$$



$$= \left[x - \frac{x^3}{3} - \frac{3}{2}.\frac{x^2}{2}\right]_{-2}^{1/2}$$

$$= \left(\frac{1}{2} - \frac{1}{24}\right) - \left(-2 + \frac{8}{3}\right) - \frac{3}{2}\left(\frac{1}{8} - 2\right)$$

$$= \frac{11}{24} - \frac{2}{3} + \frac{45}{16} = \frac{22 - 32 + 135}{48} = \frac{125}{48} \text{ sq. unit}$$

20. Find the area bounded by the curve $y = 2x^4 - x^2$ and x-axis in between two minimima point ***[DCE-2007]***

(a) 7/120 (b) 9/120

(c) 11/120 (d) None of these

Solution

(a) We know that $y = 2x^4 - x^2$

$\therefore\ \frac{dy}{dx} = 8x^3 - 2x$ Put $\frac{dy}{dx} = 0$ $x = 0, x = \pm\frac{1}{2}$

Now $\frac{d^2y}{dx^2} = 24x^2 - 2$

We seen clearly in $x = \pm\frac{1}{2}$ is y is minimized

required area $= \left|\int_{-1/2}^{1/2} ydx\right| = \left|\int_{-1/2}^{1/2} (2x^4 - x^2)dx\right|$

$$= \left|2\int_0^1 (2x^4 - x^2)dx\right| = \left|2\left[\frac{2x^5}{5} - \frac{x^3}{3}\right]_0^{1/2}\right|$$

$$= \left|2\left[\frac{1}{80} - \frac{1}{24}\right]\right| = \frac{7}{120}$$

21. Consider the function $f : (-\infty, \infty) \to (-\infty, \infty)$ defined by $f(x) = \dfrac{x^2 - ax + 1}{x^2 + ax + 1}, 0 < a < 2$. Let $g(x) = \displaystyle\int_0^{e^x} \frac{f'(t)}{1+t^2}dt$. Which of the following is true? ***[IIT JEE-2008]***

(a) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$

(b) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$

(c) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$

(d) $g'(x)$ is does not change sign on $(-\infty, \infty)$.

Solution

(b) We have $g'(x) = \dfrac{f'(e^x)e^x}{1+e^{2x}}$

Now $f'(e^x) = \dfrac{2a.(e^{2x} - 1)}{(e^{2x} + ae^x + 1)^2}$

It is seen from the above that $f'(e^x)$ and so $g'(x)$ is positive on $(0, \infty)$ and negative on $(-\infty, 0)$.

22. The area bounded by the curves y $\cos x$ and y $\sin x$ between the ordinates x 0 and $x = \dfrac{3\pi}{2}$ is ***[AIEEE-2010]***

(a) $4\sqrt{2} + 2$ (b) $4\sqrt{2} - 1$

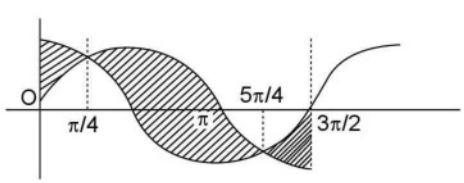
(c) $4\sqrt{2} + 1$ (d) $4\sqrt{2} - 2$

Solution

(d) Required area

$$= \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)dx +$$

$$\int_{5\pi/4}^{3\pi/2} (\cos x - \sin x)dx$$



$$= 2[\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$$

$$= 4\sqrt{2} - 2$$

Hence correct option is (d)

23. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is

(a) $e - 1$ (b) $\displaystyle\int_1^e \ln(e + 1 - y)\,dy$

(c) $\displaystyle e - \int_0^1 e^x dx$ (d) $\displaystyle\int_1^e \ln y\,dy$

Solution

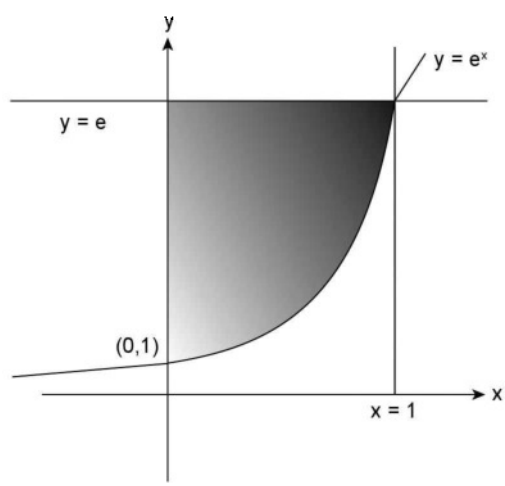
(b, c, d)

Required area

$$= \int_1^e \ln y\,dy = (y\ln y - y)_1^e = (e - e) - \{-1\} = 1$$

Also $\displaystyle\int_0^1 \ln y dy = \int_1^e \ln(e + 1 - y)\,dy$

Further the required area $= e \times 1 - \displaystyle\int_0^1 e^x dx$



24. Interval containing the value of the integral

$$\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$$

[IIT-2009]

Solution

Put $x = t + 3$

$$\int_{2}^{2}(t+2)(t+1)t(t-1)(t-2)\,dt$$

$$= \int_{-2}^{2} t(t^2-1)(t^2-4)\,dt = 0$$

(being odd function)

(p) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(l) $(-\pi, \pi)$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. Area inclined between the two curves $y^2 = 4ax$ and $x^2 = 4ay$, is: ***[SCRA-86; Roorkee-84; RPT-99; Kerala Eng.-2002, 05]***

(a) $\frac{32}{3}a^2$ sq. units (b) $\frac{16}{3}$ sq. units

(c) $\frac{32}{3}$ sq. units (d) $\frac{16}{3}a^2$ sq. units

2. The area enclosed by the parabola $y^2 = 4ax$ and the straight line $y = 2ax$, is ***[MPPET-1993]***

(a) $a^2/3$ sq. units (b) $1/3a^2$ sq. units

(c) $1/3a$ sq. units (d) $2/3a$ sq. units

3. Area bounded by lines $y = 2 + x$, $y = 2 - x$ and $x = 2$ is ***[MPPET-1996]***

(a) 3 (b) 4

(c) 8 (d) 16

4. The ratio of the areas bounded by the curves $y = \cos x$ and $y = \cos 2x$ between $x = 0$, $x = \pi/3$ and x-axis is ***[MPPET-97]***

(a) $\sqrt{2}:1$ (b) 1 : 1

(c) 1 : 2 (d) 2 : 1

5. The area in the first quadrant between $x^2 + y^2 = \pi^2$ and $y = \sin x$ is ***[MPPET-97]***

(a) $\frac{(\pi^3 - 8)}{4}$ (b) $\frac{\pi^3}{4}$

(c) $\frac{(\pi^3 - 16)}{4}$ (d) $\frac{(\pi^3 - 8)}{2}$

6. The area of the region bounded by $y = |x - 1|$ and $y = 1$ is ***[IIT Screening-1994]***

(a) 2 (b) 1

(c) 1/2 (d) None of these

7. The area bounded by the circle $x^2 + y^2 = 4$, line $x = \sqrt{3}y$ and x-axis lying in the first quadrant, is ***[RPET-1997; Kurukshetra CEE-1998]***

(a) $\pi/2$ (b) $\pi/4$

(c) $\pi/3$ (d) π

8. The area formed by triangular shaped region bounded by the curves $y = \sin x$, $y = \cos x$ and $x = 0$ is ***[MPPET-2000]***

(a) $\sqrt{2} - 1$ (b) 1

(c) $\sqrt{2}$ (d) $1 + \sqrt{2}$

9. The area of region $\{(x, y) : x^2 + y^2 \le 1 \le x + y\}$ is ***[Kerala (Engg.)-2002]***

(a) $\pi^2/5$ (b) $\pi^2/2$

(c) $\pi^2/3$ (d) $\frac{\pi}{4} - \frac{1}{2}$

10. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is ***[IIT Screening-2002]***

(a) 1 (b) 2

(c) $2\sqrt{2}$ (d) 4

11. The area of the region bounded by the curves $y = x^2$ and $y = |x|$ is ***[Roorkee-1999]***

(a) 1/6 (b) 1/3

(c) 5/6 (d) 5/3

12. The area enclosed by the parabolas $y = x^2 - 1$ and $y = 1 - x^2$ is ***[AMU-1999]***

(a) 1/3 (b) 2/3

(c) 4/3 (d) 8/3

13. For $0 \le x \le \pi$, the area bounded by $y = x$ and $y = x + \sin x$, is ***[Roorkee Qualifying-1998]***

(a) 2 (b) 4

(c) 2π (d) 4π

14. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is ***[AIEEE-2007]***

(a) 1 (b) 1/6

(c) 1/3 (d) 2/3

15. The area of the loop of the curve $y^2 = x^2(1-x)$ is *[MP PET-2007]*

(a) 15/8 (b) 4/15

(c) 8/15 (d) None of these

16. The area bounded by the curves $y = \ln x$, $y = \ln|x|$, $y = |\ln x|$ and $y = |\ln||x||$ is *[AIEEE-2002]*

(a) 4 sq. units (b) 6 sq. units

(c) 10 sq. units (d) None of these

17. The area of the region bounded by the curves $y = |x-1|$ and $y = 3 - |x|$, is *[AIEEE-2003; Orissa JEE-2003]*

(a) 2 sq. units (b) 3 sq. units

(c) 4 sq. units (d) 1 sq. units

18. The area of the circle $(x-2)^2 + (y-3)^2 = 32$ below the line $y = x + 1$ is

(a) $\int_{-2}^{6}[(x+1)+\sqrt{32-(x-2)^2}+3]dx$

(b) 27π

(c) 16π

(d) 8π

19. If area of the region bounded by the curve $xy = 1$, x-axis and ordiantes $x = 1$ and $x = 2$ is A_1 and A_2 be the area bounded by the curve $xy = 1$, x-axis and ordinates $x = 2$, $x = 4$, then

(a) $A_1 = A_2$ (b) $A_1 = 3A_2$

(c) $A_1 = \frac{A_2}{2}$ (d) $A_1 = 2A_2$

20. Area bounded by the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ ($x, y > 0$) and coordinate axes is

(a) $a^2/3$ (b) $a^2/8$

(c) $a^2/6$ (d) $2a^2/3$

21. Area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and line $\frac{x}{a} + \frac{y}{b} = 1$ is

(a) $\frac{1}{2}\pi$ (b) $\frac{1}{2}ab$

(c) $\frac{\pi ab}{4} - \frac{ab}{2}$ (d) $\frac{1}{4}ab$

22. Area bounded by the Parabola $y = x^2 + 1$ and straight line $x + y = 3$ is

(a) 45/7 (b) 25/4

(c) $\pi/18$ (d) 9/2

23. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to *[AIEEE-2008]*

(a) $\frac{4}{3}$ (b) $\frac{5}{3}$

(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

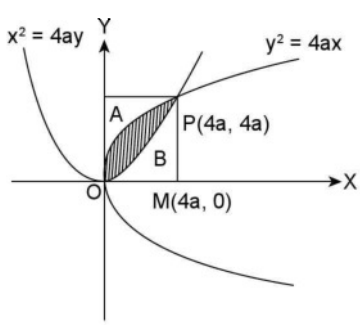
24. The area of the region bounded by the parabola $(y-2)^2 = x - 1$, the tangent to the parabola at the point (2, 3) and the x-axis is *[AIEEE-2009]*

(a) 3 (b) 6

(c) 9 (d) 12

SOLUTIONS

1. (d) Step 1: Bounded Area $= \int_0^{4a}\left[\sqrt{4ax} - \frac{x^2}{4a}\right]dx$



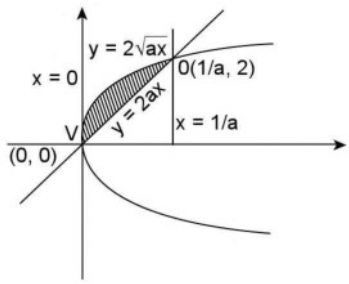
(Limit of the definite integral corresponding to the area is obtained by point of intersection of two curves)

$$= \left(2\sqrt{a}\frac{x^{3/2}}{3/2} - \frac{x^3}{12a}\right)\Bigg|_0^{4a} = \frac{16}{3}a^2$$

2. (c) Limit of the definite integral is obtained by point of intersection of parabola and straight line as follows

$$(2ax)^2 = 4ax \Rightarrow x = 0, \frac{1}{a}$$

Hence area bounded between Curves $y = 2\sqrt{ax}$ and $y = 2ax$ lies between ordinate x 0 and $x = \frac{1}{a}$



$$= \int_0^{1/a} \left(2\sqrt{ax} - 2ax\right) dx$$

$$= \frac{1}{3a} \text{ sq. units}$$

NOTES

1. Area bounded by parabola $y^2 = 4ax$ and its chord $y = mx$ $\frac{8a^2}{3m^3}$ sq. units

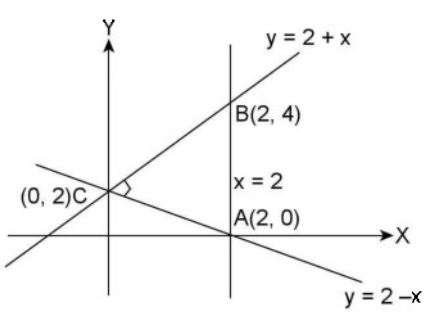
2. Area bounded by two curve $y = f(x)$ and $y = g(x)$ and between the ordinate $x = a$ and $x = b$.

$$= \int_a^b [f(x) - g(x)]\, dx$$

Here by changing order of $f(x)$ and g(x) sign of the area only changes but its value is taken always positive.

Hence Required area $= \left| \int_a^b (f(x) - g(x))\, dx \right|$

3. (b) Obviously, triangle ACB is right angled at C.



$\therefore$ Required area $= \frac{1}{2} \times AC \times BC$

$$= \frac{1}{2} \times 2\sqrt{2} \times 2\sqrt{2} = 4 \text{ sq. unit}$$

4. (d) Step 1: According to defination

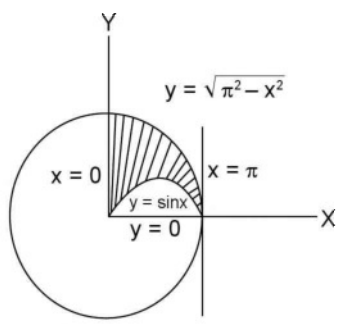
Step 2: Area bounded by the curve $y = \cos x$ and x-axis lies between ordinate $x = 0$ and $x = \frac{\pi}{3}$ is $\int_0^{\pi/3} \cos x\, dx = A_1$ (say)

Step 3: Area bounded by the curve y cos $2x$ and x-axis lies between ordinates $x = 0$ and $x = \frac{\pi}{3}$ is

$$\int_0^{\pi/3} \cos 2x\, dx = A_2 \text{ (say)}$$

Step 4: $\frac{A_1}{A_2} = \frac{(\sin x)\Big|_0^{\frac{\pi}{3}}}{\frac{\sin 2x}{2}\Big|_0^{\frac{\pi}{3}}} = \frac{\frac{\sqrt{3}}{2} - 0}{\frac{1}{2}\left(\frac{\sqrt{3}}{2} - 0\right)} = \frac{2}{1}$

5. (a) Step 1:



Step 2: Area bounded by the curves $y = \sqrt{\pi^2 - x^2}$ and y sin x lies between ordinates x 0 and x π is

$$\int_0^{\pi} \left[\sqrt{\pi^2 - x^2} - \sin x\right] dx$$

$$= \left[\frac{x}{2}\sqrt{\pi^2 - x^2} + \frac{\pi^2}{2}\sin^{-1}\frac{x}{\pi} + \cos x\right]_0^{\pi}$$

$$= \left[0 + \frac{\pi^2}{2} \times \frac{\pi}{2} + (-1) - (0 + 1)\right]$$

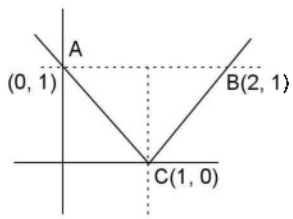
$$= \frac{\pi^3}{4} - 1 - 1 = \frac{\pi^3 - 8}{4}$$

NOTE

$$\int\sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}$$

6. (b) For A and B

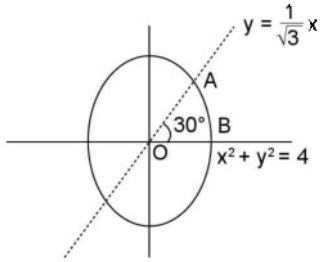
$|x-1|=1\Rightarrow x=0,2$



Required Area = Area of ΔABC

$$=\frac{1}{2}\times 2\times 1=1$$

7. (c) Area of sector $OAB=\dfrac{\angle BOA}{360^\circ}\times\pi r^2$

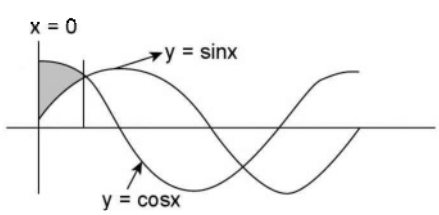


$$=\frac{1}{12}\times\pi\times 2^2=\frac{\pi}{3}$$

8. (a) At point of intersection

y sin x and y cos x

or sin x cos x or tan x 1



Required area = $\int_0^{\pi/4}\cos x\,dx-\int_0^{\pi/4}\sin x\,dx$

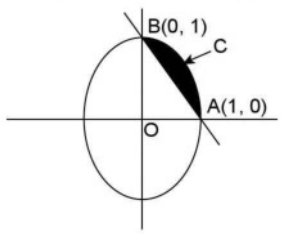
$$=[\sin x]_0^{\pi/4}-[-\cos x]_0^{\pi/4}$$

$$=\sin\frac{\pi}{4}-0+\cos\frac{\pi}{4}-\cos 0$$

$$=\frac{2}{\sqrt{2}}-1=\sqrt{2}-1$$

9. (d) $x^2+y^2=1$ and $x+y\geq 1$

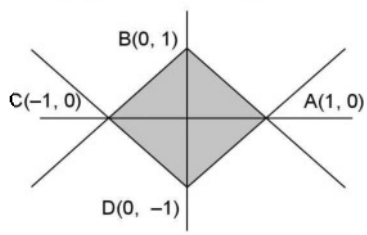
Required region is shown in figure



Area = (Area of quadrant ΛACB) – Area (ΛOAB)

$$=\frac{1}{4}\times\pi(1)^2-\frac{1}{2}\times 1\times 1=\left(\frac{\pi-2}{4}\right)=\frac{\pi}{4}-\frac{1}{2}$$

10. (b) Area between curves y $|x|-1$ and y $1-|x|$ is shown in figure



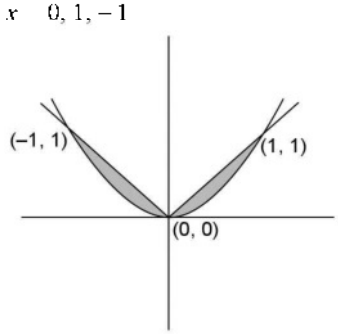
Area = Area of square $ABCD$

$$=\frac{1}{2}\times AC\times BD=\frac{1}{2}\times 2\times 2=2$$

11. (b) $y=x^2$ and $y=|x|$ are curve for point of intersection

$x^2=|x|$

$\therefore$ x 0, 1, –1

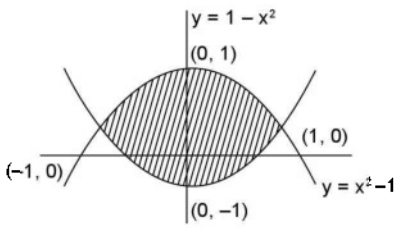


Required Area $\int_{-1}^{1}(|x|-x^2)dx=2\int_0^1(x-x^2)dx$

$$= 2\left(\frac{x^2}{2} - \frac{x^3}{3}\right)_0^1$$

$$= 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}$$

12. (d) $y = x^2 - 1;\ y = 1 - x^2$



Required Area $= 2\int_{-1}^{1}(x^2 - 1)dx$

$$= 4\int_0^1 (x^2 - 1)dx$$

$$= 4\left(\frac{x^3}{3} - x\right)_0^1$$

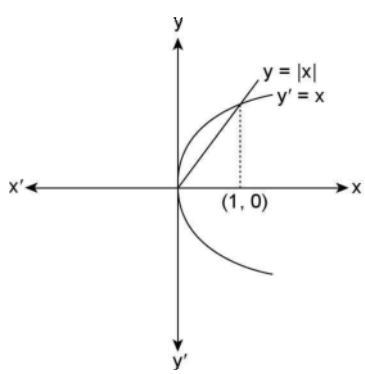
$$= 4\left(\frac{1}{3} - 1\right)$$

$$= \frac{-8}{3}$$

$\because$ Area is positive. 8/3 is Answer.

13. (a) Area $= \int_0^{\pi}(x + \sin x - x)dx = -\cos x\big|_0^{\pi} =$ 2 sq. units

14. (b) Required area $A = \int_0^1\left(\sqrt{x} - x\right)dx$



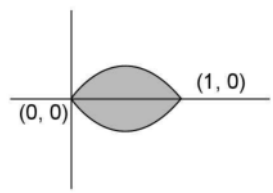
$$A = \int_0^1\left(\sqrt{x} - x\right)dx$$

$$= \left[\frac{2}{3}x^{3/2} - \frac{x^2}{2}\right]_0^1$$

$$= \frac{2}{3} - \frac{1}{2}$$

$$= \frac{1}{6} \text{ sq. unit}$$

15. (c) Curve $y = x\sqrt{1-x}$



$$\text{Area} = 2\int_0^1 x\sqrt{1-x}dx = 2\int_0^1 (1-x)\sqrt{x}dx$$

$$= 2\left(\frac{2x^{3/2}}{3} - \frac{2x^{5/2}}{5}\right)_0^1 = 2\left(\frac{2}{3} - \frac{2}{5}\right) = \frac{8}{15}$$

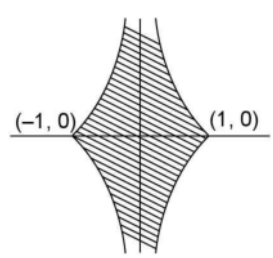
16. (a) $y = \ln x,\ y = \ln |x|;\ y = |\ln x|;\ y = |\ln |x||$

Bounded area is shown in figure

Area = 4 (Area in 4th quadrant)

$$= 4\int_0^1 \ln x dx \text{ or } 4\int_{-\infty}^0 e^y dy$$

$$= 4e^{-y}\Big|_{-\infty}^0$$



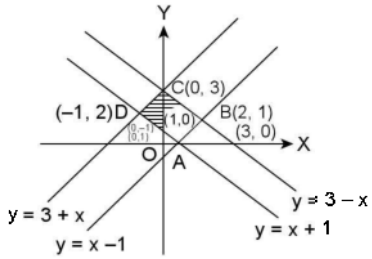
$= 4\,[1 - 0] = 4$ sq. units

$\therefore$ Area = 4

17. (c) If $x \geq 1$, then $3 - x = x - 1 \Rightarrow x = 2$

If $0 \leq x \leq 1$ then $3 - x = 1 - x$, which is not possible .

Also if $x < 0$, then $3 + x \quad 1 - x$ i.e., $x \quad -1$



Thus required area $\int_{-1}^{2}(3-|x|-|x-1|)\,dx$

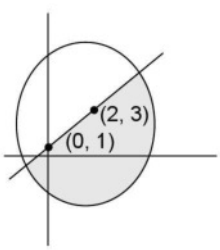
$=\int_{-1}^{0}[3+x-(1-x)]\,dx+\int_{0}^{1}[(3-x)-(1-x)\,dx$

$+\int_{1}^{2}[(3-x)-(x-1)]\,dx$

$= 1 + 2 + 1 = 4$ sq. unit.

18. (c) Circle $(x-2)^2+(y-3)^2=(4\sqrt{2})^2$

$y \quad x+1$



$\because$ (2, 3) lies on $y \quad x+1$

It is diameter of circle, Area

$=\frac{\pi r^2}{2}=\frac{\pi(32)}{2}=16\pi$

19. (a) $A_1=\int_1^2\frac{1}{x}dx=\ln x\Big|_1^2=\ln 2$

$A_2=\int_2^4\frac{1}{x}dx=\ln x\Big|_2^4=\ln 2$

$A_1=A_2$

20. (c) $\sqrt{x}+\sqrt{y}=\sqrt{a}$

Area $=\int_0^a\left(\sqrt{a}-\sqrt{x}\right)^2dx$

$=\int_0^a(a+x-2\sqrt{a}\sqrt{x})dx$

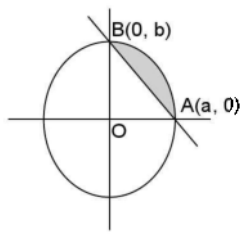
$=ax+\frac{x^2}{2}-2\sqrt{a}\frac{2}{3}x^{3/2}\Big|_0^a$

$=a^2+\frac{a^2}{2}-\frac{4}{3}\sqrt{a}.a^{3/2}$

$=\frac{3a^2}{2}-\frac{4a^2}{3}$

$=\frac{a^2}{6}$ sq. units.

21. (c) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1;\frac{x}{a}+\frac{y}{b}=1$



Required Area $=\frac{\pi ab}{4}-$ Area (ΔOAB)

$=\frac{\pi ab}{4}-\frac{1}{2}ab$

($\because$ Area covered by ellipse in each quadrant

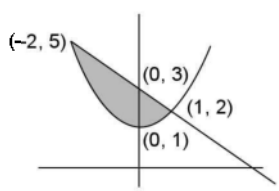
$=\frac{\text{Total Area}}{4}=\frac{\pi ab}{4}$

22. (d) For point of intersection

$x^2+1=3-x$

$x^2+x-2=0$

$x \quad -2, 1$



$\therefore$ Area $=\int_{-2}^{1}\left[3-x-(x^2+1)\right]dx$

$=\int_{-2}^{1}(2-x-x^2)dx$

$=2x\Big|_{-2}^{1}-\frac{x^2}{2}\Big|_{-2}^{1}-\frac{x^3}{3}\Big|_{-2}^{1}$

$=6+\frac{3}{2}-\frac{9}{3}$

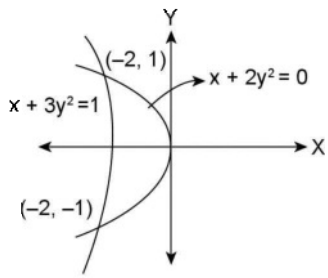
$=\frac{15}{2}-3=\frac{9}{2}$ sq. units

23. (a) Given curves are

$x + 3y^2 = 1$ ..... (i)

and $x + 2y^2 = 0$ ..... (ii)

On solving equation (i) and (ii), we get



$y \quad \pm 1$ and $x \quad -2$

$\therefore$ Required area $= \left|\int_{-1}^{1}(x_1 - x_2)dy\right|$

$= \left|\int_{-1}^{1}(1 - 3y^2 + 2y^2)dy\right|$

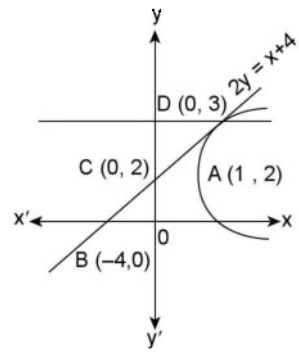
$= 2\int_{0}^{1}(1 - y^2)dy$

$= \left|2\left[y - \frac{y_3}{3}\right]_0^1\right| \quad = \left|2\left(1 - \frac{1}{3}\right)\right| = \frac{4}{3}$ sq. unit

24. (c) Equation of tangent at (2, 3) to $(y - 2)^2 = x - 1$ is

$S_1 = 0 \Rightarrow x - 2y + 4 = 0$

Required area = Area of ΔOCB + Area of OAPD – Area of ΔPCD



$= \frac{1}{2}(4 \times 2) + \int_0^3 (y^2 - 4y + 5)dy - \frac{1}{2}(1 \times 2)$

$= 4 + \left[\frac{y^3}{3} - 2y^2 + 5y\right]_0^3 - 1 = 4 - 9 - 18 + 15 - 1$

= 28 –19 = 9 sq. units.

Alternate:

Area $= \int_0^3 (2y - 4 - y^2 + 4y - 5)dy$

$= \int_0^3 (-y^2 + 6y - 5)dy$

$= -\int_0^3 (3 - y)^2 dy = \left[\frac{(y-3)^3}{3}\right]_0^3$

$= \frac{27}{3} = 9$ sq. units.

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The area within the curve $|x| + |y| = 4$ is

(a) 4 sq. units (b) 8 sq. units

(c) 16 sq. units (d) 32 sq. units

2. If area bounded by the curves $y^2 = 4ax$ and $y = mx$ is $a^2/3$, then the vlaue of m is

(a) 2 (b) –2

(c) 1/2 (d) None of these

3. The area of the region bounded by the limits $x = 0$, $x = \frac{\pi}{2}$ and $f(x) = \sin x$, $g(x) = \cos x$ is

(a) $2(\sqrt{2} + 1)$ (b) $\sqrt{3} - 1$

(c) $2(\sqrt{3} - 1)$ (d) $2(\sqrt{2} - 1)$

4. The area of the triangle formed by the tangent to the hyperbola $xy = a^2$ and coordinate axes is

(a) a^2 (b) $2a^2$

(c) $3a^2$ (d) $4a^2$

5. Area bounded by the curves $x = 1$ and $x = 3$, $xy = 1$and x-axis is

(a) log 2 (b) log 3

(c) log 4 (d) None of these

6. The area between the curve $y = 1 - |x|$ and x-axis is
(a) 2 (b) 1
(c) 5 (d) 7

7. Area bounded by the curves $y = 2 - x^2$ and $y = x$ is
(a) 7/2 (b) 8/7
(c) 9/2 (d) 7/17

8. Area bounded by the curve $y = x|x|$, x-axis and ordinates $x = 1$ and $x = -1$ is
(a) 0 (b) 1/3
(c) 2/3 (d) 1

9. Calculate the area bounded by the x-axis and parabola $y = 6 - x - x^2$ is
(a) $4\frac{5}{6}$ (b) $12\frac{3}{4}$
(c) $20\frac{5}{6}$ (d) None of these

10. Find the area of the region bounded by the curve $y = x^2 + 1$, the lines $y = x + 1$, the x-axis and the ordinates $x = 0$ and $x = 2$
(a) 5/6 sq. units (b) $5\frac{3}{6}$ sq. units
(c) $3\frac{3}{6}$ sq. units (d) $3\frac{5}{6}$ sq. units

11. The area common to parabolas $y = 5x^2$ and $y = 2x^2 + 9$ is
(a) $12\sqrt{3}$ sq. units (b) $36\sqrt{3}$ sq. units
(c) 108 sq. units (d) $144\sqrt{3}$ sq.units

12. Find the area of the region bounded by the x-axis and the curves defined by $y = \tan x$; $-\frac{\pi}{3} \le x \le \frac{\pi}{3}$ and $y = \cot x$; $\frac{\pi}{6} \le x \le \frac{3\pi}{2}$
(a) $2\log 2$ (b) $\sqrt{2}\log 2$
(c) $2\sqrt{2}\log 2$ (d) $\log(3/2)$

13. The area in square units bounded by the curves $y = x^3$, $y = x^2$ and the ordinates $x = 1$, $x = 2$ is
(a) 17/12 (b) 12/13
(c) 2/7 (d) 7/2

14. The area common to the parabolas $y = 2x^2$ and $y = x^2 + 4$ is
(a) 2/3 sq. units (b) 3/2 sq. units
(c) 32/3sq. units (d) 3/32 sq. units

15. The area in the first quadrant enclosed by the x-axis, the line $y = \frac{1}{\sqrt{3}}x$ and circle $x^2 + y^2 = 4$, is given by
(a) $\int_0^{\sqrt{3}} (x + \sqrt{4 - x^2})dx$
(b) $\int_0^{2} (x + \sqrt{4 - x^2})dx$
(c) $\int_0^{2} (\frac{x}{\sqrt{3}} + \sqrt{4 - x^2})dx$
(d) $\int_0^{\sqrt{3}} \frac{x}{\sqrt{3}}dx + \int_{\sqrt{3}}^{2} \sqrt{4 - x^2}dx$

16. Which one of the following definite integrals represents the area included between the parabola $4y = 3x^2$ and the straight line $2y = 3x + 12$?
(a) $\int_{-2}^{4} \frac{3x^2}{4}dx$
(b) $\int_0^{4} \left(\frac{3x+12}{2} - \frac{3}{4}x^2\right)dx$
(c) $\int_{-2}^{4} \left(\frac{3x+12}{2} - \frac{3}{4}x^2\right)dx$
(d) $\int_{-2}^{2} \left(\frac{3x+12}{2} - \frac{3}{4}x^2\right)dx$

17. The area of the region bounded by the curves $y = |x - 2|$, $x = 1$, $x = 3$ and the x-axis is
(a) 3 (b) 2
(c) 1 (d) 4

18. Area enclosed by the curve $\pi[4(x - \sqrt{2})^2 + y^2] = 8$ is ***[VITEEE-2008]***
(a) π (b) 2
(c) 3π (d) 4

19. The area of the upper half of the circle whose equation is $(x - 1)^2 + y^2 = 1$ is given by
(a) $\int_0^{2} \sqrt{2x - x^2}dx$ (b) $\int_0^{1} \sqrt{2x - x^2}dx$
(c) $\int_1^{2} \sqrt{2x - x^2}dx$ (d) $\pi/4$

20. The ratio in which the area between the parabola $y = 4x - x^2$ and $y = x^2 - x$ is divided by x-axis is ***[IIT-94]***
(a) 125/4 (b) 1/6
(c) 121/4 (d) None of these

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 15 minutes.
3. The worksheet consists of 15 questions. The maximum marks are 45.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. The area of the region bounded by the curve $y^2 = 2y - x$ and the y-axis is

(a) $\frac{1}{3}$ sq. units (b) $\frac{2}{3}$ sq. units

(c) $\frac{4}{3}$ sq. units (d) $\frac{5}{3}$ sq. units

2. The area bounded by the curve $xy = 4$ the x-axis and the lines $x = 1$ and $x = 3$ is

(a) 2 log 3 sq. units
(b) 4 log 3 sq. units
(c) 3 log 3 sq. units
(d) None of these

3. Area bounded by the parabola $y^2 = x$ and straight line $y = \frac{x}{2}$ is

(a) 4/3 (b) 1
(c) 2/3 (d) 1/3

4. The area between te hyperbola $xy = c^2$, then x-axis and the ordinates at a and b with $a > b$ is ***[UPSEAT-99; AMU-91]***

(a) $c^2 \log\left(\frac{a}{b}\right)$

(b) $c^2 \log\left(\frac{b}{a}\right)$

(c) $c^2 \log (a\ \ b)$
(d) None of these

5. The area enclosed by the curves $x^2 = y$, $y = x + 2$ and x-axis is ***[Roorkee-1997]***

(a) 5/6 (b) 5/4
(c) 5/2 (d) None of these

6. If the area enclosed by $y^2 = 4ax$ and the line $y = ax$ is 1/3 sq. units, then the area enclosed by $y = 4x$ with the same parabola is ***[Pb. CET-2006]***

(a) 8 sq. units (b) 4 sq. units
(c) 4/3 sq. units (d) 8/3 sq. units

7. The area bounded by the curve $y = x^2$, the normal at (1, 1) and x-axis is ***[Him. CET-2006]***

(a) 4/3 (b) 2/3
(c) 1/3 (d) None of these

8. Make a rough sketch of the graph of the function $y = 2 + \cos x$; $0 \le x \le \frac{\pi}{2}$ and find the area between the curve x-axis and the ordinates $x = 0$ and $x = \frac{\pi}{2}$

(a) π unit^2
(b) $(\pi + 1)$ sq. unit
(c) $(\pi^2 + 1)$ sq. units
(d) None of these

9. Area bounded by the curve $y = x^3$, x-axis and two ordinates $x = -2$ and $x = 1$ is

(a) -9 (b) $-15/4$
(c) 15/4 (d) 17/4

10. Area lying between the curves $y = \tan x$, $y = \cot x$ and x-axis $x \in [0, \pi/2]$

(a) $\frac{1}{2}\log 2$ (b) $\frac{1}{3}\log 2$

(c) $\log 2$ (d) $\log 3$

11. The area of the region bounded by the curves $y = \cos\left(\frac{\pi}{2}x\right)$ and $y = x^2 - 1$ between $x = 0$ and $x = 2$ is ***[Kerala PET-2008]***

(a) $\frac{4}{\pi} - 2$ (b) $\frac{4}{\pi} + 2$

(c) $\frac{\pi - 3}{\pi}$ (d) $\frac{\pi + 3}{\pi}$

12. Area bounded by the curve in between $y = x^2$ and $y = x^3$ in (square unit) ***[EAMCET-2007]***

(a) 1/12 (b) 1/6
(c) 1/3 (d) 1

13. The area enclosed between the curves $y = ax^2$ and $x = ay^2 (a > 0)$ is 1 sq. unit, then the value of a is

(a) $1/\sqrt{3}$ (b) 1/2
(c) 1 (d) 1/3

14. Curve $xy = 16$, X-axis and lines $x = 4$ and $x = 8$ is area bounded by the curve region?
[Gujarat CET-2007]

(a) $2 \log_e 16$ sq. unit (b) $16 \log_e 2$ sq. unit
(c) $\log_e 4$ sq. unit (d) $\log_e 16$ sq. unit

15. Find the area of the region bounded by the curves $y = x^2 + 2$, $y = x$, $x = 0$ and $x = 3$

(a) $5\frac{1}{2}$ (b) $10\frac{1}{2}$
(c) 15 (d) None of these

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	**6.** ⓐ ⓑ ⓒ ⓓ	**11.** ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	**7.** ⓐ ⓑ ⓒ ⓓ	**12.** ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	**8.** ⓐ ⓑ ⓒ ⓓ	**13.** ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	**9.** ⓐ ⓑ ⓒ ⓓ	**14.** ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	**10.** ⓐ ⓑ ⓒ ⓓ	**15.** ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

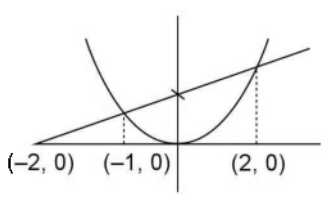
1. (c) For y-axis x $0, y^2 = 2y \Rightarrow y$ $0, 2$

$$\text{Area} = \int_0^2 (2y - y^2)dy = y^2 - \frac{y^3}{3}\Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3} \text{ sq. units}$$

5. (a) For point of intersection of x^2 y; y $x + 2$

$x^2 = x + 2 \Rightarrow x = 2, -1$



$$\text{Required Area} = \int_{-2}^{-1} (x + 2)dx + \int_{-1}^{0} x^2 dx$$

$$= \left(\frac{x^2}{2} + 2x\right)\Big|_{-2}^{-1} = \frac{x^3}{3}\Big|_{-1}^{0} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \text{ sq. units}$$

6. (d) Point of intersection of y ax and $y^2 = 4ax$

$(ax)^2 = 4ax \Rightarrow x = 0; \frac{4}{a}$

Given Area $= \frac{1}{3}$ i.e., $\int_0^{4/a} \left(ax - 2\sqrt{ax}\right)dx = \frac{1}{3}$

$$\frac{ax^2}{2} - \frac{2\sqrt{a}x^{3/2}}{3/2}\Big|_0^{4/a} = \pm\frac{1}{3}$$

$$\Rightarrow \frac{8}{a} - \frac{4}{3}, \frac{8}{a} \pm \frac{1}{3}$$

$$\frac{24 - 32}{3a} = \pm\frac{1}{3} \Rightarrow a = \pm 8$$

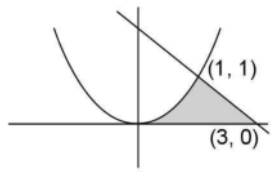
$\therefore$ Required Area $= \int_0^8 \left(\frac{y^2}{32} - \frac{y}{4}\right)dy = \frac{8}{3}$

(Point of intersection are obtained by

$\frac{y^2}{32} = \frac{y}{4}$; y $0, 8$)

7. (a) Equation of normal is

$$y-1=\frac{-1}{\left(\frac{dy}{dx}\right)_{(1,1)}}(x-1)$$



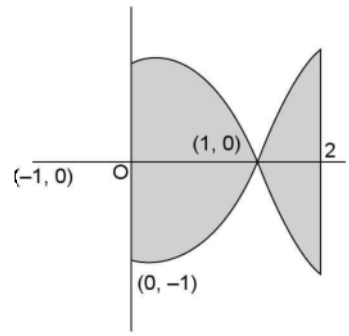
$$y-1=\frac{-1}{2}(x-1)\Rightarrow 2y-2+x-1=0$$

$x + 2y = 3$; A (3, 0)

Required Area $= \int_0^1 x^2dx+\int_1^3\left(\frac{3-x}{2}\right)dx$

$$=\frac{1}{3}+1=\frac{4}{3}$$

11. (c) $y=\cos\left(\frac{\pi}{2}x\right)$; x 0 to 2



$$=\int_0^1\left[\cos\frac{\pi}{2}-\left(x^2-1\right)\right]dx+$$

$$\int_1^2\left(x^2-1-\cos\frac{\pi}{2}x\right)dx=\frac{\pi-3}{\pi}$$

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LECTURE



8

Test Your Skills

ASSERTION/REASONING

Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.

(a) **Assertion** is True, **Reason** is True and **Reason** is a correct explanation for **Assertion**
(b) **Assertion** is True, **Reason** is True and **Reason** is NOT a correct explanation for **Assertion**
(c) **Assertion** is True and **Reason** is False
(d) **Assertion** is False and **Reason** is True

1. **Assertion:** $F(x)=\int_0^x \ln\left(\frac{1-t}{1+t}\right)dt$ is an even function,
Reason: $\ln\left(\frac{1-t}{1+t}\right)$ is an odd function.

2. **Assertion:** $\int_0^1 e^{-x}\cos^2 x\,dx < \int_a^b e^{-x^2}\cos^2 x\,dx$
Reason: $\int_a^b f(x)dx < \int_a^b g(x)dx \forall f(x) \geq g(x)$

3. **Assertion:** $\int_0^{\pi/2}\frac{\sin x}{x}dx < \frac{\pi}{2}$
Reason: $\lim_{x\to\infty}\frac{\sin x}{x}=1$

4. **Assertion:** $\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\frac{1}{n+3}+....+\frac{1}{2n}\right)=\ln 3$
Reason: $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}f\left(\frac{r}{n}\right)=\int_0^1 f(x)dx$, symbols have their usual meaning.

5. **Assertion:** $\int_{\pi/6}^{\pi/3}\frac{(\sin x)^{\sqrt{2}}}{(\sin x)^{\sqrt{2}}+(\cos x)^{\sqrt{2}}}dx=\frac{\pi}{12}$
Reason: $\int_{\pi/6}^{\pi/3}\frac{f(x)dx}{f(x)+f\left(\frac{\pi}{2}-x\right)}dx=\frac{\pi}{12}$

6. **Assertion:** If $\int_0^1 e^{\sin x}dx=\lambda$, then $\int_0^{200} e^{\sin x}dx = 200\lambda$
Reason: $\int_0^{na} f(x)dx = n\int_0^a f(x)dx$, $n \in I$ and $f(a+x)=f(x)$

7. **Assertion:** $\int_1^6 \{x+5\}^2 dx = 41$, where $\{.\}$ denotes the fractional part function.
Reason: $\{x+5\}$ is a periodic function.

8. **Assertion:** $\int_0^{2\pi}\sin^3 x dx = 0$
Reason: $\sin^3 x$ is an odd function.

9. **Assertion:** $\int_{-\pi/2}^{\pi/2}|\sin x|\,dx = 2$
Reason: $\int_a^b f(x)dx=\int_a^c f(x)dx+\int_c^b f(x)dx$, where $c \in (a, b)$

10. **Assertion:** $\int_{\pi/2}^{3\pi/2}[2\sin x]dx = 0$, where $[.]$ denotes the greatest integer function.

Reason: 2 sin x decreasing function in $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

11. **Assertion:** $\int_0^t \cos x dx = \sin t$

Reason: cos x is continuous in any closed interval $[0, t]$

12. **Assertion:** $\int_0^{1/4} \frac{dx}{1+(\cot 2\pi x)^{\sqrt{2}}} = \frac{1}{4}$

Reason: If $a + b = \frac{\pi}{2}$, then

$$\int_a^b \frac{f(\sin x)}{f(\sin x)+f(\cos x)}dx = \frac{(b-a)}{2}$$

13. **Assertion:** $\int_{-1}^{1} \frac{|x|}{x}dx = 2$

Reason: $\frac{|x|}{x}$ is undefined at $x = 0$

14. **Assertion:** $\int_0^2 f(x)dx = \frac{4(\sqrt{2}-1)}{3}$

where $f(x) = \begin{cases} x^2, & \text{for } 0 \le x < 1 \\ \sqrt{x}, & \text{for } 1 \le x \le 2 \end{cases}$

Reason: $f(x)$ is continuous in $[0, 2]$

15. **Assertion:** $\int_0^{\pi/2} (\sin^6 x + \cos^6 x)dx$ lies in the interval $\left(\frac{\pi}{8}, \frac{\pi}{2}\right)$.

Reason: $\sin^6 x + \cos^6 x$ is periodic with period $\pi/2$.

16. **Assertion:** If $f(x) = x - x^2 + 1$ and $g(x) = \max \{f(t): 0 \le t \le x\}$, then $\int_0^1 g(x)dx = \frac{29}{24}$

Reason: $f(x)$ is increasing in (0, 1/2) and decreasing in (1/2, 1).

17. **Assertion:** Area enclosed by the curve $y = e^{x^3}$ between the lines $x = a$, $x = b$ and x-axis is $\int_a^b e^{x^3}dx$.

Reason: e^{x^3} is an increasing function.

18. **Assertion:** The area of the function $y = \sin^2 x$ from 0 to π will be more than that of curve $y = \sin x$ from 0 to π.

Reason: $t^2 < t$, if $0 < t < 1$.

19. **Assertion:** The area bounded by the curves $y = x^2 + 2x - 3$ and the line $y = \lambda x + 1$ is least, If $\lambda = 2$.

Reason: The area bounded by the curve $y = x^2 + 2x - 3$ and $y = \lambda x + 1$ is $\{(\lambda - 2)^2 + 16)^{3/2}\}$ sq. unit.

20. **Assertion:** Let f be a real values function satisfying $f\left(\frac{x}{y}\right) = f(x) - f(y)$ and $\lim_{x \to 0} \frac{f(1+x)}{x} = 3$,. Then the area bounded by the curve $y = f(x)$, the y-axis and the line $y = 3$ is $3e$ sq. unit

Reason: The function $f(x)$ is concave down

21. **Assertion:** If $f(x) = (x - 1)(x - 2)(x - 3)$, then area enclosed by $|f(x)|$ between the lines $x = 2.2$, $x = 2.8$ and x-axis is equal to $\int_{2.2}^{2.8} (x-1)(x-2)(x-3)dx$

Reason: $(x - 1)(x - 2)(x - 3) \le 0 \ \forall \ x \in [2.2, 2.8]$

22. **Assertion:** The area of the ellipse $2x^2 + 3y^2 = 6$ will be more than the area bounded by $2|x| + 3|y| \le 6$.

Reason: The length of major axis of the ellipse $2x^2 + 3y^2 = 6$ is less than the distance between the points of $2|x| + 3|y| \le 6$ on x-axis.

ASSERTION/REASONING: SOLUTIONS

1. (a) $\because \ F(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right)dt$

$$F(x) = \ln\left(\frac{1-x}{1+x}\right) \Rightarrow F'(-x) = \ln\left(\frac{1+x}{1-x}\right)$$

$$= -\ln\left(\frac{1-x}{1+x}\right) = -F'(x)$$

$\Rightarrow F'(x)$ is an odd function.

Hence, $F'(x)$ is an even function.

2. (c) $\because\ 0 < x < 1$

Then $x > x^2$

$\Rightarrow\ -x < -x^2 \Rightarrow e^{-x} < e^{-x^2}$

$\Rightarrow\ e^{-x}\cos^2 x < e^{-x^2}\cos^2 x$

Then, $\int_0^1 e^{-x}\cos^2 x\,dx < \int_0^1 e^{-x^2}\cos^2 x\,dx <$

and if $f(x) \geq g(x)$

Then, $\int_a^b f(x)dx \geq \int_a^b g(x)dx$

3. (b) For $x > 0$

$\sin x < x$

$\Rightarrow\ \dfrac{\sin x}{x} < 1$

$\Rightarrow\ \int_0^{\pi/2} \dfrac{\sin x}{x}dx < \int_0^{\pi/2} 1.dx$

$\Rightarrow\ \int_0^{\pi/2} \dfrac{\sin x}{x}dx < \dfrac{\pi}{2}$ and $\lim\limits_{x\to 0}\dfrac{\sin x}{x} = 1$

4. (d) $\because\ \lim\limits_{n\to\infty}\left(\dfrac{1}{n+1}+\dfrac{1}{n+2}+\dfrac{1}{n+3}+.....+\dfrac{1}{2n}\right)$

$= \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{(n+r)}\ = \lim\limits_{x\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{n}.\dfrac{1}{\left(1+\dfrac{r}{n}\right)}$

$= \int_0^1 \dfrac{dx}{1+x} = [\ln(1+x)]_0^1$

$= \ln 2 - \ln 1 = \ln 2 - 0 = \ln 2$

5. (a) Let $I = \int_{\pi/6}^{\pi/3} \dfrac{f(x)dx}{f(x)+f\left(\dfrac{\pi}{2}-x\right)}$

$= \int_{\pi/6}^{\pi/3} \dfrac{f\left(\dfrac{\pi}{3}+\dfrac{\pi}{6}-x\right)dx}{f\left(\dfrac{\pi}{3}+\dfrac{\pi}{6}-x\right)+f\left(\dfrac{\pi}{2}-\left(\dfrac{\pi}{3}+\dfrac{\pi}{6}-x\right)\right)}$

(by property)

$= \int_{\pi/6}^{\pi/3} \dfrac{f\left(\dfrac{\pi}{2}-x\right)dx}{f\left(\dfrac{\pi}{2}-x\right)+f(x)}$ ..... (ii)

Adding Equation (i) and (ii), then

$2I = \int_{\pi/6}^{\pi/3} 1.dx = \dfrac{\pi}{3}-\dfrac{\pi}{6} = \dfrac{\pi}{6}\ \therefore\ I = \dfrac{\pi}{12}$

Also, for $f(x) = (\sin x)^{\sqrt{2}}$, $I = \dfrac{\pi}{12}$

6. (d) $\because$ Period of $e^{\sin x}$ is 2π

$\therefore\ \int_0^{200} e^{\sin x}dx \neq 200\lambda$

7. (d) $\because\ \int_1^6 \{x+5\}^5 dx = \int_0^5 \{x+6\}^2 dx$

(by property)

$= \int_0^5 \{x\}^2 dx = 5\int_0^1 \{x\}^2 dx$ ($\because$ $\{.\}$ is periodic with period 1) $= 5\int_0^1 x^2 dx = \dfrac{5}{3}$

8. (b) $\because\ I = \int_0^{2\pi} \sin^3 x dx$

$= \int_0^{2\pi} (1-\cos^2 x)\sin x\,dx$

Put $\cos x = t \Rightarrow \sin x\,dx = -dt$

Then, $I = \int_1^1 (1-t^2)(-dt) = 0$

9. (b) $\because$ $|\sin x|$ is an even function

$\therefore\ \int_{-\pi/2}^{\pi/2} |\sin x|\,dx$

$= 2\int_0^{\pi/2} |\sin x|\,dx = 2\int_0^{\pi/2} \sin x\,dx$

$= -2(\cos x)_0^{\pi/2} = -2(0-1) = 2$

10. (d) $\because\ \int_{\pi/2}^{3\pi/2} [2\sin x]dx$

$= \int_{\pi/2}^{5\pi/6} [2\sin x]dx + \int_{5\pi/6}^{\pi} [2\sin x]dx$

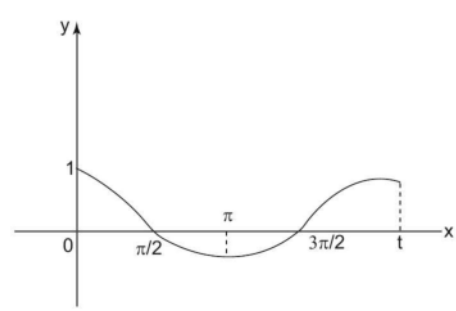
$+\int_{\pi}^{7\pi/6} [2\sin x]dx + \int_{7\pi/6}^{3\pi/2} [2\sin x]dx$

$= \int_{\pi/2}^{5\pi/6} 1.dx + 0 - \int_{\pi}^{7\pi/6} 1.dx - 2\int_{7\pi/6}^{3\pi/2} 1.dx$

$= \dfrac{\pi}{3}-\dfrac{\pi}{6}-\dfrac{2\pi}{3} = -\dfrac{\pi}{2}$

($\because$ $2\sin x$ is decreasing function in $(\pi/2, 3\pi/2)$)

11. (a) $\because$ $\cos x$ is continuous in $[0, t]$



$\therefore\ \int_0^t \cos x\,dx = [\sin x]_0^t = \sin t - \sin 0 = \sin t$

12. (d)Let $I = \int_0^{1/4} \frac{dx}{1+(\cot 2\pi x)^{\sqrt{2}}}$

$$= \frac{1}{2\pi}\int_0^{\pi/2} \frac{dx}{1+(\cot x)^{\sqrt{2}}}$$

$$= \frac{1}{2\pi}\int_0^{\pi/2} \frac{(\sin x)^{\sqrt{2}}}{(\sin x)^{\sqrt{2}}+(\cos x)^{\sqrt{2}}}dx$$

$$= \frac{1}{2\pi}\left\{\frac{\frac{\pi}{2}-0}{2}\right\} = \frac{1}{8} \text{ and let}$$

$$I_1 = \int_a^b \frac{f(\sin x)}{f(\sin x)+f(\cos x)}dx \quad \text{..... (i)}$$

$$I_1 = \int_a^b \frac{f(\sin(a+b-x))}{f(\sin(a+b-x))+f(\cos(a+b-x))}dx$$

$$= \int_a^b \frac{f(\cos x)}{f(\cos x)+f(\sin x)}dx \quad \text{..... (ii)}$$

Adding equations (i) and (ii), then

$$2I_1 = \int_a^b dx = b-a \ \therefore \ I_1 = \frac{(b-a)}{2}$$

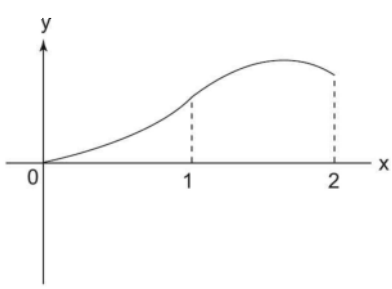
13. (d) The function $\frac{|x|}{x}$ is undefined at $x = 0$

$$\therefore \ \int_{-1}^{1}\frac{|x|}{x}dx = \int_{-1}^{0}\frac{|x|}{x}dx + \int_0^1 \frac{|x|}{x}dx$$

$$= \int_{-1}^{0}\frac{-x}{x}dx + \int_0^1 \frac{x}{x}dx = \int_{-1}^{0}dx + \int_0^1 dx$$

$$= -(0-(-1)) + (1-0) = 0$$

14. (d) $\because$ $f(x)$ is continuous in [0, 2]



$$\therefore \ \int_0^2 f(x)dx = \int_0^1 f(x)dx + \int_1^2 f(x)dx$$

$$= \int_0^1 x^2 dx + \int_1^2 \sqrt{x}dx = \frac{1}{3} + \frac{2}{3}\left(2^{3/2}-1\right)$$

$$= \frac{1}{3} + \frac{4\sqrt{2}}{3} - \frac{2}{3} = \left(\frac{4\sqrt{2}-1}{3}\right)$$

15. (b) $\because$ $\sin^6 x + \cos^6 x = (\sin^2 x)^3 + (\cos^2 x)^3$
$= (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x) = 1 - 3\sin^2 x \cos^2 x = 1 - \frac{3}{4}\sin^2 2x$
$\left(\therefore \text{period} \frac{\pi}{2}\right)$

$\therefore$ Least and greatest value of $\sin^6 x + \cos^6 x$ are $\frac{1}{4}$ and 1.

Hence, $\left(\frac{\pi}{2}-0\right)\times\frac{1}{4}$

$$< \int_0^{\pi/2}(\sin^6 x + \cos^6 x)dx < \left(\frac{\pi}{2}-0\right)\times 1$$

$$\Rightarrow \frac{\pi}{8} < \int_0^{\pi/2}(\sin^6 x + \cos^6 x)dx < \frac{\pi}{2}$$

16. (a) $\because$ $f(x) = x - x^2 + 1 \ \therefore f'(x) = 1 - 2x$

$f'(x) > 0 \Rightarrow 1 - 2x > 0 \Rightarrow x < \frac{1}{2}$ and $f'(x) < 0$

$\Rightarrow 1 - 2x < 0 \Rightarrow x > \frac{1}{2}$

$\Rightarrow f(x)$ is increasing in $\left(0, \frac{1}{2}\right)$ and decreasing in $\left(\frac{1}{2}, 1\right)$

Now, $g(x) = \max\{f(t), 0 \le t \le x\}$

$$= \begin{cases} x - x^2 + 1, & 0 \le x \le \frac{1}{2} \\ \frac{1}{2} - \frac{1}{4} + 1, & \frac{1}{2} \le x \le 1 \end{cases}$$

$$= \begin{cases} x - x^2 + 1, & 0 \le x \le \frac{1}{2} \\ \frac{5}{4}, & \frac{1}{2} \le x \le 1 \end{cases}$$

$$\therefore \ \int_0^1 g(x)dx = \int_0^{1/2}(x - x^2 + 1)dx + \int_{1/2}^1 \frac{5}{4}dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} + x\right]_0^{1/2} + \frac{5}{4}[x]_{1/2}^1$$

$$= \left(\frac{1}{8} - \frac{1}{24} + \frac{1}{2}\right) + \frac{5}{4}\left(1 - \frac{1}{2}\right)$$

$$= \frac{1}{8} - \frac{1}{24} + \frac{1}{2} + \frac{5}{8}$$

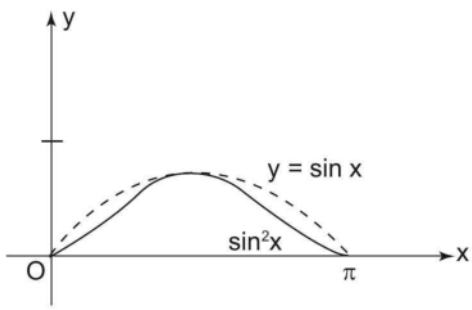
$$= \frac{3 - 1 + 12 + 15}{24} = \frac{29}{24}$$

17. (b) $\because\ y = e^{x^3} \ \therefore\ \dfrac{dy}{dx} = e^{x^3}.3x^2 > 0$

$\Rightarrow$ y is an increasing function. And area enclosed by the curve $y = e^{x^3}$ between the lines $x = a, x = b$ and x-axis is $\int_a^b e^{x^3}dx$

18. (d) For $0 < t < 1\ t^2 < t$

$\therefore\ \sin^2 x < \sin x \Rightarrow \int_0^{\pi}\sin^2 x dx < \int_0^{\pi}\sin x dx$



19. (c) The given curves are $y = x^2 + 2x - 3$ ..... (i)

and $y = \lambda x + 1$ ..... (ii)

Solving equations (i) and (ii), we get $x^2 + (2 - \lambda)x - 4 = 0$

α, β are the roots of the quadratic, then

$\alpha + \beta = \lambda - 2, \alpha\beta = -4$

Hence, required area

$$S(\lambda) = \left|\int_{\alpha}^{\mu}(\lambda x + 1) - (x^2 + 2x - 3))dx\right|$$

$$= \left|\left\{4x + (\lambda - 2)\frac{x^2}{2} - \frac{x^3}{3}\right\}_{\alpha}^{\mu}\right|$$

$$= \left|4(\beta - \alpha) + \frac{(\lambda - 2)}{2}(\beta^2 - \alpha^2) - \frac{1}{3}(\beta^3 - \alpha^3)\right|$$

$$= \sqrt{(\beta - \alpha)^2 - 4\beta\alpha}\left|\left\{4 + \frac{(\lambda - 2)}{2}(\beta + \alpha)\right.\right.$$

$$\left.\left. - \frac{1}{3}\{(\alpha + \beta)^2 - \alpha\beta\}\right\}\right|$$

$$= \frac{1}{6}\{(\lambda - 2)^2 + 16\}^{3/2}$$

For least value of $S(\lambda)$, $\lambda - 2 = 0 \ \therefore\ \lambda = 2$

20. (b) Given, $f\left(\dfrac{x}{y}\right) = f(x) - f(y)$ ..... (i)

Putting $x = y$, then $\Rightarrow f(1) = 0$

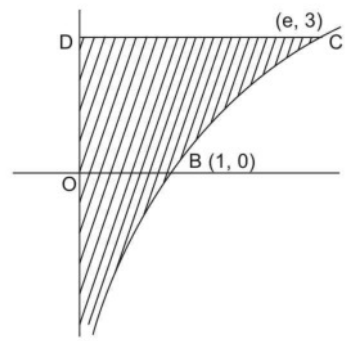
$$\therefore\ f'(x) = \lim_{h \to 0}\frac{f(x + h) - f(x)}{h}$$

$$= \lim_{h \to 0}\frac{f\left(1 + \dfrac{h}{x}\right)}{h} \text{ [from equation (i)]}$$

$$= \lim_{h \to 0}\frac{f\left(1 + \dfrac{h}{x}\right)}{x.\dfrac{h}{x}} = \frac{3}{x}\left\{\because \lim_{h \to 0}\frac{f(1 + x)}{x} = 3\right\}$$

$\therefore\ f(x) = 3 \ln x + c$

Putting $x = 1$, then $f(1) = 0 + c = 0$



$\Rightarrow f(x) = 3 \ln x = y$ (say)

$\therefore\ x = e^{y/3}$

$\therefore$ Required area $= \int_{-\infty}^{3} x dy = \int_{-\infty}^{3} e^{y/3}dy$

$= 3\left\{e^{y/3}\right\}_{-\infty}^{3} = 3(e - 0) = 3e$ sq. unit

$\because\ f''(x) = -\dfrac{3}{x^2} < 0$

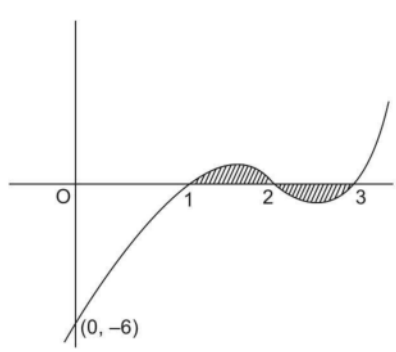
$\Rightarrow f(x)$ is concave down.

21. (d) It is clear from the figure for $x \in [2.\ 2, 2.8]$

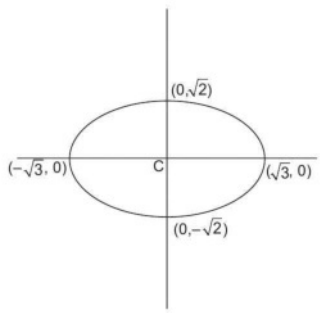
$\Rightarrow (x - 1)(x - 2)(x - 3) \le 0$

Required area

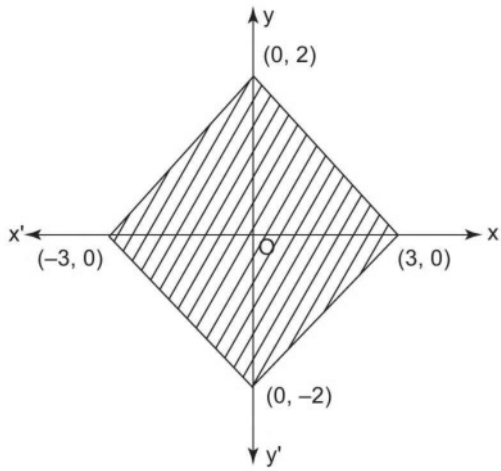
$$= \left|\int_{2.2}^{2.8} f(x)dx\right| = \left|\int_{2.2}^{2.8}(x - 1)(x - 2)(x - 3)dx\right|$$



22. (d) Area of ellipse $\frac{x^2}{3}+\frac{y^2}{2}=1$



is $\pi\sqrt{3}\sqrt{2}=3.14\times\sqrt{6}=7.8$ (approx.)sq. unit, the area bounded by $2\,|x|+3\,|y|\le 6$ is $4\times\frac{1}{2}\times 3\times 2=12$ sq unit.



and length of major axis $=2\sqrt{6}<3+3$.

MENTAL PREPARATION TEST

1. Prove that $\int_0^{2\pi}|\sin x|\,dx=4$ ***[MP-2001]***

2. Evaluate $\int_0^1 x\sqrt{\frac{1-x^2}{1+x^2}}dx$ ***[PSB-93]***

3. Evaluate $\int_{-\pi/4}^{\pi/4}x^3\sin^4 x\,dx$ ***[CBSE-95(C)]***

4. Evaluate $\int_{-3}^{3}|x|\,dx$ ***[MP-2001]***

5. Prove that $\int_0^{\pi/2}\frac{\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx=\frac{\pi}{4}$
[MP-98, 99, 2001, 2008; CBSE-92, 93(C), 96; PSB-92]

6. Prove that $\int_0^{\pi/4}\log_e(1+\tan x)dx=\frac{\pi}{8}\log_e 2$
[MP-97, 98, 99, 2000, 2002, 2003, 2008; CBSE-2002(C), 2003, 2004; PSB-93, 97, 2001; HSB-93, 97; DSB-91; HPSB-2001 (C)]

7. Prove that $\int_0^{\pi/2}\frac{dx}{1+\cot x}=\frac{\pi}{4}$
[MP-96, 99, 2008; CBSE-92; Roorkee-90]

8. Evaluate $\int_0^1\sin^{-1}\left(\frac{2x}{1+x^2}\right)dx$
[CBSE-92(C), 2002; HPB-2002]

9. Evaluate $\int_0^{\pi/2}\frac{\sin\theta}{\sqrt{1+\cos\theta}}d\theta$
[HSB-88; AISSE-85]

10. Evaluate $\int_0^{\pi}\frac{x\tan x}{\sec x\,\operatorname{cosec} x}dx$ ***[CBSE-2001]***

11. Find the area enclosed between the parabola $y^2=4ax$ and the line $y=mx$.
[MP-96, 99, 2000, 2001, 2002]

12. Evaluate the definite integral as limit of sums $\int_a^b\sin^2 x\,dx$

13. Evaluate $\int_1^3\frac{\cos(\log x)}{x}dx$ ***[MP-95]***

14. Evaluate $\int_{-\infty}^{\infty}\frac{dx}{x^2+4x+5}$ ***[MP-99]***

15. Evaluate $\int_0^{\pi}\frac{x\tan x}{\sec x+\tan x}dx$
[CBSE-95(C), 2008; MLNR-84]

16. Prove that $\int_0^{2a} \frac{f(x)}{f(x)+f(2a-x)}dx = a$

[IIT-88]

17. Evaluate $\int_0^{\pi/2} \left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$

[CBSE-2002, 2003, 2005 (outside Delhi)-III; DSB-2002]

18. Evaluate $\int_0^{\pi} \frac{dx}{6-\cos x}$

[MP-2005 (B); CBSE-97]

19. Evaluate $\int_0^{\pi/2} \sin^3 x dx$ ***[HPB-2002]***

20. Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum.

[MP-95, 99, 2001; CBSE-94; PSB-99; HPSB-99(C)]

21. For $0 \le x \le \pi$, find the area bounded by $y = x$ and $y = x + \sin x$. ***[Roorkee qualifying-98]***

22. Find the area of the smaller region bounded by the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and the straight line $\frac{x}{4}+\frac{y}{3}=1$ ***[CBSE-Sample Paper-I]***

23. Prove that $\int_0^1 (\cos^{-1} x)^2 dx = \pi - 2$

[MP-2002]

24. Prove that $\int_0^{\pi/2} \log_e(\tan x)dx = 0$

[MP-98, 2000; CBSE-90]

25. Prove that $\int_0^{\pi} \frac{x}{a^2\cos^2 x + b^2 \sin^2 x}dx = \frac{\pi^2}{2ab}$

[MP-99, 2001, 2005(C)]

26. Evaluate the integral $\int_0^{1/2} \frac{\sin^{-1} x}{(1-x^2)^{3/2}}$

[AISSE-87]

27. Find the area of circle $x^2 + y^2 = a^2$

[MP-96, 98, 2002, 2004(A), 2008; CBSE-92(C); PSB-99]

28. Find the area included between two parabolas $y^2 = 4ax$ and $x^2 = 4ay$, $a > 0$.

[MP-93, 94, 95, 97, 2000, 2001, 2004-(B), (C); CBSE-2003, 2004; AISSE-90]

29. Using integration, find the area of the region $\{(x, y) : |x| \le y \le \sqrt{4-x^2}\}$

[NMOC-98]

30. Evaluate the integral as limit of sums $\int_1^3 (2x+1)dx$ ***[CBSE-93 (C), 95 (C)]***

31. Evaluate: $\int_0^{\pi/2} \log \sin x \, dx$ ***[CBSE-2008]***

32. Evaluate: $\int_0^{\pi/2} \frac{x}{\sin x + \cos x}dx$

[CBSE-2008]

33. Using integration, find the area of the triangular region whose vertices are (1, 0), (2, 2) and (3, 1). ***[CBSE-2008]***

34. Evaluate: $\int_0^2 x\sqrt{2-x}dx$ ***[MP-2000, 2008]***

35. Find the area of the region enclosed by curve $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. ***[MP-2008]***

TOPIC WISE WARMUP TEST: 1

1. If $f(x) = \frac{x^9 - 3x^5 + 7x^3 - x + 1}{\cos^2 x}$, then $\int_{-\pi/4}^{\pi/4} f(x)dx$ is equal to

(a) 0 (b) $\pi/2$
(c) 2 (d) 1

2. $\int_{50}^{100} \frac{\ln x}{\ln x + \ln(150-x)}dx =$

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) 25 (d) 50

3. Area enclosed by the curve $|x-2|+|y+1|=1$ is equal to
   (a) 4 sq. units (b) 6 sq. units
   (c) 2 sq. units (d) 8 sq. units

4. Area bounded by the curves $y=e^x$, $y=2x-x^2$ and the line $x=0$, $x=1$; is equal to
   (a) $\frac{3e-2}{3}$ sq. units (b) $\frac{4e-5}{4}$ sq. units
   (c) $\frac{4e-7}{4}$ sq. units (d) $\frac{3e-5}{3}$ sq. units

5. If $g(x)=\frac{f(x)-f(-x)}{2}$ defined over $[-3, 3]$, and $f(x)-2x^2-4x+1$, then $\int_{-3}^{3} g(x)dx$ is equal to ***[Kerala (CEE)-2003]***
   (a) 0 (b) 4
   (c) –4 (d) 8

6. If $f(t)$ is an odd function, then $\int_0^x f(t)dt$ is
   (a) an odd function
   (b) an even function
   (c) neither even nor odd
   (d) 0

7. The value of integral $\int_1^9 e^{\sqrt{x}}dt =$ ***[VIT-2006]***
   (a) e^3 (b) $4e^3$
   (c) $4(e^3-e)$ (d) $4e^3-2e$

8. What is the area under the curve $y=|x|+|x-1|$ between $x=0$ and $x=1$?
   (a) 1/2 (b) 1
   (c) 3/2 (d) 2

9. If f and g are continuous function on $[0, a]$ satisfying $f(x)=f(a-x)$ and $g(x)-g(a-x)-2$, then $\int_0^a f(x)g(x)dx =$
   (a) $\int_0^a f(x)dx$ (b) $\int_a^0 f(x)dx$
   (c) $2\int_0^a f(x)dx$ (d) None of these

10. The value of $\int_0^{\pi} e^{\cos^2 x}\cos^5 3x dx$ ***[Bihar (CEE)-1994]***
    (a) 1 (b) –1
    (c) 0 (d) None of these

11. $\int_{-\pi}^{\pi}(\cos px-\sin qx)^2\ dx$ is equal to (where p and q are integers) ***[IIT (Screening)-92]***
    (a) $-\pi$ (b) 0
    (c) π (d) 2π

12. $\int_0^{\infty}\frac{dx}{(x+\sqrt{x^2+1})^3}=$
    (a) $\frac{3}{8}$ (b) $\frac{1}{8}$
    (c) $-\frac{3}{8}$ (d) None of these

13. If $f(0)=f'(0)=0$ and $f''(x)=\tan^2 x$, then $f(x)=$
    (a) $\log\sec x-\frac{1}{2}x^2$
    (b) $\log\cos x+\frac{1}{2}x^2$
    (c) $\log\sec x+\frac{1}{2}x^2$
    (d) None of these

14. $\int_0^1 \sin\left(2\tan^{-1}\sqrt{\frac{1+x}{1-x}}\right)dx$ is equal to ***[EAMCET-2003]***
    (a) $\pi/6$ (b) $\pi/4$
    (c) $\pi/2$ (d) π

15. $\int_{-2}^{2}|[x]|dx$ is equal to ***[EAMCET-2003]***
    (a) 1 (b) 2
    (c) 3 (d) 4

16. If $y=f(x)$ makes positive intercept of 2 and 1 unit with x and y-axis and enclosed an area of 3/4 square unit with the axes, then $\int_0^2 xf'(x)dx$ is
    (a) 3/2 (b) 1
    (c) 5/4 (d) –3/4

17. The area of the region (in square units) bounded by the curve $x^2=4y$, line $x=2$ and x-axis is
    (a) 1 (b) 2/3
    (c) 4/3 (d) 8/3

18. The area of one loop of the curve $y^2=x^2(9-x^2)$ is

(a) 18 sq. units (b) $2\sqrt{3}$ sq. units
(c) 9 sq. units (d) $4\sqrt{3}$ sq. units

19. In $(-4, 4)$ the function $f(x) = \int_{-10}^{x} (t^4 - 4)e^{-4t}dt$, it has
(a) no extrema (b) one extremum
(c) two extrema (d) four extrema

20. $\int_0^{2\pi} (\sin x + |\sin x|)dx =$ ***[CET (Karnataka)-2003]***
(a) 4 (b) 0
(c) 1 (d) 8

21. The value of $I = \int_0^{\pi/2} \frac{(\sin x + \cos x)^2}{\sqrt{1 + \sin 2x}}dx$ is ***[AIEEE-2004]***
(a) 2 (b) 1
(c) 0 (d) 3

22. The area bounded by the angle bisectors of lines $x^2 - y^2 + 2y = 1$ and the line $x + y = 3$ is
(a) 2 (b) 3
(c) 4 (d) 6

23. $\int_0^1 \frac{xdx}{[x + \sqrt{1-x^2}]\sqrt{1-x^2}} =$
(a) 0 (b) 1
(c) $\frac{\pi}{4}$ (d) $\frac{\pi^2}{2}$

24. What is value of $\int_0^2 \min(x - [x], -x - [-x])dx$?
(a) 0 (b) 1/2
(c) 1 (d) 2

25. $\int_0^{\pi/2} \frac{200\sin x + 100\cos x}{\sin x + \cos x}dx =$
(a) 50π (b) 25π
(c) 75π (d) 150π

26. Consider the following
1. $\int_0^a f(x)dx = \int_0^a f(x - a)dx$
2. $\int_{-a}^a f(x)dx = \int_0^a [f(x) + f(-x)]dx$
3. $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a - x)dx$
4. $\int_a^b f(x)dx = \int_a^b f(x - a + b)dx$

which of the above is/are correct?
(a) 2 only (b) 3 only
(c) 2 and 3 only (d) 1 and 4 only

27. $\frac{dy}{dx} = \frac{y + x\tan\frac{y}{x}}{x} \Rightarrow \sin\frac{y}{x} =$
(a) cx^2 (b) cx
(c) cx^3 (d) cx^4

28. $\int_0^{\pi/2} \frac{\sin x\cos x}{\cos^2 x + 3\cos x + 2}dx$ equals ***[PET (Raj.)-90]***
(a) log (9/8) (b) log (8/9)
(c) log (9/2) (d) log (3/4)

29. If $f(x) = \begin{cases} e^{\cos x}\sin x, & |x| \le 2 \\ 2, & \text{otherwise} \end{cases}$, then $\int_{-2}^{3} f(x)dx$ is equal to ***[IIT(Screening)-2002]***
(a) 0 (b) 1
(c) 2 (d) 3

30. $\int_0^1 \frac{\log x}{\sqrt{1-x^2}}dx$ equals ***[BIT Ranchi-1984]***
(a) $\pi\log(1/2)$
(b) $-\pi\log(1/2)$
(c) $(\pi/2)\log(1/2)$
(d) $(-\pi/2)\log(1/2)$

31. $\int_0^9 [\sqrt{x} + 2]dx$ (where [ ] is greatest integer function) is equal to ***[UPSEAT-2002]***
(a) 22 (b) 23
(c) 31 (d) 32

32. $\int_0^{\pi/2} \frac{\sin 8x\log(\cot x)}{\cos 2x}dx$ is equal to ***[Roorkee-1995]***
(a) 0 (b) $\pi/4$
(c) $\pi/2$ (d) None of these

33. $\int_{-\pi}^{\pi} (x^3 + x\cos x + \tan^5 x + 2)dx$ is equal to ***[Roorkee-88]***
(a) π (b) 2π
(c) 4π (d) 0

34. If $a < b$, then $\int_a^b (|x-a| + |x-b|)dx$ is equal to *[ICS (Pre.)-2004]*
(a) $(b-a)^2/2$ (b) $(b^2-a^2)/2$
(c) $(a^3-b^3)/2$ (d) $(b-a)^2$

35. If $f(x) = \int_x^{x^2} (t-1)dt$, $1 \le x \le 2$, then global maximum value of $f(x)$ is
(a) 1 (b) 2
(c) 3 (d) 4

36. The area bounded by the curves $y^2 = 8x$ and $x^2 = 8y$ is ... *[Orissa JEE-2008]*
(a) 64/3 sq. units (b) 16/3 sq. units
(c) 32/3 sq. units (d) None of these

37. If $\int_{[x]}^{[x]+1} f(t)dt = [x]$, then the value of $\int_{-2}^{4} f(x)dx$ is equal to
(a) 1 (b) 2
(c) –2 (d) 3

TOPIC WISE WARMUP TEST: 2

1. $\int_0^{n\pi+v} |\sin x|\,dx =$ where n is a positive integer and $0 \le v < \pi$ *[IIT-94; AMU-2003]*
(a) $2n + 1 - \cos v$
(b) $2n - 1 - \sin v$
(c) $\cos v - 2n - 1$
(d) $2n + 1$

2. The area bounded by the curve $y = \lambda x - x^2$ and the x-axis is 32/3 sq. units $\lambda =$ *[MP(PET)-1999]*
(a) 1 (b) 4
(c) 2 (d) 3

3. The area bounded by the curve $y = x^3$ x-axis and two ordinates $x = 1$ to $x = 3$ is
(a) 20 (b) 40
(c) 60 (d) 15

4. The area enclosed by the parabola $y^2 = 4ax$ and the straight line $y = \lambda x$ is $\frac{1}{3a}$, then $\frac{\lambda}{2} =$ *[Karnataka CET-1999]*
(a) a (b) $\frac{1}{2a}$
(c) $2a$ (d) $\frac{1}{a}$

5. What is the area bounded by the curves $x^2 + y^2 = 9$, $y^2 = 8x$
(a) 0
(b) $\frac{2\sqrt{2}}{3} + \frac{9\pi}{2} - 9\sin^{-1}\left(\frac{1}{3}\right)$
(c) 16π
(d) None of these

6. Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then the value of the integral $\int_0^1 f(x)g(x)dx$ is *[AIEEE-2003]*
(a) $e + \frac{e^2}{2} + \frac{5}{2}$ (b) $e - \frac{e^2}{2} - \frac{5}{2}$
(c) $e + \frac{e^2}{2} - \frac{3}{2}$ (d) $e - \frac{e^2}{2} - \frac{3}{2}$

7. The integral $\int_{-1/2}^{1/2} \left\{[x] + \ln\left(\frac{1+x}{1-x}\right)\right\}dx$ equal (where [.] is the greatest integer function) *[IIT (Screening)-2002]*
(a) –1/2 (b) 0
(c) 1 (d) 2 ln (1/2)

8. Let $T > 0$ be a fixed number. Suppose f is a continuous function such that for all $x \in R$, $f(x + T) = f(x)$. If $I = \int_0^T f(x)dx$, then the value of $\int_3^{3+3T} f(2x)dx$, is *[IIT (Screening)-2002]*
(a) $\frac{3}{2}I$ (b) $2I$
(c) $3I$ (d) $6I$

9. Let $f(x) = \int_1^x \sqrt{2-t^2}\,dt$, then the real roots of the equation $x^2 - f'(x) = 0$ are *[IIT (Screening)-2002]*
(a) ± 1 (b) $\pm 1/\sqrt{2}$
(c) ± 1/2 (d) 0 and 1

10. If $f(x)$ satisfies the conditions of Rolle's theorem in [1, 2] and $f(x)$ is continuous in [1, 2], then $\int_1^2 f'(x)dx$ is equal to

[DCE-2004]

(a) 3 (b) 0
(c) 1 (d) 2

11. $\int_0^{\pi/4} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x} =$

[Karnataka CET-1997]

(a) $\frac{1}{ab}\tan^{-1}\frac{b}{a}; a>0, b>0$

(b) $\frac{1}{ab}\tan^{-1}\frac{b}{a}; a<0, b<0$

(c) $\pi/4; a=1, b=1$

(d) $\frac{1}{ab}\tan^{-1}\frac{b}{a}$

12. The area bounded by the curve $y = x^3$, the y- axis, the lines $y = 1$ and $y = 8$ is given by

(a) 12 (b) 6
(c) 20 (d) 45/4

13. $\lim_{x\to\infty}\left[\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+.....+\frac{1}{n}\right]$ is equal to

(a) $\frac{\pi}{4}+\frac{1}{2}\log 2$ (b) $\frac{\pi}{4}-\frac{1}{2}\log 2$

(c) $\frac{\pi}{4}-2\log\frac{1}{2}$ (d) None of these

14. The points of extrema of $f(x)=\int_0^x \frac{\sin t}{t}dt$ in the domain $x > 0$ are

(a) $(2n+1)\frac{\pi}{2}, n=1,2...$

(b) $(4n+1)\frac{\pi}{2}, n=1,2...$

(c) $(2n+1)\frac{\pi}{4}, n=1,2...$

(d) $n\pi, n = 1, 2 ...$

15. If $f : R \to R$ be a differentiable function and $f(1) = 4$, then the value of $\lim_{x\to 1}\int_4^{f(x)} \frac{2t}{x-1}dt =$

[IIT-1990]

(a) $8f'(1)$ (b) $4f'(1)$
(c) $2f'(1)$ (d) $f'(1)$

16. The function $f(x)=\int_{-1}^{x} t\,(e^t-1)(t-1)(t-2)^3(t-3)^5\,dt$ has a local minimum at $x =$

[IIT-1999]

(a) 0 (b) 1
(c) 2 (d) 4

17. If $f(x)=\frac{\int_{\pi/2}^{x} tdt}{\sin(2x-\pi)}; x\neq\frac{\pi}{2}$ is continuous at $x=\frac{\pi}{2}$, then $f(\pi/2) =$

(a) 0 (b) $\pi/8$
(c) $\pi/6$ (d) $\pi/4$

18. $\lim_{n\to\infty}\frac{1}{n}\left\{\frac{\sqrt[n]{e}+\sqrt[n]{e^2}+\sqrt[n]{e^3}+.....+\sqrt[n]{e^n}}{n}\right\}$ is equal to

(a) $e + 1$ (b) $e - 1$
(c) e (d) None of these

19. Let $f(x)=\int_0^x t\sin\frac{1}{t}dt$. Then the number of points of discontinuity of the function $f(x)$ in the open interval $(0, \pi)$ is

(a) 0 (b) 1
(c) 2 (d) Infinite

20. Area bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is

(a) 8/9 sq. unit (b) 9/8 sq. unit
(c) 4/3 sq. unit (d) None of these

21. The area bounded by the curve $y = (x + 1)^2$, $y = (x - 1)^2$ and the line $y = 1/4$ is

(a) 1/6 (b) 2/3
(c) 1/4 (d) 1/3

22. Let y be the function which passes through $(1, 2)$ having slope $(2x + 1)$. The area bounded between the curve and x-axis is

(a) 6 sq. unit (b) 5/6 sq. unit
(c) 1/6 sq. unit (d) None of these

23. For which of the following values of m, is the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals 9/2

[IIT-1999]

(a) -4 (b) -2
(c) 2 (d) 9

24. The area enclosed between the curves $y = \log_e(x + e)$, $x = \log_e(1/y)$ and the x-aixs is ***[Roorkee-1990]***

(a) 2 (b) 1
(c) 4 (d) None of these

25. The area bounded by $y = 1 + \frac{8}{x^2}$ and the ordinates $x = 2$ and $x = 4$ is ***[IIT-1983]***

(a) 2 (b) 4
(c) log 2 (d) log 4

26. The area (in square units) bounded by the curve $y = x^3$, the x-axis and the ordinates at $x = -2$ and $x = 1$ is ***[Roorkee (Screening)-92; MNR-98]***

(a) 9/2 (b) 15/2
(c) 15/4 (d) 17/4

27. The area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = 1$ and $x = b$ is $(b - 1)\sin(3b + 4)$. Then $f(x)$ is ***[IIT-86; PET (Raj.)-2000]***

(a) $3(x - 1)\cos(3x + 4)$
(b) $\sin(3x + 4)$
(c) $\sin(3x + 4) + 3(x - 1)\cos(3x + 4)$
(d) None of these

28. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is ***[IIT(Screening)-2002]***

(a) 1 (b) 2
(c) $2\sqrt{2}$ (d) 4

29. The area bounded by the parabola $y^2 = 4ax$ and its latus rectum is given by ***[PET (Raj.)-84, 92, 94, 97, 2000, 2002; Kerala(CEE)-2003]***

(a) $\int_0^a y\,dx$

(b) $2\int_0^a 4ax\,dx$

(c) $2\int_0^a \sqrt{4ax}\,dx$

(d) $2\int_{-a}^a \sqrt{4ax}\,dx$

30. The ratio of the areas bounded by the curves $y = \cos 2x$ and $y = \cos x$ between $x = 0$, $x = k$ is 1 : 2, then $k =$ ***[MPPET-1997]***

(a) $\pi/6$ (b) $\pi/3$
(c) $\pi/2$ (d) $2\pi/3$

31. Area bounded by lines $y = 2 + x$, $y = 2 - x$ and $x = 2$ is ***[MPPET-1996]***

(a) 3 (b) 4
(c) 8 (d) 16

32. Area bounded by the parabola $y^2 = 4ax$ and straight line $y = mx$ is pa^2/qm^3, then

(a) $p = 8, q = 3$ (b) $p = 3, q = 8$
(c) $p = 5, q = 8$ (d) $p = 8, q = 5$

33. The area bounded by the curve $y = \log x$, the x-axis and the ordinate $x = e^2$ is ***[MPPET-1994; Pb. CET-2000]***

(a) e (b) $e^2 + 1$
(c) $e + 1$ (d) ∞

34. Area of the region bounded by the curve $y^2(2a - x) = x^3$ and straight line $x = 2a$ is ***[PET (Raj.)-91; MP-2001]***

(a) πa^2 (b) $3\pi a^2/2$
(c) $2\pi a^2$ (d) $3\pi a^2$

TOPIC WISE WARMUP TEST: 3

1. If $f(x) = A\sin\left(\frac{\pi x}{2}\right) + B$, $f'(1/2) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2A}{\pi}$, then values of A and B are

(a) $\pi/2, \pi/2$
(b) $2/\pi, 3/\pi$
(c) $4/\pi, 0$
(d) $0, -4/\pi$

2. $\int_1^{e^{37}} \frac{\pi\sin(\pi\ln x)}{x}dx$ is equal to

(a) 1 (b) 2
(c) e (d) 37

3. If $f(x) = A\,2^x + B$, $f'(1) = 2$ and $\int_0^3 f(x)dx = 7$, then values of A, B are

(a) $\frac{1}{\log 2}, \frac{7}{3}\left(1+\frac{1}{(\log 2)^2}\right)$

(b) $\frac{1}{\log 2}, \frac{7}{3}\left(1-\frac{1}{(\log 2)^2}\right)$

(c) $\frac{1}{\log 2}, \frac{1}{\log 3}$

(d) None of these

4. Let $f: (0, \infty) \to R$ and $F(x) = \int_0^x f(t)dt$. If $F(x^2) = x^2(1 + x)$, then $f(4)$ is equal to

(a) 5/4 (b) 7

(c) 4 (d) 2

5. The value of integral $\int_{\frac{1}{2n}}^{\left(\frac{an-1}{n}\right)} \frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}dx$ is ***[AMU-02]***

(a) $\frac{a}{2}$

(b) $\frac{na+2}{2n}$

(c) $\frac{na-2}{2n}$

(d) None of these

6. If for a real number y, $[y]$ is the greatest integer less than or equal to y, then the value of the integral $\int_{\pi/2}^{3\pi/2} [2\sin x]dx$ is ***[IIT-1999]***

(a) $-\pi$ (b) 0

(c) $-\pi/2$ (d) $\pi/2$

7. The area in the first quadrant enclosed by the x-axis, the line $y = \frac{1}{\sqrt{3}}x$ and circle $x^2 + y^2 = 4$, is given by ***[PET (Raj.)-90, 97; Kurukshetra CEE-98]***

(a) $\int_0^{\sqrt{3}} (x + \sqrt{4-x^2})dx$

(b) $\int_0^{2} (x + \sqrt{4-x^2})dx$

(c) $\int_0^{2} \left(\frac{x}{\sqrt{3}} + \sqrt{4-x^2}\right)dx$

(d) $\int_0^{\sqrt{3}} \frac{x}{\sqrt{3}}dx + \int_{\sqrt{3}}^{2} \sqrt{4-x^2}dx$

8. $\int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x}+e^{-\cos x}}dx =$ ***[IIT-99]***

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$

(c) $-\frac{\pi}{2}$ (d) 0

9. $\int_{1/e}^{e^2} \left|\frac{\log x}{x}\right|dx =$ ***[IIT(Screening)-2000]***

(a) $\frac{3}{2}$ (b) $\frac{5}{2}$

(c) 3 (d) 5

10. $\int_{\pi/4}^{3\pi/4} \frac{dx}{1+\cos x} =$ ***[IIT-99]***

(a) 2 (b) −2

(c) 1/2 (d) −1/2

11. $\int_0^{2\pi} |\sin^3 0| d0 =$ ***[Roorkee Qualifying-1998]***

(a) 0 (b) 3/8

(c) 8/3 (d) π

12. For $0 \le x \le \pi$, the area bounded by $y = x$ and $y = x + \sin x$ is ***[Roorkee Qualifying-1998]***

(a) 2 (b) 4

(c) 2π (d) 4π

13. Let a, b, c be non-zero real numbers such that $\int_0^3 (3ax^2 + 2bx + c)dx = \int_1^3 (3ax^2 + 2bx + c)dx$, then ***[BIT Ranchi-91]***

(a) $a + b + c = 3$ (b) $a + b + c = 1$

(c) $a + b + c = 0$ (d) $a + b + c = 2$

14. The value of $\int_0^{\pi/2} \frac{e^{x^3}}{e^{x^3}+e^{\left(\frac{\pi}{2}-x\right)^3}}dx$ is ***[AMU-99]***

(a) π/4 (b) π/2

(c) $e^{\pi^2/16}$ (d) $e^{\pi^3/4}$

15. $\int_0^{2n\pi} \left(|\sin x| - \left|\frac{1}{2}\sin x\right|\right)dx$ equals ***[Orissa (JEE)-2005]***

(a) n (b) $2n$

(c) $-2n$ (d) None of these

16. $\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}$ equals

[JEE(WB)-89; Kurukshetra (CEE)-98]

(a) e (b) $1/e$
(c) $\pi/4$ (d) $4/\pi$

17. If $f(t)=\int_{-t}^{t}\frac{dx}{1+x^2}$, then $f'(1)$ is

(a) Zero (b) 2/3
(c) -1 (d) 1

18. $\int_{-\pi/2}^{\pi/2}\sin^4 x\cos^6 x dx =$

(a) $\frac{3\pi}{64}$ (b) $\frac{3\pi}{572}$
(c) $\frac{3\pi}{256}$ (d) $\frac{3\pi}{128}$

19. If $2f(x)-3f\left(\frac{1}{x}\right)=x$, then $\int_1^2 f(x)dx$ is equal to

(a) $\frac{3}{5}\ln 2$ (b) $\frac{-3}{5}(1+\ln 2)$
(c) $\frac{-3}{5}\ln 2$ (d) None of these

20. Let $g(x)=\int_0^x f(t)dt$ where $\frac{1}{2}\le f(t)\le 1$, $t\in[0,1]$ and $0\le f(t)\le\frac{1}{2}$ for $t\in[1,2]$. Then

[IIT(Screening)-2000]

(a) $\frac{-3}{2}\le g(2)<\frac{1}{2}$ (b) $0\le g(2)<2$
(c) $\frac{3}{2}<g(2)\le\frac{5}{2}$ (d) $2<g(2)<4$

21. $\lim_{n\to\infty}\left(\frac{1}{\sqrt{4n^2-1}}+\frac{1}{\sqrt{4n^2-2^2}}+..+\frac{1}{\sqrt{3n^2}}\right)$ is equal to ***[Roorkee-97]***

(a) 0 (b) 1
(c) $\pi/3$ (d) $\pi/6$

22. $Lt_{n\to\infty}\sum_{k=1}^{n}\frac{k}{n^2+k^2}$ is equal to

[Roorkee (Screening)-1999]

(a) $\frac{1}{2}\log 2$ (b) $\log 2$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

23. If $\int_0^1 e^{x^2}(x-\alpha)dx=0$, then

[MNR-94; UPSEAT-2000; Bihar(CEE)-03]

(a) $1<\alpha<2$ (b) $\alpha<0$
(c) $0<\alpha<1$ (d) $\alpha=0$

24. If $I_n=\int_0^{\pi/2}x^n\sin x dx$, then I_4+12I_2 is equal to ***[DCE-97]***

(a) 4π (b) $3(\pi/2)^3$
(c) $(\pi/2)^2$ (d) $4(\pi/2)^3$

25. The value of $\int_{-2}^{2}\left(p\ln\left(\frac{1+x}{1-x}\right)+q\ln\left(\frac{1-x}{1+x}\right)^{-2}+r\right)dx$ depends on

[Orissa (JEE)-03]

(a) the value of p
(b) the value of q
(c) the value of r
(d) the value of p and q

26. If $I(m,n)=\int_0^1 t^m(1+t)^n dt\, m,n\in R$ then $I(m,n)$ is ***[IIT(SC)-03]***

(a) $\frac{n}{1+m}I[(m+1,n-1)]$

(b) $\frac{m}{n+1}I[(m+1,n-1)]$

(c) $\frac{2^n}{1+m}-\frac{n}{1+m}I[m+1,n-1]$

(d) $\frac{2^n}{1+m}-\frac{n}{1+n}I[m+1,n-1]$

27. $\int_0^{1000}e^{x-[x]}dx$ is ***[AMU-2002]***

(a) $e^{1000}-1$

(b) $\frac{e^{1000}-1}{e-1}$

(c) $1000(e-1)$

(d) $(e-1)/1000$

28. The value of $\int_{-1/2}^{1/2}(\cos x)\left[\log\left(\frac{1-x}{1+x}\right)\right]dx$ is ***[MNR-98; UPSEAT-2002]***

(a) $2e^{1/2}$ (b) 1
(c) $e^{1/2}$ (d) 0

29. $\int_0^{\pi}\sqrt{\frac{1+\cos 2x}{2}}dx =$

[DCE-95; JEE(Orissa), 2000; CET (Pb.) 2000]

(a) -2 (b) 2

(c) 0 (d) -3

30. $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$ equals ***[IIT-1997]***

(a) $1+\sqrt{5}$ (b) $-1+\sqrt{5}$

(c) $-1+\sqrt{2}$ (d) $1+\sqrt{2}$

31. Determine the value of $\int_{-\pi}^{\pi}\frac{2x(1+\sin x)}{1+\cos^2 x}dx$

[IIT-97; AIEEE-2002]

(a) $\pi/2$ (b) π^2

(c) $\pi/4$ (d) $\pi^2/4$

32. $\int_0^1 \log\left(\frac{1}{x}-1\right)dx =$ ***[PET (Raj.)-88]***

(a) $\frac{\pi}{2}+1$ (b) $\frac{\pi}{2}-1$

(c) $2-\frac{\pi}{2}$ (d) 0

33. The value of $\int_2^3\frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$ is ***[IIT-1994]***

(a) 6 (b) 1

(c) 1/2 (d) 2

34. If $\int_{\sin x}^{1} t^2 f(t)dt = (1-\sin x)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to ***[IIT (Screening)-2005]***

(a) 1/3 (b) $1/\sqrt{3}$

(c) 3 (d) $\sqrt{3}$

TOPIC WISE WARMUP TEST 1: SOLUTIONS

1. (c) $f(x)=\frac{x^9-3x^5+7x^3-x}{\cos^2 x}+\sec^2 x$

$=\sec^2 x\,(x^9-3x^5+7x^3-x)+\sec^2 x$

$\Rightarrow \int_{-\pi/4}^{\pi/4} f(x)dx=\int_{-\pi/4}^{\pi/4}\sec^2 x dx$

$=2\int_0^{\pi/4}\sec^2 x dx = 2\tan x\Big|_0^{\pi/4}=2$

2. (c) $\int_{50}^{100}\frac{\ln x}{\ln x+\ln(150-x)}dx$

$=\int_{50}^{100}\frac{\ln(150-x)}{\ln(150-x)+\ln x}dx$

$\Rightarrow 2I=\int_{50}^{100}dx=50 \Rightarrow I=25$

3. (c) Shifting the origin at the point, the equation of the curve becomes $|x|+|y|=1$whose area is $2a^2=2$

4. (d) $\int_0^1(e^x-(2x-x^2))dx=\frac{3e-5}{3}$

5. (a) $g(x)=\frac{1}{2}[(2x^2-4x+1)-(2x^2+4x+1)]$

$=-4x$ which is odd, So $I=0$ by property.

6. (b) By property of integration.

7. (b) Put $t=x^2$, then $dt=2x\,dx$

$\therefore\ I=2\int_1^3 xe^x dx=2[xe^x-e^x]_1^3=4e^3$

8. (b) $y=\begin{cases}1, & 0\le x\le 1\\ 2x-1, & x>1\end{cases}$ So required area is square area of side 1. So its area = 1.

9. (a) $\int_0^a f(x)g(x)dx=\int_0^a f(a-x)g(a-x)dx$

$=\int_0^a f(x)[2-g(x)]dx$

$\Rightarrow 2\int_0^a f(x)g(x)dx=2\int_0^a f(x)dx$

10. (c) Let $I=\int_0^{\pi}e^{\cos^2 x}\cos^5 3x dx$

$=\int_0^{\pi}e^{\cos^2 x}\cos^5 3(\pi-x)dx,$

$[\because \int_0^a f(x)dx=\int_0^a f(a-x)dx]$

$I=-\int_0^{\pi}e^{\cos^2 x}\cos^5 3x dx=-I$

$\Rightarrow 2I=0\Rightarrow I=0.$

11. (d) $I = 2\int_0^{\pi} \cos^2 px\,dx + 2 \int_0^{\pi} \sin^2 qx\,dx + 0$

$= \int_0^{\pi} (2 + \cos 2px - \cos 2qx)dx = 2\pi.$

12. (a) Putting $x = \tan 0$, we get

$\int_0^{\infty} \frac{dx}{(x + \sqrt{x^2+1})^3}$

$\int_0^{\pi/2} \frac{\sec^2 0 d0}{(\tan 0 + \sec 0)^3} = \int_0^{\pi/2} \frac{\cos 0}{(1+\sin\theta)} d0$

$= \left[-\frac{1}{2(1+\sin 0)^2} \right]_0^{\pi/2} = -\frac{1}{8} + \frac{1}{2} = \frac{3}{8}$

13. (a) $f''(x) = \tan^2 x \Rightarrow \int f''(x) = \int \tan^2 x dx$

$\Rightarrow \quad f'(x) = \int (\sec^2 x - 1)dx \ = \tan x - x + c_1$

$\Rightarrow f'(0) = \tan 0 - 0 + c_1$

$\Rightarrow \quad c_1 = 0$ Hence $f'(x) = \tan x - x$

$\Rightarrow \int f'(x)dx = \int \tan x\,dx - \int x\,dx$

$\Rightarrow f(x) = \log \sec x - \frac{x^2}{2} + c_2$

$\Rightarrow f(0) = \log \sec 0 - 0 + c_2$

$\Rightarrow \quad c_2 = 0$. Hence $f(x) = \log \sec\ x - \frac{x^2}{2}$.

14. (b) In the integrand put $x = \cos t$, then integrand = $\sin t = \sqrt{1-x^2}$

$\therefore \quad I = \int_0^1 \sqrt{1-x^2}dx$

$= \left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x \right]_0^1 = \frac{\pi}{4}.$

15. (d) $I = 2\int_0^2 |[x]|dx = 2\int_0^1 0\,dx + 2\int_1^2 1\,dx$

$= 2(2) = 4.$

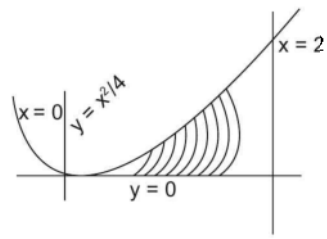
16. (d) $\frac{x}{2} + \frac{y}{1} = 1$

$\Rightarrow \quad y = 1 - \frac{x}{2} = f(x)$:

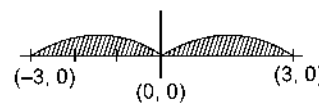
$\int_0^2 xf''(x)dx = (x)f'(x)\,|_0^2 - \int_0^2 f'(x)dx$

$\Rightarrow \quad (2\,f'(2) - 0) - \frac{3}{4}.$

17. (b) $\int_0^2 \frac{x^2}{4}dx = \frac{x^3}{12}\Big|_0^2 = \frac{8}{12} = \frac{2}{3}.$



18. (a) $2\int_0^3 x\sqrt{9-x^2}dx = 18$



(put $9 - x^2 = t^2$, symmetry).

19. (c) $f'(x) = (x^4 - 4)\,e^{-4x} = 0 \Rightarrow x^4 = 4 \Rightarrow x^2 = 2$

$\Rightarrow \quad x = \pm\sqrt{2}.$

20. (a) $\int_0^{2\pi} |\sin x|\,dx = 4\int_0^{\pi/2} \sin x\,dx$

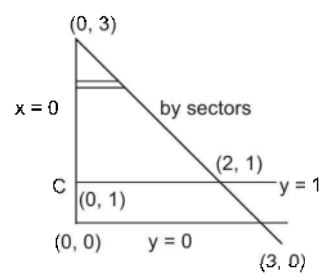
$= -4\cos x\,|_0^{\pi/2} = -4(0-1) = 4.$

21. (a) $\int_0^{\pi/2} (\sin x + \cos x)dx$

$= (-\cos x + \sin x)|_0^{\pi/2} = 2.$

22. (a) Area of region bounded by angle $y = 1$ and $x = 0$ and straight line $x + y = 3$.

$\frac{1}{2} \times 2 \times 2 = 2$ sq. units.

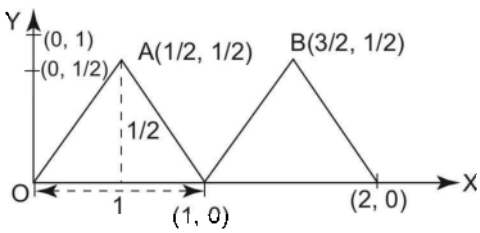


23. (c) Put $x = \sin\theta$ so that

$\int_0^{\pi/2} \frac{\sin\theta\cos\theta d\theta}{\cos 0[\sin 0 + \cos 0]}$

$= \int_0^{\pi/2} \frac{\sin\theta}{\sin 0 + \cos 0} d\theta = \frac{\pi}{4}$

24. (b) Area $= 2\int_0^1 f(x)dx = 2\left\{\frac{1}{2}\times 1\times\frac{1}{2}\right\} = \frac{1}{2}$.



25. (c) $200\times\frac{\pi}{4}+100\times\frac{\pi}{4} = 50\pi + 25\pi = 75\pi$.

26. (c) Property.

27. (b) $\frac{dy}{dx}-\frac{y}{x} = \tan\left(\frac{y}{x}\right)$

$\Rightarrow v + x\frac{dv}{dx} - v = \tan v$

$\Rightarrow \frac{dv}{\tan v} = \frac{dx}{x} \Rightarrow \log\sin v = \log x + \log c$

$\Rightarrow \sin\left(\frac{y}{x}\right) = cx.$

28. (a) $t = \cos x$

$\Rightarrow \int_0^1 \frac{tdt}{(t+1)(t+2)} = \int_0^1\left(\frac{-1}{t+1}+\frac{2}{t+2}\right)$.

29. (c) $|x| \le 2 \Rightarrow -2 \le x \le 2$

$\Rightarrow \int_{-2}^{2} \underset{\text{(even)}}{(e^{\cos x})} \quad \underset{\text{(odd)}}{(\sin x)dx} + \int_2^3 2dx$ = odd

$\Rightarrow 0 + 2x|_2^3 = 2(3-2) = 2.$

30. (c) $\int_0^{\pi/2}\log\sin 0 d0 = -\frac{\pi}{2}\log 2 = \frac{\pi}{2}\log\left(\frac{1}{2}\right)$

$[x = \sin\theta]$.

31. (c) $\int_0^1 2dx + \int_1^4 3dx + \int_4^9 4dx = 2 + 3\times 3 + 4\times 5 = 2 + 9 + 20 = 31.$

32. (a) $\int_0^{\pi/2} f(x)dx = \int_0^{\pi/2} f\left(\frac{\pi}{2}-x\right)dx = 0 \Rightarrow I = -I \Rightarrow 2I = 0 \Rightarrow I = 0.$

33. (c) Using property, $I = 2\int_0^{\pi} 2dx = 4\pi$

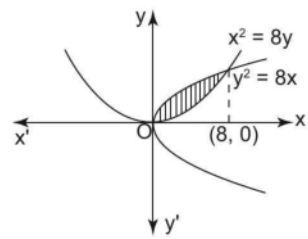
34. (d) $I = \int_a^b (x-a)dx - \int_a^b (x-b)dx$ $[\because a < b]$

35. (d) $f'(x) = (x^2-1)(2x) - (x-1) = (x-1)(2x^2 + 2x - 1)$ which is positive for $x > 1$. Hence, f increases in $[1, 2]$

Hence, global maximum of f is $f(2) = 4$.

36. (a) First curve is $y^2 = 8x$ ..... (i)

Second curve is $x^2 = 8y \Rightarrow y = \frac{x^2}{8}$



Putting the value of y in (i)

$\left(\frac{x^2}{8}\right)^2 = 8x \Rightarrow x^4 = 512\,x$

$\Rightarrow x(x^3 - 512) = 0 \Rightarrow x = 0, x = 8$

$\therefore$ Area under curve $= \int_0^8\left(\sqrt{8}x - \frac{x^2}{8}\right)dx$

$= \sqrt{8}.\frac{2}{3}[x^{3/2}]_0^8 - \frac{1}{24}[x^3]_0^8$

$= \sqrt{8}.\frac{2}{3}.8.\sqrt{8} - \frac{1}{24}.8.8.8 = \frac{128}{3} - \frac{64}{3} = \frac{64}{3}$

37. (d) $\int_{[x]}^{[x]+1} f(t)dt = [x]$

$\Rightarrow \int_{-2}^{-2+1} f(x)dx = -2, \int_{-1}^{0} f(x)dx = -1, \ldots$

$\int_3^4 f(x)dx = 3$

$\therefore \int_{-2}^{4} f(x)dx = -2 - 1 + 1 + 2 + 3 = 3$

TOPIC WISE WARMUP TEST 2: SOLUTIONS

1. (a) $I = \int_0^{n\pi} |\sin x|\,dx + \int_{n\pi}^{n\pi+v} |\sin x|\,dx$

$= n\int_0^{\pi} |\sin x|\,dx + \int_0^{v} \sin(n\pi + 0)d0$ where $x = n\pi + 0$

$= n\int_0^{\pi} \sin x dx + \int_0^{v} \sin 0 d0 = 2n + 1 - \cos v$

2. (b) For x-axis, $y = 0 \therefore \lambda x - x^2 = 0 \Rightarrow x = 0$ or $x = \lambda$

Hence required area

$$= \int_0^{\lambda} y\,dx = \int_0^{\lambda} (\lambda x - x^2)\,dx$$

$$= \left[\frac{\lambda}{2}x^2 - \frac{x^3}{3}\right]_0^{\lambda} = \frac{\lambda^3}{2} - \frac{\lambda^3}{3}$$

$$\Rightarrow \frac{3\lambda^3 - 2\lambda^3}{6} = \frac{32}{3}$$

$\Rightarrow \lambda^3 = 64 \Rightarrow \lambda = 4.$

3. (a) required area $= \int_1^0 x^3 dx + \int_0^3 x^3 dx$

$$= \left(\frac{x^4}{4}\right)_1^0 + \left(\frac{x^4}{4}\right)_0^3$$

$$= \frac{-1}{4} + \frac{81}{4} = \frac{-1+81}{4} = \frac{80}{4} = 20.$$

4. (a) Finding the area between parabola $y^2 = 4ax$ and straight line

$y = \lambda x$, we get $\frac{8}{3}\frac{a^2}{\lambda^3}$ given area is $\frac{1}{3a}$.

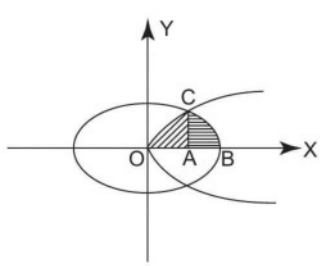
Hence $\frac{8}{3}\frac{a^2}{\lambda^3} = \frac{1}{3a} \Rightarrow \frac{8}{3}\cdot\frac{3a^3}{\lambda^3} = 1$

$\Rightarrow \lambda^3 = 8a^3 \Rightarrow \lambda = 2a \Rightarrow \frac{\lambda}{2} = \frac{2a}{2} \Rightarrow \frac{\lambda}{2} = a.$

5. (b) Solving the equations,

$x^2 + y^2 = 9$ .....(i),

$y^2 = 8x$ .....(ii)



Put $y^2 = 8x$ in (i)

$x^2 + 8x - 9 = 0 \Rightarrow (x + 9)(x - 1) = 0$

i. e., $x = -9$ or 1, $x = -9$ gives imaginary value of y for equation (ii)

hence neglected

$\therefore$ $A \equiv (1, 0)$ and $B \equiv (3, 0)$

$\therefore$ Required area = 2 times the hatched areas

$$= 2\left[\int_0^1 y dx \text{ for (ii)} + \int_1^3 y dx \text{ for (i)}\right]$$

$$= 2\left[\int_0^1 2\sqrt{2}(x)^{1/2}dx + \int_1^3 \sqrt{3^2 - x^2}dx\right]$$

$$= 2\left[2\sqrt{2}\times\left(\frac{x^{3/2}}{3/2}\right)_0^1 + \left(\frac{x\sqrt{9-x^2}}{2} + \frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right)\right)_1^3\right]$$

$$= 2\left[\frac{4\sqrt{2}}{3} + \frac{9}{2}\times\frac{\pi}{2} - \frac{\sqrt{8}}{2} - \frac{9}{2}\sin^{-1}\left(\frac{1}{3}\right)\right]$$

$$= \frac{2\sqrt{2}}{3} + \frac{9\pi}{2} - 9\sin^{-1}\left(\frac{1}{3}\right).$$

6. (d) $\because$ $f'(x) = f(x) \Rightarrow \frac{f'(x)}{f(x)} = 1$

$\Rightarrow \log f(x) = x$ (on integration) $\Rightarrow f(x) = e^x$

Now $\int_0^1 f(x)g(x)dx = \int_0^1 f(x)(x^2 - f(x))dx$

$$= \int_0^1 e^x(x^2 - e^x)dx = \int_0^1 (x^2e^x - e^{2x})dx$$

$$= [x^2e^x - 2xe^x + 2e^x - e^{2x}/2]_0^1$$

$$= e - \frac{e^2}{2} - \frac{3}{2}$$

7. (a) $I = \int_{-1/2}^{1/2} [x]dx + \int_{-1/2}^{1/2} \ln\left(\frac{1+x}{1-x}\right)dx$

$$= \int_{-1/2}^{0} (-1)dx + \int_0^{1/2} 0\,dx + I_1$$

$$= -\frac{1}{2} + I_1 \quad \text{..... (1)}$$

Now the integrand of I_1 is an odd function because $f(x)=\ln\left(\frac{1+x}{1-x}\right)$

$\Rightarrow f(-x)=\ln\left(\frac{1-x}{1+x}\right)=-\ln\left(\frac{1+x}{1-x}\right)=-f(x)$

Therefore by property of Integral $I_1=0$

$\therefore \quad I=-\frac{1}{2}+0=-\frac{1}{2}$

8. (c) Since $f(x)$ is a periodic function and $f(x+T)=f(x)\ \forall\ x\in R$, So

$$\int_a^{a+T} f(x)dx=\int_a^{T} f(x)dx+\int_T^{T-a} f(x)dx$$

$$=\int_a^{T} f(x)dx+\int_a^{a} f(x)dx=\int_0^{T} f(x)dx \qquad \ldots(1)$$

$$\therefore \int_3^{3+3T} f(2x)dx=\frac{1}{2}\int_6^{6+6T} f(t)dt \text{ where } t=2x$$

$$=\frac{1}{2}\int_0^{6T} f(t)dt \text{ [using (1)]} =\frac{1}{2}.6\int_0^{T} f(t)dt$$

[By property] $=3I$

9. (a) $\because\ f'(x)=\sqrt{2-x^2}, 1-0=\sqrt{2-x^2}$

$\therefore \quad x^2-f'(x)=0\Rightarrow x^2-\sqrt{2-x^2}=0$

$\Rightarrow x^4+x^2-2=0$

$\therefore \quad x^2=\frac{-1\pm 3}{2}=-2, 1$. Hence real values of x are ± 1

10. (b) Let $f(x)=(x-1)(x-2)=x^2-3x+2$ $f(x)$ being an algebraic function is continuous in $[1, 2]$

$\therefore$ conditions of Rolle's theorem are satisfied.

Now $f'(x)=2x-3$

Then $\int_1^2 (2x-3)dx=\left[\frac{2x^2}{2}-3x\right]_1^2$

$=[4-6]-[1-3]=-2+2=0$

11. (d) Divide the num. and denom by $\cos^2 x$

$$\therefore \int_0^{\pi/4}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$$

$$=\int_0^{\pi/4}\frac{\sec^2 x dx}{a^2+b^2\tan^2 x}=\frac{1}{b^2}\int_0^{\pi/4}\frac{\sec^2 x dx}{(a/b)^2+\tan^2 x}$$

Put $\tan x=t\Rightarrow \sec^2 x\,dx=dt$, $x=0\Rightarrow t=0$;

$x=\frac{\pi}{4}\Rightarrow t=1$

$$=\frac{1}{b^2}\int_0^1\frac{dt}{(a/b)^2+t^2}=\frac{1}{b^2}\frac{1}{a/b}\left[\tan^{-1}\left(\frac{t}{a/b}\right)\right]_0^1$$

$$=\frac{1}{ab}[\tan^{-1}(b/a)-\tan^{-1}(0)]=\frac{1}{ab}\tan^{-1}\left(\frac{b}{a}\right)$$

12. (d) given $y=x^3\Rightarrow x=y^{1/3}$

$\therefore$ Required area

$$=\int_1^0 y^{1/3}dy+\int_0^8 y^{1/3}dy=\left[\frac{y^{4/3}}{4/3}\right]_1^0+\left[\frac{y^{4/3}}{4/3}\right]_0^8$$

$$=\frac{3}{4}[-1+(2^3)^{4/3}]=\frac{3}{4}[-1+16]=\frac{3}{4}\times 15=\frac{45}{4}$$

13. (a) $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{n+r}{n^2+r^2}$

$$=\lim_{n\to\infty}\frac{n}{n^2}\sum_{r=1}^{n}\frac{1+\frac{r}{n}}{1+(r/n)^2}=\int_0^1\frac{1+x}{1+x^2}$$

14. (d) $f'(x)=\frac{\sin x}{x}, f''(x)=\frac{x\cos x-\sin x}{x^2}$

Now $f'(x)=0\Rightarrow \sin x=0\Rightarrow x=n\pi$,

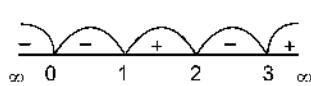
$n=1, 2, 3,\ldots$ ($\because x>0$)

15. (a) $\lim_{x\to 1}\int_4^{f(x)}\frac{2t}{t-1}dt=\lim_{x\to 1}\frac{\int_4^{f(x)}2tdt}{x-1}$ {using L – Hospital rule}

$$=\lim_{x\to 1}\frac{2f(x).f'(x)}{1}=2f(1).f'(1), \text{where} f(1)=4$$

$=8f'(1)$.

16. (b) $f(x)=\int_{-1}^{x} t(e^t-1)\ (t-1)(t-2)^3\ (t-3)^5 dt$.



$$f'(x)=\frac{d}{dx}\int_{-1}^{x} t(e^t-1)\ (t-1)(t-2)^3\ (t-3)^5 dt$$

$=x\ (e^x-1)\ (x-1)\ (x-2)^3\ (x-3)^5\times 1$

$-x\ (e^x-1)\ (x-1)\ (x-2)^3\ (x-3)^5\times 0$

$\frac{d}{dx}\int_{\phi(x)}^{\psi(x)} f(t)dt=f(\psi(x))\ \psi'(x)-f(\phi(x))\ \phi'(x)$

Formula

17. (d) $\lim_{x\to\frac{\pi}{2}} f(x) = \lim_{x\to\frac{\pi}{2}} \frac{\int_{\pi/2}^{x} t\,dt}{\sin(2x-\pi)} \Rightarrow f\left(\frac{\pi}{2}\right)$

$$= \lim_{x\to\frac{\pi}{2}} \frac{\left[\frac{t^2}{2}\right]_{\pi/2}^{x}}{\sin(2x-\pi)}$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{\frac{t^2}{2}-\frac{\pi^2}{8}}{\sin(2x-\pi)} = \lim_{x\to\frac{\pi}{2}} \frac{\frac{1}{8}(4x^2-\pi^2)}{\sin(2x-\pi)}$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{\frac{1}{8}(2x+\pi)(2x-\pi)}{\sin(2x-\pi)}$$

$$= \frac{1}{8}\left(\frac{2x}{2}+\pi\right).1$$

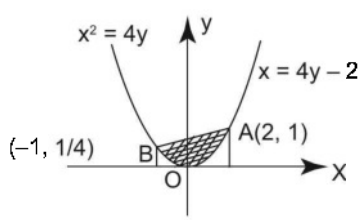
$$\left[\lim_{\theta\to 0} \because \frac{\sin\theta}{\theta} = 1\right] = \frac{\pi}{4}$$

18. (b) Limit $= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}.e^{\frac{r}{n}} = \int_0^1 e^x dx = [e^x]_0^1$

$= e - 1$

19. (a) $f'(x) = x\sin\frac{1}{x}$. At all points in $(0, \pi)$, $f'(x)$ has a definite finite value. $\therefore$ $f(x)$ is differentiable finitely in $(0, \pi)$. As a finitely differentiable function is also continuous, we get $f(x)$ is continuous in $(0, \pi)$.

20. (b) $x^2 = 4y$ and $x = 4y - 2 \Rightarrow A(2, 1)$ and $B(-1, 1/4)$ $\therefore$ required area = shaded area



$$= \left[\int_{-1}^{2} y\,dx \text{ (for } x = 4y-2)\right]$$

$$-\left[\int_{-1}^{2} y\,dx \text{ (for } x^2 = 4y)\right]$$

$$= \int_{-1}^{2} \frac{1}{4}(x+2)dx - \int_{-1}^{2} \frac{1}{4}x^2 dx$$

$$= \frac{1}{4}\left[\frac{x^2}{x}+2x\right]_{-1}^{2} - \frac{1}{4}\left[\frac{x^3}{3}\right]_{-1}^{2}$$

$$= \frac{9}{8} \text{ sq. units}$$

21. (d) Required area $= 2\left|\int_{1/4}^{1} (\sqrt{y}-1)dy\right| = \frac{1}{3}$ sq. units.

22. (c) $\frac{dy}{dx} = 2x + 1 \Rightarrow y = x^2 + x + c$

$\Rightarrow y = x^2 + x$ [$\because c = 0$ by putting $x = 1, y = 2$]

$\Rightarrow \left(x+\frac{1}{2}\right)^2 = y + \frac{1}{4}$, which of the equation of parabola whose vertices is $V(-1/2, -1/4)$.

$\therefore$ Required area

$$= \left|\int_{-1}^{0} (x^2+x)dx\right| = \left(\frac{x^3}{3}+\frac{x^2}{2}\right)_{-1}^{0} = \frac{1}{6} \text{ sq. units}$$

23. (b) For the points of intersection of the given curves $mx = x - x^2 \Rightarrow x^2 = x(1-m) \Rightarrow x = 0, 1-m$

Now as given bounded area

$$= \int_{0}^{1-m} [(x-x^2) - mx]dx = \frac{9}{2}$$

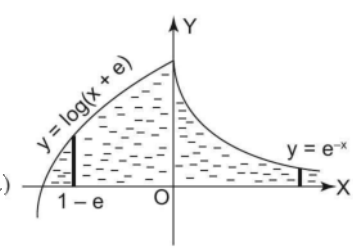
$$\Rightarrow \left[(1-m)\frac{x^2}{2} - \frac{x^3}{3}\right]_0^{1-m} = \frac{9}{2}$$

$$\Rightarrow (1-m)^3\frac{1}{6} = \begin{cases} 9/2 & \text{when } 1-m > 0 \\ -9/2 & \text{when } 1-m < 0 \end{cases}$$

$\therefore$ when $m < 1$, $(1-m)^3 = 27 \Rightarrow m = -2$

when $m > 1$, $(1-m)^3 = -27 \Rightarrow m = 4$

24. (a)



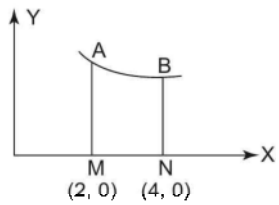
Two given curves meet at $(0, 1)$. Also $y = \log(x+e)$ meets x-axis at $(1-e, 0)$. Required area is as shown in the diagram and area

$$= \int_{1-e}^{0} \log(x+e)dx + \int_{0}^{\infty} e^{-x}dx$$

$= [x\log(x+e) - x + e\log(x+e)]_{1-e}^{0} - [e^{-x}]_0^{\infty}$

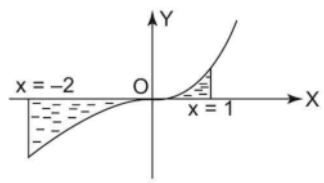
$= [e + (1-e)] - (0-1) = 2$

25. (b)



$$\text{Required area} = \int_2^4 \left(1 + \frac{8}{x^2}\right)dx = \left[x - \frac{8}{x}\right]_2^4 = 4$$

26. (d)



Obviously when $-2 \le x < 0$, then $y < 0$ and when $0 < x \le 1$, then $y > 0$.

Hence area between $x = -2$ and $x = 0$ lies below x-axis and area between $x = 0$ and $x = 1$ lies above x-axis. So required area

$$= \left|\int_{-2}^{0} x^3 dx\right| + \int_0^1 x^3 dx$$

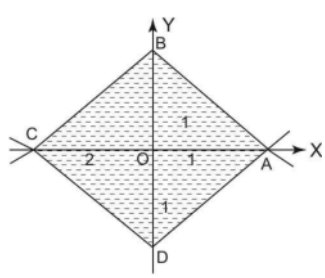
$$= \left|\left[\frac{x^4}{4}\right]_{-2}^{0}\right| + \left[\frac{x^4}{4}\right]_0^{-1} = 4 + \frac{1}{4} = \frac{17}{4}$$

27. (c) As given $\int_1^b f(x)dx = (b-1)\sin(3b+4)$

Now differentiating with respect to b (using property from definite integral) $f(b)$. $1 = \sin(3b+4) + 3(b-1)\cos(3b+4)$

$\therefore f(x) = \sin(3x+4) + 3(x-1)\cos(3x+4)$

28. (b)



$$y = |x| - 1 \Rightarrow \begin{cases} y = -x - 1, & x < 0 \\ y = x - 1, & x > 0 \end{cases};$$

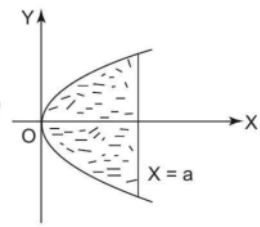
$$y = -|x| + 1 \Rightarrow \begin{cases} y = x + 1, & x < 0 \\ y = -x + 1, & x > 0 \end{cases}$$

$\therefore$ $x + y = -1, x - y = -1$, when $x < 0$; $x - y = 1$, $x + y = 1$, when $x > 0$

Obviously all these four lines from square ABCD whose side $= \sqrt{2}$.

Hence required area = area of the square = 2

29. (c)



Required area is symmetrical about x-axis and it lies between $x = 0$ and $x = a$ (latus rectum). Hence

$$\text{area} = 2\int_0^a y dx = 2\int_0^a 2\sqrt{a}\sqrt{x}dx = 2\int_0^a \sqrt{4ax}dx$$

30. (b) $A_1 = \int_0^k \cos x dx = (\sin x)_0^k = \sin k$,

$$A_2 = \int_0^k \cos 2x dx = \left(\frac{\sin 2x}{2}\right)_0^k = \frac{\sin 2k}{2} \text{ given}$$

ratio is 2 : 1, hence $A_1 : A_2 = 2 : 1$

$$\Rightarrow \frac{\sin k}{\frac{\sin 2k}{2}} = \frac{2}{1}$$

$$\Rightarrow \frac{2\sin k}{2\sin k\cos k} = \frac{2}{1}$$

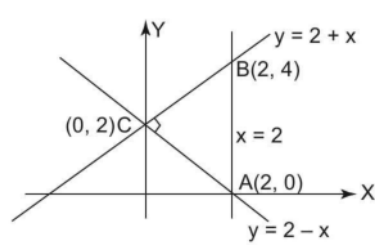
$$\Rightarrow \cos k = \frac{1}{2}$$

$$\Rightarrow k = \frac{\pi}{3}$$

31. (b) Obviously, triangle ACB is right angled at C.

$$\therefore \text{Required area} = \frac{1}{2} \times AC \times BC$$

$$= \frac{1}{2} \times 2\sqrt{2} \times 2\sqrt{2} = 4 \text{ sq. unit}$$



32. (a) The line meets the parabola where $(mx)^2 = 4ax$ or $x(m^2x - 4a) = 0$

$\therefore \quad x = 0, x = \dfrac{4a}{m^2} = h$ (say). The required

area $= \int_0^h (y_1 - y_2)\,dx$ ..... (1)

where y_1 is the ordinate of the upper curve i.e., parabola $y^2 = 4ax$

$\therefore \quad y_1 = 2\sqrt{a}\sqrt{x}$, y_2 is the ordinate of the lower curve i.e., straight line $y = mx$ $\therefore y_2 = mx$

Putting for y_1 and y_2 in (1), we get

$$A = \int_0^h 2\sqrt{a}\sqrt{x}\,dx - \int_0^h mx\,dx$$

$$= 2\sqrt{a}\frac{2}{3}[x^{3/2}]_0^h - m\left[\frac{x^2}{2}\right]_0^h$$

$$= \frac{4\sqrt{a}}{3}h^{3/2} - \frac{m}{2}h^2$$

Now put $h = \dfrac{4a}{m^2}$

$\therefore \quad A = \dfrac{8}{3}\dfrac{a^2}{m^3}$ hence $p = 8$, $q = 3$

33. (b) We have $y = \log x$ .... (i) and ordinate $x = e^2$

Now substitute $y = 0$ in equation (i), we get $= \log x$ or $x = 1$.

$\therefore$ Area bounded by the curve between $x = 1$ and $x = e^2$ is

$$A = \int_{x_1}^{x_2} y\,dx = \int_1^{e^2} \log x\,dx$$

$$A = [x\log x]_1^{e^2} - \int_1^{e^2} \frac{1}{x}.x\,dx$$

$$= e^2 \log e^2 - e^2 + 1 \Rightarrow 2e^2 - e^2 + 1$$

$$= e^2 + 1$$

34. (b) Area $= \int_0^{2a} \sqrt{\dfrac{x^3}{2a - x}}\,dx$ Now put $x = 2a\sin^2$

$$t = \int_0^{\pi/2} 8a^2 \sin^4 t\,dt$$

$$= 8a^2\left[\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}\right] \text{ (Applying gamma function)}$$

$$= \frac{3\pi a^2}{2}$$

TOPIC WISE WARMUP TEST 3: SOLUTIONS

1. (c) $f'(x) = A.\dfrac{\pi}{2}\cos\left(\dfrac{\pi}{2} - x\right)\Big|_{x=\frac{1}{2}}$

$$= A \times \frac{\pi}{2} \times \frac{1}{\sqrt{2}} = \sqrt{2} \Rightarrow A = \frac{4}{\pi}.$$

2. (b) $t = \log_e x \Rightarrow dt = \dfrac{dx}{x} \Rightarrow dx = e^t dt$:

$$\pi\int_0^{37} \sin(\pi t) = -\cos\pi t\Big|_0^{37} = -(\cos 37\pi - \cos 0)$$

$= (-1 - 1) = 2.$

3. (b) $f'(x) = A \times 2^x \log 2, f(1) = 2A \log 2 = 2$

$$\Rightarrow \quad A = \frac{1}{\log 2}\int_0^3 \left(\frac{1}{\log 2}2^x + B\right)dx = 7$$

$$\Rightarrow \quad \frac{2^x}{(\log 2)^2} + Bx\Big|_0^3 = 7.$$

4. (c) $F(x^2) = \int_0^{x^2} f(t)dt = x^2 + x^3$; $F'(x) = f(x^2)$

$(2x) = 2x + 3x^2$

5. (c) Apply property $\int_a^b f(x)dx$

$= \int_a^b f(a + b - x)dx$. Hence $A + B = a$ etc.

6. (c) If $f(x)$ be the integrand, then

$$I = \int_{\pi/2}^{5\pi/6} f(x)dx + \int_{5\pi/6}^{\pi} f(x)dx$$

$$+ \int_{\pi}^{7\pi/6} f(x)dx + \int_{7\pi/6}^{3\pi/2} f(x)dx$$

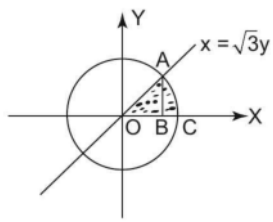
Obviously in above integration intervals $f(x) = [2\sin x] = 1, 0, -1, -2$

$$\therefore\ I = \int_{\pi/2}^{5\pi/6} 1\,dx + \int_{5\pi/6}^{\pi} 0\,dx + \int_{\pi}^{7\pi/6} (-1)dx + \int_{7\pi/6}^{3\pi/2} (-2)dx$$

$$= \frac{\pi}{3} + 0 - \frac{\pi}{6} - \frac{2\pi}{3} = -\frac{\pi}{2}.$$

7. (d) Solving the given equations for x, we have

$$x^2 + \frac{x^3}{3} = 4$$



$\Rightarrow x^2 = 3 \Rightarrow x = \pm\sqrt{3}$ $\therefore$ required area

$$= \int_0^{\sqrt{3}} \frac{x}{\sqrt{3}}dx + \int_{\sqrt{3}}^{2} \sqrt{4 - x^2}\,dx$$

8. (a) Let $I = \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}}dx$ .....(1)

$$\int_0^{\pi} \frac{e^{\cos(\pi - x)}}{e^{\cos(\pi - x)} + e^{-\cos(\pi - x)}}dx$$

$$\left[\because \int_0^a f(x)dx = \int_0^a f(a - x)dx\right]$$

$$\Rightarrow I = \int_0^{\pi} \frac{e^{-\cos x}}{e^{-\cos x} + e^{\cos x}}dx \quad \text{.....(2)}$$

Adding (1) and (2)

$$2I = \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}}dx + \int_0^{\pi} \frac{e^{-\cos x}}{e^{\cos x} + e^{-\cos x}}dx$$

$$= \int_0^{\pi} \frac{e^{\cos x} + e^{-\cos x}}{e^{\cos x} + e^{-\cos x}}dx = \int_0^{\pi} 1\,dx$$

$$= [x]_0^{\pi} = \pi$$

$\Rightarrow I = \pi/2.$

9. (b) Put $\log x = z$ $\therefore x = e^z$ $\therefore dx = e^z dz$ and limits are adjusted as -1 to 2.

$\therefore\ I = \int_{-1}^{2} \left|\frac{z}{e^z}\right| e^z dz = \int_{-1}^{2} |z|\,dz$ e^z is always + ive

$$= \int_{-1}^{0} -z\,dz + \int_0^2 z\,dz$$

$$= \left[-\frac{z^2}{2}\right]_{-1}^{0} + \left[\frac{z^2}{2}\right]_0^2 = \frac{1}{2} + 2 = \frac{5}{2}.$$

10. (a) Use property

$$\int_a^b f(x)dx = \int_a^b f(a + b - x)dx$$

$$I = \int_{\pi/4}^{3\pi/4} \frac{dx}{1 - \cos x} \quad \because \cos(\pi - x) = -\cos x$$

$$\therefore\ 2I = \int_{\pi/4}^{3\pi/4} \frac{2}{1 - \cos^2 x}dx$$

$$= 2\int_{\pi/4}^{3\pi/4} \text{cosec}^2 x\,dx = -2[\cot x]_{\pi/4}^{3\pi/4} = 4$$

$\therefore\ I = 2.$

11. (c) We have

$$2\int_0^{\pi} \sin^3\theta d\theta = 2\int_0^{\pi} \frac{(3\sin 0 - \sin 30)}{4}, d\theta$$

$$= \frac{1}{2}\left[-3\cos\theta + \frac{\cos 0}{3}\right]_0^{\pi}$$

$$= \frac{1}{2}\left[-3(-1 - 1) + \frac{(-1 - 1)}{3}\right] = \frac{8}{3}.$$

12. (a) The curves $y = x$ and $y = x + \sin x$ intersect at $(0, 0)$ and (π, π). Hence area bounded by the two curves.

$$= \int_0^{\pi} (x + \sin x)dx - \int_0^{\pi} x\,dx = \int_0^{\pi} \sin x\,dx$$

$$= [-\cos x]_0^{\pi} = -\cos\pi + \cos 0$$

$$= -(-1) + (1) = 2.$$

13. (c) $\int_0^3 (3ax^2 + 2bx + c)dx$

$$= \int_1^3 (3ax^2 + 2bx + c)dx$$

$$\Rightarrow \int_0^1 (3ax^2 + 2bx + c)dx + \int_1^3 (3ax^2 + 2bx + c)dx$$

$$\Rightarrow \int_1^3 (3ax^2 + 2bx + c)dx$$

$$\Rightarrow \int_0^1 (3ax^2 + 2bx + c)dx = 0$$

$$\Rightarrow \left[\frac{3ax^3}{3} + \frac{2bx^2}{2} + cx\right]_0^1 = 0$$

$$\Rightarrow a + b + c = 0.$$

14. (a) $I = \int_0^{\pi/2} \frac{e^{x^2}}{e^{x^2} + e^{\left(\frac{\pi}{2}-x\right)^2}}$ and

$$I = \int_0^{\pi/2} \frac{e^{\left(\frac{\pi}{2}-x\right)^2}}{e^{\left(\frac{\pi}{2}-x\right)^2} + e^{x^2}} dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} 1\,dx = (x)_0^{\pi/2} \Rightarrow I = \frac{\pi}{4}.$$

$$\left[\because \int_0^a f(x)dx = \int_0^a f(a-x).dx\right.$$

15. (b) $\int_0^{2n\pi} \left(|\sin x| - \frac{1}{2}|\sin x| \right) dx$

$$= \frac{1}{2}\int_0^{2n\pi} |\sin x|\, dx = \frac{2n}{2} \times 2\int_0^{\pi/2} \sin x\, dx$$

$$= 2n[-\cos x]_0^{\pi/2} = 2n.$$

16. (b) Let $P = \lim_{n\to\infty} \left(\frac{n!}{n^n} \right)^{1/n}$

$$= \lim_{n\to\infty} \left(\frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \ldots\ldots, \frac{n}{n} \right)^{1/n}$$

$\therefore$ log

$$P = \frac{1}{n}\lim_{n\to\infty} \left(\log\frac{1}{n} + \log\frac{2}{n} + \ldots\ldots\ldots + \log\frac{n}{n} \right)$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} \log\frac{r}{n}$$

$$\Rightarrow \log P = \int_0^1 \log x\, dx = (x\log x - x)_0^1 = -1$$

$$\Rightarrow P = \frac{1}{e}.$$

17. (d) Given $f(t) = \int_{-t}^{t} \frac{dx}{1+x^2}$

$$= [\tan^{-1} t]_{-t}^{t} = 2\tan^{-1} t$$

Differentiating with respect to t, we get

$$f'(t) = \frac{2}{1+t^2} \Rightarrow f'(t) = \frac{2}{2} = 1.$$

18. (c) $I = \int_{-\pi/2}^{\pi/2} \sin^4 x \cos^6 x\, dx$

$$= 2\int_0^{\pi/2} \sin^4 x \cos^6 x\, dx$$

Applying Gamma function, we get

$$I = \frac{2\sqrt{5}/2\sqrt{7}/2}{2\sqrt{6}} = \frac{3\pi}{256}.$$

19. (b) $2f(x) - 3f(1/x) = x$ ....(1)

Replacing x by $1/x$ in equation (1) we get

$2f(1/x) - 3f(x) = 1/x$ .....(2)

Eliminating $f(1/x)$ from (1) and (2), we get

$$-5f(x) = 2x + \frac{3}{x} = \frac{2x^2+3}{x}$$

$$\Rightarrow f(x) = -\left(\frac{2x^2+3}{5x} \right)$$

$$\int_1^2 f(x)dx = -\frac{1}{5}[x^2 + 3\log_e x]_1^2 = -\frac{3}{5}[1 + \ln 2].$$

20. (b) $g(2) = \int_0^2 f(t)dt = \int_0^1 f(t)dt + \int_1^2 f(t)dt$

$$= I_1 + I_2 \text{ (say)}$$

Now since $t \in [0, 1] \Rightarrow \frac{1}{2} \le f(t) \le 1$

$$\therefore \frac{1}{2}(1-0) \le \int_0^1 f(t)dt \le 1\,(1-0)$$

(by property)

$\Rightarrow 1/2 \le I_1 < 1$ .....(1)

Similarly, $t \in [1, 2] \Rightarrow 0 \le f(t) \le 1/2$

$$\therefore 0\,(2-1) \le \int_1^2 f(t)dt \le \frac{1}{2}(2-1)$$

$$\Rightarrow 0 \le I_2 \le \frac{1}{2} \quad \ldots(2)$$

$$(1) + (2) \Rightarrow \frac{1}{2} + 0 \le I_1 + I_2 \le 1 + \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} \le g(2) \le \frac{3}{2}$$

$$\therefore 0 < g(2) < 2$$

21. (d) Limit $= \lim_{n\to\infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{4n^2 - r^2}}$

$$\lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n-1} \frac{1}{\sqrt{4 - (r/n)^2}}$$

$$= \int_0^1 \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1}\frac{x}{2} \right]_0^1 = \frac{\pi}{6}$$

22. (a) Limit $= \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \frac{k/n}{1 + (k/n)^2}$

$$= \int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2}[\log(1+x^2)]_0^1$$

$$= \frac{1}{2}\log 2$$

23. (c) $I = 0 \Rightarrow \int_0^1 e^{x^2} x\,dx = \int_0^1 e^{x^2} \alpha\,dx$

But $\int_0^1 e^{x^2} x\,dx = \frac{1}{2}(e-1)$ and $\int_0^1 \alpha e^{x^2} dx$ has max. value ac and min. value α, So its value $= \frac{1}{2}(e-1)$ only when $0 < \alpha < 1$.

24. (d) $I_n = \int_0^{\pi/2} x^n \sin x\,dx$

$= [x^n(-\cos x)]_0^{\pi/2} - n\int_0^{\pi/2} x^{n-1}(-\cos x)dx$

$= n\left[(x^{n-1}\sin x)_0^{\pi/2} - (n-1)\int_0^{\pi/2} x^{n-2}\sin x\,dx \right]$

$= n\left[\left(\frac{\pi}{2}\right)^{n-1} - (n-1)I_{n-2} \right]$

$\therefore I_4 = 4(\pi/2)^3 - 12 I_2$

25. (c) $\because P\log\left(\frac{1+x}{1-x}\right) + q\log\left(\frac{1-x}{1+x}\right)^{-2}$ is odd function, so

$I = 0 + \int_{-2}^{2} r dx = \int_{-2}^{2} r dx$

26. (c) $I_{(m,n)} = \int_0^1 t^m (1+t)^n dt$

$= \left[(1+t)^n \frac{t^{m+1}}{m+1} \right]_0^1 - \int_0^1 n(1+t)^{n-1} \frac{t^{m+1}}{m+1} dt$

$= \frac{2^n}{m+1} - \frac{n}{m+1} I_{(m+1,n-1)}$

27. (c) $\int_0^{1000} e^{x-[x]} dx$

$= \int_0^1 e^{x-[x]} dx + \int_1^2 e^{x-[x]} dx + \ldots\ldots + \int_{999}^{1000} e^{x-[x]}\,dx$

$= 1000\int_0^1 e^{x-[x]} dx$ [$\because x - [x]$ is periodic with period = 1]

$= 1000\int_0^1 e^x dx = 1000(e-1)$.

28. (d) $\because f(-x) = \cos(-x)\log\left(\frac{1+x}{1-x}\right)$

$= \cos x \log\left(\frac{1-x}{1+x}\right) = -f(x)$

$\Rightarrow$ By property, $I = 0$

29. (b) $I = \int_0^{\pi} |\cos x|\,dx$

$= \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} (-\cos x) dx$

$= (\sin x)_0^{\pi/2} - (\sin x)_{\pi/2}^{\pi} = 1 - (-1) = 2.$

30. (b) Limit $= \int_0^2 \frac{x dx}{\sqrt{1+x^2}}$

$= (\sqrt{1+x^2})_0^2 = \sqrt{5} - 1$

31. (b) $I = 0 + 2\int_0^{\pi} \frac{2x\sin x}{1+\cos^2 x} dx$

$\Rightarrow I = 4\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x} dx$ .....(i)

$\Rightarrow I = 4\int_0^{\pi} \frac{(\pi - x)\sin x}{1+\cos^2 x} dx$ .....(ii)

$(\because \int_0^a f(x)dx = \int_0^a f(a-x)dx)$

Adding equation (i) and (ii)

$\Rightarrow 2I = 4\int_0^{\pi} \frac{\pi \sin x}{1+\cos^2 x} dx$

$\Rightarrow I = 2\pi\int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx$

Put $\cos x = t \Rightarrow -\sin x\,dx = dt$,

$I = 2\pi\int_1^{-1} \frac{-dt}{1+t^2}$

$\Rightarrow I = -2\pi[\tan^{-1} t]_1^{-1}$

$\Rightarrow I = -2\pi\left(\frac{-\pi}{4} - \frac{\pi}{4}\right) = \pi^2$

32. (d) $I = \int_0^1 \log\left(\frac{1-x}{x}\right) dx$

$\Rightarrow I = \int_0^1 \log\left[\frac{1-(1-x)}{1-x}\right] dx$ (by property)

$= \int_0^1 \log\left(\frac{x}{1-x}\right) dx$

$= -\int_0^1 \log\left(\frac{1-x}{x}\right) dx$

$= I \Rightarrow 2I = 0 \Rightarrow I = 0.$

33. (c) $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx$ .....(1)

$\Rightarrow I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}}$

(By property) $= \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}}dx$

$\therefore \quad 2I = \int_2^3 1\,dx = [x]_2^3 = 1$

$\Rightarrow \quad I = 1/2.$

34. (c) Differentiating both sides with respect to x, we have $0 - \sin^2 x\, f(\sin x).\ \cos x = -\cos x$

$\Rightarrow \quad f(\sin x) = \frac{1}{\sin^2 x}$

[$\because$ $\cos x \neq 0$ as $t \in (\sin x, \sin \pi/2)$ so $x \leq \pi/2$]

$\Rightarrow \quad f(x) = \frac{1}{x^2}$

$\therefore \quad f\left(\frac{1}{\sqrt{3}}\right) = 3$

QUESTION BANK: SOLVE THESE TO MASTER

1. The value of $\int_{-\pi/2}^{\pi/2}\left(\sin^2 x - \sin^4 x\right)(\sin x - \cos x)dx$ is

(a) 4/15 (b) 0

(c) –4/15 (d) 2/15

2. Let $I_1 = \int_0^1 \cot^{-1}\left(1 - x + x^2\right)dx$ and $I_2 = \int_0^1 \tan^{-1} x dx$ then the value of I_1/I_2 is

(a) 2 (b) 1

(c) 1/2 (d) None of these

3. The difference between the greatest and the least value of the function $f(x) = \int_0^x \left(6t^2 - 24\right)dt$ on [1, 3] is

(a) 14 (b) 10

(c) 4 (d) None of these

4. If $\int_{-1}^{-4} f(x)dx = 4$ and $\int_2^{-4}(3 - f(x))dx = 7$ then the value of $\int_{-2}^{1} f(-x)dx$ is

(a) 30 (b) 29

(c) 28 (d) None of these

5. If (–1, 2) and (2, 4) are two points on the curve y $f(x)$ and if $g(x)$ is the gradient of the curve at the point (x, y), then $\int_{-1}^{2} g(x)dx$ is

(a) 2 (b) – 2

(c) 0 (d) 1

6. If $\int_0^{\pi/2} \log \sin x dx = a$, then the value of $\int_0^{\infty} \frac{\log(1+x^2)}{1+x^2}dx$ in terms of a is

(a) $2a$ (b) $-2a$

(c) $a/2$ (d) $a \log 2$

7. The value of the integral $\int_0^{2[x]} (x - [x])dx$ is

(a) $[x]$ (b) $\frac{1}{2}[x]$

(c) $3[x]$ (d) $2[x]$

8. The area bounded by the $y = |\sin x|$, x-axis and the lines $|x| = \pi$ is

(a) 2 (b) 1

(c) 4 (d) None of these

9. The area bounded by y tan x, y cot x, x-axis in $0 \leq x \leq \frac{\pi}{2}$ is

(a) 3log 2 (b) log 2

(c) 2 log 2 (d) None of these

10. If $I_1 = \int_0^{3\pi} f(\cos^2 x)dx$ and $I_2 = \int_0^{\pi} f(\cos^2 x)dx$, then

(a) $I_1 = 5I_2$ (b) $I_1 = I_2$

(c) $I_1 = 3I_2$ (d) None of these

11. The value of

$$\lim_{n \to \infty} n\left[\frac{1}{(n+1)(n+2)} + \frac{1}{(n+2)(n+4)} + \ldots + \frac{1}{6n^2}\right]$$ is

(a) log 6 (b) –log 6
(c) log (3/2) (d) log (2/3)

12. The area of the region bounded by the curves $y = |x - 2|$, $x = 1$, $x = 3$ and the x-axis is
(a) 3 (b) 2
(c) 1 (d) 4

13. If $f(x)$ is differentiable and $\int_0^t xf(x)dx = \frac{2}{5}t^5$, then $f\left(\frac{4}{25}\right)$ equals
(a) 2/5 (b) –5/2
(c) 1 (d) 5/2

14. The area bounded by the curve $y \quad 2x - x^2$ and the straight line $y = -x$ is given by
(a) $\frac{9}{2}$ (b) $\frac{43}{6}$
(c) $\frac{35}{6}$ (d) None of these

15. Let $f : R \to R$ be a differentiable function having $f(2) = 6$, $f'(2) = \frac{1}{48}$. Then $\lim_{x\to 2}\int_6^{f(x)} \frac{4t^3}{x-2}$ dt equals
(a) 36 (b) 24
(c) 18 (d) 12

16. $\int_{-3\pi/2}^{-\pi/2}\left[(x+\pi)^3 + \cos^2(x+3\pi)\right]dx$ is equal to
(a) $\frac{\pi^4}{32}$ (b) $\frac{\pi^4}{32}+\frac{\pi}{2}$
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{2}-1$

17. Let $F(x)=f(x) \quad f\left(\frac{1}{x}\right)$, where $f(x)= \int_1^x \frac{\log t}{1+t}dt$. Then, $F(e)$ equals
(a) 0 (b) 1
(c) 2 (d)1/2

18. Let f be a positive function. Let $I_1 = \int_{1-K}^{K} xf(x(1-x))dx$ and $I_2 = \int_{1-K}^{K} f(x(1-x))dx$ where $2K - 1 > 0$. Then $\frac{I_1}{I_2}$ is
(a) 2 (b) K
(c) 1/2 (d) 1

19. If $\int_0^{n\pi - t} |\sin x|\,dx = K - \cos t \quad (0 \le t \le \pi)\ n \in I$, then $K =$
(a) $2n$ (b) $4n + 1$
(c) $3n + 1$ (d) $2n + 1$

20. Let $\int_0^a f(x)dx = \lambda$ and $\int_0^a f(2a-x)dx = \mu$ then $\int_0^{2a} f(x)dx =$
(a) $\lambda + \mu$ (b) $\lambda - \mu$
(c) $2\lambda - \mu$ (d) $\lambda - 2\mu$

21. If $f(\pi) = 2$ and $\int_0^{\pi} f(x) + f''(x))\sin x dx = 5$ then $f(0)$ is equal to, (it is given that $f(x)$ is continuous in $[0, \pi]$)
(a) 7 (b) 3
(c) 5 (d) 1

22. If $f(x) = \int_0^x \sin^4 t dt$ then $f(x + \pi)$ is equal to
(a) $f(\pi)$ (b) $f(x)$
(c) $f(x) \quad f(\pi)$ (d) $f(x).f(\pi)$

23. $\int_{-3}^{3} x^8\{x^{11}\}dx$ is equal to (where {.} is fractional part of x)
(a) 3^8 (b) 3^7
(c) 3^9 (d) None of these

24. If $[x]$ denotes the greatest integer less than or equal to x, then the value of the integral $\int_0^2 x^2[x]dx$ equals
(a) $\frac{5}{3}$ (b) $\frac{7}{3}$
(c) $\frac{8}{3}$ (d) $\frac{4}{3}$

25. The value of $\int_0^{16\pi/3} |\sin x|\,dx$
(a) 21 (b) 21/2
(c) 10 (d) 11

26. The value of the integral $\int_1^{\infty} \frac{x\log x}{(1+x^2)^2}dx$
(a) 0 (b) log 7
(c) 5 log 13 (d) None of these

27. If $f(x)$ is differentiable and $\int_0^{f(x)} xf(x)dx = \frac{2}{5}t^5$, then $f\left(\frac{4}{25}\right) =$

(a) 2/5 (b) −5/2
(c) 1 (d) 5/2

28. If $f(x)\begin{cases} e^{\sin x} \\ 2 \end{cases}$ for $|x| \le 2$ then $\int_{-2}^{3} f(x)dx =$

(a) 0 (b) 1
(c) 2 (d) 3

29. $\int_{-\frac{3\pi}{2}}^{-\pi/2}\left[(x+\pi)^3 + \cos^2(x+3\pi)\right]dx =$

(a) $\frac{\pi^4}{32}$ (b) $\frac{\pi^4}{32}+\frac{\pi}{2}$
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{2}-1$

30. Assertion: The value of $\int_{-4}^{-5}\sin\left(x^2-3\right)dx + \int_{-2}^{-1}\sin(x^2+12x+33)dx$ is zero.

Reason: $\int_{-a}^{a} f(x)dx = 0$ if $f(x)$ is odd function

31. Assertion: On the interval $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$, the least value of the function $f(x) = \int_{5\pi/4}^{x}(3\sin t + 4\cos t)dt$ is zero

Reason: If $f(x)$ is decreasing function on the interval $[a, b]$, then least value of $f(x)$ is $f(b)$.

32. Assertion: The value of $\int_0^1 \tan^{-1}\frac{2x-1}{(1+x-x^2)}dx = 0$

Reason: $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

33. Assertion: The value of $\int_0^{2\pi}\cos^{99}x dx$ is 0

Reason: $\int_0^{2a} f(x)dx = 2\int_0^{2a} f(x)dx$, if $f(2a - x) = f(x)$

ANSWERS

LECTURE 1

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (d)	**8.** (b)	**15.** (b)	**22.** (b)
2. (a)	**9.** (a)	**16.** (d)	**23.** (a)
3. (d)	**10.** (c)	**17.** (c)	**24.** (c)
4. (a)	**11.** (c)	**18.** (d)	**25.** (c)
5. (b)	**12.** (b)	**19.** (d)	
6. (b)	**13.** (d)	**20.** (a)	
7. (a)	**14.** (c)	**21.** (c)	

Worksheet: To Check the Preparation Level

1. (d)	**6.** (a)	**11.** (b)	**16.** (d)
2. (a)	**7.** (b)	**12.** (c)	**17.** (c)
3. (c)	**8.** (c)	**13.** (b)	**18.** (c)
4. (d)	**9.** (b)	**14.** (b)	
5. (b)	**10.** (b)	**15.** (a)	

LECTURE 2

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (a)	**6.** (c)	**11.** (c)	**16.** (b)
2. (a)	**7.** (c)	**12.** (b)	**17.** (a)
3. (a)	**8.** (d)	**13.** (a)	**18.** (d)
4. (a)	**9.** (c)	**14.** (b)	**19.** (d)
5. (a)	**10.** (a)	**15.** (b)	**20.** (d)

Worksheet: To Check the Preparation Level

1. (b)	**6.** (a)	**11.** (a)	**16.** (b)
2. (a)	**7.** (d)	**12.** (d)	**17.** (a)
3. (d)	**8.** (b)	**13.** (b)	
4. (d)	**9.** (c)	**14.** (d)	
5. (b)	**10.** (d)	**15.** (a)	

LECTURE 3

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (d)	**9.** (a)	**17.** (b)	**25.** (b)
2. (b)	**10.** (a)	**18.** (c)	**26.** (d)
3. (b)	**11.** (c)	**19.** (a)	**27.** (b)
4. (c)	**12.** (b)	**20.** (b)	**28.** (a)
5. (a)	**13.** (a)	**21.** (a)	**29.** (d)
6. (d)	**14.** (b)	**22.** (a)	
7. (b)	**15.** (a)	**23.** (c)	
8. (d)	**16.** (a)	**24.** (c)	

Worksheet: To Check the Preparation Level

1. (c)	**7.** (a)	**13.** (c)	**19.** (c)
2. (b)	**8.** (c)	**14.** (c)	**20.** (c)
3. (b)	**9.** (d)	**15.** (d)	**21.** (c)
4. (a)	**10.** (b)	**16.** (a)	**22.** (b)
5. (b)	**11.** (a)	**17.** (a)	
6. (c)	**12.** (a)	**18.** (a)	

LECTURE 4

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (c)	**6.** (b)	**11.** (a)	**16.** (a)
2. (b)	**7.** (a)	**12.** (b)	**17.** (a)
3. (c)	**8.** (c)	**13.** (b)	**18.** (c)
4. (d)	**9.** (a)	**14.** (a)	**19.** (a)
5. (c)	**10.** (a)	**15.** (c)	**20.** (c)

Worksheet: To Check the Preparation Level

1. (a)	**5.** (c)	**9.** (a)	**13.** (a)
2. (a)	**6.** (b)	**10.** (d)	**14.** (c)
3. (d)	**7.** (b)	**11.** (d)	**15.** (b)
4. (d)	**8.** (c)	**12.** (c)	**16.** (b)

LECTURE 5

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (a)	**4.** (a)	**7.** (a)	**10.** (c)
2. (a)	**5.** (a)	**8.** (a)	**11.** (a)
3. (b)	**6.** (d)	**9.** (b)	

Worksheet: To Check the Preparation Level

1. (b)	**3.** (a)	**5.** (d)	**7.** (a)
2. (a)	**4.** (d)	**6.** (a)	

LECTURE 5

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (b)	**6.** (b)	**11.** (b)	**16.** (c)
2. (c)	**7.** (a)	**12.** (a)	**17.** (d)
3. (b)	**8.** (a)	**13.** (b)	
4. (a)	**9.** (a)	**14.** (c)	
5. (d)	**10.** (a)	**15.** (b)	

Worksheet: To Check the Preparation Level

1. (d)	**5.** (d)	**9.** (c)	**13.** (a)
2. (a)	**6.** (b)	**10.** (c)	
3. (d)	**7.** (a)	**11.** (b)	
4. (d)	**8.** (a)	**12.** (b)	

LECTURE 5

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (d)	**6.** (b)	**11.** (a)	**16.** (c)
2. (a)	**7.** (c)	**12.** (d)	**17.** (c)
3. (d)	**8.** (c)	**13.** (a)	**18.** (d)
4. (b)	**9.** (c)	**14.** (c)	**19.** (a)
5. (b)	**10.** (d)	**15.** (d)	**20.** (c)

Worksheet: To Check the Preparation Level

1. (c)	**5.** (a)	**9.** (d)	**13.** (a)
2. (b)	**6.** (d)	**10.** (c)	**14.** (b)
3. (a)	**7.** (a)	**11.** (b)	**15.** (b)
4. (b)	**8.** (b)	**12.** (a)	

LECTURE 8

Mental Preparation Test

2. $\frac{\pi}{4}-\frac{1}{2}$

3. 0

4. 9

8. $\frac{\pi}{2}-\log 2$

9. $2(\sqrt{2}-1)$

10. $\frac{\pi^2}{4}$

11. $\frac{8a^2}{3m^2}$ s.q.

12. $\frac{1}{2}(b-a)+\frac{1}{2}$ (sin a cos a-sin b cos b).

13. sin (log 3).

14. π

15. $\pi\left(\frac{\pi}{2}-1\right)$

17. $\sqrt{2\pi}$

18. $\frac{\pi}{\sqrt{35}}$

19. 2/3

20. $\frac{8}{3}a^2$ sq. unit, $a>0$

21. $\frac{1}{3}$ sq. unit

22. 2 sq. unit

26. $3(\pi - 2)$ sq. units

27. $\frac{\pi}{6\sqrt{3}} + \log\frac{\sqrt{3}}{2}$

28. πa^2 sq. unit

29. $\frac{16}{3}a^2$ sq. units

30. $\frac{3}{2}(\pi - 2)$ sq. units

31. π sq. units

32. 10

33. $\frac{16\sqrt{2}}{15}$

34. πab sq. unit

35. $8a^2/3$

Objective Question Bank Topic Wise:
Solve these to master

1. (c)	**10.** (c)	**19.** (d)	**28.** (c)
2. (a)	**11.** (c)	**20.** (a)	**29.** (c)
3. (a)	**12.** (c)	**21.** (b)	**30.** (b)
4. (b)	**13.** (a)	**22.** (c)	**31.** (d)
5. (a)	**14.** (a)	**23.** (b)	**32.** (a)
6. (b)	**15.** (c)	**24.** (b)	**33.** (a)
7. (a)	**16.** (c)	**25.** (b)	
8. (c)	**17.** (d)	**26.** (a)	
9. (b)	**18.** (c)	**27.** (a)	

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PART C

Differential Equations

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LECTURE



1 Order and Degree of the Differential Equation

BASIC CONCEPTS

DEFINITIONS

1. **Differential Equations**

   An equation $F\left(x, y, \frac{dy}{dx}, \frac{dy}{dx^2} \ldots\ldots \frac{d^n y}{dx^n}\right) = 0$

   Which expresses the relation between dependent and independent variable and their derivatives of any order is called a differential equation

   OR

   An equation involving dependent variable and their derivatives w.r.t. one or more independent variable.

2. **Ordinary Differential Equation**

   An ordinary differential equation is a differential equation in which derivatives are w.r.t. a single independent variable i.e., number of independent variable is only one.

3. **Partial Differential Equation**

   A partial differential equation is a differential equation which contains two or more independent variable and partial derivatives w.r.t. them,

   example $\frac{x\partial z}{dx} + \frac{y\partial z}{dy} = 2x$

4. **Order**

   The order of a differential equation is the order of the highest order derivative appearing in it

   example, $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^7 + 8\frac{d^4y}{dx^4} = 0$ is a differential equation of order = 4 (Order: Always a natural number)

5. **Degree**

   The degree of a differential equation is the degree of highest order derivative appearing in it. When the derivatives are free of radicals or fractional powers.

   (i) $\sqrt[3]{x\frac{dy}{dx} + 2y} = \frac{d^2y}{dx^2}$ Order (2), Degree (3)

   $x\frac{d^4y}{dx^4} + 2y = \left(\frac{d^3y}{dx^3}\right)^3$ Order (4), Degree (1)

   (ii) $\sqrt{x+y} = \frac{d^4y}{dx^4}$ Order (4), Degree (1)

   (Degree: Always a Natural number)

6. **Formation of Differential Equation**

   Differential equation corresponding to given algebraic or transcendental equation or Elimination of constants from given algebraic equation

Rule to Form a Differential Equation From its Solution

Differential given equation or solution as many times as equal to the number of arbitary constants in given equation. By eleminating the arbitary constants from given equation and equations obtained in step 1 required differential equation is obtained.

NOTES

1. If the given equation or solution of differential equation contains n arbitrary constant then the resulting differential equation is a differential equation of order n.
2. Differential equation corresponding to $y = Ae^{px} + Be^{qx}$ is $y_2 - (p + q)\, y_1 + pqy \quad 0$
3. Differential equation corresponding to $y \quad Ae^{px} + Be^{qx} + Ce^{rx}$ is $y_3 - (p + q + r)\, y_2 + (pq + qr + rp)\, y_1 - pqr\, y = 0$.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Form the differential equation corresponding to $y^2 = a(b - x)\,(b + x)$ by eliminating parameters a and b.

[CBSE-95 C; 2004]

Solution

We have $y^2 = a(b^2 - x^2)$ ..... (i)

Since there are two arbitrary constants in the given equation, so we shall differentiate it two times and we shall get a differential equation of second order.

Differentiating (i) w.r.t. x, we get $2y.\ \frac{dy}{dx} = -2ax$

$$\Rightarrow y\frac{dy}{dx} = -ax \qquad \text{..... (ii)}$$

Differentiating (ii) w.r.t. x, we get

$$y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = -a \qquad \text{.... (iii)}$$

from (ii) and (iii), we get

$$x\left\{y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right\} = y\frac{dy}{dx}$$

2. Represent the following family of curves by forming the corresponding differential equations (a, b are parameters)

[CBSE-2005]

(i) $\frac{x}{a} + \frac{y}{b} = 1$

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(iii) $(y - b)^2 = 4(x - a)$

Solution

(i) We have $\frac{x}{a} + \frac{y}{b} = 1$ ..... (i)

where, a, b are parameters, Differentiating (i) with respect to x,

we get $\frac{1}{a} + \frac{1}{b}\frac{dy}{dx} = 0$ ..... (ii)

Differentiating (ii) with respect to x, we get

$\frac{1}{b}\frac{d^2y}{dx^2} = 0 \Rightarrow \frac{d^2y}{dx^2} = 0$ this is the required differential equation.

(ii) We have $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ..... (i)

$$\Rightarrow \frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{x}{a^2} + \frac{y}{b^2}\frac{dy}{dx} = 0 \qquad \text{..... (ii)}$$

Differentiating (ii) with respect to x, we get

$$\frac{1}{a^2} + \frac{1}{b^2}\left(\frac{dy}{dx}\right)^2 + \frac{y}{b^2}\frac{d^2y}{dx^2} = 0$$

Multiplying both sides by x, we get

$$\frac{x}{a^2} + \frac{x}{b^2}\left(\frac{dy}{dx}\right)^2 + \frac{xy}{b^2}\frac{d^2y}{dx^2} = 0 \qquad \text{..... (iii)}$$

Subtracting (ii) from (iii), we get

$$\frac{1}{b^2}\left\{x\left(\frac{dy}{dx}\right)^2 + xy\frac{d^2y}{dx^2} - y\frac{dy}{dx}\right\} = 0$$

$$\Rightarrow x\left(\frac{dy}{dx}\right)^2 + xy\frac{d^2y}{dx^2} - y\frac{dy}{dx} = 0$$

This is the required differential equation.

(iii) We have $(y - b)^2 = 4\,(x - a)$ ..... (i)

Differentiating with respect to x, we get

$$2\,(y - b)\,\frac{dy}{dx} = 4 \Rightarrow (y - b)\frac{dy}{dx} = 2 \qquad \text{..... (ii)}$$

Differentiating with respect to x, we get

$$(y-b)\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0 \quad \text{..... (iii)}$$

Eliminating $(y-b)$ from (ii) and (iii), we get

$$\frac{d^2y}{dx^2}\Big/\frac{dy}{dx}+\left(\frac{dy}{dx}\right)^2=0 \Rightarrow 2\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3=0$$

This is the required differential equation.

3. Form the differential equation of the family of curves represented by the equation (a being the parameter)

(i) $(2x+a)^2+y^2=a^2$ ***[CBSE-98 C]***

(ii) $(2x-a)^2-y^2=a^2$ ***[CBSE-98 C]***

Solution

(i) $(2x+a)^2+y^2=a^2$ ..... (i)

differentiating (i) w.r.t. x we get $2(2x+a)\times 2 + 2y\frac{dy}{dx}=0$

$$\Rightarrow 2(2x+a)+y\frac{dy}{dx}=0$$

$$\Rightarrow 2x+a=\frac{y}{2}\frac{dy}{dx}$$

$$\Rightarrow a=\frac{-y}{2}\frac{dy}{dx}-2x \quad \text{..... (ii)}$$

Substituting the value of a in equation (i) we get

$$\left(\frac{-y}{2}\frac{dy}{dx}\right)^2+y^2=\left(\frac{-y}{2}\frac{dy}{dx}-2x\right)^2$$

$$\Rightarrow \frac{y^2}{4}\left(\frac{dy}{dx}\right)^2+y^2=\frac{y^2}{4}\left(\frac{dy}{dx}\right)^2+4x^2+\frac{4xy}{2}\frac{dy}{dx}$$

$$\Rightarrow y^2-4x^2-2xy\frac{dy}{dx}=0$$

This is the required differential equation.

(ii) $(2x-a)^2-y^2=a^2$ ..... (i)

differentiating (i) w.r.t. x we get $2(2x-a)\times 2 - 2y\frac{dy}{dx}=0$

$$\Rightarrow 2(2x-a)=y\frac{dy}{dx} \Rightarrow (2x-a)=\frac{y}{2}\frac{dy}{dx}$$

$$\Rightarrow a=2x-\frac{y}{2}\frac{dy}{dx} \quad \text{..... (ii)}$$

Substituting the value of a in equation (i) we get

$$\left(\frac{y}{2}\frac{dy}{dx}\right)^2-y^2=\left(2x-\frac{y}{2}\frac{dy}{dx}\right)^2$$

$$\Rightarrow \frac{y^2}{4}\left(\frac{dy}{dx}\right)^2-y^2=4x^2+\frac{y^2}{4}\left(\frac{dy}{dx}\right)^2-2xy\frac{dy}{dx}$$

$\Rightarrow 2xy\frac{dy}{dx} = 4x^2 + y^2$ this is the required differential equation.

4. Show that the function $y = (A \quad Bx)e^{3x}$ is a solution of the equation $\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=0$

[CBSE-93; PSB-93, 99]

Solution

We have $y \quad (A + Bx)\,e^{3x}$ ..... (i)

Differentiating (i) w.r.t. x, we get

$$\frac{dy}{dx} = 3(A \ + Bx)\,e^{3x} + Be^{3x} \quad \text{..... (ii)}$$

Differentiating (ii) w.r.t. x, we get

$$\frac{d^2y}{dx^2} = 6\,Be^{3x} + 9\,e^{3x}(A \quad Bx) \quad \text{..... (iii)}$$

$$\therefore \frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 6Be^{3x} + 9(A \ + Bx)\,e^{3x} -$$

$18(A - Bx)\,e^{3x} - 6Be^{3x} + 9(A + Bx)\,e^{3x} = 0$

Thus $y \quad (A + Bx)\,e^{3x}$ satisfies the given differential equation.

Hence, it is a solution of the given differential equation.

5. Form the differential equation representing the family of curves $y = \tan^{-1}x + c\,e^{\tan^{-1}x}$ where c is a arbitrary constant.

[CBSE-94C]

Solution

Given $y \quad \tan^{-1}x + c\,e^{\tan^{-1}x}$ ..... (i)

differentiating equation (i) w.r.t. x, we get

$$\frac{dy}{dx}=\frac{1}{1+x^2}+\frac{ce^{\tan^{-1}x}}{1+x^2}$$

$$\Rightarrow (1+x^2)\frac{dy}{dx}=1+ce^{\tan^{-1}x}$$

$$\Rightarrow (1+x^2)\frac{dy}{dx}-1=ce^{\tan^{-1}x}$$

Put $ce^{\tan^{-1}x}=(1+x^2)\frac{dy}{dx}-1$ in equation (i) we get $y=\tan^{-1}x+(1+x^2)\frac{dy}{dx}-1$

$(1+x^2)\frac{dy}{dx} - y + \tan^{-1}x - 1 = 0$ is the required differential equation.

6. Show that $y = \cos(\cos x)$ is a solution of the differential equation $\frac{d^2y}{dx^2} - \cot x \frac{dy}{dx} + y\sin^2 x = 0$ ***[CBSE-(SP)-2006]***

Solution

Given $y = \cos(\cos x)$ ..... (i)

differentiating equation (i), we get

$$\frac{dy}{dx} = -\sin(\cos x)(-\sin x) = \sin x \sin(\cos x)$$

..... (ii)

differentiating equation (ii) w.r.t. x, we get

$\frac{d^2y}{dx^2} = \sin x \cos(\cos x)(-\sin x) + \sin(\cos x)(\cos x)$

$\Rightarrow \frac{d^2y}{dx^2} = -\sin^2 x \cos(\cos x) + \cos x \sin(\cos x)$

$\Rightarrow \frac{d^2y}{dx^2} = -\sin^2 x \cos(\cos x) + \cos x \frac{(dy/dx)}{\sin x}$ {using (ii)}

$\Rightarrow \frac{d^2y}{dx^2} = -y\sin^2 x + \cot x \frac{dy}{dx}$ {using (i)}

$\Rightarrow \frac{d^2y}{dx^2} - \cot x \frac{dy}{dx} + y\sin^2 x = 0$

$\therefore y = \cos(\cos x)$ is a solution of

$\frac{d^2y}{dx^2} - \cot x \frac{dy}{dx} + y\sin^2 x = 0$

7. If $y = Ae^{mx} + Be^{nx}$, prove that $\frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny = 0$. ***[CBSE-2007]***

Solution

$y = Ae^{mx} + Be^{nx}$

$\frac{dy}{dx} = mAe^{mx} + nBe^{nx}$ ..... (i)

$\frac{d^2y}{dx^2} = m^2Ae^{mx} + n^2Be^{nx}$ ..... (ii)

L.H.S. $= \frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny = m^2Ae^{mx} + n^2Be^{nx} - m^2Ae^{mx} - mnBe^{nx} - nmAe^{mx} - n^2Be^{nx} + mnAe^{mx} + mnBe^{nx} = 0$.

8. Obtain the differential equation of the family of all circles in a plane. ***[NCERT-Book]***

Solution

The equation of the family of all circles in a plane can be written as $x^2 + y^2 + 2gx + 2fy + c = 0$, g, f, c arbitrary constants. ..... (1)

Differentiating, we get $2x + 2y\frac{dy}{dx} + 2g + 2f\frac{dy}{dx} = 0 \Rightarrow x + y\frac{dy}{dx} + g + f\frac{dy}{dx} = 0$ ..... (2)

Differentiating again, we get

$$1 + y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 0 + f\frac{d^2y}{dx^2} = 0$$

$$\Rightarrow 1 + y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = -f\frac{d^2y}{dx^2}$$

..... (3)

Differentiating once again, we get

$$y\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2}\cdot\frac{dy}{dx} + 2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} = -f\frac{d^3y}{dx^3}$$

..... (4)

Dividing (4) by (3), we get,

$$\frac{yy_3 + y_1y_2 + 2y_1y_2}{1 + yy_2 + y_1^2} = \frac{y_3}{y_2}$$

$\Rightarrow yy_2y_3 + y_1y_2^2 + 2y_1y_2^2 = y_3 + yy_2y_3 + y_1^2y_3$

$\Rightarrow (1 + y_1^2)y_3 - 3y_1y_2^2 = 0$, which is the required differential equation.

9. Prove that $y = x^3 + ax^2 + bx + c$ is a solution of the differential equation $\frac{d^3y}{dx^3} = 6$. ***[MP-94, 98, 99, 2002]***

Solution

Given $y = x^3 + ax^2 + bx + c$ ..... (i)

This contains three arbitrary constants a, b and c.

Hence it will be differentiated thrice:

Differentiating (i) w.r.t. x, we get $\frac{dy}{dx} = 3x^2 + 2ax + b$ .... (ii)

Differentiating again w.r.t. x, $\frac{d^2y}{dx^2} = 6x + 2a$ ..... (iii)

Differentiating again w.r.t. x, $\frac{d^3y}{dx^3} = 6$ ..... (iv)

This shows that $y = x^3 + ax^2 + bx + c$ is a solution of the given differential equation.

Proved.

10. If $y = e^{a\sin^{-1}x}$, $-1 \le x \le 1$, then show that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y$ ***[CBSE-2010]***

Solution

$\frac{dy}{dx} = e^{a\sin^{-1}x}.\frac{a}{\sqrt{1-x^2}}$, squaring we get

$\left(\frac{dy}{dx}\right)^2 = \frac{y^2a^2}{1-x^2}$

$\left(\because y = e^{a\sin^{-1}x}\right)$

$(1-x^2)\left(\frac{dy}{dx}\right)^2 = a^2y^2 \Rightarrow$ differentiating we get

$(1-x^2)2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = a^2\left(2y\frac{dy}{dx}\right)$

$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = a^2y$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Find the differential equation of all the circles in the first quadrant which touch the co-ordinate axes.

   [HSB-95C, 2001]

2. Form the differential equation corresponding to $ay^2 = (x - c)^3$ by eliminating c.

3. Find the differential equation of all those parabolas whose latus rectum is $4a$ and axes are parallel to x-axis.

4. Form the differential equation of the family of curves y a sin $(bx + c)$, a and c being parameters.

   [PSB-92]

5. Show that the differential equation that represents the family of all parabolas having their axis of symmetry coincident with the axis of x is $yy_2 + y_1^2 = 0$.

   [NCERT-Book]

6. Find the differential equation which will represent the pair of lines y $2x + 3$ and y $4x + 5$.

   [MP-93]

7. Show that $y = Ax + \frac{B}{x}, x \ne 0$ is a solution of the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.

   [CBSE-93, 94, 97C; PSB-96]

8. Show that y $e^{-x} + ax + b$ is solution of the differential equation $e^x\frac{d^2y}{dx^2} = 1$.

   [CBSE-93, 97C]

9. Show that the function $y = A\cos x - B$ sin x is a solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$.

   [CBSE-93, 2006]

10. Construct the differential equation of the family of curve $x^2 + y^2 = a^2$.

    [MP-2007]

11. Prove that $y = e^{2x}$ is a solution of differential equation $\frac{d^2y}{dx^2} - 4y = 0$.

    [MP-2007]

ANSWERS

1. $8x\left(\frac{dy}{dx}\right)^3 - 12y\left(\frac{dy}{dx}\right)^2 = 27x$

2. $(x-y)^2\left\{1+\left(\frac{dy}{dx}\right)^2\right\} = \left\{x+y\frac{dy}{dx}\right\}^2$

3. $8a\left(\frac{dy}{dx}\right)^3 = 27y$

4. $2a\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$

5. $\frac{d^2y}{dx^2} + b^2y = 0$

7. $\left(\frac{dy}{dx}-2\right)\left(\frac{dy}{dx}-4\right) = 0$

11. $y\frac{dy}{dx} + x = 0$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. The differential equation representing the set of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c$, ($a$, b are parameters) is ***[EAMCET-2000]***

(a) $\frac{y''}{y'} - \frac{y'}{y} = \frac{1}{x}$ (b) $\frac{y''}{y'} - \frac{y'}{y} = \frac{1}{x}$

(c) $\frac{y''}{y'} + \frac{y'}{y} + \frac{1}{x} = 0$ (d) $\frac{y''}{y'} + \frac{y'}{y} = \frac{1}{x}$

Solution

(d) Differentiating $\frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$ ..... (1)

$\Rightarrow \frac{1}{a^2} + \frac{yy'' + y'^2}{b^2} = 0$ ..... (2)

Eliminating a^2, b^2, required differential equation will be

$yy' = x(yy'' + y'^2) \Rightarrow \frac{y''}{y'} + \frac{y'}{y} - \frac{1}{x} = 0$

2. The differential equation of curves y $ax\cos\left(\frac{1}{x}+b\right)$, where a, b are parameters is ***[MP-2003]***

(a) $x^2y_2 + y$ 0 (b) $x^4y_2 + y$ 0

(c) $xy_2 - y$ 0 (d) $x^3y_2 - y$ 0

Solution

(b) Differentiating twice y $ax\cos\left(\frac{1}{x}+b\right)$, we have

$y_2 = \frac{a}{x^3}\cos\left(\frac{1}{x}+b\right)$

$\Rightarrow x^4y_2 = -ax\cos\left(\frac{1}{x}+b\right)$

$\Rightarrow x^4y_2 = -y$ [from given equation]

$\Rightarrow x^4y_2 + y$ 0

3. The degree of differential equation $\left(\frac{d^3y}{dx^3}\right)^{2/3} + \left(\frac{d^3y}{dx^3}\right)^{3/2} = 0$ is ***[ICS (Pre.)-2004]***

(a) 3 (b) 5

(c) 4 (d) 9

Solution

(d) $\left(\frac{d^3y}{dx^3}\right)^{2/3} = -\left(\frac{d^3y}{dx^3}\right)^{3/2} \Rightarrow \left(\frac{d^3y}{dx^3}\right)^6 = -\left(\frac{d^3y}{dx^3}\right)^9$ which shows that its degree = 9.

4. The differential equation corresponding to the family of curves y $c(x-c)^2$, where c is a constant, is ***[ICS (Pre.)-2004]***

(a) $4y^2 = 8xyy' - (y')^2$

(b) $8y^2 = 4xyy' - (y')^3$

(c) $8y^2 = 8xy' + (y')^3$

(d) $y^2 = xyy' + (y')^2$

Solution

(b) $y = c(x - c)^2$ ..... (1)

$y' = 2c(x - c)$

$\Rightarrow y'^2 = 4c^2(x - c)^2$ ..... (2)

(1), (2) $\Rightarrow c = \dfrac{y'^2}{4y}$

Putting this value of c in (1) we get the required equation.

5. The differential equation that represents all parabolas each of which has a latus rectum $4a$ and whose axes are parallel to x-axis is

[VIT-2006]

(a) $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} = 0$

(b) $\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^3 = 0$

(c) $a\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^3 = 0$

(d) $2a\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^3 = 0$

Solution

(d) Let equation of the given family of parabolas be $(y + \alpha)^2 = 4a(x + \beta)$ ..... (1)

where α and β are parameters. Then $(y + \alpha)$

$\dfrac{dy}{dx} = 2a$ ..... (2)

and $(y + \alpha)\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2 = 0$ ..... (3)

(2), (3) $\Rightarrow \dfrac{2a}{\left(\dfrac{dy}{dx}\right)} \cdot \dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2 = 0$

So required equation is $2a\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^3 = 0$

6. The degree of the differential equation $\dfrac{d^2y}{dx^2} + 3\left[\dfrac{dy}{dx}\right]^2 = x^2 \log\left[\dfrac{d^2y}{dx^2}\right]$ is

[Pb. CET-2004]

(a) 1 (b) 2

(c) 3 (d) None of these

Solution

(d) Since the equation is not a polynomial in all differential coefficients, so its degree is not defined.

7. The differential equation which represents the three parameter. Family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$ is

(a) $y''' = \dfrac{3y'y''^2}{1 + y'^2}$ (b) $y''' = \dfrac{3y''^2}{1 + y'^2}$

(c) $y''' = \dfrac{3y'}{1 + y'^2}$ (d) $y''' = \dfrac{3y'}{1 - y'^2}$

Solution

(a) $x^2 + y^2 + 2gx + 2fy + c = 0$

$\Rightarrow x + yy' + g + fy' = 0$ ..... (1)

$\Rightarrow 1 + y'^2 + yy'' + fy'' = 0$ ..... (2)

$\Rightarrow 3y'y'' + yy''' + fy''' = 0$ ..... (3)

(1) y''' – (2) y'' gives (a)

8. Differential equation of all conics of the form $ax^2 + by^2 = 1$, a and b being parameters is

(a) $xy\dfrac{d^2y}{dx^2} + x\left(\dfrac{dy}{dx}\right)^2 - y\dfrac{dy}{dx} = 0$

(b) $xy\dfrac{d^2y}{dx^2} - x\left(\dfrac{dy}{dx}\right)^2 + y\dfrac{dy}{dx} = 0$

(c) $xy\dfrac{d^2y}{dx^2} - x\left(\dfrac{dy}{dx}\right)^2 - y\dfrac{dy}{dx} = 0$

(d) None of these

Solution

(a) We have $ax^2 + by^2 = 1$

Differentiating $2ax + 2by\dfrac{dy}{dx} = 0$ ..... (1)

Differential equation

$a + b\left(y\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2\right) = 0$ ..... (2)

from (1) and (2), we get $a = -\dfrac{by}{x}\dfrac{dy}{dx}$

$= -b\left(y\dfrac{d^2y}{dX^2} + \left(\dfrac{dy}{dx}\right)^2\right)$

$\Rightarrow xy\dfrac{d^2y}{dx^2} + x\left(\dfrac{dy}{dx}\right)^2 - y\dfrac{dy}{dx} = 0$

9. If $y \quad e^{4x} + 2e^{-x}$ satisfies the relation $\dfrac{d^3y}{dx^3} + A\dfrac{dy}{dx} + By = 0$ then values of A and B respectively are

(a) –13, 14 (b) –13, –12
(c) –13, 12 (d) 12, –13

Solution

(b) $(64 + 4A + B)\, e^{4x} + (-2 - 2A + 2B)\, e^{-x} = 0$
$\Rightarrow 64 + 4A + B = 0 \Rightarrow -2 - 2A + 2B = 0$
$\therefore A = -13, B = -12$

10. The differential equation of the family of curves represented by $c\,(y + c)^2 = x^3$ is

(a) $y\dfrac{d^2y}{dx^2} - y^2\left(\dfrac{dy}{dx}\right)^2 = 27x$

(b) $12y\left(\dfrac{dy}{dx}\right)^2 = 8x\left(\dfrac{dy}{dx}\right)^3 - 27x$

(c) $8y\left(\dfrac{dy}{dx}\right)^3 = 12x\left(\dfrac{dy}{dx}\right)^2 - 27x$

(d) None of these

Solution

(b) We have $c\,(y + c)^2 = x^3$ differentiating 2 (c)

$(y + c)\dfrac{dy}{dx} = 3x^2$ ..... (1)

Squaring both sides

$4c^2\,(y + c)^2\left(\dfrac{dy}{dx}\right)^2 = 9x^4 \Rightarrow c = \dfrac{9x}{4\left(\dfrac{dy}{dx}\right)^2}$

Put the value of 'c' in (1), we get

$$2\left(\frac{9x}{4(dy/dx)^2}\right)\left(y + \frac{9x}{4(dy/dx)^2}\right)\frac{dy}{dx} = 3x^2$$

$$\Rightarrow 12y\left(\frac{dy}{dx}\right)^2 = 8x\left(\frac{dy}{dx}\right)^3 - 27x$$

11. If $f(x)$ and $g\,(x)$ are two solutions of differential equation $a\dfrac{d^2y}{dx^2} + x^2\dfrac{dy}{dx} + y = e^x$, then $f(x) - g(x)$ is the solution of ***[AMU-1991]***

(a) $a^2\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} + y = e^x$

(b) $a^2\dfrac{d^2y}{dx^2} + y = e^x$

(c) $a\dfrac{d^2y}{dx^2} + y = e^x$

(d) $a\dfrac{d^2y}{dx^2} + x^2\dfrac{dy}{dx} + y = 0$

Solution

(d) By the given condition $af''(x) + x^2 f'(x) + f(x) = e^x$ and $ag''(x) + x^2 g'(x) + g(x) = e^x$
Subtracting, we get $a(f''(x) - g''(x)) + x^2(f'(x) - g'(x)) + f(x) - g(x) = 0$
This shows that $f(x) - g(x)$ is the solution of

$$a\frac{d^2y}{dx^2} + x^2\frac{dy}{dx} + y = 0$$

12. Let F denotes the family of ellipses whose centre is at the origin and major axis is y axis. The equation of this family F is ***[MPPET-2005; [Kerala (CEE)-2003]]***

(a) $\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx}\left(x\dfrac{dy}{dx} - y\right) = 0$

(b) $xy\dfrac{d^2y}{dx^2} - \dfrac{dy}{dx}\left(x\dfrac{dy}{dx} - y\right) = 0$

(c) $xy\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx}\left(x\dfrac{dy}{dx} - y\right) = 0$

(d) $\dfrac{d^2y}{dx^2} - \dfrac{dy}{dx}\left(x\dfrac{dy}{dx} - y\right) = 0$

Solution

(c) Let equation of the given family of ellipses be

$\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ ..... (1)

where a and b are parameters.
Differentiating (1) with respect to x, we have

$$\frac{x}{a^2} + \frac{y}{b^2}\frac{dy}{dx} = 0 \Rightarrow \frac{x}{a^2} = -\frac{y}{b^2}\frac{dy}{dx} \quad \text{..... (2)}$$

$$\Rightarrow \frac{1}{a^2} + \frac{y}{b^2}\frac{d^2y}{dx^2} + \frac{1}{b^2}\left(\frac{dy}{dx}\right)^2 = 0$$

$$\Rightarrow \frac{1}{a^2} = -\frac{1}{b^2}\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] \quad \text{..... (3)}$$

Dividing (2) by (3), we get

$$x = \frac{y\left(\dfrac{dy}{dx}\right)}{y\left(\dfrac{d^2y}{dx^2}\right) + \left(\dfrac{dy}{dx}\right)^2}$$

$$\Rightarrow xy\frac{d^2y}{dx^2} + \frac{dy}{dx}\left(x\frac{dy}{dx} - y\right) = 0$$

which is the required equation.

13. The differential equation obtained by eliminating the arbitrary constants a and b from $xy = ae^x + be^{-x}$ is ***[EAMCET-2007]***

(a) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy = 0$

(b) $\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} - xy = 0$

(c) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy = 0$

(d) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - xy = 0$

Solution

(a) $xy = ae^x + be^{-x}$

Differentiating with respect to x, 2 times

$$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = -(ae^x + be^{-x}) = -xy$$

or $$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + xy = 0$$

14. The differential equation of the family of straight lines whose slope is equal to y-intercept is ***[Karnataka CET-2007]***

(a) $(x+1)\frac{dy}{dx} - y = 0$

(b) $(x+1)\frac{dy}{dx} + y = 0$

(c) $\frac{dy}{dx} = \frac{x-1}{y-1}$

(d) $\frac{dy}{dx} = \frac{x+1}{y+1}$

Solution

(a) Equation of line whose slope = y intercept is $y \quad cx + c$

$\Rightarrow y \quad c(x+1)$ ..... (i)

Now differentiating w.r.t. x we get $\frac{dy}{dx} = c$ ..... (ii)

$\Rightarrow$ required differential equation (By substituting (ii) in (i))

$$y = \frac{dy}{dx}(x+1) \Rightarrow \frac{dy}{dx}(x+1) - y = 0$$

15. The differential equation of the system of circles touching y-axis at the origin is given by ***[NDA-2003, 2004, 2006]***

(a) $x^2 + y^2 - 2xy\frac{dy}{dx} = 0$

(b) $x^2 + y^2 + 2xy\frac{dy}{dx} = 0$

(c) $x^2 - y^2 + 2xy\frac{dy}{dx} = 0$

(d) $x^2 - y^2 - 2xy\frac{dy}{dx} = 0$

Solution

(c) Given system of circles is $x^2 + y^2 - 2ax = 0$ ..... (1)

where a is parameter.

Differentiating it, we get $2x + 2y\frac{dy}{dx} - 2a = 0$ ..... (2)

Eliminating a from (1) and (2), we get required equation

$$x^2 - y^2 + 2xy\frac{dy}{dx} = 0$$

16. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where c is a positive parameter, is of ***[IIT-1999; AIEEE-2005; MP PMT-2002; UPSC-2007]***

(a) Order 1 (b) Order 2

(c) Degree 3 (c) Degree 4

Solution

(a, c) Given curve is $y^2 = 2c(x + \sqrt{c})$

Differentiate w.r.t. x, $2y\frac{dy}{dx} = 2c \Rightarrow c = y\frac{dy}{dx}$

Hence differential equation is

$$y^2 = 2y\frac{dy}{dx}\left(x + \sqrt{y\frac{dy}{dx}}\right)$$

$$\Rightarrow \frac{y}{2dy/dx} - x = \sqrt{y\frac{dy}{dx}}$$

Squaring and multiplying by $\left(\frac{dy}{dx}\right)^2$

$$y\left(\frac{dy}{dx}\right)^3 - x^2\left(\frac{dy}{dx}\right)^2 + xy\left(\frac{dy}{dx}\right) - \frac{y^2}{4} = 0$$

Hence order is 1 and degree is 3.

17. The order and degree of the differential equation

$x\left(\frac{d^3y}{dx^3}\right)^2 + y\left(\frac{dy}{dx}\right)^4 + y^2 = 0$ is

[MPPET-2010]

(a) order 2, degree 3
(b) order 3, degree 2
(c) order 2, degree 2
(d) None of these

Solution

(b) Step 1: Order of the differential equation is order of the heighest order derivative occuring in it and degree of the differential equation is the highest power of the highest order derivative when derivatives are free of radical signs or fractional powers.
Step 2: Order of differential equation is order of $\frac{d^3y}{dx^3}$ i.e. 3 and its power 2 is defined as degree.

18. The differential equation of all conics whose centre lie at the origin is of order

[MPPET-2010]

(a) 2 (b) 3
(c) 4 (d) None of these

Solution

(a) $\because$ Centre is origin, equation of ellipse can be

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$,

No. of parameter = 2
$\therefore$ Order = 2

19. The differential equation which represents the family of curves $y = c_1e^{c_2x}$, where c_1 and c_2 are arbitrary constants, is

(a) $y' = y^2$ (b) $y'' = y'y$
(c) $yy'' = y'$ (d) $yy'' = (y')^2$

[AIEEE-2009]

Solution

(d) $y = c_1e^{c_2x}$ .....(1)

$y' = c_2c_1e^{c_2x} = \Rightarrow y' = c_2y$ .....(2)

$y'' = c_2y'$

From eqn. (2), $c_2 = \frac{y'}{y}$

So, $y'' = \frac{(y_1')^2}{y} \Rightarrow yy'' = (y')^2$

20. The differential equation, whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants, is of:

(a) Second order and second degree
(b) Second order and first degree
(c) First order and second degree
(d) First order and first degree

[MPPET-2009]

Solution

(b) Given, $Ax^2 + By^2 = 1$
On differentiating two times, we get

$2Ax + 2By\frac{dy}{dx} = 0$ ..... (i)

and $2A + 2B\left[\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right] = 0$

$\Rightarrow -2\frac{By}{x}\frac{dy}{dx} + 2B\left[\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right] = 0$

$\Rightarrow y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 - \frac{y}{x}\left(\frac{dy}{dx}\right) = 0$

It is of second order and first degree differential equation.

21. The order and degree of the differential equation

$\left(1 + 4\frac{dy}{dx}\right)^{2/3} = 4\frac{d^2y}{dx^2}$ are respectively:

[MPPET-2009]

(a) 1, 2/3 (b) 3, 2
(c) 2, 3 (d) 2, 2/3

Solution

(c) Given differential equation is

$\left(1 + 4\frac{dy}{dx}\right)^{2/3} = 4\frac{d^2y}{dx^2}$

$\Rightarrow \left(1 + 4\frac{dy}{dx}\right)^2 = 4^3\left(\frac{d^2y}{dx^2}\right)^3$

Here, highest order is 2 and degree is 3.

22. If $y = \left[x + \sqrt{1 + x^2}\right]^n$, then the value of $(1 + x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is:

[MPPET-2009]

(a) n^2y (b) $-n^2y$
(c) ny (d) $-ny$

Solution

(a) Given, $y = \left[x + \sqrt{1+x^2}\right]^n$

On differentiating w.r.t. x, we get

$$\frac{dy}{dx} = n\left[x + \sqrt{1+x^2}\right]^{n-1}\left[1 + \frac{2x}{2\sqrt{1+x^2}}\right]$$

$$= \frac{n\left[x + \sqrt{1+x^2}\right]^n}{\sqrt{1+x^2}} = \frac{ny}{\sqrt{1+x^2}}$$

$$\Rightarrow (1+x^2)\left(\frac{dy}{dx}\right)^2 = n^2y^2$$

Again, differentiating w.r.t. x, we get

$$\Rightarrow 2(1+x^2)\frac{dy}{dx}\frac{d^2y}{dx^2} + 2x\left(\frac{dy}{dx}\right)^2 = 2n^2y\frac{dy}{dx}$$

$$\Rightarrow (1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = n^2y$$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. $\frac{d^3y}{dx^3} + 2\left[1 + \frac{d^2y}{dx^2}\right] = 1$ has degree and order is ***[UPSEAT-2003]***

(a) 1, 3 (b) 2, 3
(c) 3, 2 (d) 3, 1

2. The order and degree of the diff. equ. $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 - xy = 0$ are respectively ***[MP PET-03]***

(a) 2 and 4 (b) 3 and 2
(c) 4 and 5 (d) 2 and 3

3. The order and degree of the differential equation $\left(1 + 3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are ***[AIEEE-2002]***

(a) 1, 2/3 (b) 3, 1
(c) 3, 3 (d) 1, 2

4. The diff. equation $\left(\frac{d^2y}{dx^2}\right)^2 - \left(\frac{dy}{dx}\right)^{1/2} = y^3$ has the degree ***[Roorkee-1999]***

(a) 1/2 (b) 2
(c) 3 (d) 4

5. Family $y \quad Ax + A^3$ of curve represented by the diff. equation of degree ***[MP PET-1999]***

(a) Three (b) Two
(c) One (d) None of these

6. The order of the diff. equation $y\left(\frac{dy}{dx}\right) = x \Big/ \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^3$ is ***[MP PET-1994]***

(a) 1 (b) 2
(c) 3 (d) 4

7. The order and the degree of differential equation $\frac{d^4y}{dx^4} - 4\frac{d^3y}{dx^3} + 8\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 4y = 0$ are respectively

(a) 4, 1 (b) 1, 4
(c) 1, 1 (d) None of these

8. The order of the differential equation whose general solution is given by $y = c_1e^{2x+c_2} + c_3e^x + c_4 \sin(x + c_5)$ is ***[AMU-2000]***

(a) 5 (b) 4
(c) 3 (d) 2

9. The degree and order of the differential equation of the family of all parabolas whose axis is x-axis, are respectively ***[AIEEE-2003]***

(a) 2, 1 (b) 1, 2
(c) 3, 2 (d) 2, 3

10. The order and degree of the diff. equation $\frac{d^2y}{dx^2} = \sin\left(\frac{dy}{dx}\right) + xy$ are respectively

(a) 2, 1 (b) 2, infinite
(c) 2, 0 (d) 2, not defined

11. The order and degree of the differentail equation $\sqrt{\frac{dy}{dx}} - 4\frac{dy}{dx} - 7x = 0$ are ***[PET-93]***

(a) 1 and 1/2 (b) 2 and 1
(c) 1 and 1 (d) 1 and 2

12. The order and degree of the differential equation $x\left(\frac{dy}{dx}\right)^3+2\left(\frac{d^2y}{dx^2}\right)^2+3y+x=0$ are respectively

[MPPET-2004]

(a) 3, 2 (b) 2, 1
(c) 2, 2 (d) 2, 3

13. The differential equation of the family of curves given by $c(y+c)^2=x^2$ is

(a) $\left(2x\frac{dy}{dx}-y\right)^2\left(\frac{dy}{dx}\right)^2=1$

(b) $\left(y\frac{dy}{dx}-x\right)^2\left(\frac{dy}{dx}\right)^2=1$

(c) $\left(x\frac{dy}{dx}-y\right)\left(\frac{dy}{dx}\right)^2=1$

(d) None of these

14. The diff. equation obtained on eliminating A and B from the equation $y=A\cos\omega t+B\sin\omega t$ is

[Karnataka CET-2000; Pb. CET-2001]

(a) $y''=-\omega^2y$ (b) $y''+y=0$
(c) $y''+y'=0$ (d) $y''-\omega^2y=0$

15. The differential equation of all circles passing through origin and having their centres on the x-axis is

[Kurukshetra CEE-1998; Karnataka CET-2005; AIEEE-2007]

(a) $\frac{dy}{dx}=\frac{y^2+x^2}{2xy}$ (b) $\frac{dy}{dx}=\frac{y^2+x^2}{2x}$
(c) $\frac{d^2y}{dx^2}=\frac{y^2-x^2}{2xy}$ (d) $\frac{dy}{dx}=\frac{y^2-x^2}{2xy}$

16. Which one of the following represents the differential equation of all parabolas having the axes of symmetry coincident with the axis of x?

(a) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$

(b) $y\frac{d^3y}{dx^3}+\left(\frac{d^2y}{dx^2}\right)^2=0$

(c) $y\frac{dy}{dx}+\left(\frac{d^2y}{dx^2}\right)^2=0$

(d) $y\frac{dy}{dx}+\left(\frac{d^3y}{dx^3}\right)^2=0$

17. The differential equation of the family of curves $y\quad a\cos(x+b)$ is ***[MP PET-1993]***

(a) $\frac{d^2y}{dx^2}-y=0$ (b) $\frac{d^2y}{dx^2}+y=0$
(c) $\frac{d^2y}{dx^2}+2y=0$ (d) None of these

18. The differential equation corresponding to primitive $y\quad e^{cx}$ is or The elimination of the arbitrary constant m from the equation $y=e^{mx}$ gives the differential equation

[MP PET-1995, 2000; Pb. CET-2000]

(a) $\frac{dy}{dx}=\left(\frac{y}{x}\right)\log x$ (b) $\frac{dy}{dx}=\left(\frac{x}{y}\right)\log y$
(c) $\frac{dy}{dx}=\left(\frac{y}{x}\right)\log y$ (d) $\frac{dy}{dx}=\left(\frac{x}{y}\right)\log x$

19. The differential equation of the family of curves $y^2=4a(x+a)$, where a is an arbitrary constant, is

(a) $y\left[1+\left(\frac{dy}{dx}\right)^2\right]=2x\frac{dy}{dx}$

(b) $y\left[1-\left(\frac{dy}{dx}\right)^2\right]=2x\frac{dy}{dx}$

(c) $\frac{d^2y}{dx^2}+2\frac{dy}{dx}=0$

(d) $\left(\frac{dy}{dx}\right)^3+3\frac{dy}{dx}+y=0$

20. The differential equation for all the straight lines which are at a unit distance from the origin is ***[PET-93]***

(a) $\left(y-x\frac{dy}{dx}\right)^2=1-\left(\frac{dy}{dx}\right)^2$

(b) $\left(y+x\frac{dy}{dx}\right)^2=1+\left(\frac{dy}{dx}\right)^2$

(c) $\left(y+x\frac{dy}{dx}\right)^2=1+\left(\frac{dy}{dx}\right)^2$

(d) $\left(y+x\frac{dy}{dx}\right)^2=1-\left(\frac{dy}{dx}\right)^2$

21. The differential equation whose solution is $y\quad cx+c-c^3$, is ***[PET-1997]***

(a) $\frac{dy}{dx}=c$

(b) $y=x\frac{dy}{dx}+\frac{dy}{dx}-\left(\frac{dy}{dx}\right)^3$

(c) $\frac{dy}{dx} = c - 3c^2$

(d) None of these

22. The diff. equation for which $\sin^{-1}x + \sin^{-1}y = c$ is given by ***[Karnataka CET-2003]***

(a) $\sqrt{1-x^2}\,dx + \sqrt{1-y^2}\,dy = 0$

(b) $\sqrt{1-x^2}\,dy + \sqrt{1-y^2}\,dx = 0$

(c) $\sqrt{1-x^2}\,dy - \sqrt{1-y^2}\,dx = 0$

(d) $\sqrt{1-x^2}\,dx - \sqrt{1-y^2}\,dy = 0$

23. The differential equation of all straight lines passing through origin is ***[MPPET-2006]***

(a) $y = \sqrt{x}\frac{dy}{dx}$ (b) $\frac{dy}{dx} = y + x$

(c) $\frac{dy}{dx} = y - x$ (d) none of these

SOLUTIONS

1. (a) Degree = 1; Order = 3

2. (d) Order = 2; degree = Power of $\frac{d^2y}{dx^2} = 3$

3. (c) ∵ Power on derivative is fraction,

$$\therefore \text{Cubing } \left(1 + \frac{3dy}{dx}\right)^2 = 4^3\left(\frac{d^3y}{dx^3}\right)^3$$

∴ Order = 3, degree = 3

4. (d) $\left(\frac{d^2y}{dx^2}\right)^2 - y^3 = \left(\frac{dy}{dx}\right)^{1/2}$

Squaring both sides $\left[\left(\frac{d^2y}{dx^2}\right)^2 - y^3\right]^2 = \frac{dy}{dx}$

order = 2, degree = 4

5. (a) $y = Ax + A^3$; A = arbitrary consant

differentiating both sides $\frac{dy}{dx} = A$

Put in the given equation, $y = \left(\frac{dy}{dx}\right)x + \left(\frac{dy}{dx}\right)^3$

order = 1, degree = 3

6. (a) $y\left(\frac{dy}{dx}\right) = \frac{x}{\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^3}$

$$y\left(\frac{dy}{dx}\right)^2 + y\left(\frac{dy}{dx}\right)^4 = x$$

Order = 1, degree = 4

7. (a) Order = 4, degree = 1

8. (b) $y = C_1e^{2x}.e^{C_2} + C_3e^x + C_4(\sin x\cos C_5 + \sin C_5\cos x)$

$= \left(C_1.e^{C_2}\right)e^{2x} + C_3e^x + (C_4\cos C_5)\sin x + (C_4\sin C_5)\cos x$

$= \lambda_1e^{2x} + C_3e^x + \lambda_2\sin x + \lambda_3\cos x$

∴ Order of equation = no. of arbitrary constant = 4

9. (a) Here, equation of parabola will be $y^2 = 4a(x-h)$

differentiating $2y\frac{dy}{dx} = 4a$ or $y\frac{dy}{dx} = 2a$

Again differentiating $y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$

Order = 2, degree = 1

10. (d) Degree is only defined for polynomial equation of $\frac{dy}{dx}$

∴ degree is not defined, order = 2

11. (d) $\sqrt{\frac{dy}{dx}} = \left(4\frac{dy}{dx} + 7x\right)$

Squaring both sides $\frac{dy}{dx} = \left(4\frac{dy}{dx} + 7x\right)^2$

∴ Order = 1, degree = 2

12. (c) Order = 2, degree = 2

13. (c) $c(y+c)^2 = x^2$ .....(1)

differentiating both sides, $2c(y+c)\frac{dy}{dx} = 2x$

$c(y+c)\frac{dy}{dx} = x$; squaring both sides

$c^2(y+c)^2\left(\frac{dy}{dx}\right)^2 = x^2$; dividing by (1)

$c\left(\frac{dy}{dx}\right)^2 = 1$

Put $c = \frac{1}{\left(\frac{dy}{dx}\right)^2}$ in equation

$$\frac{1}{\left(\frac{dy}{dx}\right)^2}\left[y + \frac{1}{\left(\frac{dy}{dx}\right)}\right]^2 = x^2$$

taking squares roots

$$y + \frac{1}{\left(\frac{dy}{dx}\right)^2} = \pm x\frac{dy}{dx}$$

$$\left(\pm x\frac{dy}{dx} - y\right)\left(\frac{dy}{dx}\right)^2 = 1$$

14. (a) $y = A\cos\omega t + B\sin\omega t$

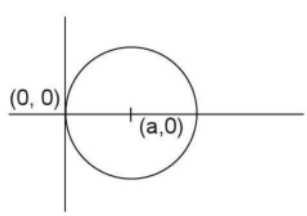
$\frac{dy}{dx} = \omega[-A\sin\omega t + B\cos\omega t]$

Again differentiating

$y'' = \omega^2(-A\cos\omega t - B\sin\omega t) = -\omega^2 y$

15. (d) Equation of circle will be:

$(x-a)^2 + y^2 = a^2$ or $x^2 + y^2 - 2ax = 0$ .....(1)



differentiating w.r.t x

$x + y\frac{dy}{dx} = a$

Put in eqn. (1) to get

$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

16. (a) Let the equation of parabola be

$y^2 = 4a(x-h)$

differentiating w.r.t x

$2y\frac{dy}{dx} = 4a$

or $y\frac{dy}{dx} = 2a$; again differentaing, we get

$y\frac{d^2y}{dx} + \left(\frac{dy}{dx}\right)^2 = 0$

17. (b) $y = a\cos(x+b) \Rightarrow$ differentiating we get

$\frac{dy}{dx} = -a\sin(x+b)$

again differentiating we get $\frac{d^2y}{dx^2} = -a\cos(x+b)$

$\frac{d^2y}{dx^2} = -y$ ($\because$ Order = No. of arbitrary constant = 2

$\therefore$ we will have to differentiate twice)

18. (a) $y = e^{mx}$ .....(1)

differentiating, $\frac{dy}{dx} = me^{mx} \Rightarrow \frac{dy}{dx} = my$

$\frac{1}{y}\frac{dy}{dx} = m$; put this value in equation of curve

$y = e^{mx}$ or $mx = \log_e y$

$\frac{x}{y}\frac{dy}{dx} = \log_e y \Rightarrow \frac{dy}{dx} = \frac{y}{x}\log y$

19. (b) $y^2 = 4a(x+a)$; differentiating twice, we get

$y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$

20. (c) Let equation of straight line be $y = mx + c$

Here $m = \frac{dy}{dx}$, Also distance from origin = 1

$\therefore \quad \left|\frac{c}{\sqrt{1+m^2}}\right| = 1 \Rightarrow c^2 = (1+m^2)$

$(y - mx)^2 = (1+m^2)$

or $\left(y - x\frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$

21. (b) $y = cx + c - c^3$, differentiating we get

$\frac{dy}{dx} = c$, differential equation is

$y = x\frac{dy}{dx} + \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^3$

22. (b) Differentiating the equation,

$$\frac{1}{\sqrt{1-x^2}}dx+\frac{1}{\sqrt{1-y^2}}dy=0$$

$$\sqrt{1-y^2}\,dx+\sqrt{1-x^2}\,dy=0$$

23. (d) Equation of straight line will be

$y = mx$

$\therefore$ Differential Equation is $y=\frac{dy}{dx}x\left(\because m=\frac{dy}{dx}\right)$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The order and degree of the diff. equation $\rho=\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ are respectively

[MP PET-2001; UPSEAT-2002]

(a) 2, 2 (b) 2 ,3
(c) 2 ,1 (d) None of these

2. Degree of the given diff. equation $\left(\frac{d^2y}{dx^2}\right)^3=\left(1+\frac{dy}{dx}\right)^{1/2}$ is ***[MP PET-1997]***

(a) 2 (b) 3
(c) 1/2 (d) 6

3. Order and degree of diff. equation $\frac{d^2y}{dx^2}=\left\{y+\left(\frac{dy}{dx}\right)^2\right\}^{1/4}$ are ***[MP PET-1996]***

(a) 4 and 2 (b) 1 and 2
(c) 1 and 4 (d) 2 and 4

4. The order of the differential equation whose solution is $x^2 + y^2 + 2gx + 2fy + c = 0$, is

[MP PET-1995]

(a) 1 (b) 2
(c) 3 (d) 4

5. The order and degree of the differential equation $\frac{d^2y}{dx^2}=\sqrt{1+\left(\frac{dy}{dx}\right)^2}$ is ***[DCE-2002]***

(a) 4, 2 (b) 1, 2
(c) 2, 2 (d) 2 , 1/2

6. The order and degree of the differential equation $\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+y^2=0$ are respectively

(a) 2 and 2 (b) 1 and 1
(c) 2 and 1 (d) 1 and 2

7. The order of the diff. equation of the family of curves $y = (c_1 + c_2)\, e^x + c_3\, e^{x+c}4$ is

(a) 1 (b) 2
(c) 3 (d) 4

8. The order and degree of the diff. equation $\sqrt{\frac{d^2y}{dx^2}}=3\sqrt{\frac{dy}{dx}+3}$ are respectively

(a) 2 and 3 (b) 2 and 2
(c) 2 and 6 (d) 2 and 1

9. If m and n are the order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^5+4\frac{\left(\frac{d^2y}{dx^2}\right)^3}{\left(\frac{d^3y}{dx^3}\right)}+\frac{d^3y}{dx^3}=x^2-1$ then

[KCET-1999]

(a) $m = 3$ and $n = 5$
(b) $m = 3$ and $n = 1$
(c) $m = 3$ and $n = 3$
(d) $m = 3$ and $n = 2$

10. Order of the differential equation of the family of all concentric circles centred at (h, k) is

[EAMCET-2002]

(a) 1 (b) 2
(c) 3 (d) 4

11. The order of differential equation of all parabolas having directrix parallel to x-axis is ***[Orissa JEE-2004]***

(a) 3 (b) 1

(c) 4 (d) 2

12. The differential equation for all the straight lines which are at a distance of 4 units from the origin is

(a) $\left(y+x\frac{dy}{dx}\right)^2=4\left\{1+\left(\frac{dy}{dx}\right)^2\right\}$

(b) $\left(y-x\frac{dy}{dx}\right)^2=16\left\{1+\left(\frac{dy}{dx}\right)^2\right\}$

(c) $\left(y-x\frac{dy}{dx}\right)^2=1+\left(\frac{dy}{dx}\right)^2$

(d) $\left(y-x\frac{dy}{dx}\right)^2=4\left\{1+\left(\frac{dy}{dx}\right)^2\right\}$

13. The differential equation for the curves $y \quad a\sin(x+b)$, where a and b are arbitrary constants is

(a) $\frac{d^2y}{dx^2}+y=0$

(b) $\frac{d^3y}{dx^3}=0$

(c) $\frac{d^2y}{dx^2}+\cos(x+b)=0$

(d) $\frac{dy}{dx}-y\tan(x+b)=0$

14. The differential equation whose solution is $y = c_1\cos ax + c_2\sin ax$ is (where c_1, c_2 are arbitrary constants) ***[MP PET-1996]***

(a) $\frac{d^2y}{dx^2}+y^2=0$ (b) $\frac{d^2y}{dx^2}+a^2y \quad 0$

(c) $\frac{d^2y}{dx^2}+ay^2=0$ (d) $\frac{d^2y}{dx^2}-a^2y \quad 0$

15. Differential equation, whose solution is $(x-h)^2+(y-k)^2=a^2$, is

(a) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3=a^2\frac{d^2y}{dx^2}$

(b) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3=a^2\left(\frac{d^2y}{dx^2}\right)^2$

(c) $\left[1+\left(\frac{dy}{dx}\right)\right]^3=a^2\left(\frac{d^2y}{dx^2}\right)^2$

(d) None of these

16. The diff. equation for the family of curves $x^2+y^2-2ay \quad 0$, where a is an arbitrary constant is ***[AIEEE-2004]***

(a) $(x^2-y^2)y'=2xy$ (b) $2(x^2+y^2)y'=xy$

(c) $2(x^2-y^2)y'=xy$ (d) $(x^2+y^2)y'=2xy$

17. The differential equation of the family of curves $v=\frac{A}{r}+B$, where A and B are arbitrary constants, is

(a) $\frac{d^2v}{dr^2}+\frac{1}{r}\frac{dv}{dr}=0$ (b) $\frac{d^2v}{dr^2}-\frac{2}{r}\frac{dv}{dr}=0$

(c) $\frac{d^2v}{dr^2}+\frac{2}{r}\frac{dv}{dr}=0$ (d) None of these

18. The differential equation found by the elimination of the arbitrary constant K from the equation $y=(x+K)e^{-x}$ is

(a) $\frac{dy}{dx}-y=e^{-x}$ (b) $\frac{dy}{dx}-ye^x=1$

(c) $\frac{dy}{dx}+ye^x=1$ (d) $\frac{dy}{dx}+y=e^{-x}$

19. The differential equation of the family of curve $y=Ae^{3x}+Be^{5x}$ where A, B are arbitrary constants is ***[MNR-1988]***

(a) $\frac{d^2y}{dx^2}+8\frac{dy}{dx}+15y=0$

(b) $\frac{d^2y}{dx^2}-8\frac{dy}{dx}+15y=0$

(c) $\frac{d^2y}{dx^2}-\frac{dy}{dx}+y=0$

(d) None of these

20. The differential equation for the line $y=mx+c$ (where c is arbitrary constant)

(a) $\frac{dy}{dx}=m$ (b) $\frac{dy}{dx}+m=0$

(c) $\frac{dy}{dx}=0$ (d) None of these

21. Family of curves $y \quad e^x(A\cos x+B\sin x)$ represents the differential equation ***[MP PET-1999]***

(a) $\frac{d^2y}{dx^2}=2\frac{dy}{dx}-y$ (b) $\frac{d^2y}{dx^2}=2\frac{dy}{dx}-2y$

(c) $\frac{d^2y}{dx^2}=\frac{dy}{dx}-2y$ (d) $\frac{d^2y}{dx^2}=2\frac{dy}{dx}+y$

22. The general solution of the differential equation $e^y \frac{dy}{dx} + (e^y + 1)\cot x = 0$ is

(a) $(e^y + 1)\cos x = K$
(b) $(e^y + 1)\operatorname{cosec} x = K$
(c) $(e^y + 1)\sin x = K$
(d) None of these

23. The differential equation of all straight lines touching the circle $x^2 + y^2 = a^2$ is

[VITEEE-2008]

(a) $\left(y - \frac{dy}{dx}\right)^2 = a^2\left[1 + \left(\frac{dy}{dx}\right)^2\right]$

(b) $\left(y - x\frac{dy}{dx}\right)^2 = a^2\left[1 + \left(\frac{dy}{dx}\right)^2\right]$

(c) $\left(y - x\frac{dy}{dx}\right) = a^2\left[1 + \frac{dy}{dx}\right]$

(d) $\left(y - \frac{dy}{dx}\right) = a^2\left[1 - \frac{dy}{dx}\right]$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 16 minutes.
3. The worksheet consists of 16 questions. The maximum marks are 48.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. The order of the differential equation whose general solution is $y = c_1e^x + c_2e^{2x} + c_3e^{3x} + c_4e^x + c_5$ where c_1, c_2, c_3, c_4, c_5 are arbitrary constants is

(a) 5 (b) 4
(c) 3 (d) None of these

2. The order and degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{1/3} + x^{1/4} = 0$ are respectively ***[PET-98]***

(a) 2, 3 (b) 3, 3
(c) 2, 6 (d) 2, 4

3. The order and degree of the differential equation $x\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + y^2 = 0$ are respectively ***[KCET-2001]***

(a) 2 and 2 (b) 1 and 1
(c) 2 and 1 (d) 1 and 2

4. The degree of the differential equation $\frac{d^2y}{dx^2} - \sqrt{\frac{dy}{dx} - 3} = x$ is ***[Orissa JEE-2004]***

(a) 2 (b) 1
(c) 1/2 (d) 3

5. The degree of the differential equation $y(x) = 1 + \frac{dy}{dx} + \frac{1}{1.2}\left(\frac{dy}{dx}\right)^2 + \frac{1}{1.2.3}\left(\frac{dy}{dx}\right)^3 + \ldots$ is ***[Orissa JEE-2005]***

(a) 2 (b) 3
(c) 1 (d) None of these

6. The differential equation that represents all parabolas each of which has a latus rectum $4a$ and whose axes are parallel to the x-axis is

(a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$

(b) $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$

(c) $a\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$

(d) $2a\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$

7. The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 - xy = 0$ are respectively

(a) 2 and 4 (b) 3 and 2
(c) 4 and 5 (d) 2 and 3

8. In the following, differential equation with equal degree and order is

(a) $\frac{d^4y}{dx^4} + 8\left(\frac{dy}{dx}\right)^8 + 5y = e^x$

(b) $5\left(\frac{d^3y}{dx^3}\right)^4 + 8\left(1 + \frac{dy}{dx}\right)^2 + 5y = x^8$

(c) $\left[1 + \left(\frac{dy}{dx}\right)^3\right]^{2/3} = 4\frac{d^3y}{dx^3}$

(d) $y = x^2\frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

9. The differential equation of all circles which passes through the origin and whose centre lies on y-axis is ***[MNR-86, DCE-2000]***

(a) $(x^2 - y^2)\frac{dy}{dx} - 2xy = 0$

(b) $(x^2 - y^2)\frac{dy}{dx} + 2xy = 0$

(c) $(x^2 - y^2)\frac{dy}{dx} - xy = 0$

(d) $(x^2 - y^2)\frac{dy}{dx} + xy = 0$

10. The order and degree of the differential equation $\left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = \frac{d^2y}{dx^2}$ are

(a) 2, 2 (b) 3, 3
(c) 2, 3 (d) 3, 2

11. The differential equation whose solution is $y = A \sin x + B \cos x$ is *[CEE-93; KCET-2002]*

(a) $\frac{d^2y}{dx^2} + y = 0$ (b) $\frac{d^2y}{dx^2} - y = 0$

(c) $\frac{dy}{dx} + y = 0$ (d) None of these

12. The differential equation representing the family of curves $y^2 = 2c\,(x + \sqrt[3]{c})$, where c is a positive parameter, is of *[Kerala PET-2008]*

(a) order 1, degree 1
(b) order 1, degree 2
(c) order 1, degree 3
(d) order 1, degree 4

13. The differential equation representing the family of curves $y = x\,e^{cx}$ (c is a constant) is *[Kerala PET-2008]*

(a) $\frac{dy}{dx} = \frac{y}{x}\left(1 - \log\frac{y}{x}\right)$

(b) $\frac{dy}{dx} = \frac{y}{x}\log\left(\frac{y}{x}\right) + 1$

(c) $\frac{dy}{dx} = \frac{y}{x}\left(1 + \log\frac{y}{x}\right)$

(d) $\frac{dy}{dx} + 1 = \frac{y}{x}\log\left(\frac{y}{x}\right)$

14. What are the order and degree, respectively of the differential equation $\left(\frac{d^2y}{dx^2}\right)^{\frac{5}{6}} = \left(\frac{dy}{dx}\right)^{3}$? *[NDA-2008]*

(a) 2, 1 (b) 2, 5
(c) 2, 5/6 (d) 1, 1/3

15. The differential equation of all Coaxal parabolas $y^2 = 4a\,(x - b)$, where a and b are arbitrary constants, is *[MP PET-2008]*

(a) $y\frac{d^2y}{dx^2} + \frac{dy}{dx} = 1$

(b) $y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1$

(c) $y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$

(d) $y\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

16. The degree and order of the differential equation of all tangent lines to the parabola $x^2 = 4y$ is: *[AMU (Engg.)-2007]*

(a) 2, 1 (b) 2, 2
(c) 1, 3 (d) 1, 4

ANSWER SHEET

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)
6. (a) (b) (c) (d)
7. (a) (b) (c) (d)
8. (a) (b) (c) (d)
9. (a) (b) (c) (d)
10. (a) (b) (c) (d)
11. (a) (b) (c) (d)
12. (a) (b) (c) (d)
13. (a) (b) (c) (d)
14. (a) (b) (c) (d)
15. (a) (b) (c) (d)
16. (a) (b) (c) (d)

HINTS AND EXPLANATIONS

1. (c) $y = c_1e^x + c_2e^{2x} + c_3e^{3x} + c_4e^x - e^{C_3}$

$= e^x(c_1 + c_4e^{c_5}) + c_2e^{2x} + c_3e^{3x}$

$= e^x\lambda + c_2e^{2x} + c_3e^{3x}$

$\therefore$ No. of arbitrary constant = 3 = order of Diff. Equation.

2. (a) Degree is defined only for polynomial differential equation

$$\therefore \left(\frac{d^2y}{dx^2} + x^{1/4}\right) = -\left(\frac{dy}{dx}\right)^{1/3}$$

$$\text{or } \left(\frac{d^2y}{dx^2} + x^{1/4}\right)^3 = -\frac{dy}{dx} \Rightarrow \text{ order} = 2,$$

degree = 3

5. (c) We know that

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots$$

$\therefore$ Differential Equation will become

$$y = e^{\frac{dy}{dx}} \Rightarrow \log y = \frac{dy}{dx}; \therefore \text{degree} = 1$$

6. (d) Equation of parabola will be

$(y-k)^2 = 4a(x-h)$

(No. of arbitrary constant = 2, $\because$ a is given it is not an arbitrary constant)

differentiating $2(y-k)\frac{dy}{dx} = 4a$ or $y - k = \frac{2a}{\frac{dy}{dx}}$

differentiating again,

$$\frac{dy}{dx} = \frac{-2a}{\left(\frac{dy}{dx}\right)^2}\frac{d^2y}{dx^2} \Rightarrow \left(\frac{dy}{dx}\right)^3 + 2a\frac{d^2y}{dx^2} = 0$$

12. (d) $y^2 = 2c\left(x + \sqrt[3]{c}\right) \Rightarrow$ differentiating

$$2y\frac{dy}{dx} = 2c$$

$$y\frac{dy}{dx} = c, \text{ put in the equation of curve}$$

$$y^2 = 2y\frac{dy}{dx}\left(x + \left(y\frac{dy}{dx}\right)^{1/3}\right)$$

$$y^2 = 2xy\frac{dy}{dx} + 2\left(y\frac{dy}{dx}\right)^{4/3}$$

$$\text{or } \left(y^2 - 2x\frac{dy}{dx}\right)^3 = 8\left(y\frac{dy}{dx}\right)^4$$

$\therefore$ Order = 1, degree = 4

13. (c) $y = xe^{cx}$, differentiating

$$\frac{dy}{dx} = e^{cx} + cxe^{cx}$$

$$\text{but } c = \frac{1}{x}\log\frac{y}{x} \Rightarrow \frac{dy}{dx} = e^{cx}(1 + cx)$$

$$\frac{dy}{dx} = \frac{y}{x}\left(1 + \log\frac{y}{x}\right) \quad \left(\because e^{cx} = \frac{y}{x}\right)$$

16. (a) Equation of tangent at $(2t, t^2)$ to parabola $x^2 = 4y$ is

$tx = y + t^2$, differentiating we get $\frac{dy}{dx} = t$

$\therefore$ Differential equation is

$$\left(\frac{dy}{dx}\right)x = y + a\left(\frac{dy}{dx}\right)^2,$$

order = 1, degree = 2

LECTURE



2

Solution of a Differential Equation

BASIC CONCEPTS

SOLUTION OF A DIFFERENTIAL EQUATION

A solution or integral or primitive of a differential equation is a relation free of derivatives between the variables involved which satisfies the given differential equation.

1. General solution or complete primitive or complete integral of a differential equation

The solution of a differential equation which contains as many arbitary constants as equal to the order of the differential equation is said to be general solution. The number of arbitary constant in the solution of a differential equation depends upon the order of a differential equation.

NOTE

It is important to note that the constant of integration is introduced immediately after the integration is carried

2. Particular solution

The particular solution of a differential equation is that which is obtained from the general solution by giving particular values to arbitary constants.

3. Singular solution

A solution which satisfy the given differential equation but not obtainable from the general solution is called the singular solution. It does not contain any arbitary constant.

NOTE

If $\frac{dy}{dx}$ occurs only in the first degree in the differential equation there will be no singular solution.

4. Various standard and special methods

Applied to different type of the differential equation of the first order and first degree are as follows:

(i) Method of seperation of variables: $f(x)\,dx = g(y)\,dy$

(ii) Reduciable to variable separable from: $\frac{dy}{dx} = f(ax + by + c)$

Method of substitution: put $ax + by + c = V$

(iii) Homogeneous differential equation

$$\frac{dy}{dx} = \frac{f_1(x,y)}{f_2(x,y)} = \frac{x^n f_1(y/x)}{x^n f_2(y/x)}$$

where, f_1, f_2 are the polynomial in x, y

put $\frac{y}{x} = v$ or $y = vx$ so that $\left\{\frac{dy}{dx} = v + x\frac{dv}{dx}\right\}$

with these two substitution homogenious differential equations reduce to variable separable form.

(iv) Non-homogeneous differential equation:

$$\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c}$$

Case-I: If $\frac{a}{a'}=\frac{b}{b'}=\frac{1}{k}$, then

$a'=ka, b'=kb$ and $\frac{dy}{dx}=\frac{ax+by+c}{k(ax+by)+c}$

$\left[\begin{array}{l}\text{put: } ax+by+c=v.....(i)\\ a+b\frac{dy}{dx}=\frac{dv}{dx}.....(ii)\end{array}\right]$ with these two substitutions this differential equation will reduce to variables separable form.

Case-II: $\frac{a}{a'}\neq\frac{b}{b'}$ $\left.\begin{array}{l}\text{put } x=X+h\\ y=Y+k\end{array}\right\}$ .....(i)

so that $\frac{dY}{dX}=\frac{dy}{dx}$ .....(ii)

with above two substitutions differential equation under consideration will reduce to

$\frac{dY}{dX}=\frac{aX+bY}{a'X+b'Y}$ and $\left.\begin{array}{l}ah+bk+c=0\\ a'h+b'k+c'=0\end{array}\right\}$

which is homogeneous differential equation.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. $\frac{dy}{dx}=\frac{e^x(\sin^2 x+\sin 2x)}{y(2\log y+1)}$ ***[CBSE-90]***

Solution

Given $\frac{dy}{dx}=\frac{e^x(\sin^2 x+\sin 2x)}{y(2\log y+1)}$

$y\,(2\log y+1)\,dy \quad e^x(\sin^2 x+\sin 2x)$

Integrating both the sides we get

$\int 2y\log dy+\int ydy=\int e^x(\sin^2 x+\sin 2x)dx+c$

$\int \log y.2(y^2/2)-\int\frac{1}{y}.\frac{2y^2}{2dy}+\frac{y^2}{2}=e^x\sin^2 x+c$

using $\int e^x[f(x)+f'(x)]dx=e^x f(x)+c$

$y^2\log y-\frac{y^2}{2}+\frac{y^2}{2}=e^x\sin^2 x+c$

$y^2\log y \quad e^x\sin^2 x+c$

2. $(x+y)^2\frac{dy}{dx}=a^2$

[CBSE-96C; PSB-98, 2001]

Solution

Given $(x \quad y)^2\frac{dy}{dx}=a^2$ ..... (i)

Putting $x+y=v$ so that $1+\frac{dy}{dx}=\frac{dv}{dx}$

$\Rightarrow \frac{dy}{dx}=\frac{dv}{dx}-1$

so the equation (i) becomes $v^2\left(\frac{dv}{dx}-1\right)=a^2$

$\Rightarrow \frac{dv}{dx}=\frac{a^2}{v^2}+1$

$\Rightarrow \frac{dv}{dx}=\frac{a^2+v^2}{v^2}\Rightarrow\frac{v^2}{v^2+a^2}\,dv \quad dx$

$\Rightarrow\left(1-\frac{a^2}{v^2+a^2}\right)dv=dx$ integrating both sides, we get $\int dv-\int\frac{a^2}{v^2+a^2}dv=\int dx+c$

$\Rightarrow v-a^2.\ \frac{1}{a}\tan^{-1}\left(\frac{v}{a}\right)=x+c$

$\Rightarrow x+y-a\tan^{-1}\left(\frac{x+y}{a}\right)=x+c$

3. $x^2ydx-(x^3+y^3)\,dy=0$

[CBSE-92, 2002; HSB-93; PSB-98]

Solution

$(x^3+y^3)\,dy \quad x^2y\,dx$

$\Rightarrow \frac{dy}{dx}=\frac{x^2y}{x^3+y^3}$ ..... (i)

since each of the functions x^2y and x^3+y^3 is a homogeneous function of degree 3, so the given differential equation is homogeneous.

Putting $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ in (i), we get

$$v + x\frac{dv}{dx} = \frac{vx^3}{x^3 + v^3x^3}$$

$$\Rightarrow v + x\frac{dv}{dx} = \frac{v}{1+v^3}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v}{1+v^3} - \frac{v}{1} \quad \Rightarrow x\frac{dv}{dx} = -\frac{v^4}{1+v^3}$$

$$\Rightarrow \frac{1+v^3}{v^4}dv = \frac{-dx}{x} \Rightarrow \left(\frac{1}{v^4} + \frac{1}{v}\right)dv = \frac{-dx}{x}$$

Integrating both sides

$$\Rightarrow \frac{v^{-3}}{-3} + \log|v| = -\log(x) + c$$

$$\Rightarrow \frac{-1}{3v^3} + \log|v| + \log(x) = c$$

$$\Rightarrow -\frac{1}{3}\frac{x^3}{y^3} + \log\left|\frac{y}{x}.x\right| = c$$

$$\Rightarrow \frac{-x^3}{3y^3} + \log|y| = c$$ which is the required solution.

4. $y\left\{x\cos\frac{y}{x} + y\sin\frac{y}{x}\right\}dx - x\left\{y\sin\frac{y}{x} - x\cos\frac{y}{x}\right\}dy = 0$

[NCERT-Book]

Solution

The given differential equation can be written as $\frac{dy}{dx} = \frac{y\left\{x\cos\frac{y}{x} + y\sin\frac{y}{x}\right\}}{x\left\{y\sin\frac{y}{x} - x\cos\frac{y}{x}\right\}}$ ..... (i)

Putting $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ in (i), we get

$$v + x\frac{dv}{dx} = \frac{vx(x\cos v + vx\sin v)}{x(vx\sin v - x\cos v)}$$

$$\Rightarrow v + x\frac{dv}{dx} = \frac{v(\cos v + v\sin v)}{(v\sin v - \cos v)}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v\cos v + v^2\sin v - v^2\sin v + v\cos v}{v\sin v - \cos v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{2v\cos v}{v\sin v - \cos v}$$

$$\Rightarrow \frac{v\sin v - \cos v}{v\cos v}dv = 2\frac{dx}{x}$$

$$\Rightarrow -\left(\frac{\cos v - v\sin v}{v\cos v}\right)dv = 2\frac{dx}{x}$$

Integrating both sides, we get

$$\Rightarrow -\int\frac{\cos v - v\sin v}{v\cos v}dv = 2\int\frac{dx}{x}$$

$$\Rightarrow -\log|v\cos v| = 2\log|x| + \log c$$

$$\Rightarrow \log\frac{1}{|v\cos v|} = \log|x^2| + \log c$$

$$\Rightarrow \frac{1}{v\cos v} = cx^2 \Rightarrow \frac{\sec v}{v} = cx^2$$

$$\Rightarrow \sec\left(\frac{y}{x}\right) = cx^2\frac{y}{x}$$

$$\Rightarrow \sec\left(\frac{y}{x}\right) = cyx$$

which is the required solution.

5. $\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}$ ***[CBSE-2000 S]***

Solution

Given $\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}$ ..... (i)

Put $x = X + h$ and $y = Y + k$ in equation (i) we get $\frac{dY}{dX} = \frac{Y - X + (k - h + 1)}{Y + X + (k + h + 5)}$ ..... (ii)

Choosing h and k such that $\left.\begin{matrix} k - h + 1 = 0 \\ k + h + 5 = 0 \end{matrix}\right\}$ ..... (iii)

Solving equation (iii), we get $h = -2$, $k = -3$ ..... (iv)

Therefore equation (ii) will take the form $\frac{dY}{dX} = \frac{Y - X}{Y + X}$ .... (v)

Put $Y = vX$ then $\frac{dY}{dX} = v + X\frac{dv}{dX}$

$\therefore$ equation (v) will become

$$v + X\frac{dv}{dX} = \frac{vX - X}{vX + X} = \frac{v - 1}{v + 1}$$

$$\Rightarrow X\frac{dv}{dX} = \frac{v - 1 - v^2 - v}{v + 1} \Rightarrow \frac{1 + v}{1 + v^2} = \frac{-dX}{X}$$

(variables separable)

Integrating, we get $\int \frac{1}{1+v^2}dv + \int \frac{v}{1+v^2}dv \quad \int$

$= -\int \frac{dX}{X}$

$\tan^{-1} v + \frac{1}{2}\log(1+v^2) = -\log X + C$

$\Rightarrow 2 \tan^{1} v + \log (1 + v^2) + 2 \log X = 2C$

$\Rightarrow 2 \tan^{-1} v + \log [X^2 (1 + v^2)] = 2C$

$\Rightarrow \frac{1}{2} \log [X^2 (1 + v^2)] + \tan^{1} v = C$

$\Rightarrow \frac{1}{2} \log [X^2 + Y^2] + \tan^{-1}\left(\frac{Y}{X}\right) = C$

$\Rightarrow \frac{1}{2}\log[(x+2)^2 + (y+3)^2] + \tan^{-1}$

$\left(\frac{y+3}{x+2}\right) = C$

6. $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$

[CBSE-90, 2002; PSB-91; HSB-95]

Solution

Given equation is $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$

$\Rightarrow \frac{dy}{dx}(a+x) = y(1-ay)$

$\Rightarrow \frac{dy}{y(1-ay)} = \frac{dx}{a+x}$ (variable separable)

$\Rightarrow \left(\frac{1}{y} + \frac{a}{1-ay}\right)dy = \frac{dx}{a+x}$

Integrating both the sides, we get

$\log_e y - \log_e (1 - ay) = \log_e (a + x) + \log_e c$

$\Rightarrow \log_e y - \log_e (1 - ay) = \log_e [(a + x) c]$

$\Rightarrow \log_e\left(\frac{y}{1-ay}\right) = \log_e [(a+x)c]$

$\Rightarrow \frac{y}{1-ay} \; c\,(a+x)$

$\Rightarrow y \quad c\,(a+x)\,(1-ay)$ which is the required solution.

7. $\frac{dy}{dx} = \frac{x^2 + 5xy + 4y^2}{x^2}$

[MP-96, 2000, 2001, 2008]

Solution

Given equation is $\frac{dy}{dx} = \frac{x^2 + 5xy + 4y^2}{x^2}$

(homogeneous) ....(1)

Put $y \quad vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Hence equation (1) becomes

$v + x\frac{dv}{dx} = \frac{x^2 + 5vy + 4y^2}{x^2} = 1 + 5v + 4v^2$

$\Rightarrow x\,\frac{dv}{dx} = 1 + 4v + 4v^2 = (2v + 1)^2$

$\Rightarrow \frac{dv}{(2v+1)^2} = \frac{dx}{x}$

Integrating, we get $-\frac{1}{2(2v+1)} = \log_e x + c$

$\Rightarrow \frac{-1}{2\left(2\frac{y}{x}+1\right)} = \log_e x + c$

$\Rightarrow \frac{-x}{2(x+2y)} = \log_e x + c$ is the required solution.

8. Solve the following differential equation $y(1 + x^2)\frac{dy}{dx} = x(1 + y^2)$ ***[CBSE-1996, 2007]***

Solution

We have $y\,(1 + x^2)\frac{dy}{dx} = x\,(1 + y^2)$

$\Rightarrow x\,(1 + y^2)dx = y\,(1 + x^2)\,dy$

$\Rightarrow \frac{ydy}{1+y^2} = \frac{xdx}{1+x^2}$

$\Rightarrow \int \frac{y}{1+y^2}dy = \int \frac{x}{1+x^2}dx$

$\Rightarrow \frac{1}{2}\int \frac{2y}{1+y^2}dy = \frac{1}{2}\int \frac{2x}{1+x^2}dx$

$\Rightarrow \frac{1}{2}\log|1+y^2| = \frac{1}{2}\log|1+x^2| + \frac{1}{2}\log|c|$

$\Rightarrow \log|1 + y^2| = \log|1 + x^2| + \log|c|$

$\Rightarrow 1 + y^2 = c\,(1 + x^2)$

9. Solve the following differential equation

$(x^3 + x^2 + x + 1)\dfrac{dy}{dx} = 2x^2 + x.$

[CBSE-2010]

Solution

Step 1: Separating the variables above differential equation can be written as

$$\frac{2x^2 + x}{(1+x)(1+x^2)}dx = dy$$

Step 2: On Integrating by partial fractions we find

$$\int dy = \int \frac{A}{1+x}dx + \int \frac{Bx + C}{1 + x^2}dx + \lambda$$

$$\int dy = \frac{1}{2}\int \frac{dx}{1+x} + \int \frac{\frac{3}{2}x - \frac{1}{2}}{1 + x^2}dx + \lambda$$

$$y = \frac{1}{2}\log(1 + x) - \frac{3}{2} \times \frac{1}{2}\log(1 + x^2) - \frac{1}{2}\tan^{-1}x + \lambda$$

10. Show that the differential equation $(x - y)\dfrac{dy}{dx} = x + 2y$, is homogeneous and solve it.

[CBSE-2010]

Solution

Step 1: $$\frac{dy}{dx} = \frac{x + 2y}{x - y} = \frac{1 + \frac{2y}{x}}{1 - \frac{y}{x}}$$

$$\frac{dy}{dx} = xf\left(\frac{y}{x}\right)$$

which is homogeneous differential equation.

Step 2: Put $y = Vx$ so that $\dfrac{dy}{dx} = V + x\dfrac{dv}{dx}$ and given differential equation becomes:

$$V + x\frac{dv}{dx} = \frac{1 + 2v}{1 - v}$$

$$x\frac{dv}{dx} = \frac{1 + 2v}{1 - v} - v = \frac{1 + 2v - v + v^2}{1 - v}$$

or separating the variables we find:

$$\frac{1 - v}{v^2 + v + 1}dv = \frac{dx}{x}$$

or $$\frac{2v + 1 - 3}{v^2 + v + 1}dv = \frac{-dx}{x}$$

or $$\int \frac{2v + 1}{v^2 + v + 1}dv - \int \frac{3dv}{\left(v + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = -\int \frac{dx}{x} + C$$

or $$\log(v^2 + v + 1) - 3 \times \frac{1}{\sqrt{3}}\tan^{-1}\frac{v + \frac{1}{2}}{\frac{\sqrt{3}}{2}} = -\log x + C$$

or $$\log\left(\frac{y^2}{x^2} + \frac{y}{x} + 1\right) - 2\sqrt{3}\tan^{-1}\frac{2y + x}{x\sqrt{3}} = -\log x + C$$

or $$\log(y^2 + x^2 + xy) - 2\sqrt{3}\tan^{-1}\left(\frac{2y + x}{x\sqrt{3}}\right) = c$$

11. Solve the following differential equation

$\sqrt{1 + x^2 + y^2 + x^2y^2} + xy\dfrac{dy}{dx} = 0.$

[CBSE-2010]

Solution

$$\sqrt{(1 + x^2)(1 - y^2)} + xy\frac{dy}{dx} = 0$$

$$\int \frac{\sqrt{1 + x^2}}{x}dx + \int \frac{y}{\sqrt{1 + y^2}}dy = 0$$

$$\Rightarrow \int \frac{1 + x^2}{x\sqrt{1 + x^2}}dx + \int \frac{y}{\sqrt{1 + y^2}}dy = 0$$

$$\int \frac{1}{x\sqrt{1 + x^2}}dx + \int \frac{x}{\sqrt{1 + x^2}}dx + \int \frac{y}{\sqrt{1 + y^2}}dy = C$$

Consider $$\int \frac{1}{x^2} \cdot \frac{1}{\sqrt{1 + \frac{1}{x^2}}}dx;$$

$$t = \frac{1}{x};\ dt = -\frac{1}{x^2}dx$$

$$\int \frac{-dt}{\sqrt{1+t^2}} = -\log(t+\sqrt{t^2+1})$$

$$= -\log\left(\frac{1+\sqrt{x^2+1}}{x}\right)$$

$\therefore$ Differential equation becomes

$$-\log\left(\frac{1+\sqrt{x^2+1}}{x}\right) + \sqrt{1+x^2} + \sqrt{1+y^2} = C$$

12. Show that the following differential equation is homogeneous, and then solve it.

$$ydx + x\log\left(\frac{y}{x}\right)dy - 2xdy = 0 \quad \textit{[CBSE-2010]}$$

Solution

$$ydx + x\log\left(\frac{y}{x}\right)dy - 2xdy = 0$$

$$\Rightarrow ydx = dy\left(2x - x\log\left(\frac{y}{x}\right)\right)$$

$$\frac{dy}{dx} = \frac{y}{2x - x\log(y/x)}$$

$$\frac{dy}{dx} = \frac{y/x}{2-\log(y/x)} = f(y/x)$$

$\therefore$ Equation is homogeneous, put

$$y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = \frac{v}{2-\log v} \Rightarrow x\frac{dv}{dx} = \frac{-v + v\log v}{2-\log v}$$

$$\int \frac{2-\log v}{v(-1+\log v)}dv = \frac{dv}{x}$$

$$\Rightarrow \int \frac{-1}{v}dv + \int \frac{1}{v(\log v - 1)}dv - \frac{dx}{x}$$

$$-\log v + \log(\log v - 1) = lnx + lnc$$

$$\log\left(\log\frac{y}{x} - 1\right)$$

$$= ln\left(\frac{y}{x}\right) + lnx + lnc\left(\because v = \frac{y}{x}\right)$$

$$\left(\log\left(\frac{y}{x}\right) - 1\right) = cy$$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

Directions: Questions 1 to 12, solve these differential equations.

1. $\frac{dy}{dx} = 4y$ ***[MP-93, 96, 98, 99, 2002, 2004]***

2. $(e^x + e^{-x})\frac{dy}{dx} = e^x - e^{-x}$

[MP-2001; HSB-93; HPSB-91; PSB-90]

3. $(x^2 - yx^2)\,dy + (y^2 + xy^2)\,dx = 0$

[MP-93S, 97; CBSE-89, 93]

4. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$.

[PSB-99; CBSE-92C; HSB-91, 2003; MP-99, 2005]

5. $3e^x \tan y\,dx + (1 - e^x)\sec^2 y\,dy = 0$

[HPB-2000; PSB-96; CBSE-91, 93, 99; Kerala-95]

6. $x\cos y\,dy$ $(xe^x \log x + e^x)\,dx$

[PSB-91 C]

7. $\frac{dy}{dx} = \frac{3e^{2x} + 3e^{4x}}{e^x + e^{-x}}$ ***[MP-2005]***

8. $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

[MP-95, 98, 99, 2002, 2004; CBSE-92C]

9. $\cos(x \quad y)\,dy = dx$ ***[MP-2001]***

10. $\tan y\,dx + \sec^2 y \tan x\,dy \quad 0$ ***[CBSE-2007]***

11. $\frac{dy}{dx}=\frac{x+y+1}{2x+2y+3}$ ***[MP-2001]***

12. $(x+y+1)\ \frac{dy}{dx}=1$ ***[MP-2000, 2002, 2004, 2007]***

13. Show that $y + x + 1 = 0$ is a solution of the differential equation $(y - x)\ dy - (y^2 - x^2)\ dx = 0$. ***[HPSB-95; NCERT-Book]***

ANSWERS

1. $y \quad ce^{4x}$

2. $y \quad \log(e^x + e^{-x}) + c$

3. $\log_e\left(\frac{x}{y}\right)=\frac{1}{x}+\frac{1}{y}+c$

4. $e^y=e^x+\frac{x^3}{3}+c$

5. $\tan y \quad c\,(1-e^x)^3$

6. $\sin y \quad e^x \log x + c$

7. $y = e^{3x} + c$

8. $y - x = c\,(1 + xy)$

9. $\tan\left(\frac{x+y}{2}\right)=y+c$

10. $\tan y \sin x = c$

11. $6y - 3x + \log_e(3x + 3y + 4) = 9c$

12. $x + y + 2 = ce^x$.

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. Solution of the differential equation $\frac{dy}{dx}=\frac{y}{x}+\tan\left(\frac{y}{x}\right)$ is

(a) $x = c \sin(y/x)$
(b) $y = c \sin(y/x)$
(c) $y = c \cos(y/x)$
(d) $x = c \cos(y/x)$

[Roorkee-82; CET (Pb.)-2000; ICS-2001]

Solution

(a) Given equation is homogeneous,
So let $y = vx$ and transformed equation will be

$$\cos v\, dv = \frac{dx}{x}$$

On integration $\sin v = x\,c \Rightarrow x = c \sin(y/x)$

2. Solution of the differential equation $\frac{dy}{dx}=\frac{x\log x^2+x}{\sin y+y\cos y}$ is ***[EAMCET-2003]***

(a) $y \sin y = x^2 \log x + c$
(b) $y \sin y = x^2 + c$
(c) $y \sin y = x^2 + \log x + c$
(d) $y \sin y = x \log x + c$

Solution

(a) $\frac{dy}{dx}=\frac{x\log x^2+x}{\sin y+y\cos y}$

$\Rightarrow \int(\sin y + y\cos y)dy = \int(x\log x^2 + x)dx$

$\Rightarrow -\cos y + y \sin y + \cos y$

$= \frac{x^2}{2}\log x^2 - \int x dx + \int x dx + c$

$\Rightarrow y \sin y = x^2 \log x + c$

3. The general solution of $y^2\,dx + (x^2 - xy + y^2)\,dy = 0$ is ***[EAMCET-2003]***

(a) $\tan^{-1}\left(\frac{x}{y}\right)+\log y+c=0$

(b) $2\tan^{-1}\left(\frac{x}{y}\right)+\log x+c=0$

(c) $\log\left(y+\sqrt{x^2+y^2}\right)+\log y+c=0$

(d) $\sin h^{-1}\left(\frac{x}{y}\right)+\log y+c=0$

Solution

(a) $\frac{dx}{dy}+\frac{x^2-xy+y^2}{y^2}=0$ Putting $x = vy$

$\Rightarrow \frac{dx}{dy}=v+y\frac{dv}{dy}$

So the transformed equation will be $v+y\frac{dv}{dy}$

$+v^2-v+1=0$

$\Rightarrow \quad \frac{dv}{v^2+1}+\frac{dy}{y}=0$

$\Rightarrow \quad \tan^{-1}v+\log y+c=0$

[on integration]

$\Rightarrow \quad \tan^{-1}\left(\frac{x}{y}\right)+\log y+c=0$

4. If $y = y(x)$ satisfies $\left(\frac{2+\sin x}{1+y}\right)\frac{dy}{dx}=\cos x$ such that $y(0) = 1$, then $y(\pi/2)$ is equal to ***[IIT (Screening)-2004]***

(a) 2 (b) 3/2

(c) 2/3 (d) 1

Solution

(a) From given equation $\frac{dy}{1+y}=\frac{\cos x}{2+\sin x}dx$

$\Rightarrow \quad \log(1+y)=\log(2+\sin x)+\log c$ (on integration)

$\Rightarrow \quad 1+y=c(2+\sin x)$

But when $x = 0, y = 1$, so $2 = c(2 + 0)$

$\Rightarrow \quad c=1$

$\therefore \quad y=1+\sin x \Rightarrow y(\pi/2)=1+1=2$

5. Solution of the differential equation $(2x - y + 2)\,dx + (4x - 2y - 1)\,dy = 0$ is ***[JEE (WB)-2005]***

(a) $2x-y=c\,e^{(x-2y)}$

(b) $2x+y=c\,e^{(2x-y)}$

(c) $x-2y=c\,e^{-(x-2y)}$

(d) None of these

Solution

(a) Given equation is

$\frac{dy}{dx}=-\left(\frac{(2x-y)+2}{2(2x-y)-1}\right)$

$\Rightarrow \quad \frac{dv}{dx}-2=\frac{v+2}{2v-1}$ where $2x - y = v$

$\Rightarrow \quad \frac{dv}{dx}=\frac{5v}{2v-1} \Rightarrow \frac{2v-1}{5v}dv=dx$

$\Rightarrow \quad \frac{2}{5}v-\frac{1}{5}\log v=x+\log c'$

(on integrating)

$\Rightarrow \quad 2(2x-y)-\log(2x-y)=5x+5\log c'$

$\Rightarrow \quad -(x+2y)=\log(2x-y)-\log c$

$\Rightarrow \quad 2x-y=c\,e^{-(x+2y)}$

6. The solution of $\left[x\operatorname{cosec}\left(\frac{y}{x}\right)-y\right]dx+x\,dy=0$ is

(a) $\log|x|-\cos\left(\frac{x}{y}\right)=c$

(b) $\log|x|-\cos\left(\frac{y}{x}\right)=c$

(c) $\log|x|-\sin\left(\frac{x}{y}\right)=c$

(d) $\log|x|-\sin\left(\frac{y}{x}\right)=c$ ***[VIT-2006]***

Solution

(b) $\frac{dy}{dx}+\operatorname{cosec}\left(\frac{y}{x}\right)-\frac{y}{x}$ = [homogenous equation]

Put $y = vx$, then transformed equation will be

$\frac{dv}{dx}+\frac{\operatorname{cosec}v}{x}=0$

$\Rightarrow \quad \sin v\,dv+\frac{1}{x}\,dx=0$

$\Rightarrow \quad -\cos v+\log|x|=c$

$\Rightarrow \quad \log|x|-\cos\left(\frac{y}{x}\right)=c$

7. The solution of $e^{dy/dx}=(x+1), y(0)=3$ is ***[Kerala (Engg.)-2005]***

(a) $y=x\log x-x+2$

(b) $y=(x+1)\log|x+1|-x+3$

(c) $y=(x+1)\log|x+1|+x+3$

(d) $y=x\log x+x+3$

Solution

(b) $\frac{dy}{dx} = \log(x+1) \Rightarrow dy = \log(x+1)\,dx$

$y = \int \log(x+1)dx = x.\log(x+1) - \int \frac{x}{x+1}dx$

$= x \,.\, \log(x+1) - \int\left(1 - \frac{1}{x+1}\right)dx$

$= x \,.\, \log(x+1) - x + \log(x+1) + c =$
$(x+1)\log(x+1) - x + c$

at $x = 0$, $y = 3$

$3 = (1)\log(1) - 0 + c \Rightarrow 3 = 0 + c \Rightarrow c = 3$

$\therefore \quad y = (x+1)\log|x+1| - x + 3.$

8. The general solution of the differential equation $(2x - y + 1)dx + (2y - x + 1)dy = 0$ is

[Karnataka CET-2005]

(a) $x^2 + y^2 + xy - x + y = c$

(b) $x^2 + y^2 - xy + x + y = c$

(c) $x^2 - y^2 + 2xy - x + y = c$

(d) $x^2 - y^2 - 2xy + x - y = c$

Solution

(b) $(2x - y + 1)\,dx + (2y - x + 1)dy = 0$

$\frac{dy}{dx} = \frac{2x - y + 1}{x - 2y - 1}$, put $x = X + h$, $y = Y + k$

$\frac{dY}{dX} = \frac{2X - Y + 2h - k + 1}{X - 2Y + h - 2k - 1}$

$2h - k + 1 = 0 \Rightarrow h - 2k - 1 = 0$

On solving $h = -1$, $k = -1$;

$\therefore \quad \frac{dY}{dX} = \frac{2X - Y}{X - 2Y}$

Put $Y = \upsilon X$; $\therefore \quad \frac{dY}{dX} = \upsilon + X\frac{dv}{dX}$

$\upsilon + X\frac{d\upsilon}{dX} = \frac{2X - \upsilon X}{X - 2\upsilon X} = \frac{2 - \upsilon}{1 - 2\upsilon}$

$X\frac{d\upsilon}{dX} = \frac{2 - 2\upsilon + 2\upsilon^2}{1 - 2\upsilon} = \frac{2(\upsilon^2 - \upsilon + 1)}{1 - 2\upsilon}$

$\therefore \quad \frac{dX}{X} = \frac{(1 - 2\upsilon)}{2(\upsilon^2 - \upsilon + 1)}d\upsilon$

Put $\upsilon^2 - \upsilon + 1 = t \Rightarrow (2\upsilon - 1)\,d\upsilon = dt$

$\therefore \quad \frac{dX}{X} = -\frac{dt}{2t} \quad \therefore \log X = \log t^{-1/2} + \log c$

$\therefore \quad X = t^{-1/2}c \Rightarrow X = (\upsilon^2 - \upsilon + 1)^{-1/2}.\, c$

$X^2(\upsilon^2 - \upsilon + 1) = \text{constant}$

$(x+1)^2\left(\frac{(y+1)^2}{(x+1)^2} - \frac{(y+1)}{x+1} + 1\right) = \text{constant}$

$(y+1)^2 - (y+1)(x+1) + (x+1)^2 = c$

$y^2 + x^2 - xy + x + y = c.$

9. The solution of the differential equation $x\frac{dy}{dx} = y(\log y - \log x + 1)$ is

[IIT-1986; AIEEE-2005]

(a) $y = xe^{cx}$ (b) $y + xe^{cx} = 0$

(c) $y + e^x = 0$ (d) None of these

Solution

(a) Here $\frac{dy}{dx} = \frac{y}{x}\left(\log\frac{y}{x} + 1\right)$ ..... (i)

It is homogeneous equation

so now put $y = \upsilon x$ and $\frac{dy}{dx} = \upsilon + x\frac{d\upsilon}{dx}$, then the equation (i)

reduces to $\frac{d\upsilon}{\upsilon\log\upsilon} = \frac{dx}{x}$

On integrating, we get log (log υ) = log x + log c

$\Rightarrow \log\left(\frac{y}{x}\right) = cx \Rightarrow y = xe^{cx}$

10. Solve the following differential equation: $(x^2 + 4y^2 + 4xy)\,dy = (2x + 4y + 1)\,dx$

[Roorkee-1999]

Solution

The given equation can be written as

$(x + 2y)^2\frac{dy}{dx} - 2(x + 2y) = 1$

Put $x + 2y = v$ $\quad \therefore \; 1 + 2\frac{dy}{dx} = \frac{dv}{dx}$

$\therefore \quad \frac{v^2}{2}\left[\frac{dv}{dx} - 1\right] = 1 + 2v$

$\Rightarrow \quad \frac{dv}{dx} = \frac{2 + 4v}{v^2} + 1 = \frac{v^2 + 4v + 2}{v^2}$

$\therefore$ We can use variable separable method

$\frac{v^2}{v^2 + 4v + 2}dv = dx$

$v^2 = l(v^2 + 4v + 2) + m(2v + 4) + n$

Comparing coefficients $l = 1$, $m = -2$, $n = 6$

$$\left[1 - 2\frac{(2v+4)}{v^2+4v+2} - 6.\frac{1}{(v+2)^2-(\sqrt{2})^2}\right]dv = dx$$

Integrating

$$v - 2\log(v^2+4v+2) - 6.\frac{1}{2\sqrt{2}}\log\left[\frac{v+2-\sqrt{2}}{v+2+\sqrt{2}}\right] = x + c$$

where $v = x + 2y$.

11. The solution of $(x + y + 1)\dfrac{dy}{dx} = 1$ ***[EAMCET-2007]***

(a) $y = (x + 2) + ce^x$
(b) $y = -(x + 2) + ce^x$
(c) $x = -(y + 1) + ce^y$
(d) $x = (y + 2)^2 + ce^y$

Solution

(c) Given differential equation is

$\dfrac{dx}{dy} + (-1)x = y + 1$ is linear

I.F. is e^{-y}.

Solution is $xe^{-y} = \int (y+1)e^{-y}dy$

or $x = -(y + 1) + ce^y$.

12. The solution of $\dfrac{dy}{dx} = \dfrac{y^2}{xy - x^2}$ is ***[EAMCET-2007]***

(a) $e^{y/x} = kx$ (b) $e^{y/x} = ky$
(c) $e^{-y/x} = kx$ (d) $e^{-y/x} = ky$

Solution

(b) Given differential equation is homogeneous

Putting $y = vx$ $\dfrac{dv}{dx} = v + x\dfrac{dv}{dx}$

$\therefore$ $\dfrac{v-1}{v}dv = \dfrac{dx}{x}$

or $v = \log(vx \,.\, k)$

$\therefore$ $e^{y/x} = ky$, k being a constant.

13. The differential equation $\dfrac{dy}{dx} = \dfrac{\sqrt{1-y^2}}{y}$ determines a family of circles with ***[IIT-JEE-2007]***

(a) variable radii and a fixed centre at (0, 1)
(b) variable radii and fixed centre at (0, −1)
(c) fixed radius 1 and variable centres along the x-axis.
(d) fixed radius 1 and variable centres along the y-axis

Solution

(c) $\dfrac{dy}{dx} = \dfrac{\sqrt{1-y^2}}{y}$

$$\int \frac{y}{\sqrt{1-y^2}}dy = \int dx$$

or $-\dfrac{1}{2}.2\sqrt{1-y^2} = x + c$

$\Rightarrow$ $-\sqrt{1-y^2} = x + c$ or $1 - y^2 = (x + c)^2$

or $(x + c)^2 + y^2 = 1$

Therefore, the differential equation represents a circle of fixed radius 1 and variable centres along x-axis.

14. Solution of the differential equation $\dfrac{dy}{dx} = \cos(x + y)$ is ***[MPPET-2010]***

(a) $\tan\dfrac{(x+y)}{2} = x + c$

(b) $\tan\dfrac{(x+y)}{2} = -x + c$

(c) $y = -x + c$

(d) None of these

Solution

(a) Step 1: Method of substitution:

Put $x + y = z$ so that $1 + \dfrac{dy}{dx} = \dfrac{dz}{dx}$

Step 2: $\dfrac{dz}{dx} - 1 = \cos z$

$$\frac{dz}{dx} = 1 + \cos z$$

$$\frac{dz}{1+\cos z} = dx \text{ (variables separable)}$$

$$\sec^2\frac{z}{2}dz = 2dx$$

On integrating we get

$$2\tan\frac{z}{2} = 2x + c'$$

$$\tan\left(\frac{x+y}{2}\right) = x + c$$

15. If $\frac{dy}{dx} = \sec y$, then ***[MPPET-2010]***

(a) $x = \sin y + c$ (b) $x = \cos y + c$

(c) $x = -\sin y + c$ (d) None of these

Solution

(a) Step 1: Variable separable

Step 2: $\cos y\, dy = dx$

On Integrating we get, $\sin y = x + c$

or $x = \sin y + c$

16. Determine the equation of the curve passing through the origin, in the form $y = f(x)$, which satisfies the differential equation $\frac{dy}{dx} = \sin(10x + 6y)$ ***[IIT-1996]***

Solution

Put $10x + 6y = v$ $\quad \therefore 10 + 6\frac{dy}{dx} = \frac{dv}{dx}$

$$\therefore \quad \frac{dv}{dx} - 10 = 6\sin v$$

$$\therefore \quad \frac{dv}{6\sin v + 10} = dx$$

$$\text{or} \quad \frac{dv}{12\sin\frac{v}{2}\cos\frac{v}{2} + 10} = dx$$

divide above and below by $\cos^2(v/2)$ and put $\tan(v/2) = t$

$$\therefore \quad \frac{2dt}{12t + 10(1+t^2)} = dx$$

$$\text{or} \quad \frac{dt}{5t^2 + 6t + 5} = dx$$

$$\text{or} \quad \frac{dt}{\left(t + \frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 5dx$$

$$\frac{5}{4}\tan^{-1}\frac{5t+3}{4} = 5x + 5k$$

$$\text{or} \quad \tan^{-1}\frac{5t+3}{4} = 4x + c$$

At origin $x = 0, y = 0$ $\quad \therefore v = 0$

$$\therefore \quad t = \tan\frac{v}{2} = 0$$

Hence from above $\tan^{-1}\frac{3}{4} = c$

$$\therefore \quad \tan^{-1}\frac{5t+3}{4} - \tan^{-1}\left(\frac{3}{4}\right) = 4x$$

$$\text{or} \quad \frac{\frac{5t+3}{4} - \frac{3}{4}}{1 + \frac{5t+3}{4}\cdot\frac{3}{4}} = \tan 4x$$

$$\text{or} \quad \frac{20t}{25 + 15t} = \tan 4x$$

$$\text{or} \quad 4t = (5 + 3t)\tan 4x$$

$$\text{or} \quad t(4 - 3\tan 4x) = 5\tan 4x$$

$$\text{or} \quad \tan\frac{v}{2} = \frac{5\tan 4x}{4 - 3\tan 4x}$$

$$\text{or} \quad \tan(5x + 3y) = \frac{5\tan 4x}{4 - 3\tan 4x}$$

17. What is the solution of the differential equation $\frac{dy}{dx} = e^{x-y}(e^{y-x} - e^y)$? ***[NDA-2009]***

(a) $y = x - e^x + c$

(b) $y = x + e^x + c$

(c) $y = e^{x-y} - e^y + c$

(d) None of the above

Solution

(a) $\because \frac{dy}{dx} = e^{x-y}(e^{y-x} - e^y)$

$= e^{-y} \cdot e^x (e^x \cdot e^{-x} - e^x)$

$$\Rightarrow \quad \int 1dy = \int (1 - e^x)dx \Rightarrow y = x - e^x + c$$

18. If $\frac{dy}{dx} = 1 + x + y + xy$ and $y(-1) = 0$, then what is $y(x)$ equal to? ***[NDA-2009]***

(a) $e^{\frac{(1+x)^2}{2}} - 1$ (b) $e^{\frac{(1-x)^2}{2}}$

(c) $\ln(1 + x) - 1$ (d) $\ln(1 - x)$

Solution

(a) $\frac{dy}{dx} = 1 + x + y + xy$

$$\Rightarrow \quad \frac{dy}{dx} = (1 + x)(1 + y)$$

$$\Rightarrow \quad \int \frac{1}{1+y}dy = \int (1 + x)dx$$

$$\Rightarrow \quad \log(1 + y) = x + \frac{x^2}{2} + c$$

At $x = -1, y = 0$

$\Rightarrow \quad \log 1 = -1 + \frac{1}{2} + c \qquad \Rightarrow \quad c = \frac{1}{2}$

$\therefore \quad \log(1+y) = x + \frac{x^2}{2} + \frac{1}{2} = \frac{(1+x)^2}{2}$

$\Rightarrow \quad y = e^{\frac{(1+x)^2}{2}} - 1$

19. The solution of differential equation $\frac{dy}{dx} = e^{x-y}$ is ***[MPPET-2009]***

(a) $e^x + e^y = c$ (b) $e^x - e^y = c$
(c) $e^x + e^{-y} = c$ (d) $e^x - e^{-y} = c$

Solution

(c) Given, $\frac{dy}{dx} = e^{x+y}$

$\Rightarrow \quad \int e^{-y} dy = \int e^x dx$

$\Rightarrow \quad -e^{-y} = e^x - c$

$\Rightarrow \quad e^x + e^{-y} = c$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. Check the solution of the differential equation $\sqrt{(a+x)}dy + xdx = 0$ at $x = 2a$. ***[MP PET-2007]***

(a) It is imaginary (b) It is constant
(c) It does not exist (d) None of these

2. The diff. equation $\cot y \, dx = x \, dy$ has a solution of the form ***[Orissa JEE-2002]***

(a) $y = \cos x$ (b) $x = \operatorname{cosec} y$
(c) $x = \sin y$ (d) $y = \sin x$

3. The solution of the equation $(2y - 1)\,dx - (2x + 3)\,dy = 0$ is ***[Kerala (Engg.)-2002]***

(a) $\frac{2x-1}{2y+3} = c$ (b) $\frac{2y+1}{2x-3} = c$
(c) $\frac{2x+3}{2y-1} = c$ (d) $\frac{2x-1}{2y-1} = c$

4. The differential equation of all straight lines passing through the origin is ***[DCE-2002; Kerala (Engg.)-2002; UPSEAT-2004]***

(a) $y = \sqrt{x\frac{dy}{dx}}$ (b) $\frac{dy}{dx} = y + x$
(c) $\frac{dy}{dx} = \frac{y}{x}$ (d) None of these

5. If $y' = \frac{x-y}{x+y}$, then its solution is ***[MP PET-2000]***

(a) $y^2 + 2xy - x^2 = C$
(b) $y^2 + 2xy + x^2 = C$
(c) $y^2 - 2xy - x^2 = C$
(d) $y^2 - 2xy + x^2 = C$

6. The solution curve of the differential equation $x + y\,(dy/dx) = 2y$ makes intercept on y-axis of length ***[MP PET-2007]***

(a) 0 (b) C
(c) ∞ (d) None of these

7. The differential equation $dy/dx = (x + 2y - 3)/(2x + y - 3)$ has a solution ***[MP PET-2007]***

(a) $x + y - 2 = c\,(x - y)^3$
(b) $x + y - 2 + c\,(x - y)^2 = 0$
(c) $x + y \quad c\,(x - y - 2)^3$
(d) None of these

8. The solution of the diff. equation $x^2\frac{dy}{dx} = x^2 + xy + y^2$ is

(a) $\tan^{-1}(y/x) = \log x + c$
(b) $\tan^{-1}(y/x) = -\log x + c$
(c) $\sin^{-1}(y/x) = \log x + c$
(d) $\tan^{-1}(x/y) = \log x + c$

9. If $y \quad c\,e^{\sin^{-1}x}$, then corresponding to this the differential equation is

(a) $\frac{dy}{dx} = \frac{y}{\sqrt{1-x^2}}$ (b) $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$
(c) $\frac{dy}{dx} = \frac{x}{\sqrt{1-x^2}}$ (d) None of these

10. The solution of the differential equation $\frac{dy}{dx} = \sec x (\sec x + \tan x)$ is
(a) $y \quad \sec x + \tan x + c$
(b) $y = \sec x + \cot x + c$
(c) $y = \sec x - \tan x + c$
(d) None of these

11. The solution of the differential equation $\frac{dy}{dx} + \sin^2 y = 0$ is ***[MP PET-1994]***
(a) $y + \cos y \quad c$
(b) $y - 2 \sin y \quad c$
(c) $x = \cot y + c$
(d) $y = \cot x + c$

12. The solution of the differential equation $\sin x \sin y \, dx + \cos x \cos y \, dy = 0$ is
(a) $\sin y = c \cos x$
(b) $\sin x = c \cos y$
(c) $\sin x \cos y \quad c$
(d) $\sin y \cos x = c$

13. The solution of $\frac{dy}{dx} = x \log x$ is ***[MP-93S, 95, 97, 99; MP PET-2003]***
(a) $y \quad x^2 \log x - \frac{x^2}{2} + c$
(b) $y \quad \frac{x^2}{2} \log x - x^2 + c$
(c) $y \quad \frac{1}{2}x^2 + \frac{1}{2}x^2 \log x + c$
(d) None of these

14. The solution of the equation $\sin^{-1}(dy/dx) = x + y$ is
(a) $\tan(x + y) + \sec(x + y) = x + c$
(b) $\tan(x + y) - \sec(x + y) = x + c$
(c) $\tan(x + y) + \sec(x + y) - x + c = 0$
(d) None of these

15. The solution of the equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$ is ***[Orissa JEE-03; DCE-99]***
(a) $x\sqrt{1-y^2} - y\sqrt{1-x^2} = c$
(b) $x\sqrt{1-y^2} + y\sqrt{1-x^2} = c$
(c) $x\sqrt{1+y^2} + y\sqrt{1+x^2} = c$
(d) None of these

16. The solution of the diff. equation $(1 + x^2)\frac{dy}{dx} = x(1 + y^2)$ is
(a) $2\tan^{-1} y = \log(1 + x^2) + c$
(b) $\tan^{-1} y = \log(1 + x^2) + c$
(c) $2\tan^{-1} y + \log(1 + x^2) + c = 0$
(d) None of these

17. The solution of the diff. equation $\frac{dy}{dx} = \frac{x-y+3}{2(x-y)+5}$ is
(a) $2(x - y) + \log(x - y) = x + c$
(b) $2(x - y) - \log(x - y + 2) = x + c$
(c) $2(x - y) + \log(x - y + 2) = x + c$
(d) None of these

18. The solution of the equation $\frac{dy}{dx} = \frac{y^2-y-2}{x^2+2x-3}$ is
(a) $\frac{1}{3}\log\left|\frac{y-2}{y+1}\right| = \frac{1}{4}\log\left|\frac{x+3}{x-1}\right| + c$
(b) $\frac{1}{3}\log\left|\frac{y+1}{y-2}\right| = \frac{1}{4}\log\left|\frac{x-1}{x+3}\right| + c$
(c) $4\log\left|\frac{y-2}{y+1}\right| = 3\log\left|\frac{x-1}{x+3}\right| + c$
(d) None of these

19. Solution of the diff. equation $\sin(dy/dx) = a$ with $y(0) = 1$ is ***[Kurukshetra CEE-1998]***
(a) $\sin^{-1}\frac{(y-1)}{x} = a$ (b) $\sin\frac{(y-1)}{x} = a$
(c) $\sin\frac{(1-y)}{(1+x)} = a$ (d) $\sin\frac{y}{(x+1)} = a$

20. For solving $\frac{dy}{dx} = (4x + y + 1)$, suitable substitution is ***[MP PET-1999]***
(a) $y \quad vx$
(b) $y = 4x + v$
(c) $y = 4x$
(d) $y + 4x + 1 = v$

21. Solution of differential equation $\frac{dy}{dx} = 2xy$ is ***[MP PET-1997]***
(a) $y \quad ce^{x^2}$
(b) $y^2 = 2x^2 + c$
(c) $y \quad e^{-x^2} + c$
(d) $y = x^2 + c$

22. The solution of differential equation $dy - \sin x \sin y\, dx = 0$ is ***[MP PET-96]***

(a) $e^{\cos x} \tan \frac{y}{2} = C$ (b) $e^{\cos x} \tan y = C$

(c) $\cos x \tan y = C$ (d) $\cos x \sin y = C$

23. If $\frac{dy}{dx} = e^{2y}$ and $y = 0$ when $x = 5$, the value of x for $y \quad 3$ is ***[MP PET-2001]***

(a) e^5 (b) $e^6 + 1$

(c) $(e^6 + 9)/2$ (d) $\log_e 6$

24. The general solution of the differential equation $ydx + (1 + x^2) \tan^{-1} x\, dy \quad 0$, is ***[MP PET-1995]***

(a) $y \tan^{1} x = C$

(b) $x \tan^{-1} y \quad C$

(c) $y + \tan^{-1} x = C$

(d) $x + \tan^{1} y = C$

25. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is ***[AIEEE-2008]***

(a) $y = x \ln x + x$ (b) $y = \ln x + x$

(c) $y = x \ln x + x^2$ (d) $y = x e^{(x \quad 1)}$

26. The solution of the differential equation $x(x - y)\frac{dy}{dx} = y(x - y)$, is ***[MP PET-2008]***

(a) $\frac{x}{y} + \log(xy) = C$

(b) $\frac{y}{x} + \log(xy) = C$

(c) $\frac{x}{y} + y \log x = C$

(d) $\frac{y}{x} + x \log y \quad C$

27. The solution of the differential equation $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$, is ***[MP PET-2008]***

(a) $e^{2y} \quad e^{3x} + x^3 + k$

(b) $\frac{1}{2}e^{2y} = \frac{1}{3}(e^{3x} + x^3) + k$

(c) $\frac{1}{2}e^{2y} = \frac{1}{3}(e^{3x} - x^3) + k$

(d) $e^{2y} \quad e^{3x} + x^2 + k$

28. The solution of the diff. equation $(1 + \cos x)\, dy \quad (1 - \cos x)\, dx$ is ***[MP-98, 99, 2002; HSB-95; PSB-89C; CBSE-2002C, 84]***

(a) $y \quad 2 \tan \frac{x}{2} - x + c$

(b) $y = 2 \tan x + x + c$

(c) $y = 2 \tan \frac{x}{2} + x + c$

(d) $y = x - 2 \tan \frac{x}{2} + c$

29. $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$ ***[MP-98, 99; HSB-93; Kerala-94; CBSE-92C, 93; MP PET-2006]***

(a) $\tan x \tan y = c$ (b) $\frac{\tan y}{\tan x} = c$

(c) $\frac{\tan^2 x}{\tan y} = c$ (d) None of these

SOLUTIONS

1. (b) $\sqrt{a+x}dy + xdx = 0$

or $dy + \frac{xdx}{\sqrt{a+x}} = 0$

$\Rightarrow dy + \left(\frac{x+a}{\sqrt{a+x}} - \frac{a}{\sqrt{a+x}}\right)dx = 0$

Integrating $y + \frac{(x+a)^{3/2}}{3/2} - 2a\sqrt{a+x} = c$

$y + \left(\sqrt{x+a}\right)\left(\frac{2(x+a)}{3} - 2a\right) = c$

$y + 2\sqrt{x+a}\left(\frac{x-2a}{3}\right) = c$

at $x = 2a \Rightarrow y = c$ where c is constant.

2. (b) $\cot y\,dx = x dy \Rightarrow \dfrac{dx}{x} = \tan y dy$

Integrating we get $\ln x = \log\sec y + \ln c$

$x = c\sec y$

3. (c) $(2y-1)\,dx - (2x+3)\,dy = 0$

$$\Rightarrow \frac{dx}{2x+3} - \frac{dy}{2y-1} = 0$$

$$\frac{1}{2}\ln(2x+3) - \frac{1}{2}\ln(2y-1) = \ln c$$

$$\ln\left(\frac{2x+3}{2y-1}\right) = 2\ln c \Rightarrow \frac{2x+3}{2y-1} = c^2 \text{ or } k$$

4. (c) Equation of straight line passing through (0, 0) is $y = mx$ differentiating we get $m = \dfrac{dy}{dx}$,

$\therefore$ differential equation is $\dfrac{y}{x} = m = \dfrac{dy}{dx}$

5. (a) $y' = \dfrac{x-y}{x+y}$ put $y = vx$ (equation is homogeneous)

$$\frac{dy}{dx} = v + x\frac{dv}{dx};\ v + x\frac{dv}{dx} = \frac{1-v}{1+v}$$

$$\frac{xdv}{dx} = \frac{1-2v-v^2}{1+v} \Rightarrow \left(\frac{1+v}{1-2v-v^2}\right)dv = \frac{dx}{x}$$

$$-\frac{1}{2}\ln(1-2v-v^2) + \ln c = \ln x$$

$$\ln c = \frac{2\ln x + \ln(1-2v-v^2)}{2};\ \text{put } v = \frac{y}{x}$$

$$2\ln c = \ln(x^2 - 2xy - y^2)$$

$$\Rightarrow x^2 - 2xy - y^2 = c^2$$

or $y^2 + 2xy - x^2 = -c^2$ or k

6. (b) $y\dfrac{dy}{dx} = 2y - x \Rightarrow \dfrac{dy}{dx} = \dfrac{2y-x}{y}$

put $y = vx;\ \dfrac{dy}{dx} = v + x\dfrac{dv}{dx};\ v + x\dfrac{dv}{dx} = \dfrac{2v-1}{v}$

$$x\frac{dv}{dx} = \frac{2v-1-v^2}{v} \Rightarrow \frac{v}{2v-v^2-1}dv = \frac{dx}{x}$$

$$\frac{vdv}{-(v-1)^2} = \frac{dx}{x}$$

$$\int\left[\frac{v-1}{(v-1)^2} + \frac{1}{(v-1)^2}\right]dv = \frac{-dx}{x}$$

$$\int\left[\frac{1}{v-1} + \frac{1}{(v-1)^2}\right]dv = -\frac{dv}{x}$$

$\Rightarrow$ Integrating $\ln(v-1) - \dfrac{1}{v-1} = -\ln x + \ln c$

$$\ln\left(\frac{y}{x}-1\right) + \ln x$$

$$= \frac{1}{\frac{y}{x}-1} + \ln c \qquad \left(\because v = \frac{y}{x}\right)$$

$$\ln(y-x) = \frac{x}{y-x} + \ln c,$$

for y intercept put $x = 0$

$\ln y = \ln c \Rightarrow y = c$, where c is constant.

7. (a) $\dfrac{dy}{dx} = \dfrac{x+2y-3}{2x+y-3} = \dfrac{x-1+2(y-1)}{2(x-1)+y-1}$

put $X = x - 1 \Rightarrow dX = dx$

$Y = y - 1 \Rightarrow dY = dy$

$$\frac{dY}{dX} = \frac{X+2Y}{2X+Y},$$

put $Y = vX, v + X\dfrac{dv}{dX}$

$$= \frac{1+2v}{2+v}$$

$$X\frac{dv}{dX} = \frac{1-v^2}{2+v} \Rightarrow \frac{2+v}{1-v^2}dv = \frac{dX}{X}$$

$$\frac{2\ln\left(\frac{1+v}{1-v}\right)}{2} - \frac{1}{2}\ln(1-v^2) = \ln X + \ln c$$

$$\ln\left(\frac{1+v}{1-v} \times \frac{1}{\sqrt{1-v^2}}\right) = \ln cX$$

$$\ln\left(\frac{(1+v)^{1/2}}{(1-v)^{3/2}}\right) = \ln cX \Rightarrow \frac{\left(1+\frac{Y}{X}\right)^{1/2}}{\left(1-\frac{Y}{X}\right)^{3/2}} = cX$$

$$\Rightarrow X\frac{(X+Y)^{1/2}}{(X-Y)^{3/2}} = cX$$

Squaring $(X+Y) = c^2(X-Y)^3$

$\Rightarrow$ $x - 1 + y - 1 = c^2(x - 1 - (y-1))^3$

$x + y - 2 = c\,(x-y)^3$

8. (a) $x^2\frac{dy}{dx} = x^2 + xy + y^2 \Rightarrow \frac{dy}{dx} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2$

put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}; v = \frac{y}{x}$

$v + x\frac{dv}{dx} = 1 + v + v^2 \Rightarrow \frac{dv}{1+v^2} = \frac{dx}{x}$

$\tan^{-1}(v) = \ln x + c$

$\tan^{-1}\left(\frac{y}{x}\right) = \ln x + c$

9. (a) $y = ce^{\sin^{-1}x}, \frac{dy}{dx} = ce^{\sin^{-1}x}.\frac{1}{\sqrt{1-x^2}}$

But $ce^{\sin^{-1}x} = y, \therefore \frac{dy}{dx} = \frac{y}{\sqrt{1-x^2}}$

10. (a) $\frac{dy}{dx} = \sec x(\sec x + \tan x) \Rightarrow \frac{dy}{dx} = \sec^2 x +$ sec x tan x

$\int dy = \int(\sec^2 x + \sec x \tan x)dx \Rightarrow y$

$= \tan x + \sec x + c$

11. (c) $\frac{dy}{dx} = -\sin^2 y \Rightarrow \int -\operatorname{cosec}^2 y dy = \int dx$

$-(-\cot y) = x \pm c \Rightarrow x = \cot y \pm c$

12. (a) $\sin x \sin y dx + \cos x \cos y dy = 0$

dividing by $\sin y \cos x$ on both sides

$\tan x dx + \cot y dy = 0$, Integrating we get

$\log\sec x + \log\sin y = \log c$

$\log(\sec x.\sin y) = \log c \Rightarrow \frac{\sin y}{\cos x} = c$

$\sin y = c\cos x$

13. (d) $\frac{dy}{dx} = x\log x \Rightarrow \int dy = \int x\log x dx$

$y = \log x\int x dx - \int\frac{1}{x}.\frac{x^2}{2}dx$

(using integrating by parts)

$y = (\log x)\frac{x^2}{2} - \frac{x^2}{4} + c$

14. (b) $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y \Rightarrow \frac{dy}{dx} = \sin(x+y)$

put $x + y = t \Rightarrow 1 + \frac{dy}{dx} = \frac{dt}{dx} \Rightarrow \frac{dy}{dx} = \frac{dt}{dx} - 1$

Substituting we get $\frac{dt}{dx} - 1 = \sin t$

$\frac{dt}{1+\sin t} = dx \Rightarrow \int\frac{1-\sin t}{1-\sin^2 t}dt = \int dx$

$\int(\sec^2 t - \sec t\tan t)dt$

$= x + c \Rightarrow \tan t - \sec t = x + c$

$\tan(x+y) - \sec(x+y) = x + c$

15. (b) $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

$\Rightarrow \int\frac{dy}{\sqrt{1-y^2}} + \int\frac{dx}{\sqrt{1-x^2}} = 0$

$\sin^{-1} y + \sin^{-1} x = 0$

$\sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) = \sin c$

$\Rightarrow x\sqrt{1-y^2} + y\sqrt{1-x^2} = c$

16. (a) $(1+x^2)\frac{dy}{dx} = x(1+y^2)$, then

Integrating both sides, we get

$\int\frac{x}{1+x^2}dx = \int\frac{dy}{1+y^2}$

$\frac{1}{2}\log(1+x^2) + c = \tan^{-1} y$

$2\tan^{-1} y = \log(1+x^2) + 2c$, c is constant

17. (c) Step 1: This is non-homogeneous differential equation of the type $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ where $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

Step 2: Put $x - y = V$

So that $1 - \frac{dy}{dx} = \frac{dV}{dx}$

$\frac{dy}{dx} = 1 - \frac{dV}{dx}$

Step 3: with above substitutions the problem becomes:

$1 - \frac{dV}{dx} = \frac{V+3}{2V+5}$

$$\frac{dV}{dx}=1-\frac{V+3}{2V+5}$$

$$\frac{dV}{dx}=\frac{(2V+5)-(V+3)}{2V+5}$$

$$\frac{dV}{dx}=\frac{V+2}{2V+5}$$

$$\frac{2V+5}{V+2}dV=dx \qquad \ldots\ldots (1)$$

$$\frac{2(V+2)}{V+2}dV+\frac{1}{V+2}dV=dx$$

On integrating we get

$$2V+\log(V+2)=x+c$$

$$2(x-y)+\log(x-y+2)=x+c$$

18. (c) Step 1: Using following formula

$$\int\frac{dx}{(x-a)(x-b)}=\frac{1}{a-b}\log\frac{x-a}{x-b}$$

Given differential equation can be solved easily as follows by the method of variable separable.

Step 2: $\int\frac{dy}{y^2-y-2}=\int\frac{dx}{x^2+2x-3}$

$$\int\frac{dy}{(y-2)(y+1)}=\int\frac{dx}{(x+3)(x-1)}$$

$$\frac{1}{3}\log\frac{y-2}{y+1}=\frac{1}{4}\log\frac{x-1}{x+3}$$

$$3\log\frac{x-1}{x+3}=4\log\left(\frac{y-2}{y+1}\right)$$

19. (b) Step 1: Given that $\frac{dy}{dx}=\sin^{-1}(a),\ y(0)=1$

$$dy=\sin^{-1}a\,dx \qquad \ldots\ldots (1)$$

Step 2: On integrating we find

$$y=(\sin^{-1}a)x+c \qquad \ldots\ldots (2)$$

Now applying initial condition $y = 1$ at $x = 0$ we get

$$1=0+c\Rightarrow c=1$$

Step 3: $y=\sin^{-1}ax+1$

$$\therefore x=\frac{1}{a}(y-1)$$

20. (d) Given differential equation is based on method of substitution. Clearly we should put $y + 4x + 1 = v$

So that $\frac{dy}{dx}+4=\frac{dv}{dx}$

Step 2: $\frac{dv}{dx}-4=v$

$$\frac{dv}{dx}=4+v$$

$$\frac{dv}{4+v}=dx$$

Which can easily be solved by variables separable method.

$\therefore$ Suitable substitution is (d)

21. (a) Separate the variables $\frac{dy}{dx}=2xy$

On integration $\int\frac{dy}{y}=\int 2x\,dx$

$$\log y=x^2+c$$

taking antilog both sides

$$y=e^{x^2+c} \text{ or } y=e^c x^{x^2}$$

$$y=ce^{x^2}\cdot\ \because e^c=c \text{ (constant)}$$

22. (a) Separate the variables of given equation

$$\frac{dy}{\sin y}-\sin x\,dx=0$$

On integration, we get

$$\int\frac{dy}{\sin y}-\int\sin x\,dx=0$$

$$\Rightarrow\int\text{cosec } y\,dy-\int\sin x\,dx=0$$

$$\log\tan\left(\frac{y}{2}\right)+\cos x=\log c$$

$$\log\left(\frac{\tan\frac{y}{2}}{c}\right)=-\cos x$$

$$\text{or } \tan\frac{y}{2}=ce^{-\cos x}$$

Hence required solution is $\tan\frac{y}{2}e^{\cos x}=c$

23. (c) From given equation $\frac{dy}{dx} = e^{-2y}$

integrating both side, we get

$$\frac{e^{2y}}{2} = x + c \qquad \ldots\ldots\ldots (1)$$

For $y = 0, x = 5$ (given)

So put $y = 0, x = 5$ in equation (1)

$$\frac{1}{2} = 5 + c \Rightarrow c = \frac{-9}{2}; \quad \frac{e^{2y}}{2} = x - \frac{9}{2}$$

and for $y = 3$, we get

$$\frac{e^6}{2} + \frac{9}{2} = x \Rightarrow x = \frac{e^6 + 9}{2}$$

24. (a) On sepration of variables the given equation will become

$$\frac{1}{(1+x^2)\tan^{-1}x}dx + \frac{dy}{y} = 0$$

Integrating both sides, we get

$$\int \frac{1}{\frac{1+x^2}{\tan^{-1}x}}dx + \int \frac{dy}{y} = \int 0\,dx$$

$$\log \tan^{-1}x + \log y = \log c$$

Hence general solution of $y(\tan^{-1}x) = c$

25. (b) $\frac{dy}{dx} = \frac{x+y}{x} = 1 + \frac{y}{x}$

Putting $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$

we get

$$v + x\frac{dv}{dx} = 1 + v \Rightarrow \int \frac{dx}{x} = \int dv$$

$$v = \ln x + c \Rightarrow y = x\ln x + cx$$

as $y(1) = 1 \qquad \therefore \; c = 1$

so solution is $y = x\ln x + cx$

26. (a) Given equation

$\frac{dy}{dx} = \frac{y(x+y)}{x(x-y)}$ is homogeneous

so put $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{vx(x+vx)}{x(x-vx)}$$

$$x\frac{dv}{dx} = \frac{v(1+v)}{1-v} - v \quad x\frac{dv}{dx} = \frac{v + v^2 - v + v^2}{1-v}$$

$$x\frac{dv}{dx} = \frac{2v^2}{1-v} \Rightarrow \frac{1-v}{v^2}dv = \frac{2}{x}dx$$

Integrating both sides, we get

$$\int \frac{1-v}{v^2}dv = 2\int \frac{dx}{x} \Rightarrow \int\left(\frac{1}{v^2} - \frac{1}{v}\right)dv = \int \frac{2}{v}dx$$

$$c - \frac{1}{v} - \log v = 2\log x$$

or $\quad \frac{1}{v} + \log x^2 v = c \qquad \frac{x}{y} + \log xy = c$

27. (b) $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y} \qquad \frac{dy}{dx} = e^{-2y}(e^{3x} + x^2)$

or $\quad e^{2y}dy = (e^{3x} + x^2)dx$

Integrating both sides, we get

$$\int e^{2y}dy = \int (e^{3x} + x^2)dx$$

$$\frac{e^{2y}}{2} = \frac{e^{3x}}{3} + \frac{x^3}{3} + k$$

$$\frac{1}{2}e^{2y} = \frac{1}{3}(e^{3x} + x^3) + k$$

28. (a) $\frac{dy}{dx} = \frac{1-\cos x}{1+\cos x}$

$$\frac{dy}{dx} = \frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}}$$

$$\frac{dy}{dx} = \tan^2\frac{x}{2}$$

Integrating both sides, we get

$$y = \int\left(\sec^2\frac{x}{2} - 1\right)dx$$

$$y = 2\tan\frac{x}{2} - x + c$$

29. (a) Given equation

$$\sec^2 x\tan y dx + \sec^2 y\tan x dy = 0$$

Integrating both sides, we get

$$\int \frac{\sec^2 x}{\tan x}dx + \int \frac{\sec^2 y}{\tan y}d = 0$$

$$\log\tan x + \log\tan y = \log c$$

$$c = \tan x\tan y$$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The solution of $\frac{dy}{dx}=\left(\frac{y}{x}\right)^{1/3}$ is ***[EAMCET-2002]***

(a) $x^{2/3}+y^{2/3}=C$ (b) $x^{1/3}+y^{1/3}=C$
(c) $y^{2/3}-x^{2/3}=C$ (d) $y^{1/3}-x^{1/3}=C$

2. The solution of the differential equation $x(e^{2y}-1)\,dy+(x^2-1)\,e^y\,dx=0$ is ***[AISSE-90]***

(a) $e^y+e^{-y}=\log x-\frac{x^2}{2}+c$

(b) $e^y-e^{-y}=\log x-\frac{x^2}{2}+c$

(c) $e^y+e^{-y}\quad\log x+\frac{x^2}{2}+c$

(d) None of these

3. The solution of the differential equation $(3xy+y^2)dx+(x^2+xy)\,dy\quad 0$ is ***[AISSE-1990]***

(a) $x^2(2xy+y^2)=c^2$ (b) $x^2(2xy-y^2)=c^2$
(c) $x^2(y^2-2xy)=cx^2$ (d) None of these

4. The solution of the differential equation $x^2\,dy=-2xy\,dx$ is ***[SCRA-1990]***

(a) $xy^2=c$ (b) $x^2y^2=c$
(c) $x^2y\quad c$ (d) $xy\quad c$

5. Solution of the diff. equation $\frac{dx}{x}+\frac{dy}{y}=0$ is ***[Karnataka CET-2002]***

(a) $xy\quad c$ (b) $x+y\quad c$
(c) $\log x\log y\quad c$ (d) $x^2+y^2=c$

6. The solution of the differential equation $\frac{dy}{dx}=$ $e^x+\cos x+x+\tan x$ is

(a) $y=e^x+\sin x+\frac{x^2}{2}+\log\cos x+c$

(b) $y\quad e^x+\sin x+\frac{x^2}{2}+\log\sec x+c$

(c) $y\quad e^x-\sin x+\frac{x^2}{2}+\log\cos x+c$

(d) $y\quad e^x-\sin x+\frac{x^2}{2}+\log\sec x+c$

7. The solution of the equation $\frac{dy}{dx}=y\,(e^x+1)$ is ***[AISSE-1986; AICBSE-1984]***

(a) $y+e^{(x+x+c)}=0$
(b) $\log y\quad e^x+x+c$
(c) $\log y+e^x=x+c$
(d) None of these

8. Solution of the equation $(e^x+1)\,y\,dy\quad(y+1)$ $e^x\,dx$ is ***[AISSE-1988]***

(a) $c(y+1)(e^x+1)+e^y=0$
(b) $c(y+1)(e^x-1)+e^y\quad 0$
(c) $c(y+1)(e^x-1)-e^y\quad 0$
(d) $c(y+1)(e^x+1)=e^y$

9. The solution of $x\sqrt{1+y^2}\,dx+(y\sqrt{1+x^2}\,)dy\quad 0$ is

(a) $\sqrt{1+x^2}\;+\sqrt{1+y^2}\;=c$
(b) $\sqrt{1+x^2}\;-\sqrt{1+y^2}\;=c$
(c) $(1+x^2)^{3/2}+(1+y^2)^{3/2}=c$
(d) None of these

10. The solution of $\frac{dy}{dx}=2^{y\;x}$ is ***[Karnataka CET-2000]***

(a) $2^x+2^y=C$ (b) $2^x-2^y=C$
(c) $\frac{1}{2^x}-\frac{1}{2^y}=C$ (d) $\frac{1}{2^x}+\frac{1}{2^y}=C$

11. The general solution of the diff. equation $(x+y)\,dx+x\,dy=0$ is ***[MP PET-1994, 95]***

(a) $x^2+y^2=c$ (b) $2x^2-y^2=c$
(c) $x^2+2xy\quad c$ (d) $y^2+2xy\quad c$

12. The general solutionof the diff. equation $\frac{dy}{dx}$ $=\cot x\cot y$ is ***[AISSE-1983; MP PET-1994]***

(a) $\cos x=c\,\mathrm{cosec}\,y$ (b) $\sin x=c\sec y$
(c) $\sin x=c\cos y$ (d) $\cos x=c\sin y$

13. What is the solution of the differential equation $\frac{dy}{dx}=\sec(x+y)$? ***[NDA-2007]***

(a) $y+\tan\left\{\frac{(x+y)}{2}\right\}=c$

(b) $y+\tan\left\{\frac{(x-y)}{2}\right\}=c$

(c) $x + \tan(x \quad y) = c$

(d) $y - \tan\left\{\frac{(x+y)}{2}\right\} = c$

14. The solution of the equation $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ is ***[MP PET-2004]***

(a) $e^y \quad e^x + \frac{x^3}{3} + c$

(b) $e^y = e^x + 2x + c$

(c) $e^y \quad e^x + x^3 + c$

(d) $y \quad e^x + c$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 16 minutes.
3. The worksheet consists of 16 questions. The maximum marks are 48.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. Differential equation of $y = \sec(\tan^{-1}x)$ is *[UPSEAT-2002]*

(a) $(1+x^2)\dfrac{dy}{dx} = y + x$

(b) $(1+x^2)\dfrac{dy}{dx} = y - x$

(c) $(1+x^2)\dfrac{dy}{dx} = xy$

(d) $(1+x^2)\dfrac{dy}{dx} = \dfrac{x}{y}$

2. When $x \to \infty$, solution of the differential equation $\dfrac{dy}{dx} - ky = 0, y(0) = 1$ tends to zero if

(a) $k = 0$ (b) $k > 0$
(c) $k < 0$ (d) never

3. Solution of the differential equation $\dfrac{dy}{dx} = \sqrt{1 - x^2 - y^2 + x^2y^2}$ is

(a) $\sin^{-1}y = \sin^{-1}x + c$

(b) $\sin^{-1}y \quad \dfrac{1}{2}\sqrt{1-x^2} + \dfrac{1}{2}\sin^{-1}x + c$

(c) $\sin^{-1}y \quad \dfrac{1}{2}x\sqrt{1-x^2} + \dfrac{1}{2}\sin^{-1}x + c$

(d) $\sin^{-1}y = \dfrac{1}{2}x\sqrt{1-x^2} + \dfrac{1}{4}\cos^{-1}x + c$

4. Solution of the differential equation $\dfrac{dy}{dx} = \sin x + 2x$ at (0, 1)

(a) $y \quad x^2 - \cos x - 2$
(b) $y \quad \cos x + x^2 + 2$
(c) $y \quad 2x^2 - \cos x + 4$
(d) $y \quad x^2 - \cos x + 2$

5. Solution of the differential equation $\dfrac{dy}{dx} = \dfrac{y-x}{y+x}$ at (0, 1)

(a) $y^2 + 2xy - 2x + x^2 + 1 = 0$

(b) $\left(1 + \dfrac{x}{y}\right)y = \left(1 - \dfrac{x}{y}\right)x + c$

(c) $\log(x^2 + y^2) + 2\tan^{-1}\dfrac{y}{x} = \pi$

(d) None of these

6. Solution of differential equation $x\,dx - y\,dy = 0$ represents

(a) rectangular hyperbola
(b) straight line passing through origin
(c) Parabola whose centre is at origin
(d) Circle whose centre is at origin

7. The differential equation of the family of parabolas with focus at the origin and the x-axis as axis is

(a) $y\left(\dfrac{dy}{dx}\right)^2 + 4x\dfrac{dy}{dx} = 4y$

(b) $-y\left(\dfrac{dy}{dx}\right)^2 = 2x\dfrac{dy}{dx} - y$

(c) $y\left(\dfrac{dy}{dx}\right)^2 + y = 2xy\dfrac{dy}{dx}$

(d) $y\left(\dfrac{dy}{dx}\right)^2 + 2xy\dfrac{dy}{dx} + y = 0$

8. The solution of $\cos(x - y)\,dy \quad dx$ is

(a) $y = \tan\left(\dfrac{x+y}{2}\right) + C$

(b) $y + \cos^{-1}\left(\dfrac{y}{x}\right) = C$

(c) $y \quad x\sec\left(\dfrac{y}{x}\right) + C$

(d) None of these

9. The third order differential equations is

(a) $y'^2 + x = y^2$
(b) $y'y'' + y \quad \sin x$
(c) $y''' + y'' + y \quad 0$
(d) $y' = y$

10. Solution of $(x + y - 1)\,dx + (2x + 2y - 3)\,dy$ 0 is when $x = 2$, y 1 is
(a) $2y + x + \log(x + y - 2) = 4$
(b) $2y + 2x + \log(x + y - 2) = 6$
(c) $y + x + \log(x + y - 2) = 3$
(d) $y + 2x + \log(x + y - 2) = 5$

11. The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2 \log\left(\frac{d^2y}{dx^2}\right)$ is ***[NDA-2001]***
(a) 1 (b) 2
(c) 3 (d) None of these

12. The solution of the differential equation $(1 + x^2)\frac{dy}{dx} + 1 + y^2 = 0$ is ***[DCE-1996]***
(a) $\tan^{-1}x + \tan^{-1}y = \tan^{-1}c$
(b) $\tan^{-1}y - \tan^{-1}x = \tan^{-1}c$
(c) $\tan^{-1}y + \tan^{-1}x = \tan^{-1}c$
(d) $\tan^{-1}y + \tan^{-1}x = \tan^{-1}c$

13. The differential equation $dy/dx = [x(1 + y^2)]/[y(1 + x^2)]$ represents a family of ***[Orissa-JEE-2007]***
(a) parabola
(b) hyperbola
(c) circle
(d) ellipse

14. Solution of differential equation $\frac{dy}{dx} = \sin x + 2x$, is ***[MP PET-1997]***
(a) $y \quad x^2 - \cos x + c$
(b) $y = \cos x + x^2 + c$
(c) $y = \cos x + 2$
(d) $y \quad \cos x + 2 + c$

15. The general solution of $x^2\frac{dy}{dx} = 2$ is ***[AISSE-1984]***
(a) $y \quad c + \frac{2}{x}$ (b) $y \quad c - \frac{2}{x}$
(c) $y \quad 2cx$ (d) $y \quad c - \frac{3}{x^3}$

16. The solution of $\frac{dy}{dx} + 1 = e^{x+y}$ is ***[EAMCET-2007]***
(a) $e^{-(x+y)} + x + c = 0$
(b) $e^{(x+y)} - x + c = 0$
(c) $e^{x+y} + x + c = 0$
(d) $e^{x+y} - x + c = 0$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	7. ⓐ ⓑ ⓒ ⓓ	13. ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	8. ⓐ ⓑ ⓒ ⓓ	14. ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	9. ⓐ ⓑ ⓒ ⓓ	15. ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	10. ⓐ ⓑ ⓒ ⓓ	16. ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	11. ⓐ ⓑ ⓒ ⓓ	
6. ⓐ ⓑ ⓒ ⓓ	12. ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

2. (c) $\frac{dy}{dx} = ky \Rightarrow \frac{dy}{y} = kdx$

Integrating we get $\ln y = kx + c$

Also $y(0) = 1,\ 0 = 0 + c \Rightarrow c = 0$

$\therefore$ $\ln y = kx \Rightarrow y = e^{cx}$

at $x \to \infty,\ y \to 0$; is possible iff $k < 0$.

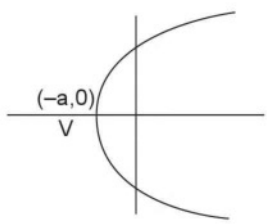
3. (c) $\frac{dy}{dx} = \sqrt{1.(1-x^2) - y^2(1-x^2)}$

$$\frac{dy}{dx} = \sqrt{(1-x^2)(1-y^2)}$$

$$\int \frac{dy}{\sqrt{1-y^2}} = \int \sqrt{1-x^2}dx$$

$$\sin^{-1} y = \frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x + c$$

7. (b)



$\therefore$ equation of parabola

$\Rightarrow$ $y^2 = 4a(x + a)$

differentiating $2y\frac{dy}{dx} = 4a$

or $\frac{1}{2}y\frac{dy}{dx} = a$

$\therefore$ Differential equation

$$y^2 = 2y\frac{dy}{dx}\left(x + \frac{y}{2}\frac{dy}{dx}\right)$$

$$y^2 = 2x\frac{dy}{dx} + y\left(\frac{dy}{dx}\right)^2$$

$$\Rightarrow -y\left(\frac{dy}{dx}\right)^2 = -y + 2x\frac{dy}{dx}$$

10. (a) $\frac{dy}{dx} + \frac{x + y - 1}{2x + 2y - 3} = 0$

put $x + y = t;\ \frac{dy}{dx} + 1 = \frac{dt}{dx}$

Now differential equation is,

$$\frac{dt}{dx} - 1 + \frac{t-1}{2t-3} = 0$$

$$\frac{dt}{dx} = 1 - \frac{t-1}{2t-3} \Rightarrow \frac{dt}{dx} = \frac{t-2}{2t-3}$$

$$\frac{2t-3}{t-2}dt = dx \Rightarrow \int\left(2 + \frac{1}{t-2}\right)dt = \int dx$$

$$2t + \log(t-2) = x + c$$

$$2x + 2y + \log(x + y - 2) = x + c$$

or $x + 2y + \log(x + y - 2) = c$,

also $x = 2,\ y = 1$

$$2 + 2 + \log 1 = c \Rightarrow c = 4$$

$\therefore$ $x + 2y + \log(x + y - 2) = 4$

13. (b) $\frac{dy}{dx} = \frac{x(1+y^2)}{y(1+x^2)}$

$$\int \frac{y}{1+y^2}dy = \int \frac{x}{1+x^2}dx$$

$$\frac{1}{2}\log(1+y^2) = \frac{1}{2}\log(1+x^2) + \log c$$

$$\frac{1+y^2}{1+x^2} = c^2 \Rightarrow -c^2x^2 + y^2 = c^2 - 1$$

which is hyperbola

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LECTURE



3 Various Types of Differential Equations

BASIC CONCEPTS

LINEAR DIFFERENTIAL EQUATION

A differential equation is said to be linear if the unknown and its derivatives which occur in the equation occur in first degree.

NOTE

A linear diferential equation is always of first degree but its converse is not always true i.e., every equation of 1st degree is not linear differential equation

Example $\left(\frac{dy}{dx}\right)^2 + \frac{d^2y}{dx^2} + 7y = 10x$, order (2), degree (1)

Following types of term do not occur in a linear differential equation $y^2, y\frac{dy}{dx}, \left(\frac{dy}{dx}\right)^n, y^n$

(A) Linear Differential Equation in y: $\frac{dy}{dx} + Py = Q$, Dependent variable $= y$, Independent variable $= x$

P and Q are constants or functions of x only.

(i) **Integrating factor:** By integrating factor we mean those functions of x and y which when multiplied with differential equation will make it exact. Infinite many differential equations can have the same integrating factor.

(ii) **Solution of linear differential equation in y:**

$$y \times (I.F.) = \int Q.(I.F.)dx + c, \; I.F. = e^{\int P dx}$$

(B) Linear Differential Equation in (x): $\frac{dx}{dy} + Px = Q$

Independent varialbe $= y$, independent variable $= x$

I.F. $= e^{\int P dy}$

Here P and Q are constants or functions of y only.

Solution is: $x\, e^{\int P dy} = \int Q.(e^{\int P dy})dy + c$

BERNOULLI'S DIFFERENTIAL EQUATION

or (Extended form of linear differential equation) $\frac{dy}{dx} + Py = Qy^n$, (Dividing through out by y^n and substituting V for $y^{(n\ \ 1)}$ we get linear differential equation in V as follows: $\frac{dV}{dx} + P(1-n)V = Q(1-n)$ whose solution is $y^{1-n} \times e^{\int (1-n)pdx} = \int (1-n)Q.e^{\int (1-n)pdx} + C$

DIFFERENTIAL EQUATION OF 1ST ORDER AND HIGHER DEGREE

It's general form is $P^n\lambda_0 + \lambda_1 P^{n-1} + \lambda_2 P^{n-2} + ... + \lambda n = 0$ where $P = \frac{dy}{dx}$ and $\lambda_0, \lambda_1, \lambda_2, \lambda_3$ etc are constants or functions of x. It is mainly of 4 types:

(i) differential equations solvable for P.
(ii) differential equation solvable for y.
(iii) differential equation solvable for x.

CLAIRAUTS DIFFERENTIAL EQUATION

$y = x\frac{dy}{dx} + f\left(\frac{dy}{dx}\right)$ Solution of clairauts differential equation is obtained by substituting c for $\frac{dy}{dx}$

i.e., solution is $y = cx + f(c)$

EXACT DIFFERENTIAL EQUATION

(i) **General form of exact differential equation:**

$Mdx + Ndy = 0$

$\frac{\partial m}{\partial y} = \frac{\partial N}{\partial x}$ (Condition of Exact ness)

(ii) **Solution of exact differential equation:** Solution of exact differential equation is obtained in 4 steps:

1. Integrate M with respect to x treating y as constant.
2. Integrate those terms of N which are free from x with respect to y.
3. Sum of step (1) + step (2) is a solution of exact differential equation, when equated to some other constant.
4. If there is no term of N which is free from x then its solution is given by step (1) only out.

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. $x\frac{dy}{dx} + y = x\log x;\ y(1) = \frac{1}{4}$ ***[NCERT-Book]***

Solution

We have $x\frac{dy}{dx} + y = x\log x$

$\Rightarrow \frac{dy}{dx} + \frac{1}{x}y = \log x$ ..... (i)

This is linear differential equation of the form $\frac{dy}{dx} + Py = Q$ with

$P = \frac{1}{x}$ and $Q = \log x$.

$\therefore\ I.F. = e^{\int \frac{1}{x}dx} = e^{\log x} = x$

Hence its solution is $y.\ I.F. = \int Q.I.F.dx + C$

$\Rightarrow\ yx = \int x\log x dx + C$

$\Rightarrow\ xy = \frac{x^2}{2}(\log x) - \frac{1}{2}\int x dx + C$

$\Rightarrow\ xy = \frac{x^2}{2}(\log x) - \frac{x^2}{4} + C$ ..... (ii)

It is given that $y(1) = 1/4$ i.e., $y = 1/4$ when $x = 1$

$\frac{1}{4} = 0 - \frac{1}{4} + C \Rightarrow C = \frac{1}{2}$

Putting $C = 1/2$ in (ii), we get

$xy = \frac{x^2}{2}(\log x) - \frac{x^2}{4} + \frac{1}{2}$

$\Rightarrow\ y = \frac{1}{2}x\log x - \frac{x}{4} + \frac{1}{2x}$

2. $\frac{dy}{dx} + y = e^x$ ***[MP-96, 99, 2002; CBSE-90C]***

Solution

The given differential equation is $\frac{dy}{dx} + y = e^x$, (linear) ..... (1)

Here, $P = 1$ and $Q = e^x$

$\therefore$ I.F. $= e^{\int P dx} = e^{\int dx} = e^x$

Hence, the solution of differential equation (1) is $y.e^{\int P dx} = \int Q.e^{\int P dx} dx + c$, where c is constant of integration

$\Rightarrow y \,.\, e^x = \int e^x.e^x dx + c = \int e^{2x} dx + c$

$\Rightarrow ye^x = \frac{e^{2x}}{2} + c$

$\Rightarrow y = \frac{e^x}{2} + ce^{-x}$.

Which is the required solution.

3. $\frac{d^2y}{dx^2} + \sec^2 x = e^x$, given that $\frac{dy}{dx} = 1$ and $y = 1$ when $x = 0$ ***[CBSE-2001]***

Solution

$\frac{d^2y}{dx^2} = e^x - \sec^2 x$ integrating w.r.t x

$\frac{dy}{dx} = e^x - \tan x + c$ ..... (i)

$y = e^x - \log \sec x + c\,x + c_1$ ..... (ii)

Put $\frac{dy}{dx} = 1$ and $x = 0$ in equation (i)

$1 = 1 - 0 + c \Rightarrow c = 0$

Put $x = 0, y = 1, c = 0$ in equation (ii)

$c_1 = 0$

$\therefore y = e^x - \log \sec x$

4. Solve the following differential equation $4\frac{dy}{dx} + 8y = 5e^{-3x}$. ***[CBSE-2007]***

Solution

$\frac{dy}{dx} + 2y = \frac{5}{4}e^{-3x}$ $I.F. = e^{\int 2dx} = e^{2x}$

its solution is

$y \times e^{2x} = \int \frac{5}{4}e^{-3x} \times e^{2x} dx + c$

$y \times e^{2x} = -\frac{5}{4}e^{-x} + c$

$y = \frac{-5}{4}e^{-3x} + ce^{-2x}$

5. Solution of the following differential equation $(x^2+1)\frac{dy}{dx} + 2xy = \sqrt{x^2+4}$ ***[CBSE-2010]***

Solution

Rearrange as linear differential equation in x as follows:

$\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{\sqrt{x^2+4}}{1+x^2}$

Clearly here $P = \frac{2x}{1+x^2}$ and $Q = \frac{\sqrt{x^2+4}}{1+x^2}$

$\therefore$ Integrating factor $= I.F. = e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx}$

$= e^{\log(1+x^2)} = 1 + x^2$

Step 2: Its solution is

$(y) \times I.F. = \int Q \times I.F. dx + C$

$(y)(1+x^2) = \int \frac{\sqrt{x^2+4}}{1+x^2}(1+x^2)dx + C$

$(y)(1+x^2)$

$= \frac{x}{2}\sqrt{x^2+4} + \frac{4}{2}\log\left[x + \sqrt{x^2+4}\right] + C$

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. Verify that $y = (x^3 - x)\log(cx)$ is a solution of the differential equation $(x^3 - x)\frac{dy}{dx} - (3x^2 - 1)y = x^5 - 2x^3 + x$.

Solve the following differential equations:

2. $(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2$

[MP-97, 98, 99, 2000, 2001, 2004, 2008]

3. $x\frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x.$

[MP-2000, 2001, 2002]

4. $(1 + y^2)\,dx = (\tan^{-1} y - x)\,dy$

[MP-2000; PSB-98 C; CBSE-2002]

5. $\cos x\frac{dy}{dx} + y = \sin x$, given that $y = 2$ when $x = 0$

[MP-98, 2000, 2002, 2005; CBSE-92, 94, 2004]

6. $\frac{dy}{dx} = \frac{y + x + 1}{x + 1}$

7. $\sqrt{1 - y^2}\,dx = (\sin^{-1} y - x)dy, y(0) = 0$

8. $\frac{dy}{dx} + \frac{4x}{x^2 + 1}y + \frac{1}{(x^2 + 1)^2} = 0$ ***[CBSE-2005]***

9. $\sin x\frac{dy}{dx} + \cos x\, y = \cos x \cdot \sin^2 x.$

[CBSE-2006]

10. $x\,dy - y\,dx = \sqrt{x^2 + y^2}\,dx$

[PSB-95; CBSE-2005]

11. $\frac{dy}{dx} + 2y = \sin x.$

[MP-99; CBSE-91C, 96; HPSB-96]

12. $(x + 2y^3)\frac{dy}{dx} = y$

[PSB-2001, 2002]

13. $\frac{dy}{dx} + y\tan x = 2x + x^2 \tan x, y(0) = 1.$

[NCERT-Book]

ANSWERS

2. $y(1 + x^2) = \frac{4x^3}{3} + C$
3. $y = x \log_e \tan x + cx$
4. $x = \tan^{-1} y - 1 + c\,e^{-\tan^{-1} y}$
5. $y(\sec x + \tan x) = \sec x + \tan x - x + c$ or $y = \tan x - x + \sec x + 1.$
6. $\frac{y}{1 + x} = \log(1 + x) + c$
7. $x - \sin^{-1} y + 1 = e^{\sin^{-1} y}$
8. $(x^2 + 1)^2 y + x = c$
9. $y \sin x = \frac{\sin^3 x}{3} + c$
10. $\left\{y + \sqrt{x^2 + y^2}\right\}^2 = c^2 x^4$
11. $y = \frac{1}{5}[2\sin x - \cos x] + ce^{-2x}$
12. $x = y^3 + cy$
13. $y = x^2 + \cos x$

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{1}{x}$ is ***[NDA-2003]***

(a) $y = \frac{1}{2}\log x + c\,(\log x)^{-1}$

(b) $y = \log x + c\,(\log x)^{-1}$

(c) $y = \frac{1}{2}\log x + \frac{c}{(\log x)^2}$

(d) $y = \frac{1}{3}\log x - c\,(\log x)^{-1}$

Solution

(a) It is linear in y, So $I.F. = e^{\int 1/x \log x dx} = \log x$

$\therefore$ Solution is $y(\log x) = \int \frac{1}{x} \log x dx + c$

$\Rightarrow y \log x = \frac{1}{2} (\log x)^2 + c$

$\Rightarrow y = \frac{1}{2} \log x + c (\log x)^{-1}$

2. The differential equation $\sin 2y \frac{dy}{dx} + 2 \tan x \cos^2 y = 2 \sec x \cos^3 y$ is reduced to the linear form $\frac{dv}{dx} + Pv = Q$, where P and Q are functions of x alone. The substitution for this reduction is: ***[ICS-2006]***

(a) $\tan y = v$ (b) $\sec y = v$

(c) $\cot y = v$ (d) $\operatorname{cosec} y = v$

Solution

(b) Given equation can be written as

$\tan y \sec y \frac{dy}{dx} + \tan x \sec y = \sec x$

Now on putting $\sec y = v$, it reduces to

$\frac{dv}{dx} + (\tan x) v = \sec x$

3. Solution of the differential equation $(1 + \tan y)(dx - dy) + 2x\, dy = 0$ is: ***[Roorkee-98]***

(a) $x(\sin y + \cos y) = \sin y + ce^{-y}$

(b) $x(\sin y + \cos y) = \sin y + ce^{y}$

(c) $y(\sin x + \cos x) = \sin x + ce^{x}$

(d) None of these

Solution

(a) Given equation can be written as

$(1 + \tan y)\, dx = (1 + \tan y - 2x)\, dy$

$\Rightarrow \frac{dx}{dy} + \frac{2}{1 + \tan y} x = 1$ which is linear in x, so

$I.F. = e^{\int 2/(1+\tan y) dy} = e^{\int [1+(\cos y - \sin y)/(\cos y + \sin y)] dy}$

$= e^{y} (\cos y + \sin y)$

$\therefore$ solution is $xe^{y}(\cos y + \sin y) = \int e^{y}(\cos y + \sin y) dy + c$

$\Rightarrow xe^{y}(\cos y + \sin y) = e^{y} \sin y + c$

$\Rightarrow x(\sin y + \cos y) = \sin y + ce^{-y}$

4. If $y + \frac{d}{dx}(xy) = x(\sin x + \log x)$, then ***[Roorkee-95]***

(a) $y = \cos x + \frac{2}{x} \sin x + \frac{2}{x^2} \cos x + \frac{x}{3} \log x - \frac{x}{9} + \frac{c}{x^2}$

(b) $y = -\cos x + \frac{2}{x} \sin x + \frac{2}{x^2} \cos x - \frac{x}{3} \log x - \frac{x}{9} + \frac{c}{x^2}$

(c) $y = -\cos x + \frac{2}{x} \sin x + \frac{2}{x^2} \cos x + \frac{x}{3} \log x - \frac{x}{9} + \frac{c}{x^2}$

(d) None of these

Solution

(c) $y + \frac{d}{dx}(xy) = x(\sin x + \log x)$

$\Rightarrow x \frac{dy}{dx} + 2y = x(\sin x + \log x)$

$\Rightarrow \frac{dy}{dx} + \frac{2}{x} y = \sin x + \log x$

which is linear in y, so $I.F. = e^{\int 2/x dx} = x^2$

$\therefore$ solution is $yx^2 = \int x^2 (\sin x + \log x) dx + c$

$= -x^2 \cos x + 2x \sin x + 2 \cos x + \frac{x^3}{3} \log x - \frac{x^3}{9} + c$

$\Rightarrow y = -\cos x + \frac{2}{x} \sin x + \frac{2}{x^2} \cos x + \frac{x}{3} \log x - \frac{x}{9} + \frac{c}{x^2}$

5. The solution of $\frac{dy}{dx} = \frac{e^{x}(\sin^2 x + \sin 2x)}{y(2 \log y + 1)}$ is ***[AISSE-1990]***

(a) $y^2 (\log y) - e^{x} \sin^2 x + c = 0$

(b) $y^2 (\log y) - e^{x} \cos^2 x + c = 0$

(c) $y^2(\log y)+e^x\cos^2 x+c=0$

(d) None of these

Solution

(a) $\dfrac{dy}{dx}=\dfrac{e^x(\sin^2 x+\sin 2x)}{y(2\log y+1)}$

$\Rightarrow \int(2y\log y+y)dy=\int e^x(\sin^2 x+\sin 2x)dx$

On integrating by parts, we get $y^2(\log y)=e^x\sin^2 x+c$.

6. The solution of differential equation. $y\,dx+(x+x^2y)dy=0$ is: ***[AIEEE-2004]***

(a) $\log y=cx$ (b) $-\dfrac{1}{xy}+\log y=c$

(c) $-\dfrac{1}{xy}+\log y=c$ (d) $-\dfrac{1}{xy}+\log y=c$

Solution

(b) $ydx+xdy=-x^2ydy \Rightarrow \dfrac{1}{(xy)^2}dxy=-\dfrac{dy}{y}$

Integrating both side we get

$$-\frac{1}{xy}=-\log y+c\Rightarrow -\frac{1}{xy}+\log y+c$$

7. If c is any arbitrary constant, then the general solution of the differential equation $ydx-xdy=xydx$ is given by: ***[J & K-2005]***

(a) $y=cxe^{-x}$ (b) $x=cye^{-x}$

(c) $y+e^x=cx$ (d) $ye^x=cx$

Solution

(d) Given $ydx-x\,dy=xy\,dx$

$$\Rightarrow \frac{ydx-xdy}{xy}=dx\Rightarrow d\left[\ln\left(\frac{x}{y}\right)\right]=dx$$

Integrating both sides, we get $\ln\dfrac{x}{y}+\ln c=x$

$\Rightarrow ye^x=cx.$

8. An integrating factor of the differential equation $x\dfrac{dy}{dx}+y\log x=xe^x x^{2^{\frac{1}{\log x}}},(x>0)$ is: ***[Kerala (Engg.)-2005]***

(a) $x^{\log x}$ (b) $\left(\sqrt{x}\right)^{\log x}$

(c) $\left(\sqrt{e}\right)^{\log x}$ (d) e^{x^2}

Solution

(b) $\dfrac{dy}{dx}+\left(\dfrac{\log x}{x}\right)y=e^x x^{\frac{1}{2}\log x}$

$$I.F.=e^{\int\frac{\log x}{x}dx}=e^{\frac{1}{2}(\log x)^2}=\left(e^{\frac{1}{2}(\log x)}\right)^{\log x}$$

$$=\left(e^{\log\sqrt{x}}\right)^{\log x}=\left(\sqrt{x}\right)^{\log x}$$

9. If integrating factor of $x(1-x^2)dy+(2x^2y-y-ax^3)dx=0$ is $e^{\int pdx}$, then P is equal to: ***[MP PET-1999]***

(a) $\dfrac{2x^2-ax^3}{x(1-x^2)}$ (b) $(2x^2-1)$

(c) $\dfrac{2x^2-1}{ax^3}$ (d) $\dfrac{(2x^2-1)}{x(1-x^2)}$

Solution

(d) $x(1-x^2)\,dy+(2x^2y-y-ax^3)\,dx=0$

$$\frac{dy}{dx}+\frac{(2x^2-1)}{x(1-x^2)}y=\frac{ax^2}{(1-x^2)},\therefore\ p=\frac{2x^2-1}{x(1-x^2)}$$

10. Find the value of differential equation

$y\cos\dfrac{y}{x}(xdy-y\,dx)+x\sin\dfrac{y}{x}(x\,dy+y\,dx)=$ ***[Roorkee-1997]***

(a) $\left(1+\dfrac{2}{\pi}\right)$ (b) $\left(1,\dfrac{\pi}{2}\right)$

(c) $\left(1-\dfrac{\pi}{2}\right)$ (d) $\left(1+\dfrac{\pi}{2}\right)$

Solution

(b) The given equation can be written as

$$\frac{xdy-ydx}{xdy+ydx}=-\frac{x}{y}\tan\frac{y}{x}$$

or $$\frac{(xdy-ydx)/x^2}{xdy+ydx}=-\frac{1}{xy}\tan\frac{y}{x}$$

or $$\cot\left(\frac{y}{x}\right)d\left(\frac{y}{x}\right)+\frac{d(xy)}{(xy)}=0$$

Integrating $\log \sin \frac{y}{x} + \log (xy) = \log k$

$\therefore \quad xy \sin \frac{y}{x} = k$

or $\quad xy \sin \frac{y}{x} = \frac{\pi}{2}$

$\because \quad k = \frac{\pi}{2}$ at $\left(1, \frac{\pi}{2}\right)$

11. Find the value of the differential equation $\cos^2 x \frac{dy}{dx} - (\tan 2x)\, y = \cos^4 x$,

[Roorkee-1996]

(a) $\frac{1}{2}\sin 2x + c$

(b) $\frac{1}{2}\cos 2x + c$

(c) $\frac{2}{3}\sin 2x + c$

(d) $\frac{2}{3}\cos 2x + c$

Solution

(a) On dividing by $\cos^2 x$, we have

$$\frac{dy}{dx} - \frac{2\tan x}{1-\tan^2 x}\sec^2 x.y = \cos^2 x$$

$$\int P dx = \int -\frac{2t}{1-t^2} dt \qquad \text{where } t = \tan x$$

$$\int P dx = \log(1-t^2)$$

$\therefore \quad I.F. = 1 - t^2 = 1 - \tan^2 x$

Multiplying both sides by *I.F.* and integrating, we get

$$y(1-\tan^2 x) = \int (1-\tan^2 x)\cos^2 x dx + c$$

$$= \int \cos 2x dx + c = \frac{1}{2}\sin 2x + c$$

12. Solution of the differential equation $2y\sin x \frac{dy}{dx} = 2 \sin x \cos x - y^2 \cos x$ satisfying $y(\pi/2) = 1$ is given y:

(a) $y^2 = \sin x$

(b) $y = \sin^2 x$

(c) $y^2 = \cos x + 1$

(d) $y^2 \sin x = 4\cos^2 x$

Solution

(a) $\frac{d}{dx}(y^2 \sin x) = \sin 2x,$

$$y^2 \sin x = \frac{-\cos 2x}{2} + c$$

13. Solution of the differential equation $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$ is ***[ICS-2006]***

(a) $e^{e^y}(e^x - e^y + 1) = c$

(b) $e^{e^y}(e^y - e^x + 1) = c$

(c) $e^{e^x}(e^x - e^y + 1) = c$

(d) $e^{e^x}(e^y - e^x + 1) = c$

Solution

(d) Given equation can be written as

$$e^y \frac{dy}{dx} + e^x . e^y = e^{2x}$$

Now let $e^y = v$, then $e^y \frac{dy}{dx} = \frac{dv}{dx}$, so transformed equation will be $\frac{dv}{dx} + e^x v = e^{2x}$ which is linear in v. So its $I.F. = e^{\int e^x dx} e^{e^x}$

$\therefore$ solution is $ve^{e^x} = \int e^{e^x} . e^{2x} dx + c$

$$\Rightarrow e^y . e^{e^x} = e^{e^x}(e^x - 1) + c$$

$$\Rightarrow e^{e^y}(e^y - e^x + 1) = c$$

14. The solution of the differential equation $\frac{dy}{dx} - \frac{\tan y}{x} = \frac{\tan y \sin y}{x^2}$ is: ***[DCE-2006]***

(a) $\frac{x}{\sin y} + \ln x = c$ (b) $\frac{y}{\sin x} + \ln x = c$

(c) $\ln y + x = c$ (d) $\ln x + y = c$

Solution

(a) Given equation is $\frac{dy}{dx} - \frac{\tan y}{x} = \frac{\tan y \sin y}{x^2}$

$$\Rightarrow \cot y \operatorname{cosec} y \frac{dy}{dx} - \frac{\operatorname{cosec} y}{x} = \frac{1}{x^2}.$$

Let $-\operatorname{cosec} y = t$

$$\Rightarrow \cot y \operatorname{cosec} y \frac{dy}{dx} = \frac{dt}{dx} \quad \therefore \quad \frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

On comparing this equation with $\frac{dt}{dx}+Pt=Q$, we get

$$P=\frac{1}{x}, Q=\frac{1}{x^2}$$

$$\therefore \quad I.F.=e^{\int Pdx}=e^{\int \frac{1}{x}dx}=e^{\log x}=x$$

$$\therefore \quad \text{Solution is } tx=\int x.\frac{1}{x^2}dx+c$$

$$\Rightarrow -\text{cosec } y.x=\log x+c \Rightarrow \frac{x}{\sin y}+\log x=c.$$

15. The solution of differential equation

$$(1+y^2)+(x-e^{\tan^{-1}y})\frac{dy}{dx}=0 \text{ is:} \quad \textit{[DCE-2005]}$$

(a) $2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+k$

(b) $xe^{2\tan^{-1}y}=e^{\tan^{-1}y}+k$

(c) $xe^{\tan^{-1}y}=e^{\tan^{-1}y}+k$

(d) $(x-2)=ke^{-\tan^{-1}y}$

Solution

(a) Given equation can be rewritten as

$$(1+y^2)\frac{dx}{dy}+(x-e^{\tan^{-1}y})=0$$

or $$\frac{dx}{dy}+\frac{1}{1+y^2}x=\frac{e^{\tan^{-1}y}}{1+y^2}$$

This is a linear differential equation,

where $P=\frac{1}{1+y^2}, Q=e^{\tan^{-1}y}$

$$I.F.=e^{\int Pdy}=e^{\int 1/(1+y^2)dy}=e^{\tan^{-1}y}$$

$\therefore$ Complete solution is

$$xe^{\tan^{-1}y}=\int\frac{e^{\tan^{-1}y}e^{\tan^{-1}y}}{1+y^2}dy$$

$$=\frac{1}{2}e^{2\tan^{-1}y}+c$$

$$\therefore \quad 2xe^{\tan^{-1}y}=e^{2\tan^{-1}y}+k \text{ [where } k=2c]$$

16. What is the solution of the differential equation $(1+e^{x/y})dx+e^{x/y}\left(1-\frac{x}{y}\right)dy=0$?

[UPSC-2007]

(a) $x+ye^{x/y}=c$ (b) $y+xe^{x/y}=c$

(c) $x-ye^{x/y}=c$ (d) None of these

Solution

(a) $(1+e^{x/y})dx+e^{x/y}\left(1-\frac{x}{y}\right)dy=0$

$$\Rightarrow dx+e^{x/y}dy+e^{x/y}\left(dx-\frac{x}{y}dy\right)=0$$

$$\Rightarrow dx+e^{x/y}dy+e^{x/y}.\left(\frac{ydx-xdy}{y}\right)=0$$

Multiplying throughout by $1/y$

$$\frac{1}{y}dx+\frac{1}{y}e^{x/y}dy+e^{x/y}\left(\frac{ydx-xdy}{y^2}\right)=0$$

$$\Rightarrow \quad \frac{1}{y}dx+\frac{1}{y}e^{x/y}dy+e^{x/y}d\left(\frac{x}{y}\right)=0$$

$$\Rightarrow \quad \frac{1}{y}dx+\frac{1}{y}e^{x/y}dy+d\left(e^{x/y}\right)=0$$

$$\Rightarrow \quad dx+\underbrace{e^{x/y}dy+yd(e^{x/y})}=0$$

$$\Rightarrow \quad dx+d(y.e^{x/y})=0$$

on integrating we have $x+ye^{x/y}=c$

NOTE

We have made use of $\frac{ydx-xdy}{y^2}=d\left(\frac{x}{y}\right)$ and $xdy+ydx=d(xy)$

17. Solution of the differential equation $x^2y^2\,dy = (1 - xy^3)\,dx$ is:

(a) $2x^3y^3=3x^2+c$

(b) $x^3y^3=x^2+c$

(c) $x^3y^3=x^2+x+c$

(d) $x^3y^3=3x^2+c$

Solution

(a) $$\frac{dy}{dx}+\frac{y}{x}=\frac{1}{x^2y^2}\Rightarrow y^2\frac{dy}{dx}+\frac{y^3}{x}=\frac{1}{x^2}.$$

Put $z=y^3$ so that $\frac{dz}{dx}=3y^2\frac{dy}{dx}$

$$\Rightarrow \frac{dz}{dx}+\frac{3z}{x}=\frac{3}{x^2} \text{ (linear in } z)$$

$$I.F.=e^{\int Pdx}=e^{\int\frac{3}{x}dx}=e^{3\log x}=e^{\log x^3}=x^3;$$

$$(x^3).(z)=\int\frac{3}{x^2}\times x^3dx\Rightarrow(x^3)(y^3)=\frac{3x^2}{2}+c$$

18. The general solution of the differential equation $(x + \tan y)\,dy + \tan y\,dx = 0$, is:
(a) $x \cdot \cos y + \sin y = c$
(b) $x \cdot \cos y - \sin y = c$
(c) $x \cdot \sin y + \cos y = c$
(d) $x \sin y - \cos y = c$

Solution

(d) $x \cos y\,dy + \sin y\,dx + \sin y\,dy = 0$;
$d(x \sin y) - d(\cos y) = 0 \Rightarrow x \sin y - \cos y = c$

19. Solve $x(1 - x^2)\,dy + (2x^2y - y - 5x^3)dx = 0$ ***[Roorkee-1994]***

Solution

$$\frac{dy}{dx} + \frac{2x^2 - 1}{x(1 - x^2)}y = \frac{5x^3}{x(1 - x^2)}$$

$$\int P dx = \int \frac{2x^2 - 1}{x^2(1 - x^2)}x dx \text{ . Put } x^2 = t$$

$$\int P dx = \frac{1}{2}\int \frac{2t - 1}{t(1 - t)}dt = \frac{1}{2}\int\left(\frac{1}{1 - t} - \frac{1}{t}\right)dt$$

$$= -\frac{1}{2}\log(1 - x^2)x^2 = \log\frac{1}{x\sqrt{(1 - x^2)}}$$

$$I.F. = \frac{1}{x\sqrt{(1 - x^2)}}$$

Multiplying both sides by $I.F.$ and integrating

$$y.\frac{1}{x\sqrt{(1 - x^2)}} = \int \frac{5x^3}{x(1 - x^2)}.\frac{1}{x\sqrt{1 - x^2}}dx$$

Put $1 - x^2 = z$

$$= -\frac{5}{2}\int \frac{dz}{z^{3/2}} = -\frac{5}{2}.\frac{(-2)}{\sqrt{z}} = \frac{5}{\sqrt{(1 - x^2)}}$$

$$\therefore \quad y.\frac{1}{x\sqrt{(1 - x^2)}} = \frac{5}{\sqrt{1 - x^2}} + c$$

or $y = 5x + c\ x\ \sqrt{1 - x^2}$

20. The gradient of a curve passing through (4, 0) is given by $\frac{dy}{dx} - \frac{y}{x} + \frac{5x}{(x + 2)(x - 3)} = 0$. Determine the equation of the curve. Also find y when $x = 5$.

Solution

$$\frac{y}{x} = \log\frac{x + 2}{x - 3}.k$$

where $0 = \log 6k \quad \therefore \quad 6k = 1$

Also $\frac{y}{5} = \log\frac{7}{2}k$

$$\therefore \quad y = 5\log\frac{7}{2}.\frac{1}{6} = 5\log\frac{7}{12}$$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. The differential equation $dy/dx = y \tan x - 2 \sin x$ has a solution ***[MP PET-2007]***
(a) $y \cos x = c \sec x$ (b) $y = \cos x + c \sec x$
(c) $y \cos x = c + \sec x$ (d) None of these

2. The differential equation $\frac{dy}{dx} + \frac{y}{x} = y^2$ has a solution ***[MP PET-07]***
(a) $xy \cdot \log_e(cx) = 1$ (b) $xy \cdot \log_e x = c$
(c) $xy \cdot \log_e(c/x) = 1$ (d) None of these

3. Which of the following equation is linear
(a) $\frac{dy}{dx} + xy^2 = 1$ (b) $x^2\frac{dy}{dx} + y = e^x$
(c) $\frac{dy}{dx} + 3y = xy^2$ (d) $x\frac{dy}{dx} + y^2 = \sin x$

4. Solution of the equation $y\,dx - x\,dy + \log x\,dx = 0$ is
(a) $y = cx - (1 + \log x)$
(b) $y = cx + (1 + \log x)$
(c) $y + cx + (1 + \log x) = 0$
(d) None of these

5. A solution of differential equation $(\sec^2 y)\,dy/dx + 2x \tan y = x^3$ is ***[MP PET-2007]***
(a) $2 \tan y = ce^{-x^2} + x^2 - 1$
(b) $\tan y = ce^{-x^2} + x^2 - 1$
(c) $\tan y = ce^{x^2} + x^2 - 1$
(d) None of these

6. What is the only solution of the initial value problem $y' = t(1 + y)$, $y(0) = 0$? ***[NDA-2007]***
   (a) $y = -1 + e^{t^2}$ (b) $y = 1 + e^{t^2}$
   (c) $y = -t$ (d) $y = t$

7. Differential equation $(2x - 10y^3)\dfrac{dy}{dx} + y = 0$ is having solution
   (a) $x + y = ce^{2x}$
   (b) $y^2 = 2x^3 + c$
   (c) $xy^2 = 2y^5 + c$
   (d) $x(y^2 + xy) = 0$

8. The integrating factor of the differential equation $\dfrac{dy}{dx} = y \tan x - y^2 \sec x$, is
   [MP PET-1995; Pb. CET-2002]
   (a) $\tan x$ (b) $\sec x$
   (c) $-\sec x$ (d) $\cot x$

9. The solution of the differential equation $x \log x \dfrac{dy}{dx} + y = 2 \log x$ is
   (a) $y = \log x + c$
   (b) $y = \log x^2 + c$
   (c) $y \log x = \log x^2 + c$
   (d) $y = x \log x + c$

10. Integrating factor of $\dfrac{dy}{dx} + \dfrac{y}{x} = x^3 - 3$ is
    [MP PET-1999]
    (a) x (b) $\log x$
    (c) $-x$ (d) e^x

11. Solution of the differential equation $y' = y \tan x - 2 \sin x$, is ***[AMU-1999]***
    (a) $y = \tan x + 2C \cos x$
    (b) $y = \tan x + C \cos x$
    (c) $y = \tan x - 2C \cos x$
    (d) None of these

12. Find the integral factor of equation $(x^2 + 1)\dfrac{dy}{dx} + 2xy = x^2 - 1$ ***[UPSEAT-2002]***
    (a) $x^2 + 1$ (b) $2x/x^2 + 1$
    (c) $x^2 - 1/x^2 + 1$ (d) None of these

13. $y + x^2 = \dfrac{dy}{dx}$ has the solution
    (a) $y + x^2 + 2x + 2 = Ce^x$
    (b) $y + x + x^2 + 2 = Ce^{2x}$
    (c) $y + x + 2x^2 + 2 = Ce^x$
    (d) $y^2 + x + x^2 + 2 = Ce^x$

14. The solution of $\dfrac{dy}{dx} + y = e^{-x}$, $y(0) = 0$
    [Kerala (Engg.)-2002]
    (a) $y = e^{-x}(x - 1)$ (b) $y = xe^x$
    (c) $y = xe^{-x} + 1$ (d) $y = xe^{-x}$

15. The general solution of the differential equation $\dfrac{dy}{dx} + \sin(x + y) = \sin(x - y)$ is
    (a) $\log \tan y + \sin x = C$
    (b) $\log \tan \dfrac{y}{2} + 2 \sin x = C$
    (c) $\tan \dfrac{y}{2} + \log \sin x = C$
    (d) None of these

16. Integrating factor of the diff. equation $(1 - x^2)\dfrac{dy}{dx} + xy = x(1 - x^2)^{1/2}$ is
    (a) $\sqrt{1 - x^2}$ (b) $1/\sqrt{1 - x^2}$
    (c) $\ln \sqrt{1 - x^2}$ (d) $\ln |x|$

17. If the integrating factor of the differential equation $\dfrac{dy}{dx} + P(x)y = (x)$ is x, then $P(x)$ is
    [Kerala PET-2007]
    (a) x (b) $x^2/2$
    (c) $1/x$ (d) $1/x^2$

18. The integrating factor of the differential equation $\dfrac{dy}{dx} + \dfrac{1}{x}y = 3x$ is ***[MPPET-2006]***
    (a) x (b) $\ln x$
    (c) 0 (d) ∞

19. The solution of the differential equation $\dfrac{dy}{dx} = (4x + y + 1)^2$, is ***[MP PET-2008]***
    (a) $(4x + y + 1) = \tan(2x + k)$
    (b) $(4x + y + 1)^2 = 2 \tan(2x + k)$
    (c) $(4x + y + 1)^3 = 3 \tan(2x + k)$
    (d) $(4x + y + 1) = 2 \tan(2x + k)$

20. Solution of $\cos x \dfrac{dy}{dx} + y \sin x = 1$ is
    [MP PET-1999]
    (a) $y \sec x \tan x = C$
    (b) $y \sec x = \tan x + C$
    (c) $y \tan x = \sec + C$
    (d) $y \tan x = \sec x \tan x + C$

SOLUTIONS

1. (b) $\frac{dy}{dx} - y\tan x = -2\sin x$

$\therefore$ Equation is in form of $\frac{dy}{dx} + yP(x) + Q(x)$

$\therefore$ Integrating factor $= e^{\int P(x)dx} = e^{\int -\tan x dx}$

$= e^{-\log_e \sec x} = e^{\log_e (\sec x)^{-1}} = \frac{1}{\sec x} = \cos x$

$\therefore$ Solution is y (I.F.) $= \int (I.F)Q(x)dx$

$y.\cos x = \int \cos x.(-2\sin x)dx = \frac{2(\cos x)^2}{2} + c$

$y.\cos x = \cos^2 x + c$ or $y = \cos x + c\sec x$

2. (c) $\frac{dy}{dx} + \frac{y}{x} = y^2$ or $\frac{1}{y^2}\frac{dy}{dx} + \left(\frac{1}{y}\right).\frac{1}{x} = 1$

Put $\frac{1}{y} = t; \frac{-1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

$\left(-\frac{dt}{dx}\right) + \frac{t}{x} = 1 \Rightarrow \frac{dt}{dx} - \frac{t}{x} = -1$

$I.F. = e^{\int \frac{-1}{x}dx} = \frac{1}{x}$

$\therefore$ Solution is $t.\frac{1}{x} = \int \frac{1}{x}(-1)dx$

$\frac{1}{xy} = -\log x + \log c \Rightarrow \frac{1}{xy} = \log\frac{c}{x}$

$(xy)\log\frac{c}{x} = 1$

3. (b) By definition ans. is (b) ($\because$ Power of y must be linear)

4. (a) $ydx - xdy + \log x dx = 0$ or

$dx(y + \log x) = xdy$

$\frac{dy}{dx} - \frac{y}{x} = \frac{\log x}{x}$ which is linear in x.

$I.F = \int \frac{-1}{x}dx = \frac{1}{x}$

Solution is $y.\frac{1}{x} = \int \frac{1}{x}.\frac{\log x}{x}dx \Rightarrow \frac{y}{x} = \int \frac{\log x}{x^2}dx$

$\frac{y}{x} = \log x\left(\frac{-1}{x}\right) - \int \frac{1}{x}\left(\frac{-1}{x}\right)dx$

(Using by parts on R.H.S.)

$\frac{y}{x} = \frac{-\log x}{x} - \frac{1}{x} + c$ or $y = -\log x - 1 + cx$

5. (b) $(\sec^2 y)\frac{dy}{dx} + 2x\tan y = x^3$; put $\tan y = t$ or

$\sec^2 y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} + 2xt = x^3 \Rightarrow I.F. = e^{\int 2xdx} = e^{x^2}$

solution is $te^{x^2} = \int e^{x^2}.x^3dx$

Considering $\int e^{x^2}.x^3dx = \int e^u x^3.\frac{du}{2x}$

Put $x^2 = u; 2xdx = du$

$= \frac{1}{2}\int e^u udu = \frac{1}{2}[ue^u - e^u]$

$= \frac{1}{2}e^{x^2}(x^2 - 1)$

= Solution is $\tan y.e^{x^2} = \frac{1}{2}e^{x^2}(x^2 - 1) + c$

6. (a) $\frac{dy}{dt} = t(1 + y) \Rightarrow \int \frac{dy}{1 + y} = \int tdt$

$\log(1 + y) = \frac{t^2}{2} + c; \because y(0) = 0$

$$\log 1 = \frac{0^2}{2} + c \Rightarrow c = 0$$

$$\therefore \log(1+y) = \frac{t^2}{2}$$

$$\Rightarrow y = e^{t^2/2} - 1$$

7. (c) $(2x - 10y^3)\frac{dy}{dx} + y = 0$

On rearranging we get $y\frac{dx}{dy} - 10y^3 + 2x = 0$

$$\frac{dx}{dy} + \frac{2x}{y} = 10y^2, I.F. = e^{\int \frac{2}{y} dy} = e^{2\log y} = y^2$$

Solution is $x.y^2 = \int y^2.10y^2 dy$

consider $xy^2 = \frac{10y^5}{5} + c \Rightarrow xy^2 = 2y^5 + c$

8. (b) $\frac{dy}{dx} - y\tan x = -y^2 \sec x$

$$-\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\tan x = \sec x$$

Put $\frac{1}{y} = t; \frac{-1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

$\therefore$ Equation becomes $\frac{dt}{dx} + t\tan x = \sec x$

$$I.F. = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

9. (c) Equation is $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{2}{x}$

$$I.F. = \int e^{\frac{1}{x\log x}} dx = e^{\log(\log x)} = \log x$$

Solution is $y\log x = \int \log x.\frac{2}{x} dx$

$$y.\log x = 2\frac{(\log x)^2}{2} + c \Rightarrow y\log x = (\log x)^2 + c$$

10. (a) $I.F. = e^{\int \frac{1}{x} dx} = e^{\log_e x} = x$

12. (a) $\frac{dy}{dx} + \frac{2x}{x^2+1}y = x^2 - 1$

Integrating Factor $= e^{\int \frac{2x}{x^2+1} dx} = e^{\log_e (x^2+1)} =$

$x^2 + 1$

13. (a) $\frac{dy}{dx} - y = x^2, I.F = e^{\int -dx} = e^{-x}$

Solution is $y.e^{-x} = \int e^{-x}.x^2 dx$

$$y.e^{-x} = -e^{-x}x^2 - (e^{-x}2x) + (-e^{-x}2)$$

(Using by parts)

$$y.e^{-x} = e^{-x}(-x^2 - 2x - 2) + c$$

$$y = -x^2 - 2x - 2 + ce^x$$

14. (d) $\frac{dy}{dx} + y(1) = e^{-x} \Rightarrow I.F. = e^{\int 1dx} = e^x$

Solution is $y.e^x = \int e^x.e^{-x} dx$

$y.e^x = x + c$, but at $x = 0, y = 0$

$0 = 0 + c \Rightarrow c = 0 \Rightarrow y.e^x = x$

or $y = xe^{-x}$

15. (c) $\frac{dy}{dx} = \sin(x-y) - \sin(x+y) \Rightarrow \frac{dy}{dx} =$

$-2\sin y \cos x$

$\operatorname{cosec} y dy = -2\cos x dx$, Integrating we get,

$$\log\left(\tan\frac{y}{2}\right) = -2\sin x + c$$

16. (b) Differential Equation is

$$(1-x^2)\frac{dy}{dx} + xy = x(1-x^2)^{1/2}$$

$$\frac{dy}{dx} + \frac{x}{1-x^2}y = \frac{x}{\sqrt{1-x^2}}, I.F. = e^{\int \frac{x}{1-x^2} dx}$$

$$= e^{-\frac{1}{2}\log(1-x^2)} = e^{\log(1-x^2)^{-1/2}} = \frac{1}{\sqrt{1-x^2}}$$

17. (c) Here $e^{\int P(x)dx} = x$

$\int P(x)dx = \log x$; differentiating we get

$$P(x) = \frac{1}{x}$$

18. (a) $I.F. = e^{\int \frac{1}{x} dx} = e^{\log_e x} = x$

19. (d) $\frac{dy}{dx} = (4x + y + 1)^2$, put $4x + y + 1 = t$

differentiating $4x + y + 1 = t$, Substituting in D.E.

$\frac{dt}{dx} - 4 = t^2 \Rightarrow \frac{dt}{dx} = t^2 + 4$

$\frac{dt}{t^2+4} = dx$, Integrating $\frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right) = x + \frac{k}{2}$

$\therefore \quad \tan^{-1}\left(\frac{4x+y+1}{2}\right) = 2\left(x + \frac{k}{2}\right)$

$4x + y + 1 = 2\tan(2x + k)$

20. (b) $\cos x \frac{dy}{dx} + y\sin x = 1$

$\frac{dy}{dx} + y\tan x = \sec x$

$I.F. = e^{\int \tan x dx} = \sec x$

$\therefore$ Solution is $y.\sec x = \int \sec x.\sec x dx$

$y\sec x = \tan x + c$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. The expression satisfying the differential equation $(x^2 - 1)\frac{dy}{dx} + 2xy = 1$ is

(a) $x^2y - xy^2 = c$ (b) $(y^2 - 1)x = y + c$

(c) $(x^2 - 1)y = x + c$ (d) None of these

2. Which of the following equation is non-linear

(a) $\frac{dy}{dx} + \frac{y}{x} = \log x$ (b) $y\frac{dy}{dx} + 4x = 0$

(c) $dx + dy = 0$ (d) $\frac{dy}{dx} = \cos x$

3. The solution of diff. equation $\frac{dy}{dx} + y = e^x$ is

(a) $y = e^x + ce^{-x}$ (b) $y = e^{-x} + ce^x$

(c) $y = \frac{1}{2}e^x + ce^{-x}$ (d) $y = \frac{1}{2}e^{-x} + ce^x$

4. The solution of the differential equation $\frac{dy}{dx} + y = \cos x$ is ***[AISSE-1990]***

(a) $y = \frac{1}{2}(\cos x + \sin x) + ce^{-x}$

(b) $y = \frac{1}{2}(\cos x - \sin x) + ce^{-x}$

(c) $y = \cos x + \sin x + ce^{-x}$

(d) None of these

5. The solution of the given differential equation $\frac{dy}{dx} + 2xy = y$ is ***[Roorkee-1995]***

(a) $y = ce^{x-x^2}$ (b) $y = ce^{x^2-x}$

(c) $y = ce^x$ (d) $y = ce^{x^2}$

6. A radioactive element disintegrates at a rate proportional to the quantity of substance Q present at any time t. What is the differential equation of the disintegration? ***[NDA-2007]***

(a) $\frac{dQ}{dt} = -Q$

(b) $\frac{dQ}{dt} = -kQ, k < 0$

(c) $\frac{dQ}{dt} = -kQ, k > 0$

(d) $\frac{dQ}{dt} = Q$

7. In order to solve the differential equation $x\cos x\frac{dy}{dx} + y(x\sin x + \cos x) = 1$ the integrating factor is

(a) $x\cos x$ (b) $x\sec x$

(c) $x\sin x$ (d) $x\operatorname{cosec} x$

8. Solution of the differential equation $\frac{dy}{dx} + ay = e^{mx}$ is ***[MP PET-1996]***

(a) $(a + m)y = e^{mx} + C$

(b) $ye^{ax} = me^{mx} + C$

(c) $y = e^{mx} + Ce^{-ax}$

(d) $(a + m)y = e^{mx} + Ce^{-ax}(a + m)$

9. The solution of $\frac{dy}{dx} + \frac{y}{3} = 1$ is ***[EAMCET-2002]***

(a) $y = 3 + Ce^{x/3}$

(b) $y = 3 + Ce^{-x/3}$

(c) $3y = C + e^{x/3}$

(d) $3y = C = e^{-x/3}$

10. The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi(y/x)}{\phi'(y/x)}$ is

[Kerala PET-2007]

(a) $x\phi(y/x) = k$ (b) $\phi(y/x) = kx$

(c) $y\phi(y/x) = k$ (d) $\phi(y/x) = ky$

11. The integrating factor of the differential equation $\frac{dy}{dx} + \frac{y}{(1-x)\sqrt{x}} = 1 - \sqrt{x}$ is

[Kerala PET-2008]

(a) $\frac{1-\sqrt{x}}{1+\sqrt{x}}$ (b) $\frac{1+x}{1-x}$

(c) $\frac{1+\sqrt{x}}{1-\sqrt{x}}$ (d) $\frac{\sqrt{x}}{1-\sqrt{x}}$

12. Integrating factor of the differential equation $\frac{dy}{dx} + y\tan x - \sec x = 0$ is ***[MP PET-2002]***

(a) $e^{\sin x}$ (b) $1/\sin x$

(c) $1/\cos x$ (d) $e^{\cos x}$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 16 minutes.
3. The worksheet consists of 16 questions. The maximum marks are 48.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. In order to solve the differential equation $x\cos x\frac{dy}{dx} + y\,(x\sin x + \cos x) = 1$ the integrating factor is ***[NDA-2003]***

(a) $x\cos x$ (b) $x\sec x$
(c) $x\sin x$ (d) $x\operatorname{cosec} x$

2. Solution of the differential equation $\frac{dy}{dx}\tan y = \sin(x+y) + \sin(x-y)$ is ***[Kerala Engg.-2005]***

(a) $\sec y + 2\cos x = c$ (b) $\sec y - 2\cos x = c$
(c) $\cos y - 2\sin x = c$ (d) $\tan y - 2\sec y = c$

3. Integrating factor of differential equation $\sin x\frac{dy}{dx} + y\cos x = 1$

(a) $\cos x$ (b) $\tan x$
(c) $\sec x$ (d) $\sin x$

4. Integrating factor of $x\frac{dy}{dx} + y = x^4 - 3x$ is

(a) $\log x$ (b) $-x$
(c) e^x (d) None of these

5. The solution of the differential equation $\frac{dy}{dx} = \frac{y^2}{1-3xy}$ is given by ***[NDA-2002]***

(a) $y^3x = \frac{y^2}{2} + c$ (b) $y^3 = \frac{xy^2}{2} + c$

(c) $x = \frac{(1+2cy)}{y^3}$ (d) $x = \frac{c}{y^3}$

6. The solution of $\frac{dy}{dx} + 2y\tan x = \sin x$ is ***[DCE-1999]***

(a) $y\sec^3 x = \sec^2 x + c$
(b) $y\sec^2 x = \sec x + c$
(c) $y\sin x = \tan x + c$
(d) None of these

7. Solution of the differential equation $ydx - 2xdy = 0$ is a family of parabolas whose common axis and tangent at common vertex are respectively

(a) $x = c, y = 0$ (b) $y = 0, x = 0$
(c) $x = 0, y = 0$ (d) $x = c, y = c$

8. Solution of differential equation $\frac{dy}{dx} + y\sec^2 x = \tan x\sec^2 x$ is

(a) $y = \tan x - 1 + ce^{\tan x}$
(b) $y^2 = \tan x - 1 + ce^{\tan x}$
(c) $ye^{\tan x} = \tan x - 1 + c$
(d) $ye^{\tan x} = \tan x - 1 + c$

9. The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{1}{x}$ is

(a) $y = \frac{1}{2}\log x + c\,(\log x)^{-1}$
(b) $y = \log x + c\,(\log x)^{1}$
(c) $y = \frac{1}{2}\log x + \frac{c}{(\log x)^2}$
(d) $y = \frac{1}{3}\log x - c\,(\log x)^{-1}$

10. The solution of $\frac{dy}{dx} + p(x)\,y = 0$ is ***[Kerala (Engg.)-2002]***

(a) $y = ce^{\int Pdx}$ (b) $x = ce^{-\int Pdx}$
(c) $y = ce^{-\int Pdx}$ (d) $x = ce^{\int Pdx}$

11. The solution of differential equation $\frac{dy}{dx} = \frac{x(2\log x + 1)}{\sin y + y\cos y}$, is ***[Orissa-JEE-2007]***

(a) $y\sin y = x^2\log x + \frac{x^2}{2} + C$
(b) $y\cos y = x^2(\log x + 1) + C$
(c) $y\cos y = x^2\log x + \frac{x^2}{2} + C$
(d) $y\sin y = x^2\log x + C$

12. The general solution of the differential equation $x \cdot \cos\frac{y}{x}(ydx + xdy) = y \cdot \sin\frac{y}{x}(xdy - y\,dx)$ is

(a) $xy = c.\sec(x/y)$

(b) $y = x.\ c \cdot \sec(x/y)$

(c) $xy = c.\sec(y/x)$

(d) None of these

13. The solution of $(x \operatorname{cosec}(y/x) - y)\,dx + xdy = 0$ is

(a) $\log|x| - \cos(x/y) = c$

(b) $\log|x| - \cos(y/x) = c$

(c) $\log|x| - \sin(x/y) = c$

(d) $\log|x| - \sin(y/x) = c$

14. Integrating factor of differential equation $\cos x\,\frac{dy}{dx} + y\sin x = 1$ is

(a) $\cos x$ (b) $\tan x$

(c) $\sec x$ (d) $\sin x$

15. Integrating factor of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ is

(a) $\int p dx$ (b) $\int Q dx$

(c) $e^{\int r\,dx}$ (d) $e^{\int Q dx}$

16. To reduce the differential equation $\frac{dy}{dx} + P(x)y = Q(x).y^n$ to the linear form, the substitution is ***[MPPET-2006]***

(a) $v = \frac{1}{y^n}$ (b) $v = \frac{1}{y^{n-1}}$

(c) $v = y^n$ (d) $v = y^{n-1}$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	**7.** ⓐ ⓑ ⓒ ⓓ	**13.** ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	**8.** ⓐ ⓑ ⓒ ⓓ	**14.** ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	**9.** ⓐ ⓑ ⓒ ⓓ	**15.** ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	**10.** ⓐ ⓑ ⓒ ⓓ	**16.** ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	**11.** ⓐ ⓑ ⓒ ⓓ	
6. ⓐ ⓑ ⓒ ⓓ	**12.** ⓐ ⓑ ⓒ ⓓ	

HINTS AND EXPLANATIONS

1. (b) Differential equation is

$$x\sec x\frac{dy}{dx} + y(x\sin x + \cos x) = 1$$

$$\frac{dy}{dx} + y\left(\tan x + \frac{1}{x}\right) = \frac{1}{x\cos x},\ I.F = e^{\int\left(\tan x + \frac{1}{x}\right)dx}$$

$$= e^{\log\sec x + \log x}$$

$$= e^{\log_e(x\sec x)} = x\sec x$$

2. (a) $\frac{dy}{dx}\tan y = \sin(x+y) + \sin(x-y)$

$$\frac{dy}{dx}\tan y = 2\sin x\cos y$$

$$\Rightarrow \frac{dy}{dx}\frac{\sin y}{\cos y} = 2\sin x\cos y$$

$$\sec x\tan y\,dy = 2\sin x\,dx$$

Integrating we get $\sec y = -2\cos x + c$

6. (b) It's linear equation in y, so $I.F. = e^{2\int \tan x dx} = \sec^2 x$

$\therefore$ Solution is $y\sec^2 x = \int \sin x \sec^2 x dx$

$= \sec x + c \quad \Rightarrow \quad y\sec^2 x = \sec x + c$

7. (b) $ydx - 2xdy = 0$

$$\frac{dy}{dx} = \frac{y}{2x} \Rightarrow \frac{dy}{dx} - \frac{y}{2x} = 0$$

$$\frac{dy}{y} = \frac{dx}{2x}$$

$$2\ln y = \ln x + \ln c$$

$y^2 = cx$, it is parabola whose axis if $y = 0$ and tangent at vertex is $x = 0$.

10. (c) $\frac{dy}{dx} + P(x)y = 0$; $I.F. = e^{\int P(x)dx}$

Solution is $y.e^{\int P(x)dx} = \int 0 dx$

$$y.e^{\int P(x)dx} = c \Rightarrow y = ce^{-\int P(x)dx}$$

11. (d) $dy(\sin y + y\cos y) = (2x\log x + x)dx$

Integrating both side, we get

$$\int \sin y dy + \int y\cos y dy = \int \{2x\log x\}dx + \sqrt{x}dx$$

$$-\cos y + y\sin y - \int \sin y dy$$

$$= x^2 \log x - \frac{x^2}{2} + \frac{x^2}{2} + c$$

$$y\sin y = x^2 \log x + c$$

12. (c) $x\cos\frac{y}{x}(ydx + xdy) = y\sin\frac{y}{x}(xdy - ydx)$

$$x\cos\frac{y}{x}d(xy) = y\sin\frac{y}{x}\left(x^2 d\left(\frac{y}{x}\right)\right)$$

$$\Rightarrow \frac{d(xy)}{xy} = \tan\frac{y}{x}d\left(\frac{y}{x}\right)$$

Integrating we get

$$\log(xy) = \log\sec\left(\frac{y}{x}\right) + \log c$$

$$xy = c\sec\frac{y}{x}$$

13. (b) $x\,\text{cosec}(y/x)dx - ydx + xdy = 0$

$$x\,\text{cosec}\left(\frac{y}{x}\right)dx + x^2\left(\frac{xdy - ydx}{x^2}\right) = 0$$

$$\frac{dx}{x} + \sin\frac{y}{x}d\left(\frac{y}{x}\right) = 0$$

Integrating we get

$$\log|x| - \cos\frac{y}{x} = c$$

16. (b) Dividing both sides by y^n

$$\frac{1}{y^n}\frac{dy}{dx} + \frac{1}{y^{n-1}}P(x) = Q(x)$$

$\therefore$ we will have to substitute $\frac{1}{y^{n-1}} = v$

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LECTURE



4 Solutions of Exact Differential Equations

BASIC CONCEPTS

RULE TO FIND INTEGRATING FACTOR

(i) When $Mx + Ny \neq 0$ and differential equation is homogeneous then one integrating factor of $Mdx + Ndy \quad 0$ is $\frac{1}{Mx + Ny}$.

(ii) When $Mx - Ny \neq 0$ and in the differential equation $Mdx + Ndy = 0$, $M = yf_1(x, y)$ kand $N = xf_2(x, y)$, then integrating factor of differential equation is $= \frac{1}{Mx - Ny}$

(iii) If in a differential equation $Mdx + Ndy \quad 0$, $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = h(x)$ i.e., function of x only then integrating factor of differential equation is $e^{\int h(x)dx}$.

(iv) If in a differential equation $Mdx + Ndy \quad 0$, $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = k(y)$ i.e., function of y only then integrating factor of the differential equation $Mdx + Ndy = 0$ is: $e^{\int k(y)dy}$.

(v) **Integrating Factor (I.F.)**: For the differential equation $x^a y^b (mydx + n\,xdx) + x^r y^s (pydx + qxdy) = 0$ where a, b, m, n, r, s, p and q are constants, is $x^h.y^k$ where values of h and k are suitably selected so that given differential equation reduces to exact when it is multiplied by $x^h y^k$.

DIFFERENTIAL EQUATIONS OF THE FORM $\frac{d^2y}{dx^2} = F(X)$

For finding solution of such type of differential equation, we integrate twice with respect to x and find general solution with two arbitrary constants.

$\frac{d^2y}{dx^2} = f(x)$

SOLUTION BY INSPECTION METHOD

The following derivatives must be remembered as they are very useful in solving some differential equations directly.

(i) $d(x + y) = dx + dy$

(ii) $d(xy) = ydx + xdy$

(iii) $d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$

(iv) $d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}$

(v) $d(\log xy) = \frac{ydx + xdy}{xy}$

(vi) $d\left(\log\frac{y}{x}\right) = \frac{xdy - ydx}{xy}$

(vii) $d(x^m y^n) = x^{m-1} y^{n-1}(my\,dx + nx\,dy)$

(viii) $d\left(\frac{1}{2}\log(x^2+y^2)\right) = \frac{xdx - ydy}{x^2+y^2}$

(ix) $d\left(\frac{1}{2}\log\frac{x+y}{x-y}\right) = \frac{xdy - ydx}{x^2-y^2}$

(x) $d\left(\tan^{-1}\frac{y}{x}\right) = \frac{xdy - ydx}{x^2+y^2}$

(xi) $\frac{d[f(x,y)]^{1-n}}{1-n} = \frac{f'(x,y)}{[f(x,y)]^n}$

(xii) $d(\sqrt{x^2+y^2}) = \frac{xdx + ydy}{\sqrt{x^2+y^2}}$

(xiii) $d\left(\frac{x^2}{y}\right) = \frac{2yxdx - x^2dy}{y^2}$

(xiv) $d\left(\frac{y^2}{x}\right) = \frac{2xydy - y^2dx}{x^2}$

(xv) $d\left(\frac{y^2}{x^2}\right) = \frac{2x^2ydy - 2xy^2dy}{x^4}$

(xvi) $d\left(\log\left(\frac{x}{y}\right)\right) = \frac{ydx - xdy}{xy}$

(xvii) $d\left(\frac{e^x}{y}\right) = \frac{ye^xdx - e^xdy}{y^2}$

SOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): FOR BETTER UNDERSTANDING AND CONCEPT BUILDING OF THE TOPIC

1. Assume that a spherical rain drop evaporates at a rate proportional to its surface area. Form a differential equation involving the rate of change of the radius of the rain drop.

[IIT-1997]

Solution

Let r denote the radius (in mm) of the rain drop after t minutes. Since the radius is decreasing as t increases, the rate of change of r must be negative. If V denote the volume of the rain drop and S its surface area, we have $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$. It is also given that $\frac{dv}{dt} \propto S$.

Hence $\frac{4}{3}\pi . 3r^2 \frac{dr}{dt} = -K . 4\pi r^2$

$\Rightarrow \frac{dr}{dt} = -K$, which is the required differential equation.

2. The marginal cost of manufacturing a certain article is given by $c'(x) = \frac{dc}{dx} = 2 + 0.15x$. Find the total cost $c(x)$, given that $c(0) = 100$.

[MP-1993]

Solution

Given $c'(x) = \frac{dc}{dx} = 2 + 0.15x$ ..... (i)

integrating the equation

$c(x) = 2x + \frac{0.15x^2}{2} + p$ ..... (ii)

Now if $x = 0$, then

$c(0) = 2\times 0 + \frac{0.15}{2}\times 0^2 + p$

$\Rightarrow c(0) = p = 100$ [$\because c(0) = 100$]

Hence $p = 100$

Substituting the value of p in (ii) we get

$c(x) = 2x + 0.075x^2 + 100$

3. Solve the differential equation $\cos^2 x \frac{d^2y}{dx^2} = 1$.

[MP-95, 98, 99, 2004]

Solution

Given differential equation is $\cos^2 x \frac{d^2y}{dx^2} = 1$.

$\Rightarrow \frac{d^2y}{dx^2} = \frac{1}{\cos^2 x} \Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \sec^2 x$

Integrating w.r.t. x, we get

$$\int \frac{d}{dx}\left(\frac{dy}{dx}\right)dx = \int \sec^2 x\,dx \Rightarrow \frac{dy}{dx} = \tan x + c_1$$

Integrating again w.r.t. x, we get

$$\int \frac{dy}{dx}dx = \int \tan x\,dx + c_1\int dx + c_2$$

$\Rightarrow y = \log_e \sec x + c_1 x + c_2$

This is the required solution.

4. Solve the differential equation $\frac{d^2y}{dx^2} = xe^x$ ***[MP-99, 2000, 2004]***

Solution

Given differential equation is $\frac{d^2y}{dx^2} = xe^x$

$$\Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = xe^x$$

Integrating w.r.t. x,

we get $\int \frac{d}{dx}\left(\frac{dy}{dx}\right)dx = \int xe^x dx \Rightarrow \frac{dy}{dx}$

$= xe^x - e^x + c_1$

Integrating again w.r.t. x, we get

$$\int \frac{dy}{dx}dx = \int xe^x dx - \int e^x dx + c_1\int dx$$

$\Rightarrow y = xe^x - e^x - e^x + c_1 x + c_2 \Rightarrow y$
$xe^x - 2e^x + c_1 x + c_2$

This is the required solution.

5. Solve the differential equation $\frac{d^2y}{dx^2} = \frac{x-1}{(2x-x^2)^{3/2}}$ ***[CBSE-03]***

Solution

Given $\frac{d^2y}{dx^2} = \frac{x-1}{(2x-x^2)^{3/2}}$ it can be written as

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{x-1}{(2x-x^2)^{3/2}}$$

on integrating, we get

$$\frac{dy}{dx} = \int \frac{x-1}{(2x-x^2)^{3/2}}dx + c_1$$

$$= \frac{-1}{2}\int 2x-x^2)^{-3/2}(2-2x)dx + c_1$$

$$= \frac{-1}{2}\frac{(2x-x^2)^{-1/2}}{-1/2} + c_1 = \frac{1}{\sqrt{2x-x^2}} + c_1$$

On integrating once again, we get

$$y = \int\left(\frac{1}{\sqrt{2x-x^2}} + c_1\right)dx + c_2$$

$$\Rightarrow y = \int \frac{1}{\sqrt{2x-x^2}}dx + c_1 x + c_2$$

$$\Rightarrow y = \int \frac{1}{\sqrt{1^2-(x-1)^2}}dx + c_1 x + c_2$$

$$\Rightarrow y = \sin^{-1}\left(\frac{x-1}{1}\right) + c_1 x + c_2$$

$\Rightarrow y$ $\sin^{-1}(x-1) + c_1 x + c_2$ where c_1, c_2 are arbitrary constants.

6. The surface area of a balloon being inflated changes at a constant rate. If initially, its radius is 3 units and after 2 seconds it is 5 units. Find the radius after t seconds. ***[CBSE-1994]***

Solution

Let S be the surface area of a balloon after t seconds. Then

$$S = 4\pi r^2 \Rightarrow \frac{dS}{dt} = 8\pi r\frac{dr}{dt} \quad \text{..... (i)}$$

It is given $\frac{dS}{dt}$ = constant = k (say)

$$\Rightarrow 8\pi r\frac{dr}{dt} = k \text{ [using (i)]} \Rightarrow 8\pi r\,dr = k\,dt$$

Integrating both sides $\int 8\pi r dr = \int k dt$

$$\Rightarrow \frac{8\pi r^2}{2} = kt + c \Rightarrow 4\pi r^2 = kt + c \quad \text{..... (ii)}$$

Now $r = 3$, when $t = 0$ and $r = 5$ when $t = 2$

$\therefore$ $36\pi = k(0) + c$ and $100\pi = 2k + c$

$\Rightarrow c = 36\pi$ and $k = 32\pi$

Put these values of c and k in (ii), we obtain

$4\pi r^2 = 32\pi t + 36\pi$

$\Rightarrow r^2 = 8t + 9 \Rightarrow r = \sqrt{8t+9}$

7. A wet porous substance in the open air loses its moisture at a rate proportional to the moisture contents. If a sheet hung in the open loses half the moisture during the first hour, when will it has lost 95%, weather condition remaining the same. ***[CBSE-1998]***

Solution

Let m_0 be the moisture content initially and m be the moisture content after t hours.

Given $\frac{dm}{dt} = -km$, where k is constant of proportionality ($k > 0$).

$\Rightarrow \frac{dm}{dt} = -kdt$ ..... (i)

Integrating both sides, we get

$$\int \frac{dm}{m} = -k\int dt \Rightarrow \log m = -kt + c \quad \text{..... (ii)}$$

when $t = 0$, $m = m_0$ $\therefore$ $\log m_0 = -k(0) + c$

$\Rightarrow$ $c = \log m_0$. Put this value of c in (ii), we get $\log m = -kt + \log m_0$

$\Rightarrow \log\left(\frac{m}{m_0}\right) = -kt$ ..... (iii)

when $t = 1$, $m = \frac{1}{2}m_0 \Rightarrow \frac{m}{m_0} = \frac{1}{2}$

$\therefore$ from (iii), we get $\log\left(\frac{1}{2}\right) = -k(1)$

$\Rightarrow$ $k = \log 2$ put this value in (iii), we get

$$\log\left(\frac{m}{m_0}\right) = -\log 2.t \quad \text{..... (iv)}$$

Again when the sheet loses 95% of the moisture i.e., when $m = \frac{5}{100}m_0$, from equation (iv), we get $\log\left(\frac{5}{100}\right) = -\log 2.t$

$\Rightarrow$ $\log 1 - \log 20 = -\log 2.t$

$\Rightarrow$ $-\log 20 = -\log 2.t$

$\Rightarrow$ $t = \frac{\log 20}{\log 2}$ hours.

8. The line normal to a given curve at each point (x, y) on the curve passes through the point (2, 0). If the curve contains the point (2, 3), find its equation. ***[CBSE-1999]***

Solution

The equation of the normal to a curve at point (x, y) is $Y - y \quad \frac{-1}{(dy/dx)}(X - x)$

If it passes through (2, 0), then

$$0 - y = \frac{-1}{\left(\frac{dy}{dx}\right)}(2 - x) \Rightarrow y\,\frac{dy}{dx} = 2 - x$$

$\Rightarrow y\,dy = (2 - x)\,dx$

Integrating both sides $\int y dy = \int (2 - x)dx$

$\Rightarrow \frac{y^2}{2} = 2x - \frac{x^2}{2} + C$ ..... (i)

Since the curve passes through (2, 3),

So putting $x = 2$, $y = 3$ in (i), we obtain

$$\frac{9}{2} = 4 - 2 + C \Rightarrow C = \frac{5}{2}$$

Putting $C = \frac{5}{2}$ in (i), we get $y^2 = 4x - x^2 + 5$

Which is the required equation of the curve.

9. A population grows at the rate of 5% per year. How long does it take for the population to double? ***[CBSE-97]***

Solution

Let P_0 be the initial popilation and let the population after t years be P. Then, $\frac{dp}{dt} = \frac{5p}{100}$ (Given)

$\Rightarrow \frac{dp}{dt} = \frac{5p}{20}$

$\Rightarrow \frac{dp}{p} = \frac{1}{20}dt$ [By seprating the variables]

$\Rightarrow \int \frac{1}{p}dp = \frac{1}{20}\int 1.dt$ [Integrating both sides]

$\Rightarrow \log p = \frac{1}{20}\,t + c$ ..... (1)

At $t = 0$, we have $p = p_0$

$\therefore$ $\log p_0 = \frac{1\times 0}{20} + C$ [Puting $t = 0$ and $p = p_0$ In (1)] $\Rightarrow C = \log p_0$

Subst. $C = \log p_0$ in (1)

$\log p = \frac{1}{20}\,t + \log p_0$

$\Rightarrow \log\frac{p}{p_0} = \frac{1}{20}t \Rightarrow t = 20\log\left(\frac{p_0}{p}\right)$

When $p = 2p_0$ we have $t = 20\log\left(\frac{p_0}{2p_0}\right)$

$= 20\log\left(\frac{1}{2}\right)$

10. The surface area of a balloon, being inflated, changes at a rate proportional to the time t, if initially its radius is 1 unit and after 1 second it is 3 units, find the radius after t seconds.

[CBSE-94]

Solution

Let s be the surface area and r the radius of the balloon after t seconds. According to the given condition, $\frac{ds}{dt} \propto t \Rightarrow \frac{ds}{dt} = kr$, where k is the constant of proportionality

$$\Rightarrow \frac{d}{dt}(4\pi r^2) = kt \Rightarrow 8\pi r\frac{dr}{dt} = kt$$

$$\Rightarrow 8\pi r dr = kt dt \quad \Rightarrow \quad r dr = \frac{k}{8\pi} t dt$$

$$\Rightarrow \int r dr = \frac{k}{8\pi}\int t dt \quad \Rightarrow \quad \frac{r^2}{2} = \frac{k}{8\pi}.\frac{t^2}{2} + c$$

Now, $r = 1$ when $t = 0$ and $r = 3$ when $t = 1$

$$\therefore \quad \frac{1}{2} = \frac{k}{8\pi} \times 0 + c \Rightarrow c = \frac{1}{2} \text{ and}$$

$$\frac{9}{2} = \frac{k}{4\pi} \times \frac{1}{2} + c$$

$$\Rightarrow \quad \frac{9}{2} = \frac{k}{16\pi} + \frac{1}{2}, \left(\because c = \frac{1}{2}\right)$$

$$\Rightarrow k = 64\pi$$

Putting the values of k and c in equation (1), we get $\frac{r^2}{2} = \frac{64\pi}{8\pi} \times \frac{t^2}{2} + \frac{1}{2} \Rightarrow r^2 = 8t^2 + 1 \Rightarrow \sqrt{8t^2+1}$

$$\Rightarrow r = \sqrt{8t^2 + 1}$$

Hence, the required radius is $\sqrt{8t^2+1}$ units.

11. $\frac{d^2y}{dx^2} = \log x$, given that $x = 1, y$ 1 and $\frac{dy}{dx} = 1$.

[CBSE-2003]

Solution

Given $\frac{d^2y}{dx^2} = \log x$

Integrating w.r.t. x, we get $\frac{dy}{dx} = x\log x - x + c$ ..... (i)

Again, integrating the w.r.t. x

$$y = \int x\log x dx - \int x dx + \int c dx + c_1$$

$$= \frac{x^2}{2}\log x - \frac{x^2}{4} - \frac{x^2}{2} + cx + c_1 \quad \text{..... (ii)}$$

Putting $x = 1$, $\frac{dy}{dx} = 1$ in equation (i), we get $c = 2$

Again putting $x = 1$ and y 1 and $c = 2$ in equaion (ii) we get $c_1 = 1/2$

$$\therefore y = \frac{x^2}{2}\log x - \frac{x^2}{4} - \frac{x^2}{2} + 2x + \frac{1}{2}$$

12. If $y = 3e^{2x} + 2e^{3x}$, prove that $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

[CBSE-2007]

Solution

Given y $3e^{2x} + 2e^{3x}$ ..... (i)

differentiating w.r.t. x

$$\frac{dy}{dx} = 6e^{2x} + 6e^{3x} \quad \text{..... (ii)}$$

$$\frac{d^2y}{dx^2} = 12e^{2x} + 18e^{3x} \quad \text{..... (iii)}$$

$$\therefore \text{L.H.S.} = \frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y$$

$$= 12e^{2x} + 18e^{3x} - 30e^{2x} - 30e^{3x} + 18e^{2x} + 12e^{3x} = 0$$

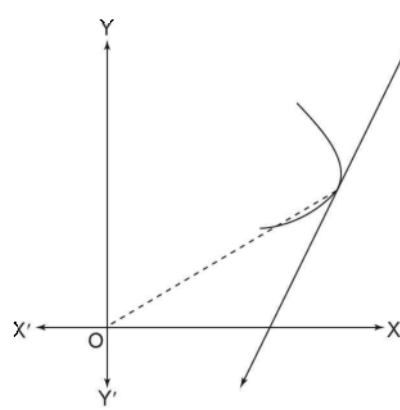
Proved.

13. The slope of the tangent at any point of a curve is λ times the slope of the straight line joining the point of contact to the origin. Form the differential equation representing the problem and hence find the equation of the curve.

[CBSE-97, 99]

Solution

Let $P(x, y)$ be any point on the given curve. Then,



Slope of the tangent at $P = \lambda$ (Slope of the line OP)

$$\Rightarrow \frac{dy}{dx} = \lambda\left(\frac{y-0}{x-0}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\lambda y}{x}$$

This is the required differential equation.

Now, $$\frac{dy}{dx} = \frac{\lambda y}{x} \Rightarrow \frac{dy}{y} = \frac{\lambda dx}{x}$$

Integrating on both sides

we get $\log y = \log_e x + \log c$

$\Rightarrow y = cx^{\lambda}$ is required equation of curve

UNSOLVED SUBJECTIVE PROBLEMS (XII BOARD (C.B.S.E./STATE)): TO GRASP THE TOPIC SOLVE THESE PROBLEMS

1. $e^{(dy/dx)} = x + 1; y(0) = 5$ ***[Bihar CEE-98]***

2. $(1 + e^{2x})\, dy + (1 + y^2)\, e^x\, dx = 0$ given that $x = 0, y = 1$ ***[CBSE-96, 2004, 2005]***

3. $(1 + y^2)\,(1 + \log x)\, dx + x\, dy = 0$ given that when $x = 1, y = 1$. ***[CBSE-1996]***

4. $\frac{dy}{dx} = y\cot 2x, y\left(\frac{\pi}{4}\right) = 2$ ***[CBSE-93]***

5. Solve: $\frac{d^2y}{dx^2} = x$ ***[MP-2007]***

6. $(y^3 + 1)\,(e^x + xe^x)\, dx - xe^x y^2\, dy = 0$ ***[HSB-2000]***

7. $(3xy + y^2)\, dx + (x^2 + xy)\, dy = 0$ ***[HPSB-95; MP-2000; CBSE-90C; HSB-97, 2001]***

8. $(x^2 - y^2)dx + 2xy\, dy = 0$ given that $y = 1$, when $x = 1$. ***[CBSE-92, 95]***

9. Solve the differential equation $\frac{d^2y}{dx^2} + \sec^2 x$ $= e^x$ given that $\frac{dy}{dx} = 1$ and y 1 when $x = 0$. ***[CBSE-2001]***

10. $\frac{d^2y}{dx^2} + \sin x = x$ given that $\frac{dy}{dx} = 1$ and $y = 1$ when $x = 0$. ***[CBSE-2001]***

11. $\frac{d^2y}{dx^2} = x\cos x$ ***[CBSE-2002C]***

12. The slope of a curve at any point is half of the reciprocal of the ordinate at that point and it passes through the point (4, 3). Prove that the equation of the curve is $y^2 = x + 5$. ***[CBSE-99]***

13. If $y = x\tan x$, prove that $\cos^3 x\frac{d^2y}{dx^2} - 2y + 2x = 0$ ***[CBSE-2007]***

14. The surface area of a balloon, being inflated, changes at a rate proportional to the time t, if initially its radius is 1 unit and after 1 second it is 3 units, find the radius after t seconds. ***[CBSE-94]***

ANSWERS

1. $y = x \log (x + 1) - x + \log (x + 1) + 5$
2. $y = \frac{1}{e^x}$
3. $y = \tan\left\{\frac{\pi}{4} + \frac{1}{2} - \frac{1}{2}(1 + \log x)^2\right\}$
4. $y = \pm 2\sqrt{\sin 2x}$
5. $y = \frac{x^3}{6} + cx + c_1$
6. $x + \log|x| - \frac{1}{3}\log|y^3 + 1| = c$
7. $x^2 y (2x + y) = c$
8. $x^2 + y^2 = 2x$
9. $y \quad e^x - \log \sec x$
10. $y \quad \frac{x^3}{6} + \sin x + 1$
11. $y \quad -x \cos x + 2 \sin x + c_1 x + c_2$
14. $\sqrt{8t^2 + 1}$ units.

SOLVED OBJECTIVE PROBLEMS: HELPING HAND

1. Solution of the equation $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$, when $y(0) = 1$, is

[UPSEAT-96; MNR-96]

(a) $x^3 = y^3 \sin x$
(b) $x^3 = 3 y^3 \sin x$
(c) $y^3 = 3x^3 \sin x$
(d) None of these

Solution

(b) Given equation is $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$

$\Rightarrow x^3 \frac{dy}{dx} - x^2 y = -y^4 \cos x$

$\Rightarrow -\frac{1}{y^4}\frac{dy}{dx} + \frac{1}{y^3}\frac{1}{x} = \frac{1}{x^3}\cos x$

Now on putting $\frac{1}{y^3} = v$, it becomes $\frac{1}{3}\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x^3}\cos x$

$\Rightarrow \frac{dv}{dx} + \frac{3}{x}v = \frac{3}{x^3}\cos x$

This is linear in v, so I.F. $= e^{\int 3/x dx} = x^3$

$\therefore$ solution is $v(x^3) = \int x^3 \frac{3}{x^3}\cos x dx + c$

$\Rightarrow \frac{x^3}{y^3} = 3\sin x + c$

But when $x = 0, y \quad 1$ so $c = 0$. Hence particular solution is $x^3 = 3y^3 \sin x$.

2. The differential equation of all non-vertical lines in a plane is

[AIEEE-2002]

(a) $\frac{d^2 y}{dx^2} = 0$ (b) $\frac{d^2 x}{dy^2} = 0$
(c) $\frac{dy}{dx} = 0$ (d) $\frac{dx}{dy} = 0$

Solution

(a) In a plane general solution of all non-vertical lines is $ax + by = 1$ $(b \neq 0)$

$\Rightarrow a + b\frac{dy}{dx} = 0 \Rightarrow b\frac{d^2 y}{dx^2} = 0 \Rightarrow \frac{d^2 y}{dx^2} = 0$

3. The differential equation $3\frac{dy}{dx}+\frac{3y}{x}=2x^4y^4$ is not linear. The integrating factor to solve this equation is ***[NDA-2003]***

(a) $1/x^2$ (b) $1/x^3$

(c) $1/x^4$ (d) x^4

Solution

(b) Equation can be written as

$$3y^{-4}\frac{dy}{dx}+3\frac{y^{-3}}{x}=2x^4$$

Now put $y^{-3}=v \Rightarrow -3y^{-4}\frac{dy}{dx}=\frac{dv}{dx}$

So transformed equation will be $-\frac{dv}{dx}+3\frac{v}{x}=2x^4$

$\Rightarrow \frac{dv}{dx}-\frac{3}{x}v=-2x^4$ which is linear equation.

So I.F. $= e^{\int -\frac{3}{x}dx}=\frac{1}{x^3}$

4. Solution of the equation $x\left(\frac{dy}{dx}\right)^2+(y-x)\frac{dy}{dx}-y=0$ is ***[UPSEAT-1997]***

(a) $(x-y+c)(2xy-c)=0$

(b) $(x-y+c)(xy-c)=0$

(c) $(x+y+c)(xy-c)=0$

(d) $(x+y+c)(2xy-c)=0$

Solution

(b) Given equation can be written as $\left(\frac{dy}{dx}-1\right)\left(x\frac{dy}{dx}+y\right)=0$

$\Rightarrow \frac{dy}{dx}=1$ or $x\frac{dy}{dx}+y \quad 0$

Now $\frac{dy}{dx}=1 \Rightarrow y=x+c$ and $x\frac{dy}{dx}+y=0$

$\Rightarrow \frac{dy}{y}+\frac{dx}{x}=0$

$\Rightarrow xy \quad c$

Hence solution is $(x-y+c)(xy-c)=0$.

5. A normal is drawn at a point $P(x, y)$ of a curve. It meets x-axis at Q. If PQ is of constant length k, then the differential equation describing such a curve is ***[IIT-94; Pb. (CET)-2004]***

(a) $y\frac{dy}{dx}=\pm\sqrt{k^2-y^2}$

(b) $x\frac{dy}{dx}=\pm\sqrt{k^2-x^2}$

(c) $y\frac{dy}{dx}=\pm\sqrt{y^2-k^2}$

(d) $x\frac{dy}{dx}=\pm\sqrt{x^2-k^2}$

Solution

(a) Given length of the normal $= k$. So

$$k=y\sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

$$\Rightarrow \frac{k^2}{y^2}=1+\left(\frac{dy}{dx}\right)^2$$

$$\Rightarrow y\frac{dy}{dx}=\pm\sqrt{k^2-y^2} \text{ or } y^2\left(\frac{dy}{dx}\right)^2=k^2-y^2$$

which is the required differential equation.

6. The solution of $dy=\cos x(2-y\ \text{cosec}\ x)dx$ where $y=2$ when $x=\frac{\pi}{2}$ is ***[J & K-2005]***

(a) $y=\sin x+\text{cosec}\ x$

(b) $y=\tan\frac{x}{2}+\cot\frac{x}{2}$

(c) $y=\frac{1}{\sqrt{2}}\sec\frac{x}{2}+\sqrt{2}\cos\frac{x}{2}$

(d) None of these

Solution

(a) $\frac{dy}{dx}=2\cos x-y\cot x \Rightarrow \frac{dy}{dx}+y\cot x=2\cos x$

I.F. $= e^{\int \cot x dx}=\sin x$

$y\,.\,\sin x=\int 2\cos x.\sin x dx+c$

$y\sin x=\sin^2 x+c$ at $y=2$ and $x=\frac{\pi}{2}$, $c=1$;

$y\sin x=\sin^2 x+1$

or $y=\sin x+\text{cosec}\ x$

7. Equation of curve passing through (3, 9) which satisfies the differential equation $\frac{dy}{dx} = x + \frac{1}{x^2}$, is ***[WB JEE-1986]***

(a) $6xy = 3x^2 - 6x + 29$

(b) $6xy = 3x^3 - 29x + 6$

(c) $6xy = 3x^3 + 29x - 6$

(d) None of these

Solution

(c) $\frac{dy}{dx} = x + \frac{1}{x^2}$

$\Rightarrow \int dy = \int \left(x + \frac{1}{x^2} \right) dx \Rightarrow y = \frac{x^2}{2} - \frac{1}{x} + c$

Since it passes through (3, 9), therefore

$9 = \frac{9}{2} - \frac{1}{3} + c \Rightarrow c = \frac{29}{6}$

$\therefore \quad y = \frac{x^2}{2} - \frac{1}{x} + \frac{29}{6}$

$\Rightarrow \; 6xy = 3x^3 + 29x - 6.$

8. A function $y = f(x)$ has a second order derivatives $f''(x) = 6(x-1)$. If its graph passes through the point (2, 1) and at that point the tangent to the graph is $y = 3x - 5$, then the function is ***[AIEEE-2004]***

(a) $(x+1)^3$ (b) $(x-1)^3$

(c) $(x+1)^2$ (d) $(x-1)^2$

Solution

(b) Given $f''(x) = 6(x-1)$

$f'(x) = 3(x-1)^2 + c_1$ ..... (i)

But at point (2, 1) the line $y \quad 3x - 5$ is tangent to the graph $y \quad f(x)$. Hence $\left.\frac{dy}{dx}\right|_{x=2} = 3$ or $f'(2) = 3$

Then from (i) $f'(2) = 3(2-1)^2 + c_1$

$3 = 3 + c_1 \Rightarrow c_1 = 0$ i.e., $f'(x) = 3(x-1)^2$

Given $f(2) = 1$

$f(x) = (x-1)^3 + c_2 \Rightarrow f(2) = 1 + c_2$

$\Rightarrow \; 1 = 1 + c_2$

$\Rightarrow \; c_2 = 0$

Hence $f(x) = (x-1)^3$.

9. The solution of the differential equation $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ is ***[MP PET-2003]***

(a) $y = \log x + c_1 x + c_2$

(b) $y = -\log x + c_1 x + c_2$

(c) $y = -\frac{1}{x} + c_1 x + c_2$

(d) None of these

Solution

(a) $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$. Now integrating both sides, we get

$\frac{dy}{dx} = \frac{1}{x} + c_1 \Rightarrow y = \log x + c_1 x + c_2$.

10. The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$ is ***[AIEEE-2002]***

(a) $\frac{1}{4} e^{2x}$

(b) $\frac{1}{4} e^{-2x} + cx + d$

(c) $\frac{1}{4} e^{-2x} + cx^2 + d$

(d) $\frac{1}{4} e^{-2x} + c + d$

Solution

(b) $\frac{d^2y}{dx^2} = e^{-2x}$

Integrating both sides, we get $\frac{dy}{dx} = \frac{e^{-2x}}{-2} + c$

Again integrate, we get $y = \frac{e^{-2x}}{4} + cx + d$

11. A slope of the tangent at (x, y) to a curve passing through $\left(1, \frac{\pi}{4}\right)$ is given by $\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$, then the equation of the curve is ***[Kurukshetra CEE-2002]***

(a) $y = \tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$

(b) $y = x\tan^{-1}\left[\log\left(\frac{x}{e}\right)\right]$

(c) $y = x\tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$

(d) None of these

Solution

(c) We have $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Putting y υx so that $\frac{dy}{dx} = \upsilon + x\frac{d\upsilon}{dx}$, we get

$$\upsilon + x\frac{d\upsilon}{dx} = \upsilon - \cos^2\upsilon \text{ or } \frac{d\upsilon}{\cos^2\upsilon} = -\frac{dx}{x}$$

On integrating, we get $\tan\upsilon = -\log x + \log c$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log C$$

This passes through $\left(1, \frac{\pi}{4}\right)$, therefore $1 = \log c$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log e$$

$$\Rightarrow y = x\tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$$

12. The rate of increase of bacteria in a certain culture is proportional to the number present. If it double in 5 hours then in 25 hours, its number would be

[Pb. CET-2004]

(a) 8 times the original
(b) 16 times the original
(c) 32 times the original
(c) 64 times the original

Solution

(c) Let P_0 be the initial population and let the population after t years be P. Then $\frac{dP}{dt} = kP \Rightarrow \frac{dP}{P} = kdt$

On integrating, we have $\log P = kt + c$

At $t = 0$, $P = P_0$

$\therefore \log P_0 = 0 + c;$

$\therefore \log P = kt + \log P_0$

$\Rightarrow \log \frac{P}{P_0} = kt$

when $t = 5$ hrs, $P = 2P_0$

$\therefore \log \frac{2P_0}{P_0} = 5k \Rightarrow K = \frac{\log 2}{5};$

$\therefore \log\frac{P}{P_0} = \frac{\log 2}{5}t$

when $t = 25$ hours, we have

$$\log\frac{P}{P_0} = \frac{\log 2}{5} \times 25 = 5\log 2 = \log 32;$$

$\therefore P = 32P_0$

13. The solution of the differential equation $x^4\frac{dy}{dx} + x^3y + \text{cosec}(xy) = 0$ is equal to

[Pb. CET-2004]

(a) $2\cos(xy) + x^{-2} = c$

(b) $2\cos(xy) + y^{-2} = c$

(c) $2\sin(xy) + x^{-2} = c$

(d) $2\sin(xy) + y^{-2} = c$

Solution

(a) $x^4\frac{dy}{dx} + x^3y + \text{cosec}\,(xy) = 0$

$x^4\,dy + x^3y\,dx + \text{cosec}\,(xy)\,dx = 0$

$x^3\,(xdy + y\,dx) + \text{cosec}\,(xy)\,dx = 0$

$x^3\,d\,(xy) + \text{cosec}\,(xy)\,dx = 0$

$$\frac{d(xy)}{\text{cosec}(xy)} + \frac{dx}{x^3} = 0$$

Integrating both sides, $\int\frac{d(xy)}{\text{cosec}(xy)} + \int\frac{dx}{x^3} = 0$

$$\int\sin(xy)d(xy) + \int x^{-3}dx = 0$$

$$-\cos\,(xy) + \left(\frac{x^{-2}}{-2}\right) = c;\ 2\cos\,(xy) + x^{-2} = c.$$

14. The solution of the equation $x^2\frac{d^2y}{dx^2} = \ln x$, when $x = 1$, $y = 0$ and $\frac{dy}{dx} = -1$ is

[Orissa JEE-2003]

(a) $\frac{1}{2}(\ln x)^2 + \ln x$ (b) $\frac{1}{2}(\ln x)^2 - \ln x$

(c) $-\frac{1}{2}(\ln x)^2 + \ln x$ (d) $-\frac{1}{2}(\ln x)^2 - \ln x$

Solution

(d) $\frac{d^2y}{dx^2}=\frac{\log x}{x^2}\Rightarrow\frac{dy}{dx}=\frac{-(\log x+1)}{x}+c$

At $\frac{dy}{dx}=-1, x=1, y=0, \therefore c=0$

$\Rightarrow y=-\int\frac{\log x+1}{x}dx=-\frac{1}{2}(\log x)^2-\log x$

15. If $y(t)$ is solution of $(t+1)\frac{dy}{dt}-ty \quad 1, y(0)$ $=-1$. At $t=1$ the solution is

[IIT-Screening-2003]

(a) $e+\frac{1}{2}$ (b) $-\frac{1}{2}$

(c) $\frac{1}{2}$ (d) $e-\frac{1}{2}$

Solution

(b) $\frac{dy}{dt}-\frac{t}{t+1}y=\frac{1}{t+1}$, which is linear in y

$\int Pdt=\int-\frac{t}{t+1}dt=\int\left(-1+\frac{1}{t+1}\right)dt$

$=-t+\log(t+1)$

I.F. $=e^{\int Pdt}=e^{-t}.(t+1)$.

Multiplying by I.F. and integrating

$ye^{-t}(t+1)=\int e^{-t}(t+1)\frac{1}{(t+1)}dt+c=-e^{-t}+c$

$\therefore y(t+1)=-1+ce^{t}$. Given at $t=0, y \quad -1$

$\therefore -1=-1+c$ or $c=0$

$\therefore y=\frac{-1}{t+1}+0 \quad \therefore$ at $t=1, y \quad -\frac{1}{2}$

16. Solve $(1+x^2)\frac{dy}{dx}+xy \quad x^3y^3$

[Bihar C.E.E.-1999]

Solution

$\frac{1}{y^3}\frac{dy}{dx}+\frac{x}{1+x^2}.\frac{1}{y^2}=\frac{x^3}{1+x^2}$

Put $-\frac{1}{2}.\frac{1}{y^2}=v; \therefore \frac{1}{y^3}\frac{dy}{dx}=\frac{dv}{dx}$

$\therefore \quad \frac{dv}{dx}-\frac{2x}{1+x^2}.v=\frac{x^3}{1+x^2}$ I.F. $=e^{\int\frac{-2x}{1+x^2}dx}$

$=e^{-\log_e(1+x^2)}=\frac{1}{1+x^2}$

$\therefore$ D.E. is $v.\frac{1}{(1+x^2)}=\int\frac{x^3}{(1+x^2)^2}dx+c$

Put $1+x^2=t \therefore v.\frac{1}{t}=\frac{1}{2}\left[\log t+\frac{1}{t}\right]+c$ etc.

17. Solution of $\left(\frac{dy}{dx}\right)^2-x\frac{dy}{dx}+y=0$ is

[IIT-1999]

(a) $y=2$ (b) $y=2x$

(c) $y \quad 2x-4$ (d) $y \quad 2x^2-4$

Solution

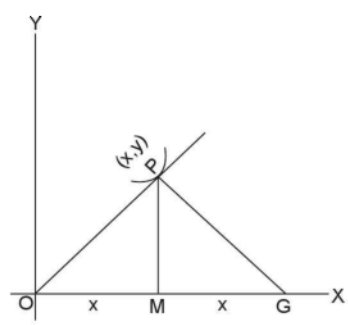
(c) Putting $\frac{dy}{dx}=p$, the above equation is $y=px-p^2$. It is Clairaut form.

Hence its solution is $y \quad cx-c^2$ where c is any arbitrary constant.

Choosing $c=2$, the required solution is $y=2x-4$

18. The normal PG to a curve meets the axis in G. If the distance of G from the origin is twice the abscissa of P, prove that the curve is a rectangular hyperbola.

Solution



$OG=2x$ or $OM+MG=2x$

or $x+MG=2x; \therefore MG=x$

But MG is sub-normal $=y\frac{dy}{dx} \therefore y\frac{dy}{dx}=x$

or $y\,dy=x\,dx$. Integrate.

$\therefore \quad x^2-y^2=a^2$ which is rectangular hyperbola.

19. A particle p moves so that its velocities parallel to the axes of x and y are respectively $-ky$ and kx where k is a constant different from zero. Find the path of the particle, if it passes through the point (3, 4).

Solution

$$\frac{dy}{dt} = kx \text{ and } \frac{dx}{dt} = -ky \text{ given}$$

$$\therefore \quad \frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = -\frac{x}{y}$$

$\therefore$ $y\,dy + x\,dx = 0$. Integrating, we get

$$\frac{y^2}{2} + \frac{x^2}{2} = c$$

or $x^2 + y^2 = 2c$

It passes through (3, 4)

$\therefore$ $9 + 16 = 2c$ or $2c = 25$

Hence the path is $x^2 + y^2 = 25$.

20. Find the equation of the curve which passes through the origin and the tangent to which at every point (x, y) has slope equal to $\dfrac{x^4 + 2xy - 1}{1 + x^2}$ ***[Roorkee-2001]***

Solution

$$\tan\psi = \frac{dy}{dx} = \frac{x^4 + 2xy - 1}{1 + x^2} = x^2 - 1 + \frac{2xy}{x^2 + 1}$$

$$\therefore \quad \frac{dy}{dx} - \frac{2x}{1 + x^2}\,y = x^2 - 1; \text{ Linear}$$

$$\text{I.F.} = -\log(1 + x^2) = \frac{1}{1 + x^2}$$

Multiplying both sides by I.F. and integrating

$$y.\frac{1}{x^2 + 1} = \int \frac{x^2 - 1}{x^2 + 1}\,dx + c = y.\frac{1}{x^2 + 1} = \int \frac{x^2 - 1}{x^2 + 1}$$

$dx + c$

$$\text{or } \int\left(1 - \frac{2}{x^2 + 1}\right) = x - 2\tan^{-1}x + c$$

Since it passes through (0, 0) $\therefore$ $c = 0$. Hence y $(x^2 + 1)(x - 2\tan^{-1}x)$.

21. A curve $y = f(x)$ passes through the point $P(1, 1)$. The normal to the curve at P is $a(y - 1) + (x - 1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, then the equation of curve is ***[IIT-1996]***

(a) $y \quad e^{k(x-1)}$ (b) $y \quad e^{kx}$

(c) $y = e^{k(x\ \ 2)}$ (d) None of these

Solution

(a) Let $a \neq 0$ Slope of the normal at (1, 1)

$$= -\frac{1}{a}$$

Slope of the tangent at (1, 1) = a i.e.,

$$\left(\frac{dy}{dx}\right)_{(1,1)} = a$$

Since $\dfrac{dy}{dx}$ is proportional to y

$$\Rightarrow \quad \frac{dy}{dx} = ky \Rightarrow \frac{dy}{y} = kdx$$

$$\Rightarrow \quad \log y = kx + c \Rightarrow y = e^{kx\ 1\ c} = Ae^{kx}$$

It passes through (1, 1)

$$\therefore \quad 1 = Ae^{k} \therefore A = e^{-k} \therefore y = e^{-k}.\,e^{kx} = e^{k(x-1)}$$

22. The acceleration after t sec. of a particle which starts from rest and moves in a straight lines is $\left(5 - \dfrac{t}{4}\right)$ cm/s². The velocity of the particle at the instant when the acceleration is zero is

(a) 20 cm/sec (b) 40 cm/sec

(c) 50 cm/sec (d) 55.5 cm/s

Solution

(c) $$\frac{dv}{dt} = 5 - \frac{t}{4} \Rightarrow v = 5t - \frac{t^2}{8} + c$$

$$(t = 0, v = 0, \text{then } c = 0) \Rightarrow \frac{dv}{dt} = 0 \Rightarrow t = 20 \text{ sec}$$

$$v\text{/at } t = 20 \text{ sec} = 5 \times 20 - \frac{400}{8} = 100 - 50$$

$= 50$ cm/sec

23. The degree of the differential equation satisfying the relation

$$\sqrt{1 + x^2} + \sqrt{1 + y^2} = \lambda\left(x\sqrt{1 + y^2} - y\sqrt{1 + x^2}\right) \text{is:}$$

[DCE-2007]

(a) 1 (b) 2

(c) 3 (d) None of these

Solution

(a) Putting $x = \tan A$, and $y = \tan B$ in the given relation

we get $\cos A + \cos B = \lambda(\sin A - \sin B)$

$$2\cos\left(\frac{A + B}{2}\right)\cos\left(\frac{A - B}{2}\right)$$

$$= \lambda\left(2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)\right)$$

$$\Rightarrow \tan\left(\frac{A-B}{2}\right) = \frac{1}{\lambda}$$

$$\Rightarrow \tan^{-1}x - \tan^{-1}y = 2\tan^{-1}\left(\frac{1}{\lambda}\right)$$

Differentiating w.r.t. x, we get

$$\frac{1}{1+x^2} - \frac{1}{1+y^2}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{1+y^2}{1+x^2}$$

Clearly, it is a differential equation of degree 1.

24. The value of $ydx + (x - y^3)dy = 0$ is: ***[MPPET-2010]***

(a) $xy = \frac{y^4}{4} + c$ (b) $x = \frac{y}{4} + c$

(c) $x = \frac{y^2}{4} + c$ (d) None of these

Solution

(a) Step 1: $(x - y^3)dy = -ydx$

$$\frac{dx}{dy} = -\frac{x}{y} + y^2$$

$$\frac{dx}{dy} + \frac{x}{y} = y^2 \quad \text{..... (i)}$$

Linear differential equation in x

$\therefore$ Integrating factor = I.F. = $e^{\int P dy}$

$$= e^{\int \frac{1}{y}dy} = e^{\log y} = y$$

Step 2: Solution of (i) is given by

$$(x)(I.F.) = \int Q \times I.F.dy + c$$

$$xy = \int y^2 \times ydy + c$$

$$xy = \frac{y^4}{4} + c$$

25. (a) The slope of the curve at any point is the reciprocal of twice the ordinate at the point. The curve also passes through the point (4, 3). Find the equation.

Solution

Slope of the curve = slope of tangent

$$= \tan\psi = \frac{dy}{dx} = \frac{1}{2y} \text{ (given)}$$

$\therefore$ $2y\,dy = dx$ i.e., $y^2 = x + c$.

It passes through (4, 3).

$\therefore$ $9 = 4 + c$ i.e., $c = 5$

$\therefore$ Equation is $y^2 = x + 5$.

(b) Show that the equation of the curve whose slope at any point is equal to $y + 2x$ and which passes through origin is y 2 $(e^x - x - 1)$.

Solution

$\frac{dy}{dx}$ = y + $2x$ or $\frac{dy}{dx}$ − y $2x$, Linear,

I.F. = $e^{\int dx} = e^{-}$

$ye^{-x} = \int e^{-x}.2xdx$

$ye^{-x} = -2\,e^{-x}(x + 1) + c$,

It passes through (0, 0)

$\therefore$ $c = 2, y = 2(e^x - x - 1)$.

(c) The gradient of a curve passing through (4, 0) is given by $\frac{dy}{dx} - \frac{y}{x} + \frac{5x}{(x+2)(x-3)} = 0$.

Determine the equation of the curve. Also find y when $x = 5$.

Solution

$$\text{I.F} = e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$$

required D.E is $y.\frac{1}{x} = \int \frac{1}{x}.\frac{3x}{(x+2)(x-3)}dx$

$$\frac{y}{x} = \int\left(\frac{1}{x-3} - \frac{1}{x+2}\right)dx$$

$$\Rightarrow \frac{y}{x} = \log(x-3) - \log(x+2) + \log k$$

$\frac{y}{x} = \log\frac{x+2}{x-3}.k$ where $0 = \log 6k$ $\therefore$ $6k = 1$

Also $\frac{y}{5} = \log\frac{7}{2}k$

$\therefore$ $y = 5\log\frac{7}{2}.\frac{1}{6} = 5\log\frac{7}{12}$

26. A curve passing through the point (1, 1) has the property that the perpendicular distance of the origin from the normal at any point p of

the curve is equal to the distance of p from the x-axis.Determine the equation of the curve. ***[IIT-1999]***

Solution

Normal is $Y - y = -\frac{1}{y'}(X - x)$ or $y'Y + X = yy' + x$ perp. from $(0, 0) = y$ (given)

$$\therefore \frac{yy' + x}{\sqrt{(1 + y'^2)}} = y \text{ or } x^2 + 2xy\frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2 = y^2 + y^2\left(\frac{dy}{dx}\right)^2$$

or $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$ Homogeneous. Put $y = vx$

$$\therefore v + x\frac{dv}{dx} = \frac{v^2 - 1}{2v} \text{ or } x\frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = -\frac{1 + v^2}{2v}$$

$$\therefore \frac{2v}{1 + v^2}dv = -\frac{dx}{x} \Rightarrow \log(1 - v^2) - \log x - \log k$$

or $x\left(1 + \frac{y^2}{x^2}\right) = k$ or $x^2 + y^2 = kx$. It passes through $(1, 1)$ $\therefore 2 = k$

$\therefore x^2 + y^2 = 2x$ is the required equation.

27. A curve passes through $(2, 0)$ and the slope of tangent at point $P(x, y)$ equals $\frac{1}{y - \sqrt{x}}$. Find the equation of the curve and area enclosed by the curve and the x-axis in the fourth quadrant. ***[IIT-2004]***

Solution

Given slope of tangent at $(x, y) = \frac{(x+1)^2 + y - 3}{(x+1)}$

Put $x + 1 = X$, $y - 3 = Y$ $\therefore \frac{dy}{dx} = \frac{dY}{dX}$

$$\therefore \frac{dY}{dX} = X + \frac{Y}{X} \text{ or } \frac{dY}{dX} - \frac{Y}{X} = X$$

Above is linear differential equation.

$$\therefore \text{I. F.} = e^{\int -\frac{1}{X}dX} = e^{-\log X} = \frac{1}{X}$$

Multiplying both sides by I.F. and integrating

$$Y \cdot \frac{1}{X} = \int X \cdot \frac{1}{X}dX + c = X + c \text{ or } \frac{y - 3}{x + 1} = (x + 1) + c.$$

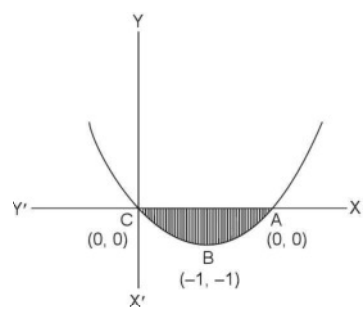
The curve passes through the point $(2, 0)$

$$\therefore -\frac{3}{3} = 3 + c \text{ or } c = -4.$$ Hence the required equation is

$y - 3 = (x + 1)^2 - 4(x + 1)$

or $y = x^2 - 2x$ or $y + 1 = (x - 1)^2$

Above represents a parabola with vertex at $(1, -1)$ which meets the x-axis at the points $(0, 0)$ and $(2, 0)$. Its shape is as shown in the figure below.



Required area in fourth quadrant is

$$\int_0^2 y\,dx = \int_0^2 (x^2 - 2x)dx = \left(\frac{x^3}{3} - x^2\right)_0^2 = \frac{4}{3}$$

sq. units.

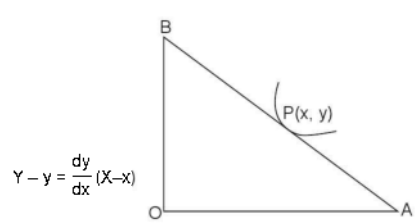
28. A curve C has the property that if the tangent drawn at any point P on C meets the co-ordinate axis at A and B, then P is the mid-point of AB. If the curve passes through the point $(1, 1)$ then equation of curve is ***[IIT-1998]***

(a) $xy = 2$ (b) $xy = 3$

(c) $xy = 1$ (d) None of these

Solution

(c) Tangent at $P(x, y)$ is



It meets the x-axis at A and y-axis at B.

$\therefore A$ is $\left(x - y\frac{dx}{dy}, 0\right)$ and B is $\left(0, y - x\frac{dy}{dx}\right)$.

Since P is the mid point of AB,

$\therefore \quad 2x = x - y\frac{dx}{dy}$

$\Rightarrow \quad -x = y\frac{dx}{dy} \quad \Rightarrow \quad \frac{dx}{dy} = -\frac{y}{x}$

$\Rightarrow \quad \frac{dy}{y} + \frac{dx}{x} = 0 \quad \Rightarrow \log y + \log x = \log c$

$\Rightarrow \log(xy) = c \Rightarrow xy = c$

It passes through $(1, 1) \therefore c = 1$

$\therefore$ Equation of curve is $xy \quad 1$

29. The solution of the differential equation $ydx + (x + x^2y)dy = 0$ is: ***[MPPET-2009]***

(a) $\frac{1}{xy} + \log x = c$ (b) $-\frac{1}{xy} + \log x = c$

(c) $\frac{1}{xy} + \log y = c$ (d) $-\frac{1}{xy} + \log y = c$

Solution

(d) Given, $ydx + (x + x^2y)dy = 0$

$\Rightarrow \quad \frac{ydx + xdy}{(xy)^2} + \frac{1}{y}dy = 0$

$\Rightarrow \quad \frac{d}{dx}\left(-\frac{1}{xy}\right) + \frac{1}{y}dy = 0$

On integrating, we get

$$-\frac{1}{xy} + \log y = c$$

30. Solution of the differential equation $\cos x dy = y(\sin x - y)dx, 0 < x < \frac{\pi}{2}$ is:

[AIEEE-2010]

(a) $y \sec x = \tan x + c$ (b) $y \tan x = \sec x + c$

(c) $\tan x = (\sec x + c)y$ (d) $\sec x = (\tan x + c)y$

Solution

(d) $\cos x dy - y \sin x dx = -y^2 dx$

$\cos x dy + yd(\cos x) = -y^2 dx$

$\frac{d(y\cos x)}{y^2 \cos^2 x} = -\frac{dx}{\cos^2 x}$

$\frac{1}{-y\cos x} = -\tan x + c \quad -\sec x = y(-\tan x + c)$

$\sec x = y(\tan x + k)$ (Here $k = -c$)

Hence correct option is (d).

31. Let f be a real-valued differentiable function on R (the set of all real numbers) such that $f(1) = 1$. If the y-intercept of the tangent any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P, then the value of $f(-3)$ is equal to. ***[IIT (1)-2010]***

(a) 3 (b) 0

(c) 1 (d) 9

Solution

(d) $Y - y = m(X - x)$

y-intercept $(X = 0)$

$y = y - mx$

Given that $y - mx = x^3 \Rightarrow x\frac{dy}{dx} - y = -x^3$

$\Rightarrow \quad \frac{dy}{dx} - \frac{y}{x} = -x^2$

Integrating factor $e^{-\int \frac{1}{x}dx} = \frac{1}{x}$

$y.\frac{1}{x} = \int \frac{1}{x}.(-x^2)dx \Rightarrow f(x) = y = -\frac{x^3}{2} + cx$

Given $f(1) = 1 \Rightarrow c = \frac{3}{2}$

$\therefore \quad f(x) = -\frac{x^2}{2} + \frac{3x}{2} \Rightarrow f(-3) = 9$

32. If $xdy = y(dx + ydy)$, $y > 0$ and $y(1) = 1$, then $y(-3)$ is equal to ***[IIT Screening-2005]***

(a) 1 (b) 3

(c) 5 (d) -1

Solution

(b) $xdy = y(dx + y\,dy)$

$\Rightarrow \quad \frac{xdy - ydx}{y^2} = dy \Rightarrow -d\left(\frac{x}{y}\right) = dy$

Integrating both sides, we get $\frac{x}{y} + y = c$

$\because \quad y(1) = 1$

$\Rightarrow \quad c = 2;$

$\therefore \quad \frac{x}{y} + y = 2$

For $x = -3$, $y^2 - 2y - 3 = 0$

$\Rightarrow \quad y = -1$ or $3 \Rightarrow y = 3 \ (\because y > 0)$

OBJECTIVE PROBLEMS: IMPORTANT QUESTIONS WITH SOLUTIONS

1. The solution of the diff. equation $x\frac{d^2y}{dx^2} = 1$, given that $y = 1$, $\frac{dy}{dx} = 0$ when $x = 1$, is
   (a) $y \quad x \log x + x + 2$
   (b) $y = x \log x - x + 2$
   (c) $y = x \log x + x$
   (d) $y \quad x \log x - x$

2. The solution of the equation $\frac{dy}{dx} = \frac{(1+x)y}{(y-1)x}$ is
   [AISSE-1986; AICBSE-1982; MPPET-2004]
   (a) $\log xy + x + y \quad c$
   (b) $\log (x/y) + x - y \quad c$
   (c) $\log xy + x - y = c$
   (d) None of these

3. The solution of $\frac{d^2y}{dx^2} = \cos x - \sin x$ is
   (a) $y = -\cos x + \sin x + c_1x + c_2$
   (b) $y = -\cos x - \sin x + c_1x + c_2$
   (c) $y \quad \cos x - \sin x + c_1x^2 + c_2x$
   (d) $y \quad \cos x + \sin x + c_1x^2 + c_2x$

4. Equation of curve passing through (3, 9) which satisfies the differential equation $\frac{dy}{dx} = x + \frac{1}{x^2}$, is ***[W.B. JEE-1986]***
   (a) $6xy = 3x^2 - 6x + 29$
   (b) $6xy = 3x^3 - 29x + 6$
   (c) $6xy \quad 3x^3 + 29x - 6$
   (d) None of these

5. The equation of the curve which passes through the point (1, 1) and whose slope is given by $2y/x$, is
   [Roorkee-1987]
   (a) $y \quad x^2$ (b) $x^2 - y^2 = 0$
   (c) $2x^2 + y^2 = 3$ (d) None of these

6. For what value of k, does the differential equation $\frac{dy}{dx} = ky$ represent the law of natural decay? ***[NDA-2007]***
   (a) 0.01 (b) $(10)^{-1}$
   (c) -5 (d) 0

7. What is the degree of the differential equation $k\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^3\right]^{3/2}$, where k is a constant?
   [NDA-2007]
   (a) 3 (b) 4
   (c) 1 (d) 2

8. The differential equation of all circles in the first quadrant which touch the coordiante axes is of order
   (a) 1 (b) 2
   (c) 3 (d) None of these

9. The differential equation of all parabolas whose axes are parallel to y-axis is
   (a) $\frac{d^3y}{dx^3} = 0$ (b) $\frac{d^2x}{dy^2} = c$
   (c) $\frac{d^2y}{dx^2} + \frac{d^2x}{dy^2} = 0$ (d) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = c$

10. A particle moves in a straight line with a velocity given by $\frac{dx}{dt} = x + 1$ (x is the distance described). The time taken by a particle to traverse a distance of 99 metres is
    (a) $\log_{10}e$ (b) $2\log_e 10$
    (c) $2\log_{10}e$ (d) $\frac{1}{2}\log_{10}e$

11. The differential equation $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \sin y + x^2 = 0$ is of the following type
    (a) Linear
    (b) Homogeneous
    (c) order two
    (d) Degree one

12. The second order differential equation is
    (a) $y'^2 + x = y^2$
    (b) $y'y'' + y = \sin x$
    (c) $y''' + y'' + y \quad 0$
    (d) $y' = y$

13. The solution of $y' - y = 1$, $y(0) = -1$ is given by $y(x) =$ ***[MP PET-2000]***
    (a) $-\exp(x)$ (b) $-\exp(-x)$
    (c) -1 (d) $\exp(x) - 2$

14. If $x^2 + y^2 = 1$ then $\left(y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}\right)$

[IIT Screening-2000]

(a) $y'' - 2(y')^2 + 1 = 0$
(b) $yy'' + (y')^2 + 1 = 0$
(c) $yy'' - (y')^2 - 1 = 0$
(d) $yy'' + 2(y')^2 + 1 = 0$

15. The differential equation $(x^2y^3 + xy^2 + y)dx + (x^3y^2 - x^2y + x)dy = 0$ is not exact The integrating factor in order to convert it into exact from is ***[NDA-2003]***

(a) x^2y^2 (b) $1/x^2y^2$
(c) $2x^2y^2$ (d) $1/2x^2y^2$

16. If $\frac{d^2y}{dx^2} = 4x - 3x^2$ and $\frac{dy}{dx} = 2$ and $y = 3$. when $x = 0$, then the value of y when $x = 1$ is

(a) 5 (b) 5/12
(c) $5\frac{5}{12}$ (d) $5\frac{7}{12}$

17. The solution of the differential equation $(xy^2 + x)\,dx + (yx^2 + y)\,dy \quad 0$ is

(a) $(x^2 + 1)(y^2 + 1) = c$
(b) $e^{x^2} + e^{y^2} = c$
(c) $(y^2 + 1) = c(x^2 + 1)$
(d) None of these

18. The equation of the curve through the point (1, 0) and whose slope is $\frac{y-1}{x^2+x}$ is

[Orissa JEE-2007]

(a) $2x + (y - 1)(x + 1) = 0$
(b) $2x - (y - 1)(x + 1) = 0$
(c) $2x + (y - 1)(x - 1) = 0$
(d) None of these

19. Solution of the equation $x\left(\frac{dy}{dx}\right)^2 + 2\sqrt{xy}\frac{dy}{dx} + y = 0$ is

[MPPET-2005]

(a) $x + y = a$ (b) $\sqrt{x} - \sqrt{y} = \sqrt{a}$
(c) $x^2 + y^2 = a^2$ (d) $\sqrt{x} + \sqrt{y} = \sqrt{a}$

20. The solution of $\frac{dy}{dx} + 1 = \text{cosec}(x + y)$ is

[MPPET-2005]

(a) $\cos(x \quad y) + x = c$
(b) $\cos(x + y) = c$
(c) $\sin(x - y) + x = c$
(d) $\sin(x \quad y) + \sin(x \quad y) = c$

21. $(x^2 + y^2)dy \quad xydx$. If $y(x_0) = e$, $y(1) = 1$, then value of $x_0 =$ ***[IIT-Screening-2005]***

(a) $\sqrt{3}e$ (b) $\sqrt{e^2 - \frac{1}{2}}$
(c) $\sqrt{\frac{e^2 - 1}{2}}$ (d) $\sqrt{\frac{e^2 + 1}{2}}$

22. The first integral of the differential equation $x\,dx + y\,dy = a^2(x\,dy - y\,dx)/(x^2 + y^2)$ is

[MP PET-2007]

(a) $x^2 + y^2 = a^2 \cdot \tan^{1}(x/y) + c$
(b) $x^2 + y^2 + 2a^2 \cdot \tan^{1}(x/y) = c$
(c) $a^2(x^2 + y^2) \cdot \tan^{-1}(x/y) = c$
(d) None of these

23. The solution of the differential equation $(x + y)(dx - dy) = dx + dy$ is

[MPPET-2005; NDA-2007]

(a) $x - y = ke^{x-y}$ (b) $x + y = ke^{x+y}$
(c) $x + y = k(x - y)$ (d) $x + y = ke^{x-y}$

SOLUTIONS

1. (b) $\frac{d^2y}{dx^2} = \frac{1}{x}$, integrating, we get $\frac{dy}{dx} = \ln x + c_1$

at $x = 1; \frac{dy}{dx} = 0 \Rightarrow 0 = \ln_1 + c_1 \Rightarrow c_1 = 0$

Now, $\frac{dy}{dx} = \ln x, \int dy = \int \ln x dx$

$y = x\ln x - x + c_2$, at $x = 1; y = 1$

$1 = 0 - 1 + c_2 \Rightarrow c_2 = 2; \therefore y = x\ln x - x + 2$

2. (b) Rearranging equation and integrating, we get

$$\int \frac{y-1}{y}dy = \int \frac{1+x}{x}dx$$

$$\int\left(1-\frac{1}{y}\right)dy=\int\left(\frac{1}{x}+1\right)dx$$

$$\Rightarrow y-\ln y=\ln x+x+c$$

$$\ln x+\ln y+x-y=-c\Rightarrow \ln xy+x-y=k \qquad (k=-c)$$

3. (a) Integrating we get $\frac{dy}{dx}=\sin x-(-\cos x)+c_1$

Now $\int dy=\int(\sin x+\cos x+c_1)dx$

$$y=-\cos x+\sin x+c_1x+c_2$$

4. (b) $dy=\left(x+\frac{1}{x^2}\right)dx$, integrating we get

$$y=\frac{x^2}{2}-\frac{1}{x}+c$$

at $x=3;\ y=9;\ 9=\frac{3^2}{2}-\frac{1}{3}+c\Rightarrow c=\frac{29}{6}$

$$\therefore\quad y=\frac{x^2}{2}-\frac{1}{x}+\frac{29}{6}\Rightarrow 6xy=3x^3-6+29x$$

5. (a) Slope $=\frac{dy}{dx}=\frac{2y}{x},\int\frac{dy}{y}=\int 2\frac{dx}{x}$

$$\ln y=2\ln x+\ln c\Rightarrow \ln y=\ln(x^2c)$$

$y=x^2c$, $\because$ it passes through (1, 1)

$$\therefore\quad 1=1^2c\Rightarrow c=1,\therefore y=x^2$$

6. (c) For decay, $\frac{dy}{dx}$ must be –ve, $\therefore k<0$

Only option (c) is possible.

7. (d) Squaring we get $\left(k\frac{d^2y}{dx^2}\right)^2=\left(1+\left(\frac{dy}{dx}\right)^3\right)^3$

$\therefore$ degree = 2

($\because$ degree is defined for polynomial differential equation, power on derivative must be natural no.)

8. (a) Centre will be (r, r), radius r

$\therefore$ equation will be

$$(x-r)^2+(y-r)^2=r^2;\ r>0$$

$\because$ No. of parameter = 1 = order of D.E.

9. (a) Equation of parabola will be $y=ax^2+bx+c$

($\because$ Order = No. of arbitrary const. = 3)

Differentiating thrice we get $\frac{d^3y}{dx^3}=0$

10. (b) $\frac{dx}{x+1}=dt$ is Differential Equation

Integrating, $\ln(x+1)\Big|_0^{99}=t\Big|_0^1$

$$\ln(100)=t\Rightarrow t=2\log_e^{10}$$

11. (c, d) It is obvious

12. (b) By definition

13. (c) $\frac{dy}{dx}-y=1\Rightarrow\frac{dy}{dx}=y+1$

$$\int\frac{dy}{y+1}=\int dx,\quad \ln(y+1)=x+c$$

or $y+1=e^{(x+c)}$

at $x=0=e^{0+c}$

$\therefore\quad y+1=0\Rightarrow y=-1$

14. (b) Differentiating, $2x+2y\frac{dy}{dx}=0$

or $x+y\frac{dy}{dx}=0$

Again differentiating, we get

$$1+\left(\frac{dy}{dx}\right)^2+y\frac{d^2y}{dx^2}=0$$

$$1+(y')^2+y(y'')=0$$

15. (b) $(x^2y^3+xy^2+y)dx+(x^3y^2-x^2y+x)dy=0$

$$x^2y^2(ydx+xdy)+xy(ydx-xdy)+ydx+xdy=0$$

$$\Rightarrow\quad ((xy)^2+1)d(xy)+xy^3\left(\frac{ydx-xdy}{y^2}\right)=0$$

$\Rightarrow$ Dividing both sides by $(xy)^2$

$$\left(1+\frac{1}{(xy)^2}d(xy)\right)+\frac{y}{x}d\left(\frac{x}{y}\right)=0$$

$$\therefore\quad \text{I.F.}=\frac{1}{x^2y^2}$$

16. (c) $\frac{d^2y}{dx^2} = 4x - 3x^2$

Integrating we get $\frac{dy}{dx} = \frac{4x^2}{2} - \frac{3x^2}{3} + c_1$

$$\frac{dy}{dx} = 2x^2 - x^3 + c_1$$

at $x = 0, \frac{dy}{dx} = 2$

$\therefore$ $2 = 0 - 0 + c_1 \Rightarrow c_1 = 2$

$$\int dy = \int (2x^2 - x^3 + 2)dx$$

$$y = \frac{2x^3}{3} - \frac{x^4}{4} + 2x + c_2$$

at $x = 0;\ y = 3$

$$3 = C_2;\ y = \frac{2x^3}{3} - \frac{x^4}{4} + 2x + 3$$

$\therefore$ at $x = 1;\ y = \frac{2}{3} - \frac{1}{4} + 2 + 3 = \frac{65}{12}$ or $5\frac{5}{12}$

17. (a) $(xy^2 + x)dx + (yx^2 + y)dy = 0$

$$xy(ydx + xdy) + xdx + ydy = 0$$

$$xyd(xy) + xdx + ydy = 0$$

Integrating we get $\frac{(xy)^2}{2} + \frac{x^2}{2} + \frac{y^2}{2} = c$

$$x^2y^2 + x^2 + y^2 = 2c$$

Adding 1 on both sides

$$1 + x^2 + y^2 + x^2y^2 = 2c + 1$$

$$(1 + x^2)(1 + y^2) = \text{constant}$$

18. (a) slope $= \frac{dy}{dx} = \frac{y-1}{x^2 + x}$

$$\frac{dy}{y-1} = \frac{dx}{x(x+1)}$$

Integrating, $\ln(y-1) = \ln\left(\frac{x}{x+1}\right) + \ln c$

$$y - 1 = \frac{cx}{x+1}$$

$\because$ It passes through (1, 0)

$$-1 = \frac{c}{2} \Rightarrow c = -2$$

$\therefore$ D.E. is $(x+1)(y-1) = -2x$

19. (d) Dividing both sides by x

$$\left(\frac{dy}{dx}\right)^2 + 2\sqrt{\frac{y}{x}}.\frac{dy}{dx} + \left(\sqrt{\frac{y}{x}}\right)^2 = 0$$

$$\left(\frac{dy}{dx} + \sqrt{\frac{y}{x}}\right)^2 = 0 \Rightarrow \frac{dy}{dx} = \frac{-\sqrt{y}}{\sqrt{x}}$$

$\frac{dy}{\sqrt{y}} = \frac{-dx}{\sqrt{x}}$, integrating

$$2\sqrt{y} = -2\sqrt{x} + c$$

$$\sqrt{x} + \sqrt{y} = \frac{c}{2} = \sqrt{a}$$

20. (a) $\frac{dy}{dx} + 1 = \text{cosec}(x + y)$

Put $x + y = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$\therefore$ $\frac{dt}{dx} = \text{cosec}\, t$

$$\sin t dt = dx$$

integrating we get

$$c - \cos t = x \quad \because \quad x + \cos(x + y) = c$$

21. (a) Differential Equation is $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$

Which is homogeneous

$\therefore$ Put $y = vx$

Also $\frac{dy}{dx} = v + x\frac{dv}{dx}$

Substituting in equation, we get

$$v + x\frac{dv}{dx} = \frac{v}{1 + v^2}$$

$$x\frac{dv}{dx} = \frac{v}{1 + v^2} - v$$

$$x\frac{dv}{dx} = \frac{-v^3}{1 + v^2}$$

$$\left(\frac{1 + v^2}{v^3}\right)dv = \frac{-dx}{x}$$

Integrating $\int\left(\frac{1}{v^3} + \frac{1}{v}\right)dv = -\ln x + c$

$$\frac{-1}{2v^2} + \ln v = -\ln x + c$$

$$\frac{-x^2}{2y^2}+\ln\left(\frac{y}{x}\right)=-\ln x+c$$

$$\frac{-x^2}{2y^2}+\ln y=c$$

also at $x=1, y=1$

$$\Rightarrow \quad c=\frac{-1}{2}$$

$$\frac{-x^2}{2y^2}+\ln y=-\frac{1}{2}$$

at $x=x_0, y=e$

$$\Rightarrow \quad \frac{-x_0^2}{2e^2}+\ln e=-\frac{1}{2}$$

$$\frac{x_0^2}{2e^2}=\frac{3}{2}\Rightarrow x_0=\sqrt{3e}$$

22. (b) $xdx+ydy=a^2\dfrac{(xdy-ydx)}{x^2+y^2}$

$$\frac{d(x^2+y^2)}{2}=a^2y^2\left(\frac{(xdy-ydx)}{1+\left(\frac{x}{y}\right)^2}\right)$$

$$\frac{d(x^2+y^2)}{z}=a^2y^2\frac{-d\left(\frac{x}{y}\right)}{1+\left(\frac{x}{y}\right)^2}$$

$\therefore$ Integrating we get

$$\frac{x^2+y^2}{2}=-a^2\tan^{-1}\left(\frac{y}{x}\right)+c$$

$$x^2+y^2+2a^2\tan^{-1}\left(\frac{y}{x}\right)=2c=\text{constant}$$

23. (d) $(x+y)(dx-dy)=dx+dy$

$dx-dy=\dfrac{d(x+y)}{x+y}$, Integrating $x-y=\log$

$(x-y)+\log c$

$x-y=\log(c(x+y))$

$\Rightarrow \quad c(x+y)=e^{x-y}$

$$x+y=\frac{1}{c}e^{x-y}$$

or $\quad x+y=ke^{x-y}$

$\left(\because k=\dfrac{1}{c}\right)$

UNSOLVED OBJECTIVE PROBLEMS (IDENTICAL PROBLEMS FOR PRACTICE): FOR IMPROVING SPEED WITH ACCURACY

1. What is the differential equation of the curve $y = ax^2 + bx$? ***[NDA-2007]***

(a) $x^2\dfrac{d^2y}{dx^2}-2x\dfrac{dy}{dx}+2y=0$

(b) $x^2\dfrac{d^2y}{dx^2}-y\left(\dfrac{dy}{dx}\right)^2+2=0$

(c) $(1-x^2)\dfrac{d^2y}{dx^2}-\left(y\dfrac{dy}{dx}\right)^2=0$

(d) None of the above

2. If $\dfrac{d^2y}{dx^2}+\sin x=0$, then the solution of the diff. equation is ***[Pb. CET-2001]***

(a) $\sin x$ (b) $\cos x$

(c) $\tan x$ (d) $\log \sin x$

3. Equation of curve through point (1, 0) which satisfies the differential equation $(1 + y^2)dx - xy\,dy \quad 0$, is ***[W.B. JEE-1986]***

(a) $x^2+y^2=1$

(b) $x^2-y^2=1$

(c) $2x^2+y^2=2$

(d) None of these

4. What is the degree of the differential equation $\left(\dfrac{d^3y}{dx^3}\right)^{2/3}-3\dfrac{d^2y}{dx^2}+5\dfrac{dy}{dx}+4=0$?

(a) 3 (b) 2/3

(c) 2 (d) 6

5. A particle starts at the origin and moves along the x-axis in such a way that its velocity at the point $(x, 0)$ is given by the formula $\frac{dx}{dt} = \cos^2 \pi x$. Then the particle never reaches the point on

[AMU-2000]

(a) $x = 1/4$ (b) $x = 3/4$
(c) $x = 1/2$ (d) $x = 1$

6. The solution of the differential equation $(x^2 + 3xy + y^2)\,dx - x^2\,dy \quad 0$ is
(a) $x + \log x = c\,(x \quad y)$
(b) $x + y + \log x = cx$
(c) $x \log x + x = cy$
(d) None of these

7. The diff. equation $\frac{dy}{dx} = f(x, y)$ is a homogeneous equation if $f(x, y)$ is of degree
(a) -1 (b) 0
(c) 1 (d) $n \in N$

8. If c_1, c_2, c_3, c_4, c_5 and c_6 are constants, then the order of the differential equation whose general solution is given by $y = c_1 \cos(x + c_2) + c_3 \sin(x + c_4) + c_5 e^x + c_6$ ***[Kerala PET-2007]***
(a) 6 (b) 5
(c) 4 (d) 3

9. The solution of $\frac{dy}{dx} = 1 + y + y^2 + x + xy + xy^2$ is ***[Kerala PET-2008]***

(a) $\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right) = x + x^2 + c$

(b) $4 \tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right) = \frac{\sqrt{3}}{2}(2x + x^2) + c$

(c) $\sqrt{3} \tan^{-1}\left(\frac{3y+1}{3}\right) = 4(1 + x + x^2) + c$

(d) $4 \tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right) = \sqrt{3}(2x + x^2) + c$

10. The general solution of $ydx - xdy - 3x^2y^2\ e^{x^3} dx = 0$ is
(a) $\frac{x}{y} = e^{x^3} + c$ (b) $\frac{y}{x} = e^{x^3} + c$
(c) $xy = e^{x^3} + c$ (d) $xye^{x^3} + c$

WORKSHEET: TO CHECK THE PREPARATION LEVEL

Important Instructions

1. The answer sheet is immediately below the worksheet.
2. The test is of 15 minutes.
3. The worksheet consists of 15 questions. The maximum marks are 45.
4. Use Blue/Black Ball point pen only for writing particulars/marking responses. Use of pencil is strictly prohibited.

1. If $\frac{dy}{dx}=\frac{xy+y}{xy+x}$, then the solution of the differential equation is ***[SCRA-80]***
   (a) $y = xe^x + c$ (b) $y = e^x + c$
   (c) $y \quad Axe^{x-y}$ (d) $y \quad x + A$

2. $y = 2e^{2x} - e^{-x}$ is a solution of the differential equation ***[Kerala PET-2007]***
   (a) $y_2 + y_1 + 2y = 0$
   (b) $y_2 - y_1 + 2y = 0$
   (c) $y_2 + y_1 = 0$
   (d) $y_2 - y_1 - 2y \quad 0$

3. If differential equation $x(1 - x^2)dy + (2x^2y - y - ax^3)dx = 0$ is linear in y of the type $\frac{dy}{dx} + Py \quad Q$, then $Q =$ ***[PET-98]***
   (a) $\frac{ax^3}{x(1-x^2)}$ (b) $\frac{x(1-x^2)}{2x^2-1}$
   (c) $\frac{2x^2-ax^3}{x(1-x^2)}$ (d) $\frac{2x^2-1}{1-x^2}$

4. The differential equation $\frac{dy}{dx} + Py = Qy^n, n >$ 2 can be reduced to linear from by substituting ***[DCE-1996]***
   (a) $z = y^{n-1}$ (b) $z = y^n$
   (c) $z = y^{n+1}$ (d) $z = y^{1-n}$

5. Under which one of the following conditions does the solution of $\frac{dy}{dx}=\frac{ax+b}{cy+d}$ represent a parabola? ***[NDA-2007]***
   (a) $a = 0, c \neq 0, b \neq 0$
   (b) $a = 1, c = 1$
   (c) $a = 0, c = 0$
   (d) $a = 1, b = 2, c \neq 0$

6. The degree of the differential equation $2\left(\frac{d^2y}{dx^2}\right)+3\left(\frac{dy}{dx}\right)^2+4y^3=x$, is ***[Orissa-JEE-2007]***
   (a) 0 (b) 1
   (c) 2 (d) 3

7. What type of the differential equation $\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2+\sin y+x^2=0$ is? ***[Gujarat CET-2007]***
   (a) whose order is 2 (b) whose degree is 2
   (c) linear (d) homogeneous

8. If $x = \sin t, y \quad \cos pt$, then
   (a) $(1-x^2)y_2 + xy_1 + p^2y = 0$
   (b) $(1-x^2)y_2 + xy_1 - p^2y = 0$
   (c) $(1+x^2)y_2 - xy_1 + p^2y = 0$
   (d) $(1-x^2)y_2 - xy_1 + p^2y = 0$

9. The solution of the equation $x^2\frac{d^2y}{dx^2} = \ln x$, when $x = 1, y = 0$ and $\frac{dy}{dx} = -1$ is ***[Orissa JEE-2003]***
   (a) $\frac{1}{2}(\ln x)^2 + \ln x$ (b) $\frac{1}{2}(\ln x)^2 - \ln x$
   (c) $-\frac{1}{2}(\ln x)^2 + \ln x$ (d) $-\frac{1}{2}(\ln x)^2 - \ln x$

10. The differential equation $\left|\frac{dy}{dx}\right| + |y| + 3 = 0$ admits ***[VITEEE-2008]***
    (a) infinite number of solutions
    (b) no solution
    (c) a unique solution
    (d) many solutions

11. The differential equation $y\frac{dy}{dx} + x = c$ represents ***[Karnataka CET-2008]***

(a) a family of parabolas
(b) a family of circles whose centres are on the x-axis
(c) a family of hyperbolas
(d) a family of circles whose centres are on the y-axis.

12. The solution of the differential equation $x\frac{dy}{dx} = 2y + x^3e^x$, where $y = 0$ when $x = 1$, is *[MP PET-2008]*

(a) $y = x^3(e^x - e)$ (b) $y = x^3(e - e^x)$
(c) $y \quad x^2(e^x - e)$ (d) $y \quad x^2(e - e^x)$

13. The order and degree of the differential equation $5\left(\frac{d^2y}{dx^2}\right)^5 + 4\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{dy}{dx}\right)^3 + 2y + x^3 = 0$ are respectively *[MP PET-2008]*

(a) (2, 5) (b) (3, 2)
(c) (1, 3) (d) (2, 3)

14. $\frac{d^2y}{dx^2} = 0$, then *[UPSEAT-1999]*

(a) $y \quad ax + b$ (b) $y^2 = ax + b$
(c) $y = \log x$ (d) $y = e^x + C$

15. The solution of the differential equation $xdy - ydx = \sqrt{x^2 + y^2}\,dx$ is *[MP PET-2004]*

(a) $x + \sqrt{x^2 + y^2} = cx^2$

(b) $y - \sqrt{x^2 + y^2} = cx$

(c) $x - \sqrt{x^2 + y^2} = cx$

(d) $y + \sqrt{x^2 + y^2} = cx^2$

ANSWER SHEET

1. ⓐ ⓑ ⓒ ⓓ	**6.** ⓐ ⓑ ⓒ ⓓ	**11.** ⓐ ⓑ ⓒ ⓓ
2. ⓐ ⓑ ⓒ ⓓ	**7.** ⓐ ⓑ ⓒ ⓓ	**12.** ⓐ ⓑ ⓒ ⓓ
3. ⓐ ⓑ ⓒ ⓓ	**8.** ⓐ ⓑ ⓒ ⓓ	**13.** ⓐ ⓑ ⓒ ⓓ
4. ⓐ ⓑ ⓒ ⓓ	**9.** ⓐ ⓑ ⓒ ⓓ	**14.** ⓐ ⓑ ⓒ ⓓ
5. ⓐ ⓑ ⓒ ⓓ	**10.** ⓐ ⓑ ⓒ ⓓ	**15.** ⓐ ⓑ ⓒ ⓓ

HINTS AND EXPLANATIONS

2. (d) Differentiating we get

$$y_1 = 2e^{2x}(2) - e^{-x}(-1)$$
$$= 4e^{2x} + e^{-x}$$
$$y_2 = 8e^{2x} - e^{-x}$$

Subtracting y_1 from y_2

$$y_2 - y_1 = 4e^{2x} - 2e^{-x}$$
$$= 2(2e^{2x} - e^{-x})$$
$$y_2 - y_1 = 2y$$
$$\Rightarrow y_2 - y_1 - 2y = 0$$

4. (a) Dividing both sides by $x(1 - x^2)\,dx$

$$\Rightarrow \frac{dy}{dx} + \frac{(2x^2 - 1)y - ax^3}{x(1 - x^2)} = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{(2x^2 - 1)y}{x(1 - x^2)} = \frac{ax^3}{x(1 - x^2)}$$

$$\Rightarrow \frac{dy}{dx} + \frac{2x^2 - 1}{x(1 - x^2)}y = \frac{ax^2}{1 - x^2}$$

$$\therefore Q = \frac{ax^2}{1 - x^2} \quad \text{or} \quad \frac{ax^3}{x(1 - x^2)}$$

6. (a) Differential equation is $(cy + d)\,dy = (ax + b)\,dx$

Integrating we get $\frac{cy^2}{2} + dy = \frac{ax^2}{2} + bx + k$

(k = arbitrary constant)

$ax^2 - cy^2 + 2bx - 2dy + 2k = 0$

It will be a parabola if either $a = 0$ or $c = 0$;

($\because h^2 = ab$ for parabola)

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$$

$\therefore$ From option (a) is correct.

10. (c) $\frac{d^2y}{dx^2} = \frac{\ln x}{x^2}$

Integrating, we get $\frac{dy}{dx} =$

$$\ln x\left(\frac{-1}{x}\right) - \int \frac{1}{x}\left(-\frac{1}{x}\right)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\ln x}{x} + \int \frac{1}{x^2}dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\ln x}{x} - \frac{1}{x} + c$$

at $x = 1$; $\frac{dy}{dx} = -1$

$$-1 = 0 - 1 + c \Rightarrow c = 0$$

$$\frac{dy}{dx} = \frac{-\ln x}{x} - \frac{1}{x}$$

$$\int dy = -\int\left(\frac{\ln x}{x} + \frac{1}{x}\right)dx$$

$$y = \frac{-(\ln x)^2}{2} + \ln x + c$$

at $x = 1$, $y = 0$, $c = 0$

$$\therefore y = \frac{-\ln^2 x}{2} + \ln x$$

12. (b) $y\frac{dy}{dx} + x = c$ or $y\frac{dy}{dx} = c - x$

$$\therefore \int y\,dy = \int (c - x)dx \Rightarrow \frac{y^2}{2} = cx - \frac{x^2}{2} + k,$$

k is arbitrary constant

$x^2 + y^2 - 2cx - 2k = 0$, equation is a circle with centre $(c, 0)$

$\therefore$ For all values of k, centre of circle will remain same,

$\therefore$ Equation represents family of circles whose centre are on x axis.

13. (c) Differential equation is $\frac{dy}{dx} - \frac{2y}{x} = x^2e^x$

$$I.F. = e^{\int \frac{-2}{x}dx} = e^{-2\log_e x} = e^{\log_e x^{-2}} = \frac{1}{x^2}$$

$\therefore$ Solution is $y.\frac{1}{x^2} = \int x^2e^x.\frac{1}{x^2}dx$

$\frac{y}{x^2} = e^x + c$, also $y = 0$ when $x = 1$

$\therefore 0 = e^1 + c \Rightarrow c = -e$

$\therefore$ Equation is $y = x^2(e^x - e)$

16. (d) $xdy - ydx = \sqrt{x^2 + y^2}\,dx$

dividing both sides by x^2, $\frac{xdy - ydx}{x^2} =$

$$\sqrt{1 + \left(\frac{y}{x}\right)^2}\frac{dx}{x}$$

$\frac{d(y/x)}{\sqrt{1 + \left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$, Integrating we get

$$\log\left(\frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}\right) = \ln x + \ln c$$

$$\log\left[\frac{y + \sqrt{x^2 + y^2}}{x}\right] = \ln cx$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = cx^2$$

LECTURE



5

Test Your Skills

ASSERTION/REASONING

Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.

(a) **Assertion** is True, **Reason** is True and **Reason** is a correct explanation for **Assertion**

(b) **Assertion** is True, **Reason** is True and **Reason** is NOT a correct explanation for **Assertion**

(c) **Assertion** is True and **Reason** is False

(d) **Assertion** is False and **Reason** is True

1. **Assertion:** The elimination of four arbitrary constants in $y \quad (c_1 + c_2 + c_3 e^{c_4}) x$ results into a differential equation of the first order $x \frac{dy}{dx} = y$.

   Reason: Elimination of n arbitrary constants requires, in general, a differential equation of the nth order.

2. **Assertion:** $y = a \sin x + b \cos x$ is a general solution of $y'' + y \quad 0$

   Reason: $y \quad a \sin x + b \cos x$ is a trigonometric function.

3. **Assertion:** Order of the differential equation whose solution is $y = c_1 e^{x+c_2} + c_3 e^{x+c_4}$ is 4.

   Reason: Order of the differential equation is equal to the number of independent arbitrary constant mentioned in the solution of differential equation.

4. **Assertion:** The degree of the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} = \ln\left(\frac{d^2y}{dx^2}\right)$ is 2.

   Reason: The degree of a differential equation which can be written as polynomial in the derivatives is the degree of the derivative of the highest order occuring in it.

5. **Assertion:** The differential equation of all circles in a plane must be of order 3.

   Reason: If three points are non collinear, then only one circle always passing through these points.

6. **Assertion:** The differential equation of the form $xf(xy)\,dx + x\phi(xy)\,dy \quad 0$ can be converted to homogeneous forms by substitution $xy = v$.

   Reason: All differential equation of first order and first degree become homogeneous, if we put $y = vx$

7. **Assertion:** The differential equation $\frac{dy}{dx} = \frac{2xy}{x^2 + y^2}$ can be solved by putting $y \quad vx$

   Reason: Since the given differential equation is homogeneous.

8. **Assertion:** The differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$ can be solved by finding I.F. $= e^{\int P\,dx} = e^{\int \frac{1}{x}\,dx} = e^{\log x} = x$. Then solution is $y.x. = \int x^3 dx + c$

Reason: Since the given differential equation is of the form $\frac{dy}{dx} + Py \quad Q$ where P, Q are functions of x.

9. **Assertion:** The differential equation $(xy \sin xy + \cos xy)\, y\, dx + (xy \sin xy - \cos xy)\, x\, dy \quad 0$ can be solved by putting $xy \quad z$
Reason: Since the given differential equation is of the type $f_1(xy)\, y\, dx + f_2(xy)\, x\, dy = 0$

10. **Assertion:** The differential equation $\frac{dy}{dx} - xy = xy^3$ can be solved by dividing it through out by y^3 then by putting $y^{-2} = z$.
Reason: The given differential equation is of the form $\frac{dy}{dx} + Py = Qy^n$ is Bernoulli's form.

11. **Assertion:** Integrating factor of $\frac{dy}{dx} + y = x^2$ is e^x
Reason: Integrating factor of $\frac{dy}{dx} + P(x)y = Q(x)$ is $e^{\int P(x)dx}$

12. **Assertion:** The general solution of $\frac{dy}{dx} + y$ 1 is $ye^x = e^x + c$
Reason: The number of arbitrary constants in the general solution of the differential equation is equal to the order of differential equation.

13. **Assertion:** If the lengths of subtangent and subnormal at point (x, y) on $y \quad f(x)$ are respectively 9 and 4. Then $x = \pm 6$.
Reason: Product of sub tangent and sub normal is square of the ordinate of the point.

14. Let a solution $y = y(x)$ of the differential equation $x\sqrt{x^2-1}dy - y\sqrt{y^2-1}dx = 0$ satisfy $y(2) = \frac{2}{\sqrt{3}}$
Assertion: $y(x) = \sec\left(\sec^{-1} x - \frac{\pi}{6}\right)$
Reason: $y(x)$ is given by $\frac{1}{y} = \frac{2\sqrt{3}}{x} - \sqrt{1-\frac{1}{x^2}}$

ASSERTION/REASONING: SOLUTIONS

1. (a) Let $c_1 + c_2 + c_3 e^{c_4} = A$ (constant)
Then, $y = Ax$
$\Rightarrow \frac{dy}{dx} = A$ or $y = x\frac{dy}{dx}$ or $x\frac{dy}{dx} = y$

2. (b) $\because y = a \sin x + b \cos x$ ..... (i)
$\therefore y' = a \cos x - b \sin x$
$\Rightarrow y'' = -a \sin x - b \cos x = -y$
[from equation (i)]
or $y'' + y \quad 0$

3. (d) $\therefore \quad y = (c_1 e^{c_2} + c_3 e^{c_4})e^x = ce^x$ (say)
$\therefore \frac{dy}{dx} = ce^x = y$
$\therefore$ Order is 1.

4. (d) $\because$ The given equation cannot be written as a polynomial in all the differentials.
$\because$ Degree of the equation is not defined.

5. (a) Let $x^2 + y^2 + 2gx + 2fy + c = 0$
Here three constants.
$\therefore$ Order = 3
$\because$ Circle passes through three noncollinear points, then we get three constants g, f, c.

6. (d) $xy \quad v$
$\therefore \quad x\frac{dv}{dx} + y = \frac{dv}{dx}$
Then, the given equation reduces to
$\frac{v}{x} f(v) + x\phi(v)\left(\frac{1}{x}\left(\frac{dv}{dx} - y\right)\right) = 0$
$\Rightarrow \frac{v}{x} f(v) + \phi(v)\frac{dv}{dx} - y\phi(v) = 0$
$\Rightarrow \left\{\frac{v(f(v) - \phi(v))}{x}\right\} + \phi(v)\frac{dv}{dx} = 0$
$\Rightarrow \frac{dx}{x} + \frac{\phi(v)dv}{v(f(v)) - \phi(v)} = 0$
which is variable seperable form.

7. It is obvious

8. It is obvious

9. It is obvious

10. It is obvious

11. (a) I.F. of $\frac{dy}{dx} + y = x^2$ is $e^{\int 1\,dx}$.

12. (b) $\frac{dy}{dx} + y = 1$

$\Rightarrow \frac{dy}{1-y} = dx$

$\int \frac{dy}{1-y} = \int dx = -\log(1-y) = x$

$1 - y = e^{-x}$; $ye^x = e^x + c$

Order of differential equation is the number of orbitarary constants. Both are true but reason is not correct reason.

13. (d) $\left|\frac{y_1}{m}\right| = 9$ and $|y_1 m| = 4 \Rightarrow |y_1|^2 = 36$

$\Rightarrow y_1 = \pm 6$

Product of subtangent and sub normal is y_1^2. Assertion is false reason is true.

14. (c) It is obvious.

MENTAL PREPARATION TEST

1. Solve $\frac{dy}{dx} = \sec x\,(\sec x + \tan x)$.

[MP-94, 96, 98, 2003]

2. Solve $\frac{dy}{dx} = 1 + y + x + xy$

[MP-98; CBSE-90, 90 C; HSB-92, 97 C]

3. Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$

[MP-97, 2000, 2001; PSB-93 S, 2005]

4. Solve $\frac{dy}{dx} = y \sin 2x$, it being given that $y(0) = 1$. ***[CBSE-97, 2004]***

5. Solve $(x^2 + y^2)\,dx - 2xy\,dy = 0$

[MP-2001, 2006; CBSE-93]

6. Solve $\frac{dy}{dx} + y = \sin x$

[HSB-92, 2001; HPSB-99]

7. Show that the differential equation of which $y \quad 2(x^2 - 1) + c\,e^{-x^2}$ is a solution of $\frac{dy}{dx} + 2xy = 4x^3$. ***[PSB-99 C, 2001; CBSE-94]***

8. Solve $(y + xy)dx + (x - xy^2)\,dy \quad 0$

[CBSE-2002]

9. Solve $\frac{d^2y}{dx^2} = x^2 + \sin 3x$ given that $\frac{dy}{dx} = 1 = y$, when $x = 0$. ***[CBSE-2004]***

10. Solve $\frac{dy}{dx} = y \tan x - 2 \sin x$

[MP-96; CBSE-92; PSB-99]

11. Find the differential equation corresponding to the family of curves $y = Ae^{2x} + Be^{3x}$ where A and B are arbitrary constants.

[CBSE-2000]

12. Solve $\frac{dy}{dx} + \frac{4x}{x^2+1} y = \frac{-1}{(x^2+1)^{3/2}}$

[PSB-97, 99 C, 2001]

13. Solve $\frac{dy}{dx} = \sec^2 x + 3x^2$

[MP-95, 99, 2002]

14. Solve $x \log_e x\,dy - y\,dx = 0$ ***[MP-99]***

15. Show that the function $y = A \cos 2x + B \sin 2x$ is a solution of the differential equation $\frac{d^2y}{dx^2}$ $+ 4y = 0$ ***[CBSE-93, 94 (C)]***

16. Solve $\frac{dy}{dx} - 4y \quad e^x$ ***[MP-98, 2002]***

17. Solve $\frac{dy}{dx} = e^{y+x} + e^{y}x^{2}$

[PSB-91C; CBSE-2006]

18. Solve $\frac{d^2y}{dx^2} + \cos x = x$, given that $y = 1$ and $\frac{dy}{dx} = 0$ when $x = 0$ ***[CBSE-2001]***

19. Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$

[MP-97, 2000, 2001; Kerala-93C; HSB-93, 2001; PSB-94; CBSE-96, 2000, 2008]

20. A population grows at the rate of 10% per year. How long will it take for the population to double?

21. Solve $\frac{dy}{dx} = \frac{x+y+1}{2x+2y+1}$ ***[PSB-97]***

22. Solve $\frac{dy}{dx} + y\cot x = x^2\cot x + 2x$

[CBSE-2005]

23. Form the differential equation representing the family of curves $y \quad A\cos(x + B)$, where A and B are parameters.

[CBSE-94; HSB-97; HPSB-2001]

24. Solve $\frac{dy}{dx} + y = \cos x$

[HSB-97; HPSB-93; CBSE-90, 94]

25. $\frac{dy}{dx} = \frac{y}{x} + \sin\left(\frac{y}{x}\right)$ ***[CBSE-2003]***

26. Show that $y \quad be^{x} + ce^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

[CBSE-93, 2003]

27. Solve $\frac{d^2y}{dx^2} = e^{x} + \cos x$, given that $\frac{dy}{dx} = 1 = y$, when $x = 0$

[CBSE-2005]

28. Solve $(1 + x^2)\frac{dy}{dx} - 2xy \quad (x^2 + 2)(x^2 + 1)$

[CBSE-2005]

29. Solve $y^2 + x^2\frac{dy}{dx} = xy\frac{dy}{dx}$ ***[MP-96, 2008]***

30. Solve $(1 + x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}$

[CBSE-2002; Kerala-95]

31. Solve $\frac{dy}{dx} + y = \frac{1+\sin x}{1+\cos x}$

[HSB-94, 2001]

32. Solve $\frac{dy}{dx} = \cos(x - y) + \sin(x + y)$

[HB-2002]

33. Solve $(x^2 + xy)dy = (x^2 + y^2)\,dx$

[CBSE-94, 2005]

34. Solve $\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$ ***[MP-2008]***

35. Solve $\cos x\frac{dy}{dx} + y \quad \sin x$ ***[MP-2008]***

36. Solve $\frac{dy}{dx} + ay \quad e^{mx}$ ***[MP-2008]***

37. Solve $x^2\frac{dy}{dx} = y^2 + 2xy$. Given that $y = 1$, when $x = 1$. ***[CBSE-2008]***

TOPIC WISE WARMUP TEST

1. The differential equation $3\frac{dy}{dx} + \frac{3y}{x} = 2x^4y^4$ is not linear. The integrating factor to solve this equation is

(a) $1/x^2$ (b) $1/x^3$

(c) $1/x^4$ (d) x^4

2. Solution of differential equation $2xy\frac{dy}{dx} = x^2 + 3y^2$ is

(a) $x^3 + y^3 = px^2$

(b) $\frac{x^2}{2} + \frac{y^3}{x} = y^2 + p$

(c) $x^2 + y^3 = px^2$
(d) $x^2 + y^2 = px^3$

3. The solution of the differential equation $x \frac{dy}{dx} = y(\log y - \log x + 1)$ is
(a) $y = xe^{cx}$ (b) $y + xe^{cy} = 0$
(c) $y + e^x = 0$ (d) None of these

4. Solution of the differential equation $\sqrt{4+x}\frac{dy}{dx} + xy = 0$ is
(a) $y = Ae^{\frac{2}{3}(8-x)\sqrt{4+x}}$
(b) $y = Ae^{\frac{2}{3}(x-8)\sqrt{4+x}}$
(c) $y = Ae^{\frac{2}{3}(x+8)\sqrt{4+x}}$
(d) $y = Ae^{-\frac{2}{3}(8-x)\sqrt{4+x}}$

5. Solution of differential equation $\frac{dy}{dx} + py$ e^{qx} is $(\lambda)y$ $e^{qx} + c\lambda e^{-px}$ then $\lambda =$
(a) $p - q$ (b) $2p + q$
(c) $p + q$ (d) $p + 2q$

6. Solution of differential equation $\frac{dy}{dx} = 2xy$ at $(1, 1)$ $x = 1$, $y = 1$
(a) $y = 2e^{x^2}$ (b) $y = e^{x^2 - 1}$
(c) $y^2 = 2x^2 + 1$ (d) y $x^2 + 1$

7. Solution of the differential equation $3e^x \tan y\, dx + 2(1 - e^x)\sec^2 y\, dy$ 0 is
(a) $(1 - e^x)\tan y$ c
(b) $(1 - e^x)\tan^3 y = c$
(c) $\tan y = c(1 - e^x)^3$
(d) $(\tan y)^2 = c(1 - e^x)^3$
where c is an arbitrary constant

8. An integrating factor for the differential equation $(1 + y^2)\,dx - (\tan^{-1} y - 2xy)\,dy$ 0 is
(a) $(1 + y^2)$ (b) $\tan^{-1} y$
(c) $1/i + y^2$ (d) $e^{\tan^{-1} y}$

9. The solution of the differential equation $(1 + y^2) + (x - e^{\tan^{-1} y})\frac{dy}{dx} = 0$, is
(a) $(x - 2) = ke^{\tan^{-1} y}$
(b) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + k$
(c) $xe^{\tan^{-1} y} = \tan^{-1} y + k$
(d) $xe^{2\tan^{-1} y} = e^{\tan^{-1} y} + k$

10. Solution of differential equation $(1 - x^2)\frac{dy}{dx} - xy = 1$ is
(a) $y\sqrt{1 - x^2} = \sin^{-1} x + c$
(b) $y(1 - x^2) = -\cos^{-1} x + c$
(c) $y(1 + x^2) = \tan^{-1} x + c$
(d) $y\sqrt{1 - x^2} = \cos^{-1} x + c$

11. The solution of $y' = \frac{y+1}{x-1}$, at $x = 1$ is
(a) 2 (b) 1
(c) 0 (d) –1

12. A continuously differentiable function $\phi(x) \in (0, \pi) - \{\pi/2\}$ satisfying $y' = 1 + y^2$, $y(0) = 0 = y(\pi)$ is
(a) $\tan x$ (b) $x(x - \pi)$
(c) $(x - \pi)(1 - e^x)$ (d) Not possible

13. The differential equation which represents the family of plane curves y $\exp(px)$ is
(a) $x \ln y = y\frac{dy}{dx}$
(b) $y \ln y$ $x \cdot \frac{dy}{dx}$
(c) $x\frac{dy}{dx} - \ln y$ 0
(d) $\frac{dy}{dx} = py$

14. Integral curve satisfying $y' = \frac{x^2 + y^2}{x^2 - y^2}$; $y(1) =$ 2 has the slope at the point (4, 3) of the curve, equal to
(a) 25/7 (b) –25/7
(c) 7/25 (d) –7/25

15. The solution of $y' - y = 1$ is
(a) $1 + y = e^x$ (b) $1 + y = ce^x$
(c) $y - 1 = e^x$ (d) $y - 1 = ce^x$

16. The solution of $y' = 1 + x + y^2 + xy^2$ is
(a) $\tan^{-1} y$ $x + \frac{x^2}{2} + c$
(b) $y^2 = \exp\left(x + \frac{x^2}{2}\right) - 1$
(c) $y^2 = 1 + c\exp\left(x + \frac{x^2}{2}\right)$
(d) y $\tan(c + x + x^2)$

17. The slope of the tangent at (x, y) to a curve passing through $(1, \pi/4)$ is given by $\frac{y}{x} - \cos^2\frac{y}{x}$, then the equation of the curve is

(a) $y = \tan^{-1}\left(\log\left(\frac{x}{e}\right)\right)$

(b) $y = x\tan^{-1}\left(\log\left(\frac{x}{e}\right)\right)$

(c) $y = x\tan^{-1}\left(\log\left(\frac{e}{x}\right)\right)$

(d) None of these

18. The solution of the differential equation $9y\frac{dy}{dx} + 4x = 0$ is

[Orissa JEE-2008]

(a) $\frac{y^2}{9} + \frac{x^2}{4} = c$ (b) $\frac{y^2}{4} + \frac{x^2}{9} = c$

(c) $\frac{y^2}{9} - \frac{x^2}{4} = c$ (d) $\frac{y^2}{4} - \frac{x^2}{9} = c$

19. The solution of the differential equation $\frac{d^2y}{dx^2} = e^{-2x}$ is $y = c_1e^{-2x} + c_2x + c_3$ where c_1 is

[Orissa JEE-2008]

(a) 4 (b) 1/2

(c) 1/4 (d) None of these

20. The degree of differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + 6y = 0$ is

[Kerala (Engg.)-2002]

(a) 1 (b) 3

(c) 2 (d) 5

21. The order and degree of the differential equation whose solution is $y = cx + c^2 - 3c^{3/2} + 2$, where c is a parameter, are respectively

(a) 1 and 4 (b) 1 and 3

(c) 1 and 2 (d) None of these

22. $y \quad ae^{mx} + be^{-mx}$ satisfies which of the following differential equations

[Karnataka CET-2002]

(a) $\frac{dy}{dx} - my \quad 0$ (b) $\frac{dy}{dx} + my \quad 0$

(c) $\frac{d^2y}{dx^2} + m^2y \quad 0$ (d) $\frac{d^2y}{dx^2} - m^2y \quad 0$

23. The Elimination of the arbitrary constants A, B and C from $y \quad A + B\,x + Ce^{-x}$ leads to the differential equation ***[AMU-1999]***

(a) $y''' - y' = 0$ (b) $y''' - y'' + y' = 0$

(c) $y''' + y'' = 0$ (d) $y'' + y''' - y' = 0$

24. The general solution of the differential equation $\log(dy/dx) = x + y$ is

[MP PET-1994; DSSE-1984]

(a) $e^x + e^y \quad c$ (b) $e^x + e^{-y} \quad c$

(c) $e^{-x} + e^y \quad c$ (d) $e^{-x} + e^{-y} \quad c$

25. The solution of the differential equation $(x - y^2x)dx = (y - x^2y)\,dy$ is

[AMU-2001; DSSE-1984]

(a) $(1 - y^2) = c^2(1 - x^2)$

(b) $(1 + y^2) = c^2(1 - x^2)$

(c) $(1 + y^2) = c^2(1 + x^2)$

(d) None of these

26. The solution of the differential equation $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ is ***[SCRA-1986]***

(a) $1 + xy + c\,(y - x) = 0$

(b) $x + y = c\,(1 - xy)$

(c) $y - x = c\,(1 + xy)$

(d) $1 + xy \quad c\,(x + y)$

27. The solution of the differential equation $\frac{dy}{dx} = x^2 + \sin 3x$ is ***[DSSE-1981]***

(a) $y = \frac{x^3}{3} + \frac{\cos 3x}{3} + c$

(b) $y = \frac{x^3}{3} - \frac{\cos 3x}{3} + c$

(c) $y = \frac{x^3}{3} + \sin 3x + c$

(d) None of these

28. The solution of the differential equation $\frac{dy}{dx} + \frac{1+\cos 2y}{1-\cos 2x} = 0$

[AISSE-1982; Karnataka CET-2004; AISSE-1983]

(a) $\tan y + \cot x = c$ (b) $\tan y \,.\, \cot x = c$

(c) $\tan y - \cot x = c$ (d) None of these

29. Solution of the equation $\cos x \cos y \frac{dy}{dx} = -\sin x \sin y$ is ***[DSSE-1987]***

(a) $\sin y + \cos x = c$ (b) $\sin y - \cos x = c$

(c) $\sin y\, c \cos x = 0$ (d) $\sin y \quad c \cos x$

30. The solution of the differential equation $(x + y)^2 \frac{dy}{dx} = a^2$ is

(a) $(x \quad y)^2 = \frac{a^2}{2}x + C$

(b) $(x - y)^2 = a^2 x + C$

(c) $(x + y)^2 = 2a^2 x + C$

(d) $y = a\tan^{-1}\left(\frac{x+y}{a}\right) + C$

31. If $y' = \frac{x-y}{x+y}$, then its constant of integration c is when $x = 1, y = 2$

(a) 3 (b) 4

(c) 7 (d) 0

32. What is the equation of the curve pa ssing through the origin and satisfying the differential equation $dy \quad (y\tan x + \sec x)\,dx$?

[NDA-07]

(a) $xy = \cos x$ (b) $y \sin x = x$

(c) $y \quad x\cos x$ (d) $y\cos x = x$

33. Solution of differential equation $x\frac{dy}{dx} = y + x^2$ is

[MP PET-1997]

(a) $y = \log_e x + \frac{x^2}{2} + a$

(b) $y = \frac{x^3}{3} + \frac{a}{x}$

(c) $y \quad x^2 + ax$

(d) None of these

TOPIC WISE WARMUP TEST: SOLUTIONS

1. (b) $\frac{3}{y^4}\frac{dy}{dx} + \frac{3}{y^3}\frac{1}{x} = 2x^4$ Let $v = \frac{1}{y^3} \to \frac{dv}{dx} - \frac{3}{x}v = -2x^4$

$= -2x^4$

I.F. $= e^{\int \frac{-3}{x}dx} = e^{-3\log x} = \frac{1}{x^3}$

2. (d) Putting $y \quad vx \Rightarrow \frac{dy}{dx} = v + x\frac{dV}{dx}$

So, we get $x\frac{dV}{dx} = \frac{1+V^2}{2V} \Rightarrow \frac{2VdV}{1+V^2} = \frac{dx}{x}$

On integrating, we get $x^2 + y^2 = px^3$.

3. (a) From given equation $\frac{dy}{dx} = \frac{y}{x}\left(\log\frac{y}{x} + 1\right)$ which is a homogeneous equation.

So putting $y = Vx$, $\frac{dy}{dx} = V + x\frac{dV}{dx}$ and transformed equation will be $V + x\frac{dV}{dx} = V(\log V + 1)$

$\Rightarrow \frac{dV}{dx} = \frac{V\log V}{x} \Rightarrow \frac{dV}{V\log V} = \frac{dx}{x}$

On integration $\log(\log v) = \log x + \log c$

$\Rightarrow \log(y/x) = cx \Rightarrow y \quad xe^{cx}$

4. (a) Given $\frac{dy}{dx} + \frac{xy}{\sqrt{4+x}} = 0 \Rightarrow \frac{dy}{y} = \frac{-xdx}{\sqrt{4+x}}$

Integrating both sides, $\int \frac{dy}{y} = \int \frac{-x}{\sqrt{4+x}}dx$

$\log y = -\int \frac{x+4-4}{\sqrt{x+4}}dx = -\int\sqrt{4+x}\,dx + \int \frac{4}{\sqrt{4+x}}dx$

$\Rightarrow \log y = -\frac{2}{3}(4+x)^{3/2} + 8\sqrt{4+x} + \log A$

$y = Ae^{-2/3(4+x)^{3/2}+8\sqrt{4-x}} = Ae^{\left[\sqrt{4-x}\left(\frac{-2}{3}(4+x)+8\right)\right]}$

$= Ae^{\left[\sqrt{4+x}\left(\frac{-2x-8+24}{3}\right)\right]} = Ae^{\left[\frac{-2}{3}\sqrt{4+x}(x-8)\right]}$

$\Rightarrow y = Ae^{\left[\frac{2}{3}\sqrt{4-x}(8+x)\right]}$

5. (c) I.F. $= e^{\int pdx} = e^{px}$ $\therefore$ Required solution is given by $y.e^{px} = \int e^{qx}.e^{px}dx = \frac{e^{(p-q)x}}{p+q} + c$

$\Rightarrow y = \frac{e^{qx}}{p+q} + ce^{-px}$

$\Rightarrow y(p+q) = e^{qx} + c(p+q)e^{-px}$ compaired by given solution $\lambda = p + q$.

6. (b) $\int \frac{dy}{y} = \int 2x dx \Rightarrow \log_e y \quad x^2 + C$ at $x = 1, y = 1 \therefore c = -1$

Hence $\log_e y \quad x^2 - 1 \Rightarrow y \quad e^{x^2-1}$.

7. (d) From given equation $\frac{3}{2}\frac{e^x}{1-e^x}dx + \frac{\sec^2 y}{\tan y}dy = 0$

$\Rightarrow \frac{3}{2}\int \frac{e^x}{1-e^x}dx + \int \frac{\sec^2 y}{\tan y}dy = 0$

$\Rightarrow -\frac{3}{2}\log(1-e^x) + \log(\tan y) = \log C$

$\Rightarrow \log\left\{(1-e^x)^{-3/2}\tan y\right\} = \log C$

$\Rightarrow \frac{\tan y}{(1-e^x)^{3/2}} = C$

$\Rightarrow \tan y = C(1-e^x)^{3/2} \Rightarrow (\tan y)^2 = C(1-e^x)^3$ where C is an arbitrary constant.

8. (a) $\frac{dx}{dy} - \frac{\tan^{-1} y - 2xy}{1+y^2} = 0 \Rightarrow \frac{dx}{dy} + \frac{dy}{1+y^2}x = \frac{\tan^{-1} y}{1+y^2}$

which is linear equation. So its $IF = e^{\int \frac{2y}{1+y^2}dy} = 1 + y^2$.

9. (b) From given equation $\frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{e^{\tan^{-1} y}}{1+y^2}$

which is linear in x. So its I.F. $= e^{\int 1/(1+y^2)dy} = e^{\tan^{-1} y}$

$\therefore$ Solution is $x \cdot e^{\tan^{-1} y} = \int\left(\frac{e^{\tan^{-1} y}}{1+y^2}\right)e^{\tan^{-1} y}dy + k$

$\Rightarrow xe^{\tan^{-1} y} = \frac{1}{2}e^{2\tan^{-1} y} + k$

10. (a) Given equation is $\frac{dy}{dx} - \left(\frac{x}{1-x^2}\right)y = \frac{1}{1-x^2}$

which is linear in y, So, I.F. $= e^{-\int x/1-x^2 dx} = e^{\frac{1}{2}\log(1-x^2)} = \sqrt{1-x^2}$

Hence $y(\sqrt{1-x^2}) = \int \sqrt{1-x^2}.\frac{1}{1-x^2}dx$

$\Rightarrow y\sqrt{1-x^2} = \int \frac{1}{\sqrt{1-x^2}}dx$

$\Rightarrow y\sqrt{1-x^2} = \sin^{-1}x + C$.

11. (d) Given $y' = \frac{y+1}{x-1}$

$\Rightarrow \frac{dy}{dx} = \frac{y+1}{x-1} \Rightarrow \frac{dy}{y+1} = \frac{dx}{x-1}$

$\Rightarrow \int \frac{dy}{y-1} = \int \frac{dx}{x-1}$

$\Rightarrow \log(y+1) = \log(x-1) + \log C$

$\Rightarrow \log(y+1) = \log\{C(x-1)\} \Rightarrow y + 1 = C(x-1)$ at $x = 1 \Rightarrow y + 1 = 0 \Rightarrow y \quad -1$.

12. (a) Given $y' = 1 + y^2 \Rightarrow \frac{dy}{dx} = 1 + y^2$

$\Rightarrow \frac{dy}{1+y^2} = dx \Rightarrow \int \frac{dx}{1+y^2} = \int dx$

$\Rightarrow \tan^{-1} y = x + C$ when $x = 0, y = 0$

$\Rightarrow C = 0$ when $x = \pi, y = 0$

$\Rightarrow C = -\pi \therefore \tan^{-1} y \quad x$ or $\tan^{-1} y \quad x - \pi$

Hence $\tan^{-1} y = x \Rightarrow y \quad \tan x$.

13. (b) Given $y = e^{px} \Rightarrow y' = e^{px}.\ p = py$

$\Rightarrow xy' = pxy \Rightarrow xy' = y\log y$ [$\because y = e^{px} \Rightarrow px = \log y$]

14. (a) Given $y' = \frac{x^2+y^2}{x^2-y^2} \Rightarrow \frac{dy}{dx} = \frac{x^2+y^2}{x^2-y^2}$

e

$\therefore$ Slope of the tangent at $(4, 3) = \frac{16+9}{16-9} = \frac{25}{7}$.

15. (b) Given $y' - y = 1 \Rightarrow \frac{dy}{dx} = 1 + y \Rightarrow \frac{dy}{1+y} = dx$

$\Rightarrow \int \frac{dx}{1+y} = \int dx \Rightarrow \log(1+y) = x + \log C$

$\Rightarrow \log(1+y) - \log C = x$

$\Rightarrow \log\left(\frac{1+y}{C}\right) = x$

$\Rightarrow \frac{1+y}{C} = e^x \Rightarrow 1 + y \quad Ce^x$.

16. (a) from given equation $\frac{dy}{dx} = (1+x)(1+y^2)$

$\Rightarrow (1+x)\,dx = \frac{1}{1+y^2}\,dy \Rightarrow \int (1+x)dx$

$= \int \frac{1}{1+y^2}dy \Rightarrow x + \frac{x^2}{2} + C = \tan^{-1} y.$

17. (c) We have $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Putting $y = vx$, so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$, we get

$v + x\frac{dv}{dx} = v - \cos^2 v$ or $\frac{dv}{\cos^2 v} = -\frac{dx}{x}$ on integrating, we get $\tan v = -\log x + \log C$

$\Rightarrow \tan(y/x) = -\log x + \log C$

This passes through $\left(1, \frac{\pi}{4}\right)$, therefore $1 = \log C$

$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log e \Rightarrow y = x\tan^{-1}\left[\log\left(\frac{e}{x}\right)\right]$

18. (b) $9y\frac{dv}{dx} + 4x = 0$

$\Rightarrow 9ydy = -4xdx$ separating variables

$\Rightarrow 9\int ydy = -4\int xdx$ Integrating both sides

$\Rightarrow \frac{9y^2}{2} = -\frac{4x^2}{2} + c_1 \Rightarrow 2x^2 + \frac{9}{2}y^2 = c_1$

$\Rightarrow \frac{x^2}{9} + \frac{y^2}{4} = \frac{c_1}{18} = c \Rightarrow \frac{x^2}{9} + \frac{y^2}{4} = c$

19. (c) $\frac{d^2y}{dx^2} = e^{-2x}$

Integrating both side w.r.t. x, we get

$\frac{dy}{dx} = \int e^{-2x}dx = -\frac{1}{2}e^{-2x} + c_2$

Again integrating, we get

$y = \int\left(-\frac{1}{2}e^{-2x} + c_2\right)dx = \frac{1}{4}e^{-2x} + c_2x + c_3$

Comparing with $y \quad c_1 e^{-2x} + c_2 x + c_3$

$c_1 = 1/4$

20. (a) Clearly, degree = 1.

21. (a) Differentiating, $\frac{dy}{dx} = c$

$\therefore$ Differential equation is

$y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 - 3\left(\frac{dy}{dx}\right)^{3/2} + 2$

$\therefore \left[y - x\frac{dy}{dx} - \left(\frac{dy}{dx}\right)^2 - 2\right] - 3\left(\frac{dy}{dx}\right)^{3/2}$

$\therefore$ squaring we get degree = 4

22. (d) $y = ae^{mx} + be^{-mx}$. Differentiating, we get

$\frac{dy}{dx} = mae^{mx} - mbe^{-mx}$.

Differentiating again, we get

$\frac{d^2y}{dx^2} = m^2ae^{mx} + m^2be^{-mx}$

$= m^2(ae^{mx} + be^{-mx}) = m^2y$ or $\frac{d^2y}{dx^2} - m^2y = 0$.

23. (c)

24. (b) $\log\left(\frac{dy}{dx}\right) = x + y \Rightarrow e^{x+y} = \frac{dy}{dx}$

$\Rightarrow e^x e^y = \frac{dy}{dx}$

$\Rightarrow \int e^x dx = \int \frac{dy}{e^y} \Rightarrow e^x = -e^{-y} + c$

$\Rightarrow e^x + e^{-y} = c.$

25. (a) Given equation can be written as

$\frac{x}{1-x^2}dx = \frac{y}{1-y^2}dy$

On integrating we get

$-\frac{1}{2}\log(1-x^2) = -\frac{1}{2}\log(1-y^2) + \log c$

$\Rightarrow \log(1-x^2) - \log(1-y)^2 = -\log c$

$\Rightarrow \frac{1-x^2}{1-y^2} = c^{-2}$ Hence $(1-y^2) = c^2(1-x^2)$.

26. (c) $\frac{dy}{dx} = \frac{1+y^2}{1+x^2} \Rightarrow \frac{1}{1+y^2}dy = \frac{1}{1+x^2}dx$

Now on integrating both sides, we get

$\tan^{-1} y \quad \tan^{-1} x + \tan^{-1} c$

$\Rightarrow \tan^{-1} y = \tan^{-1}\left(\frac{x+c}{1-cx}\right)$

$\Rightarrow y = \frac{x+c}{1-cx} \Rightarrow y - x = c(1+xy)$.

27. (b) $\frac{dy}{dx} = x^2 + \sin 3x.$

On integrating, $y = \frac{x^3}{3} - \frac{\cos 3x}{3} + c.$

28. (c) $\frac{dy}{dx} = -\frac{1+\cos 2y}{1-\cos 2x} \Rightarrow \frac{dy}{dx} = -\frac{2\cos^2 y}{2\sin^2 x}$

$\Rightarrow \sec^2 y\, dy \quad -\operatorname{cosec}^2 x\, dx$

On integrating both sides, we get

$\tan y \quad \cot x + c \Rightarrow \tan y - \cot x = c.$

29. (d) $\cos x \cos y \frac{dy}{dx} = -\sin x \sin y$

$\Rightarrow \frac{\cos y}{\sin y} dy = -\frac{\sin x}{\cos x} dx$

$\Rightarrow \cot y\, dy = -\tan x\, dx$

On integrating, we get

$\log \sin y = \log \cos x + \log c$

$\Rightarrow \sin y = c \cos x.$

30. (d) Put $x + y = u$

$\Rightarrow 1 + \frac{dy}{dx} = \frac{du}{dx} \Rightarrow \frac{dy}{dx} = \frac{du}{dx} - 1$

$\therefore u^2\left(\frac{du}{dx} - 1\right) = a^2$

$\Rightarrow \frac{du}{dx} = \frac{a^2}{u^2} + 1 = \frac{a^2 + u^2}{u^2} \Rightarrow \frac{u^2}{a^2 + u^2} du = dx$

$\Rightarrow \left(1 - \frac{a^2}{a^2 + u^2}\right) du = dx \Rightarrow u - a \tan^{-1} \frac{u}{a} =$

$= x + c$

$\Rightarrow \quad y = a \tan^{-1}\left(\frac{x+y}{a}\right) + c.$

31. (a) Given $\frac{dy}{dx} = \frac{x-y}{x+y}$. Put $y = ux \Rightarrow \frac{dy}{dx}$

$= u + x\frac{du}{dx} \quad \therefore \quad u + x\frac{du}{dx} = \frac{x - ux}{x + y}$

$\Rightarrow u + x\frac{du}{dx} = \frac{1-u}{1+u} \Rightarrow \frac{1+u}{2-(1+u)^2} du = \frac{dx}{x}$

Integrating both sides, $\int \frac{1+u}{2-(1+u)^2} du = \int \frac{dx}{x}$

Put $(1+u)^2 = t \Rightarrow 2(1+u)du = dt$

$\Rightarrow \frac{1}{2}\int \frac{dt}{2-t} = \int \frac{dx}{x} \Rightarrow -\frac{1}{2}\log(2-t) = \log xc$

$\Rightarrow -\frac{1}{2}\log[2-(1+u)^2] = \log xc$

$\Rightarrow -\frac{1}{2}\log[-u^2 - 2u + 1] = \log xc$

$\Rightarrow \log \frac{1}{\sqrt{1-2u-u^2}} = \log xc$

$\Rightarrow x^2 c^2 (1 - 2u - u^2) = 1$

$\Rightarrow y^2 + 2xy \quad x^2 = c_1.$

32. (d) $\frac{dy}{dx} - y \tan x = \sec x$ is linear differential equation,

Here I.F. $= e^{\int -\tan x dx} = e^{-\log \sec x} = (\sec x)^{-1}$

33. (c) $\frac{dy}{dx} - \frac{y}{x} = x$; I.F. $= e^{\int -\frac{1}{x} dx} = \frac{1}{x}$

$\therefore$ Solution is $y.\frac{1}{x} = \int x.\frac{1}{x} dx \Rightarrow \frac{y}{x} = x + a$

$\Rightarrow y \quad x^2 + ax.$

QUESTION BANK: SOLVE THESE TO MASTER

1. Solution of the differential equation $\frac{dy}{dx} + \frac{3x + 2y - 5}{2x + 3y - 5} = 0$ are
   (a) $3x^2 + 4xy + 3y^2 - 10x - 10y = c$
   (b) $x^2 + 4xy - y^2 - 4x + 6y = c$
   (c) $(x + 2y)^2 + 3y = c$
   (d) None of these

2. The solution of $dy/dx = \sqrt{y - x}$ is
   (a) $x = 2\left(\sqrt{y-x} - \ln\left|\sqrt{y-x} - 1\right|\right) + c$
   (b) $y = 2\left(\sqrt{y-x} + \ln\left|\sqrt{y-x} - 1\right|\right) + c$
   (c) $y = 2\left(\sqrt{y-x} - \ln\left|\sqrt{y-x} - 1\right|\right) + c$
   (d) $x = 2\left(\sqrt{y-x} + \ln\left|\sqrt{y-x} - 1\right|\right) + c$

3. The equation of the curve through the point (1, 1) and whose slope is $\frac{2ay}{x(y-a)}$, is

(a) $y^a \cdot x^{2a} = e^{x-1}$ (b) $y^a \cdot x^{2a} = e^x$
(c) $y^{2a} \cdot x^a = e^{x-1}$ (d) none of these

4. Solution of the differential equation

$(x + 2y^3)\frac{dy}{dx} = y$ is

(a) $x = y^2(c + y^2)$ (b) $x = y(c - y^2)$
(c) $x = 2y(c - y^2)$ (d) $x = y(c + y^2)$

5. The order of the differential equation whose general solution is given by $y = c_1 \cos(2x + c_2) - (c_3 + c_4)\, a^{x+c_5} + c_6 \sin(x - c_7)$ is

(a) 3 (b) 4
(c) 5 (d) 2

6. The solution of $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ satisfying $y(1) = 1$ is given by

(a) A system of parabolas
(b) A system of circles
(c) $y^2 = x(1 + x) - 1$
(d) $(x - 2)^2 + (y - 3)^2 = 5$

7. The degree of the differential equation satisfying the relation $\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda\left(x\sqrt{1+y^2} - y\sqrt{1+x^2}\right)$ is

(a) 1 (b) 2
(c) 3 (d) None of these

8. If $y = y(x)$ and $\frac{2 + \sin x}{y+1}\left(\frac{dy}{dx}\right) = -\cos x, y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ equals

(a) 1/3 (b) 2/3
(c) –1/3 (d) 1

9. For the primitive integral equation $ydx + y^2dy = xdy$; $x \in R$, $y > 0$, $y = (x)$, $y(1) = 1$, then $y(-3)$ is

(a) 3 (b) 2
(c) 1 (d) 5

10. Solution of differential equation

$\frac{dy}{dx} = \sqrt{1 - x^2 - y^2 + x^2y^2}$ is

(a) $\sin^{-1} y = \sin^{-1} x + c$

(b) $\sin^{-1} y = \frac{1}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x + c$

(c) $\sin^{-1} y = \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x + c$

(d) $\sin^{-1} y = \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{4}\cos^{-1} x + c$

11. The differential equation of family of ellipse with foci. Let the design and major axis at y axis, then equation of I^r is

(a) $\frac{d^2y}{dx^2} + \frac{dy}{dx}\left(x\frac{dy}{dx} - y\right) = 0$

(b) $xy\frac{d^2y}{dx^2} - \frac{dy}{dx}\left(x\frac{dy}{dx} - y\right) = 0$

(c) $xy\frac{d^2y}{dx^2} + \frac{dy}{dx}\left(x\frac{dy}{dx} - y\right) = 0$

(d) $\frac{d^2y}{dx^2} - \frac{dy}{dx}\left(x\frac{dy}{dx} - y\right) = 0$

12. The solution of the equation $x^2y^2\, dy = (1 - xy^3)\, dx$ is

(a) $2x^3y^3 = 3x^2 + c$ (b) $x^3y^3 = x^2 + c$
(c) $x^3y^3 = x^2 + x + c$ (d) $x^3y^3 = 3x^2 + c$

13. General solution of differential equation $x \cdot \cos\frac{y}{x}(ydx + xdy) = y \cdot \sin\frac{y}{x}(xdy - y\,dx)$ is:

(a) $xy = c \cdot \sec(x/y)$ (b) $y = x \cdot c \cdot \sec(x/y)$
(c) $xy = c \cdot \sec(y/x)$ (d) None of these

14. Order and degree of differential equation

$\left(\frac{d^2y}{dx^2}\right)^5 + 4\frac{\left(\frac{d^2y}{dx^2}\right)^3}{\left(\frac{d^3y}{dx^3}\right)} + \frac{d^3y}{dx^3} = x^2 - 1$ are m and n respectively, then

(a) $m = 3$ and $n = 5$ (b) $m = 3$ and $n = 1$
(c) $m = 3$ and $n = 3$ (d) $m = 3$ and $n = 2$

15. What is the solution of differential equation $\frac{dy}{dx} = \frac{y}{(x + 2y^3)}$ is

(a) $y(1 - xy) = cx$ (b) $y^3 - x = cy$
(c) $x(1 - xy) = cy$ (d) $x(1 + xy) = cy$

16. Solution of differential equation $x\frac{dy}{dx} = y(\log y - \log x + 1)$ is

(a) $y = xe^{cx}$ (b) $y + xe^{cx} = 0$
(c) $y + e^x = 0$ (d) None of these

17. The solution of differential equation (x cosec $(y/x) - y$) $dx + xdy = 0$ is
(a) $\log|x| - \cos(x/y) = c$
(b) $\log|x| - \cos(y/x) = c$
(c) $\log|x| - \sin(x/y) = c$
(d) $\log|x| - \sin(y/x) = c$

18. Curve $y = e^x (A\cos x + B\sin x)$ respresents the differential equation $\frac{d^2y}{dx^2} + l\frac{dy}{dx} + my = 0$, then $l + m =$
(a) 1 (b) 0
(c) -1 (d) 2

19. The solution of differential equation $\sqrt{4+x}\frac{dy}{dx} + xy = 0$ is
(a) $y = Ae^{\frac{2}{3}(8-x)\sqrt{x+4}}$
(b) $y = Ae^{-\frac{2}{3}(4-x)\sqrt{x+4}}$
(c) $y = Ae^{\frac{2}{3}(8+x)\sqrt{x+4}}$
(d) $y = Ae^{-\frac{2}{3}(8-x)\sqrt{x+4}}$

20. The solution of differential equation $\frac{dy}{dx} + py = e^{qx}$ is $(\lambda)y = e^{qx} + c\lambda e^{px}$, then $\lambda =$
(a) $p - q$ (b) $2p + q$
(c) $p + q$ (d) $p + 2q$

21. Integrating factor of $\sin x \frac{dy}{dx} + y\cos x = 1$
(a) $\cos x$ (b) $\tan x$
(c) $\sec x$ (d) $\sin x$

22. The order of the differential equation, of which $xy = ce^x + be^{-x} + x^2$ is a solution, is
(a) 1 (b) 2
(c) 3 (d) None of these

23. The differential equation of family of parabolas with foci at the origin and axis along the x-axis is
(a) $y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} - y = 0$
(b) $x\left(\frac{dy}{dx}\right)^2 + 2y\frac{dy}{dx} - y = 0$
(c) $y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} + y = 0$
(d) None of these

24. The solution of $\frac{dy}{dx} = \frac{x^2+y^2+1}{2xy}$, satisfying $y(1) = 0$, is given by
(a) a hyperbola
(b) a circle
(c) $y^2 = x(1+x) - 10$
(d) $(x-2)^2 + (y-3)^2 = 5$

25. The order and degree of the differential equation $\left(1 + 3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are
(a) $\left(1, \frac{2}{3}\right)$ (b) (3, 1)
(c) (3, 3) (d) (1, 2)

26. The particular solution of $\cos y dx + (1 + 2e^x)\sin y dy = 0$, when $x = 0$, $y = \frac{\pi}{4}$ is
(a) $e^x - 2 = 3\sqrt{2}\cos y$
(b) $e^x + 2 = \sqrt{2}\cos y$
(c) $e^x + 2 = 3\sqrt{2}\cos y$
(d) None of these

27. The solution of the equation $\left(xy^2 - e.\frac{1}{x^3}\right)dx - x^2ydy = 0$, given $y = 0$, when $x = 1$ is
(a) $y^2 = \frac{e}{2}\left(\frac{1}{x^4} - x^2\right)$ (b) $y^2 = \frac{e}{3}\left(\frac{1}{x^4} - x^2\right)$
(c) $y = \frac{e}{3}\left(\frac{1}{x^4} + x^2\right)$ (d) None of these

28. The solution of the differential equation $\frac{dy}{dx} = \frac{y(2y-x)}{x(2y+x)}$, which satisfies $y = 1$ at $x = 1$, is
(a) $x\log(xy) = 2(x - y)$
(b) $y\log(xy) = 2(y - x)$
(c) $x\log(xy) = 2(x + y)$
(d) None of these

29. The differential equation of all non-vertical lines in a palne is
(a) $\frac{d^2y}{dx^2} = 0$ (b) $\frac{d^2x}{dy^2} = 0$
(c) $\frac{dy}{dx} = 0$ (d) $\frac{dx}{dy} = 0$

30. Solution of the equation $x\left(\frac{dy}{dx}\right)^2+(y-x)\frac{dy}{dx}-y=0$ is
(a) $(x-y+c)(xy-c)=0$
(b) $(x+y+c)(xy-c)=0$
(c) $(x-y+c)(2xy-c)=0$
(d) $(y-x+c)(xy-c)=0$

31. A solution of the differential equation $\left(\frac{dy}{dx}\right)^2-x\frac{dy}{dx}+y=0$ is
(a) $y=2$
(b) $y=2x$
(c) $y=2x-4$
(d) $y=2x^2-4$

32. The degree of the differential equation $\frac{d^3y}{dx^3}+x\left(\frac{dy}{dx}\right)^4=4\log\left(\frac{d^4y}{dx^4}\right)$ is
(a) 1 (b) 3
(c) 4 (d) None of these

33. A population grows at the rate of 5% per year. Then the population will be doubled in
(a) 10 log2 years (b) 20 log2 years
(c) 30 log2 years (d) None of these

ANSWERS

LECTURE 1

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (a)	8. (a)	15. (c)	22. (c)
2. (d)	9. (d)	16. (a)	23. (b)
3. (d)	10. (a)	17. (c)	
4. (c)	11. (a)	18. (d)	
5. (c)	12. (b)	19. (b)	
6. (c)	13. (a)	20. (a)	
7. (a)	14. (b)	21. (b)	

Worksheet: To Check the Preparation Level

1. (c)	5. (c)	9. (a)	13. (c)
2. (a)	6. (d)	10. (c)	14. (b)
3. (c)	7. (d)	11. (a)	15. (c)
4. (a)	8. (c)	12. (d)	16. (a)

LECTURE 2

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (c)	5. (a)	9. (a)	13. (d)
2. (a)	6. (b)	10. (c)	14. (a)
3. (a)	7. (b)	11. (c)	
4. (c)	8. (d)	12. (b)	

Worksheet: To Check the Preparation Level

1. (c)	5. (c)	9. (c)	13. (b)
2. (c)	6. (a)	10. (a)	14. (a)
3. (c)	7. (b)	11. (a)	15. (b)
4. (d)	8. (a)	12. (d)	16. (a)

LECTURE 3

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (c)	4. (a)	7. (b)	10. (b)
2. (b)	5. (a)	8. (d)	11. (c)
3. (c)	6. (c)	9. (b)	12. (c)

Worksheet: To Check the Preparation Level

1. (b)	5. (a)	9. (a)	13. (b)
2. (a)	6. (b)	10. (c)	14. (c)
3. (d)	7. (b)	11. (d)	15. (c)
4. (d)	8. (a)	12. (c)	16. (b)

LECTURE 4

Unsolved Objective Problems (Identical Problems for Practice): For Improving Speed with Accuracy

1. (a)	**4.** (c)	**7.** (b)	**10.** (a)
2. (a)	**5.** (c)	**8.** (d)	
3. (b)	**6.** (d)	**9.** (d)	

Worksheet: To Check the Preparation Level

1. (c)	**5.** (a)	**9.** (d)	**13.** (b)
2. (d)	**6.** (b)	**10.** (b)	**14.** (a)
3. (a)	**7.** (a)	**11.** (b)	**15.** (d)
4. (d)	**8.** (d)	**12.** (c)	

LECTURE 5

Mental Preparation Test

1. $y = \tan x + \sec x + c$

2. $\log (1 + y) = x + \frac{x^2}{2} + c$

3. $4xy = x^4 + 4c$

4. $y = e^{\sin^2 x}$

5. $c (x^2 - y^2) = x$

6. $y = ce^{-x} + \frac{1}{2} (\sin x - \cos x)$

8. $\log x + x + \log y - \frac{y^2}{2} = c$

9. $y = \frac{x^4}{12} - \frac{1}{8}\sin 3x + \frac{4x}{3} + 1$

10. $y \cos x = \frac{1}{2}\cos 2x + c$

11. $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$

12. $y(x^2 + 1)^2 = -\left[\frac{x\sqrt{x^2+1}}{2} + \frac{1}{2}\log\left|x + \sqrt{x^2+1}\right|\right] + c$

13. $y = \tan x + x^3 + c$

14. $y = c \log_e x$

16. $ye^{4x} = \frac{-1}{3}e^{3x} + c$

17. $e^x + e^y + \frac{x^3}{3} + c = 0$

18. $y = \frac{x^3}{6} + \cos x$

19. $y = \tan x - 1 + ce^{-\tan x}$

20. 10 log 2 years

21. $6y - 3x - \log (3x + 3y + 2) = A$

22. $y = x^2 + c \operatorname{cosec} x$

23. $\frac{d^2y}{dx^2} + y = 0$

24. $y = ce^{-x} + \frac{1}{2}(\cos x + \sin x)$

25. $\tan\left(\frac{y}{2x}\right) = cx$

27. $y = e^x - \cos x + 1$

28. $y = x (x^2 + 1) + (x^2 + 1) \tan^{-1} x + c (x^2 + 1)$

29. $\frac{y}{x} - \log_e y = c$

30. $2ye^{\tan^{-1} x} = e^{2\tan^{-1} x} + c$

31. $y = \tan\frac{x}{2} + ce^{-x}$

32. $\log\left|1 + \tan\left(\frac{x+y}{2}\right)\right| = x + c$

33. $c (x - y)^2 = |x| e^{y/x}$

34. $\sin (y/x) = cx$

35. $y (\sec x + \tan x) = \sec x + \tan x - x + c$

36. $\frac{1}{a+m}e^{mx} + ce^{ax}$

37. $2y = x (x + y)$

Question Bank

1. (d)	**10.** (c)	**19.** (a)	**28.** (a)
2. (d)	**11.** (c)	**20.** (c)	**29.** (a)
3. (a)	**12.** (a)	**21.** (d)	**30.** (a)
4. (d)	**13.** (c)	**22.** (b)	**31.** (c)
5. (c)	**14.** (d)	**23.** (a)	**32.** (d)
6. (c)	**15.** (b)	**24.** (a)	**33.** (b)
7. (a)	**16.** (a)	**25.** (c)	
8. (a)	**17.** (b)	**26.** (c)	
9. (a)	**18.** (b)	**27.** (b)	